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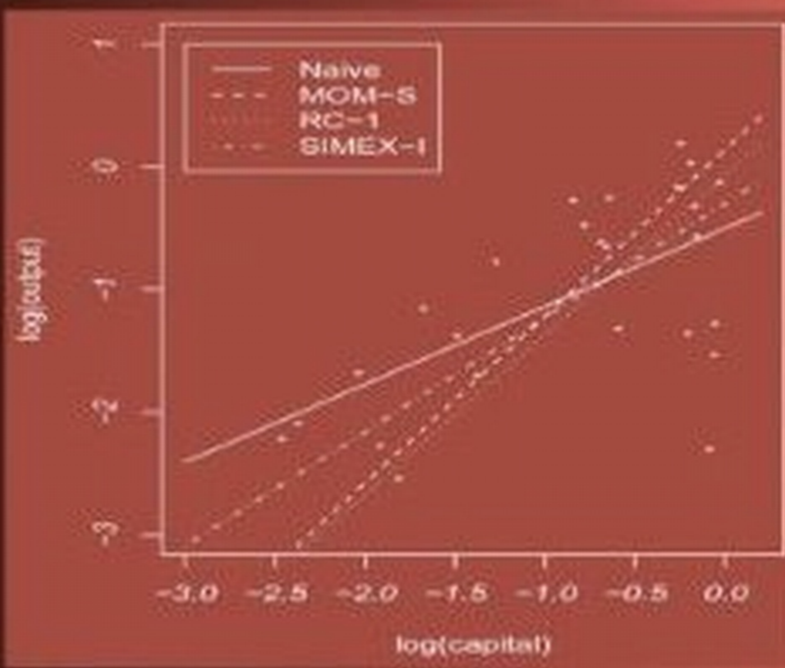

John P. Buonaccorsi

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Taylor & Francis Group
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CRC Press
Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business
A CHAPMAN & HALL BOOK

Chapman & Hall/CRC
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

Printed in the United States of America on acid-free paper
10 9 8 7 6 5 4 3 2 1

International Standard Book Number: 978-1-4200-6656-2 (Hardback)

Library of Congress Cataloging-in-Publication Data

Buonaccorsi, John P.
Measurement error : models, methods, and applications / John P. Buonaccorsi.
p. cm. -- (Chapman & Hall/CRC interdisciplinary statistics series)
Includes bibliographical references and index.
ISBN 978-1-4200-6656-2 (hardcover : alk. paper)
1. Error analysis (Mathematics) I. Title. II. Series.

QA275.B965 2010
511'.43--dc22 2009048849

Visit the Taylor & Francis Web site at
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DEDICATION

In memory of my parents, Eugene and Jeanne, and to my wife, Elaine, and my children, Jessie and Gene.

Contents

Preface **xix**

List of Examples **xxv**

1 Introduction **1**

1.1 What is measurement error? 1

1.2 Some examples 1

1.3 The main ingredients 4

1.4 Some terminology 5

1.4.1 Measurement versus Berkson error models 6

1.4.2 Measurement error models for quantitative values 7

1.4.3 Nondifferential/differential measurement error, conditional independence and surrogacy 7

1.5 A look ahead 8

2 Misclassification in Estimating a Proportion **11**

2.1 Motivating examples 11

2.2 A model for the true values 14

2.3 Misclassification models and naive analyses 14

2.4 Correcting for misclassification 17

2.4.1 Ignoring uncertainty in the misclassification rates 17

2.4.2 Using external validation data and misclassification rates 19

2.4.3 Internal validation data and the use of reclassification rates 22
2.5 Finite populations 25
2.6 Multiple measures with no direct validation 27
2.7 The multinomial case 28
2.8 Mathematical developments 30

3 Misclassification in Two-Way Tables 33
3.1 Introduction 33
3.2 Models for true values 35
3.3 Misclassification models and naive estimators 38
3.4 Behavior of naive analyses 40
3.4.1 Misclassification in X only 40
3.4.2 Misclassification in Y only 46
3.4.3 Misclassification in X and Y both 47
3.5 Correcting using external validation data 48
3.5.1 Misclassification in X only 49
3.5.2 Misclassification in Y only 55
3.5.3 Misclassification in X and Y both 57
3.6 Correcting using internal validation data 58
3.6.1 Misclassification in X only 60
3.6.2 Misclassification in Y only 65
3.6.3 Misclassification in X and Y both 65
3.7 General two-way tables 66
3.8 Mathematical developments 68
3.8.1 Some expected values 68
3.8.2 Estimation using internal validation data 69
3.8.3 Results for covariance matrices 69

4 Simple Linear Regression **73**

4.1 Introduction 73

4.2 The additive Berkson model and consequences 76

4.3 The additive measurement error model 77

4.4 The behavior of naive analyses 79

4.5 Correcting for additive measurement error 83

4.5.1 Moment-based corrected estimators 84

4.5.2 Inferences for regression coefficients 86

4.5.3 Replication 89

4.6 Examples 90

4.6.1 Nitrogen-yield example 90

4.6.2 Defoliation example with error in both variables 93

4.7 Residual analysis 94

4.8 Prediction 96

4.9 Mathematical developments 102

5 Multiple Linear Regression **105**

5.1 Introduction 105

5.2 Model for true values 106

5.3 Models and bias in naive estimators 107

5.4 Correcting for measurement error 114

5.4.1 Moment corrected estimators 115

5.4.2 Sampling properties and approximate inferences 116

5.4.3 Replication 118

5.4.4 Correction for negative estimates 121

5.4.5 Residual analysis and prediction 122

5.5 Weighted and other estimators 122

5.6 Examples 124

5.6.1 Defoliation example revisited 124

5.6.2 LA data with error in one variable 126

5.6.3 House price example 129

5.7 Instrumental variables 130

5.7.1 Example 135

5.8 Mathematical developments 136

5.8.1 Motivation for moment corrections 136

5.8.2 Defining terms for general combinations of predictors 138

5.8.3 Approximate covariance of estimated coefficients 139

5.8.4 Instrumental variables 141

6 Measurement Error in Regression: A General Overview 143

6.1 Introduction 143

6.2 Models for true values 144

6.3 Analyses without measurement error 148

6.4 Measurement error models 149

6.4.1 General concepts and notation 149

6.4.2 Linear and additive measurement error models 151

6.4.3 The linear Berkson model 152

6.4.4 Nonlinear measurement error models 154

6.4.5 Heteroscedastic measurement error 154

6.4.6 Multiplicative measurement error 158

6.4.7 Working with logs 159

6.4.8 Misclassification from categorizing a quantitative variable 160

6.5 Extra data 163

6.5.1 Replicate values 163

6.5.2 External replicates: Are reliability ratios exportable? 168

6.5.3 Internal validation data 169

6.5.4 External validation data 171

6.5.5 Other types of data 172

6.6 Assessing bias in naive estimators 173

6.7 Assessing bias using induced models 174
6.7.1 Linear regression with linear Berkson error 175
6.7.2 Second order models with linear Berkson error 176
6.7.3 Exponential models with normal linear Berkson error 177
6.7.4 Approximate induced regression models 178
6.7.5 Generalized linear models 178
6.7.6 Binary regression 179
6.7.7 Linear regression with misclassification of a binary predictor 182
6.8 Assessing bias via estimating equations 186
6.9 Moment-based and direct bias corrections 189
6.9.1 Linearly transforming the naive estimates 190
6.10 Regression calibration and quasi-likelihood methods 191
6.11 Simulation extrapolation (SIMEX) 194
6.12 Correcting using likelihood methods 196
6.12.1 Likelihoods from the main data 198
6.12.2 Likelihood methods with validation data 200
6.12.3 Likelihood methods with replicate data 203
6.13 Modified estimating equation approaches 204
6.13.1 Introduction 204
6.13.2 Basic method and fitting algorithm 206
6.13.3 Further details 208
6.14 Correcting for misclassification 209
6.15 Overview on use of validation data 211
6.15.1 Using external validation data 211
6.15.2 Using internal validation data 213
6.16 Bootstrapping 215
6.16.1 Additive error 216
6.16.2 Bootstrapping with validation data 218
6.17 Mathematical developments 219

6.17.1 Justifying the MEE fitting method 219
6.17.2 The approximate covariance for linearly transformed coefficients 220
6.17.3 The approximate covariance of pseudo-estimators 221
6.17.4 Asymptotics for ML and pseudo-ML estimators with external validation. 222

7 Binary Regression 223
7.1 Introduction 223
7.2 Additive measurement error 224
7.2.1 Methods 224
7.2.2 Example: Cholesterol and heart disease 231
7.2.3 Example: Heart disease with multiple predictors 241
7.2.4 Notes on ecological applications 243
7.2.5 Fitting with logs 243
7.3 Using validation data 247
7.3.1 Two examples using external validation and the Berkson model 248
7.3.2 Fitting with internal validation data and the Berkson model 250
7.3.3 Using external validation data and the measurement error model 252
7.4 Misclassification of predictors 256

8 Linear Models with Nonadditive Error 259
8.1 Introduction 259
8.2 Quadratic regression 260
8.2.1 Biases in naive estimators 261
8.2.2 Correcting for measurement error 265
8.2.3 Paper example 267
8.2.4 Additive error in the response 272
8.2.5 Quadratic models with additional predictors 272

8.3 First order models with interaction 275
8.3.1 Bias in naive estimators 277
8.3.2 Correcting for measurement error 279
8.3.3 Example 281
8.3.4 More general interaction models 284
8.4 General nonlinear functions of the predictors 286
8.4.1 Bias of naive estimators 287
8.4.2 Correcting for measurement error 288
8.4.3 Linear regression in log(x) 291
8.5 Linear measurement error with validation data 298
8.5.1 Models and bias in naive estimators 298
8.5.2 Correcting with external validation data 301
8.5.3 External validation example 303
8.5.4 Correcting with internal validation 304
8.5.5 Internal validation example 306
8.6 Misclassification of a categorical predictor 309
8.6.1 Introduction and bias of naive estimator 309
8.6.2 Correcting for misclassification 312
8.6.3 Further details 314
8.7 Miscellaneous 315
8.7.1 Bias expressions for naive estimators 315
8.7.2 Likelihood methods in linear models 317

9 Nonlinear Regression 319
9.1 Poisson regression: Cigarettes and cancer rates 319
9.2 General nonlinear models 322

10 Error in the Response **325**

10.1 Introduction 325

10.2 Additive error in a single sample 325

10.2.1 Estimating the mean and variance 327

10.2.2 Estimating the mean-variance relationship 329

10.2.3 Nonparametric estimation of the distribution 333

10.2.4 Example 338

10.3 Linear measurement error in the one-way setting 341

10.3.1 One measuring method per group 345

10.3.2 General designs 349

10.4 Measurement error in the response in linear models 350

10.4.1 Models 351

10.4.2 Correcting for measurement error 353

10.4.3 Example 356

10.4.4 Further detail 358

11 Mixed/Longitudinal Models **361**

11.1 Introduction, overview and some examples 361

11.2 Berkson error in designed repeated measures 366

11.2.1 Bias in naive estimators 370

11.2.2 Correcting for measurement error 375

11.3 Additive error in the linear mixed model 377

11.3.1 Naive estimators and induced models 377

11.3.2 Correcting for measurement error with no additional data 378

11.3.3 Correcting for measurement error with additional data 382

12 Time Series **385**

12.1 Introduction 385

12.2 Random walk/population viability models 387

12.2.1 Properties of naive analyses 389

12.2.2 Correcting for measurement error 389

12.2.3 Example 395

12.3 Linear autoregressive models 398

12.3.1 Properties of naive estimators 399

12.3.2 Correcting for measurement error 401

12.3.3 Examples 405

13 Background Material **409**

13.1 Notation for vectors, covariance matrices, etc. 409

13.2 Double expectations 409

13.3 Approximate Wald inferences 410

13.4 The delta-method: Approximate moments of nonlinear functions 410

13.5 Fieller's method for ratios 411

References **413**

Author index **429**

Subject index **435**

Preface

This book is about measurement error, which includes misclassification. This occurs when some variables in a statistical model of interest cannot be observed exactly, usually due to instrument or sampling error. It describes the impacts of these errors on "naive" analyses that ignore them and presents ways to correct for them across a variety of statistical models, ranging from the simple (one-sample problems) to the more complex (some mixed and time series models) with a heavy focus on regression models in between.

The consequences of ignoring measurement error, many of which have been known for some time, can range from the nonexistent to the rather dramatic. Throughout the book attention is given to the effects of measurement error on analyses that ignore it. This is mainly because the majority of researchers do not account for measurement error, even if they are aware of its presence and potential impact. In part, this is due to a historic lack of software, but also because the information or extra data needed to correct for measurement error may not be available. Results on the bias of naive estimators often provide the added bonus of suggesting a correction method.

The more dominant thread of the book is a description of methods to correct for measurement error. Some of these methods have been in use for a while, while others are fairly new. Both misclassification and the so-called "errors-in-variables" problem in regression have a fairly long history, spanning statistical and other (e.g., econometrics and epidemiology) literature. Over the last 20 years comprehensive strategies for treating measurement error in more complex models and accounting for the use of extra data to estimate measurement error parameters have emerged. This book provides an overview of some of the main techniques and illustrates their application across a variety of models. Correction methods based on the use of known measurement error parameters, replication, internal or external validation data, or (in the case of linear models) instrumental variables are described. The emphasis in the book is on the use of some relatively simple methods: moment corrections, regression calibration, SIMEX, and modified estimating equation methods. Likelihood techniques are described in general, but only implemented in some of the examples.

There are a number of excellent books on measurement error, including

Fuller's seminal 1987 book, mainly focused on linear regression models, and the second edition of Carroll et al. (2006). The latter provides a comprehensive treatment of many topics, some of which overlap with the coverage in this book. Other, more specialized books include Cheng and Van Ness (1999) and Gustafson (2004). The latter provides Bayesian approaches for dealing with measurement error and misclassification in numerous settings. This book (i.e., the one in your hands), which is all non-Bayesian, differs from these others both in the choice of topics and in being more applied. The goal is to provide descriptions of the basic models and methods, and associated terminology, and illustrate their use. The hope is that this will be a book that is accessible to a broad audience, including applied statisticians and researchers from other disciplines with prior exposure to statistical techniques, including regression analysis. There are a few places (e.g., the last two chapters and parts of Chapters 5 and 6) where the presentation is a bit more advanced out of necessity.

Except for Chapter 6, the book is structured around the model for the true values. In brief, the strategy is to first discuss some simpler problems including misclassification in estimating a proportion and in two-way tables, as well as additive measurement error in simple and multiple regression. We then move to a broad treatment of measurement error in Chapter 6, and return to specific models for true values in the later chapters. A more detailed overview of the chapters and the organization of the book is given in Section 1.5. The layout is designed to allow a reader to easily locate a specific problem of interest. A limited number of mathematical developments, based on relatively basic theory, are isolated into separate sections for interested readers. However, I have made no attempt to duplicate the excellent theoretical coverage of certain topics in the aforementioned texts.

The measurement error literature has been growing at a tremendous rate recently. I have read a large number of other papers that have influenced my thinking but do not make their way explicitly into this book. I have tried to be careful to provide a direct reference whenever I use a result from another source. Still, given the more applied focus of the book, the desire to limit the coverage in certain areas, and size restrictions, means that many important papers are left unrecognized. I apologize beforehand to anyone who may feel slighted in any way by my omission of their work. In addition, while some very recent work is referenced in certain places, since parts of the book were completed up to two years before publication date, I have not tried to incorporate recent developments associated with certain topics. I also recognize that in the later stages of the book the coverage is based heavily on my own work (with colleagues). This is especially true in the last two chapters of the book. This is largely a matter of writing about what I know best, but it also happens to coincide with coverage of some of the more basic settings within some broader topics. When I set out, I had planned to provide more discussion about the connections between measurement error models and structural equation and latent variable models, as well as related issues with missing data and omitted covariates. There are a few side comments on this, but a fuller discussion was omitted for reasons of both time and space.

In a way this book began in the spring of 1989, when I was on my first sabbatical. In the years right before that I had wandered into the "errors-in-variables", or measurement error, world through an interest in calibration and ratios. Being a bit frustrated by the disconnect between much of the traditional statistics literature and that from other areas, especially econometrics, I began to build a systematic review and a bibliography. Fortuitously, around this time Jim Ware invited me to attend a conference held at NIH in the fall of 1989 on measurement error problems in epidemiology. I also had the opportunity to attend the IMS conference on measurement error at Humboldt State University the following summer and listen to many of the leading experts in the field. This period marked the beginning of an explosion of new work on measurement error that continues to this day. I am extremely grateful to Jim for the invitation to the NIH conference, getting me involved with the measurement error working group at Harvard and supporting some of my early work in this area through an EPA-SIMS (Environmental Protection Agency and the Societal Institute of the Mathematical Sciences) grant. I benefited greatly from interactions with some of the core members of the Harvard working group including Ernst Linder, Bernie Rosner, Donna Spiegelman and Tor Tosteson. This led to an extensive and fruitful collaboration with Tor and Eugene Demidenko (and many pleasant rides up the Connecticut River Valley to Dartmouth to work with them, and sometimes Jeff Buzas from the University of Vermont).

By the early 90s I had a preliminary set of notes and programs, portions of which were used over a number of years as part of a topics in regression course at the University of Massachusetts. The broader set became the basis for a short course taught for the Nordic Region of the Biometrics Society in Ås, Norway in 1998. These were updated and expanded for a short course taught in 2005 at the University of Oslo Medical School, and then a semester long course at the University of Massachusetts in the fall of 2007. I appreciate the feedback from the students in those various courses. The material from those courses made up the core of about half of the book. I am extremely grateful to Petter Laake for helping to arrange support for my many visits to the Medical School at the University of Olso. I have enjoyed my collaborations with him, Magne Thoresen, Marit Veirod and Ingvild Dalen and greatly appreciate the hospitality shown by them and the others in the biostatistics department there.

I'm also grateful to Ray Carroll for sharing some of the Framingham Heart Study data with me, Walt Willet for generously allowing me access to the the external validation data from the Nurses Health Study, Rob Greenberg for use of the beta-carotene data, Sandy Liebhold for the egg mass defoliation data, Joe

Elkinton for the mice density data, Paul Godfrey and the Acid Rain monitoring project for the pH and alkalinity data, and Matthew DiFranco for the urinary neopterin/HIV data.

A number of other people contributed to this book, directly or indirectly. Besides sharing some data, Ray Carroll has always been willing over the years to answer questions and he also looked over a portion of Chapter 10. My measurement error education owes a lot to reading the works of, listening to many talks by, and having conversations with Ray, Len Stefanski, and David Ruppert. After years of a lot of traveling to collaborate with others, I certainly appreciate having John Staudenmayer, and his original thinking, here at the University of Massachusetts. Our joint work forms the basis of much of Chapter 12. I can't thank Meng-Shiou Shieh enough for her reading of the manuscript, double checking many of the calculations, picking up notational inconsistencies and helping me clean up the bibliography. Magne Thoresen generously provided comments on the first draft of Chapter 7 and Donna Spiegelman advised on the use of validation data. I am grateful to Graham Dunn for his reviews of Chapters 2–5 and Gene Buonaccorsi for his assistance with creating the index. Of course, none of those mentioned are in any way responsible for remaining shortcomings or errors. I want to thank Hari Iyer for being my adviser, mentor, collaborator, and friend over the years. On the lighter side, the Coffee Cake Club helped keep me sane (self report) over the years with the Sunday long runs.

Over the years I have been fortunate to receive support for some of my measurement error work from a number of sources. In addition to EPA-SIMS, mentioned earlier, this includes the National Cancer Institute, the U.S. Department of Agriculture and the Division of Mathematical Sciences at the NSF. The University of Massachusetts, in general, and the Department of Mathematics and Statistics in particular, has provided continuing support over the years, including in the computational area. I especially appreciate the assistance of the staff in the departmental RCF (research computing facility). Finally, I am extremely grateful to the Norwegian Research Council for supporting my many visits to collaborate with my colleagues in the Medical School at the University of Oslo.

I appreciate the guidance and patience of Rob Calver of Taylor & Francis throughout this project as well as the production assistance from Karen Simon and Marsha Pronin. I am grateful to John Wiley and Sons, the American Statistical Association, the Biometrics Society, Blackwell Publishing, Oxford University Press, Elsevier Publishing and Taylor & Francis for permissions to use tables and figures from earlier publications. Finally, I want to thank my wife, Elaine Puleo, for her constant support, editing Chapter 1 and serving as a sounding board for my statistical ideas throughout the process of writing of the book.

Computing

Computing was occasionally an obstacle. While there are a number of programs, described below, which handle certain models and types of measurement errors, there were many techniques that I wanted to implement in the examples for which programs were not available. While I had built up some SAS programs to handle measurement error over the years, many of the examples in the book required new, and extensive, programming. There are still examples where I would have liked to have illustrated more methods, but I needed to stop somewhere.

The software used in the book, and some associated programs, include:

- STATA routines rcal (regression calibration) and simex. These, along with the related qvf routine, are available through http://www.stata.com/merror/.
- STATA gllamm routines based on Skrondal and Rabe-Hesketh (2004). Available at http://www.gllamm.org/. This includes the cme command for parametric maximum likelihood used in a couple of places in the book, as well as some other measurement error related programs.
- SAS macro Blinplus from Spiegelman and colleagues at Harvard University. Available at http://www.hsph.harvard.edu/faculty/spiegelman/blinplus.html. This corrects for measurement error using external and internal validation data, based on a linear Berkson model.
- SAS macro from Spiegelman and colleagues at Harvard University which correct for measurement error under additive error with the use of replicate values. This is available at http://www.hsph.harvard.edu/faculty/spiegelman/relibpls8.html. This was not used in the book, since the rcal command in STATA was used instead.

The remainder of the computing was carried out using my own SAS programs. Any of these available for public use can be found at
http://www.math.umass.edu/~johnpb/meprog.html.

Although not spelled out in any real detail in the book (except for a few comments here and there), there are other programs within the standard software packages that will handle certain analyses involving measurement errors. This includes programs that treat structural equation or latent variables models, instrumental variables and mixed models.

John Buonaccorsi
Amherst, Massachusetts
November, 2009

List of Examples

- Proportions
  Estimating HIV prevalence, 11, 21
  Estimating the proportion of smokers, 12, 24
  Estimating abortion rates, 13, 18
  Estimating land use proportions, 13, 29
- Two-way tables
  Antibiotic use and sudden infant death syndrome (SIDS), 34, 51, 63
  Accident injury and seat belt use, 34, 55, 57, 66
  Marijuana use of parents and students, 44, 62
- Simple linear regression
  Soil nitrogen level and corn yield, 74, 90
  Gypsy moth egg mass density and defoliation rate, 74, 93, 95, 100
- Multiple linear regression
  Defoliation rate, egg mass density and forest stand, 124
  LA data: Cholesterol levels and age, 126
  House price data: Sale price, square footage and tax rate, 126
  Beta carotene data: Diet and serum beta-carotene and fat index, 135
- Logistic regression
  Framingham Heart Study: 231, 241, 246, 255
  Nurses Health Study: Heart disease, dietary intakes and age, 248
  Harvard Six Cities Study: Respiratory disease and pollution exposure, 250
  Framingham Heart Study: Hypertension and categorized blood pressure, 258
- Linear models with nonadditive error
  Paper strength and hardwood concentration, 267
  LA data: Cholesterol measures and weight, with interaction, 281
  Output and capital expenditure, 294
  Water pH and alkalinity, 298
  Wheat data: Water and protein content with infrared measurements, 306
  Rehabilitation rates and fitness category, 309
- Poisson regression
  Lung cancer rates and cigarette consumption, 319

- Response error
  Beta carotene data: Estimating the mean-variance relationship and distribution, 326, 338
  Water pH over multiple districts, 343,347
  Urinary neopterin for HIV positive and negative individuals, 351,356
- Mixed/Longitudinal models
  Diet intervention study, 365
  Beta carotene data: Serum beta-carotene and diet intake, 380
- Time series
  Whooping crane abundance, 395
  Mouse abundance, 390, 405
  Childhood respiratory disease study, 406

Other applications not in worked examples (in order of appearance)

Miscellaneous examples, 2

Copenhagen Male Study, 45

Lichen and caribou habitat, 28

Verbal autopsy, 28

Nitrogen intake and balance, 75, 367

Economic consumption, 145

Water quality (National Eutrophication Survey), 145

Framingham Heart Study, 160

Dietary intake, 110, 172,

Body mass index, 160

Harvard Six Cities Study, 171

Physical activity, 172

Birthweight, smoking and weight, 182, 184

Bird breeding and habitat variables, 243

Knapweed establishment and environmental factors, 243

Israeli Glucose Intolerance, Obesity and Hypertension (GOH) Study, 275

Seychelle's Study 275

Cobb-Douglas production model 286

Consumer expenditure survey, 286

Windmill power, 286

Medical indices, labor statistics and retail sales, 385,

Vole populations, 403

CHAPTER 1

Introduction

1.1 What is measurement error?

This is a book about measurement error in statistical analyses; what it is, how to model it, what the effects of ignoring it are and how to correct for it. In some sense, all statistical problems involve measurement error. For the purposes here, measurement error occurs whenever we cannot exactly observe one or more of the variables that enter into a model of interest. There are many reasons such errors occur, the most common ones being instrument error and sampling error. In this chapter we provide a collection of examples of potentially mismeasured variables, followed by a brief overview of the general structure and objectives in a measurement error problem, along with some basic terminology that appears throughout the book. Finally, we provide a road map for the rest of the book.

1.2 Some examples

Measurement error occurs in nearly every discipline. A number of examples are given here to illustrate the variety of contexts where measurement error can be a concern. Some of these examples appear later in the book. Any of the variables described below could play the role of either an outcome or a predictor, but there is no need for notational distinctions at this point. Where any notation is used here, the true value is denoted x and the variable observed in place of x, by w. The latter can go by many names, including the observed or measured value, the error-prone measurement, a proxy variable or a surrogate measure. The term surrogate will carry a more precise meaning later. When the true and observed values are both categorical then measurement error is more specifically referred to as **misclassification**. Our first set of examples fall under this heading.

Misclassification examples

- Disease status. In epidemiology, the outcome variable is often presence or absence of a disease, such as AIDS, breast cancer, hepatitis, etc. This is often assessed through an imperfect diagnostic procedure, such as a blood test or an imaging technique, which can lead to either false positives or false negatives.
- Exposure. The other primary variable of interest in epidemiologic studies is "exposure," used in a broad sense. This is typically measured with self report or in some other manner which is subject to error. Potentially misclassified categorical values of this type include antibiotic use during pregnancy or not, whether an individual is a heavy smoker or not, a categorized measure of physical activity, dietary intake, tanning activity level (used in skin cancer studies), drug use, adherence to prescribed medication, etc.
- Acceptance Sampling. In quality control, a product can be classified by whether it meets a certain standard or tolerance. Obtaining an error-free determination can be an expensive proposition and cheaper/faster tests, subject to inspection errors, may be used instead.
- Auditing can involve assessing whether a transaction/claim is fraudulent or contains a mistake of some type. The item of interest may be complex and misclassification can occur from making a decision on partial information (e.g., sampling) or simply from errors of judgment in the evaluation.
- Satellite images are now routinely used for categorizing land usage or habitat type. These categorizations are often prone to some errors, but with some validation data available based on determining the truth on the ground.

Mismeasurement of a quantitative variable

- Measuring water or blood chemistry values typically involves some error. Laboratories routinely calibrate their measuring instruments using standards and values are returned based on the use of the resulting calibration curve. These will still contain some errors unless there is a deterministic relationship between the true and measured values, rarely the case. Examples in Chapters 8 and 10 use some of the data from the acid rain monitoring project (Godfrey et al., 1985). This involved water samples from approximately 1800 water bodies with analyses for ph, alkalinity and other quantities carried out by 73 laboratories. In this case, each lab here was calibrated by being sent blind samples with "known" values.
  Another application of this type is with immunoassays which measure concentrations of antigens or antibodies. The measured value is a radioactive count, or standardized count. Similar to the water chemistry examples, the assay is calibrated using standards and the resulting curve here is usually

nonlinear. One of the examples in Chapter 10 utilizes a four-parameter logistic model for measuring urinary neopterin with a radioimmunoassay.

- The Harvard Six Cities Study (see Section 7.3.1) examined the impact of exposure to certain pollutants on a child's respiratory status. An individual's actual exposure was difficult to measure. In its place observations were made in the home, in different rooms at different times of year. These values served as surrogates for exposure. The measures were validated in studies where individuals wore a lapel monitor that measures exposure continuously.
- Instead of categorized exposure, as discussed in the misclassification examples, quantitative measures of "exposure" are often preferred. The notion of true exposure can be a bit elusive here, especially when it is a dietary intake, but could be defined as a total or average intake over a certain time period. In that case, the measurement error can come from two sources: sampling over time and/or from the use of a fallible instrument, such as a food frequency questionnaire. The same comments apply in measuring physical activity and other quantities.
- In many designed experiments, there is some target value that an experimenter wants to apply to a "unit," but the actual delivered "dose" may differ from the target value. Examples include temperature and pressure in industrial experiments, fertilizer or watering levels in agricultural settings, protein levels in a diet in balance/intake studies to determine nutritional requirements, speed on a tread mill, etc. In this case the "measured" value is the fixed target dose, but the true value of interest is random.
- Many ecological problems strive to model the relationship among variables, over different spatial locations. Costs often make it impossible to measure the variables of interest exactly and instead some spatial sampling is used to obtain estimated values. An example of this appears in Chapters 4 and 5, where gypsy moth egg mass densities and defoliation rates are estimated over stands 60 hectares in size, based on sampling a small fraction of the total area.
- Many financial and economic variables, whether measured at an individual, company or other aggregate level (e.g., state, country, etc.), are subject to measurement error. The error may be due to self-reporting, sampling (similar to the ecological problems in the previous item), the construction of variables from other data, etc.
- The U.S. and many other governments carry out continuous surveys to measure unemployment, wages, and other labor and health variables. These are often then modeled over time. The measurement error here is sampling error arising from estimation using a subsample of individuals, households, etc. An interesting feature of these problems is that the measurement errors

can be correlated as a result of block resampling, in which individuals or households stay in the survey for some amount of time and are then rotated out.

- Another important ecological problem is the modeling of population abundance over time and exploring the relationship of abundance to other factors such as weather or food abundance. It is impossible to ever obtain an exact abundance, or population density, but instead it is estimated. This is well known to be a challenging problem, especially for mobile populations, and can involve the use of capture/recapture methods or other techniques, such as aerial surveillance. Once again the measurement error here is sampling error with the added complication that the accuracy and precision of the estimate may be changing over time as a result of changes in the population being sampled and/or changes in the sampling effort or technique. These same issues arise in modeling temperature and its relationship to human activities. In this latter application the quality of the data (i.e., the nature of the measurement error) has changed considerably over time, with both changes in instruments and enlargement of the grid of monitoring stations.
- As a final illustration, we consider an example discussed by Tosteson et al. (2005). The goal was to use vascular area, obtained from an image, to classify breast disease. One measure of the effectiveness of a classification technique is the receiver operating characteristic (ROC) curve. This depends on the distribution of the variable in each of the disease and nondisease groups or, under normality, the mean and variance. There is measurement error at the individual level however as the image has to be spatially subsampled and the vascular area for the whole image estimated. The induced measurement error will distort the estimated ROC curve.

1.3 The main ingredients

There are typically three main ingredients in a measurement error problem.

1. A Model for the True Values.
   This can be essentially any statistical model. See the overview below for where the emphasis is in this book.
2. A Measurement Error Model.
   This involves specification of the relationship between the true and observed values. This can be done in a couple of different ways as outlined in the next section and described in fuller detail in later chapters.
3. Extra data, information or assumptions that may be needed to correct for measurement error. This may not always be available, in which case one has to be satisfied with an assessment of the impacts of the measurement error. This extra "information" is typically:

(a) Knowledge about some of the measurement error parameters or functions of them.
(b) Replicate values. This is used mainly, but not exclusively, for additive error in quantitative variables.
(c) Estimated standard errors attached to the error prone variables.
(d) Validation data (internal or external) in which both true and mismeasured values are obtained on a set of units.
(e) Instrumental variables.

Some discussion of validation data and replication appear throughout Chapters 2 to 5, and instrumental variables are discussed in Section 5.7 in the context of multiple linear regression. A broader discussion of extra data is given in Section 6.5.

There are two general objectives in a measurement error problem.

- What are the consequences for **naive analyses** which ignore the measurement error?
- How, if at all, can we correct for measurement error?
  With some exceptions (see parts of Chapters 11 and 12), correcting for measurement error requires information or data as laid out in item 3 above. Myriad approaches to carrying out corrections for measurement error have emerged, a number of which will be described in this book. These include direct bias corrections, moment based approaches, likelihood based techniques, "regression calibration," SIMEX and techniques based on modifying estimating equations.

1.4 Some terminology

This section provides a brief introduction to measurement error models and some associated terminology. These models are revisited and expanded on in later chapters. Chapters 2 and 3 spell out misclassification models while additive error models arise in treating linear regression problems in Chapters 4 and 5. Section 6.4 then provides a more comprehensive and in depth look at these and other models. Still, a preliminary overview is helpful at this point. For convenience the discussion here is in terms of a univariate true value and its mismeasured version, but this can, and will be, extended to handle multivariate measures later.

In some places it is necessary to distinguish a random variable from its realized value. When necessary, we follow the usual convention of a capital letter (e.g., X) for the random variable and a small letter (e.g., x) for the realized value. Throughout the notation $|x$ is a shorthand for the more precise $|X = x$ which is read "given X is equal to x."

When the x is a predictor in a regression type problem a distinction is made between the **functional case** where the x's are treated as fixed values and the **structural case** where X's are random. This distinction, and its implications, is spelled out carefully in Chapters 4 and 5, with that discussion carrying over into later chapters. Some authors use structural to refer just to the case where a distribution is specified for a random X. The term functional model has also sometimes been used to refer to the case where there is a deterministic relationship among true variables. Our usage of the terminology will be as defined above. We also note that a combination of fixed and random predictors is possible. Chapter 5 illustrates this and adopts a strategy that handles these mixed cases.

1.4.1 Measurement versus Berkson error models

A fundamental issue in specifying a measurement error model is whether we make an assumption about the distribution of the observed values given the true values or vice versa. The **classical measurement error** model specifies the former, while the latter will be referred to as **Berkson error** model. The latter was initially introduced by Berkson (1950) in cases where an experimenter is trying to achieve a target value w but the true value achieved is X. Examples of this were presented earlier. In this case there is no random variable W but just a targeted fixed value w. In other problems X and W can both be random and the phrase "Berkson error model" has been broadened to include referring to the conditional distribution of $X|w$.

If the x's are fixed or we condition on them, the model for $W|x$ should be used. If the observed w is fixed then the Berkson model applies. With random X and random W given x, then a choice can sometimes be made between which model to use. This choice will depend in part on the nature of the main study (are the X's a random sample?) and the nature of any validation data. See Heid et al. (2004) for an example discussing the use of classical and Berkson error models in the measurement of radon exposure.

A special case is when both X and W are categorical. For classical measurement error, the model is given by the probability function $P(W = w|x)$ and these quantities are referred to as **misclassification probabilities** or misclassification rates. The Berkson error model, on the other hand, specifies $P(X = x|w)$, which we refer to as **reclassification probabilities** or reclassification rates. Both of these are discussed in Chapters 2 and 3.

1.4.2 Measurement error models for quantitative values

For a quantitative variable, any type of regression model can be used for how W and X are related. Historically, the most commonly used model is the **additive measurement error** model where

$$W|x = x + u \tag{1.1}$$

where u is a random variable with $E(u|x) = 0$. Equivalently $E(W|x) = x$, so W is unbiased for the unobserved x.

Nonadditive measurement error includes everything else, where $E(W|x) = g(\boldsymbol{\theta}, x) \neq x$. This means there is some form of bias present. Examples include constant bias, $E(W|x) = \theta + x$ and linear measurement error, $E(W|x) = \theta_0 + \theta_1 x$, while with nonlinear measurement error models $g(x, \boldsymbol{\theta})$ is nonlinear in the θ's. Nonadditive models are particularly prevalent when the measurement error is instrument error.

A **constant variance/homoscedastic** measurement error model refers to the case where the variance of W given x is constant. A **heteroscedastic** model allows the measurement error variance to possibly change. While historically, constant measurement error variance has been used, there are many situations where changing variances need to be allowed. This is discussed in some detail in Section 6.4.5. Heteroscedastic measurement error models are accommodated throughout many parts of this book.

1.4.3 Nondifferential/differential measurement error, conditional independence and surrogacy

To this point the discussion has been limited to how a single measured W relates to a true value x. More generally, we need to accommodate the fact that the measurement error may depend on other variables, which themselves may or may not be measured with error. General developments of this type are accommodated in Chapter 6. Here we introduce a few important concepts where W (measured) and X (true) are both univariate as is a third variable Y, which is, usually, a response variable. That designation is not essential to the discussion below, nor is the fact that W and X are univariate.

- The measurement error model for W given x is said to be **nondifferential** (with respect to Y) if the distribution of $W|x, y$ (W given $X = x$ and $Y = y$) equals the distribution of $W|x$. That is, the measurement error model does not depend on the value y.
  The measurement error model is said to be **differential** if the distribution of $W|x, y$ changes with y.

- W is a **surrogate** for X (with respect to Y) if the distribution of $Y|x, w$ equals that of $Y|x$. This says that given x, W contains no information about the distribution of Y beyond that which is contained in x.
- **Conditional independence** states that Y and W are independent given $X = x$. Mathematically, $f(y, w|x) = f(y|x)f(w|x)$, where we have used the convenient, but obviously imprecise, device of denoting the distribution of interest by the symbols in the argument. So $f(y|x)$ is the conditional density or mass function of Y given $X = x$, $f(w|x)$ is that of W given $X = x$, and $f(y, w|x)$ is the joint density or mass function of Y and W given $X = x$.

An interesting, and very useful, result is the following:

The three concepts of surrogacy, conditional independence and nondifferential measurement error are equivalent. That is, any one implies the other two.

1.5 A look ahead

The guiding principles in laying out the book were 1) have it be introductory and cover some of the more basic statistical models in detail and 2) to allow the reader who is interested in a particular type of problem to jump into the book at various places and be able to at least get a feel for how to handle that particular problem. The second emphasis means the book is based primarily around the model for the true values, rather than around the measurement error correction technique. The exception to this is the somewhat monolithic Chapter 6. This chapter provides a fairly wide ranging overview and details on the various components of the problem: models for true values, various measurement error models, types of additional data and correction techniques. Rather than jump into this first, which would make for fairly boring reading without a lot of excursions into particular problems, some simple problems are discussed first.

Misclassification of categorical data is treated first in the contexts of estimating a single proportion and in two-way tables, Chapters 2 and 3, respectively. This provides a good look in particular at the various uses of validation data. We then consider additive error, in the predictor(s) and/or the response, for simple and multiple linear regression. This is in Chapters 4 and 5. Once again, we follow the principle of looking at a simple setting first (simple linear models) in which to illustrate a number of basic ideas in explicit detail before moving on to the more complex setting of multiple predictors. To a large extent, these four chapters go over somewhat old ground, but hopefully in a way that provides a unified summary in an applied manner. The multiple linear regression chapter has a few useful features, including allowing error in either predictors or response with possibly changing measurement error variances or (if applicable) covariances, from the beginning, and a formulation within which a number of results can be expressed which cover random or fixed predictors, or a combination thereof.

After this is the aforementioned Chapter 6. Some readers may chose a selective read through this depending on what they are after, and return to it later as needed. It does contain all of the key concepts and methods used elsewhere in the book. At the least, it is suggested that the reader look over the descriptions of measurement error models, the use of extra data, the descriptions of regression calibration, SIMEX and modified estimating equation correction methods (which are used in most examples), as well as the overviews on using validation data and bootstrapping.

The subsequent chapters cover specific models. Chapter 7 illustrates most of the key elements of Chapter 6 in the context of binary regression. This provides an important setting in which to examine many of the correction techniques for nonlinear models in general, and generalized linear models, in particular. Chapter 8 then returns to linear models but now with nonadditive error in certain predictors or functions of them. Among other things, this chapter includes how to handle quadratic and interaction terms as well as the use of external validation data for measurement error models that have systematic biases. Chapter 9, which is fairly short, reinforces a few ideas from the binary chapter and expands a bit on handling nonlinear models that are not generalized linear models. Chapter 10 isolates some additional questions when dealing with error in the response only. Among other things, this chapter includes some discussion of nonparametric estimation of a distribution in the presence of measurement error and fitting mean-variance relationships. Finally, the last two chapters cover mixed/longitudinal and time-series models. These are a bit different than the earlier chapters in a couple of ways. They both provide fairly general surveys and then focus on a few specific problems within the much larger class of problems. They also assume more prior knowledge of the topics than some of the earlier chapters.

See the preface for additional comments about coverage, emphasis and computational aspects.

CHAPTER 2

Misclassification in Estimating a Proportion

This chapter treats the problem of estimating a proportion when the true value is subject to misclassification. After some motivating examples, Sections 2.2 - 2.4 examine the problem under the assumption that the true values are a random sample in the sense of having independent observations. We first examine bias in the naive estimator which ignore the misclassification and then show how to correct for the misclassification using both internal and external validation data or with known misclassification probabilities. As part of this discussion, Section 2.3 delineates the difference between misclassification and reclassification probabilities. Section 2.5 extends the results to allow for sampling from a finite population, while Section 2.6 provides some brief comments on how to account for misclassification with repeated or multiple measures without validation data. Finally, Section 2.7 provides an overview of the case where there are more than two categories for the true values and Section 2.8 provides a few mathematical developments.

2.1 Motivating examples

Example 1. Knowing the prevalence of the HIV virus in a population is critical from a public health perspective. This is of special interest for the population of women of child bearing age. For example, Mali et al. (1995) estimated the prevalence of women with HIV-1 attending a family planning clinic in Nairobi to be 4.9% from a sample of 4404 women. Assays that test for the presence or absence of the virus are often subject to misclassification. What effect do the misclassification errors have on the estimated prevalence of 4.9%? If we had estimates of the misclassification rates how can we correct for the misclassification? As an example of validation data, Table 2.1 shows some data, based on Weiss et al. (1985) and also used by Gastwirth (1987), for an HIV test on 385 individuals. The error-prone w is based on ELISA (enzyme-linked immuonabsorbent assay). There are two false negatives. If what Weiss et al.

(1985) considered borderline are classified as negative then there are only 4 false positives, while if they are considered positive there are 22 false positives. The latter case is given by the parenthetical values in the first line of the table. Later, for illustration, we use this data as external validation data in analyzing the data from Mali et al. (1995).

Table 2.1 *External validation data for the ELISA test for presence of HIV. x is the true value, with indicating being HIV positive, and w is the error prone measure from the assay.*

		ELISA(w) 0	1	
Truth(x)	0	293 (275)	4 (22)	297
	1	2	86	88

Example 2. Here we use an example from Fleiss (1981, p. 202) where the goal was to estimate the proportion of heavy smokers in a population. The "truth" is determined by a blood test while the error-prone measure w is based on a self assessment by the individual. Some level of misclassification is often present with self reporting. Table 2.2 shows data for a random sample of 200 individuals of which 88 reported themselves to be heavy smokers. A random subsample of 50 from the 200 were subjected to a blood test which determined the true x. The values of w and x for these 50 subjects are given in the top portion of the table. The other 150 have a self reported value (w), but the truth is unknown. This differs from the first example in that the individuals for which we obtain true values are a subset of the main study. This is an example of internal validation.

Table 2.2 *Smoking example: x = true status with w = self assessment. From Fleiss (1981). Used with permission of John Wiley and Sons.*

		w 0	1	
x =	0	24	2	26
	1	6	18	24
		30	20	50
	?	82	68	150
		112	88	200

Example 3. Lara et al. (2004) use what is known as a randomized response technique to estimate the proportion of induced abortions in Mexico. In their implementation, with probability 1/2 the woman answered the question “Did you ever interrupt a pregnancy?” and with probability 1/2 answered the question “Were you born in April?” Only the respondent knew which question they answered. Of 370 women interviewed, 56 answered yes. The objective is to estimate the proportion of women who interrupted a pregnancy. The randomized response technique (RRT) intentionally introduces misclassification but with known misclassification probabilities. In this example, the sensitivity and specificity (defined in the next section) are 13/24 and 23/24, respectively. See Lara et al. (2004) and van den Hout and van der Heijden (2002) for further discussion of the RRT method. The latter paper also describes the situation where misclassification is intentionally introduced to protect privacy, again with known probabilities.

Example 4. The use of satellite imagery to classify habitat types has become widespread. Table 2.3 shows validation data taken from an online document at the Canada Centre for Remote Sensing for five classes of land-use/land-cover maps produced from a Landsat Thematic Mapper image of a rural area in southern Canada. In this case, each unit is a pixel. The goal would be to use this validation data to obtain corrected estimates for the proportions for an area on which only the Landsat classifications are available. This is an example with multiple categories, which is touched on in Section 2.7.

Table 2.3 *Validation data for remote sensing.*

			LANDSAT		
	Water	BareGround	Deciduous	Coniferous	Urban
TRUTH					
Water	367	2	4	3	6
BareGround	2	418	8	9	17
Deciduous	3	14	329	24	25
Coniferous	12	5	26	294	23
Urban	16	26	29	43	422

There are two questions considered in the following sections.

- What if we ignore the misclassification?
- How do we correct for the bias induced by misclassification and obtain corrected standard error and/or confidence intervals for the proportion of interest?

2.2 A model for the true values

We begin by assuming that we have a "random sample" of size n where each observation falls into one of two categories. In terms of random variables, $X_1, \ldots, X_n$ are assumed to be *independent and identically distributed* (i.i.d.), where X_i = 1 if the ith observation is a "success" and = 0 if a "failure," and

$$\pi = P(X_i = 1).$$

The objective is estimation of π, which can be the proportion of successes in a finite population or, as above, the probability of success on repeated trials. The assumption of independence is only approximately true when sampling from a finite population, but generally acceptable unless the sample size is "large" relative to the population size. Explicit modifications for sampling from a finite population appear in Section 2.5.

With no measurement error, $T = \sum_{i=1}^{n} X_i$ = number of successes in the sample is distributed Binomial(n, π) and

$$p = T/n,$$

the proportion of successes in the sample, is an unbiased estimator of π, with $E(p) = \pi$ and $V(p) = \pi(1-\pi)/n$. The estimated standard error of p is $SE(p) = [p(1-p)/n]^{1/2}$. An approximate large sample confidence interval for π is given by $p \pm z_{\alpha/2} SE(p)$ and approximate large sample tests of hypotheses about π, such as $H_0 : \pi = \pi_0$, are based on either $Z = (p - \pi_0)/SE(p)$ or $(p - \pi_0)/(\pi_0(1-\pi_0)/n)^{1/2}$. These large sample inferences for a proportion can be found in most introductory statistics books. For small to moderate samples, "exact" confidence intervals and test of hypotheses are available based on the Binomial distribution; see for example Agresti and Coull (1998) or Casella and Berger (2002, Exercise 9.21).

2.3 Misclassification models and naive analyses

The fallible/error-prone measure is W, which is assumed to also be binary. It is the outcome of W, denoted by w, that is obtained on the n observations making up the main study. The measurement error model, which specifies the behavior of W given $X = x$ ($x = 0$ or 1), is specified by the misclassification probabilities,

$$\theta_{w|x} = P(W = w|X = x),$$

with

sensitivity: $P(W = 1|X = 1) = \theta_{1|1}$

specificity: $P(W = 0|X = 0) = \theta_{0|0}$.

We only need to specify two probabilities since $P(W = 0|X = 1) = \theta_{0|1} = 1 - \theta_{1|1}$ (the probability of a false negative) and $P(W = 1|X = 0) = \theta_{1|0} = 1 - \theta_{0|0}$ (the probability of a false positive).

Since X is random, if we specify the distribution of $W|x$ we can also derive a model for the distribution of X given w. This is given by

$$\lambda_{x|w} = P(X = x|W = w).$$

We refer to these as **reclassification probabilities**, but they are also known as **predictive probabilities**. Again, we need just two probabilities

$$\lambda_{1|1} = P(X = 1|W = 1) \text{ and } \lambda_{0|0} = P(X = 0|W = 0).$$

The reclassification model is a special case of a Berkson model (see Chapter 1) in that it models the truth given the observed. The reclassification and misclassification rates are related via $\lambda_{x|w} = \theta_{w|x} P(X = x)/(\theta_{w0}(1 - \pi) + \theta_{w1}\pi)$.

Naive inferences are based on use of the W's rather than the possibly unobservable X's. The naive estimator of π is

$$p_W = \text{the proportion of the sample with } W = 1.$$

The W_i's are a random sample with (see Section 2.8)

$$\pi_W = P(W_i = 1) = \pi(\theta_{1|1} + \theta_{0|0} - 1) + 1 - \theta_{0|0}. \tag{2.1}$$

We can also reverse the roles of X and W, leading to

$$\pi = \pi_W(\lambda_{1|1} + \lambda_{0|0} - 1) + 1 - \lambda_{0|0}. \tag{2.2}$$

Since $E(p_W) = \pi_W$ rather than π, the naive estimator has a bias of

$$BIAS(p_w) = \pi_W - \pi = \pi(\theta_{1|1} + \theta_{0|0} - 2) + (1 - \theta_{0|0}).$$

For a simple problem the nature of the bias can be surprisingly complex. Figure 2.3 illustrates the bias for π = .01 and .5, displaying the bias as a function of sensitivity for different levels of specificity. Notice that the absolute bias can actually increase as the sensitivity or specificity increases with the other held fixed. For example, consider the case with $\pi = .5$ and specificity .90. At sensitivity .9, the bias is 0 with the bias then increasing as sensitivity increases with a bias of .05 at sensitivity = 1. With a rarer event ($\pi = .01$), the bias is insensitive to the level of sensitivity, but heavily sensitive to specificity with severe relative bias even at specificity of .95 where the bias is approximately .05, when estimating a true value of .01.

We could also evaluate the naive performance through the behavior of the naive confidence interval. Using the standard normal based confidence interval

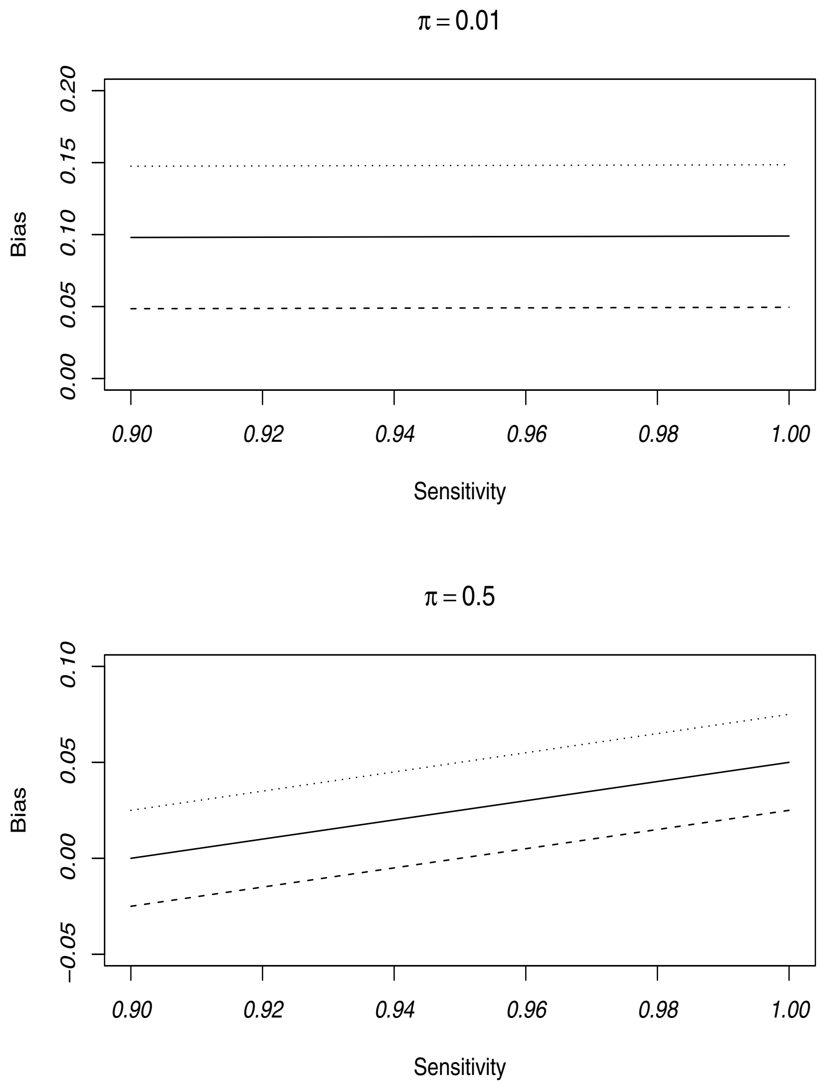


Figure 2.1 *Plot of bias in naive proportion plotted versus sensitivity for different values of specificity (dotted line = .85; solid line = .90; dashed line = .95).*

the probability that the naive interval contains an arbitrary value c is approximately

$$P\left[\frac{c-\pi_w}{(\pi_W(1-\pi_W)/n)^{1/2}}-z_{\alpha/2}\leq Z\leq\frac{c-\pi_W}{(\pi_W(1-\pi_W)/n)^{1/2}}+z_{\alpha/2}\right],$$

where Z is distributed as a standard normal. With $c=\pi$ this gives the coverage rate of the naive interval.

2.4 Correcting for misclassification

Equation (2.1) can be used to correct for misclassification using known or estimated misclassification rates. For convenience these are denoted $\widehat{\theta}_{0|0}$ and $\widehat{\theta}_{1|1}$, even if they are known. Inverting (2.1) leads to the corrected estimator

$$\hat{\pi}=\frac{p_W-(1-\widehat{\theta}_{0|0})}{\widehat{\theta}_{0|0}+\widehat{\theta}_{1|1}-1}. \tag{2.3}$$

It is possible for this estimate to be less than 0 or greater than 1. If it is, then from the point estimation perspective, the estimate would be set to 0 or 1, respectively. Notice that the estimated specificity is particularly important in this regard since if the estimated probability of a false positive, $1-\widehat{\theta}_{0|0}$, is bigger than p_w then the corrected estimate is negative (assuming $\widehat{\theta}_{0|0}+\widehat{\theta}_{1|1}-1$ is positive, which is essentially always the case). Certain of the intervals presented later bypass this problem since they don't use a point estimate.

If the sensitivity and specificity are known, then $\hat{\pi}$ is unbiased for π. Otherwise the corrected estimator is consistent but biased with decreasing bias as the size of the validation sample increases. In practice the bias of the corrected estimator could be an issue with small validation studies. This can be investigated through the bootstrap.

2.4.1 Ignoring uncertainty in the misclassification rates

As noted in the introduction, there are cases where the misclassification rates are known exactly. This happens with randomized response techniques and with intentional misclassification to protect privacy. In other cases, previously reported sensitivity and specificity may be treated as known. In this case, the estimated standard error of $\hat{\pi}$ is

$$SE(\hat{\pi})=\left[\frac{p_W(1-p_W)}{n(\theta_{1|1}+\theta_{0|0}-1)^2}\right]^{1/2}$$

and an approximate large sample confidence interval for π is

$$\hat{\pi}\pm z_{\alpha/2}SE(\hat{\pi}). \tag{2.4}$$

As an alternative to the Wald interval in (2.4), an exact confidence interval can be obtained. First, an exact confidence interval (L_W, U_W) can be obtained for π_W using the error-prone values. This would be based on the Binomial distribution. Then, assuming $\widehat{\theta}_{1|1} + \widehat{\theta}_{0|0} - 1$ is positive (it usually is) and using the fact that π is a monotonic function of π_W, a confidence interval for π is given by

$$[L, U] = \left[\frac{(L_W - (1 - \widehat{\theta}_{0|0}))}{(\widehat{\theta}_{1|1} + \widehat{\theta}_{0|0} - 1)}, \frac{(U_W - (1 - \widehat{\theta}_{0|0}))}{(\widehat{\theta}_{1|1} + \widehat{\theta}_{0|0} - 1)} \right]. \tag{2.5}$$

If $\widehat{\theta}_{1|1} + \widehat{\theta}_{0|0} - 1$ is not positive then some modification is needed, but this would almost never occur. The value of this approach is that it allows one to handle small sample problems exactly, while at the same time bypassing any potential problems with point estimates of π that are outside of the interval $[0, 1]$. The same approach could be carried out by constructing the interval (L_W, U_W) with the method proposed by Agresti and Coull (1998) as an alternative to the binomial based interval.

Hypothesis tests about π can be carried out using confidence intervals for π. Alternatively, an hypothesis about π can be expressed in terms of π_W and tested directly based on the W data.

Example 3 (continued). Table 2.4 provides the analysis for the third example in the introduction, estimating the proportion of interrupted unwanted pregnancies, using a randomized response technique, which has known sensitivity $\theta_{1|1} = 13/24$ and specificity $\theta_{0|0} = 23/24$. Lara et al. (2004) were interested in comparing the RRT method to other techniques, rather than in providing inferences for the population proportion, as we do here. The naive p_W, which is the proportion that answered yes to whichever question they received, is approximately .15. The corrected estimate is $\widehat{\pi} = (.22 - (1/24))/((36/24) - 1) = .22$ with standard errors and confidence intervals calculated as described above. The correction is about 50 percent of the original estimate and the corrected confidence intervals are wider and shifted towards larger values, with fairly close agreement between the Wald and exact intervals.

Table 2.4 *Analysis of abortion example*

Method	Estimate	SE	Wald Interval	Exact Interval
Naive	.1514	.0186	(.1148, .1878)	(.1161, .1920)
Corrected	.2194	.0373	(.1463, .2924)	(.1488, .3007)

2.4.2 Using external validation data and misclassification rates

Here we account for uncertainty from estimating the misclassification rates assuming those estimates come from external validation data, represented generically in Table 2.5.

Table 2.5 *Representation of external validation data*

		W: 0	W: 1	
X	0	n_{V00}	n_{V01}	$n_{V0.}$
	1	n_{V10}	n_{V11}	$n_{V1.}$
		$n_{V.0}$	$n_{V.1}$	n_V

There are n_V observations for which W and X are both observed. The . notation indicates summation over the missing index; e.g., $n_{V0.} = n_{V00} + n_{V01}$. These data are assumed to be independent of the main body of data. The W's entering into the external validation study also must be random (so we can estimate the probabilities for W given X) and therefore the validation study can't be obtained using stratification based on W. To illustrate this point, suppose we are validating reported smoking status versus true smoking status. If this external validation design used stratified sampling from reported smokers and non-smokers then this data only allows estimation of reclassification rates and so could not be used in the manner described in this section. The other important assumption is that the misclassification model is assumed to be the same for this data as the main data; that is, the measurement error model is exportable from the validation data to the main study. It is less common to use the reclassification rates from external validation data. If, however, this is done then the estimation would proceed as when using internal validation; see (2.7) and the discussion following it.

The validation data yields estimated specificity and sensitivity

$$\widehat{\theta}_{0|0} = n_{V00}/n_{V0.} \qquad \text{and} \qquad \widehat{\theta}_{1|1} = n_{V11}/n_{V1.}$$

leading to $\widehat{\pi}$ as in (2.3). If we condition on the x values in the external validation data, this is in fact the maximum likelihood estimator (MLE) of π, as long as $0 \leq \hat{\pi} \leq 1$.

Delta and Fieller intervals.

The estimate $\widehat{\pi}$ is a ratio Z_1/Z_2, where $Z_1 = p_W - (1 - \widehat{\theta}_{0|0})$ and $Z_2 = \widehat{\theta}_{0|0} + \widehat{\theta}_{1|1} - 1$. The estimated variances of Z_1 and Z_2 and their covariance are given by

$$\hat{V}_1 = \frac{p_W(1-p_W)}{n} + \frac{\widehat{\theta}_{0|0}(1-\widehat{\theta}_{0|0})}{n_{V0.}}, \qquad \hat{V}_2 = \frac{\widehat{\theta}_{0|0}(1-\widehat{\theta}_{0|0})}{n_{V0.}} + \frac{\widehat{\theta}_{1|1}(1-\widehat{\theta}_{1|1})}{n_{V1.}}$$

and $\hat{V}_{12} = \widehat{\theta}_{0|0}(1-\widehat{\theta}_{0|0})/n_{V0.}$, respectively.

Using the approximate variance of a ratio and estimating it (see Section 2.8) leads to an estimated standard error $SE(\widehat{\pi}) = \hat{V}(\hat{\pi})^{1/2}$, where

$$\hat{V}(\hat{\pi}) = \frac{1}{(\widehat{\theta}_{0|0} + \widehat{\theta}_{1|1} - 1)^2}\left[\hat{V}_1 - 2\widehat{\pi}\hat{V}_{12} + \hat{\pi}^2\hat{V}_2\right] \tag{2.6}$$

$$= \frac{p_W(1-p_W)}{n(\widehat{\theta}_{0|0} + \widehat{\theta}_{1|1} - 1)^2} + \frac{1}{(\widehat{\theta}_{0|0} + \widehat{\theta}_{1|1} - 1)^2}\left[\frac{\widehat{\theta}_{0|0}(1-\widehat{\theta}_{0|0})}{n_{V0.}} - 2\widehat{\pi}\hat{V}_{12} + \hat{\pi}^2\hat{V}_2\right].$$

The expression for the estimated variance following (2.6) is written as the variance if the misclassification parameters are assumed known plus a piece which accounts for estimation from the validation data. Notice that as the validation sample sizes ($n_{V0.}$ and $n_{V1.}$) increase the second term goes to 0.

An approximate Wald confidence interval for π, based on the delta method approximation to the variance of $\widehat{\pi}$, is given by $\hat{\pi} \pm z_{\alpha/2}SE_{EV}(\hat{\pi})$.

A better option when working with ratios is to use Fieller's method, which is discussed in Section 13.5. The use of this method in the current context was recently investigated by Shieh (2009) and Shieh and Staudenmayer (2009). With the definitions above the Fieller interval for π can be calculated using (13.1) in Section 13.5. This interval holds as long as Z-test of $\theta_{0|0}+\theta_{1|1}-1=0$ is signficant at size α. This is almost always the case unless the error-prone measure is particularly bad. When this test is highly significant, the Fieller and Wald intervals are very similar.

Bootstrapping.

Another approach to inference is to use the bootstrap, discussed in a bit more detail in Section 6.16. Here, this is implemented as follows:

For the bth bootstrap sample, $b = 1, \ldots, B$, where B is large:

1. Generate $p_{wb} = T_b/n$, $\widehat{\theta}_{0|0b} = n_{V00b}/n_{V0.}$ and $\widehat{\theta}_{1|1b} = n_{V11b}/n_{V1.}$, where T_b, n_{V00b} and n_{V11b} are generated independently as Binomial(n, p_w), Binomial$(n_{V0.}, \widehat{\theta}_{0|0})$ and Binomial $(n_{V1.}, \widehat{\theta}_{1|1})$, respectively.
2. Use the generated quantities to obtain $\widehat{\pi}_b = (p_{wb} - (1 - \widehat{\theta}_{0|0b}))/(\widehat{\theta}_{0|0b} + \widehat{\theta}_{1|1b} - 1)$ (truncated to 0 or 1 if necessary).

3. Use the B bootstrap values $\widehat{\pi}_1, \ldots, \widehat{\pi}_B$ to obtain bootstrap estimates of bias and standard error for $\widehat{\pi}$, and to compute a bootstrap confidence interval. Here, simple bootstrap percentile intervals will be used.

An exact confidence interval.

An "exact" approach can also be considered. First obtain confidence intervals for each π_W, $\theta_{0|0}$ and $\theta_{1|1}$, denoted by $[L_w, U_w]$, $[L_{00}, U_{00}]$ and $[L_{11}, U_{11}]$, respectively, where each of these are exact intervals, assumed to each have confidence coefficient $1 - \alpha^*$. A confidence set for π is constructed by finding the minimum and maximum value of $(\pi_W - (1 - \theta_{0|0}))/(\theta_{0|0} + \theta_{1|1} - 1)$ as the three values vary over their respective intervals. This can be done numerically. Often the interval can be obtained by just looking at the minimum and maximum values over the eight combinations of the three sets of interval endpoints, but there are some exceptions depending on the signs involved. This is discussed by Shieh(2009). Using independence, the confidence coefficient for the interval for π is $\geq (1 - \alpha^*)^3$, so if each of the intervals is computed using $\alpha^* = 1 - (1 - \alpha)^{1/3}$, the overall coverage rate is $\geq 1 - \alpha$. This interval does offer protection with smaller sample sizes but it can be conservative.

Example 1 (continued). Here we apply the methods above to the HIV example that opened this chapter. The main sample from Mali et al. (1995) has a sample of $n = 4404$ with a naive estimator of the proportion of women with HIV of $p_w = .049$. We run two analyses using the data in Table 2.1 as external validation data, for illustration. In each $n_V = 385$, $n_{V10} = 2, n_{V11} = 86$, $n_{V1.} = 88$ and $n_{V0.} = 297$. In the first analysis $n_{V00} = 293$ and $n_{V01} = 4$ (so four false positives) leading to estimated specificity and sensitivity of $\widehat{\theta}_{0|0}$ = .9864 and $\widehat{\theta}_{1|1} = .9773$, respectively. In the second case, $n_{V00} = 275$ and $n_{V01} = 22$, with $\widehat{\theta}_{1|1}$ unchanged and estimated specificity $\widehat{\theta}_{0|0}$ = .9259.

The analysis appears in Table 2.6. Ignoring the uncertainty in the estimated misclassification rates leads to standard errors and confidence intervals that are much too small. The Wald, Fieller and bootstrap intervals are all similar and the bootstrap standard errors are close to those obtained using an analytical approximation. The more conservative "exact" intervals are somewhat wider, as expected. A comparison of the bootstrap mean to the corrected estimate indicates that bias is not an issue. In the first case the correction to the estimated proportion is moderate, a change from 4.9% to 3.7%. In the second case, with more false positives, and so lower specificity, the correction is dramatic, leading to an estimate of -.028. Clearly, as a point estimate this would be rounded to 0. The Fieller, bootstrap and exact intervals do not rely directly on the use of the negative point estimate.

Table 2.6 *Analysis of HIV example. Cor-K refers to corrected estimates and intervals treating the misclassification rates as known, while Cor-U accounts for uncertainty in the estimated misclassification rates. Confidence intervals are 95%. Boot indicates the bootstrap analysis with (M) indicating the bootstrap mean and PI the 95% bootstrap percentile confidence interval.*

Method	Est.	SE	CI	Fieller	Exact
Naive	.049	.0033	(.043,.055)		(.043, .056)
			4 false positives		
Cor-K	.0369	.0033	(.030,.043)		(.030,.044)
Cor-U	.0369	.0075	(.022,.052)	(.022,.052)	(.002,.047)
Boot	.0369(M)	.0075	PI: (.021,.050)		
			22 false positives		
Cor-K	-.0278	.0036	(-.035,-.021)		(-.035,-.020)
Cor-U	-.0278	.0177	(-.062,.007)	(-.064,.006)	(-.098, .004)
Boot	-.0274 (M)	.0179	PI: (-.065,.006)		

2.4.3 Internal validation data and the use of reclassification rates

We turn now to the case with internal validation data, where W is observed for all n units, while X is only observed on a subset of size n_V. This is the case for the second example where the goal is to estimate the proportion of heavy smokers. This is also called **double sampling** and leads naturally to the use of reclassification rates. The full data can be represented as in Table 2.7, where n is the overall sample size, $n_{.0}$ and $n_{.1} = n - n_{.0}$ are the number of observations with $W = 0$ and $W = 1$, respectively, $n_I = n - n_V$ (I for incomplete) is the number of observations with W only and the other notation should be clear from the table. As before, a . indicates summation over an index. For the smoking example in Table 2.2, $n_{I0} = 82$, $n_{.0} = 112$, $n_{V01} = 2$, etc.

The validated units are often a simple random sample of size n_V from the n units in which case the number of cases in the subsample with W equal 0 ($n_{V.0}$) and $W = 1$ ($n_{V.1}$) are random. An alternate design is where the W_i's are observed first, and the validation sample is chosen by sampling $n_{V.0}$ from the $n_{.0}$ cases with $W = 0$ and $n_{V.1}$ from the $n_{.1}$ cases with $W = 1$. This is referred to as a **two-phase design** (in the sampling terminology) or **designed double sampling.** In the smoking example this would occur if specified numbers were independently sampled from among the reported heavy smokers and nonheavy smokers. The use of a two-phase design may be particularly helpful

Table 2.7 *Representation of main study and internal validation data.*

		W = 0	W = 1	
X =	0	n_{V00}	n_{V01}	$n_{V0.}$
	1	n_{V10}	n_{V11}	$n_{V1.}$
		$n_{V.0}$	$n_{V.1}$	n_V
	?	n_{I0}	n_{I1}	n_I
		$n_{.0}$	$n_{.1}$	n

if the proportion of interest is near 0 or 1 since just random sampling for validation may result in few, or no, observations for the value of W associated with the small probability. This would be the case for estimating the prevalence of a rare disease or other infrequently occurring characteristics.

Random subsamples.

We first treat the case where the internal validation data arises from a random sample of n_V out of the original sample of size n; see Tenenbein (1970). One approach to correcting for misclassification would be to mimic what was done with external validation data, using the estimated misclassification rates from the internal validation data. This turns out to be inefficient and also leads to additional considerations in obtaining the standard error since the naive proportion obtained from all units uses some of the data entering into the estimation of the misclassification rates. An easier and more efficient approach is to use the estimated reclassification rates

$$\hat{\lambda}_{11} = n_{V11}/n_{V.1} \text{ and } \hat{\lambda}_{00} = n_{V00}/n_{V.0}$$

along with the naive proportion $\hat{\pi}_W = n.1/n$ and form the estimator

$$\widehat{\pi}_r = p_W(\widehat{\lambda}_{1|1} + \widehat{\lambda}_{0|0} - 1) + 1 - \widehat{\lambda}_{0|0}. \tag{2.7}$$

This follows from (2.2) with the r subscript indicating the use of the reclassification rates.

The estimator $\widehat{\pi}_r$ is, in fact, the maximum likelihood estimator, as long as $0 < \widehat{\pi}_r < 1$. The estimated standard error of $\widehat{\pi}_r$ is

$$SE(\widehat{\pi}_r) = \left[p_W^2 \widehat{V}(\widehat{\lambda}_{1|1}) + (p_W - 1)^2 \widehat{V}(\widehat{\lambda}_{0|0}) + (\widehat{\lambda}_{0|0} + \widehat{\lambda}_{1|1} - 1)^2 \widehat{V}(p_w)\right]^{1/2} \tag{2.8}$$

where $\widehat{V}(\widehat{\lambda}_{1|1}) = \widehat{\lambda}_{1|1}(1 - \widehat{\lambda}_{1|1})/n_{V.1}$, $\widehat{V}(\widehat{\lambda}_{0|0}) = \widehat{\lambda}_{0|0}(1 - \widehat{\lambda}_{0|0})/n_{V.0}$, and $\widehat{V}(p_W) = p_W(1 - p_W)/n$. A Wald confidence interval for π is given by $\widehat{\pi}_r \pm z_{\alpha/2} SE(\widehat{\pi}_r)$.

Bootstrapping. With the validation sample chosen as a random sample from the main study units, the bootstrap can be carried out as follows. For the bth bootstrap sample, $b = 1, \ldots, B$:

1. For $i = 1$ to n generate W_{bi} distributed Bernoulli (0 or 1) with $P(W_{bi} = 1) = p_w$.
2. For $i = 1$ to n_V generate X_{bi} distributed Bernoulli, with $P(X_{bi} = 1)$ equal to $1 - \widehat{\lambda}_{0|0} = \widehat{\lambda}_{1|0}$ if $W_{bi} = 0$, and equal to $\widehat{\lambda}_{1|1}$ if $W_{bi} = 1$.
3. Obtain $p_{wb} = \sum_{i=1}^{n} W_{bi}/n$, $n_{V.1b} = \sum_{i=1}^{n_V} W_{bi}$, $n_{V.0b} = n_V - n_{V.1b}$, $\widehat{\lambda}_{0|0b} = \sum_{i=1}^{n_V}(1-X_{bi})(1-W_{bi})/n_{V.0b}$ and $\widehat{\lambda}_{1|1b} = \sum_{i=1}^{n_V}(X_{bi}W_{bi})/n_{V.1b}$. Note that these calculation use the fact that $(1-X)(1-W) = 1$ only if both X and W are 0 and $XW = 1$ only if both X and W equal 1.
4. Calculate $\widehat{\pi}_{rb} = p_{wb}(\widehat{\lambda}_{1|1b} + \widehat{\lambda}_{0|0b} - 1) + 1 - \widehat{\lambda}_{0|0b}$ which is the estimate of π based on the reclassification rates for the bth bootstrap sample.
5. Use the B bootstrap values to obtain bootstrap inferences.

Exact confidence interval. Using Bonferonni's inequality, we can form exact $100(1-\alpha/3)\%$ confidence intervals for π_w, $\lambda_{1|1}$ and $\lambda_{0|0}$ based on the binomial or other alternatives. These are then used to find a $100(1-\alpha)\%$ confidence interval for π by finding the minimum and maximum value of $\pi_W(\lambda_{1|1}+\lambda_{0|0}-1)+1-\lambda_{0|0}$ as the three parameters range over their respective intervals. This results in a confidence interval for π of

$$[L_W(L_{11} + L_{00} - 1) + 1 - U_{00}, U_W(U_{11} + U_{00} - 1) + 1 - L_{00}],$$

where $[L_{jj}, U_{jj}]$ is the exact interval for λ_{jj} and $[L_w, U_w]$ is the interval for π_w.

Example 2 (continued). Table 2.8 shows the analysis of the smoking example. The estimated reclassification rates are $\widehat{\lambda}_{0|0} = 24/30 = .8$ and $\widehat{\lambda}_{1|1} = 18/20 = .9$, leading to a corrected estimator of the percentage of heavy smokers of 50.8% versus a naive estimator of 44%. The bootstrap estimate of bias (.5088 - .508) is small and the corrected Wald and bootstrap confidence intervals for π are in close agreement. The "exact" corrected interval, which is conservative, is considerably larger than the Wald and bootstrap intervals, and is not recommended given the relatively large sample sizes involved.

Designed/two-phase sampling. As noted earlier, the validation data could be obtained sequentially by taking independent samples of size $n_{V.0}$ and $n_{V.1}$ from $n_{.0}$ and $n_{.1}$ units with $W = 0$ and $W = 1$, respectively. This is an example of double sampling for stratification, which has a long history in the sampling literature (see for example Lohr, 1999, Chapter 12). Here the stratification is on the values of W. To connect to the sampling results we can view $1 - \hat{\lambda}_{00}$ and $\widehat{\lambda}_{1|1}$ as sample means, $\bar{y}_0$ and $\bar{y}_1$, of binary variables, equal to 1

Table 2.8 *Smoking example: Estimation of proportion of heavy smokers.*

Method	Estimate	SE	95% Wald Interval	95% Exact Interval
Naive	.44	.0351	(0.3712, .5088)	(.3680, 0.5118)
Corrected	.508	.0561	(.3980, .6180)	(.0982, .9502)
Bootstrap	.509(mean)	.0571	95% Bootstrap CI (.3973, .6213)	

if $X = 1$ and equal 0 otherwise, taken over the $n_{V.0}$ values with $W = 0$ the $n_{V.1}$ values with $W = 1$, respectively. The usual doubling sampling estimator of the population mean, which is π here, is a weighted average of these means $(1 - p_w)\bar{y}_0 + p_w\bar{y}_1$, which is identical to our $\widehat{\pi}_r$. If we take $n_{.0} - 1$ as approximately $n_{.0}$, $n_{.1} - 1$ as approximately $n_{.1}$ and n as approximately $n - 1$ it can be shown that the square root of the resulting estimated variance in (12.5) of Lohr, is exactly the standard error given in (2.8) under random sampling. (A more explicit connection between our notation and that in Lohr is given in Section 2.5.) The conclusion is that:

We can proceed with the correction for misclassification in the same way for designed double sampling, stratified on values of the error prone measurement, as we would with a random subsample.

This is true for estimation, standard error and Wald intervals but the bootstrap procedure needs to be modified since n_{V0} and n_{V1} are fixed beforehand. Define $a_0 = n_{V0}/n_V$ to be the proportion of the validation sample with $W = 0$. At the first stage we generate $n_{.1b}$ from a Binomial(n, p_w) and take $n_{.0b} = n - n_{.1b}$. We then take $n_{V0b} = a_0 n_V$, rounded to the nearest integer and $n_{V1b} = n - n_{V0b}$ as long as $n_{V0b} \leq n_{.0b}$ and $n_{V1b} \leq n_{.1b}$. If not then either $n_{V0b} = n_{.0b}$ or $n_{V1b} = n_{.1b}$ as needed with the other validation sample size adjusted as needed so $n_{V0b} + n_{V1b} = n_V$. Then n_{00b} is generated from a Binomial $(n_{V0b}, \widehat{\lambda}_{0|0})$ and n_{11b} from a Binomial $(n_{V1b}, \widehat{\lambda}_{1|1})$. With n and n_V fixed, the remaining quantities of the data table can be filled in and the estimate calculated using (2.7) for the bth bootstrap sample.

2.5 Finite populations

Here we have a simple random sample of size n without replacement from a population of size N. If n is quite small relative to N then there is little danger in using the procedures presented in the preceding sections. We can, however, account for the finiteness of the population if needed. It may be the case that $n = N$, so the error prone measure W is available for the whole population.

For examples, with remote sensing, error-prone satellite classifications may be available for every pixal in the area of interest, or all units in a shipment may undergo an inexpensive but error-prone evaluation for conformity.

The finite population correction factor is defined to be $f = 1 - (n/N)$. If $n = N$ then $f = 0$ while $f = 1$ corresponds to the random sample case (i.e., independent observations) treated earlier. Let p_w be the proportion of 1's observed for the W_i's, for which $E(p_w) = \pi_w$, as before. As derived in Section 2.8

$$\begin{aligned} V(p_w) &= \frac{1}{n}[f\pi(1-\pi)(\theta_{1|1}+\theta_{0|0}-1)^2 + \pi(\theta_{1|1}(1-\theta_{1|1}) \\ &\quad - \theta_{0|0}(1-\theta_{0|0})) + \theta_{0|0}(1-\theta_{0|0})]. \end{aligned} \tag{2.9}$$

Notice that $f = 0$ corresponds to a "random sample" and $V(p_w)$ above can be shown to reduce to $\pi_w(1-\pi_w)/n$, as it should.

External Validation Data. If we have external validation data the estimation and inference can proceed as in Section 2.4.2, except that in computing the estimated variance in (2.6) the quantity $p_w(1-p_w)/n$ is replaced by an estimate of $V(p_w)$ in (2.9).

Internal Validation data. With internal validation and either a random sample or a two-phase sample based on the value of W, this is an example of double sampling from a finite population. We can proceed as in Section 2.4.3 using $\widehat{\pi}_r$ as in (2.7). The standard error of $\widehat{\pi}_r$ can be obtained by dividing the quantity in equation (12.4) of Lohr by N^2 and then taking the square root. With a conversion of notation this results in an estimated standard error of $SE(\widehat{\pi}_r) = \hat{V}(\widehat{\pi}_r)^{1/2}$, where

$$\hat{V}(\widehat{\pi}_r) = \frac{N(N-1)}{N^2}\sum_{w=0}^{1}\left(\frac{n_{.w}-1}{n-1} - \frac{n_{Vw}-1}{N-1}\right)\frac{n_{.w}\widehat{\lambda}_{ww}(1-\widehat{\lambda}_{ww})}{n(n_{Vw}-1)n_{Vw}}$$
$$+\frac{1}{n-1}f\sum_{w=0}^{1}\frac{n_{.w}}{n}(\widehat{\lambda}_{ww}-\widehat{\pi}_r)^2.$$

Arbitrary validation sample. Here the n_V units making up the validation sample arise in an arbitrary manner. They may or may not have come from some type of random subsampling. We can view the problem in a different way from before. Let π_{xv} be the proportion of successes in the n_v validated units (this is observed) and $a = n_v/N$ the proportion of the population that is validated. The population proportion π is $\pi = a\pi_{xv} + (1-a)\pi_{xI}$ where π_{xI} is the proportion of successes in the collection of $n_I = n - n_v$ nonvalidated units. From this perspective the problem is simply one of estimating π_{xI}. The n_V validated units can be used to obtain estimated misclassification probabilities. From the nonvalidated units we obtain $p_{wI} = n_{I1}/n_I$, where n_I is the number of non-validated units in the sample, leading to

$$\hat{\pi}_{xI} = \frac{p_{wI} - (1 - \widehat{\theta}_{0|0})}{\widehat{\theta}_{0|0} + \widehat{\theta}_{1|1} - 1} \qquad \text{and} \qquad \widehat{\pi} = a\pi_{xv} + (1-a)\widehat{\pi}_{xI}.$$

We can treat π_{xv} as fixed and so the standard error of $\widehat{\pi}$ is $(1-a)SE(\widehat{\pi}_{xI})$, where $SE(\widehat{\pi}_{xI})$ is obtained using the methods for external validation data described above. This works since the n_v units with both X and W play the role of external validation data relative to the n_I units with just W. The finite population correction factor used in the calculation of $SE(\widehat{\pi}_{xI})$ is $1 - (n_I/(N - n_V))$.

2.6 Multiple measures with no direct validation

It is possible to correct for misclassification without ever observing the true value based on the availability of multiple measures on each unit that allow for estimation of the misclassification parameters and the underlying π. Walter and Irwig (1988) have a fairly comprehensive discussion. The multiple measures might correspond to different evaluators, as in the example in Walter and Irwig (1988), where the measures are from three different radiologists assessing whether there is pleural thickening or not. Or the multiple replicates may be replicate measures, as with multiple application of a questionnaire.

We sketch the key idea in analyzing this problem. Suppose on the ith unit instead of observing a single misclassified unit we observe J measures $W_{i1}, \ldots, W_{iJ}$. The assumption of an equal number on each unit is not essential, but used here for notational expediency. The different replicates might correspond to different observers of methods, each with their own misclassification probabilities, or they may be replicates with each measure behaving in a similar fashion. As long as $J \geq 3$, it is usually possible to estimate the misclassification rates and the true proportion π. If we allow different sensitivity $\theta_{1|1(j)}$ and specificity $\theta_{0|0(j)}$ for the jth measure then there are $2J+1$ total parameters with $P(W_{ij} = 1) = \pi_{W(j)} = \theta_{1|1(j)}\pi + (1 - \theta_{0|0(j)})(1-\pi)$ and $P(W_{ij} = 0) = 1 - \pi_{W(j)}$. We can express $P(W_{ij} = w_{ij})$ as $\pi_{W(j)}^{w_{ij}}(1 - \pi_{W(j)})^{1-w_{ij}}$ for $w_{ij} = 0$ or 1, leading to a likelihood function of

$$L(\boldsymbol{\theta}, \pi) = \prod_{i=1}^{n} \prod_{j=1}^{J} \pi_{W(j)}^{w_{ij}} (1 - \pi_{W(j)})^{1-w_{ij}}$$

where $\boldsymbol{\theta}$ contains the $2J$ unknown misclassification probabilities. From this formulation standard maximum likelihood methods can be applied. In particular the EM algorithm is convenient for obtaining estimates as described by Dawid and Skene (1979). There can be data for which these estimates cannot be obtained. The approximate covariance matrix of the estimates is obtained using an estimate of the inverse of the information matrix, or can be estimated using the bootstrap.

With replicates the sensitivity and specificity are common over j, so $P(W_{ij} = 1) = \pi_W = \theta_{1|1}\pi + (1 - \theta_{0|0})(1 - \pi)$ for each i and j. Notice that if T_i denotes the number of 1's observed from the J replicates on unit i then T_i has a binomial distribution based on J trials and probability of success π_W leading to

$$P(T_i = t) = \left(\begin{array}{c} J \\ t \end{array} \right) \pi_W^t (1 - \pi_W)^{J-t} = \eta_t,$$

say. This can be viewed as a multinomial problem where on each trial the possible values of T are $0, 1, \ldots J$ and one can work directly with the likelihood arising from the totals.

2.7 The multinomial case

In the multinomial setting the X and W take on more than two values. This was seen in the remote sensing example introduced in the first section of this Chapter. Other examples include Theau et al. (2005) where there are 10 categories involved in mapping lichen in Caribou habitat, and Chandramohan et al. (2001) who examine misclassification based on "verbal autopsy" in determination of the cause of death, which has seven possible categories.

The developments for the binary case can be generalized to handle this setting but a matrix-vector formulation is needed. Suppose there are $M \geq 2$ categories with probability π_j that X is in category x; that is $P(X = x) = \pi_x$ for $x = 1, \ldots M$ with $\sum_{x=1}^{M} \pi_x = 1$. Let

$$\boldsymbol{\pi}' = (\pi_1, \ldots \pi_M)$$

but note that $\pi_M = 1 - \pi_1 - \ldots - \pi_{M-1}$. The observed W could have a different number of categories but for discussion here we'll assume it has the same M categories as X with misclassification probabilities

$$P(W = w | X = x) = \theta_{w|x}.$$

This means $\gamma_w = P(W_i = w) = \sum_{x=1}^{M} \theta_{w|x}\pi_x$. In matrix form with $\boldsymbol{\gamma}' = (\gamma_1, \ldots, \gamma_M)$ then

$$\boldsymbol{\gamma} = \boldsymbol{\Theta\pi} \text{ and } \boldsymbol{\pi} = \boldsymbol{\Lambda\gamma},$$

where $\boldsymbol{\Theta}$ is a matrix of misclassification probabilities with $\theta_{w|1}, \ldots \theta_{w|M}$ in the wth row. Similarly, $\boldsymbol{\Lambda}$ is a matrix of reclassification probabilities, using $\lambda_{x|w} = P(X = x|W = w)$.

The vector of naive proportions, $\mathbf{p}_W$ estimate $\boldsymbol{\Theta\pi}$ rather than $\boldsymbol{\pi}$ leading to a bias of $(\boldsymbol{\Theta} - \mathbf{I})\boldsymbol{\pi}$ where $\mathbf{I}$ is an identity matrix.

If we have estimated misclassification rates from external validation data the corrected estimator is $\hat{\boldsymbol{\pi}} = \hat{\boldsymbol{\Theta}}^{-1}\mathbf{p}_W$. With internal validation data the maximum likelihood estimator (assuming all components of $\widehat{\boldsymbol{\pi}}_r$ are between 0 and 1) is $\hat{\boldsymbol{\pi}}_r = \hat{\boldsymbol{\Lambda}}\mathbf{p}$, where $\mathbf{p}$ is the vector of proportions formed using all of the observations.

Mapping example. To illustrate consider the mapping validation data given in Example 3 and suppose there is separate random sample of an area with 1000 pixels yielding proportions $\mathbf{p}'_W = (.25, .10, .15, .20, .30)$, where .25 is the proportion of pixels that are sensed as water, etc. The combined proportions for W over the main and validation data are $\mathbf{p}' = (.208, .181, .175, .183, .245)$.

From the validation data, the estimated misclassification and reclassification rates are

$$\hat{\boldsymbol{\Theta}} = \begin{bmatrix} 0.961 & 0.005 & 0.010 & 0.008 & 0.016 \\ 0.004 & 0.921 & 0.018 & 0.020 & 0.037 \\ 0.008 & 0.035 & 0.833 & 0.061 & 0.063 \\ 0.033 & 0.014 & 0.072 & 0.817 & 0.064 \\ 0.030 & 0.049 & 0.054 & 0.080 & 0.787 \end{bmatrix}$$

and

$$\hat{\boldsymbol{\Lambda}} = \begin{bmatrix} 0.918 & 0.005 & 0.008 & 0.030 & 0.040 \\ 0.004 & 0.899 & 0.030 & 0.011 & 0.056 \\ 0.010 & 0.020 & 0.831 & 0.066 & 0.073 \\ 0.008 & 0.024 & 0.064 & 0.788 & 0.115 \\ 0.012 & 0.034 & 0.051 & 0.047 & 0.856 \end{bmatrix}.$$

Treating the validation data as external and internal, respectively, leads to

$$\widehat{\pi} = \begin{bmatrix} 0.251 \\ 0.087 \\ 0.134 \\ 0.195 \\ 0.337 \end{bmatrix} \quad \text{and} \quad \widehat{\pi}_r = \begin{bmatrix} 0.209 \\ 0.185 \\ 0.181 \\ 0.191 \\ 0.243 \end{bmatrix}.$$

Treating the validation data as external, Table 2.9 provides a bootstrap analysis of the estimated proportions in each of the five categories based on 1000 bootstrap samples. In each bootstrap sample, the naive proportions are generated using a multinomial with $n = 1000$ and probabilities equal to the original naive proportions. For each value of x each row of the validation data table is generated using a multinomial with the sample size fixed to the original number of validation samples for that row and using the estimated misclassification rates for probabilities. While it looks like some of the misclassification probabilities are high (for example, the probability of misclassifying an urban pixel is 1 - .787 = .213) the corrections to the estimated proportions are relatively minor.

The mathematical developments in the next Chapter (see Section 3.8.3)

Table 2.9 *Analysis of mapping data*

TYPE	Estimate Naive	Corrected	SE	Bootstrap 90% CI
Water	0.25	0.251	0.015	(0.227, 0.275
BareGround	0.1	0.087	0.011	(0.069, 0.105)
Deciduous	0.15	0.134	0.015	(0.111, 0.158)
Coniferous	0.2	0.195	0.017	(0.167, 0.224)
Urban	0.3	0.337	0.021	(0.302, 0.371)

could be used to obtain analytical expressions for the covariance of either $\widehat{\boldsymbol{\pi}}$ or $\widehat{\boldsymbol{\pi}}_r$, but we omit any details here. See also Greenland (1988) and the discussion in Kuha et al. (1998) for some further details.

2.8 Mathematical developments

- **Proof of (2.1)**.
  Using double expectations (see Section 13.2) equation (2.1) follows from

$$\begin{aligned} P(W_i = 1) &= \pi_W \\ &= P(W_i = 1|X_i = 0)P(X_i = 0) \\ &\quad + P(W_i = 1|X_i = 1)P(X_i = 1) \\ &= (1 - \theta_{0|0})(1 - \pi) + \theta_{1|1}\pi = \pi(\theta_{1|1} + \theta_{0|0} - 1) + 1 - \theta_{0|0}. \end{aligned}$$

- **Proof of (2.6)**.

$$\hat{\pi} = Z_1/Z_2, \text{ where } Z_1 = p_W - (1 - \widehat{\theta}_{0|0}), \quad Z_2 = \widehat{\theta}_{0|0} + \widehat{\theta}_{1|1} - 1.$$

  Using the delta method for the approximate variance of a ratio (see Section 13.4)

$$V(\hat{\pi}) \approx \frac{1}{(\theta_{0|0} + \theta_{1|1} - 1)^2}\left[V(Z_1) - 2\pi cov(Z_1, Z_2) + \pi^2 V(Z_2)\right]$$

  where

$$V(Z_1) = V(p_W) + V(\widehat{\theta}_{0|0}) = \frac{\pi_W(1 - \pi_W)}{n} + \frac{\theta_{0|0}(1 - \theta_{0|0})}{n_{V0.}}$$

$$V(Z_2) = V(\widehat{\theta}_{0|0} + \widehat{\theta}_{1|1}) = \frac{\theta_{0|0}(1 - \theta_{0|0})}{n_{V0.}} + \frac{\theta_{1|1}(1 - \theta_{1|1})}{n_{V1.}}$$

$$Cov(Z_1, Z_2) = V(\widehat{\theta}_{0|0}) = \frac{\theta_{0|0}(1 - \theta_{0|0})}{n_{V0.}}$$

- **Estimation with Internal validation data.**
Write $f_W(w_i) = P(W_i = w_i)$ and

$$\begin{aligned} f_{WX}(w_i, x_i) &= P(W_i = w_i, X_i = x_i) \\ &= P(X_i = x_i | W_i = w_i) P(W_i = w_i) \\ &= f_{X|W}(x_i | w_i) f_W(w_i) \end{aligned}$$

The joint mass function of all of the data (assuming without loss of generality that the data is ordered so the first n_V have both X and W) is $\prod_{i=1}^{n_V} f_{WX}(w_i, x_i) \prod_{j=n_V}^{n} f_W(w_i)$

$$= \prod_{i=1}^{n_V} f_{X|W}(x_i | w_i; \lambda_{1|1}, \lambda_{0|0}) \prod_{i=1}^{n} f_W(w_i; \pi_W) = L(\lambda_{0|0}, \lambda_{1|1}, \pi_W),$$

where $L(\lambda_{0|0}, \lambda_{1|1}, \pi_W)$ is the **likelihood function** for in terms of $\lambda_{0|0}$, $\lambda_{1|1}$ and π_W. We can maximize the two pieces of the likelihood separately yielding $\widehat{\lambda}_{0|0}$ and $\widehat{\lambda}_{1|1}$ from the validation data and $\widehat{\pi}_w = p_w$. Since $\pi = \pi_W \lambda_{1|1} + (1 - \pi_W)(1 - \lambda_{0|0})$, the MLE of π is $p_W \widehat{\lambda}_{1|1} + (1 - p_W)(1 - \widehat{\lambda}_{0|0})$. The estimated standard error in (2.8) results from taking the square root of an estimate of the approximate variance of $\widehat{\pi}_r$. The approximate variance can be obtained in various ways. One way is to first obtain the asymptotic covariance matrix of p_w, $\widehat{\lambda}_{1|1}$ and $\widehat{\lambda}_{0|0}$ using the inverse of the information matrix (this shows these three quantities are asymptotically uncorrelated) and then use the delta method to obtain the approximate variance of $\widehat{\pi}_r$.

- **Proof of (2.9).**
Consider a simple random sample of n out of N units and let p_x be the proportion of successes in terms of the true values. We will be a bit careless with the notation here and use p_x to denote either this proportion as a random variable (random from the sampling when $n < N$) or to denote the realized value when we condition on the n selected units. The usage should be clear from the context. With misclassification p_x is not observed. Note that n could equal N in which case p_x is π, the proportion of successes in the population. From standard results $E(p_x) = \pi$ and $V(p_x) = f\pi(1 - \pi)/n$. Conditioning on the true values that occur in the n units, then $W_1, \ldots, W_n$ are independent with W_i distributed Bernoulli with $P(W_i = 1 | x_i) = \theta_{1|x_i}$ so $E(W_i = 1 | x_i) = \theta_{1|x_i}$ and $V(W_i = 1 | x_i) = \theta_{1|x_i}(1 - \theta_{1|x_i})$. Write $|\mathbf{x}$ for conditioning on the true values $x_1, \ldots, x_n$. Since $p_w = \sum_{i=1}^{n} W_i / n$,

$$E(p_w | \mathbf{x}) = \sum_{i=1}^{n} E(W_i)/n = \sum_{i=1}^{n} \theta_{1|x_i}/n = p_x \theta_{1|1} + (1 - p_x)\theta_{1|0}$$

and

$$V(p_w|\mathbf{x}) = \frac{\sum_{i=1}^n V(W_i)}{n^2} = \frac{\sum_{i=1}^n \theta_{1|x_i}(1-\theta_{1|x_i})}{n^2}$$
$$= \frac{p_x\theta_{1|1}(1-\theta_{1|1}) + (1-p_x)\theta_{0|0}(1-\theta_{0|0})}{n}.$$

Using a result from Section 13.2, $V(p_w) = E[V(p_w|p_x)] + V[E(p_w|p_x)] = (\theta_{1|1} + \theta_{0|0} - 1)^2 V(p_x) + E[p_x\theta_{1|1}(1-\theta_{1|1}) + (1-p_x)\theta_{0|0}(1-\theta_{0|0})]/n$. Upon substitution for $V(p_x)$ and $E(p_x)$, this yields (2.9).

CHAPTER 3

Misclassification in Two-Way Tables

3.1 Introduction

This chapter is primarily concerned with the case of two binary variables Y and X, with one or both subject to misclassification. This is, of course, a special case of misclassification in general two-way tables but in addition to being important in its own right, isolating this case lets us present the key ideas and methods in the least complicated manner. Section 3.7 comments briefly on the treatment of arbitrary two-way tables. The problem of misclassification in two-way tables, and the 2×2 table in particular, has received considerable attention and generated a vast literature. Some suggested starting points for access to the work on this topic are Kuha et al. (1998), Morrissey and Spiegelman (1999), Höfler (2005) and Fox et al. (2005). Throughout what follows, if one of the variables is a grouping variable and the other a response/dependent variable we use X for the former and Y for the latter.

The remainder of this section provides a couple of motivating examples. Section 3.2 provides background on defining the parameters of interest, different sampling designs and a brief review of how the analysis proceeds without any misclassification. Sections 3.3 and 3.4 define the various misclassification models, the naive estimates, and characterize the properties of naive estimators and tests that ignore the misclassification. Sections 3.5 and 3.6 provide methods to correct for misclassification using external and internal validation data, respectively. Finally Section 3.7 touches on extensions to general two-way tables, followed by some mathematical developments in 3.8.

There is a somewhat overwhelming number of special cases to consider based on the sampling design used and which variables are misclassified. While we touch on all of these to some extent, providing the ingredients needed to both assess biases in naive methods and to correct for misclassification, we do not provide full analytical details (or computational implementation) for all scenarios. In particular, while Section 3.8.3 provides what is needed to get analytical standard errors when correcting for misclassification in both variables, our examples only utilize bootstrap techniques.

Example: Antibiotics/SIDS. This example is from Greenland (1988) and involves a case-control study of the association of antibiotic use by the mother during pregnancy, X, and the occurrence of sudden infant death syndrome (SIDS), Y. The error-prone measurement of antibiotic use, W, is based on a self report from the mother. A validation study of 428 women also determined the true antibiotic use for a women as determined from medical records. The objective is to estimate and compare the probability of SIDS separately for those that did and those that did not use antibiotics during pregnancy and estimate either the odds ratio or relative risk. Notice that for the controls ($Y = 0$), the estimated specificity and sensitivity are .933 and .568 respectively while for the cases ($Y = 1$), the values are .867 and .630. This suggests that misclassification may be differential. These data are analyzed in Sections 3.5.1 and 3.6.1 treating the validation data as external and internal, respectively.

Table 3.1 *Antibiotic use and SIDS example. From Greenland (1988). Used with permission of John Wiley and Sons.*

MAIN STUDY

		Y	
		Controls (0)	Cases (1)
W	No Use (0)	479	442
	Use (1)	101	122
		580	564

VALIDATION DATA

		Control (Y=0)		Cases (Y=1)	
		X		X	
		0 (no use)	1 (use)	0 (no use)	1 (use)
W =	0 (no use)	168	16	143	17
	1 (use)	12	21	22	29
		180	37	165	46

Accident Example. The second example uses data from Hochberg (1977) who investigated misclassification associated with police reports involving accidents. We'll investigate the relationship between wearing a seat belt or not and whether the driver was injured or not for accidents in North Carolina in 1974-1975 with high damage to the car and involving male drivers. We are ignoring the low damage cases and those involving female drivers here. The police report on seat belt use and injury are subject to misclassification, while the nonpolice (NP) classifications based on more intensive inquiry are considered the truth. The validation data are given in Table 3.2 with D being the police report on injury or not (with 0 = uninjured and 1 = injured) and W the police report on seat belt use or not (with 0 = no use and 1 = use). The true values from intensive inquiry are Y and X, respectively. A naive analysis of the main data leads to an estimated proportion injured of .278 for those without seat belt use and .213 for those using a seat belt leading to an estimated difference of 6.5% more injured without seat belt use with a standard error for the difference of .8%. We analyze these data in Sections 3.5.3 and 3.6.3 treating the validation data as external and internal, respectively. We also ignore the misclassification on injury report to analyze these data in Section 3.5.1 to illustrate correcting with misclassification in X only with a random sample and external validation data.

Table 3.2 *Data from Hochberg (1977) on injury and seat belt use for males in accidents with high damage in North Carolina 1974-75. Used with permission of the American Statistical Association.*

MAIN STUDY

		D		
		Not Injured(0)	Injured(1)	
W	No Seat Belt (0)	17476	6746	2422
	Seat Belt (1)	2155	583	2738

VALIDATION DATA

		D = 0		D = 1	
Y	X	W= 0	W= 1	W = 0	W = 1
0	0	299	4	11	1
0	1	20	30	2	2
1	0	59	1	118	0
1	1	9	6	5	9

3.2 Models for true values

Without misclassification, the problem is one of handling a two-way contingency table. We provide a brief review here, but see Agresti (2002) or Fleiss et al. (2003), among many other sources, for a fuller discussion.

There are three model formulations that might be used to describe the parameters of interest, in part related to the type of sampling design used. These involve modeling either the joint behavior of the two variables or the conditional distribution of one given the other.

- A joint model for X and Y. If X and Y are both random, their joint distribution is specified by

$$\gamma_{xy} = P(X = x, Y = y)$$

represented by

		Y 0	Y 1	
X	0	γ_{00}	γ_{01}	$\gamma_{0.}$
	1	γ_{10}	γ_{11}	$\gamma_{1.}$
		$\gamma_{.0}$	$\gamma_{.1}$	1

- The conditional model for Y given X is specified by

$$\pi_x = P(Y = 1|X = x),$$

represented by

		Y 0	Y 1
x	0	$1 - \pi_0$	π_0
	1	$1 - \pi_1$	π_1

If both X and Y are random this is simply the conditional distribution of Y given $X = x$, in which case $\pi_x = \gamma_{x1}/\gamma_{x.}$. Or the x may be fixed as when it designates a group or stratum from which samples are taken. In this case π_x is simply the probability Y equals 1 when sampling from units with a value of x.

- The conditional model for X given Y is specified by

$$\alpha_y = P(X = 1|Y = y),$$

represented by

		y 0	y 1
X	0	$1 - \alpha_0$	$1 - \alpha_1$
	1	α_0	α_1

If both variables are random then $\alpha_y = \gamma_{1y}/\gamma_{.y}$.

In terms of the true values we can consider three sampling designs.

- Random Sample. The pairs (X_i, Y_i) are independent and identically distributed. Note that the independence is among pairs. Within a pair X and Y may be dependent/associated and in fact this association is the main object of interest. This sampling allows for estimation of any of the parameters and functions of them.
- "Prospective" or "cohort" study. Here we pre-stratify based on the value of x and sample independently from those with $X = 1$ and those with $X = 0$. This allows only for estimation of the π 's and functions of them including the odds ratio and relative risk (see below). The term prospective is often used when the X variable represents exposure and Y is disease status to be observed in the future.
- "Case-Control" study. Here we pre-stratify on the outcome and sample independently from the "cases" ($Y = 1$) and the "controls" ($Y = 0$). Only the α's and functions of them can be estimated.

Which parameters are of interest depends on the application. For example, we might be interested in the γ's and functions of them, including various measures of association. When X is an exposure/grouping variable, the focus is typically on the π's and their difference $\pi_1 - \pi_0$ or the ratio π_1/π_0. In epidemiologic applications where X is an exposure variable and Y a disease outcome, the interest is usually in the

$$\text{Odds Ratio} = \Psi = \frac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)} = \frac{\pi_1(1-\pi_0)}{\pi_0(1-\pi_1)}$$

or the relative risk $\pi_1/\pi_0 = \gamma_{11}(\gamma_{01} + \gamma_{00})/(\gamma_{01}(\gamma_{10} + \gamma_{11}))$. The two are related in that the odds ratio is approximately the relative risk if π_0 and π_1 are small (e.g., with rare diseases).

With X representing a grouping or exposure variable, the α's are usually not of any particular interest by themselves. However, the odds ratio can be written as

$$\frac{\pi_1(1-\pi_0)}{\pi_0(1-\pi_1)} = \frac{\gamma_{11}\gamma_{00}}{\gamma_{01}\gamma_{10}} = \frac{\alpha_1(1-\alpha_0)}{\alpha_0(1-\alpha_1)}.$$

This means the odds ratio can be expressed in terms of the joint distribution or either conditional distribution. This has implications for study design in that it means the odds ratio can be estimated when the design is stratified on Y (i.e., a case-control study).

One objective of many analyses is to assess whether the two variables are related or not, often done through a test of "independence." With X and Y both random, X and Y being stochastically independent is equivalent to $\pi_0 = \pi_1$ or $\alpha_0 = \alpha_1$. Notice that under these conditions the odds ratio and relative risk are both equal to 1. Independence is usually tested via a Pearson chi-square, likelihood ratio or Fisher's exact test, available in most statistical packages.

One can also approach the assessment of independence/dependence by using a confidence interval for a difference in proportions $\pi_1 - \pi_0$ or the odds ratio. One rejects independence if the confidence interval for the difference does not contain 0 or that for the odds-ratio does not contain 1. This confidence interval approach to testing is more informative than a straight test via a P-value.

3.3 Misclassification models and naive estimators

Specifying the misclassification model requires some care since we need to account for which variables are potentially misclassified and whether the error is differential or not. If there is misclassification in X then W denotes the error prone version of X, while if Y is subject to misclassification, D denotes the error-prone version.

Both variables misclassified.

If both variables are misclassified, the most general misclassification model specifies

$$\theta_{wd|xy} = P(W = w, D = d|X = x, Y = y). \tag{3.1}$$

To illustrate, consider the accident example introduced earlier, for which

$$\theta_{10|01} = P(W = 1, D = 0|X = 0, Y = 1) = P(\text{ police record seat belt use and no injury}|\text{no seat belt use and injury}).$$

Other θ's are similarly defined.

Only X misclassified.

With only X misclassified, as in the SIDS/antibiotic use example, we use

$$\theta_{w|xy} = P(W = w|X = x, Y = y) \tag{3.2}$$

so $\theta_{1|1y}$ and $\theta_{0|0y}$ are the sensitivity and specificity, respectively, at $Y = y$. The misclassification in X is **differential** (with respect to Y) if $\theta_{w|xy}$ changes with y, while **nondifferential misclassification** assumes that $\theta_{w|xy}$ does not depend on y.

Differential misclassification can arise for a number of reasons. This includes recall bias in a retrospective study where the subject knows their outcome status Y and the error prone W comes from some type of self reporting. Differential misclassification can also result when categories are formed by categorization of a quantitative mismeasured predictor (e.g., body mass index or amount of smoking); see, for example, Flegal et al. (1991) and Gustafson and Le (2002). This is discussed further in Section 6.4.8.

If the misclassification is nondifferential then we write simply

$$\theta_{w|x} = P(W = w|X = x, Y = y).$$

Only Y misclassified.

When only Y is misclassified, we use

$$\theta^*_{d|xy} = P(D = d|X = x, Y = y) \tag{3.3}$$

with the nondifferential case written as

$$\theta^*_{d|y} = P(D = d|Y = y).$$

If there is misclassification in both variables but the misclassification is both independent and nondifferential then

$$P(W = w, D = d|X = x, Y = y) = \theta_{w|x}\theta^*_{d|y}. \tag{3.4}$$

Note that we can also entertain independent but differential misclassification when both are measured with error.

Regardless of the sampling scheme, the naive data can be described as in Table 3.3, where n_{wd} denotes the number of observations with $W = w$ and $D = d$. It could be that $D = Y$ or $X = W$ if only one variable is misclassified. With a random sample, n is fixed, with a "case-control" study $n_{.0}$ and $n_{.1}$ are fixed, and with a "prospective" study $n_{0.}$ and $n_{1.}$ are fixed. The observed cell and marginal proportions are given in Table 3.4.

Note that p_w and π_w carry a different meaning here than in the previous chapter. Here, for example, p_w is the proportion of times D equal 1 among observations with $W = w$.

Table 3.3 *Naive data for a 2 × 2 table.*

		D 0	1	
W	0	n_{00}	n_{01}	$n_{0.}$
	1	n_{10}	n_{11}	$n_{1.}$
		$n_{.0}$	$n_{.1}$	n

The naive estimator of the odds ratio, which we will denote by Ψ, is

$$\hat{\Psi}_{naive} = \frac{a_1(1 - a_0)}{a_0(1 - a_1)} = \frac{n_{11}n_{00}}{n_{10}n_{01}} = \frac{p_1(1 - p_0)}{p_0(1 - p_1)}$$

and other naive estimators are similarly defined.

Table 3.4 *Proportions from naive 2 $\times$ 2 table.*

Proportion	Quantity	Naive Estimate of:
Overall	$p_{wd} = n_{wd}/n$	γ_{wd}
Row	$p_w = n_{w1}/n_{w.}$	π_w
Column	$a_d = n_{1d}/n_{.d}$	α_d

3.4 Behavior of naive analyses

There are numerous combinations to consider based on the sampling design and which of the two variables is potentially misclassified. The nature of the bias in the naive approaches depends on which combination is under consideration. As we will see these biases sometimes depend on the misclassification rates, sometimes on the "reclassification" rates and sometimes on both.

3.4.1 Misclassification in X only

Recall that with error in X only the misclassification model is given by $\theta_{w|xy} = P(W = w|X = x, Y = y)$ or in the case of nondifferential misclassification, simply by $\theta_{w|x}$.

Random Sampling or stratified on y (case-control study).

For either of these cases, the naive estimator of $\alpha_y = P(X = 1|Y = y)$ is a_y. Using the results from Section 2.3, $E(a_y) = \mu_y$, where

$$\mu_0 = \theta_{1|00} + \alpha_0(\theta_{1|10} + \theta_{0|00} - 1) \quad \text{and} \quad \mu_1 = \theta_{1|01} + \alpha_1(\theta_{1|11} + \theta_{0|01} - 1). \tag{3.5}$$

With nondifferential misclassification, where the sensitivity is now $\theta_{1|1}$ and the specificity $\theta_{0|0}$, these become

$$\mu_0 = 1 - \theta_{0|0} + \alpha_0(\theta_{1|1} + \theta_{0|0} - 1) \quad \text{and} \quad \mu_1 = \theta_{1|0} + \alpha_1(\theta_{1|1} + \theta_{0|0} - 1). \tag{3.6}$$

- Testing for independence.
  When the misclassification is differential, independence in the (W, Y) table ($\mu_1 = \mu_0$) is not equivalent to independence in the (X, Y) table ($\alpha_1 = \alpha_0$) and so the naive test for independence is incorrect. On the other hand, if the measurement error is nondifferential, then $\mu_1 - \mu_0 = (\alpha_1 - \alpha_0)(\theta_{1|1} - \theta_{1|0})$. Except for the unlikely scenario where $\theta_{1|1} = \theta_{1|0}$, $\mu_1 = \mu_0$ if and only if $\alpha_1 = \alpha_0$. So,
  With nondifferential misclassification in one variable and no misclassification in the other, naive tests of independence are valid in that they have the

correct size (approximately). With differential misclassification these tests are not valid.

- Bias in naive estimators of odds-ratio.
Recall that we are still in the case of random sampling or stratification based on y and that the odds ratio is denoted by Ψ. The naive estimator of the odd ratio, $\hat{\Psi}_{naive} = a_1(1 - a_0)/a_0(1 - a_1)$, is a consistent estimator of

$$\Psi_{naive} = \frac{\mu_1(1 - \mu_0)}{\mu_0(1 - \mu_1)},$$

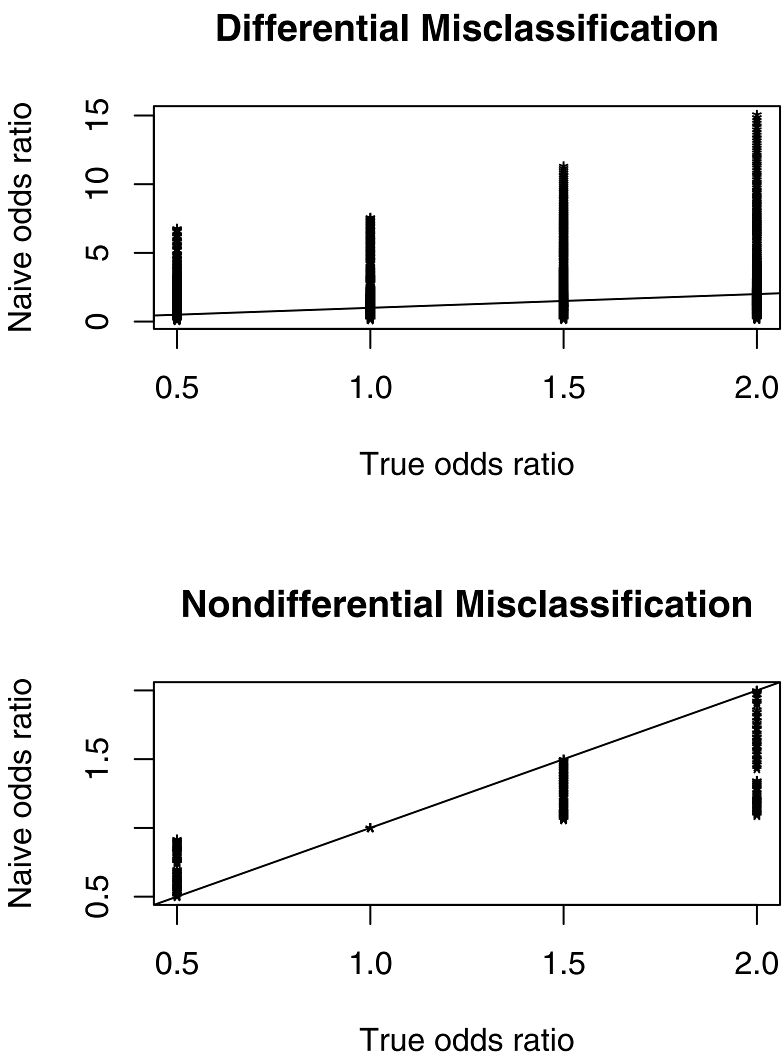
with μ_1 and μ_0 given by (3.5) or (3.6). The approximate bias in the naive estimator of the odds ratio is then $\Psi_{naive} - \Psi$.

The bias, which can be rewritten in various ways, is difficult to characterize. It may go in either direction, either over or underestimating the odds-ratio (on average) when the misclassification is differential. With nondifferential misclassification, things simplify using the expression for μ_0 and μ_1 in (3.6). As discussed in detail by Gustafson (2004, Section 3.3), in this case as long as $\theta_{1|1} + \theta_{0|0} - 1 > 0$ (which will almost always hold) the naive estimator is biased towards the value of 1 in that either $1 \leq \Psi_{naive} \leq \Psi$ or $\Psi \leq \Psi_{naive} \leq 1$. This result characterizes asymptotic bias with increasing sample sizes. It may be possible for the bias to be away from the null in small sample sizes and, of course, there is no guarantee that the actual estimate in an individual study is towards the null; see for example Jurek et al. (2005).

This bias towards the null with nondifferential misclassification is demonstrated in the lower part of Figure 3.1. This plots the limiting value of the naive estimator versus the true odds ratio in cases with nondifferential misclassification. The fact that the asymptotic bias can go in either direction with differential misclassification is seen in the top part of Figure 3.1. In both cases the true odds ratio was set to .5, 1, 1.5 or 2, α_1 ranged from .05 to .95 by .3, α_0 was chosen to yield the desired odds ratios and the sensitivity and specificity (taken to be the same for both values of y in the nondifferential case) ranged from .8 to 1 by .5. With differential misclassification, separate sensitivity and specificity are specified for each y. These ranged from .8 to 1 by .5. Only the cases where either the sensitivity or the specificity differed over y (and so the misclassification is differential) are shown in the top panel of the figure.

Whether the misclassification is differential or not, it is not possible to express the bias in terms of just the odds ratio and the misclassification rates. Further discussion on the effects of misclassification can be found in Hofler (2005) and references therein. Table 3.5 examines the bias issues further through some simulations with a case-control study based on 10000 simulations. This provides the performance of the naive estimators and also shows how well the approximate bias of the naive estimator of the odds

Figure 3.1 *Plot of limiting value of naive estimator of the odds ratio versus true odds ratio in a* 2×2 *table with either differential or nondifferential misclassification. Line indicates where there is no bias.*



ratio does compared to the simulated bias. With samples of 100 there were some outliers in the naive estimator which is why the mean and median values differ for the odds ratio and the approximate bias. This table clearly shows the dramatic difference that can arise in going from differential to nondifferential measurement error. These results can also be interpreted as resulting from a prospective study classified on a perfectly measured X with misclassification in the Y (discussed in the next section) with α_1 and α_0 replaced by π_1 and π_0, respectively.

Table 3.5 *Simulated performance of estimators with misclassification with stratification on y and misclassification in X. a_y = proportion of observations with $W = 1$ when $Y = y$; Diff = $a_1 - a_0$; OR = simulated odds ratio; biasor = simulated bias. True value next to biasor is the asymptotic bias of the naive estimator of the odds ratio. Nondifferential misclassification: sensitivity = .70 and specificity = .85 for each y. Differential misclassification: sensitivity = .70 and specificity = .85 for $Y = 1$ and sensitivity = .60 and specificity = .95 for $Y = 0$.*

n	Misclass.	Variable	True	Mean	Median	Std Dev
100	Differential	a_0	.15	0.133	0.130	0.034
		a_1	.25	0.287	0.290	0.045
		Diff	.10	0.155	0.160	0.057
		OR	1.89	2.921	2.688	1.243
		biasor	.753	1.032	0.799	1.243
100	Nondifferential	a_0	.15	0.232	0.230	0.043
		a_1	.25	0.287	0.290	0.045
		Diff	.10	0.055	0.050	0.062
		OR	1.89	1.417	1.339	0.492
		biasor	-.557	-0.472	-0.550	0.492
500	Differential	a_0	.15	0.132	0.132	0.015
		a_1	.25	0.288	0.288	0.020
		Diff	.10	0.155	0.156	0.026
		OR	1.89	2.697	2.654	0.461
		biasor	.753	0.808	0.765	0.461
500	Nondifferential	a_0	.15	0.232	0.232	0.019
		a_1	.25	0.288	0.288	0.020
		Diff	.10	0.056	0.056	0.028
		OR	1.89	1.351	1.335	0.198
		biasor	-.557	-0.537	-0.554	0.198

- Bias in other quantities for an overall random sample.
  The naive estimator of the cell probability γ_{wy} is p_{wy} = the proportion of

observations with $W = w$ and $Y = y$. As discussed in Section 3.8.1

$$E(p_{wy}) = \mu_{wy} = \sum_x \theta_{w|xy}\gamma_{xy}. \tag{3.7}$$

In the 2×2 table this leads to $\mu_{00} = \theta_{0|00}\gamma_{00} + \theta_{0|10}\gamma_{10}$, $\mu_{01} = \theta_{0|01}\gamma_{01} + \theta_{0|11}\gamma_{11}$, $\mu_{10} = \theta_{1|00}\gamma_{00} + \theta_{1|10}\gamma_{10}$ and $\mu_{11} = \theta_{1|01}\gamma_{01} + \theta_{1|11}\gamma_{11}$ (which also equals $1 - (\mu_{00} + \mu_{01} + \mu_{10})$).

The exact bias of the naive estimator of γ_{wy} is $\mu_{wy} - \gamma_{wy}$. These biases can be used to determine exact or approximate bias for other naive estimators. For example, the asymptotic bias of the naive estimate of π_x is $\mu_{x1}/(\mu_{x1} + \mu_{x0}) - \pi_x$, and of the difference, $\pi_1 - \pi_0$, is

$$\frac{\mu_{11}}{\mu_{11} + \mu_{10}} - \frac{\mu_{01}}{\mu_{01} + \mu_{00}} - (\pi_1 - \pi_0).$$

Similarly, the limiting bias of the naive estimator of relative risk is

$$\frac{\mu_{11}(\mu_{01} + \mu_{00})}{\mu_{01}(\mu_{10} + \mu_{11})} - \frac{\pi_1}{\pi_0}.$$

As before the biases are complicated in general but simplify with nondifferential misclassification.

Marijuana use in college. The biases above will be illustrated based on a study of marijuana use in college by Ellis and Stone (1979). Combining some of their original categories leads to proportions approximately as given in Table 3.6. These will be treated as true cell proportions, with $\gamma_{00} = .3$, $\gamma_{01} = .2$, etc. Also, $\pi_0 = .4$ and $\pi_1 = .6$ are the proportion of students who sometimes use marijuana for those where neither parent used and those where one or both parents used, respectively.

Table 3.6 *Hypothetical true cell proportions for marijuana use.*

		Student use (Y)		
		0 (never)	1 (some)	
Parental	0 (never)	.3	.2	.5
Use (X)	1 (some)	.2	.3	.5
		.5	.5	1

For illustration, assume there is misclassification in the parental use only, and it is nondifferential with $P(W = 1|X = 1) = \theta_{1|1}$ and $P(W = 0|X = 0) = \theta_{0|0}$. Table 3.7 shows the expected or limiting values for the naive estimators of the cell probabilities. Under each parameter is the true value in parentheses and then the corresponding expected value (e.g., μ_{00} under γ_{00}) over different scenarios.

Table 3.7 *Limiting values of naive estimators of γ's π_0, π_1 and the difference $\pi_0 - \pi_1$, with true values in parentheses.*

$\theta_{0\|0}$	$\theta_{1\|1}$	γ_{00}	γ_{01}	γ_{10}	γ_{11}	π_0	π_1	$\pi_0 - \pi_1$
		(.3)	(.2)	(.2)	(.3)	(.4)	(.6)	(-.2)
1	0.9	0.32	0.27	0.18	0.27	0.418	0.6	-.182
0.9	1	0.27	0.32	0.23	0.32	0.4	0.582	-.182
0.95	0.9	0.305	0.28	0.195	0.28	0.419	0.589	-.170
0.95	0.8	0.325	0.25	0.175	0.25	0.435	0.588	-.143

Prospective sampling based on misclassified X.

What if we pre-stratify on w, the potentially misclassified value of x? For example, Suadicani et al. (1997) report on the Copenhagen Male Study investigating the relationship of mortality and morbidity among smokers and nonsmokers, using a prospective study based on self report. The self-report smoking status was subject to misclassification as seen by measuring serum cotinine, treated as an objective marker of use of tobacco, with values greater than or equal to 100 taken as indicating an active smoker.

Recall that p_w is the proportion of observed values with $Y = 1$ and $W = w$. This is the naive estimate of $\pi_w = P(Y = 1|W = w)$. Here we need to utilize reclassification rates since the values of W are fixed. As in the preceding chapter let

$$\lambda_{x|w} = P(X = x|W = w)$$

which is the probability that the truth is x given the error prone measure is w. Assuming that X is a surrogate for W, that is $P(Y = y|x, w) = P(Y = y|x)$, then (see Section 3.8.1) the expected values of the naive estimators are given by

$$E(p_1) = \pi_1\lambda_{1|1} + \pi_0(1 - \lambda_{1|1}) \text{ and } E(p_0) = \pi_0\lambda_{0|0} + \pi_1(1 - \lambda_{0|0}). \quad (3.8)$$

From these the bias in the naive estimators of π_0 and π_1 are $(\pi_0 - \pi_1)(\lambda_{0|0} - 1)$ and $(\pi_1 - \pi_0)(\lambda_{1|1} - 1)$, respectively. These biases depend on the true difference. From these the bias for the difference $\pi_1 - \pi_0$, or the approximate bias of the naive estimator of the odds ratio or relative risk can be easily obtained.

Since $E(p_1) - E(p_0) = (\pi_1 - \pi_0)(\lambda_{1|1} + \lambda_{0|0} - 1)$, then unless $\lambda_{1|1} + \lambda_{0|0} = 1$ (which would be unusual), the naive test for equal proportions $\pi_1 = \pi_0$ is correct.

3.4.2 *Misclassification in Y only*

For this case the roles of X and Y in the previous section can simply be reversed, so there is nothing new here. For the convenience of the reader, however, we will spell out the results in the original notation for the case where X is a grouping variable with no misclassification and the misclassification is now in the response Y. The misclassification probabilities are now denoted by $\theta^*_{d|xy} = P(D = d|X = x, Y = y)$, with $\theta^*_{1|x1}$ and $\theta^*_{0|x0}$ being the sensitivity and specificity, respectively, at x. With nondifferential misclassification (with respect to X) we simply write $\theta^*_{1|1}$ and $\theta^*_{0|0}$ for the sensitivity and specificity.

Random sample.

For an overall random sample the naive cell proportion p_{xd}, the naive estimator of γ_{xd}, has expected value

$$E(p_{xd}) = \mu^*_{xd} = \sum_y \theta^*_{d|xy}\gamma_{xy}.$$

Biases can be computed as in the previous section with μ_{xy} replaced by μ^*_{xy}.

Sampling stratified on x.

The naive estimator of π_x is $p_x = n_{x1}/n_{x.}$. These have expected values

$$\begin{aligned} E(p_0) &= \mu^*_0 = \theta^*_{1|00} + \pi_0(\theta^*_{1|01} - \theta^*_{1|00}) \qquad \text{and} \\ E(p_1) &= \mu^*_1 = \theta^*_{1|10} + \pi_1(\theta^*_{1|11} - \theta^*_{1|10}). \end{aligned}$$

The naive estimate of the difference $\pi_1 - \pi_0$ estimates $\mu^*_1 - \mu^*_0 = \pi_1 - \pi_0 + b^*$, where

$$b^* = \theta^*_{1|10} - \theta^*_{1|00} + \pi_1(\theta^*_{1|11} - \theta^*_{1|10} - 1) - \pi_0(\theta^*_{1|01} - \theta^*_{1|00} - 1) \quad (3.9)$$

and the bias in the odds ratio and relative risk are as in the previous section but with the μ's replaced by μ^*'s. When the misclassification is nondifferential then $b^* = \theta^*_{1|0} - \theta^*_{1|0} + \pi_1(\theta^*_{1|1} - \theta^*_{1|0} - 1) - \pi_0(\theta^*_{1|1} - \theta^*_{1|0} - 1)$.

Similar to the preceding section, so long as $\theta^*_{1|1} - \theta^*_{1|0} \neq 1$, tests for independence are correct if the misclassification in Y is nondifferential with respect to x, but otherwise they are not.

The simulation results given earlier in Table 3.5 can also be applied here if we view them as coming from a prospective study stratified on the true X with misclassification in the response Y and with α_1 and α_0 replaced by π_1 and π_0, respectively. This demonstrates the biases in estimating π_0 and π_1 and the associated odds ratio.

"Case-control" sampling based on misclassified Y.

Here we pre-stratify on d, the potentially misclassified value of Y. Zheng and Tian (2005) provide a number of examples of case-control studies using disease diagnoses that may result in misclassification, including the diagnosis of Alzheimer's disease which can only be definitively determined through autopsy upon death.

For this situation, define the reclassification rates $\lambda^*_{1|1} = P(Y = 1|D = 1)$ and $\lambda^*_{0|0} = P(Y = 0|D = 0)$. These are taken to be free of X. The expected values of the naive estimators of α_0 and α_1 are

$$E(a_1) = \alpha_1\lambda^*_{1|1} + \alpha_0(1 - \lambda^*_{1|1}) \text{ and } E(a_0) = \alpha_0\lambda^*_{0|0} + \alpha_1(1 - \lambda^*_{0|0}).$$

With stratification on the misclassified version of a response Y it is only the odds ratios or the test for "independence" that is of real interest here. The approximate bias in the odds ratio is

$$\frac{E(a_1)(1 - E(a_0))}{E(a_0)(1 - E(a_1))} - \frac{\alpha_1(1 - \alpha_0)}{\alpha_0(1 - \alpha_1)}.$$

Since $E(a_0) - E(a_1) = (\alpha_1 - \alpha_0)(\lambda^*_{1|1} + \lambda^*_{0|0} - 1)$, the naive test for "independence" or odds-ratio equal to 1 is still correct, assuming $\lambda^*_{1|1} + \lambda^*_{0|0} - 1$ is not 0.

3.4.3 Misclassification in X and Y both

Finally, we consider possible misclassification in X and Y both. This occurs in the accident example and would also be true in epidemiologic studies with potential misclassification in both disease diagnosis and exposure. The general misclassification model is as in (3.1), where

$$\theta_{wd|xy} = P(W = w, D = d|X = x, Y = y).$$

For a **random sample**, it is relatively easy to address bias in the naive estimators since the naive estimate of γ_{wv} is p_{wd} which has expected value

$$E(p_{wd}) = \mu_{wd} = \sum_x \sum_y \theta_{wd|xy}\gamma_{xy}. \tag{3.10}$$

This is developed in the same fashion as (3.7). This immediately yields the bias in p_{wd} as an estimator of γ_{wd} and can be used to determine biases (sometimes exact, often approximate) for other quantities in the same manner as in the previous sections. For example the approximate bias in the naive estimate of $\pi_1 - \pi_0$ is

$$\frac{\mu_{11}}{\mu_{1.}} - \frac{\mu_{01}}{\mu_{0.}} - (\pi_1 - \pi_0)$$

and for the odds-ratio is

$$\frac{\mu_{11}\mu_{00}}{\mu_{01}\mu_{10}} - \frac{\gamma_{11}\gamma_{00}}{\gamma_{01}\gamma_{10}}.$$

As with our earlier discussions, the bias expressions are complicated and there is little to be provided in the way of general characterizations of the bias. A special case occurs when the misclassification is both nondifferential and independent, so $P(W = w, D = d|x, y) = \theta_{wd|xy} = \theta_{w|x}\theta^*_{d|y}$, with $\theta_{w|x} = P(W = w|X = x)$ and $\theta^*_{d|y} = P(D = d|Y = y)$.

If the design is prospective, stratified on the error prone w, then we need to utilize the reclassification probabilities for X given W and the misclassification probabilities for D given Y to get the expected value of the naive estimators p_0 and p_1 and associated biases. Similarly if the design is stratified on the error prone d then we need to use the reclassification probabilities for Y given D and the misclassification probabilities for W given X to get the expected value of a_0 and a_1. The results, which are fairly complicated, are outlined in Section 3.8.1.

3.5 Correcting using external validation data

In this section we present methods that allow us to correct for misclassification using estimated or known misclassification rates, obtained from outside the main study. As is always the case with the use of external data, the assumption is that the estimates are transportable to the main study. The nature of the external data and the misclassification rates involved depend on which variables are misclassified and whether the misclassification is differential or not. This is summarized in Table 3.8. For example, if only X is misclassified and the misclassification is nondifferential then the external validation data only needs X and W while if it is differential X, W and Y must be observed simultaneously. We do not treat the situation where there is stratification based on a misclassified variable and external data. This is because the correction needs to use reclassification rates, which would have to be transportable from the external validation data to the main study. This is typically harder to justify than transportability of the misclassification probabilities.

We first outline the general approach to making use of the estimated misclassification rates from external data and then illustrate special cases in subsequent sections. Readers more interested in the applications can skim or skip this general discussion.

The general approach uses the properties of naive proportions (based on misclassified values) from the main study along with the estimated misclassification rates from the validation data to solve for the parameters of interest. We start with a collection of proportions from the main study, captured in a vector $\mathbf{p}$, which are naive estimators of parameters contained in $\boldsymbol{\phi}$. With a random sample and consideration of all of the cell probabilities then

Table 3.8 *Layout of cases for use of external validation data for correcting for misclassification. Case-control indicates design is stratified on y and cohort that it is stratified on x.*

Sampling Design	Variable(s) misclassified	Content of External Data
Random Sample (RS)	X and Y	(X, W, Y, D)
Case- Control or RS	X	(X, W, Y) if differential
Case- Control or RS	X	(X, W) if nondifferential
Cohort or RS	Y	(X, Y, D) if differential
Cohort or RS	Y	(Y, D) if nondifferential

$\mathbf{p}' = (p_{00}, p_{01}, p_{10}, p_{11})$ and $\boldsymbol{\phi}' = (\gamma_{00}, \gamma_{01}, \gamma_{10}, \gamma_{11})$. With a random sample or cohort study with interest only in π_0 and π_1 (or functions of them) we would use $\mathbf{p}' = (p_0, p_1)$ with $\boldsymbol{\phi}' = (\pi_0, \pi_1)$, while for a case-control study $\mathbf{p}' = (a_0, a_1)$ with $\boldsymbol{\phi}' = (\alpha_0, \alpha_1)$.

For each of the settings in Table 3.8, the results in the previous section can be used to write

$$E(\mathbf{p}) = \mathbf{b} + \mathbf{B}\boldsymbol{\phi}, \tag{3.11}$$

where $\mathbf{b}$ and $\mathbf{B}$ are functions of the misclassification rates. In some cases $\mathbf{b}$ is $\mathbf{0}$. The external data are used to estimate the misclassification rates leading to estimates $\hat{\mathbf{b}}$ and $\hat{\mathbf{B}}$. Assuming $\hat{\mathbf{B}}$ is nonsingular, this leads to a corrected estimator

$$\hat{\boldsymbol{\phi}} = \hat{\mathbf{B}}^{-1}(\mathbf{p} - \hat{\mathbf{b}}).$$

As seen later, for some special cases the estimators can be written down without needing to revert to matrix inversion.

Explicit expression for the approximate covariance matrix of $\hat{\boldsymbol{\phi}}$ and functions of it (e.g., estimated odds-ratio or difference in proportions) are given for some special cases in the following section with a more general discussion in 3.8.3. For data analysis an alternative is to use the bootstrap. This needs to be done in a manner which mimics the way the original data was obtained, both in the main study and the external validation study. As in other settings, this avoids the need for analytical expressions, but deprives us of expressions that help understand the contributions to the variability.

3.5.1 Misclassification in X only

If the misclassification is differential then the external data must contain values of W and X for each value of y (0 or 1) with counts as described in Table 3.9. Since we are estimating the misclassification probabilities, the validation design cannot be pre-stratified on W.

Table 3.9 *External validation data at $Y = y$.*

.

		W		
		0	1	
X =	0	$n_{V00(y)}$	$n_{V01(y)}$	$n_{V0.(y)}$
	1	$n_{V10(y)}$	$n_{V11(y)}$	$n_{V1.(y)}$
		$n_{V.0(y)}$	$n_{V.1(y)}$	$n_{V(y)}$

From the validation data,

$$\widehat{\theta}_{1|1y} = n_{V11(y)}/n_{V1.(y)} \text{ (estimated sensitivity at} Y = y)$$

$$\widehat{\theta}_{0|0y} = n_{V00(y)}/n_{V0.(y)} \text{ (estimated specificity at} Y = y),$$

with $\widehat{\theta}_{0|1y} = 1 - \widehat{\theta}_{1|1y}$ and $\widehat{\theta}_{1|0y} = 1 - \widehat{\theta}_{0|0y}$.

If the misclassification is nondifferential then there is one set of external data. The (y) in Table 3.9 can be eliminated, leading to estimated sensitivity $\widehat{\theta}_{1|1} = n_{V11}/n_{V1.}$ and estimated specificity $\widehat{\theta}_{0|0} = n_{V00}/n_{V0.}$. As before, $\widehat{\theta}_{0|1} = 1 - \widehat{\theta}_{1|1}$ and $\widehat{\theta}_{1|0} = 1 - \widehat{\theta}_{0|0}$.

Estimation of α's and the odds ratio.

Suppose the design is either a random sample or case-control with the odds-ratio being of primary interest. Then $\mathbf{p}' = (a_0, a_1)$ and $\boldsymbol{\phi}' = (\alpha_0, \alpha_1)$, and solving (3.5) leads to

$$\hat{\alpha}_0 = (a_0 - \hat{\theta}_{1|00})/(\hat{\theta}_{1|10} - \hat{\theta}_{1|00}) \quad \text{and} \quad \hat{\alpha}_1 = (a_1 - \hat{\theta}_{1|01})/(\hat{\theta}_{1|11} - \hat{\theta}_{1|01}) \tag{3.12}$$

in the case of differential error and

$$\hat{\alpha}_0 = (a_0 - \hat{\theta}_{1|0})/(\hat{\theta}_{1|1} - \hat{\theta}_{1|0}) \quad \text{and} \quad \hat{\alpha}_1 = (a_1 - \hat{\theta}_{1|0})/(\hat{\theta}_{1|1} - \hat{\theta}_{1|0}) \tag{3.13}$$

in the case of nondifferential error. These estimators are all just like the correction in (2.3).

The estimate of the odds ratio is $\widehat{\Psi} = \widehat{\alpha}_1(1 - \widehat{\alpha}_0)/\widehat{\alpha}_0(1 - \widehat{\alpha}_1)$ and of $L = log(\Psi)$, the log-odds ratio is

$$\hat{L} = log(\hat{\alpha}_1) + log(1 - \hat{\alpha}_0) - log(\hat{\alpha}_0) - log(1 - \hat{\alpha}_1).$$

In using the delta method, we have a choice of getting the asymptotic variance of $\widehat{\Psi}$ and forming a Wald interval directly for Ψ or we can instead use the delta method to get an approximate variance for $\widehat{L}$ and a confidence interval for $L = log(\Psi)$ and then obtain the interval for Ψ by transforming the interval for L by taking exponents. Without misclassification, it has become fairly common practice to work with the log of the odds-ratio and then transform back, so we will use that approach here. The resulting interval is not the same as if we applied the delta method directly to $\widehat{\Psi}$, a deficiency of the delta method. This issue does not arise if we form bootstrap confidence intervals using the percentile method, as the intervals for Ψ and $log(\Psi)$ are just monotonic transformations of one another.

Using the delta method

$$V(\hat{L}) \approx \frac{V(\hat{\alpha}_0)}{\alpha_0^2(1-\alpha_0)^2} + \frac{V(\hat{\alpha}_1)}{\alpha_1^2(1-\alpha_1)^2} - 2\frac{cov(\hat{\alpha}_0, \hat{\alpha}_1)}{\alpha_0(1-\alpha_0)\alpha_1(1-\alpha_1)}, \quad (3.14)$$

with the form of $V(\hat{\alpha}_0)$, $V(\hat{\alpha}_1)$ and $cov(\hat{\alpha}_0, \hat{\alpha}_1)$ depending on whether the misclassification is differential or not.

- *Differential Misclassification.*
  If the misclassification is differential data then separate data is used for $y = 0$ and $y = 1$, both in the main study and the validation data. This means $cov(\widehat{\alpha}_0, \widehat{\alpha}_1) = 0$ and the approximate variance of $\widehat{\alpha}_0$ and $\widehat{\alpha}_1$ are obtained in the same way as we obtained the variance of $\widehat{\pi}$ in estimating a single proportion,

$$V(\hat{\alpha}_0) \approx \frac{1}{\Delta_0^2}\left[V(a_0) + (1-\alpha_0)^2 V(\hat{\theta}_{1|00}) + \alpha_0^2 V(\hat{\theta}_{1|10})\right]$$

$$V(\hat{\alpha}_1) \approx \frac{1}{\Delta_1^2}\left[V(a_1) + (1-\alpha_1)^2 V(\hat{\theta}_{1|01}) + \alpha_1^2 V(\hat{\theta}_{1|11})\right]$$

  where $V(a_y) = \mu_y(1-\mu_y)/n_{.y}$, $V(\hat{\theta}_{1|xy}) = \theta_{1|xy}(1-\theta_{1|xy})/n_{Vx.(y)}$, $\Delta_0 = \theta_{1|10} - \theta_{1|00}$ and $\Delta_1 = \theta_{1|11} - \theta_{1|01}$.
- *Nondifferential Misclassification.*
  When the measurement error is nondifferential then in the variances above $\Delta_0 = \Delta_1 = \Delta = \theta_{1|1} - \theta_{1|0}$, while $n_{Vx.(y)}$ is replaced by $n_{Vx.}$. In addition $\hat{\alpha}_0$ and $\hat{\alpha}_1$ are now correlated, with

$$cov(\hat{\alpha}_0, \hat{\alpha}_1)) \approx \frac{1}{\Delta^2}\left[(1-\alpha_0)(1-\alpha_1)V(\hat{\theta}_{1|0}) + \alpha_0\alpha_1 V(\hat{\theta}_{1|1})\right].$$

The standard error of $\hat{L}$, denoted $SE(\hat{L})$, is obtained by taking the square root of the estimate of $V(\hat{L})$, calculated by replacing all unknowns with estimates. The approximate Wald interval for L is $\hat{L} \pm z_{\alpha/2}SE(\hat{L}) = [a, b]$ and for the odds ratio the confidence interval is $[e^a, e^b]$.

SIDS/Antibiotic Example. This example, based on Greenland (1988), was introduced earlier. It is a case-control study, stratified on Y (presence or absence of SIDS) assumed to be determined without error. Here we treat the validation data as if it were external with primary interest in the odds ratio.

Connecting back to our general notation, the naive estimates of the α's are $a_0 = 101/580 = .174$, $a_1 = 122/564 = .216$ and the naive estimate of the odds ratio is $(122 * 479)/(101 * 422) = 1.309$. A naive confidence interval for the odds ratio is (.98,1.76).

Allowing differential misclassification,

$$\begin{aligned}
\hat{\theta}_{0|00} &= .933 = \text{estimated specificity at } y = 0, \\
\hat{\theta}_{1|10} &= .568 = \text{estimated sensitivity at } y = 0, \\
\hat{\theta}_{0|01} &= .867 = \text{estimated specificity at } y = 1, \text{ and} \\
\hat{\theta}_{1|11} &= .630 = \text{estimated sensitivity at } y = 1.
\end{aligned}$$

Corrected estimates and associated inferences are given in top part of Table 3.10. Note the change in the estimated odds ratio, from 1.309 to .7335 (from antibiotic use being a risk to being protective) when allowing for differential misclassification.

Table 3.10 *Analysis of the SIDS-Antibiotic use example with validation data treated as external. Confidence intervals are 95%. Boot SE is the bootstrap estimate of standard error while Boot CI refers to a bootstrap percentile interval.*

Differential Misclassification

Quantity	Naive	Corrected	SE	Wald CI	Boot SE	Boot CI
α_0	.174	.215				
α_1	.216	.167				
Odds ratio	1.309	.7335	.403	(.25,2.15)	.532	(.167,2.17)

Nondifferential Misclassification

Quantity	Naive	Corrected	SE	Wald CI	Boot SE	Boot CI
α_0	.174	.150				
α_1	.216	.234				
Odds ratio	1.309	1.73	.555	(.92, 3.24)	.222	(.963, 4.00)

We also proceed assuming the misclassification is nondifferential. This is for illustration, since given the estimated sensitivities and specificities, assuming these are the same for cases and controls, is not really warranted. As Greenland (1988) illustrated and discussed, this produces very different results. We can see the corrected estimate of the odds ratio is now 1.73. This is dramatically different than the correction assuming differential misclassification, going in the opposite direction. The associated confidence interval is now (.92, 3.24) compared to the (.25, 1.25) under differential misclassification.

The table also shows bootstrap standard errors and 95% percentile intervals based on 5000 bootstrap samples. Under differential misclassification the bootstrap mean is .8260 leading to a bootstrap estimate of bias in the corrected estimator of .8260 - .7335 = .0925. Under nondifferential misclassification, which is of less interest here, the bootstrap mean is 1.950 leading to a bootstrap estimate of bias .222. The confidence intervals are 95% intervals. The Wald intervals for the odds ratio are obtained by transforming the Wald interval for the log (odds ratio).

The calculations for the example were calculated using SAS-IML. Certain aspects of the analysis can also be carried out using the GLLAMM procedure in STATA; see the example in Section 14.3 of Skrondal and Rabe-Hesketh (2004).

Estimation of $\pi_x = P(Y = 1|x)$ *and the difference* $\pi_1 - \pi_0$.

We are still in the setting with misclassification in x only. Here there is a random sample with the goal of estimating and comparing π_0 and π_1 with misclassification in the grouping variable (X). With $\mathbf{p}' = (p_{00}, p_{01}, p_{10}, p_{11})$ and $\boldsymbol{\phi}' = (\gamma_{00}, \gamma_{01}, \gamma_{10}, \gamma_{11})$, $E(\mathbf{p}) = \mathbf{B}\boldsymbol{\phi}$ becomes

$$E\begin{bmatrix} p_{00} \\ p_{01} \\ p_{10} \\ p_{11} \end{bmatrix} = \begin{bmatrix} \theta_{0|00} & 0 & \theta_{0|10} & 0 \\ 0 & \theta_{0|01} & 0 & \theta_{0|11} \\ \theta_{1|00} & 0 & \theta_{1|10} & 0 \\ 0 & \theta_{1|01} & 0 & \theta_{1|11} \end{bmatrix} \begin{bmatrix} \gamma_{00} \\ \gamma_{01} \\ \gamma_{10} \\ \gamma_{11} \end{bmatrix}.$$

Substituting the estimated misclassification rates to form $\hat{\mathbf{B}}$ then the estimate cell proportions are obtained by $\hat{\boldsymbol{\gamma}} = \hat{\mathbf{B}}^{-1}\mathbf{p}$ and from these we obtain

$$\hat{\pi}_x = \hat{\gamma}_{x1}/(\hat{\gamma}_{x0} + \hat{\gamma}_{x1}).$$

Given the structure in $\mathbf{B}$, we can simplify here by isolating two sets of equations, one set for γ_{00} and γ_{10} and another set for γ_{01} and γ_{11} together. This leads to two different systems each involving two equations and two unknowns and exact solutions

$$\begin{aligned}\begin{bmatrix} \widehat{\gamma}_{0y} \\ \widehat{\gamma}_{1y} \end{bmatrix} &= \begin{bmatrix} \widehat{\theta}_{0|0y} & \widehat{\theta}_{0|1y} \\ \widehat{\theta}_{1|0y} & \widehat{\theta}_{1|1y} \end{bmatrix}^{-1} \begin{bmatrix} p_{0y} \\ p_{1y} \end{bmatrix} \\ &= \frac{1}{\widehat{\theta}_{1|1y} + \widehat{\theta}_{0|0y} - 1} \begin{bmatrix} \widehat{\theta}_{1|1y} p_{0y} - \widehat{\theta}_{0|1y} p_{1y} \\ \widehat{\theta}_{0|0y} p_{1y} - \widehat{\theta}_{1|0y} p_{0y} \end{bmatrix}.\end{aligned}$$

Recall $E(p_{wy}) = \mu_{wy}$. The variances and covariances of the observed proportions, which follow from results for multinomial sampling, are given by:

$$V(p_{wy}) = \frac{\mu_{wy}(1-\mu_{wy})}{n}, \text{ and } cov(p_{wy}, p_{w'y'}) = \frac{-\mu_{wy}\mu_{w'y'}}{n} \tag{3.15}$$

for $(wy) \neq (w'y')$. In matrix form,

$$Cov(\mathbf{p}) = [\mathbf{D}_\mu - \boldsymbol{\mu}\boldsymbol{\mu}']/n, \tag{3.16}$$

where $\mathbf{p}$ is the vector of p's, $\boldsymbol{\mu}$ is the vector of μ's and $\mathbf{D}_\mu$ is a diagonal matrix with diagonal elements μ_{00}, μ_{01}, μ_{10} and μ_{11}. With differential misclassification the eight estimated misclassification rates arise from independent binomials with $V(\hat{\theta}_{1|xy}) = \theta_{1|xy}(1-\theta_{1|xy})/n_{Vx.(y)}$ as earlier. With nondifferential misclassification there are four estimated misclassification rates which are independent with $V(\hat{\theta}_{1|x}) = \theta_{1|x}(1-\theta_{1|x})/n_{Vx.}$.

Using $Cov(\mathbf{p})$ and the variance-covariance matrix of the $\widehat{\theta}$'s, the covariance of the $\widehat{\gamma}$'s or of $\widehat{\pi}$'s and their difference can be obtained using the delta method. This is tedious but straightforward. See Section 3.8.3 for some details.

Accident Example. We consider the accident example introduced at the beginning of the chapter but assume that there is only misclassification in seat belt use (X). The injury status (Y) is taken as perfectly measured, so D in the first part of Table 3.2 is taken to be Y and the validation data in Table 3.11 is obtained by pooling some of the validation data. This will be treated here as external validation data. The goal is to estimate and compare the proportion of injuries for those using and not using seat belts. Note that if we had the same objective but with misclassification only in the injury report, then the methods used above for the SIDS/Antibiotic example could be employed but with the role of X and Y reversed. See the next section also.

Table 3.11 *Validation data on seat belt use.*

Y	X	W= 0	W= 1
0	0	310	5
0	1	22	32
1	0	177	1
1	1	14	15

Table 3.12 shows the naive estimates, corrected estimates and analytical and bootstrap standard errors and confidence intervals. The bootstrap sampling generated the validation data using independent binomials for each of the four (X, Y) combinations and with probabilities equal to the estimated misclassification rates. The main data counts were generated using a multinomial with cell probabilities equal to the proportions in the original data. This means we are resampling from a model with parameters equal to the naive estimates.

From this perspective comparing the bootstrap means there are signs of some biases in the correction methods. (This is not due to outliers as the medians are similar.) This is due to the inversion used to do the correction; specifically, the term $\widehat{\theta}_{1|1y} + \widehat{\theta}_{0|0y} - 1$ showing up in the divisor in the corrections. For example, the naive estimate of the difference is $-.086$ and the bootstrap mean is $-.022$ yielding a bootstrap estimate of bias of .064. A simple correction using this would subtract $-.064$ from the $-.086$ to yield a corrected estimate of $-.150$. See Efron and Tibshirani (1993, p. 138) for a discussion of bias correction. This means an estimated decrease of about 15% versus the straight correction which yields a decrease of about 11%. The bias issue also clouds the interpretation of the confidence intervals. Generally the bootstrap interval would be preferred here and there are cases where the percentile interval will compensate for bias. But this is certainly not always the case and is certainly questionable here with a lower bound of $-.105$ in light of the bias correction just discussed.

Table 3.12 *Analysis of accident data assuming only seat belt use is misclassified, with validation data treated as external. Cor. denotes corrected estimates. With SB and SB^c denoting seatbelt use and nonuse, respectively, and I and I^c denoting injury and no injury. $\gamma_{00} = P(SB^c \cap I^c)$, $\gamma_{01} = P(SB^c \cap I)$, $\gamma_{10} = P(SB \cap I^c)$, $\gamma_{11} = P(SB \cap I)$, $\pi_0 = P(I|SB^c)$ and $\pi_1 = P(I|SB)$. $D = \pi_1 - \pi_0$. Confidence intervals are 90%.*

	Estimates				Bootstrap Analysis		
	Naive	Cor.	SE	Wald CI	Mean	SE	CI
γ_{00}	.648	.610	.0162	(.583, .636)	.610	.0167	(.578, .633)
γ_{01}	.250	.233	.0082	(.219, .246)	.231	.0092	(.215, .244)
γ_{10}	.080	.119	.0157	(.0930, .144)	.120	.0166	(.096, .150)
γ_{11}	.022	.040	.0078	(.027, .052)	.041	.0090	(.029, .057)
π_0	.279	.294	.0089	(.279, .308)	.275	.0100	(.258, .291)
π_1	.213	.183	.0446	(.110, .256)	.254	.0473	(.185, .338)
D	$-.086$	$-.111$	.0530	$(-0.200, -0.024)$	$-.022$	.0567	$(-.105, .080)$

3.5.2 Misclassification in Y only

The developments of the preceding section apply so all that is needed is to swap the role of X and Y and convert to our earlier notation. This means replacing a_0 by p_0, a_1 by p_1, α_0 by π_0, α_1 by π_1, μ_0 by μ_0^*, μ_1 by μ_1^*, $\theta_{1|11}$ by $\theta_{1|11}^*$, etc. The final results are given below. This covers the case where the misclassification is in the response and the design is either a random sample or a cohort/prospective design stratified on the perfectly measured X.

The external validation data is relabeled as in Table 3.13. Recall that $\theta_{d|xy}^* = P(D = d|X = x, Y = y)$. At $X = x$, the estimated sensitivity is $\widehat{\theta}_{1|x1}^* =$

Table 3.13 *External Validation data with $X = x$ with misclassification in Y.*

		D 0	1	
Y =	0	$n_{V00(x)}$	$n_{V01(x)}$	$n_{V0.(x)}$
	1	$n_{V10(x)}$	$n_{V11(x)}$	$n_{V1.(x)}$
		$n_{V.0(x)}$	$n_{V.1(x)}$	$n_{V(x)}$

$n_{V11(x)}/n_{V1.(x)}$ and the estimated specificity is $\widehat{\theta}^*_{0|x0} = n_{V00(x)}/n_{V0.(x)}$. The estimates of $\pi_0 = P(Y = 1|X = 0)$ and $\pi_1 = P(Y = 1|X = 1)$ are $\hat{\pi}_0 = (p_0 - \hat{\theta}^*_{1|00})/(\hat{\theta}^*_{1|01} - \hat{\theta}^*_{1|00})$ and $\hat{\pi}_1 = (p_1 - \hat{\theta}^*_{1|10})/(\hat{\theta}^*_{1|11} - \hat{\theta}^*_{1|10})$ in the case of differential error and $\hat{\pi}_0 = (p_0 - \hat{\theta}^*_{1|0})/(\hat{\theta}^*_{1|1} - \hat{\theta}^*_{1|0})$ and $\hat{\pi}_1 = (p_1 - \hat{\theta}^*_{1|0})/(\hat{\theta}^*_{1|1} - \hat{\theta}^*_{1|0})$ in the case of nondifferential error. Notice it is the first subscript after the $|$ that is dropped. For example, $\theta^*_{1|11} = \theta^*_{1|01} = \theta^*_{1|1}$ under nondifferential misclassification. The estimated difference is $\widehat{\pi}_1 - \widehat{\pi}_0$ with variance

$$V(\hat{\pi}_1 - \hat{\pi}_0) = V(\hat{\pi}_0) + V(\hat{\pi}_1) - 2cov(\hat{\pi}_0, \hat{\pi}_1).$$

With *differential misclassification*, $cov(\widehat{\pi}_0, \widehat{\pi}_1) = 0$,

$$V(\hat{\pi}_0) \approx \frac{1}{\Delta_0^2}\left[V(p_0) + (1-\pi_0)^2 V(\hat{\theta}^*_{1|00}) + \pi_0^2 V(\hat{\theta}^*_{1|01})\right],$$

and

$$V(\hat{\pi}_1) \approx \frac{1}{\Delta_1^2}\left[V(p_1) + (1-\pi_1)^2 V(\hat{\theta}^*_{1|10}) + \pi_1^2 V(\hat{\theta}^*_{1|11})\right]$$

where $V(p_x) = \mu^*_x(1-\mu^*_x)/n_{x.}$, $V(\hat{\theta}^*_{1|xy}) = \theta^*_{1|xy}(1-\theta^*_{1|xy})/n_{Vy.(x)}$,, $\Delta_0 = \theta^*_{1|01} - \theta^*_{1|00}$ and $\Delta_1 = \theta^*_{1|11} - \theta^*_{1|10}$.

With *nondifferential misclassification*, $\Delta_0 = \Delta_1 = \theta^*_{1|1} - \theta^*_{1|0}$ while $n_{Vy.(x)}$ is replaced by $n_{Vy.}$ = the number of cases in the validation data with $Y = y$ and

$$cov(\hat{\pi}_0, \hat{\pi}_1)) \approx \frac{1}{\Delta_1^2}\left[(1-\pi_0)(1-\pi_1)V(\hat{\theta}^*_{1|0}) + \pi_0\pi_1 V(\hat{\theta}^*_{1|1})\right].$$

The estimated log odds ratio $L = log(\Psi)$ is estimated by

$$\hat{L} = log(\hat{\pi}_1) + log(1-\hat{\pi}_0) - log(\hat{\pi}_0) - log(1-\hat{\pi}_1)$$

with

$$V(\hat{L}) \approx \frac{V(\hat{\pi}_0)}{\pi_0^2(1-\pi_0)^2} + \frac{V(\hat{\pi}_1)}{\pi_1^2(1-\pi_1)^2} - 2\frac{cov(\hat{\pi}_0, \hat{\pi}_1)}{\pi_0(1-\pi_0)\pi_1(1-\pi_1)}. \quad (3.17)$$

From here we proceed as in the preceding section but now using the expressions for $V(\hat{\pi}_1)$, $V(\hat{\pi}_0)$ and $Cov(\hat{\pi}_1, \hat{\pi}_0)$ as given above.

3.5.3 Misclassification in X and Y both

The case with misclassification in both variable is the most general, and challenging, scenario. Here $E(\mathbf{p}) = \mathbf{B}\boldsymbol{\gamma}$ becomes

$$E\begin{bmatrix} p_{00} \\ p_{01} \\ p_{10} \\ p_{11} \end{bmatrix} = \begin{bmatrix} \theta_{00|00} & \theta_{00|01} & \theta_{00|10} & \theta_{00|11} \\ \theta_{01|00} & \theta_{01|01} & \theta_{01|10} & \theta_{01|11} \\ \theta_{10|00} & \theta_{10|01} & \theta_{10|10} & \theta_{10|11} \\ \theta_{11|00} & \theta_{11|01} & \theta_{11|10} & \theta_{11|11} \end{bmatrix} \begin{bmatrix} \gamma_{00} \\ \gamma_{01} \\ \gamma_{10} \\ \gamma_{11} \end{bmatrix}.$$

In the most general case there are 12 distinct misclassification rates involved, since each column of $\mathbf{B}$ sums to 1. The quantities in $\mathbf{B}$ might take a different form if there are simplifying assumptions, such as nondifferentiality or independent misclassification for X and Y. For example, if both of these hold then $\theta_{wd|xy} = \theta_{w|x}\theta^*_{d|y}$ and these would enter into $\mathbf{B}$.

The estimator of the underlying joint probabilities is

$$\hat{\boldsymbol{\gamma}} = \hat{\mathbf{B}}^{-1}\mathbf{p}. \tag{3.18}$$

The variances of the $\widehat{\theta}$'s and estimates of these variances can be obtained as in earlier cases based on binomials, while the variance-covariance structure of the naive cell proportions is given in (3.15). These can be used along with the results of Section 3.8.3 to obtain an approximate expression for $Cov(\widehat{\boldsymbol{\gamma}})$ and to obtain approximate variances for the estimated π's, their difference, the odds ratio, etc. Our example below, however, will only utilize the bootstrap for obtaining standard errors and confidence intervals.

Accident Example. Here we analyze the accident data treating the validation data as external, but now with both variables measured with error. Instead of analytical based standard errors the bootstrap is used. First a vector of main sample counts is generated using a multinomial with sample size $n_I = 26960$ and a vector of probabilities $\mathbf{p} = (0.6482, .2502, .0799, .0216)$ based on the observed cell proportions. The validation data are generated by generating a set of counts for each (x, y) combination using a multinomial with sample size equal to n_{Vxy}, the number of observations in validation data with $X = x$ and $Y = y$, and probabilities given by the estimated reclassification rates associated with that (x, y) combination. The naive estimates are $\widehat{\pi}_0 = .2785$ and $\widehat{\pi}_1 = .2129$. These are estimates of the probability of injury without and with seat belts, respectively. This leads to an estimated difference of $-.0656$, an estimated standard error of .0083 and a 90% confidence interval of $(-0.0793, -0.0518)$.

The corrected estimates of the γ's are given in Table 3.14 leading to $\widehat{\pi}_0 = .3948$, $\widehat{\pi}_1 = .3188$ with an estimated difference of -.0761. A bootstrap analysis based on 1000 bootstrap samples is also given, including 90% bootstrap percentile intervals. Notice that there is not the same kind of bias issues that arose in our previous treatment of this example, where we ignored the misclassification in injury status. With respect to the difference, the conclusion is rather different for the corrected analysis than it was for the naive analysis. The bootstrap standard error for the difference is .1474, compared to .0083 for the naive analysis, the result of uncertainty associated with the estimated misclassification rates. Also, the 90% percentile inference for the difference is $(-.252, .198)$, so we would not reject the hypothesis of equal injury rate with and without seat belts based on a test of size .10. This is contrary to the conclusion based on the naive analysis.

Table 3.14 *Analysis of accident data with validation data treated as external and both variables misclassified. Cor. denotes corrected estimate. Bootstrap analysis based on 1000 bootstrap samples.*

	Estimates		Bootstrap Analysis			
Parameter	Naive	Cor.	Mean	Median	SE	90% CI
γ_{00}	.648	.508	0.505	0.507	0.0278	(0.460, 0.547)
γ_{01}	.250	.331	0.329	0.329	0.027	(0.289, 0.369)
γ_{10}	.080	.110	0.108	0.108	0.025	(0.071, 0.148)
γ_{11}	.022	.051	0.060	0.053	0.063	(0.026, 0.104)
π_0	.279	.395	0.394	0.392	0.031	(0.350, 0.445)
π_1	.213	.319	0.344	0.333	0.135	(0.160, 0.571)
$\pi_1 - \pi_0$	−.066	−.076	−0.050	−0.070	0.147	(−0.252, 0.198)

3.6 Correcting using internal validation data

Here the validation data are internal as laid out in Table 3.15. The error-prone measure and any variable not subject to misclassification are observed on all n main study units. These may come from any of the three types of designs, with the variables obtained on all observations contained in $\mathbf{O}$. This includes true values that are not subject to misclassification. On a subsample of size n_V we observe the true values for those variables that are subject to misclassification. These quantities are denoted by $\mathbf{T}_1$. What makes up $\mathbf{O}$ and $\mathbf{T}_1$ depends on which variables are misclassified. Without loss of generality, the observations are indexed so the first n_V make up the validation data. The number of observations with $\mathbf{O}$ only is denoted $n_I = n - n_V$, where the I indicates incomplete.

For the accident data presented in Table 3.2, there is misclassification in both so we observe $\mathbf{O} = (W, D)$ only for $n_I = 26960$ observations and observe both $\mathbf{O}$ and $\mathbf{T}_1 = (X, Y)$ (in this case all of the true values) for $n_V = 576$ observations.

Table 3.15 *Layout with internal validation. - indicates missing.*

Sample number	1	$\ldots$	n_V	$n_V + 1$	$\ldots$	n
	O_1	$\ldots$	O_{n_V}	O_{n_V+1}	$\ldots$	O_n
	T_{11}	$\ldots$	T_{1,n_V}	-	$\ldots$	-

Misclassified	O_i	T_{1i}
X only	(W_i, Y_i)	X_i
Y only	(X_i, D_i)	Y_i
X and Y	(W_i, D_i)	(X_i, Y_i)

We first provide a general overview to correcting with internal validation data and then return to specific cases. We discuss three strategies that can be employed, with the later emphasis being on the use of reclassification rates.

Using misclassification rates.

If the validation sample is a random sample of the main study units then one approach is to estimate the misclassification rates and then get corrected estimators as in the Section 3.5. The naive proportions would use all the observed values in $\mathbf{O}$ from all n observations. This has been referred to as the **matrix method**; see Morrissey and Speigelman (1999) and references therein. Note that this approach would not make sense if we stratify on a mismeasured value for the second stage sampling. For example, if X is misclassified and the validation data are obtained by stratifying on the observed w, then we can't estimate the distribution of W given $X = x$. The analytical expressions for the variances given in Section 3.5 do not directly apply since the naive proportions and estimated misclassification rates may be correlated due to use of some common data. New expression can be developed but we will not do so here. The bootstrap can also be used for inference. The use of misclassification rates in this way with internal validation data is often inefficient.

A weighted averaging approach.

When the validation subsample is a random sample of the first stage sample then a second approach is the following. First, working just with the validation sample only, get an estimate for the quantity of interest (e.g., an odds-ratio), say $\widehat{\phi}_V$, and its variance as if the validation sample is all that is available. Then the misclassification rates are obtained from the validation data and used to get a correct estimator from the n_I values with only the first stage data, say $\widehat{\phi}_E$. The variance of this second estimator is obtained using variance expressions from Section 3.5 since the n_I values with O only are independent of the n_V validation samples. Finally a final estimator is formed using a weighted average of $\widehat{\phi}_V$ and $\widehat{\phi}_E$ with weights inversely proportional to their estimated variances; see Morrissey and Spiegelman (1999) and Greenland (1988) for further discussion and illustration in the case where only X is subject to misclassification in a random sample or case-control scenario.

Using reclassification rates.

To use the reclassification rates, we try to express the parameter(s) of interest, contained in $\boldsymbol{\phi}$, as a function of reclassification rates, captured in $\boldsymbol{\Lambda}$ and parameters in $\boldsymbol{\mu}$ associated with the distribution of the always observed $\mathbf{O}$. Suppose this yields $\boldsymbol{\phi} = g(\boldsymbol{\Lambda}, \boldsymbol{\mu})$. We can estimate $\boldsymbol{\mu}$ from the $\mathbf{O}$ values, which are always observed, estimate $\boldsymbol{\Lambda}$ from the internal validation data and then use $\hat{\boldsymbol{\phi}} = g(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Lambda}})$. This often leads to maximum likelihood estimators as discussed in Section 3.8.2.

With all of the approaches we note that there is the possibility that these methods will produce estimates of probabilities that are less than 0 or greater than 1, and some modifications may be necessary. While this might happen infrequently in the original data analysis, it can become more of a problem when bootstrapping.

3.6.1 Misclassification in X only

In this scenario we always observe (W, Y) while the true X is observed on a subsample of size n_V. The validation sample of size n_V can be either a random sample of n_V or a designed subsample stratified on the observed w and y. It is assumed that

$$P(X = x|W = w, Y = y) = \lambda_{x|wy}$$

where as in Chapter 2, $\lambda_{x|wy}$ is referred to as a reclassification rate.

Overall random sample.

Consider first an overall random sample where (see Section 3.8.1)

$$\gamma_{xy} = P(X = x, Y = y) = \sum_w \lambda_{x|wy}\mu_{wy},$$

with $\mu_{wy} = P(W = w, Y = y)$. Collectively we can write $\boldsymbol{\gamma} = \boldsymbol{\Lambda}\boldsymbol{\mu}$ or more explicitly

$$\begin{bmatrix} \gamma_{00} \\ \gamma_{10} \\ \gamma_{01} \\ \gamma_{11} \end{bmatrix} = \begin{bmatrix} \lambda_{0|00} & \lambda_{0|10} & 0 & 0 \\ \lambda_{1|00} & \lambda_{1|10} & 0 & 0 \\ 0 & 0 & \lambda_{0|01} & \lambda_{0|11} \\ 0 & 0 & \lambda_{1|01} & \lambda_{1|11} \end{bmatrix} \begin{bmatrix} \mu_{00} \\ \mu_{10} \\ \mu_{01} \\ \mu_{11} \end{bmatrix}.$$

The MLE of γ is given by $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\Lambda}}\mathbf{p}$. To distinguish this from the matrix method using misclassification rates this has sometimes been called (somewhat ironically since it does not involve an inverse) the **inverse matrix method.** The resulting estimators are of particularly easy form here. Using the fact that there are only four reclassification rates involved since $\lambda_{1|wy} = 1 - \lambda_{0|wy}$ and writing all results in terms of $\lambda_{1|wy}$, we have

$$\widehat{\gamma}_{00} = \widehat{\lambda}_{0|00}p_{00} + (1 - \widehat{\lambda}_{1|10})p_{10}, \quad \widehat{\gamma}_{10} = \widehat{\lambda}_{1|00}p_{00} + \widehat{\lambda}_{1|10}p_{10},$$

$$\widehat{\gamma}_{01} = \widehat{\lambda}_{0|01}p_{01} + (1 - \widehat{\lambda}_{1|11})p_{11} \text{ and } \widehat{\gamma}_{11} = \widehat{\lambda}_{1|01}p_{01} + \widehat{\lambda}_{1|11}p_{11},$$

where these estimates add to 1.

The covariance matrix of $\mathbf{p}$ is given in (3.15). It can be shown that the proportions in $\mathbf{p}$ are uncorrelated with the four estimated reclassification rates. These in turn can be treated as independent with

$$V(\widehat{\lambda}_{1|wy}) = \lambda_{1|wy}(1 - \lambda_{1|wy})/n_{Vwy} \tag{3.19}$$

where n_{Vwy} (assumed to be nonzero) is the number of observations in the validation sample with $W = w$ and $Y = y$.

Using the discussion in Section 3.8.3 the approximate covariance matrix of the estimated cell probabilities is given by

$$Cov(\hat{\boldsymbol{\gamma}}) = \boldsymbol{\Sigma}_{\widehat{\gamma}} \approx \mathbf{A} + \boldsymbol{\Lambda} Cov(\mathbf{p})\boldsymbol{\Lambda}' \tag{3.20}$$

where $\mathbf{A}$ is a 4×4 matrix with $(j, k)th$ element equal to $\boldsymbol{\mu}' C_{jk}\boldsymbol{\mu}$, with $C_{jk} = cov(\mathbf{Z}_j, \mathbf{Z}_k)$, where $\mathbf{Z}'_j$ is the jth row of the matrix $\widehat{\boldsymbol{\Lambda}}$ with $j = 1, \ldots 4$. For example $\mathbf{Z}'_1 = (1 - \widehat{\lambda}_{1|00}, 1 - \widehat{\lambda}_{1|11}, 0, 0)$ and $\mathbf{Z}'_4 = (0, 0, \widehat{\lambda}_{1|01}, \widehat{\lambda}_{1|11})$. Replacing unknowns by estimated values the estimate of $Cov(\hat{\boldsymbol{\gamma}})$ is

$$\widehat{\boldsymbol{\Sigma}}_{\widehat{\gamma}} = \widehat{\mathbf{A}} + \widehat{\boldsymbol{\Lambda}}\widehat{\boldsymbol{\Sigma}}_p\widehat{\boldsymbol{\Lambda}}',$$

where $\widehat{\boldsymbol{\Sigma}}_p = [\widehat{\mathbf{D}}_p - \mathbf{p}\mathbf{p}']/n$ (see (3.16)) and the elements of $\widehat{\mathbf{A}}$ are given by $\widehat{a}_{11} = \widehat{a}_{22} = \mathbf{p}'_0\mathbf{Q}_0\mathbf{p}_0$, $\widehat{a}_{12} = \widehat{a}_{21} = -\widehat{a}_{11}$, $\widehat{a}_{33} = \widehat{a}_{44} = \mathbf{p}'_1\mathbf{Q}_1\mathbf{p}_1$, $\widehat{a}_{34} = \widehat{a}_{43} = -\widehat{a}_{33}$, where $\mathbf{p}'_0 = (p_{00}, p_{10})$, $\mathbf{p}'_1 = (p_{01}, p_{101})$ and for $y = 0$ or 1 $\mathbf{Q}_y$ is a 2×2 diagonal matrix with $V(\widehat{\lambda}_{0|0y})$ and $\hat{V}(\widehat{\lambda}_{1|1y})$ on the diagonals. From this we can also get estimated standard errors for any estimator which is a function of the $\hat{\boldsymbol{\gamma}}$ using the delta method; see Section 3.8.3.

Marijuana Use Example. To illustrate the method, consider the marijuana use example introduced right before Table 3.6. With X denoting parental use and Y student use, as in that table, the actual cell counts from Ellis and Stone (1979) are $n_{00} = 141$, $n_{01} = 94$, $n_{10} = 85$ and $n_{11} = 141$, leading to naive estimates of the cell probabilities as given in Table 3.17. We simulated internal validation data as given in Table 3.16, with only parental use misclassified, leading to estimated reclassification rates of $\widehat{\lambda}_{0|00} = .7917$, $\widehat{\lambda}_{0|01} = .8077$, $\widehat{\lambda}_{1|10} = .9375$ and $\widehat{\lambda}_{1|11} = .9091$. From the misclassification perspective the probabilities involved were chosen to reflect a tendency of the parent to under-report their use. A naive analysis leads to an estimate of π_0 (probability of use by student given use by neither parent) of .421 and of π_1 (probability of use by a student given use by one or more parent) of .593 with a naive estimate of the difference of .172 with a standard error of .043. The corrected analysis is given in Table 3.17, with a change af about 10% in the estimated difference between π_1 and π_0. There is very close agreement between the analysis based on the analytical formulas above and the bootstrap approach. To implement the bootstrap here for each bootstrap sample we first randomly generate (W, Y) pairs (using a multinomial) independently for those that will not be validated and those that will be validated, each using the naive cell proportions. Then within each (W, Y) combination of the validation data the X's are generated using independent binomials and the estimated reclassification rates. New observed proportions and estimated reclassification rates are obtained on the bootstrap samples from which $\widehat{\boldsymbol{\gamma}}$ and other quantities are obtained.

Table 3.16 *Simulated validation data where parental use is misclassified. Y is student use; X is true parental use; W is reported parental use. ≥ 1 indicates 1 or both parents used.*

		Student Use			
		Y=0		Y=1	
		X		X	
		0 (neither)	1 (≥ 1)	0 (neither)	1 (≥ 1)
W =	0 (neither)	19	5	21	5
	1 (≥ 1)	1	15	2	20
		20	20	23	25

Case-control or interest only in odds-ratio.

If we have stratified on the perfectly measured Y, or even if there is an overall random sample but with interest only in estimation of the odds ratio, then we can focus on estimation of α_0 and α_1. Recall that $\alpha_y = P(X = 1|Y =$

Table 3.17 *Analysis of marijuana use data for a random sample of college students with internal validation data assuming parental use is misclassified. Cor. denotes corrected estimate. Confidence intervals are 95%.* $D = \pi_1 - \pi_0$.

	Estimates				Bootstrap Analysis		
	Naive	Cor.	SE	Wald CI	Mean	SE	CI
γ_{00}	.310	.257	.032	(.204, .310)	.257	.031	(.205, .308)
γ_{01}	.189	.207	.028	(.161, .253)	.207	.029	(.159, .256)
γ_{10}	.225	.242	.032	(.190, .295)	.242	.031	(.193, .294)
γ_{11}	.276	.294	.030	(.245, .343)	.294	.031	(.243, .345)
π_0	.421	.446	.048	(.367, .525)	.446	.049	(.366, .528)
π_1	.593	.548	.044	(.476, .620)	.548	.044	(.475, .623)
D	.172	.102	.103	(−.030, .235)	.102	.082	(−0.034, .237)

y), which can also be written as

$$\alpha_y = \sum_w \lambda_{1|wy}(\mu_y w + (1-\mu_y)(1-w)),$$

where $\mu_y = P(W = 1|Y = y) = E(a_y)$. This leads to $\alpha_0 = \lambda_{1|10}\mu_0 + \lambda_{1|00}(1-\mu_0)$ and $\alpha_1 = \lambda_{1|11}\mu_1 + \lambda_{1|01}(1-\mu_1)$ and estimates

$$\widehat{\alpha}_0 = \widehat{\lambda}_{1|10}a_0 + \widehat{\lambda}_{1|00}(1-a_0) \quad \text{and} \quad \widehat{\alpha}_1 = \widehat{\lambda}_{1|11}a_1 + \widehat{\lambda}_{1|01}(1-a_1).$$

At each y we are simply faced with the problem of correcting for estimation of a single proportion using internal validation data. This was covered in Chapter 2. At $Y = y$ the corrected estimator of α_y is exactly like that in (2.7). Since we are using separate data for each y, $Cov(\widehat{\alpha}_0, \widehat{\alpha}_1) = 0$, while using (2.8),

$$V(\widehat{\alpha}_y) \approx \left[\mu_y^2 V(\widehat{\lambda}_{1|1y}) + (\mu_y - 1)^2 V(\widehat{\lambda}_{0|0y}) + (\lambda_{1|1y} - \lambda_{1|0y})^2 V(a_y))\right] \tag{3.21}$$

The variance of the $\widehat{\lambda}$'s is given in (3.19) while, as noted earlier, $V(a_y) = \mu_y(1-\mu_y)/n_{.y}$. This provides what is needed to estimate $V(\widehat{\alpha}_y)$ from which standard errors for the $\widehat{\alpha}$'s and functions of them can be obtained. See equation (3.14) and following for estimation of the odds ratio.

SIDS/Antibiotic Example revisited. We return to the SIDS/antibiotic use example but now treat the validation data as internal. Greenland (1988) provides an analysis where the validation data is considered a random subsample and forms a weighted average of the estimated odds ratio from the validation data only (1.356) and the estimator formed using the main data with W and Y only and the estimated misclassification rates from the validation data. This leads to an estimated odds ratio of 1.22 and a 95% Wald interval of (.79,1.90). Here we analyze the data using the maximum likelihood approach which is based on use of the reclassification rates. This assumes random samples within the cases and controls but allows the subsampling for the validation to possibly be stratified on Y and W. However, the bootstrap analysis is based on random subsampling to obtain the validation data. At $Y = 0$ the estimated reclassification rates/predictive probabilities are $\widehat{\lambda}_{0|00} = .9130$ and $\widehat{\lambda}_{1|10} = .6363$ (the former estimating $P(X = 1|W = 0, Y = 0)$ and the latter $P(X = 1|W = 1, Y = 0)$ while at $Y = 1$ the estimated reclassification rates are $\widehat{\lambda}_{0|01} = .8938$ and $\widehat{\lambda}_{1|11} = .5686$. The resulting analysis for the odds-ratio is given in Table 3.18. The maximum likelihood approach is in almost perfect agreement with the weighted average approach and the bootstrap and analytical confidence intervals for the odds ratio are very close. Notice that the naive estimators differ here from those in Table 3.10 since here they are based on all of the observations while in Table 3.10 they are based only on the nonvalidated data. The Wald interval for the odds ratio is obtained by transforming the Wald interval for the log(odds ratio).

Table 3.18 *Partial analysis of the SIDS-Antibiotic example with the validation data treated as internal. Cor. is corrected estimate. Confidence intervals are 95%. Bootstrap based on 1000 bootstrap samples.*

	Estimates					
Quantity	Naive	Cor.	SE	Wald CI	Boot SE	95% Boot CI
α_0	.168	.179				
α_1	.223	.209				
Odds ratio	1.42	1.21	.2682	(.786, 1.871)	.280	(.790, 1.858)

Prospective study stratified on misclassified X.

Here the strategy for the analysis changes somewhat from the preceding cases since we are stratifying on an error prone W. Using (3.8) and writing $\lambda_{x|w}$ for $P(X = x|W = w)$,

$$\begin{bmatrix} E(p_1) \\ E(p_0) \end{bmatrix} = \begin{bmatrix} 1 - \lambda_{1|1} & \lambda_{1|1} \\ \lambda_{0|0} & 1 - \lambda_{0|0} \end{bmatrix} \begin{bmatrix} \pi_0 \\ \pi_1 \end{bmatrix}$$

leading to

$$\hat{\boldsymbol{\pi}} = \begin{bmatrix} \widehat{\pi}_0 \\ \widehat{\pi}_1 \end{bmatrix} = \frac{1}{1 - \widehat{\lambda}_{1|1} - \widehat{\lambda}_{0|0}} \begin{bmatrix} (1 - \widehat{\lambda}_{0|0})p_1 - \widehat{\lambda}_{1|1}p_0 \\ (1 - \widehat{\lambda}_{1|1}p_0 - \widehat{\lambda}_{0|0}p_1 \end{bmatrix}.$$

The variance of p_w is estimated by $p_w(1 - p_w)/n_{w.}$ and that of $\widehat{\lambda}_{x|w}$ by $\widehat{\lambda}_{x|w}(1 - \widehat{\lambda}_{x|w})/n_{Vw}$ where n_{Vw} is the number of validation samples with

$W = w$. The two p's and the two estimated reclassification rates can be treated as uncorrelated and so the variance-covariance matrix of $\hat{\boldsymbol{\pi}}$ can be calculated in a relatively straightforward way using the multivariate delta method.

3.6.2 *Misclassification in Y only*

If only Y is misclassified then the results of the preceding section apply but reversing the role of X and Y and replacing W by D, the misclassified version of Y. Notationally converting those results requires swapping $X = x$ with $Y = y$, replacing $W = w$ with $D = d$, replacing $\lambda_{x|wy}$ with $\lambda^*_{y|dx}$ (which defines $P(Y = y|D = d, X = x)$), replacing $\lambda_{x|w}$ with $\lambda^*_{y|d} = P(Y = y|D = d)$ for the case where we stratify on D and swapping α_y with π_x.

3.6.3 *Misclassification in X and Y both*

Finally we consider the case where both variables are misclassified and there is an overall random sample. Then

$$\gamma_{xy} = \sum_w \sum_y P(X = x, Y = y|W = w, D = d)P(W = w, D = d)$$

so

$$\begin{bmatrix} \gamma_{00} \\ \gamma_{01} \\ \gamma_{10} \\ \gamma_{11} \end{bmatrix} = \begin{bmatrix} \lambda_{00|00} & \lambda_{00|01} & \lambda_{00|10} & \lambda_{00|11} \\ \lambda_{01|00} & \lambda_{01|01} & \lambda_{01|10} & \lambda_{01|11} \\ \lambda_{10|00} & \lambda_{10|01} & \lambda_{10|10} & \lambda_{10|11} \\ \lambda_{11|00} & \lambda_{11|01} & \lambda_{11|10} & \lambda_{11|11} \end{bmatrix} \begin{bmatrix} \mu_{00} \\ \mu_{01} \\ \mu_{10} \\ \mu_{11} \end{bmatrix}$$

or $\boldsymbol{\gamma} = \boldsymbol{\Lambda}\boldsymbol{\mu}$, where $\mu_{wd} = P(W = w, D = d)$.

With $\mathbf{p} = (p_{00}, p_{01}, p_{10}, p_{11})'$, which estimates $\boldsymbol{\mu}$, the estimated cell probabilities are given by $\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\Lambda}}\mathbf{p}$. Notice that each estimated γ is a linear combination of the naive cell proportions with the coefficients being estimated reclassification rates.

The covariance matrix of $\mathbf{p}$ is given in (3.15) but with y replaced by d. Within a row of $\hat{\boldsymbol{\Lambda}}$ the estimated misclassification rates have a variance-covariance structure arising from a multinomial. So,

$$V(\widehat{\lambda}_{xy|wd}) = \lambda_{xy|wd}(1 - \lambda_{xy|wd})/n_{Vwd},$$

and

$$cov(\widehat{\lambda}_{xy|wd}, \widehat{\lambda}_{x'y'|wd}) = -\lambda_{xy|wd}\lambda_{x'y'|wd}/n_{Vwd} \text{ for} (xy) \neq (x'y'),$$

where n_{Vwd} (assumed to be nonzero) is the number of observations in the validation sample with $W = w$ and $D = d$. Estimated reclassification rates are independent across rows (i.e., for different sets of w and d). The covariance of $\widehat{\boldsymbol{\gamma}}$ can be obtained using (3.23) from the appendix. See also (3.20). As pointed out many times before the variances of functions of $\widehat{\boldsymbol{\gamma}}$ can then be obtained using the delta method.

Accident Example with internal validation data. We revisit the accident data once again this time with both variables misclassified and the validation data treated as a random internal subsample. There are $n_I = 26969$ cases with W and D only and $n_V = 576$ validated cases, also containing X and Y. Combined $n = 27536$ observations have values for W and D. The naive cell proportions are $\mathbf{p} = (0.6487, .249, .0796, .0216)$. A naive analysis of the full n observations has $\widehat{\pi}_0 = .2781$ and $\widehat{\pi}_1 = .2132$ with an estimated difference of -.0649 and an estimated standard error of .0083. The estimated reclassification rates are given by

$$\hat{\boldsymbol{\Lambda}} = \begin{bmatrix} 0.7726098 & 0.0808824 & 0.097561 & 0.0833333 \\ 0.1524548 & 0.8676471 & 0.0243902 & 0 \\ 0.0516796 & 0.0147059 & 0.7317073 & 0.1666667 \\ 0.0232558 & 0.0367647 & 0.1463415 & 0.75 \end{bmatrix}$$

where the proportions in a column are the estimated probabilities that $X = x$ and $Y = x$ given the (w, d) corresponding to that column.

The corrected estimates of the cell probabilities are

$$\begin{bmatrix} \widehat{\gamma}_{00} \\ \widehat{\gamma}_{01} \\ \widehat{\gamma}_{10} \\ \widehat{\gamma}_{11} \end{bmatrix} = \begin{bmatrix} .5310 \\ .3177 \\ .0992 \\ .0522 \end{bmatrix},$$

leading to $\widehat{\pi}_0 = .3743$, $\widehat{\pi}_1 = .3447$ with an estimated difference of -.0297. Table 3.19 provides a bootstrap analysis based on 1000 bootstrap samples. From comparing the bootstrap means (the medians were similar) there is little evidence of bias. The bootstrap standard error for the corrected difference is .0476 and a 90% bootstrap percentile interval for the difference is -.11 to .05. This ranges from an estimated decrease in the injury rate due to seat belt use of 11% to an increase of 5%.

3.7 General two-way tables

As noted in the introduction, 2×2 tables were addressed in detail both for its general usefulness and since it allowed us to present most of the results in fairly simple form. Many of the results developed to this point can be extended fairly easily to handle general two-way tables, where either of X or Y may have more than two categories; say c_x for x and c_y for y. To handle the general setting, what is needed is more liberal use of matrix formulations. It is also true

Table 3.19 *Bootstrap analysis of accident data with internal validation and both variables misclassified.*

Variable	Mean	SE	90% CI
γ_{00}	.5310	0.5819	(0.5049, 0.5573)
γ_{01}	.3175	0.3682	(0.2949, 0.3411)
γ_{10}	.0994	0.1303	(0.0836, 0.1148)
γ_{11}	.0520	0.0799	(0.0395, 0.0651)
π_0	.3742	0.4267	(0.3481, 0.4001)
π_1	.3433	0.4883	(0.2712, 0.4181)
$\pi_1 - \pi_0$	$-.0309$	0.1121	(-0.1070, 0.0489)
Odds ratio (Ψ)	.8890	1.5966	(0.6111, 1.2315)

that some of the simplified analytical expressions used in handling 2×2 tables no longer apply. Kuha et al. (1998) have some general discussion on handling two-way tables.

The case of random sampling carries over almost immediately. As before $\gamma_{xy} = P(X = x, Y = y)$ and the definitions of misclassification rates $\theta_{wd|xy}$ and reclassification rates $\lambda_{xy|wd}$ still apply with more than two categories. The expressions for the expected value of naive proportions (e.g., 3.7) carry over directly as do the methods for correcting for misclassification using estimated misclassification or reclassification rates as outlined in Sections 3.5 and 3.6. Some of the analytical expressions given however are specific to the 2×2 matrix but can be generalized using the results of Section 3.8.3.

To work with conditional models some notational generalization is needed. Consider conditioning on y, with misclassification in X but not in Y. We generalize α_0 and α_1 to vectors $\boldsymbol{\alpha}_y$, $y = 1 \ldots c_y$, where $\boldsymbol{\alpha}_y$ contains the conditional probabilities for the various values of X given $Y = y$. Similarly we generalize a_0 and a_1 to $\boldsymbol{a}_y$ (the empirical proportions for W given $Y = y$). This leads to $\boldsymbol{\mu}_y = E(\boldsymbol{a}_y) = \mathbf{b}_y + \mathbf{B}_y\boldsymbol{\alpha}_y$, where $\mathbf{b}_y$ and $\mathbf{B}_y$ are functions of the misclassification rates at $Y = y$. This leads to expressions for bias that mimic those for the 2×2 table and correction techniques using external data following the general lines presented in the early part of Section 3.5. When working with internal data we have $\boldsymbol{\alpha}_y = \boldsymbol{\Lambda}_y\boldsymbol{\mu}_y$, where $\boldsymbol{\Lambda}_y$ contains the reclassification rates at $Y = y$. This leads to the use of estimates $\widehat{\boldsymbol{\alpha}}_y = \widehat{\boldsymbol{\Lambda}}_y\boldsymbol{a}_y$. Covariance matrices for the resulting estimators, internal or external, can be derived using the general theory in Section 3.8.3. The same approach, with just a change in notation, can be employed when we condition on x with misclassification in Y but not in X, with $\boldsymbol{\alpha}_y$ replaced by $\boldsymbol{\pi}_x$, etc..

3.8 Mathematical developments

3.8.1 Some expected values

- *Proof of (3.5).*
  $E(a_y) = \mu_y = P(W = 1|Y = y) = \sum_x P(W = 1|X = x, Y = y)P(X = x|Y = y) = \sum_x \theta_{1|xy} P(X = x|Y = y)$, leading to $\mu_0 = \theta_{1|00}(1-\alpha_0) + \theta_{1|10}\alpha_0 = \theta_{1|00} + \alpha_0(\theta_{1|10} - \theta_{1|00})$, $\mu_1 = \theta_{1|01}(1-\alpha_1) + \theta_{1|11}\alpha_1 = \theta_{1|01} + \alpha_1(\theta_{1|11} - \theta_{1|01})$ and $\mu_1 - \mu_0 = \alpha_1 - \alpha_0 + b$, where $b = \theta_{1|01} - \theta_{1|00} - \alpha_0(\theta_{1|00} - \theta_{1|10} - 1) + \alpha_1(\theta_{1|11} - \theta_{1|01} - 1)$.
- *Proof of (3.7).*
  $E(p_{wy}) = \mu_{wy} = P(W = w, Y = y) = \sum_x P(W = w, X = x, Y = y) = \sum_x P(W = w|X = x, Y = y)P(X = x, Y = y) = \sum_x \theta_{w|xy}\gamma_{xy}$.
  In vector/matrix form $\boldsymbol{\mu} = \mathbf{B}\boldsymbol{\gamma}$, where $\boldsymbol{\mu}$ and $\boldsymbol{\gamma}$ are column vectors containing the μ_{wy}'s and γ_{xy}'s, respectively, and $\mathbf{B}$ is a matrix depending on the misclassification probabilities.
- *Proof of (3.8).*
  $E(p_1) = P(Y = 1|W = 1) = P(Y = 1|W = 1, X = 1)P(X = 1|W = 1) + P(Y = 1|W = 1, X = 0)P(X = 0|W = 1) = P(Y = 1|X = 1)P(X = 1|W = 1) + P(Y = 1|X = 0)P(X = 0|W = 1) = \pi_1\lambda_{1|1} + \pi_0(1 - \lambda_{1|1})$. A similar development leads to the expression for $E(p_0)$.
- *Expressions with misclassification in both and stratification on one of the mismeasured variables.*
  This was discussed at the end of Section 3.4.3 where we were concerned with the case where the sampling was stratified on a misclassified variable. Consider stratifying on the misclassified X. We assume here that the reclassification rates for X depend only on W and the misclassification of Y is nondifferential (free of X). Then $E(p_0) = P(D = 1|W = 0) =$
  $P(D = 1|W = 0, X = 0)P(X = 0|W = 0) + P(D = 1|W = 0, X = 1)P(X = 1|W = 0)$
  $= P(D = 1|X = 0)P(X = 0|W = 0) + P(D = 1|X = 1)P(X = 1|W = 0)$
  $= P(D = 1|X = 0)\lambda_{0|0} + P(D = 1|X = 1)\lambda_{10}$.
  In addition, $P(D = 1|X = 0) = P(D = 1|X = 0, Y = 1)P(Y = 1|X = 0) + P(D = 1|X = 0, Y = 0)P(Y = 0|X = 0) = \theta^*_{1|1}\pi_0 + \theta^*_{1|0}(1 - \pi_0)$ and $P(D = 1|X = 1) = P(D = 1|X = 1, Y = 1)P(Y = 1|X = 1) + P(D = 1|X = 1, Y = 0)P(Y = 0|X = 1) = \theta^*_{1|1}\pi_1 + \theta^*_{1|0}(1 - \pi_1)$.
  Together these yield
  $$E(p_0) = (\theta^*_{1|1}\pi_0 + \theta^*_{1|0}(1 - \pi_0))\lambda_{0|0} + (\theta^*_{1|1}\pi_1 + \theta^*_{1|0}(1 - \pi_1))\lambda_{10}.$$
  Similar calculation can be carried out for $E(p_1)$ when stratifying on X and

for $E(a_0)$ and $E(a_1)$ if we stratified on Y with these last two depending on reclassification rates for $Y|D$ and misclassification rates for $W|X$.

- *Derivation of γ_{xy} when X is misclassified and random sampling.*
  This result is used in Section 3.6.1. The development is very similar to that of (3.7).
  $\gamma_{xy} = P(X = x, Y = y) = \sum_w P(X = x, Y = y, W = w) =$
  $\sum_w P(X = x|Y = y, W = w)P(W = w, Y = y)$
  $= \sum_w \lambda_{x|wy}\mu_{wy}$, where $\mu_{wy} = P(W = w, Y = y)$.

3.8.2 Estimation using internal validation data

Let $f(\mathbf{o}_i, \boldsymbol{\mu})$ denote the probability mass function for $\mathbf{O}_i$, which, as indicated, involves some parameters, denoted by $\boldsymbol{\mu}$. The nature of this depends on what makes up $\mathbf{O}$ and what sampling design is used. Let $f(\mathbf{t}_{1i}|o_i, \boldsymbol{\Lambda})$ denote the conditional mass function for $\mathbf{T}_{1i}$ given $\mathbf{O}_i = \mathbf{o}_i$, which involves parameters (reclassification rates) captured in $\boldsymbol{\Lambda}$. The likelihood function is

$$L(\boldsymbol{\mu}, \boldsymbol{\Lambda}) = \prod_{i=1}^{n} f(\mathbf{o}_i, \boldsymbol{\mu}) \prod_{i=1}^{n_V} f(\mathbf{t}_{1i}|o_i, \boldsymbol{\Lambda}).$$

The quantities in $\boldsymbol{\mu}$ and $\boldsymbol{\Lambda}$ are usually unlinked so the likelihood can be maximized by maximizing the two pieces separately. In that case the MLE of $\boldsymbol{\phi} = g(\boldsymbol{\mu}, \boldsymbol{\Lambda})$ is $\hat{\boldsymbol{\phi}} = g(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Lambda}})$, as long as the estimates are in the parameter space. With a random sample this leads to $\boldsymbol{\phi} = \boldsymbol{\gamma}$ and $\widehat{\boldsymbol{\gamma}} = \widehat{\boldsymbol{\Lambda}}\mathbf{p}$, with the covariance matrix of $\widehat{\boldsymbol{\gamma}}$ discussed in the next section.

3.8.3 Results for covariance matrices

In a number of cases appearing in this chapter the estimate for a vector of parameters $\boldsymbol{\phi}$ is of the form $\widehat{\boldsymbol{\phi}} = \widehat{\mathbf{H}}\mathbf{p}$ where $\widehat{\mathbf{H}}$ is a random matrix and $\mathbf{p}$ is a vector of cell or marginal proportions. There are two important cases; one with $\widehat{\mathbf{H}} = \widehat{\boldsymbol{\Lambda}}$ (containing estimated reclassification rates), the other with $\widehat{\mathbf{H}} = \widehat{\mathbf{B}}^{-1}$, where $\widehat{\mathbf{B}}$ contains estimated misclassification rates.

Let $E(\mathbf{p}) = \boldsymbol{\mu}$, $E(\widehat{\mathbf{H}}) = \mathbf{H}$ and let $\mathbf{C}_{kj}$ denote the covariance of the kth row of $\widehat{\mathbf{H}}$ with the jth row of $\widehat{\mathbf{H}}$. Using conditioning arguments $Cov(\hat{\boldsymbol{\phi}}) = E(Cov(\widehat{\mathbf{H}}\mathbf{p}|\mathbf{p})) + Cov(E(\widehat{\mathbf{H}}\mathbf{p}|\mathbf{p}))$ where $|\mathbf{p}$ denotes conditioning on $\mathbf{p}$. The outside expectation and covariance are with respect to $\mathbf{p}$ random.

If $E(\mathbf{H}|\mathbf{p}) = \mathbf{H}$, then the second term is $Cov(\mathbf{H}\mathbf{p}|\mathbf{p}) = \mathbf{H}Cov(\mathbf{p})\mathbf{H}'$. The inside of the first term is $Cov(\widehat{\mathbf{H}}\mathbf{p}|\mathbf{p}) = \mathbf{p}' \otimes \mathbf{C} \otimes \mathbf{p}$, where $\mathbf{C}$ is a matrix with $(j,k)th$ block equal to C_{jk} as defined above and $\otimes$ denotes Kronecker product. This means that the $(j,k)th$ term of $Cov(\widehat{\mathbf{H}}\mathbf{p}|\mathbf{p})$ is given by $\mathbf{p}'\mathbf{C}_{jk}\mathbf{p}$. The expectation of $Cov(\widehat{\mathbf{H}}\mathbf{p}|\mathbf{p})$ is approximately $\boldsymbol{\mu}' \otimes \mathbf{C} \otimes \boldsymbol{\mu}$. This leads to

$$Cov(\hat{\boldsymbol{\phi}}) \approx \boldsymbol{\mu}' \otimes \mathbf{C} \otimes \boldsymbol{\mu} + \mathbf{H}Cov(\mathbf{p})\mathbf{H}'. \tag{3.22}$$

Internal validation.

When $\widehat{\mathbf{H}} = \widehat{\boldsymbol{\Lambda}}$ determining the quantities in $\mathbf{C}$ is straightforward since $\widehat{\boldsymbol{\Lambda}}$ contains estimated reclassification rates whose covariance structure is determined by either binomial or multinomial results. This was seen in the special cases in Section 3.6. Note also that $E(\widehat{\boldsymbol{\Lambda}}|\mathbf{p}) = \boldsymbol{\Lambda}$, so in this case the approximate covariance of $\widehat{\boldsymbol{\gamma}}$ is

$$Cov(\hat{\boldsymbol{\gamma}}) \approx \boldsymbol{\mu}' \otimes \mathbf{C} \otimes \boldsymbol{\mu} + \widehat{\boldsymbol{\Lambda}}Cov(\mathbf{p})\widehat{\boldsymbol{\Lambda}}'. \tag{3.23}$$

This was used in equation (3.20) and would also apply in Section 3.6.3.

When $\widehat{\mathbf{H}} = \widehat{\mathbf{B}}^{-1}$, the problem is harder. While $E(\widehat{\mathbf{B}}|\mathbf{p}) = \mathbf{B}$, $E(\widehat{\mathbf{B}}^{-1}|\mathbf{p})$ is only approximately $\mathbf{B}^{-1}$ and this approximation may be poor, in which case using the expression for the approximate covariance of $\widehat{\boldsymbol{\gamma}}$ may not work well. Even if we use the approximation, one more step is needed to get to the $\mathbf{C}$ which involves the covariances of rows of $\widehat{\mathbf{B}}^{-1}$. This can be done using the delta method and results on the derivative of an inverse. This approach is discussed, for example, in the appendices of Greenland (1988) and Rosner et al. (1992). We only sketch the general idea, assuming that $\widehat{\mathbf{B}}$ is $K \times K$ with kth row $\mathbf{b}'_k$. The covariance matrix for the components of $\widehat{\mathbf{B}}$ can be expressed as

$$\boldsymbol{\Sigma}_B = Cov\begin{pmatrix} \mathbf{b}_1 \\ \cdot \\ \cdot \\ \mathbf{b}_K \end{pmatrix}.$$

This is a $K^2 \times K^2$ matrix whose components can be obtained using results from binomial or multinomial sampling, depending on how the estimated misclassification rates arose.

If we write $\mathbf{A} = \widehat{\mathbf{B}}^{-1}$ with ith row equal to $\boldsymbol{a}'_i$, then $\mathbf{C}_{ik} = Cov(\boldsymbol{a}_i, \boldsymbol{a}_k)$ and

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} & \dots & \mathbf{C}_{1K} \\ \mathbf{C}_{21} & \mathbf{C}_{22} & \dots & \mathbf{C}_{2K} \\ \cdot & \cdot & \dots & \cdot \\ \mathbf{C}_{K1} & \cdot & \dots & \mathbf{C}_{KK} \end{bmatrix} \approx \mathbf{Q}\boldsymbol{\Sigma}_B\mathbf{Q}'.$$

The matrix $\mathbf{Q}$ is $K^2 \times K^2$ and the row of $\mathbf{Q}$ corresponding to a_{ij} consists of $(\partial a_{ij}/\partial b_{11}, \partial a_{ij}/\partial b_{12}, \dots, \partial a_{ij}/\partial b_{KK})$.

Using a result on the derivative of a matrix (see for example Fuller (1987, p. 390)) $\partial a_{ij}/\partial b_{mn} = -\boldsymbol{a}_i' E_{mn} \boldsymbol{a}_j$, where E_{mn} is a $K^2 \times K^2$ matrix with 0's everywhere except for a 1 in the (m,n)th position.

Random sampling with misclassification in one variable and external validation data.

This provides the covariance matrix for the estimators at the end of Section 3.5.1, where there is an overall random sample with misclassification in the grouping variable X, external validation data and interest in the cell probabilities or π_0, π_1 and their difference. The same method can be applied in Section 3.5.2 where it is Y that is misclassified. Let $\widehat{\boldsymbol{\gamma}}' = (\widehat{\gamma}_{00}, \widehat{\gamma}_{10}, \widehat{\gamma}_{01}, \widehat{\gamma}_{01})$, $\mathbf{p}' = (p_{00}, p_{10}, p_{01}, p_{01})$ and $\widehat{\boldsymbol{\theta}}' = (\widehat{\theta}_{0|00}, \widehat{\theta}_{1|10}, \widehat{\theta}_{0|01}, \widehat{\theta}_{1|11})$, where the ordering has been chosen for convenience in some of the expressions that follow. While this could be treated through the general theory of the preceding section, since the problem yields explicit expressions for the estimators the multivariate delta method can be applied directly. This leads to

$$Cov(\widehat{\boldsymbol{\gamma}}) \approx \mathbf{D}\boldsymbol{\Sigma}\mathbf{D}'$$

where

$$\boldsymbol{\Sigma} = Cov\begin{pmatrix} \widehat{\boldsymbol{\theta}} \\ \mathbf{p} \end{pmatrix} = \begin{bmatrix} Cov(\widehat{\boldsymbol{\theta}}) & \mathbf{0}_4 \\ \mathbf{0}_4 & Cov(\mathbf{p}) \end{bmatrix},$$

$$\mathbf{D} = \begin{bmatrix} D_{0t} & \mathbf{0}_2 & D_{0p} & \mathbf{0}_2 \\ \mathbf{0}_2 & D_{1t} & \mathbf{0}_2 & D_{1p} \end{bmatrix},$$

$\mathbf{0}_j$ denotes a $j \times j$ matrix of zeros, and with $\Delta_y = \theta_{1|1y} + \theta_{0|0y} - 1$,

$$D_{yt} = \begin{bmatrix} (p_{1y} - \theta_{1|1y}p_{.y})/\Delta_y^2 & (p_{1y} - \theta_{1|0y}p_{.y})/\Delta_y^2 \\ `(p_{0y} + (\theta_{1|1y} - 1)p_{.y})/\Delta_y^2 & (p_{0y} - \theta_{0|0y}p_{.y})/\Delta_y^2 \end{bmatrix} \text{ and}$$

$$D_{yp} = \begin{bmatrix} \theta_{1|1y}/\Delta_y & (\theta_{1|1y} - 1)/\Delta_y \\ (\theta_{0|0y} - 1)/\Delta_y & \theta_{0|0y}/\Delta_y \end{bmatrix}.$$

$Cov(\widehat{\boldsymbol{\theta}})$ is a 4×4 diagonal matrix with diagonal elements $V(\widehat{\theta}_{0|00})$, $V(\widehat{\theta}_{1|10})$,$V(\widehat{\theta}_{0|01})$, and $V(\widehat{\theta}_{1|11})$ respectively and $Cov(\mathbf{p})$ is given in (3.16).

Applying the delta method to marginal proportions.

In the cases with random sampling suppose that we have obtained an estimate $\widehat{\boldsymbol{\gamma}}$ and with covariance $\boldsymbol{\Sigma}_{\widehat{\gamma}}$. This can arise through the use of either internal or external validation data as in Sections 3.5.1 and 3.6.1. Assume the ordering has $\widehat{\boldsymbol{\gamma}} = (\widehat{\gamma}_{00}, \widehat{\gamma}_{01}, \widehat{\gamma}_{10}, \widehat{\gamma}_{11})'$ and $\widehat{\boldsymbol{\pi}} = (\widehat{\pi}_0, \widehat{\pi}_1)'$. Applying the multivariate delta method,

$$Cov(\widehat{\boldsymbol{\pi}}) \approx \mathbf{D}\boldsymbol{\Sigma}_{\widehat{\gamma}}\mathbf{D}',$$

where

$$\mathbf{D} = \left[\begin{array}{llll} -\gamma_{01}/\gamma_{0.}^2 & \gamma_{00}/\gamma_{0.}^2 & 0 & 0 \\ 0 & 0 & -\gamma_{11}/\gamma_{1.}^2 & \widehat{\gamma}_{10}/\gamma_{1.}^2 \end{array} \right].$$

CHAPTER 4

Simple Linear Regression

4.1 Introduction

Regression analysis, in one form or another, is by far the most commonly employed statistical technique. This chapter begins to explore measurement error in regression contexts. While we could proceed right to multiple regression it is instructive to treat the simple linear model by itself first. This problem is of interest in its own right, but also allows us to introduce a number of the main features in a simple context, which also allows us to present some exact results in an easy way.

The simple linear regression model assumes

$$Y_i|x_i = \beta_0 + \beta_1 x_i + \epsilon_i \tag{4.1}$$

where Y is the response, x the predictor and the ϵ_i are assumed uncorrelated with mean 0 and, unless noted otherwise, assumed to have constant variance σ^2. The term ϵ_i is referred to as the **error in the equation** to distinguish it from potential measurement error in the response. It is sometimes important to distinguish between the **functional** case, where the x_i's are treated as fixed values, and the **structural** case, where X_i is random. In the structural setting the X_i's are usually assumed to be i.i.d. with mean μ_X and variance σ^2_X, but that is not essential. The distinction between the functional and structural settings is often downplayed in standard regression problems without measurement error since under appropriate assumptions the resulting analysis for the regression coefficients is the same. In measurement error problems the distinction can be more important. With random X we might also be interested in the correlation between X and Y, $\rho = \sigma_{XY}/\sigma_X\sigma_Y = \beta_1(\sigma_X/\sigma_Y)$ where $\sigma_{XY} = cov(X, Y)$. The main focus, however, is on the regression coefficients and the error variance σ^2.

We allow for error in the predictor and/or the response. Following earlier notation, W denotes the error-prone measurement for X and, if there is error in the response, D denotes the error-prone measurement for Y. The majority of the literature on this problem has assumed no error in Y.

The **naive estimators** of the coefficients and the error variance are

$$\hat{\beta}_{1naive} = S_{WD}/S_{WW}, \qquad \hat{\beta}_{0naive} = \bar{D} - \hat{\beta}_{1naive}\bar{W},$$

and

$$\hat{\sigma}^2_{naive} = \sum_i (D_i - (\hat{\beta}_{0naive} + \hat{\beta}_{1naive}W_i))^2/(n-2),$$

where $\overline{W} = \sum_i W_i/n$, $\overline{D} = \sum_i D_i/n$,

$$S_{WD} = \frac{\sum_i (W_i - \bar{W})(D_i - \overline{D})}{n-1} \text{ and } S_{WW} = \frac{\sum_i (W_i - \bar{W})^2}{n-1}.$$

Without any measurement errors these estimators are well known to be unbiased and are maximum (or restricted maximum) likelihood estimators under normality.

The rest of this section provides some motivating examples followed by an overview of the remainder of the chapter.

Motivating examples.

Soil nitrogen content-corn yield. First discussed by DeGracie and Fuller (1972) and treated in Chapter 1 of Fuller (1987, p.11), this example examines the relationship between corn yield (Y) and soil nitrogen content (X), measured on 11 sites in Marshall County, Iowa. The yields are assumed measured without error. As described by Fuller, the nitrogen content associated with the spatial area that comprises a unit is measured with error both due to spatial subsampling and instrument error in assessing nitrogen content of the soil. The measurement error is assumed additive with a constant variance of 57, so $W_i = x_i + u_i$, with $E(u_i) = 0$ and $V(u_i) = \sigma^2_{ui} = \sigma^2_u = 57$, treated as known.

Gypsy moth egg masses and defoliation. The gypsy moth has become a pest in some parts of the U.S. and knowledge of potential defoliation by gypsy moth is important from a management perspective (Liebhold et al., 1993). In particular, if high defoliation rates are anticipated then control measures, such as aerial spraying, are employed. Table 4.1 contains data from the U.S.D.A. Forest Service, arising from 18 sixty hectare stands (hereafter referred to as units) from three different forests collected with the objective of modeling the relationship between gypsy moth egg mass densities and defoliation with the long term objective of predicting the latter from the former. Here, as in many such studies, it is prohibitively costly to get exact values on the whole unit, in this case a 60 hectare plot. Instead they are estimated through some sampling. This introduces measurement error, which in this case is sampling error. On the ith unit gypsy moth egg mass density and defoliation (expressed as a percent) were obtained on m_i circular subplots, each 0.1 hectares in size. So, the defoliation and egg mass values are paired on the m_i subplots. The table shows the observed mean egg mass density w_i and defoliation d_i, both computed by using the mean over the m_i subplots. The remaining columns contain $\widehat{\sigma}_{ui}$, an estimate of the standard deviation of the measurement error associated with w_i, $\widehat{\sigma}_{qi}$, an estimate of the standard deviation of the measurement error associated with d_i, and $\widehat{\sigma}_{uqi}$, an estimate of the covariance of the two measurement errors; see Section 4.5.3 for details. Note that the two measurement errors are correlated since defoliation and egg mass densities are measured on a common set of subplots. Also note that the estimated measurement error standard deviations vary considerably across units. Certain parts of this data were first examined by Buonaccorsi (1994).

Table 4.1 *Estimated gypsy moth egg mass (w) and defoliation (d) on 18 plots, using m_i subplots on the ith plots. See text for description of other values.*

FOREST	m_i	d_i	$\hat{\sigma}_{qi}$	w_i	$\hat{\sigma}_{ui}$	$\hat{\sigma}_{uqi}$
GW	20	45.50	6.13	69.50	12.47	13.20
GW	20	100.00	0.00	116.30	28.28	0.00
GW	20	25.00	2.67	13.85	3.64	1.64
GW	20	97.50	1.23	104.95	19.54	0.76
GW	20	48.00	2.77	39.45	7.95	−2.74
GW	20	83.50	4.66	29.60	6.47	12.23
MD	15	42.00	8.12	29.47	7.16	−5.07
MD	15	80.67	7.27	41.07	9.57	18.66
MD	15	35.33	7.92	18.20	3.88	18.07
MD	15	14.67	1.65	15.80	4.32	−2.07
MD	15	16.00	1.90	54.99	26.28	25.67
MD	15	18.67	1.92	54.20	12.98	−1.31
MD	15	13.33	2.32	21.13	5.40	−3.17
MM	10	30.50	5.65	72.00	26.53	18.22
MM	10	6.00	0.67	24.00	8.84	1.78
MM	10	7.00	0.82	56.00	14.85	−5.78
MM	10	14.00	4.46	12.00	6.11	12.44
MM	10	11.50	2.48	44.00	28.25	8.22

Nitrogen intake-nitrogen balance: Berkson error.

This example, based on Young et al. (1975) and representative of a number of similar studies, illustrates Berkson error. There are 58 individuals randomized to different levels of nitrogen intake. The intake is controlled by having the subjects eat only prepared meals with known nitrogen content. The actual intake, X, is random (and unobservable) due to variation in nutritional content, serving size, etc. The target intake is a fixed value, so this is a case of Berkson error. The goal of a study like this is to model how balance changes as a function of intake and determine at what intake the expected balance is 0. As seen in Section 4.2, this is a case where the measurement error can be ignored.

Chapter overview.

The focus in this chapter is on additive measurement error models. Section 4.2 defines the additive Berkson error model in which the error prone measure is fixed and the true value is random and establishes that for many purposes this error can be ignored. In Section 4.3 we define the classical additive measurement error model in which the error prone measures are unbiased estimates of the unobserved true values. Section 4.4 provides an assessment of the properties of naive methods that ignore measurement error. Important features of the treatment here include allowing for error in either the response or the predictor, allowing the errors to be correlated if both are measured with error, and allowing the measurement error variances and, if needed, covariance to change across observations.

Estimates that correct for measurement error and associated inferences are presented in Section 4.5. For inferences, both Wald and bootstrap techniques are presented and limitations on the use of the bootstrap are discussed. Correcting for measurement error requires estimates of the measurement error variances. Details on obtaining these in the case of replication are presented in Section 4.5.3. Section 4.6 illustrates the methodology through detailed analyses of the corn yield and defoliation examples. In Sections 4.7 and 4.8 we turn our attention to residual analysis and prediction in the presence of measurement error and illustrate the techniques with the defoliation example.

Some mathematical developments have been isolated in Section 4.9. We have somewhat limited the treatment in this chapter, especially as it is a special case of the next chapter which covers multiple regression. There are some topics (e.g., handling settings with replication on only some observation, alternative estimators, and instrumental variables) which are postponed until the next section, but are obviously applicable here. In fact, the instrumental variables example in Chapter 5 is for a simple linear regression problem.

4.2 The additive Berkson model and consequences

In the Berkson model, the "error-prone value," w_i, is fixed while the true value, X_i, is random. This occurs when w_i is a target "dose" as was the case in the nitrogen balance-intake example introduced above. Other examples include where w is the dose of a drug, a desired temperature or pressure in a machine setting, or a target speed on a treadmill. The true X_i is random and the

Berkson model specifies a model for X_i given w_i. The additive Berkson model with constant variance assumes:

$$X_i = w_i + e_i, \qquad E(e_i) = 0 \text{ and } V(e_i) = \sigma_e^2.$$

If there is measurement error in y then $D_i = y_i + q_i$ with $E(q_i) = 0$ and $V(q_i) = \sigma_q^2$. In this context it is usually reasonable to assume q_i is independent of e_i. Substitution in the original regression model leads to

$$D_i = \beta_0 + \beta_1 w_i + \eta_i,$$

where $\eta_i = \beta_1 e_i + \epsilon_i + q_i$ with

$$E(\eta_i) = 0, \qquad V(\eta_i) = \beta_1^2\sigma_e^2 + \sigma^2 + \sigma_q^2.$$

A key point here is that with w_i fixed, η_i is uncorrelated with w_i (since any random variable is uncorrelated with a constant), implying

For the additive Berkson model, naive inferences for the coefficients, based on regressing D on w, are correct. If there is no error in Y, naive predictions of Y from w, including prediction intervals, are correct. If there is measurement error in Y then the naive prediction intervals for Y from w are not correct.

The last part of the result follows from the fact that predicting Y from w should use $V(Y|w) = \beta_1^2\sigma_e^2 + \sigma^2$, while the naive MSE estimates $\beta_1^2\sigma_e^2 + \sigma^2 + \sigma_q^2$.

Notice that the conclusion that we can ignore the Berkson error in making inferences on the coefficients is dependent on both the additivity of the error and the fact that we are working with a linear model. We will see later that with additive Berkson error in nonlinear models, the naive methods no longer have this robustness. Also, for a nonadditive Berkson error model, a naive analysis is easily seen to be incorrect. For example with a linear Berkson error model $X_i = \lambda_0 + \lambda_1 w_i + \delta_i$, the naive estimate of the slope estimates $\beta_1\lambda_1$ rather than β_1.

4.3 The additive measurement error model

From the start, we allow measurement error in either the predictor or response and for possibly changing measurement error variances. The commonly treated cases with error in the predictor only and/or with constant measurement error variances will be highlighted as special cases. Broadly, the measurement error model specifies the joint behavior of W_i and D_i given x_i and y_i. The additive measurement error model assumes that given x_i and y_i,

$$D_i = y_i + q_i \qquad \text{and} \qquad W_i = x_i + u_i, \tag{4.2}$$

with $E(u_i|x_i) = 0$, $E(q_i|y_i) = 0$,

$$V(u_i|x_i) = \sigma^2_{ui}, \quad V(q_i|y_i) = \sigma^2_{qi} \quad \text{and} \quad Cov(u_i, q_i|x_i, y_i) = \sigma_{uqi}.$$

In the above, q_i is the measurement error in D_i as an estimator of y_i and u_i is the measurement error in W_i as an estimator of x_i. The measurement error is additive in the sense that these have mean 0. We have allowed the measurement error variances and covariances to change with the observation i. This is discussed a bit more below and extensively in Section 6.4.5. A final assumption is that the measurement errors (u_i, q_i) are independent over i and uncorrelated with the ϵ_i. This is also discussed in the remarks below.

Historically, the measurement error literature has concentrated heavily on the special cases where there is no measurement error in y or the measurement error variances/covariance do not change with i; i.e., $\sigma^2_{ui} = \sigma^2_u$, $\sigma^2_{qi} = \sigma^2_q$ and $\sigma_{uqi} = \sigma_{uq}$.

REMARKS:

1. Either of the variables may be observed exactly in which case the appropriate measurement error is set to 0, as is the corresponding variance and covariance. In particular with no error in y, $d_i = y_i$, q_i = 0, $\sigma^2_{qi} = 0$ and $\sigma_{uqi} = 0$.
2. We have allowed a heteroscedastic measurement error model in which the conditional measurement error variances (or covariance if applicable) may change with i. This heteroscedasticity can arise for a number of reasons including a change in sampling effort, the fact that the variance may be related to the true value or simply due to a change in variability in the measuring instrument for different observations. If the variance is related to the true value, then when the underlying true values are random (the structural case), the measurement error variances are also random, while in the functional case they are fixed. With random measurement error variances, to be precise we should distinguish between the conditional and unconditional variance. Rather than belabor this point here, we simply use σ^2_{ui} without distinguishing carefully between the two and point the interested reader to Section 6.4.5 for a fuller discussion.
3. *Measurement error uncorrelated with the error in the equation.* Here is an interesting point that is often overlooked. We have assumed that the measurement errors are uncorrelated with the error in the equation, but it is not always necessary to assume they are independent. Independence may not hold, for example, if the measurement error variance(s) are related to the true values. Interestingly, the assumption that the measurement errors have conditional mean 0 is enough to ensure that they are uncorrelated with ϵ_i; see Section 4.9.

4.4 The behavior of naive analyses

Given what happened with the Berkson model, why should we encounter a problem with naive inferences for coefficients in the presence of additive measurement error? One might assume we could rewrite the model to a linear form with the original coefficients. To see why this fails, fix x_i and substitute $Y_i = D_i - q_i$ and $x_i = W_i - u_i$ into $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$. Then,

$$D_i = \beta_0 + \beta_1 W_i + \epsilon_i^* \tag{4.3}$$

with $\epsilon_i^* = -\beta_1 u_i + \epsilon_i + q_i$. Even though this looks like a regression model with slope β_1, ϵ_i^* is correlated with W_i with $Cov(W_i, \epsilon_i^*) = \sigma_{uqi} - \beta_1 \sigma_{ui}^2$. This violates one of the standard regression assumptions. Another way to look at this is that the new error term does not have conditional mean 0 since $E(\epsilon_i^* | W_i = w_i) = -\beta_1(w_i - x_i)$. This will not be 0 unless $w_i = x_i$ or $\beta_1 = 0$.

It is difficult to state results about the exact behavior of the naive analysis, except for the following special case.

Normal structural model with normal additive measurement error and constant measurement error variances/covariance.

Assume that $X_1, \ldots, X_n$ are i.i.d. (independent and identically distributed) $N(\mu_X, \sigma_X^2)$, that the error in the equation ϵ_i is $N(0, \sigma^2)$ and that given (y_i, x_i) the measurement errors are bivariate normal. As shown in Section 4.9 this leads to

$$D_i | w_i = \gamma_0 + \gamma_1 w_i + \delta_i,$$

where δ_i is normal with mean 0 and variance σ_δ^2 with

$$\gamma_0 = \beta_0 + \frac{\mu_X}{\sigma_X^2 + \sigma_u^2}(\beta_1 \sigma_u^2 - \sigma_{uq}) \tag{4.4}$$

$$\gamma_1 = \left[\frac{\sigma_X^2}{\sigma_X^2 + \sigma_u^2}\right]\beta_1 + \frac{\sigma_{uq}}{\sigma_X^2 + \sigma_u^2} = \beta_1 - \left[\frac{\sigma_u^2}{\sigma_X^2 + \sigma_u^2}\right]\beta_1 + \frac{\sigma_{uq}}{\sigma_X^2 + \sigma_u^2} \tag{4.5}$$

and

$$\sigma_\delta^2 = \sigma^2 + \sigma_q^2 + \beta_1^2 \sigma_x^2 - (\beta_1 \sigma_x^2 + \sigma_{uq})^2 / (\sigma_x^2 + \sigma_u^2). \tag{4.6}$$

The results above give exact bias expressions for the naive estimators under the given assumptions in that

$$E(\hat{\beta}_{0naive}) = \gamma_0, \quad E(\hat{\beta}_{1naive}) = \gamma_1 \quad \text{and } E(\hat{\sigma}^2_{naive}) = \sigma_\delta^2.$$

These also provide approximate/asymptotic bias expressions under weaker assumptions, but with some modified definitions. In general, define

$$\mu_X = \sum_i E(X_i)/n \quad \text{and} \quad \sigma_X^2 = E(S_{XX}),$$

where $S_{XX} = \sum_i (X_i - \overline{X})^2/(n-1)$. This handles either the structural case (without the X_i necessarily being i.i.d.) or the functional cases. In the functional case, $X_i = x_i$ fixed, so $\sigma_X^2 = \sum_{i=1}^n (x_i - \overline{x})^2/n$ and $\mu_X = \sum_{i=1}^n x_i/n$. Also, allowing changing measurement error variances and covariances define

$$\sigma_u^2 = \sum_{i=1}^n \sigma_{ui}^2/n, \qquad \sigma_q^2 = \sum_{i=1}^n \sigma_{qi}^2/n \qquad \text{and } \sigma_{uq} = \sum_{i=1}^n \sigma_{uqi}/n.$$

With the definition above, then even without distributional assumptions and allowing for possible changes in measurement error variances/covariance, the expressions in (4.4), (4.5) and (4.6) provide the approximate/asymptotic means of the naive estimators. We haven't been very precise about the approximation here. One way to view them is as first order approximations arising by replacing S_{WW} and other quantities in the naive estimators with their expected values. More formally, the naive estimators converge in probability to (i.e., are consistent for) γ_0, γ_1 and σ_δ^2 under certain conditions. Given our focus here, we won't spell out exact conditions, but among other things we need $S_{XX} \rightarrow \sigma_X^2$, $\bar{X} \rightarrow \mu_X$, $\sum_{i=1}^n \sigma_{ui}^2/n \rightarrow \sigma_u^2$, etc. where the limits are either numerically (in the functional case) or in probability (in the structural case) as needed. Section 4.9 provides some additional discussion. Of course, aside from the exact case outlined above, (4.4), (4.5) and (4.6) only provide approximate biases and these approximations may not work very well with small sample sizes. This is examined briefly in some simulations below.

With correlated measurement errors, the biases can be complex and the bias in the estimated slope, in particular, can go in either direction. There are special cases, however, where there are more specific results.

Special case: No error in the response or uncorrelated measurement errors.

This is an important and widely studied special case, where $\sigma_{uq} = 0$, for which we have the following results, interpreted to be exact or asymptotic as the case may be.

- The naive estimate of the slope estimates

  $$\gamma_1 = \left[\frac{\sigma_X^2}{\sigma_X^2 + \sigma_u^2}\right]\beta_1 = \kappa\beta_1,$$

  where

  $$\kappa = \left[\frac{\sigma_X^2}{\sigma_X^2 + \sigma_u^2}\right]$$

  is the **reliability ratio**. When $\beta_1 \neq 0$ and $\sigma_u^2 > 0$ then $|\gamma_1| < |\beta_1|$ leading to what is known as **attenuation**. This means that the naive estimator of the slope is biased towards 0.

- $\gamma_1 = 0$ if and only if $\beta_1 = 0$.
  This implies that in general, the naive test of $H_0 : \beta_1 = 0$ is essentially correct. Under the normal structural/normal measurement error model with constant measurement error variance the naive t-test for $\beta_1 = 0$ provides an exact test.

The general statements that "the effect of measurement error is to underestimate the slope but the test for 0 slope are still okay" are not always true. They do hold when there is additive error and either there is error in X only or, if there is measurement error in both variables, the measurement errors are uncorrelated.

While the emphasis is usually on the individual coefficients, the value of the regression function $\beta_0 + \beta_1 x_0 = E(Y|x_0)$ at a specified x_0 is also often of equal, or greater, interest. The naive estimator of this quantity has an expected value (exactly or approximately) of $\gamma_0 + \gamma_1 x_0$. As was the case with individual parameters, the bias is complicated in the most general case with $\sigma_{uq} \neq 0$. In the special case with $\sigma_{uq} = 0$, the bias can be expressed as $\beta_1(\kappa - 1)(x_0 - \mu_X)$. Assuming there is error in X ($\kappa < 1$), this quantity will equal 0 when either $\beta_1 = 0$, or $x_0 = \mu_X$. With $\beta_1 \neq 0$ the bias is negative if $x_0 > \mu_X$ and is positive if $x_0 < \mu_X$.

For estimating the correlation in the structural case,

$$E(\hat{\rho}_{naive}) \approx \frac{\rho + (\sigma_{uq}/\sigma_X\sigma_Y)}{(1 + (\sigma_u^2/\sigma_X^2))^{1/2}(1 + (\sigma_q^2/\sigma_Y^2))^{1/2}}.$$

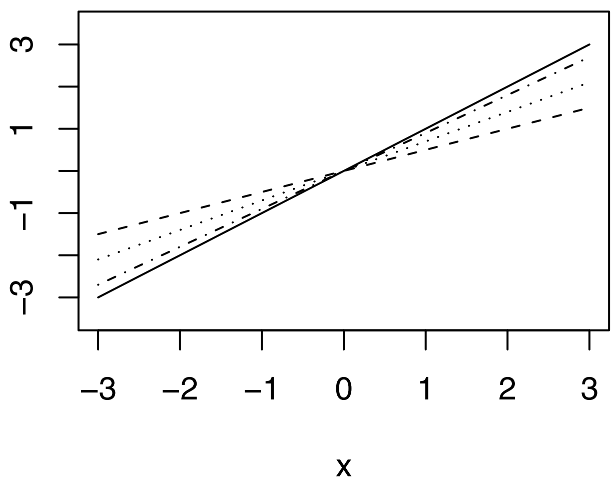
This is approximate even under normality and constant measurement error variances/covariance. In the case with $\sigma_{uq} = 0$, $\hat{\rho}_{naive}$ is estimating $\rho/[(1+(\sigma_u^2/\sigma_X^2))(1+(\sigma_q^2/\sigma_Y^2))]^{1/2}$, so with measurement error in either variable the naive estimator of the correlation is biased towards 0.

Numerical illustrations of bias.

Here we illustrate the biases in the coefficients when there is error in x only, with $\beta_0 = 0$, $\beta_1 = 1$, $\mu_X = 1$,$\sigma_X^2 = 1$ and $\sigma = 1$. In the normal structural model with normal measurement error the bias expressions are exact. With the parameters given here $\gamma_0 = \beta_0 = 0$, so there is no bias in the intercept, while $\gamma_1 = \kappa\beta_1$, where $\kappa = 1/(1 + \sigma_u^2)$. Figure 4.1 displays the true line and the estimated lines for $\kappa = .9$ ($\sigma_u = .3333$), $\kappa = .7$ ($\sigma_u = .6546$) and $\kappa = .5$ ($\sigma_u = 1$). This demonstrates both the attenuation in the slope and that a point on the regression line, $\beta_0 + \beta_1 x_0$, is underestimated if $x_0 > \mu_X = 0$ and overestimated if $x_0 < \mu_X = 0$.

As noted earlier the biases expression given are only exact under the normal structural model with normal measurement error having constant variance.

Figure 4.1 *Illustration of bias in simple linear regression with error in X only. Solid line = true regression line, with intercept* 0 *and slope* 1*. Other lines are what is estimated under reliability ratios $\kappa = .5$ (dots), $\kappa = .7$ (short dash) and $\kappa = .9$ (long dash).*



Otherwise, they are approximate and the question is how well those approximations apply. Using the same parameters as above, in addition to the normal structural case, biases were simulated for the structural model with uniform distributions (for X, the measurement error and the error in the equation) and then for the functional case using normal or uniform distributions (for the measurement error and the error in the equation). In the functional case 5 different x values were spread out uniformly with equal numbers of observations at each x such that $\sum_i (x_i - \bar{x})^2/(n-1) = 1$, to agree with $\sigma_X^2 = 1$ in the structural settings. Samples of size $n = 10, 30, 50$ and 100 were used. For each setting the average estimate of the naive estimate of the slope was obtained over 5000 simulations. The results appear in Table 4.2. The mean, minimum and maximum values for a grouping variable (e.g., type) are found using the four mean values over the other grouping variable (e.g., n). The results shows fairly strong agreement between the expression for the approximate expected value of the naive estimate ($\gamma_1 = \kappa_1 \beta_1$) and its average value over 5000 simulations. The approximation does break down a bit in the functional normal settings with a small sample size of 10, but even there it is not particularly bad.

Table 4.2 *Simulated performance of estimated slope. Approximate limiting value is γ_1. SN = structural + normal; SU = structural + uniform; FN = functional + normal; FU = functional + uniform. n = sample size. True $\beta_1 = 1$.*

Type	Mean	Min	Max	n	Mean	Min	Max
$\gamma_1 = .5$							
FN	0.509	0.502	0.525	10	0.509	0.495	0.525
FU	0.506	0.500	0.517	30	0.502	0.500	0.506
SN	0.500	0.498	0.503	50	0.503	0.501	0.504
SU	0.499	0.495	0.501	100	0.500	0.498	0.502
$\gamma_1 = .7$							
FN	0.713	0.703	0.736	10	0.719	0.699	0.736
FU	0.709	0.702	0.725	30	0.704	0.699	0.707
SN	0.699	0.699	0.699	50	0.704	0.699	0.707
SU	0.705	0.701	0.715	100	0.701	0.699	0.703
$\gamma_1 = .9$							
FN	0.905	0.902	0.914	10	0.909	0.897	0.917
FU	0.906	0.901	0.917	30	0.902	0.900	0.903
SN	0.900	0.897	0.901	50	0.902	0.901	0.903
SU	0.903	0.901	0.907	100	0.902	0.901	0.903

4.5 Correcting for additive measurement error

In order to correct for measurement error, some additional information or data is usually required. There is a long and rich history and a plethora of techniques for correcting for measurement error in simple linear regression. Our discussion of correction techniques is limited somewhat here. The interested reader can consult Fuller (1987) and Cheng and Van Ness (1999) and extensive references therein for discussion of additional methods. Some of the available techniques are based on knowledge of some of the variances or function of them, such as the reliability ratio or the "ratio of measurement error variances." The latter has been put in quotes since what is actually being assumed known in the latter case (sometimes leading to what is known as orthogonal regression) is the value of $(\sigma_q^2 + \sigma^2)/\sigma_u^2$. This is only the ratio of the measurement error variances when there is no error in the equation. It only seems reasonable to offer up a value for this ratio when there is no error in the equation ($\sigma^2 = 0$), or we know something about the ratio of the measurement error variances to the variance of the error in the equation. Neither is very feasible in practice. Assuming the reliability ratio known is more realistic (but see the discussion in Section 6.5.2 for some caveats) although in practice it also is never known exactly.

The focus here is on cases where there are data available to estimate the measurement error variance(s) and, if needed, covariance. Let $\hat{\sigma}^2_{ui}$, $\hat{\sigma}^2_{qi}$ and $\hat{\sigma}_{uqi}$ denote the estimates of the measurement error variances and covariance for the ith observation. There are many ways these quantities arise depending on the context. One common situation is where there are replicates, discussed explicitly in Section 4.5.3. In general the measurement error may be sampling error, with a possibly complex design. The $\hat{\sigma}_{ui}$, for example, is then the estimated standard error associated with W_i as an estimate of x_i. If a measurement error variance or covariance is constant, the i subscript can be dropped. Much of the traditional literature on the problem deals with error in X only with constant, known measurement error variance, resulting in just the use of σ^2_u.

4.5.1 Moment-based corrected estimators

The general idea behind the moment corrected estimators is to correct for the bias in S_{WW} and S_{WD} as estimators of σ^2_X and $\sigma_{XY} = \beta_1\sigma^2_X$, respectively. The motivation for these estimators and further details are given in Section 4.9. The corrected estimators of the coefficients are

$$\hat{\beta}_1 = \frac{S_{WD} - \hat{\sigma}_{uq}}{S_{WW} - \sigma^2_u} = \frac{\widehat{\sigma}_{XY}}{\widehat{\sigma}^2_X} \qquad \text{and } \hat{\beta}_0 = \bar{D} - \hat{\beta}_1\bar{W} \tag{4.7}$$

where

$$\widehat{\sigma}^2_u = \sum_i \hat{\sigma}^2_{ui}/n, \qquad \widehat{\sigma}_{uq} = \sum_i \hat{\sigma}_{uqi}/n, \quad \text{and} \quad \widehat{\sigma}^2_q = \sum_i \hat{\sigma}^2_{qi}/n.$$

An estimator for the variance of the error in the equation, a special case of equation (3.1.22) in Fuller (1987), is

$$\hat{\sigma}^2 = \frac{\sum_i r_i^2}{n-2} - \widehat{\sigma}^2_q - \widehat{\beta}^2_1\widehat{\sigma}^2_u + 2\widehat{\beta}_1\widehat{\sigma}_{uq}, \tag{4.8}$$

where

$$r_i = D_i - (\hat{\beta}_0 + \hat{\beta}_1 W_i)$$

is the residual using the naive measurements and the corrected coefficients. Replacing $n-2$ in (4.8) with $n-1$ leads, after some simplification, to an alternative

$$\hat{\sigma}^2 = S^2_D - \widehat{\sigma}^2_q - \widehat{\beta}^2_1\widehat{\sigma}^2_X, \tag{4.9}$$

where $\widehat{\sigma}_X^2 = S_{WW} - \widehat{\sigma}_u^2$, and $S_D^2 = \sum_i (D_i - \bar{D})^2/(n-1)$. This estimator also arises naturally from the fact that $\sigma^2 = \sigma_Y^2 - \beta_1^2 \sigma_X^2$ and that $E(S_D^2) = \sigma_Y^2 + \sigma_q^2$.

With no error in the response $\widehat{\sigma}^2$ in (4.8) is equivalent to the estimate in (1.2.3) of Fuller (1987).

A corrected estimator of ρ, for use in the structural setting, is

$$\hat{\rho} = \frac{\widehat{\sigma}_{XY}}{(\widehat{\sigma}_X^2 \widehat{\sigma}_Y^2)^{1/2}} = \frac{S_{WD} - \widehat{\sigma}_{uq}}{\left[(S_{WW} - \widehat{\sigma}_u^2)(S_{DD} - \widehat{\sigma}_q^2)\right]^{1/2}}.$$

Remarks:

- The moment based estimators above are applicable in either the functional or structural setting. The corrected estimators are not unbiased but are consistent under fairly general conditions. Recall that the naive estimators are both biased and inconsistent.
- Under the normal structural model with normal measurement errors and constant known measurement error variances/covariance, the moment estimators are essentially maximum likelihood estimators, as long as $\hat{\sigma}_X^2 = S_{WW} - \widehat{\sigma}_u^2$, $S_{DD} - \widehat{\sigma}_q^2$ and $\hat{\sigma}^2$ are all nonnegative. We say essentially because the MLEs would be computed using divisors of n in calculating S_{WW} and S_{WD}, rather than $n-1$. This comment also applies if we use estimated variances based on normal replicates with constant measurement error variances/covariance.
  The maximum likelihood approach fails in the functional case where the x_i's are viewed as unknown parameters, leading to inconsistent estimators of the coefficients. This is a result of an increasing number of nuisance parameters (the x_i's) and is known as the Neyman-Scott problem; see Fuller (1987, p. 104).
- **Special case:** $\sigma_{uq} = 0$.
  Recall that this includes the case of no error in Y or uncorrelated measurement errors. Assuming $\hat{\sigma}_X^2 = S_{WW} - \hat{\sigma}_u^2 > 0$, the corrected estimator of the slope can be rewritten as

$$\hat{\beta}_1 = \hat{\beta}_{1naive} \hat{\kappa}^{-1}, \qquad \hat{\kappa} = \frac{\hat{\sigma}_X^2}{\hat{\sigma}_X^2 + \hat{\sigma}_u^2} \tag{4.10}$$

  where $\hat{\kappa}$ is the estimated reliability ratio. So, the estimator modifies the naive estimator by scaling by the reciprocal of the estimated reliability ratio. This correction is also suggested immediately by the fact that $\hat{\beta}_{1naive}$ is an estimate of $\kappa\beta_1$ as described earlier.
- With no measurement error in y, the moment corrected estimators in (4.10) can also be obtained by regressing Y_i on

$$\hat{x}_i = \bar{W} + \hat{\kappa}(w_i - \bar{W}) \tag{4.11}$$

which is the best predictor of the unobserved x_i in the structural setting with constant measurement error variance. This is a special case of regression calibration, discussed in more detail in Section 6.10.

- **Modified estimators.** In some cases, the moment estimators above need some modification since $\widehat{\sigma}^2$, $S_{WW} - \widehat{\sigma}_u^2$ (an estimator of σ_X^2) and $S_{WW} - \widehat{\sigma}_q^2$ (an estimator of $\sigma^2 + \beta_1^2 \sigma_X^2$) should all be nonnegative. One strategy if $\widehat{\sigma}^2$ is negative is to set $\hat{\sigma}^2$ to 0 and then estimate the other quantities, assuming there is no error in the equation. This is not an altogether satisfying solution since it implies that the true Y and X are deterministically related. If there were no error in the equation, then $Y_i = \beta_0 + \beta_1 X_i$ implying $\sigma_Y^2 = \beta_1^2 \sigma_X^2$. But, we also know that $\beta_1 = \sigma_{XY}/\sigma_X^2$, implying that $\sigma_X^2 = \sigma_{XY}^2/\sigma_Y^2$. With no error in Y this leads to corrected estimators (see Fuller (1987, p 15)) of

$$\widehat{\sigma}_X^2 = S_{WY}/S_{YY} \text{ and } \widehat{\beta}_1 = S_{WY}/\widehat{\sigma}_X^2 = S_{YY}/S_{WY}.$$

  Fortunately the problem of negative estimates does not arise frequently in practice. This can be a problem when bootstrapping, however, as seen in our analysis of the yield example. When they do occur in the primary analysis, another option is to bypass point estimation completely and attack the problem through a confidence interval that doesn't use a point estimate. One way to do this is through the use of the transformation interval described at the end of the next section.

4.5.2 Inferences for regression coefficients

We describe two main approaches here: Wald-based inferences based on the approximate normality of the estimators and bootstrap based inferences. The vector of coefficients is denoted by $\boldsymbol{\beta}$.

Wald type inferences.

The Wald inferences are based on the fact that $\widehat{\boldsymbol{\beta}}$ is approximately normal with mean $\boldsymbol{\beta}$ and covariance matrix $\Sigma_{\hat{\boldsymbol{\beta}}}$, estimated by

$$\widehat{\Sigma}_{\hat{\boldsymbol{\beta}}} = \begin{bmatrix} v_{00} & v_{01} \\ v_{01} & v_{11} \end{bmatrix},$$

where v_{00} and v_{11} denote the estimated variances of $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively, and v_{01} the estimated covariance.

Fuller (1987) provides a fairly comprehensive discussion of $\Sigma_{\hat{\boldsymbol{\beta}}}$ and ways to estimate it. We defer a more complete discussion of the techniques and the motivation behind them until multiple linear regression is treated in the next chapter. For now we simply point to the robust and "normal-based" estimates of

Σ_β given in equations (5.12) and (5.13), respectively. The latter estimate actually depends on both normal measurement errors and the assumption of known measurement error variances and covariances. The estimate used in Theorem 1.2.1 of Fuller (1987) for simple linear regression is essentially (5.13), but see the corn yield example for further comment.

Using the estimated covariance matrix an approximate confidence interval for β_j is given by $\widehat{\beta}_j \pm z_{\alpha/2} v_{jj}^{1/2}$, and an interval for $E(Y|X = x_0) = \beta_0 + \beta_1 x_0$ is given by

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 \pm z_{\alpha/2}(v_{00} + 2x_0 v_{01} + x_0^2 v_{11})^{1/2}.$$

An approximate test Z-test of $H_0 : \beta_1 = 0$ uses $Z = \widehat{\beta}_1 / v_{11}^{1/2}$ with an approximate P-value obtained from the standard normal.

Bootstrapping.

The data on the ith unit is $(Y_i, W_i, \widehat{\boldsymbol{\theta}}_i)$ where $\widehat{\boldsymbol{\theta}}_i$ contains quantities associated with estimating the measurement error parameters. With error in X only then $\widehat{\boldsymbol{\theta}}_i = \widehat{\sigma}_{ui}^2$, while with correlated errors in both variables, $\widehat{\boldsymbol{\theta}}_i' = (\widehat{\sigma}_{ui}^2, \widehat{\sigma}_{qi}^2, \widehat{\sigma}_{uqi})$. In some cases $\widehat{\boldsymbol{\theta}}_i$ may be the same for all i or treated as fixed.

Some care needs to be exercised when employing the bootstrap depending on whether the structural or functional model applies and the manner in which the estimated measurement error parameters arise. A more general discussion of bootstrapping appears in Section 6.16.

In the *structural case*, one recommendation is to bootstrap by simply resampling units, which means resampling from the collection of $(y_i, x_i, \widehat{\boldsymbol{\theta}}_i)$'s with replacement. This is later referred to as a **one-stage** bootstrap method. This method is only really justified with an initial random sample of units. It also requires that the manner in which the estimated measurement error parameters $\widehat{\boldsymbol{\theta}}_i$ are obtained is either common for all units, random or attached to the unit. It would not work, for example, if there was some changing sampling effort (e.g., replication) that was attached to the position in the sample, as opposed to being attached to the unit occurring in that position.

If the initial sampling of units is stratified, then we can resample independently within each stratum as long as the way in which the $\widehat{\boldsymbol{\theta}}_i$ are obtained satisfy the conditions discussed above within each stratum.

In the *functional case* with fixed x_i's, or the structural case where we condition on the x_i's, bootstrapping is more challenging. Without measurement error this setting calls for resampling residuals; see for example Efron and Tibshirani (1993, Ch. 9). With constant measurement error variance and distributional assumptions, bootstrapping in a parametric fashion is fairly straightforward.

To illustrate, consider error in x only and suppose the error in the equation and the measurement error are both assumed to be normally distributed, each with mean 0 and variances σ^2 and σ_u^2, respectively. In the bth bootstrap sample the ith observation is (Y_{bi}, W_{bi}), where $Y_{bi} = \widehat{\beta}_0 + \widehat{\beta}_1 \widehat{x}_i + \widehat{\sigma} Z_{bi1}$ and $W_{bi} = \widehat{x}_i + \widehat{\sigma}_u Z_{bi2}$, where Z_{bi1} and Z_{bi2} are independent each distributed as a standard normal. There are various choices one could use for $\widehat{x}_i$ including W_i, the $\widehat{x}_i$ in (4.11) or an estimate of x_i^* in (4.12) arising in the discussion of residual analysis in Section 4.7. This approach is easily extended to handle concurrent measurement error in Y and can also be modified to accommodate changing measurement error variances, although this requires modeling the heteroscedasticity.

The difficult challenge in the functional case arises when we do not assume normality for the error in the equation. The bootstrap needs a nonparametric estimate of the distribution of the error in the equation in order to generate $Y_{bi} = \widehat{\beta}_0 + \widehat{\beta}_1 \widehat{x}_i + r_{bi}$ where r_{bi} is generated to reflect the distribution of ϵ_i. Without measurement error this is usually accomplished by resampling from the residuals. With measurement error, consider the residual $r_i = D_i - (\hat{\beta}_0 + \hat{\beta}_1 W_i)$. Ignoring any error in the estimated coefficients, $r_i = \epsilon_i + d_i - \beta_1 u_i$. This shows that the residual is contaminated with contributions from the measurement error and we need to untangle the distribution of ϵ_i from that of $d_i - \beta_1 u_i$. This is a type of unmixing/deconvolution problem, discussed in Section 10.2.3.

Transforming intervals.

Similar to what was suggested in some of the misclassification problems, another strategy is to bypass estimation and form a confidence interval for the quantity of interest by transforming other intervals. Consider, for example, making inferences for the slope β_1. The naive confidence interval, $[L_n, U_n]$, for the slope is actually a confidence interval for γ_1. This is exact in the case of the normal structural model with additive normal measurement and homoscedasticity. Otherwise this interval is approximate and if there is evidence of heteroscedasticity in the regression of D on w this could be accounted for in forming the interval for γ_1. If we knew $\kappa = \sigma_X^2/(\sigma_X^2 + \sigma_u^2) > 0$ and $\omega = \sigma_{uq}/(\sigma_X^2 + \sigma_u^2)$ then since $\beta_1 = (\gamma_1 - \omega)/\kappa$, a confidence interval for β_1 is given by

$$\left[\frac{L_{naive} - \omega}{\kappa}, \frac{U_{naive} - \omega}{\kappa}\right].$$

Of course, σ_X^2, σ_u^2 and σ_{uq} will usually need to be estimated leading to uncertainty in κ and ω which should be accounted for. This can be done by viewing β_1 as a function of γ_1, σ_X^2, σ_u^2 and σ_{uq} and building a confidence set for β_1 from simultaneous confidence intervals for the other components. This seems like a promising approach but at this point it is speculation as to how well it works and further investigation is warranted. One case where this approach is straightforward is under a normal model with measurement error only in X and constant known measurement error variance. Then $\sigma_W^2 = V(W_i) = \sigma_X^2 + \sigma_u^2$ and we can write $\beta_1 = \gamma_1\sigma_W^2/(\sigma_W^2 - \sigma_u^2)$. Under normality a chi-squared based confidence interval for σ_W^2 can be obtained, and using Bonferroni's inequality we can easily obtain simultaneous confidence intervals for γ_1 and σ_W^2. With σ_u^2 known, a confidence set for β_1 is obtained by computing the range of values for $\gamma_1\sigma_W^2/(\sigma_W^2 - \sigma_u^2)$ as γ_1 and σ_W^2 vary over their respective intervals. This is illustrated in the corn yield example in Section 4.6.

4.5.3 Replication

With additive error the measurement error variances and covariances are often estimated through replication. General discussions on the use of replication appear in Sections 5.4.3 and 6.5.1. Here, suppose on unit i there are m_i replicate values $W_{i1}, \ldots, W_{im_i}$ of the error-prone measure of x and, if there is error in y, there are k_i replicates $D_{i1}, \ldots, D_{ik_i}$ of the error-prone measure of y. With error in both variables the replicates do not have to be paired, unless the measurement errors are allowed to be correlated. In the egg-mass/defoliation example, the replicates are paired with m_i = 10, 15 or 20.

The replicate values are assumed independent with

$$W_{ij} = x_i + u_{ij} \qquad D_{il} = y_i + q_{il},$$

$$E(u_{ij}) = 0, \qquad E(q_{il}) = 0, \qquad V(u_{ij}) = \sigma^2_{ui(1)}, \qquad V(q_{il}) = \sigma^2_{qi(1)}.$$

The quantities $\sigma^2_{ui(1)}$ and $\sigma^2_{qi(1)}$ represent the per replicate variances.

For consistency with the notation used elsewhere, the mean values are denoted by $W_i = \sum_{j=1}^{m_i} W_{ij}/m_i$ and $D_i = \sum_{l=1}^{k_i} D_{il}/k_i$. Then, $W_i|x_i = x_i + u_i$ and $D_i = y_i + q_i$, where u_i has mean 0 and variance $\sigma^2_{ui} = \sigma^2_{ui(1)}/m_i$ and q_i has mean 0 and variance $\sigma^2_{qi} = \sigma^2_{qi(1)}/k_i$. If there is error in both variables with paired replicates then $l = j$, $k_i = m_i$, with $Cov(W_{ij}, D_{ij}) = \sigma_{uqi(1)}$ and $\sigma_{uqi} = cov(W_i, D_i) = \sigma_{uqi(1)}/m_i$.

The estimated measurement error variances and covariance for the means are $\hat{\sigma}^2_{ui} = s^2_{Wi}/m_i$, $\hat{\sigma}^2_{qi} = s^2_{Di}/k_i$ and (if the replicates are paired) $\hat{\sigma}_{uqi} = s_{WDi}/m_i$, where

$$s^2_{Wi} = \sum_{j=1}^{m_i}(W_{ij} - W_i)^2/(m_i - 1), \qquad s^2_{Di} = \sum_{j=1}^{k_i}(D_{ij} - D_i)^2/(k_i - 1)$$

and, with paired replicates, $s_{WDi} = \sum_{j=1}^{m_i}(W_{ij} - W_i)(D_{ij} - D_i)/(m_i - 1)$.

4.6 Examples

This section revisits the two examples presented in the introduction.

4.6.1 Nitrogen-yield example

The response here is corn yield, assumed measured without error and the predictor is nitrogen content, x, measured with error. The measurement error, assumed to be additive with constant variance, is $\sigma_u^2 = 57$. We'll present this example in a lot of detail, which allows us to illustrate a number of key ideas in a simple setting. The analysis is presented in Table 4.3. Computations were carried out in the SAS-IML program (the first portion of the table) as well as using the rcal function in STATA. For this example we have shown how rcal is run and have left the output as given by STATA.

- From the observed nitrogen values, $S_{WW} = 304.85$ and $\bar{W} = 70.64$ leading to $\hat{\sigma}_X^2 = 247.85$ and an estimated reliability ratio of $\hat{\kappa} = \hat{\sigma}_X^2 / S_{WW} = .8130$.
- The naive estimate of β_1 is .3440, while the corrected estimate is .4232, illustrating the correction for attenuation in the naive estimator.
- The corrected estimate of β_1 also equals $\hat{\beta}_{1,naive}/\hat{\kappa}$. Treating $\widehat{\kappa}$ as known, the estimated standard error of $\hat{\beta}_i$ is (SE of $\hat{\beta}_{1,naive})/\widehat{\kappa}$ = .1687.
- The line labeled COR-F gives corrected estimates and inferences based on Chapter 1 of Fuller (1987). The estimate of 43.291 for σ^2 corresponds to the use of (4.9).
- The line labeled COR-N has an estimate of 49.24 for σ^2, resulting from the use of (4.8), and standard errors and confidence intervals for the coefficients using the normal based estimated covariance matrix given in (5.13). The minor difference in standard errors here compared to Fuller's analysis is due to the use of slightly different $\widehat{\sigma}^2$ and Fuller's use of a divisor of $n(n-1)$ where (5.13) uses n^2. The difference is unimportant here.
- The COR-R analysis uses the robust estimate of Σ_β given in (5.13). There are different versions of this robust estimator differing by the use of n^2 or $n(n-2)$ in a divisor. The table gives the result from the use of n^2 divisor, since this matches more closely with the robust estimator in STATA. If $n(n-2)$ is used instead the standard error for the slope is .162 with an associated confidence interval of $(.106, .740)$.
- The COR-RC line comes from simply regressing Y_i on $\hat{X}_i = \bar{W} + \widehat{\kappa}(W_i - \bar{W}) = \bar{W} + .81303(W_i - \bar{W})$, which yields the regression calibration estimates. These are identical to the moment corrected estimates, with the

Table 4.3 *Analysis of the nitrogen-yield example. COR denotes the corrected estimates.* v_{01} *is the estimated covariance of* $\widehat{\beta}_0$ *and* $\widehat{\beta}_1$*. The confidence interval (CI) is for* β_1

Method	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\sigma}^2$	$SE(\hat{\beta}_0)$	$SE(\hat{\beta}_1)$	v_{01}	CI
NAIVE	73.153	.344	57.32	9.951	.137	-1.328	(.034, .654)
COR-F	67.56	.423	43.291	12.54	.174	-2.151	
COR-N	67.56	.423	49.235	12.56	.176	-2.157	(.081,.766)
COR-R	67.56	.423	49.234	10.787	.146	-1.547	(.137,.710)
COR-RC	67.56	.423	57.32	12.130	.1687		
Tran.							(-.042, .750)

Bootstrap Analysis of slope. Two runs of 5000.

	Mean	Median	SE	Percentile Interval
untrimmed	.3823	.4358	3.701	(-.046,.857)
trimmed	.4319	.4358	.2029	(.048,.816)
untrimmed	.4300	.4292	.8924	(-.067,.853)
trimmed	.4252	.4292	.2014	(.025,.799)

Using the rcal function in STATA.

```
. mat input suu=(57)
. rcal (yield=) (w:nitrogen), suuinit(suu)
-----------------------------------------------------------------
 yield |    Coef.   Std. Err.   t     P>|t|     [95% Conf. Interval]
-------+---------------------------------------------------------
     w |   .4232    .1712     2.47   0.035     .0358    .8105
 _cons |   67.56    9.951     6.79   0.000     45.053    90.075
-----------------------------------------------------------------
  rcal (yield=) (w: nitrogen), suuinit(suu) robust
-----------------------------------------------------------------
       |        Semi-Robust
 yield |  Coef.   Std. Err.  t      P>|t|     [95% Conf. Interval]
-------+---------------------------------------------------------
     w | .4232    .1491       2.84   0.019    .0860    .7604
 _cons |  67.56   8.559       7.89   0.000     48.202   86.926
-----------------------------------------------------------------
. rcal (yield = ) (w: nitrogen), suuinit(suu) bstrap brep(5000)
-----------------------------------------------------------------
       |        Bootstrap
 yield |  Coef.   Std. Err.  t     P>|t|    [95% Conf. Interval]
-------+---------------------------------------------------------
     w |.4232    .1927     2.20   0.056    -.01276    .8591
 _cons | 67.56   14.16     4.77   0.001    35.530     99.599
-----------------------------------------------------------------
```

standard error for the slope the same as treating $\widehat{\kappa}$ known above. Note that the estimate of σ^2 from this regression is not corrected.

- The confidence interval labeled Tran. comes from transforming 95% simultaneous confidence intervals for γ_1 (the slope estimated by the naive analysis) and for σ^2_W to get a confidence interval for β_1, as described at the end of Section 4.5.2. Under the normal structural model with error in x only, constant measurement error variance and known measurement error variance, this procedure has a confidence coefficient known to be greater than or equal to .95.
- The bootstrap analysis is based on resampling observations with replacement, with a focus on estimating the slope. The bootstrap is only applicable here if the units were selected randomly to start. The bootstrap was programmed to correct the estimators if either $\widehat{\sigma}^2$ or $\widehat{\sigma}^2_X$ is negative as described in Section 4.5. Even though there was no problem with negative estimates with the original data, the need to modify the estimators happens rather frequently in the bootstrap analysis. In four separate runs of the bootstrap with 5000 bootstrap samples the number of samples needing correcting were 420, 382, 416 and 366.
  For illustration, two bootstrap samples of size 5000 were obtained and for each we present the analysis using both untrimmed and trimmed bootstrap samples. The trimmed analysis drops the top and bottom 2 percent, chosen to coincide with the default for bootstrap analyses using the rcal function in STATA. The untrimmed analysis can be a bit volatile with respect to the bootstrap mean and the bootstrap estimate of standard error. There can be more dramatic differences in these quantities than we have shown in the two illustrations here. The primary cause is the extreme values of the estimated slope resulting when $\widehat{\sigma}^2$ is set to 0, although there are also outliers resulting from very large values of $\widehat{\sigma}^2$. The percentile bootstrap intervals for β_1 and the median from the untrimmed analyses stay fairly consistent over the different bootstrap samples and these are the most useful quantities from this analysis.
- In the STATA results, note that in the first two nonbootstrap analyses *the standard error for the intercept is not corrected*, although the standard error for the slope is. The first analysis uses what Hardin et al. (2003) refer to as the "traditional calculation." This is the default and the results here are close to those based on the normal based estimate of the covariance. The robust option specifies (using the language of Hardin et al. (2003)) "the Huber/White/sandwich estimator of variance is to be used," and differs in a minor way from the robust estimate of $Cov(\widehat{\boldsymbol{\beta}})$ in (5.12). The bootstrap analysis in the STATA rcal function uses a Wald type interval based on the bootstrap estimate of the standard error from the trimmed sample.

4.6.2 Defoliation example with error in both variables

Here we analyze the egg mass/defoliation data presented in Table 4.1. There is error in both variables, which is additive as a result of the subsampling used. The measurement error variances and covariances are estimated from paired replicates on each unit. (The sampling was actually systematic on each unit, but for illustration we have treated the observations as replicates. This is defensible in the absence of any periodic behavior over the sampling grid; see Cochran (1977).) There are large differences among these quantities over different observations the result of changing sample sizes (10, 15 or 20 replicates) and the fact that the per replicate variance may be changing with the mean. For example, the larger values of $\widehat{\sigma}_{ui}$, which estimate the measurement error standard deviation in estimated egg mass density, generally correspond to units with larger estimated egg mass densities.

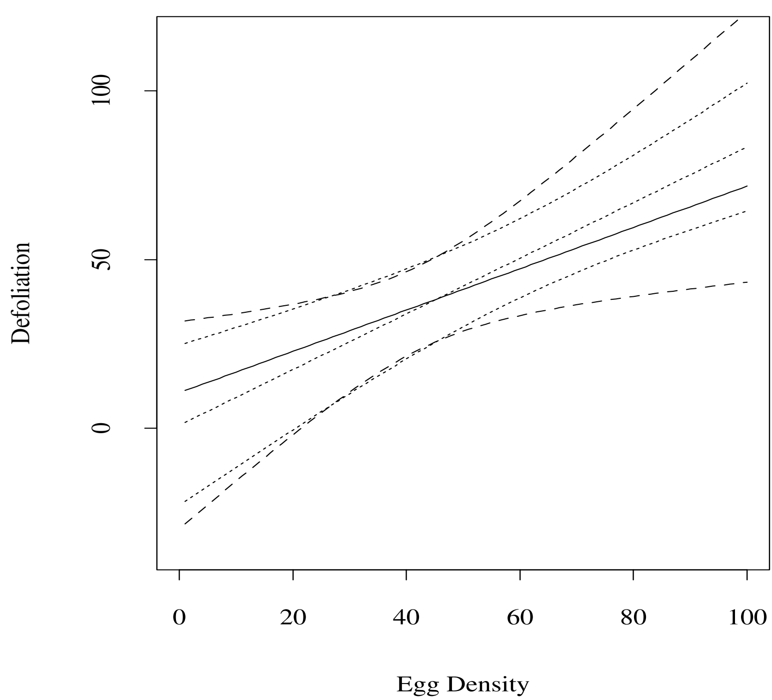
For illustration, we assume here that a common linear regression model holds over all three forests, but see Section 5.6 for further analysis. This example utilizes the more general correction techniques allowing for measurement error in the response as well as the predictor, with the two measurement errors being correlated and with the measurement error variances and covariance allowed to change over observations. The analysis is given in Table 4.4, using the same labeling in the previous example. Figure 4.2 shows the naive and corrected fits as well as confidence intervals around the corrected fit based on the robust and normal based estimates of the covariance matrix. These intervals make use of the estimated covariance, v_{01}. The standard errors and associated confidence from the robust and normal based procedures differ substantially. A number of things could contribute to this difference including the fact that the robust estimate is also accommodating possible heteroscedasticity in the error in the equation. The conservative approach is to use the normal based analysis, although some of our later examples show that this often tends to overestimate standard errors.

A bootstrap analysis is not provided here. Since the 18 stands were not an overall random sample an analysis based on resampling of observations would be misleading. If we were willing to assume that there was a random sample of stands in each forest, then we could bootstrap by resampling sets within each forest.

Table 4.4 *Analysis of gypsy moth data.*

Method	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\sigma}^2$	$SE(\hat{\beta}_0)$	$SE(\hat{\beta}_1)$	v_{01}	CI for β_1
NAIVE	10.504	.6125	696.63	11.459	.2122	-2.042	(.163,1.063)
COR-R	.8545	.8252	567.25	12.133	.182	-1.918	(.469,1.181)
COR-N	.8545	.8252	567.25	15.680	.338	-4.848	(.163,1.488)

Figure 4.2 *Analysis of defoliation data. Naive fit (solid line), corrected fit (dashed line), confidence intervals using robust covariance(dotted line) and using normal based covariance (dashed line).*



4.7 Residual analysis

In any regression problem an assessment of the model assumptions is important. The ith residual, based on the corrected estimates and observed values is

$r_i = D_i - (\widehat{\beta}_0 + \widehat{\beta}_1 w_i)$. Simply plotting these residuals versus w_i to check assumptions in is not always the right thing to do in the presence of measurement error, as discussed, for example, by Fuller (1987), Carroll and Speigelman (1992) and Buonaccorsi (1994). Even with known regression coefficients,

$$e_i = D_i - (\beta_0 + \beta_1 w_i) = \epsilon_i + q_i - \beta_1 u_i$$

is correlated with w_i due to the common measurement error u_i entering in both quantities. So, a plot of r_i (which estimates e_i) versus w_i can be misleading. It can be shown that e_i is uncorrelated with

$$x_i^* = w_i + (\beta_1 \sigma_{ui}^2 - \sigma_{uqi}) e_i / \sigma_{ei}^2, \tag{4.12}$$

where $V(e_i) = \sigma_{ei}^2 = \sigma^2 + \sigma_{qi}^2 + \beta_1^2 \sigma_{ui}^2 - 2\beta_1 \sigma_{uqi}$. The quantity x_i^* arises by considering the β's as known and obtaining a generalized least squares estimator of x_i (Fuller, 1987, p.21). Under normality, e_i and x_i^* are independent and $E(e_i|x_i^*) = E(\epsilon_i)$. This suggests a plot of r_i versus $\hat{x}_i$ to assess the linearity assumption, where $\hat{x}_i = w_i + ((\hat{\beta}_1 \hat{\sigma}_{ui}^2 - \hat{\sigma}_{uqi}) r_i)/\hat{\sigma}_{ei}^2$ and $\hat{\sigma}_{ei}^2 = \hat{\sigma}^2 + \hat{\sigma}_{qi}^2 + \hat{\beta}_1^2 \hat{\sigma}_{ui}^2 - 2\hat{\beta}_1 \hat{\sigma}_{uqi}$.

A plot of r_i versus $\hat{x}_i$, however, is not always helpful in assessing the assumption of constant variance for the error in the equation. The reason is that r_i^2 is approximately estimating σ_{ei}^2 (or $|r_i|$ is approximately estimating σ_{ei}), but changes in this quantity do not necessarily reflect changes in $V(\epsilon_i)$ when the measurement error variances or covariance change over i. In that case we consider a **modified squared residual**

$$msr_i = r_i^2 - \hat{\sigma}_{qi}^2 - \hat{\beta}_1^2 \hat{\sigma}_{ui}^2 + 2\hat{\beta}_1 \hat{\sigma}_{uqi}. \tag{4.13}$$

If β's and the σ's were known then $E(msr_i) = V(\epsilon_i)$. A trend in the plot of msr_i, or its absolute value, versus $\hat{x}_i$ is suggestive of changing variance for the error in the equation.

The modified squared residual does suffer from the possibility of being negative, in which case it can be rounded to 0. This problem is compounded by the fact that we need to estimate the measurement error variances and those estimates may themselves be noisy. Despite this, using the modified squared residuals or their absolute values is preferred to simply using the residuals when the measurement error variances are changing.

Egg mass/defoliation example. Figure 4.3 provides the residual plots for the egg mass/defoliation example. The top panel, plotting r_i versus $\widehat{x}_i$, is used to assess the linearity assumption, with violations of that assumption indicated by the residuals not randomly centering around 0. Since the response is percent defoliation, with a maximum value of 100, a nonlinear model will be needed at higher values of the egg-mass density. There is no evidence, however, that the linear model is not adequate over the range of x used here. The bottom panel plots the square root of the modified squared residual plotted versus $\widehat{x}_i$. A trend here indicates a changing variance for the error in the equation, something we might expect with the response being percent defoliation. There is some indication of increasing variance. This is offset by the value 0 at the largest value of $\widehat{x}$, but this is due to a large measurement error variance for that observation leading to a modified squared residual of 0. As noted when we analyzed this example in the preceding section, the robust estimate of the covariance of the coefficients does allow for heteroscedasticity in the error in the equation.

4.8 Prediction

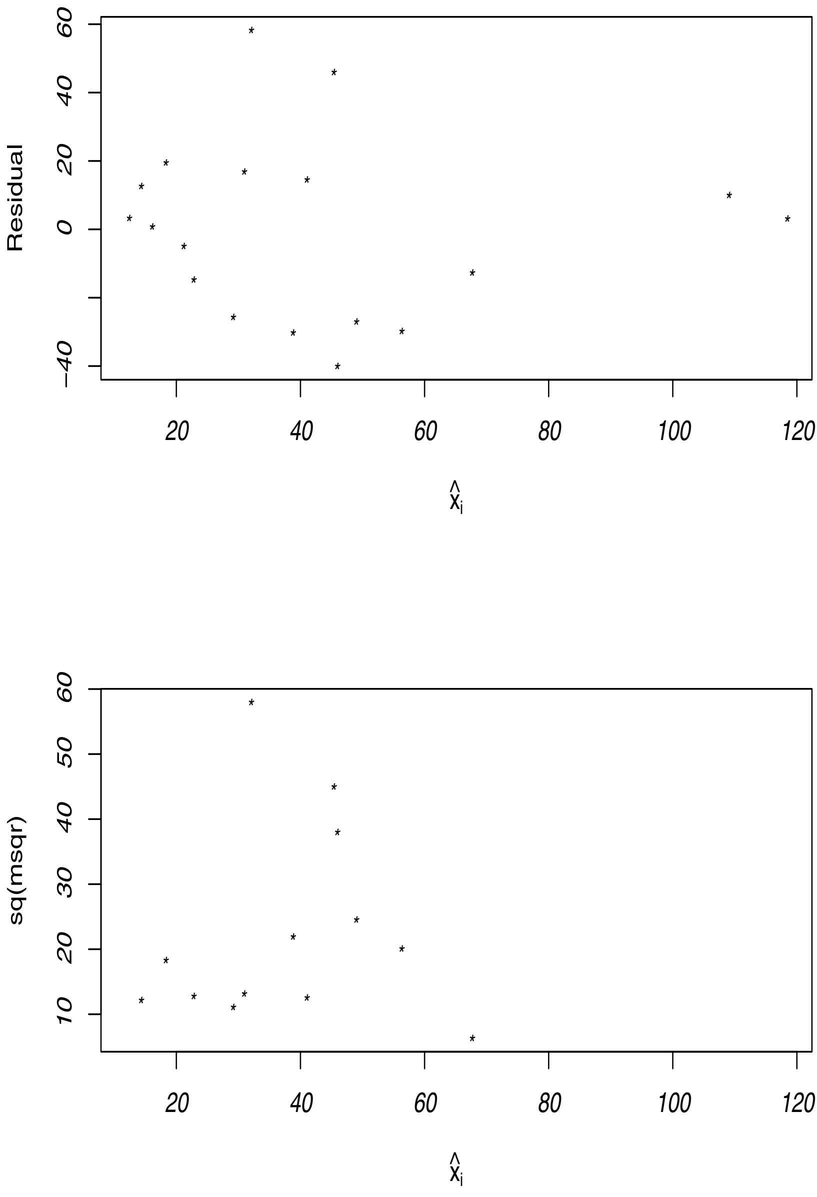
This section considers the problem of prediction in simple linear regression with additive measurement error in the regressors and/or the response. It is based mostly on Buonaccorsi (1995), with related work in Ganse, Amemiya and Fuller (1983), Fuller (1987, Section 1.6.3) and Schaalje and Butts (1992,1993). For motivation, consider the problem of predicting defoliation based on a measure of gypsy moth egg mass density in the example used earlier. Prediction was the ultimate objective in that problem. After building a model, a new stand, assumed also to be 60 ha in size, is considered in the following year on which an estimated egg mass density w_o is obtained. From this a predicted value and a prediction interval for the future defoliation is desired.

In general, let x_o be the true predictor on the future unit. Let W_o denote the estimator of x_o and let Y_o denote the random response we are trying to predict. We assume the additive error model applies on the new unit so given x_o, $W_o = x_o + u_o$ with $E(u_0|x_o) = 0$ and $V(u_o|x_o) = \sigma^2_{u_o}$. It is possible the new unit may have no measurement error so $\sigma^2_{u_o} = 0$. An estimator $\hat{\sigma}^2_{u_0}$ is assumed available, where how this is formed depends on the setting. It could be formed using data collected on the future unit (e.g., using replicates that go into constructing w_o) or, with constant measurement error variance, it might use $\widehat{\sigma}^2_u$, constructed from replicates on the original data.

Prediction with a normal structural model and normal additive measurement error can be handled using special methods. Example 1.61 in Fuller (1987) illustrates the case with unequal measurement error covariances being a function of the within unit sample size, while Ganse et al. (1983) treat a case with the marginal distribution on the x being different for the units on which prediction is to be made than it was on the training data. Reilman and Gunst (1986) consider simple linear regression with a normal structural model, normal and independent measurement errors with constant variances, and known reliability ratio.

The first question we address is, which regression should we use? Here are three options:

Figure 4.3 *Egg mass/defoliation example. Plot of residual (top panel) and square root of modified square residual versus* $\widehat{x}_i$.



1. Use the naive analysis and then predict Y_0 from w_0 in the usual way. Thus, $\hat{Y}_{0naive} = \widehat{\beta}_{0naive} + \widehat{\beta}_{1naive} w_0$ with an approximate prediction interval given by $\hat{Y}_{0naive} \pm z_{\alpha/2} SE_{pred,naive}$, where the estimated standard error for prediction is

$$SE_{pred,naive} = (\widehat{\sigma}^2_{naive}(1 + \mathbf{w}_0'(\mathbf{W}'\mathbf{W})^{-1}\mathbf{w}_0))^{1/2},$$

with $\mathbf{w}_0' = (1, w_0)$ and

$$\mathbf{W} = \begin{bmatrix} 1 & w_1 \\ 1 & w_2 \\ . & . \\ 1 & w_n \end{bmatrix}. \tag{4.14}$$

2. With error in the response, as in option 1, use the naive analysis but obtain a modified estimate for σ^2, and a modified prediction interval. This uses a modified standard error for prediction $(\widehat{\sigma}^2 + \widehat{\sigma}^2_{naive}\mathbf{w}_0'(\mathbf{W}'\mathbf{W})^{-1}\mathbf{w}_0)^{1/2}$.
3. Correct for measurement error in the main data to obtain an estimate of the regression of Y on x and an estimate of σ^2 and carry out prediction using these corrected estimates. We return to the details for this approach after some further discussion.

Option 1 (and option 2 when there is error in Y) is reasonable as long as the conditional behavior of Y_i given w_i in the main data is the same as the conditional behavior of Y_o given w_o on the new unit. One requirement for this is that *the regressor must be random both in the training set and on the new unit.* This point was made by Fuller (1987, p.76) in a setting with random regressors in the main data and a fixed regressor on the new unit. The reason is that when x is fixed, the distribution of $Y_i|w_i$ is simply the distribution of $Y_i|x_i$, which is not estimated by the naive analysis.

Even with random regressors, it is not always the case that options 1 or 2 should be automatically pursued. The conditional distribution of Y_i given w_i will depend on both the marginal distribution of X_i and the measurement error for the observation. For example, consider the normal structural case with normal measurement error in x with a variance $\sigma^2_{ui} = c/m_i$. Then the conditional distribution of $Y_i|w_i$ is normal with a mean and variance that depends on m_i as well as on distribution (and hence the moments) of X_i.

For illustrative purposes, here is a setting where option 1 or 2 would work in the context of the defoliation example. Suppose there is a general region consisting of some population of units of which the n units in the training set are a random sample. Further the subsampling scheme is always the same in any selected unit; for example, one always takes a systematic or random sample of m subunits. In addition, the new unit upon which prediction is to be made is randomly chosen from the original population of interest and the subsampling on this new unit, which yields w_o, also uses a sample of m subunits, chosen in the same way as the training data. In this case the conditional distribution of y given w is well defined and is the same for all training units and the future unit. It is worth noting that there is nothing that prohibits the conditional measurement error variance in x from being a function of the realized x. This changing variance simply gets rolled into the conditional model for $Y|x$. Since the objective here is to predict the true defoliation on the whole unit, the error in y should be accounted for, as in option 2.

The preceding discussion leads to the conclusion that:

With either fixed regressors in the main data or on the new unit, or unequal subsampling efforts, or changing means for true values in the structural case, option 3 should be used.

Option 3 is clearly the best strategy for the original defoliation data, and we now consider the details for that option. Recall that our interest is in predicting the realization of the random quantity Y_o from the observed w_o. Consider x_0 as fixed. Using the well known fact that the best linear predictor of Y_o is $E(Y_o \mid x_o)$ and the fact that $E(W_o) = x_o$, the natural candidate is $\hat{Y}_o = \hat{\beta}_0 + \hat{\beta}_1 w_o$. The prediction error is $T = \hat{Y}_o - Y_o$ with

$$V(T) = \sigma_T^2 = \sigma^2 + \mathbf{x}_o' Cov(\hat{\boldsymbol{\beta}})\mathbf{x}_o + \sigma_{u0}^2 E(\hat{\beta}_1^2). \tag{4.15}$$

The expected squared error of the predictor is $V(T) + E(T)^2$. If $\hat{\boldsymbol{\beta}}$ is consistent for $\boldsymbol{\beta}$, then $\hat{Y}_o$ is asymptotically unbiased for Y_o, in the sense that $E(T)$ converges to 0, and $V(T)$ is the approximate mean squared error for prediction.

There are three different contributions to the prediction variance in (4.15). The first, σ^2, is due to the error in the equation, the second arises from the uncertainty in $\hat{\boldsymbol{\beta}}$, while the third, containing σ_{u0}^2, results from measurement error on the new unit. Omitting uncertainty arising from estimation of $\boldsymbol{\beta}$ (i.e., setting $Cov(\hat{\boldsymbol{\beta}}) = \mathbf{0}$), (4.15) agrees with the average squared prediction error in Fuller (1987, p. 76).

For estimating σ_T^2, $\hat{\beta}_1^2\hat{\sigma}_{u0}^2$ is unbiased for $E(\hat{\beta}_1^2)\sigma_{u0}^2$, σ^2 is estimated by $\hat{\sigma}^2$, while $\mathbf{x}_o' Cov(\hat{\boldsymbol{\beta}})\mathbf{x}_o = V(\hat{\beta}_o) + 2x_o Cov(\hat{\beta}_o, \hat{\beta}_1) + x_o^2 V(\hat{\beta}_1)$. Since $E(W_o^2) = x_o^2 + \sigma_{u0}^2$ a reasonable estimate of $\mathbf{x}_o' Cov(\hat{\boldsymbol{\beta}})\mathbf{x}_o$ is $v_o + 2W_o v_{01} + (W_o^2 - \hat{\sigma}_{u0}^2)v_1$. Combined, the estimated standard deviation for prediction is

$$\hat{\sigma}_{pred} = \left[\hat{\sigma}^2 + \hat{\beta}_1^2\hat{\sigma}_{u0}^2 + v_0 + 2W_0 v_{01} + W_o^2 v_1 - \hat{\sigma}_{u0}^2 v_1\right]^{1/2}, \tag{4.16}$$

where v_j is an estimate of $V(\widehat{\beta}_j)$ and v_{01} the estimated covariance of $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

Dropping the uncertainty in $\hat{\boldsymbol{\beta}}$ yields $\hat{\sigma}_{pre}^2 = \hat{\sigma}^2 + \hat{\beta}_1^2\hat{\sigma}_{u_o}^2$ which is essentially equation (4.2) in Schaalje and Butts (1993). The connection is not exact in that here an estimate of measurement error variance ($\hat{\sigma}^2_{u_o}$) specific to the new unit is allowed, while they work with a constant measurement error variance.

The formation of prediction intervals can pose problems. The simplest approach is to treat $T/\widehat{\sigma}_{pred}$ as approximately $N(0,1)$ leading to an approximate prediction interval of $\hat{Y}_o \pm z_{\alpha/2}\widehat{\sigma}_{pred}$. While this may work in some settings, some caution should be exercised. With a suitable sample size, $\mathbf{x}'_o\hat{\boldsymbol{\beta}}$ is approximately normal based on large sample theory while normality of ϵ_o requires normality of the error in the equation for the new unit. The term $\hat{\beta}_1 u_o$ can also be troublesome since it is a product of two random variables. While "asymptotically" this will be normally distributed, the asymptotics require an increasing subsampling effort on the new unit, or that u_o be normally distributed and $\hat{\beta}_1$ be precise enough to be treated as a constant.

Defoliation example. We return to the defoliation example. After fitting using the original data, consider each of the observations as if they came from a new unit with w_o and $\widehat{\sigma}^2_{u_o}$ set equal to the corresponding w_i and $\widehat{\sigma}^2_{ui}$, respectively. Figure 4.4 displays $\hat{Y}_o \pm 1.96\hat{\sigma}_{pred}$ using the robust and the normal based $\hat{\Sigma}_\beta$. Despite what are rather large differences between the estimated variances and covariances with the two methods (see Table 4.1), there is not much difference between the two prediction intervals for most values of w. The difference does become more pronounced, however, at the two larger values of w. This difference is academic, since both intervals are uninformative due to the fact that defoliation must be between 0 and 100.

A second item of interest is how much of the prediction error is due to the three causes itemized following (4.15). To this end, the prediction intervals are displayed three ways in Figure 4.5 utilizing the normal based version of $\hat{\Sigma}_\beta$.

1. Using $\hat{\sigma}_{pred}$ in its entirety.
2. Eliminating the piece due to measurement error on the new unit, i.e., setting $\hat{\sigma}^2_{u_o} = 0$ in $\hat{\sigma}_{pred}$.
3. Eliminating the piece due to uncertainty in $\hat{\beta}$ and the piece due to measurement error in the new unit, i.e., $\hat{\sigma}_{pred} = \hat{\sigma}$.

The key contribution to the width of the prediction interval is the error in the equation, represented by $\hat{\sigma}^2$. At the larger values of w, which are accompanied by large measurement error variance, there is also a major contribution due to measurement error on the new unit, illustrated by the difference from $+$ to x. On such units, an increased sampling effort would be effective at reducing the width of the interval. The uncertainty due to the estimation of the regression coefficients, represented by the change from x to $\triangle$, is the least important of the three sources in this application.

Figure 4.4 *Egg mass/defoliation example with prediction intervals using robust (+) and the normal ($\triangle$) estimates of covariance matrix.*

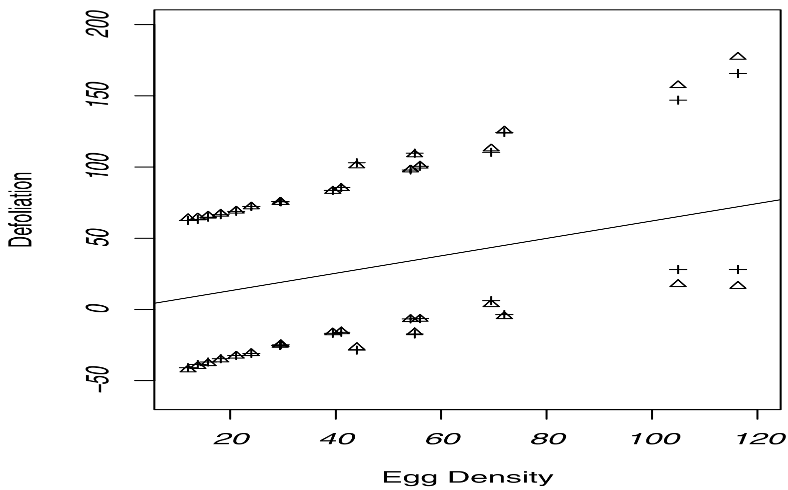
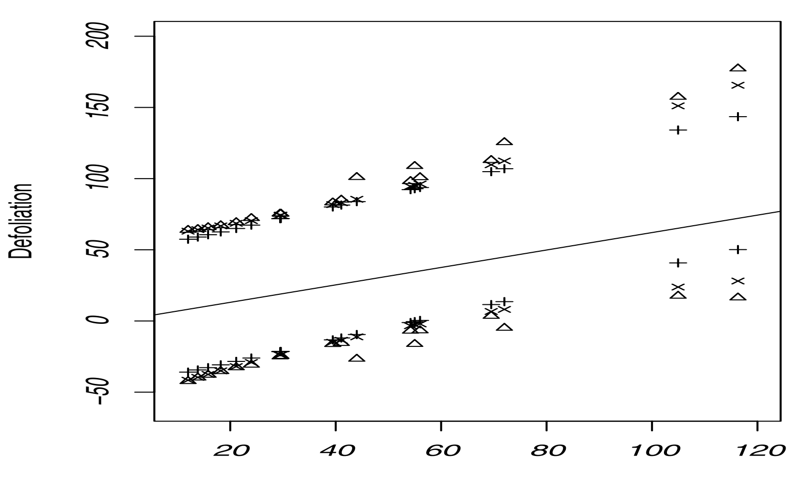


Figure 4.5 *Plot of prediction intervals based on $\hat{\sigma}_{pred}$ computed three ways; $\triangle$ uses equation (4.15); x uses (4.15) with $\hat{\sigma}_{u_o} = 0$; $+$ uses $\hat{\sigma}_{pred} = \hat{\sigma}$.*



4.9 Mathematical developments

1. *Justification for 0 covariance between measurement error and error in the equation.*
   Here we justify the fact that if the measurement errors have conditional mean 0 then they are uncorrelated with the error in the equation, even though they may be dependent. We do this for u_i with the same argument applying to error in the response. Consider u_i with $E(u_i|x_y, y_i) = 0$. Using double expectations (Section 13.2), $Cov(u_i, \epsilon_i) = E(Cov(u_i, \epsilon_i|X_i, Y_i)) + Cov(E(u_i|X_i, Y_i), E(\epsilon_i|X_i, Y_i))$, where the expectation and second covariance are over (X_i, Y_i). This does not preclude that X_i may be fixed at x_i. Since $E(u_i|X_i = x_, Y_i = y_i) = 0$ and $Cov(u_i, \epsilon_i|X_i = x_i, Y_i = y_i) = 0$ (since ϵ_i is fixed given x_i and y_i), then $Cov(u_i, \epsilon_i) = E(0) + Cov(0, \epsilon_i) = 0$.

2. *The normal structural model with normal additive measurement error and constant variance.*
   If
   $$\begin{bmatrix} Y_i \\ X_i \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_Y \\ \mu_X \end{bmatrix}, \begin{bmatrix} \sigma_Y^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_X^2 \end{bmatrix}\right)$$
   and given y_i, x_i
   $$\begin{bmatrix} D_i \\ W_i \end{bmatrix} \sim N\left(\begin{bmatrix} y_i \\ x_i \end{bmatrix}, \begin{bmatrix} \sigma_q^2 & \sigma_{uq} \\ \sigma_{uq} & \sigma_u^2 \end{bmatrix}\right),$$
   then
   $$\begin{bmatrix} D_i \\ W_i \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_Y \\ \mu_X \end{bmatrix}, \begin{bmatrix} \sigma_Y + \sigma_q^2 & \sigma_{XY} + \sigma_{uq} \\ \sigma_{XY} + \sigma_{uq} & \sigma_X^2 + \sigma_u^2 \end{bmatrix}\right)$$
   or
   $$\begin{bmatrix} D_i \\ W_i \end{bmatrix} \sim N\left(\begin{bmatrix} \beta_0 + \beta_1\mu_X \\ \mu_X \end{bmatrix}, \begin{bmatrix} \sigma^2 + \beta_1^2\sigma_X^2 + \sigma_q^2 & \beta_1\sigma_X^2 + \sigma_{uq} \\ \beta_1\sigma_X^2 + \sigma_{uq} & \sigma_X^2 + \sigma_u^2 \end{bmatrix}\right).$$
   The second version above results from the fact that the assumption that $E(Y|X = x) = \beta_0 + \beta_1 x$ and $V(Y|X = x) = \sigma^2$ implies that $\mu_Y = \beta_0 + \beta_1\mu_X$, $\sigma_Y^2 = \sigma^2 + \beta_1^2\sigma_X^2$ and $\beta_1 = \sigma_{XY}/\sigma_X^2$.
   Using the result for a conditional distribution from a bivariate normal $D_i|W_i = w_i \sim N(\gamma_0 + \gamma_1 w_i, \sigma_\delta^2)$, where γ_0, γ_1 and σ_δ^2 are as given in Section 4.4.

3. *Developing approximate bias expressions.*

   Here we can have fixed x_i's or random X_i's. For the latter, the X_i are uncorrelated but do not have to be identically distributed.
   Define $S_y^2 = (\sum_i (y_i - \bar{y})^2/(n-1)$, $S_x^2 = (\sum_i (x_i - \bar{x})^2/(n-1)$ and $S_{xy} = (\sum_i (y_i - \bar{y})(x_i - \bar{x})/(n-1)$. These involve fixed x and y values. Define S_Y^2, etc. in a similar fashion by replacing Y_i (random) for y_i and X_i

for x_i. Also, as earlier $\sigma_u^2 = \sum_{i=1}^n \sigma_{ui}^2/n$, $\sigma_q^2 = \sum_{i=1}^n \sigma_{qi}^2/n$ and $\sigma_{uq} = \sum_{i=1}^n \sigma_{uqi}/n$. Finally, let $|\mathbf{x}, \mathbf{y}$ denote conditioning on (x_i, y_i) for $i = 1$ to n and $|\mathbf{x}$ denote just conditioning on the x's.

Direct calculations yield: $E(S_{WD}|\mathbf{x}, \mathbf{y}) = S_{xy} + \sigma_{uq}$,
$E(S_{DD}|\mathbf{x}, \mathbf{y}) = S_y^2 + \sigma_q^2$, $E(S_{WW}|\mathbf{x}, \mathbf{y}) = S_x^2 + \sigma_u^2$,
$E(S_Y^2|\mathbf{x}) = \sigma^2 + \beta_1^2 S_x^2$ and $E(S_{xY}|\mathbf{x}) = \beta_1 \sigma_x^2$.

Taking double expectations of the quantities above with respect to the distribution of the Y's given $\mathbf{x}$ yields $E(S_{WD}|\mathbf{x}) = \beta_1 S_x^2 + \sigma_{uq}$, $E(S_{DD}|\mathbf{x}) = \sigma^2 + \beta_1^2 S_x^2 + \sigma_q^2$ and $E(S_{WW}|\mathbf{x}) = S_x^2 + \sigma_u^2$.

Finally, the unconditional expectations are given by replacing S_x^2 with $E(S_X^2) = \sigma_X^2$ (by definition) leading to $E(S_{WD}) = \beta_1 \sigma_X^2 + \sigma_{uq}$, $E(S_{DD}) = \sigma^2 + \beta_1^2 \sigma_X^2 + \sigma_q^2$ and $E(S_{WW}) = \sigma_X^2 + \sigma_u^2$. The approximate expected values for the naive estimators are obtained by replacing S terms in the naive estimators by their expected values. For example, $E(\widehat{\beta}_{1naive}) \approx E(S_{WD})/E(S_{WW})$.

The developments above also show that if we have unbiased estimators for σ_u^2, σ_q^2 and σ_{uq}, then $E(S_{WW} - \widehat{\sigma}_u^2) = \sigma_X^2$, $E(S_{WD} - \widehat{\sigma}_{uq}) = \beta_1 \sigma_X^2$ and $E(S_{DD} - \widehat{\sigma}_q^2) = \sigma_Y^2 = \sigma^2 + \beta_1^2 \sigma_X^2$. These lead to the corrected moment estimators.

A rigorous development of conditions under which the naive estimators are consistent, especially with changing measurement error variances, is beyond the scope of this book. Among other conditions, we need assurances that quantities involving the estimated measurement error variances are well behaved, such as assuming $\sum_{i=1}^n \widehat{\sigma}_{ui}^2/n$ converges in probability to a constant σ_u^2.

CHAPTER 5

Multiple Linear Regression

5.1 Introduction

This chapter extends the previous one by turning to multiple linear regression with error in one or more of the predictors and possible error in the response. A number of the key points from the discussion of simple linear regression carry over to here but there are some new dimensions (pun intended) to the problem with more than one predictor. Most importantly, the nature of the bias in naive estimators is more complex. The other big change is notational, as we now express all of the models and methods in matrix-vector form. There are many books on multiple linear regression that can be consulted. Some relatively applied texts include Kutner et al. (2005), Montgomery and Peck (1992) and Griffiths et al. (1993), while more advanced treatments can be found in linear models books, such as Seber and Lee (2003), Ravishankar and Dey (2000), etc.

Section 5.2 briefly summarizes the model and basic methods without measurement error. This section also addresses how to define certain mean and covariance matrices associated with the predictors when they may contain mix of random and/or fixed values. This is something that becomes important in dealing with measurement error and is also discussed in more detail in Section 5.8.2. Section 5.3 describes the multivariate additive measurement error model and the behavior of naive analyses, both analytically and through numerical illustrations. One of the key results of that section is that measurement error in one predictor can lead to bias in the naive estimates of coefficients measured without error. Given that measurement error is often ignored, bias results are important and they also play a role in leading to moment corrected estimators, which are described in Section 5.4. The relationship of these estimates to the so-called "regression calibration" method is also discussed and some variations presented in Sections 5.4.4 and 5.5. Some of the additional topics in these sections are applicable to the simple linear regression setting, although they were not discussed there. In particular, Chapter 4 omitted any details on obtaining an estimated covariance matrix for the coefficients for use in Wald type inferences. This is treated here, with the computational formulas given in Section 5.4.2 and further discussion and theoretical developments in Section 5.8.3.

In Section 5.6 we explore three examples, illustrating various features of correcting for measurement error with multiple predictors. These are: i) Fitting the defoliation data from the previous chapter allowing separate intercepts for each of the three forests; ii) Using an epidemiologic study and regressing serum cholesterol in 1962 on serum cholesterol in 1950 and age using three simulated replicate measures of cholesterol in 1950; iii) Examining data on relating the selling price of a house to its square footage and tax assessment, recognizing that one or both of the predictors will be measured with error. The reader most interested in applications may choose to go right to these examples and return to the earlier sections as needed for further understanding and details.

Section 5.7 addresses the use of instrumental variables and Section 5.8 provides some additional mathematical developments.

5.2 Model for true values

Without measurement error, the multiple linear regression model for true values assumes

$$Y_i|\mathbf{x}_i = \beta_0 + \sum_{j=1}^{p-1} \beta_j x_{ij} + \epsilon_i = \boldsymbol{\beta}'\mathbf{x}_{i*} + \epsilon_i = \beta_0 + \boldsymbol{\beta}_1'\mathbf{x}_i + \epsilon_i$$

where $\mathbf{x}_i = (x_1, \ldots, x_{p-1})'$ is a collection of predictors, $\mathbf{x}_{i*}' = (1, x_1, \ldots, x_{p-1})$,

$$\boldsymbol{\beta}' = (\beta_0, \beta_1, \ldots, \beta_{p-1}) = (\beta_0, \boldsymbol{\beta}_1'), \qquad \boldsymbol{\beta}_1' = (\beta_1, \ldots, \beta_{p-1})$$

and the ϵ_i are assumed uncorrelated with mean 0 and variance σ^2. The model allows an intercept β_0. Without it, $\mathbf{x}_i = \mathbf{x}_{i*}$ and $\boldsymbol{\beta} = \boldsymbol{\beta}_1$.

In matrix form

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_{1*}' \\ \mathbf{x}_{2*}' \\ \cdot \\ \mathbf{x}_{n*}' \end{bmatrix}$$

is an $n \times p$ matrix, $\mathbf{Y}' = (Y_1, \ldots, Y_n)$ and $\boldsymbol{\epsilon}' = (\epsilon_1, \ldots, \epsilon_n)$ with $E(\boldsymbol{\epsilon}) = \mathbf{0}$ and $Cov(\boldsymbol{\epsilon}) = \sigma^2 I$.

Define

$$\bar{\mathbf{X}} = \sum_i \mathbf{X}_i/n, \bar{Y} = \sum_i Y_i/n, S_Y^2 = \sum_i (Y_i - \bar{Y})^2/(n-1),$$

$$\mathbf{S}_{XX} = \frac{\sum_i(\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})'}{n-1} \text{ and } \mathbf{S}_{XY} = \frac{\sum_i(\mathbf{X}_i - \bar{\mathbf{X}})(Y_i - \bar{Y})}{n-1}.$$

Note that $\mathbf{S}_{XX}$ is a $(p-1) \times (p-1)$ matrix and $\mathbf{S}_{XY}$ is a $(p-1) \times 1$ vector.

Without measurement error the estimated coefficients are $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ or equivalently $\hat{\boldsymbol{\beta}}_1 = \mathbf{S}_{XX}^{-1}\mathbf{S}_{XY}$ and $\widehat{\beta}_0 = \bar{Y} - \widehat{\boldsymbol{\beta}}_1'\bar{\mathbf{X}}$. An unbiased estimator of σ^2 is $\widehat{\sigma}^2 = \sum_i(Y_i - \widehat{Y}_i)^2/(n-p)$, where $\widehat{Y}_i = \widehat{\boldsymbol{\beta}}'\mathbf{x}_{i*}$ is the ith fitted value.

Some of the predictors may be random and we will need some notation to distinguish random quantities from the actual numerical values that occur. In general, $\mathbf{X}_i$, as used above, denotes the "random" vector associated with the ith observation. Random is in quotes since we do allow that some, or even all, of the components may be fixed. If they are all fixed, then $\mathbf{X}_i = \mathbf{x}_i$. In general $\mathbf{x}_i$ denotes the realized value of $\mathbf{X}_i$. As an example of a mix of fixed and random predictors consider an industrial experiment where some inputs are fixed by design (e.g., a temperature or pressure setting) and then over the course of the experiment the outcome of other variables that are random are observed (e.g., humidity). Another example is in randomized designs where certain treatments or doses are assigned to units, with the units being random selected. Any variables going into the regression model that are associated with the unit are random, while the doses or treatment group (represented through dummy variables) would be fixed.

Mimicking what was done in the simple linear case, we define

$$\boldsymbol{\mu}_X = \sum_{i=1}^{n} E(\mathbf{X}_i) \text{ and } \boldsymbol{\Sigma}_{XX} = E(\mathbf{S}_{XX}). \tag{5.1}$$

These definitions of $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_{XX}$ accommodate any mix of random and fixed predictors. In the fully structural case with all random predictors and the $\mathbf{X}_i$ being identically distributed, then $\boldsymbol{\mu}_X = E(\mathbf{X}_i)$ and $\boldsymbol{\Sigma}_{XX} = Cov(\mathbf{X}_i)$, while in the functional case with all fixed predictors, $\boldsymbol{\mu}_X = \sum_i \mathbf{x}_i/n$ and $\boldsymbol{\Sigma}_{XX} = \mathbf{S}_{xx} = \sum_i(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})'/(n-1)$. With a combination of fixed and random predictors the expected value and covariance of X_i are conditional on any fixed components (and so can change with i). This is spelled out in detail in Section 5.8.2.

5.3 Measurement error model and biases in naive estimators

If the only measurement error is additive Berkson error, then the general discussion and conclusions in Section 4.2 for simple linear regression also apply here. So, if $\mathbf{w}_i$ is fixed and $\mathbf{X}_i = \mathbf{w}_i + \mathbf{e}_i$ where $\mathbf{e}_i$ has mean $\mathbf{0}$, then inferences for the coefficients are correct (see also Section 6.4.3).

The **additive measurement error model** assumes that given y_i and $\mathbf{x}_i$,

$$D_i = y_i + q_i, \qquad \mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i,$$

with $E(q_i|y_i, \mathbf{x}_i) = 0$, $E(\mathbf{u}_i|y_i, \mathbf{x}_i) = \mathbf{0}$,

$$V(q_i|y_i, \mathbf{x}_i) = \sigma^2_{qi}, Cov(\mathbf{u}_i|y_i, \mathbf{x}_i) = \mathbf{\Sigma}_{ui} \text{ and } Cov(\mathbf{u}_i, q_i|y_i, \mathbf{x}_i) = \mathbf{\Sigma}_{uqi}.$$

This allows error in the response as well as the predictors. If there is no error in y then $\sigma^2_{qi} = 0$ and $\mathbf{\Sigma}_{uqi} = \mathbf{0}$. If parts of $\mathbf{x}_i$ are measured without error, the appropriate components of $\mathbf{u}_i$ equal 0 and all of the components in $\mathbf{\Sigma}_{ui}$ and $\mathbf{\Sigma}_{uqi}$ involving that variable also equal 0.

The measurement error variances and covariances are allowed to change with i with average values defined by

$$\mathbf{\Sigma}_u = \sum_{i=1}^{n} \mathbf{\Sigma}_{ui}/n, \qquad \mathbf{\Sigma}_{uq} = \sum_{i=1}^{n} \mathbf{\Sigma}_{uqi}/n \quad \text{and} \quad \sigma^2_q = \sum_{i=1}^{n} \sigma^2_{qi}/n.$$

The naive estimators of the coefficients are given by

$$\hat{\boldsymbol{\beta}}_{naive} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\mathbf{D},$$

where $\mathbf{W}$ is the same as $\mathbf{X}$ but using $\mathbf{W}'_{i*} = (1, \mathbf{W}'_i)$ in place of $\mathbf{x}_{i*}$'s, $\mathbf{D}' = [D_1, \ldots D_n]$, $\mathbf{W}'\mathbf{W} = \sum_i \mathbf{W}_{i*}\mathbf{W}'_{i*}$ and $\mathbf{W}'\mathbf{D} = \sum_i \mathbf{W}_{i*}D_i$. The naive estimators can also be expressed as

$$\hat{\boldsymbol{\beta}}_{1naive} = \mathbf{S}^{-1}_{WW}\mathbf{S}_{WD}, \;\; and \;\; \hat{\beta}_{0naive} = \bar{D} - \hat{\boldsymbol{\beta}}'_{1naive}\bar{\mathbf{W}}, \tag{5.2}$$

where the definitions of $\mathbf{S}_{WW}$ and S_{WD} parallel those of $\mathbf{S}_{XX}$ and $\mathbf{S}_{XY}$. The naive estimate of the variance of the error in the equation is

$$\widehat{\sigma}^2_{naive} = \frac{\sum_i (D_i - \widehat{D}_i)^2}{n-p} = \frac{n-1}{n-p}\left[S^2_D - \hat{\boldsymbol{\beta}}'_{1naive}\mathbf{S}_{WW}\hat{\boldsymbol{\beta}}_{1naive}\right],$$

where $\widehat{D}_i$ is the fitted value using the naive estimates.

The properties of the naive estimators of the coefficients can be succinctly summarized as follows

$$E(\hat{\boldsymbol{\beta}}_{1naive}) \approx \boldsymbol{\gamma}_1 = (\mathbf{\Sigma}_{XX} + \mathbf{\Sigma}_u)^{-1}\mathbf{\Sigma}_{XX}\boldsymbol{\beta}_1 + (\mathbf{\Sigma}_{XX} + \mathbf{\Sigma}_u)^{-1}\mathbf{\Sigma}_{uq}, \tag{5.3}$$

$$E(\hat{\boldsymbol{\beta}}_{0naive}) \approx \gamma_0 = \beta_0 + (\boldsymbol{\beta}_1 - \boldsymbol{\gamma}_1)'\boldsymbol{\mu}_X \tag{5.4}$$

and

$$E(\widehat{\sigma}^2_{naive}) \approx \sigma^2 + \sigma^2_q + \boldsymbol{\beta}'_1\mathbf{\Sigma}_{XX}\boldsymbol{\beta}_1 - \boldsymbol{\gamma}'_1(\mathbf{\Sigma}_{XX} + \mathbf{\Sigma}_u)\boldsymbol{\gamma}_1,$$

where the meaning of $\mathbf{\Sigma}_{XX}$ and $\boldsymbol{\mu}_x$ are discussed following (5.1) and in Section 5.8.2.

An alternate form involving all of the coefficients, including the intercept, is

$$E(\hat{\boldsymbol{\beta}}_{naive}) \approx \boldsymbol{\gamma} = (\mathbf{M}_{XX} + \boldsymbol{\Sigma}_{u*})^{-1}(\mathbf{M}_{XX}\boldsymbol{\beta} + \boldsymbol{\Sigma}_{uq*}), \tag{5.5}$$

where $\boldsymbol{\mu}'_{X*} = (1, \boldsymbol{\mu}'_X)$,

$$\mathbf{M}_{XX} = \sum_i E(\mathbf{X}_{i*}\mathbf{X}'_{i*})/n = \boldsymbol{\Sigma}_{XX*} + \boldsymbol{\mu}_{X*}\boldsymbol{\mu}'_{X*},$$

$$\boldsymbol{\Sigma}_{XX*} = \begin{bmatrix} 0 & \mathbf{0}' \\ \mathbf{0} & \boldsymbol{\Sigma}_{XX} \end{bmatrix}, \boldsymbol{\Sigma}_{u*} = \begin{bmatrix} 0 & \mathbf{0}' \\ \mathbf{0} & \boldsymbol{\Sigma}_{u} \end{bmatrix} \text{ and } \boldsymbol{\Sigma}_{uq*} = \begin{bmatrix} 0 \\ \boldsymbol{\Sigma}_{uq} \end{bmatrix}. \tag{5.6}$$

In the fully functional case with all fixed predictors, $\mathbf{M}_{XX}$ is simply $\sum_{i=1}^n \mathbf{x}'_{i*}\mathbf{x}_{i*}/n$.

- Similar to simple linear regression the expectations are exact under the normal structural model with normal measurement error and constant measurement error covariance matrix, but otherwise they are only approximate/ asymptotic.
- If $\boldsymbol{\Sigma}_{uq} = \mathbf{0}$ (i.e., no error in the response or the error in the response is uncorrelated with any errors in the predictors) then

  $$E(\hat{\boldsymbol{\beta}}_{1naive}) = \boldsymbol{\gamma}_1 = (\boldsymbol{\Sigma}_{XX} + \boldsymbol{\Sigma}_u)^{-1}\boldsymbol{\Sigma}_{XX}\boldsymbol{\beta}_1 = \boldsymbol{\kappa}\boldsymbol{\beta}_1 \tag{5.7}$$

  where

  $$\boldsymbol{\kappa} = (\boldsymbol{\Sigma}_{XX} + \boldsymbol{\Sigma}_u)^{-1}\boldsymbol{\Sigma}_{XX} \tag{5.8}$$

  is referred to as the **reliability matrix.** See Gleser (1992) and Aickin and Ritenbaugh (1996), for example, for discussion and illustrations of the role of the reliability matrix.

As seen from these expressions the resulting biases can be rather complex. One important conclusion is that:

Measurement error in one of the variables often induces bias in the estimates of all of the coefficients, including those that are not measured with error.

We explore the nature of the biases with a collection of special cases.

Illustration 1: Two predictors with measurement error in one.

Consider two predictors x_1 and x_2 with regression function $\beta_0 + \beta_1 x_1 + \beta_2 x_2$, with x_1 subject to measurement error with variance σ_u^2, but with no measurement error in either x_2 or y. Then

$$\mathbf{X}_i = \begin{bmatrix} X_{i1} \\ X_{i2} \end{bmatrix} \text{ and } \boldsymbol{\Sigma}_{XX} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix},$$

with the components of $\boldsymbol{\Sigma}_{XX}$ interpreted accordingly depending on whether the X's are fixed or random. Since there is no error in x_2,

$$\Sigma_{ui} = \begin{bmatrix} \sigma_u^2 & 0 \\ 0 & 0 \end{bmatrix}$$

and computing γ_1 in (5.3) with $\mathbf{\Sigma}_{uq} = \mathbf{0}$ leads to

$$E(\hat{\beta}_{1naive}) \approx \left[\frac{\sigma_2^2\sigma_1^2 - \sigma_{12}^2}{\sigma_2^2(\sigma_1^2 + \sigma_u^2) - \sigma_{12}^2}\right]\beta_1$$

so

$$\text{Bias in } \hat{\beta}_{1naive} = \left[\frac{-\sigma_2^2\sigma_u^2}{\sigma_2^2(\sigma_1^2 + \sigma_u^2) - \sigma_{12}^2}\right]\beta_1$$

and

$$\text{Bias in } \hat{\beta}_{2naive} = \left[\frac{\sigma_{12}\sigma_u^2}{\sigma_2^2(\sigma_1^2 + \sigma_u^2) - \sigma_{12}^2}\right]\beta_1.$$

REMARKS:

- If $\sigma_{12} = 0$, then there is no bias in the naive estimator of β_2. If the two predictors are "correlated" ($\sigma_{12} \neq 0$) then the measurement error in x_1 induces bias in the estimated coefficient for x_2. The bias in the naive estimator of β_2 (the coefficient associated with the variable measured without error) can be either positive or negative. It does not depend on β_2, but does depend on β_1.
- If $\sigma_{12} = 0$, then $\hat{\beta}_{1naive}$ estimates $\kappa_1\beta_1$, where $\kappa_1 = \sigma_1^2/(\sigma_1^2 + \sigma_u^2)$. This means the attenuation is the same as in simple linear regression. *It is critical to note that this requires X_1 and X_2 to be uncorrelated.*

Here is a numerical illustration of the result above. It is based on values in Armstrong et.al. (1989), where x_2 = caloric intake (in 100's of grams) and x_1 = fiber intake (in gms). Suppose $V(X_2) = 33$, $V(X_1) = 72.0$, $cov(X_1, X_2) = 16.4$ (yielding a correlation of .336) and there is measurement error in the assessment of fiber with a measurement variance of $\sigma_u^2 = 70.1$. Assume there is no measurement error in the response. The true regression model with both variables is $E(Y|x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$.

- In a simple linear regression model with only x_1 in it the naive estimator of the coefficient for x_1 is estimating $\kappa\times$(true coefficient for x_1), where $\kappa = 72/(72 + 70.1) = .507$. In a model with both x_1 and x_2, then naive estimate of coefficient of x_1 is estimating $\kappa_1\beta_1$ where $\kappa_1 = ((33)(72) - 16.4^2)/(33(72 + 70.1) - 16.4^2) = .4767$. In this case, if we interpret the bias (or correct the naive estimator) by assuming the attenuation factor of .507 applies in the multiple setting, then we do not go too wrong. This is because the correlation between X_1 and X_2 is relatively low.

- If we change ρ to .8, then $\kappa = .507$, but $\kappa_2 = .270$, while with $\rho = .8$ and σ_u^2 reduced to 10, then $\kappa = .878$, but $\kappa_2 = .722$. This illustrates how it can be misleading to try and characterize the bias in β_1 using κ. This is emphasized further in Figure 5.1. This plots κ_1, the attenuation factor for the coefficient of β_1 in the model containing both variables, versus the correlation ρ for different values of σ_u^2, equivalently different values of $\kappa = \sigma_X^2/(\sigma_X^2 + \sigma_u^2)$. At $\rho = 0$, $\kappa_1 = \kappa$, which is the attenuation factor if the two variables are uncorrelated or for the coefficient in a model with the mismeasured variable only.

Figure 5.1 *Plot of attenuation factor (κ_1) in the naive estimator of the coefficient of the mismeasured predictor in a model with two variables versus ρ = correlation between the mismeasured variable and the second, perfectly measured, predictor. The reliability on the predictor by itself is .9 (solid line), .7 (dotted line) and .5 (dashed line).*

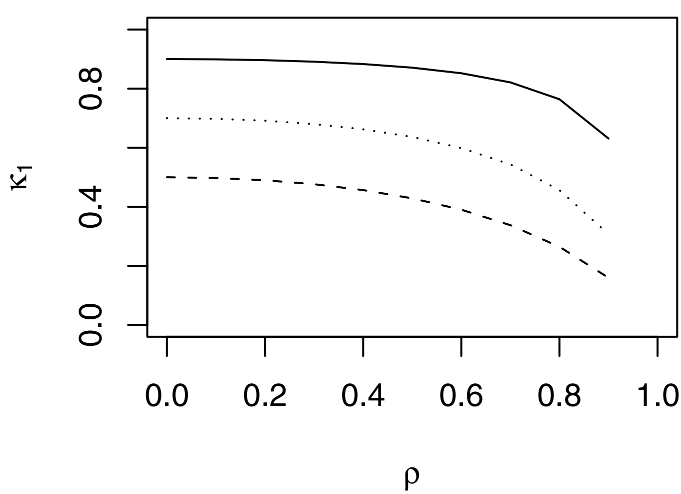


Illustration 2.

The previous result can be generalized in a multivariate way. Suppose $\mathbf{x}' = (\mathbf{x}_1', \mathbf{x}_2')$ with regression function $\beta_0 + \boldsymbol{\beta}_{11}'\mathbf{x}_1 + \boldsymbol{\beta}_{12}'\mathbf{x}_2$, so the nonintercept coefficients have partitioned as $\boldsymbol{\beta}_1' = (\boldsymbol{\beta}_{11}', \boldsymbol{\beta}_{12}')$. Assume y and $\mathbf{x}_2$ are measured perfectly but there is measurement error in $\mathbf{x}_1$, with average measurement error covariance matrix $\mathbf{\Sigma}_{u,1}$ so

$$\mathbf{\Sigma}_u = \begin{bmatrix} \mathbf{\Sigma}_{u,1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}.$$

Correspondingly $\mathbf{\Sigma}_{XX}$ is partitioned as

$$\mathbf{\Sigma}_{XX} = \begin{bmatrix} \mathbf{\Sigma}_{11} & \mathbf{\Sigma}_{12} \\ \mathbf{\Sigma}_{21} & \mathbf{\Sigma}_{22} \end{bmatrix}.$$

Then

$$E(\hat{\boldsymbol{\beta}}_{1naive}) \approx \begin{bmatrix} \mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1} & \mathbf{\Sigma}_{12} \\ \mathbf{\Sigma}_{21} & \mathbf{\Sigma}_{22} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{\Sigma}_{11} & \mathbf{\Sigma}_{12} \\ \mathbf{\Sigma}_{21} & \mathbf{\Sigma}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}_{11} \\ \boldsymbol{\beta}_{12} \end{bmatrix},$$

which, using the formula for the inverse of a partitioned matrix, reduces to

$$\begin{bmatrix} \boldsymbol{\beta}_{11} - (\mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21})^{-1}\mathbf{\Sigma}_{u,1}\boldsymbol{\beta}_{11} \\ \beta_{12} + \mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}(\mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21})^{-1}\mathbf{\Sigma}_{u,1}\boldsymbol{\beta}_{11} \end{bmatrix}.$$

This means that the bias in the naive estimator of $\boldsymbol{\beta}_{11}$ (the coefficients of the variables measured with error) is

$$-(\mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21})^{-1}\mathbf{\Sigma}_{u,1}\boldsymbol{\beta}_{11}$$

while the bias in the naive estimator of $\boldsymbol{\beta}_{12}$ (the coefficients of the variables measured without error) is

$$\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}(\mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1}^{1} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21})^{-1}\mathbf{\Sigma}_{u,1}\boldsymbol{\beta}_{11}.$$

The general conclusion is:

If $\mathbf{\Sigma}_{12} = \mathbf{0}$ ($\mathbf{X}_1$ and $\mathbf{X}_2$ are "uncorrelated") then there is no bias in the naive estimator of $\boldsymbol{\beta}_{12}$, the coefficients of the perfectly measured predictors. The naive estimator of $\boldsymbol{\beta}_{11}$ estimates $(\mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1})^{-1}\mathbf{\Sigma}_{11}\boldsymbol{\beta}_{11}$ leading to a bias of $-(\mathbf{\Sigma}_{11} + \mathbf{\Sigma}_{u,1})^{-1}\mathbf{\Sigma}_{u,1}\boldsymbol{\beta}_{11}$.

Illustration 3.

Here we consider further the setting of Illustration 2 for a special case with a slightly different formulation and a more convenient notation. Suppose X (univariate) is random and subject to measurement error with measurement error variance σ_u^2 and $\mathbf{z}$ is measured exactly and considered as fixed. The vector $\mathbf{z}$ plays the role of $\mathbf{x}_2$ in the preceding discussion. The true regression model is

$$E(Y|x, \mathbf{z}) = \beta_0 + \beta_x x + \boldsymbol{\beta}_z'\mathbf{z}.$$

Assume that conditional on the values in $\mathbf{z}$

$$E(X|\mathbf{z}) = \alpha_0 + \alpha_1'\mathbf{z} \text{ and } V(X|\mathbf{z}) = \sigma^2_{X|z}.$$

This is a case where the vector of predictors is a combination of a random predictor X and a vector of fixed predictors in $\mathbf{z}$. Defining $\mathbf{X}' = (X, \mathbf{z}')$ and using the definition of Σ_{XX} in Section 5.8.2,

$$\mathbf{\Sigma}_{XX} = \begin{bmatrix} \sigma^2_{X|z} + \boldsymbol{\alpha}_1' S_{zz}\boldsymbol{\alpha}_1 & \boldsymbol{\alpha}_1' S_{zz} \\ S_{zz}\boldsymbol{\alpha}_1 & S_{zz} \end{bmatrix}.$$

Assuming there is no error in Y, then

- The naive estimator of β_x estimates $\kappa_x \beta_x$, where
$$\kappa_x = \frac{\sigma^2_{X|z}}{\sigma^2_{X|z} + \sigma^2_u}.$$
- The naive estimator of $\boldsymbol{\beta}_z$ estimates
$$\boldsymbol{\gamma}_z = \boldsymbol{\beta}_z + \beta_x \boldsymbol{\alpha}_1 (1 - \kappa_x).$$
- If X and z are unrelated in the sense that $E(X|\mathbf{z}) = \alpha_0$ then $\boldsymbol{\alpha}_1 = \mathbf{0}$, so $\boldsymbol{\alpha}_1' \mathbf{S}_{zz} = \mathbf{0}$ and $\sigma^2_{X|z} = V(X) = \sigma^2_X$.

  In this case there is no bias in the naive estimator of β_z, and the attenuation of the naive estimator of β_x is the same as in simple linear regression.

- If the values in $\mathbf{z}$ are dummy variables representing grouping variables, then the model for true values is a linear regression with different intercepts in each group and a common slope β_x. If the expected value of X is the same for each group, then $E(X|\mathbf{z}) = \alpha_0$ and the result above says that inferences for the intercepts and functions of them are still correct even with measurement error in x. See also the analysis of covariance discussion below.

These exact same results also follow (see Carroll et al. (2006) for details) if we bypass the additive measurement error formulation and instead just assume a linear Berkson model in which $E(X|w, \mathbf{z})$ is linear in w and $\mathbf{z}$. Notice that this also allows w to be a fixed quantity. In the case above if we have normal model for $X|\mathbf{z}$ and normal additive measurement $W|x, \mathbf{z} = x + u$ then we have a linear Berkson model for $X|w, \mathbf{z}$.

The analysis of covariance.

This preceding illustration also covers what is known as the analysis of covariance; see, for example, Carroll et al. (1985) . In this model units (individuals, plants, etc.) are randomized to groups (e.g., treatments) with the objective of comparing the means of some response across the groups. With randomization the comparison of means can be carried out via a one-way analysis of variance. Often though there is an additional variable (referred to as a covariate) associated with the unit and known to potentially influence the response. The randomization averages out the influence of this variable, so it could be ignored but it may more efficient/powerful to use the covariate in the analysis. The standard analysis of covariance model assumes that for the kth unit assigned to the jth group the response Y_{jk} follows the model $Y_{jk} = \tau_j + \beta_x x_{jk} + \epsilon_{jk}$, where x_{jk} is the value of the covariate for this unit. In our earlier notation, $\mathbf{z}$ is now a collection of dummy variables that denote group membership and $\boldsymbol{\beta}'_z = (\tau_1, \ldots, \tau_J)$, where J is the number of groups involved. The population mean associated with group j is μ_j, which is the expected average response if all units in the population get treatment j. With randomization $\mu_j = \tau_j + \beta_x \mu_X$, where μ_X is the population mean of the covariate. The τ_j are the adjusted group means, i.e., $\tau_j = \mu_j - \beta_x \mu_X$. Notice that contrasts (such as differences in means) in the μ_j's are also contrasts in the τ_j's; that is $\theta = \sum_j c_j \mu_j = \sum_j c_j \tau_j$ if $\sum_j c_j = 0$. With a randomized design $E(X_{jk}|\mathbf{z}) = \mu_X$ for all observations. The result above leads to the following:

In the standard analysis of covariance model, which involves randomization to treatment groups and a constant coefficient for the covariate in each group, measurement error in the covariate does not influence inferences for the τ's and functions of them. In particular inferences for contrasts in the group means (e.g., testing equality of group means and estimating difference in group means) are correct.

It is important to notice however that the naive estimators of the group means are not unbiased since $\widehat{\tau}_{j,naive} + \widehat{\beta}_{xnaive}\bar{x}$ is a biased estimator of μ_j.

5.4 Correcting for measurement error

We assume that individual or common estimates of the measurement error variances and covariances are available. For the ith unit, these are denoted by $\widehat{\boldsymbol{\Sigma}}_{ui}$ and, if needed, $\widehat{\boldsymbol{\Sigma}}_{uqi}$ and $\widehat{\sigma}^2_{qi}$ and we define averages

$$\widehat{\boldsymbol{\Sigma}}_u = \sum_i \widehat{\boldsymbol{\Sigma}}_{ui}/n, \widehat{\boldsymbol{\Sigma}}_{uq} = \sum_i \widehat{\boldsymbol{\Sigma}}_{uqi}/n, \text{ and } \widehat{\sigma}^2_q = \sum_i \widehat{\boldsymbol{\Sigma}}_{qi}/n.$$

In other cases, the measurement error parameters may be assumed constant with a common $\widehat{\boldsymbol{\Sigma}}_u$, $\widehat{\boldsymbol{\Sigma}}_{uq}$ and $\widehat{\sigma}^2_q$ used for each observation. There may be estimated measurement error parameters available on some, but not all, observations. In that case estimated parameters need to be constructed for those observations without estimates originally. This will require some assumption about how the measurement error changes. This is discussed in detail in Section 5.4.3 when there is replication.

5.4.1 Moment corrected estimators

There are many different settings and assumptions under which estimators can be developed and various alternative estimators are available. Fairly comprehensive treatment can be found in Fuller (1987), Cheng and Van Ness (1999) and references therein. Similar to simple linear regression, the focus is on some simple moment corrected estimators, presented in this section, with some modifications in Section 5.5.

The unweighted moment corrected estimators are

$$\hat{\boldsymbol{\beta}}_1 = \widehat{\boldsymbol{\Sigma}}_{XX}^{-1}\widehat{\boldsymbol{\Sigma}}_{XY} \text{ and } \widehat{\beta}_0 = \bar{D} - \widehat{\boldsymbol{\beta}}_1'\bar{\mathbf{W}}, \tag{5.9}$$

where $\widehat{\boldsymbol{\Sigma}}_{XX} = \mathbf{S}_{WW} - \hat{\boldsymbol{\Sigma}}_u$ and $\widehat{\boldsymbol{\Sigma}}_{XY} = \mathbf{S}_{WD} - \hat{\boldsymbol{\Sigma}}_{uq}$. Notice that with no error in the response $\widehat{\boldsymbol{\Sigma}}_{XY}$ is simply $\mathbf{S}_{WD}$.

This correction is motivated by the fact that $E(\mathbf{S}_{WW}) = \boldsymbol{\Sigma}_{XX} + \boldsymbol{\Sigma}_u$ and $E(\mathbf{S}_{WD}) = \boldsymbol{\Sigma}_{XY} + \boldsymbol{\Sigma}_{uq}$. An alternate computational formula is

$$\hat{\boldsymbol{\beta}} = \widehat{\mathbf{M}}_{XX}^{-1}\widehat{\mathbf{M}}_{XY} \tag{5.10}$$

where

$$\widehat{\mathbf{M}}_{XX} = \frac{\mathbf{W}'\mathbf{W}}{n} - c\widehat{\boldsymbol{\Sigma}}_{u*} \quad \text{and} \quad \widehat{\mathbf{M}}_{XY} = \frac{\mathbf{W}'\mathbf{D}}{n} - c\widehat{\boldsymbol{\Sigma}}_{uq*},$$

where $\widehat{\boldsymbol{\Sigma}}_{u*}$ and $\widehat{\boldsymbol{\Sigma}}_{uq*}$ are formed by using estimates of $\boldsymbol{\Sigma}_u$ and $\boldsymbol{\Sigma}_{uq}$ in the definitions of $\boldsymbol{\Sigma}_{u*}$ and $\boldsymbol{\Sigma}_{uq*}$ in (5.6). The constant c can be taken to be either $c = 1$ or $c = (n-1)/n$. There is obviously little difference in the two choices of c except at very small samples sizes. Taking $c = (n-1)/n$ these estimates equal those in (5.9) and this is what is used in the examples. Sometimes $c = 1$ is used, which comes from the fact that with the $x's$ fixed, $E(\widehat{\mathbf{M}}_{XX}) = (\mathbf{X}'\mathbf{X})/n$ and $E(\widehat{\mathbf{M}}_{XY}) = (\mathbf{X}'\mathbf{X})\boldsymbol{\beta}/n$. This corrects for bias in the terms $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{Y}$, which go into the usual least squares estimators.

A corrected estimator of the variance of the error in the equation is

$$\hat{\sigma}^2 = \frac{\sum_i (D_i - (\widehat{\beta}_0 + \widehat{\boldsymbol{\beta}}_1'\mathbf{W}_i))^2}{n-p} - (\widehat{\sigma}_q^2 - 2\hat{\boldsymbol{\beta}}_1'\hat{\boldsymbol{\Sigma}}_{uq} + \hat{\boldsymbol{\beta}}_1'\hat{\boldsymbol{\Sigma}}_u\hat{\boldsymbol{\beta}}_1). \tag{5.11}$$

If a divisor of $n-1$ is used in the first piece above we have a slightly modified estimator $\hat{\sigma}^2 = S_d^2 - \widehat{\sigma}_q^2 - \hat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\Sigma}}_{XX}\hat{\boldsymbol{\beta}}_1$.

Remarks:

- Under the normal structural model with normal measurement errors and a constant measurement error covariance matrix these are essentially maximum likelihood estimators (the difference being in the use of $n-1$ divisors in forming the $\mathbf{S}$ matrices).

- With no error in Y then
$$\widehat{\beta}_1 = (\mathbf{S}_{WW} - \widehat{\mathbf{\Sigma}}_u)^{-1}\mathbf{S}_{WW}\widehat{\boldsymbol{\beta}}_{1naive} = \hat{\boldsymbol{\kappa}}^{-1}\widehat{\boldsymbol{\beta}}_{1naive},$$
where $\hat{\boldsymbol{\kappa}} = (\widehat{\mathbf{\Sigma}}_{XX} + \widehat{\mathbf{\Sigma}}_u)^{-1}\widehat{\mathbf{\Sigma}}_{XX}$ is the estimated reliability matrix.
- Still with no error in Y the estimated coefficients also result from regressing Y_i on $\hat{\mathbf{x}}_i = \bar{\mathbf{w}} + \hat{\boldsymbol{\kappa}}'(\mathbf{w}_i - \bar{\mathbf{w}})$. The estimated coefficients are equivalent to those resulting from a regression calibration approach in which Y_i is regressed on an imputed vector of predictors. In this case a common $\widehat{\boldsymbol{\kappa}}$, based on $\widehat{\mathbf{\Sigma}}_u = \sum_i \widehat{\mathbf{\Sigma}}_{ui}/n$, is used with each observation, even if there are individual estimates of the measurement error covariance matrix. In some cases, an alternative is to use $\widehat{\boldsymbol{\kappa}}_i$ specific to the ith observation; see Section 5.5.
- Similar to simple linear regression, the estimators need to be modified in certain cases. In particular some correction is needed if $\widehat{\sigma}_y^2 = S_d^2 - \widehat{\sigma}_q^2 < 0$, $\widehat{\sigma}^2 < 0$ or the matrix $\widehat{\mathbf{\Sigma}}_{XX} = \mathbf{S}_{WW} - \widehat{\mathbf{\Sigma}}_u$ is "negative." This is discussed in Section 5.4.4.

5.4.2 Sampling properties and approximate inferences

Both Wald-based and bootstrap methods can be used. The same discussion and cautions regarding bootstrapping in simple linear regression apply here; see Section 4.5.2 and also Section 6.16. We remind the reader that simply resampling the observations is only fully justified with an overall random sample and some other conditions on how the estimated measurement error parameters are obtained. Also, as in simple linear regression when the bootstrap is used we need to be careful to account for samples where modifications are needed to account for "negative" estimates of certain quantities.

The Wald based methods are based on assuming that $\hat{\boldsymbol{\beta}}$ is distributed approximately normal with mean $\boldsymbol{\beta}$ and covariance matrix $\mathbf{\Sigma}_{\boldsymbol{\beta}}$ and use of an estimate $\widehat{\mathbf{\Sigma}}_{\boldsymbol{\beta}}$. The square roots of the diagonal elements of $\widehat{\mathbf{\Sigma}}_{\boldsymbol{\beta}}$ give the estimated standard errors for the individual coefficients. The challenge here is in getting an approximate expression for $\Sigma_{\boldsymbol{\beta}}$ and then an estimate of it. There is a large number of special cases with various assumptions, extra types of data, etc. that can be considered. Among the considerations that come into play are:

- Do we have separately estimated measurement error parameters for each i?
- Are the measurement error parameters constant over i?
- Can the estimated measurement error parameters be treated as known?
- Is ϵ_i assumed normal?
- Are the measurement errors assumed normal?

- If the measurement error variances and covariances are estimated what can we say about their sampling properties?
- If replication is involved are the replicates assumed normal? If so, this provides an answer to the previous question Note that with replication one might assume that the measurement errors associated with the mean values are approximately normal but not be willing to assume normal replicates.

Section 5.8.3 provides some technical details and development for the interested reader, much of it based on the seminal work of Fuller (1987). Here we present four estimators of $Cov(\widehat{\boldsymbol{\beta}})$. First, define $r_i = D_i - \mathbf{W}'_{i*}\widehat{\boldsymbol{\beta}}$, which is the ith "residual" formed using the observed values and the corrected estimated coefficients. Also define

$$\hat{\Delta}_i = \mathbf{W}_{i*}r_i - (\hat{\Sigma}_{uqi*} - \hat{\Sigma}_{ui*}\widehat{\boldsymbol{\beta}}) \text{ and } \widehat{\mathbf{Z}}_i = \hat{\Sigma}_{uqi*} - \hat{\Sigma}_{ui*}\widehat{\boldsymbol{\beta}}.$$

The estimates are:

1. Robust Estimate.

$$\hat{\boldsymbol{\Sigma}}_{\beta,Rob} = \widehat{\mathbf{M}}_{XX}^{-1}\hat{\mathbf{H}}_R\widehat{\mathbf{M}}_{XX}^{-1}, \tag{5.12}$$

where

$$\hat{\mathbf{H}}_R = \sum_i \hat{\boldsymbol{\Delta}}_i\hat{\boldsymbol{\Delta}}_i'/n(n-p).$$

This result covers multiple predictors, allows error in all variables and changing measurement error variances and covariances with estimates available on each unit.

2. Normal based with known measurement error parameters.

$$\hat{\boldsymbol{\Sigma}}_{\beta,N} = \widehat{\mathbf{M}}_{XX}^{-1}\hat{\mathbf{H}}_N\widehat{\mathbf{M}}_{XX}^{-1}, \tag{5.13}$$

where

$$\hat{\mathbf{H}}_N = \sum_{i=1}^{n}(\mathbf{W}_{i*}\mathbf{W}'_{i*}\widehat{\sigma}^2_{ei} + \widehat{\mathbf{Z}}_i\widehat{\mathbf{Z}}'_i)/n^2 \tag{5.14}$$

with

$$\widehat{\sigma}^2_{ei} = \widehat{\sigma}^2 + \widehat{\sigma}^2_{qi} - 2\widehat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\Sigma}}_{uqi} + \widehat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\Sigma}}_{ui}\widehat{\boldsymbol{\beta}}_1.$$

As noted in Section 5.8.3 this is based on (3.1.23) in Fuller (1987). If $\widehat{\sigma}^2_{ei}$ is negative it should be set equal to 0.

3. Normal based with constant, estimated measurement error parameters.

$$\hat{\boldsymbol{\Sigma}}_{\beta,C} = \widehat{\mathbf{M}}_{XX}^{-1}\hat{\mathbf{H}}_{NC}\widehat{\mathbf{M}}_{XX}^{-1}, \tag{5.15}$$

where

$$\hat{\mathbf{H}}_{NC} = \hat{\mathbf{H}}_N + \frac{\widehat{\boldsymbol{\Sigma}}_{u*}(\widehat{\sigma}^2 - MSE_c) + \widehat{\mathbf{Z}}\widehat{\mathbf{Z}}'}{n * df}, \tag{5.16}$$

and $MSE_c = \sum_i (D_i - (\widehat{\beta}_0 + \widehat{\boldsymbol{\beta}}_1' \mathbf{W}_i))^2)/(n-p)$ is the mean square error using the corrected estimate. The quantity *df* is the degrees of freedom associated with the estimated measurement error variances and covariances; with pooling of replicates $df = \sum_i (m_i - 1)$. The additional term in (5.16) accounts for estimation of the measurement error variances, but only applies under certain conditions; see Section 5.8.3.

4. $\widehat{\boldsymbol{\Sigma}}_{RC}$ based on regression calibration.
Since the moment corrected estimates are also regression calibration estimates (see the third remark in the preceding section), the asymptotic covariance estimates described in Appendix B of Carroll et al. (2006) can be applied. Their "sandwich" estimate is essentially the robust estimate above. Their "information-type asymptotic covariance matrix" given in their equation (B.11) falls between the robust estimate and the two "normal-based" estimates given above, using more assumptions than the former and less than the latter. This estimate is available in the rcal function in STATA and is illustrated in the examples.

Remarks:

- In some versions of the robust estimate the $n(n-p)$ in $\widehat{\mathbf{H}}_R$ is replaced by n^2. It is more conservative to use $n(n-p)$ and that is the approach we take in our examples.
- If there is no measurement error then the robust estimate, computed with an n^2 divisor, reduces to $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Q}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ where $\mathbf{Q}$ is a diagonal matrix with the squared residuals as the diagonal elements. This is simply White's robust covariance estimator without measurement error (White (1980)) designed to protect against heteroscedasticity in the error in the equation.
- If there is no measurement error the "normal based" estimates $\hat{\boldsymbol{\Sigma}}_{\beta,N}$ and $\hat{\boldsymbol{\Sigma}}_{\beta,C}$ both reduce to $MSE(\mathbf{X}'\mathbf{X})^{-1}$. This is the usual estimate of the covariance of $\widehat{\boldsymbol{\beta}}$ which, without measurement error, depends only on the errors in the equation being uncorrelated with mean 0 and constant variance.

5.4.3 Replication

This section discusses some aspects of estimating the measurement error variances and covariances using replicates. Some extended discussion appears in Section 6.5.1.

Suppose on unit i there are $m_i > 1$ replicate values $\mathbf{W}_{i1}, \ldots \mathbf{W}_{im_i}$ of the error-prone measure of $\mathbf{x}$. Recall that we are not splitting out variables measured with or without error so the "replicate" measures of some components may always equal the corresponding true value. If there is error in y also, then suppose there are $k_i > 1$ replicates $\mathbf{D}_{il}, \ldots, \mathbf{D}_{ik_i}$ of the error-prone measure of y. The assumption is that given $\mathbf{x}_i$ and y_i,

$$\mathbf{W}_{ij} = \mathbf{x}_i + \mathbf{u}_{ij} \qquad D_{il} = y_i + q_{il},$$

$$E(\mathbf{u}_{ij}) = 0, \quad E(q_{il}) = 0, \quad Cov(\mathbf{u}_{ij}) = \mathbf{\Sigma}_{ui(1)}, \quad V(q_{il}) = \sigma^2_{qi(1)}.$$

The $\mathbf{\Sigma}_{ui(1)}$ represent the per-replicate measurement error covariance matrix on the ith unit, with $\sigma^2_{qi(1)}$ having a similar interpretation.

The error-prone estimates are the means

$$\mathbf{W}_i = \sum_{j=1}^{m_i} \mathbf{W}_{ij}/m_i \qquad \text{and} \qquad D_i = \sum_{l=1}^{k_i} D_{il}/k_i$$

so $\mathbf{\Sigma}_{ui} = \mathbf{\Sigma}_{ui(1)}/m_i$ and $\sigma^2_{qi} = \sigma^2_{qi(1)}/k_i$. If there is error in the response and the $\mathbf{u}_{ij}$ are independent of the q_{il}'s then $\mathbf{\Sigma}_{uq} = \mathbf{0}$. With paired replicates with $Cov(\mathbf{u}_{ij}, q_{ij}) = \mathbf{\Sigma}_{uqi(1)}$ then $\mathbf{\Sigma}_{uqi} = \mathbf{\Sigma}_{uqi(1)}/m_i$.

Assuming m_i (and if error in the response all k_i) > 1, the sample variances and covariances for the ith observation are given by

$$\mathbf{S}_{Wi} = \frac{\sum_{j=1}^{m_i}(\mathbf{W}_{ij} - \mathbf{W}_i)(\mathbf{W}_{ij} - \mathbf{W}_i)'}{m_i - 1,}, s^2_{Di} = \sum_{l=1}^{k_i}(D_{il} - D_i)^2/(k_i - 1)$$

and, if there is error in the response with paired replicates (so $m_i = k_i$),

$$\mathbf{S}_{WDi} = \frac{\sum_{j=1}^{m_i}(\mathbf{W}_{ij} - \mathbf{W}_i)(D_{ij} - D_i)'}{m_i - 1}.$$

Note that any components of $\mathbf{S}_W$ or $\mathbf{S}_{WD}$ involving predictors not measured with error will equal 0.

Estimates of the individual measurement error variances and covariances can be obtained in one of two ways.

I. Replication on all observations.

With more than one replicate on each unit (for any mismeasured variable),

$$\widehat{\mathbf{\Sigma}}_{ui} = \mathbf{S}_{Wi}/m_i, \quad \hat{\sigma}^2_{qi} = s^2_{Di}/k_i \text{ and } \hat{\mathbf{\Sigma}}_{uqi} = \mathbf{S}_{WDi}/m_i, \qquad (5.17)$$

where the expression $\hat{\mathbf{\Sigma}}_{uqi}$ only applies if there is error in the response with replicates paired with the replicates on the predictors.

II. With or without replication on all observations.

Whether all of the observations have replication or not, another option is to use

$$\widehat{\boldsymbol{\Sigma}}_{ui} = \widehat{\boldsymbol{\Omega}}_u/m_i, \quad \hat{\sigma}^2_{qi} = \widehat{\boldsymbol{\omega}}^2_q/k_i, \text{ and } \hat{\boldsymbol{\Sigma}}_{uqi} = \widehat{\boldsymbol{\Omega}}_{uq}/m_i, \tag{5.18}$$

where $\widehat{\boldsymbol{\Omega}}_u = \sum_i(m_i-1)S_{Wi}/\sum_i(m_i-1)$, $\widehat{\boldsymbol{\omega}}^2_q = \sum_i(k_i-1)S^2_{Di}/\sum_i(k_i-1)$ and, if appropriate, $\widehat{\boldsymbol{\Omega}}_{uq} = \sum_i(m_i-1)S_{WDi}/\sum_i(m_i-1)$.

Some remarks on these estimates:

- As noted, the estimates in (5.17) require more than one replicate on mismeasured values for all observations.
- The estimates in (5.18) can be used in either case, with or without replication on all observations.
- The estimates in (5.18) can be motivated by considering pooling under the assumption of a common "per-replicate variance/covariance". For example, suppose in the above discussion that $\boldsymbol{\Sigma}_{ui(1)} = Cov(\mathbf{u}_{ij}) = \boldsymbol{\Omega}_u$, which represents the common per replicate covariance matrix. Then $\boldsymbol{\Sigma}_{ui} = \boldsymbol{\Omega}_u/m_i$ and $\widehat{\boldsymbol{\Sigma}}_{ui} = \widehat{\boldsymbol{\Omega}}_u/m_i$, where $\widehat{\boldsymbol{\Omega}}_u$ given above is the pooled estimate of the common per-replicate covariance matrix.
- Using (5.18), estimates can be constructed for observations with m_i and/or k_i equal to 1, even though those observations do not contribute anything to the estimated per-replicate variances and covariances.
- The estimates in (5.18) can also be motivated in the structural model with heteroscedastic measurement error. Suppose $E(\mathbf{X}_i) = \mu_X$ and $Cov(\mathbf{X}_i) = \Sigma_{XX}$ and $Cov(\mathbf{u}_{ij}|\mathbf{x}_i) = G(\mathbf{x}_i)$, with some unspecified dependence on the underlying true values. This means that given the per replicate covariance in the measurement errors depends on the true values and the conditional covariance of $\mathbf{u}_i$ is $\boldsymbol{\Sigma}_{ui} = G(\mathbf{x}_i)/m_i$. This is a case where part of the contribution to the unequal measurement error covariance is due to the missing true values. Unconditionally, that is over random $\mathbf{X}_i$, $Cov(\mathbf{u}_{ij}) = E(G(\mathbf{X}_i)) = \boldsymbol{\Omega}_u$, say, and $Cov(\mathbf{u}_i) = \boldsymbol{\Omega}_u/m_i$. This suggests the use of $\widehat{\boldsymbol{\Sigma}}_{ui} = \widehat{\boldsymbol{\Omega}}_u/m_i$.
- The argument used in the preceding item does not carry over to the functional case. There if some observations have no replication, one must assume constant per-replicate variances/covariances or model the heteroscedasticity in some fashion.

Weighted estimates with unequal numbers of replicates.

In the structural case where the $\mathbf{X}_i$ are i.i.d. with common mean and covariance matrix one can form alternate estimates of $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_{XX}$. Using standard techniques for the multivariate one-way analysis of variance, see Carroll et al.

(2006, p. 71), suggests

$$\widehat{\boldsymbol{\mu}}_X = \frac{\sum_i m_i \mathbf{W}_i}{\sum_i m_i} \qquad \widehat{\boldsymbol{\Sigma}}_{XX} = \frac{1}{b}\left[\sum_i (\mathbf{W}_i - \widehat{\boldsymbol{\mu}}_X)(\mathbf{W}_i - \widehat{\boldsymbol{\mu}}_X) - (n-1)\widehat{\boldsymbol{\Omega}}_u\right] \tag{5.19}$$

where $b = \sum_i m_i - (\sum_i m_i^2 / \sum_i m_i)$. Similar estimates can be formed for μ_Y, $\boldsymbol{\Sigma}_{XY}$ and σ_Y^2.

Hasabelnaby, Ware, and Fuller (1989) provide an interesting example of the use of replicates to build estimated measurement error variances and covariances for measures of air pollution in assessing the relationship between pulmonary performance and pollution. Also, as noted earlier, Section 6.5.1 extends the discussion above and addresses other issues in the use of replicates.

5.4.4 Correction for negative estimates

As with simple linear regression we need to worry about certain estimators, and, here, matrices being "negative." Even if this doesn't occur in the original analysis, it can be a problem when bootstrapping so we implement the fix below, where needed, when bootstrapping in our examples. We describe the technique from Section 3.1.2 of Fuller (1987) (with similar developments elsewhere in his book) but with $\alpha = 0$. (The α is associated with a small sample modification.) The quantity c could be $(n-1)/n$ or 1. Fuller uses $c = 1$ in Section 3.1.2. First define

$$\mathbf{Q}_i = \begin{bmatrix} D_i \\ \mathbf{W}_{i*} \end{bmatrix}$$

and

$$\mathbf{Q}'\mathbf{Q} = \sum_i \mathbf{Q}_i \mathbf{Q}_i' = \begin{bmatrix} \sum_i D_i^2 & \sum D_i \mathbf{W}_{i*}' \\ \sum \mathbf{W}_{i*} D_i & \sum_i \mathbf{W}_{i*}\mathbf{W}_{i*}' = \mathbf{W}'\mathbf{W} \end{bmatrix}.$$

Also let

$$\widehat{\boldsymbol{\Sigma}}_a = \begin{bmatrix} \widehat{\sigma}_q^2 & \widehat{\boldsymbol{\Sigma}}_{qu*} \\ \widehat{\boldsymbol{\Sigma}}_{qu*}' & \widehat{\boldsymbol{\Sigma}}_{u*} \end{bmatrix}$$

and $\widehat{\mathbf{A}}_a = nc\widehat{\boldsymbol{\Sigma}}_a$.

Let λ be the smallest root solving $|\mathbf{Q}'\mathbf{Q} - \lambda \hat{\mathbf{A}}_a| = 0$. If $\lambda > 1 + n^{-1}$ the estimators and associated inferences will be as defined earlier in this section. If $\lambda \leq 1 + n^{-1}$, then define

$$\begin{bmatrix} \widehat{\mathbf{M}}_{yy} & \widehat{\mathbf{M}}_{yx} \\ \widehat{\mathbf{M}}xy & \widehat{\mathbf{M}}_{XX} \end{bmatrix} = \frac{1}{n}\left[\mathbf{Q}'\mathbf{Q} - (\lambda - n^{-1})\hat{\mathbf{A}}_a\right]$$

and

$$\widehat{\boldsymbol{\beta}} = \widehat{\mathbf{M}}_{XX}^{-1}\widehat{\mathbf{M}}_{XY}.$$

5.4.5 Residual analysis and prediction

The developments in Sections 4.7 and 4.8 can be generalized to accommodate multiple predictors. For assessing whether the errors in the equation have constant variance, generalizing (4.13), the modified squared residual is $msr_i = r_i^2 - \hat{\sigma}_{qi}^2 - \hat{\boldsymbol{\beta}}_1'\hat{\boldsymbol{\Sigma}}_{ui}\hat{\boldsymbol{\beta}}_1 + 2\hat{\boldsymbol{\beta}}_1'\hat{\boldsymbol{\Sigma}}_{uqi}$.

For prediction, consider a new unit with fixed true values $\mathbf{x}_o$ (not including the intercept) for which we observe $\mathbf{W}_o = \mathbf{x}_o + \mathbf{u}_o$, where $Cov(\mathbf{u}_o) = \boldsymbol{\Sigma}_{uo}$ with an estimate $\widehat{\boldsymbol{\Sigma}}_{uo}$. The predictor of the outcome Y_o is $\hat{Y}_o = \hat{\beta}_0 + \hat{\boldsymbol{\beta}}_1'\mathbf{W}_o$ with prediction error $T = \hat{Y}_o - Y_o$. The expected squared error of the predictor given in (4.15) generalizes to $V(T) = \sigma^2 + E(\hat{\boldsymbol{\beta}}_1'\boldsymbol{\Sigma}_{uo}\hat{\boldsymbol{\beta}}_1) + \mathbf{x}_{o*}'Cov(\hat{\boldsymbol{\beta}})\mathbf{x}_{o*}$, where $\mathbf{x}_{o*}' = (1, \mathbf{x}_o')$. Arguments similar to those used in the simple linear regression setting (see Buonaccorsi (1995)) lead to an estimated standard deviation for prediction of

$$\hat{\sigma}_{pred} = \left[\hat{\boldsymbol{\beta}}_1'\hat{\boldsymbol{\Sigma}}_{uo}\hat{\boldsymbol{\beta}}_1 + \hat{\sigma}^2 + \mathbf{W}_o'\hat{\boldsymbol{\Sigma}}_{\beta}\mathbf{W}_o - trace(\hat{\boldsymbol{\Sigma}}_{\beta}\hat{\boldsymbol{\Sigma}}_{uo})\right]^{1/2}.$$

5.5 Weighted and other estimators

There are a number of other estimates that can be used besides the simple unweighted moment corrected estimators given in (5.9). We will not try to be exhaustive here, but briefly describe some options most closely related to the approach above. Among approaches not discussed here are robust estimators (see Eltinge et al. (1993)) which build off of the estimating equations associated with the moment corrected estimates.

Using modified covariance estimates.

Use (5.9) but with modified estimators of $\boldsymbol{\Sigma}_{XX}$ and $\boldsymbol{\Sigma}_{XY}$ and $\widehat{\beta}_0 = \widehat{\mu}_Y - \widehat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\mu}}_X$, where $\widehat{\mu}_Y$ and $\widehat{\boldsymbol{\mu}}_X$ are estimators of μ_Y and $\boldsymbol{\mu}_X$, respectively. For example, with unequal numbers of replicates per observation one could use the estimates in (5.19).

General regression calibration.

When there is changing measurement error covariances an alternative way to invoke regression calibration is to regress D_i on $\widehat{\mathbf{x}}_i$ but now with $\hat{\mathbf{x}}_i = \widehat{\boldsymbol{\mu}}_X + \hat{\boldsymbol{\kappa}}_i'(\mathbf{w}_i - \widehat{\boldsymbol{\mu}}_X)$, where $\widehat{\boldsymbol{\mu}}_X$ could be $\bar{\mathbf{W}}$, or some other estimate, and $\hat{\boldsymbol{\kappa}}_i = (\widehat{\boldsymbol{\Sigma}}_{XX} + \widehat{\boldsymbol{\Sigma}}_{ui})^{-1}\widehat{\boldsymbol{\Sigma}}_{XX}$. This uses the specific estimate $\widehat{\boldsymbol{\Sigma}}_{ui}$ of the measurement error covariance for unit i and also allows for weighted estimators of $\boldsymbol{\Sigma}_{XX}$ or $\boldsymbol{\mu}_X$. See Section 6.10 for further discussion.

Small sample modifications.

Fuller (1987, chapters 2 and 3) discusses the use of a small sample correction aimed at producing estimators with better small sample properties. These involve the use of an adjustment parameter α that must be selected, and which can also be used in the correction for nonnegative matrices given in the previous section.

Weighted estimators.

As described in Section 3.1 of Fuller(1987) one can use estimated weights $\widehat{\pi}_i$ to obtain weighted estimators of the raw sums of squares and cross products, such as

$$\widehat{\mathbf{M}}_{XX\pi} = \frac{\sum_i \widehat{\pi}_i(\mathbf{W}_{i*}\mathbf{W}_{i*} - \widehat{\mathbf{\Sigma}}_{ui*})}{n} \text{ and } \widehat{\mathbf{M}}_{XY\pi} = \frac{\sum_i \widehat{\pi}_i(\mathbf{W}_{i*}D_i - \widehat{\mathbf{\Sigma}}_{uqi*})}{n}.$$

(Similar to our unweighted approach one could multiply the second terms in these expressions by $c = (n-1)/n$.) The "weighted estimator" for the coefficients is $\widehat{\boldsymbol{\beta}}_\pi = \widehat{\mathbf{M}}_{XX\pi}^{-1}\widehat{\mathbf{M}}_{XY\pi}$

To apply this approach estimated weights are needed. Fuller (1987) suggests the use of

$$\widehat{\pi}_i = 1/\widehat{\sigma}_{ei}^2$$

where $\widehat{\sigma}_{ei}^2$ is defined following (5.14). Hasabelnaby, Ware, and Fuller (1989) and Hasabelnaby and Fuller (1991) illustrate the use of these estimators and gains in efficiency in evaluating the relationship between pollution and pulmonary performance.

In practice, weighted estimators be may not possible or it may not be beneficial to weight in this manner. For example $\widehat{\sigma}_{ei}^2$ may be negative. This has occurred in some of our experiences working with some of the data from the CSFII (Continuing Survey of Food Intakes by Individuals). where there was two replicates of nutrient intake (and other variables) each year on most individuals. The magnitude of the estimated measurement error variance (due to large variability in the two measures) leads to a negative value of $\widehat{\sigma}_{ei}^2$ on many individuals. In our examples, we will see that trying to use this estimator on the LA county data produces undesirable results, while in the defoliation there are some improvements from the use of weighting.

With weights the estimates of $\mathbf{\Sigma}_\beta$ must be amended. For the robust estimate in general we can modify (5.12) to

$$\hat{\mathbf{\Sigma}}_{\beta,Rob,\pi} = \widehat{\mathbf{M}}_{XX\pi}^{-1}\hat{H}_{R\pi}\widehat{\mathbf{M}}_{XX\pi}^{-1}, \tag{5.20}$$

where$\hat{\mathbf{H}}_{R\pi} = \sum_i \hat{\mathbf{\Delta}}_i\hat{\mathbf{\Delta}}_i'/n(n-p)$ with $\hat{\Delta}_i = \widehat{\pi}_i(\mathbf{W}_{i*}r_i - \hat{\Sigma}_{uqi*} + \hat{\Sigma}_{ui*}\widehat{\boldsymbol{\beta}})$.

With weights inversely proportional to $\widehat{\sigma}^2_{ei}$, a modification to (5.13) is needed. Using equation (3.1.27) in Fuller (1987) we use

$$\hat{\mathbf{\Sigma}}_{\beta,norm,\pi} = \widehat{\mathbf{M}}^{-1}_{XX\pi}\hat{H}_{N\pi}\widehat{\mathbf{M}}^{-1}_{XX\pi}, \tag{5.21}$$

where $\hat{\mathbf{H}}_{N\pi} = \sum_{i=1}^{n}\widehat{\sigma}^{-4}_{ei}(\mathbf{W}_{i*}\mathbf{W}'_{i*}\widehat{\sigma}^2_{ei} + \widehat{\mathbf{Z}}_i\widehat{\mathbf{Z}}'_i)/n^2$.

5.6 Examples

5.6.1 Defoliation example revisited

Here we reanalyze the egg mass/defoliation example given in Table 4.1, but now allow a different intercept for each forest. The model assumes that for an observation from forest j with egg mass density x has mean $\tau_j + \beta_x x$. This can be written as $Y_i = \boldsymbol{\beta}'\mathbf{x}_i + \epsilon_i$, where $\mathbf{x}'_i = (x_{i1}, x_{i2}, x_{i3}, x_i)$, $\boldsymbol{\beta}' = (\tau_1, \tau_3, \tau_3, \beta_x)$ and x_{i1}, x_{i2} and x_{i3} are dummy variables indicating forest membership. So, $x_{ij} = 1$ if the ith observation is from forest j and equals 0 otherwise. Notice that in this formulation there is no overall intercept.

As in Chapter 4, the quantities σ^2_{ui} and σ^2_{qi} represent the measurement error variances associated with error in egg mass density (x) and defoliation (y), respectively, while σ_{uqi} is the the covariance of the two measurement errors. The estimates of these appear in Table 4.1. Since there is no overall intercept $\mathbf{\Sigma}_{u*} = \mathbf{\Sigma}_u$, so

$$\Sigma_{ui} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{ui} \end{bmatrix} \quad \text{and} \quad \Sigma_{uqi} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \sigma_{uqi} \end{bmatrix}.$$

The analysis is given in Table 5.1. The corrected estimates are based on the use of (5.9) or, equivalently, (5.10). Robust standard errors and associated Wald intervals (SE-R and CI-R) and "normal based" standard errors and associated Wald intervals (SE-N and CI-N); are based on (5.12) and (5.13), respectively.

There is a fairly substantial correction for attenuation in the estimate of β_x from .44 to a corrected estimate of .64. There are also some relatively large changes in the estimated intercepts. Recalling the discussion in illustration 3 in Section 5.3, the estimates of the intercepts will be biased if the expected value of the X variable differs across forests. The corrections for the intercepts here are related to the fact that the mean estimated egg mass density was 62.3, 33.6 and 41.6 over the three different forests, indicating "correlation" between the dummy variables indicating forest stand and the egg mass density.

The second part of the table illustrates the effects of the measurement error

Table 5.1 *Analysis of egg mass/defoliation example with changing intercepts. Cor denotes the corrected estimate; - R indicates use of robust standard error; - N indicates use of "normal" based standard error.*

	Naive (SE)	Cor	SE-R	CI-R	SE-N	CI-N
τ_1	39.21 (14.65)	26.86	21.26	$(-14.82, 68.54)$	20.66	$(-13.63, 67.35)$
τ_2	16.78 (10.26)	10.12	11.31	$(-12.05, 32.30)$	13.26	$(-15.86, 36.11)$
τ_3	-4.49 (12.37)	-12.73	12.75	$(-37.72, 12.25)$	16.31	$(-44.70, 19.24)$
β_x	0.44 (0.19)	0.638	0.239	(0.17, 1.11)	0.321	(0.01, 1.27)
σ^2	455.24	381.87				

Null Hypothesis	Naive (Pvalue)	Robust (Pvalue)	Normal (Pvalue)
$\beta_x = 0$	2.32 (.036)	2.67 (.008)	1.98 (.047)
$\tau_1 = \tau_2 = \tau_2$	5.24 (.020)	12.58 (.002)	7.29 (.026)

correction on test of hypotheses. First is a test for $\beta_x = 0$, that is of no effect of egg-mass density on defoliation, showing the naive t-test and z-test using $Z = estimate/SE$ and a P-value based on the standard normal for both types of corrected standard errors. The second line of the second part of the table provides a test of whether the three intercepts are equal. To compute this the full estimate of $\mathbf{\Sigma}_\beta$ is needed, as it is for certain other inferences, including estimating the expected defoliation at a particular density in a particular forest. Three estimates are given below, the naive covariance matrix and the robust and normal-based estimates from (5.12) and (5.13). The naive test is the standard F-test while the corrected tests are based on a chi-square statistic with 2 degrees of freedom; see Section 13.3. (The chi-square test statistics are not directly comparable to the F-statistic, unless they are divided by 2.) From the perspective of testing these hypotheses, qualitatively the conclusions are the same, whether we correct for measurement error or not.

$$\hat{\mathbf{\Sigma}}_{\beta,naive} = \begin{bmatrix} 214.75 & 74.82 & 92.77 & -2.23 \\ 74.82 & 105.34 & 49.98 & -1.20 \\ 92.77 & 49.98 & 153.02 & -1.49 \\ -2.23 & -1.20 & -1.49 & 0.04 \end{bmatrix}$$

$$\hat{\mathbf{\Sigma}}_{\beta,rob} = \begin{bmatrix} 452.17 & 122.02 & 220.91 & -4.77 \\ 122.02 & 128.03 & 66.47 & -1.31 \\ 220.91 & 66.47 & 162.52 & -2.74 \\ -4.77 & -1.31 & -2.74 & 0.06 \end{bmatrix}$$

$$\hat{\mathbf{\Sigma}}_{\beta,norm} = \begin{bmatrix} 426.75 & 194.84 & 234.53 & -5.99 \\ 194.84 & 175.77 & 132.19 & -3.37 \\ 234.53 & 132.19 & 266.06 & -4.06 \\ -5.99 & -3.37 & -4.06 & 0.10 \end{bmatrix}.$$

Lastly we implement the corrected analysis using weighting as described in the last portion of Section 5.5, with weight $1/\widehat{\sigma}^2_{ei}$ assigned to the ith observation. The weighted corrected estimates and associated inferences are given in Table 5.2. There are some changes in the weighted corrected estimates compared to the unweighted ones, but nothing very dramatic. There are indications of a gain in precision as measured by the estimated standard errors, which are smaller than in the unweighted analysis. Note also that there is generally less discrepancy between the standard errors from the robust and normal based estimates here than in the unweighted analysis.

Table 5.2 *Weighted analysis of egg mass/defoliation example with changing intercepts. Cor denotes the corrected estimate; -R indicates use of robust standard error; -N indicates use of "normal" based standard error.*

Coef.	Corrected	SE-R	CI-R	SE-N	CI-N
τ_1	30.44	19.71	$(-8.18, 69.07)$	17.69	$(-4.22, 65.11)$
τ_2	12.90	10.28	$(-7.25, 33.06)$	11.95	$(-10.51, 36.32)$
τ_3	-9.32	11.54	$(-31.94, 13.30)$	14.48	$(-37.70, 19.06)$
β_x	.58	0.23	$(0.13, 1.04)$	0.28	$(0.04, 1.12)$
σ^2	397.53				

5.6.2 LA data with error in one variable

This example uses data from Afifi and Azen (1979) based on an epidemiologic study of 200 employees from LA county. Among other variables were age and serum cholesterol levels in 1950 and 1962. We will work just with these variables for now and consider regressing serum cholesterol level in 1962 (sc62) on age and serum cholesterol in 1950 (sc50). For illustration sc62 and age are assumed to be measured without error, while sc50 is measured with error. To demonstrate some of the computational and estimation options we constructed an artificial data set by generating three replicate measures of serum cholesterol in 1950. For the ith observation three replicates were generated using $W_{ij} = x_i + u_{ij}$, $j = 1$ to 3, where u_{ij} has mean 0 and $V(u_{ij}) = (.30x_i)^2$. This is a model where the per-replicate measurement error variance has a coefficient of variation of .30. The observed value W_i is $\sum_{j=1}^3 W_{ij}/3$, which has a measurement error variance of $\sigma^2_{ui} = (.30x_i)^2/3$. This leads to relatively large measurement errors. The sc50 values ranged from 83 to 564, leading to measurement error standard deviations of σ_{ui} ranging from 14.3 to 97.65. The estimated measurement error standard deviations, the $\widehat{\sigma}_{ui}$, range from 1.3 to 125.33.

The results appear in Table 5.3. The first part of the table implements the techniques described in the preceding sections in the same manner as used for the defoliation example. Here we have run a bootstrap analysis based on resampling individuals, which is appropriate if the original sample was a random sample. There were 5000 bootstrap samples with no trimming and the bootstrap confidence intervals are percentile intervals. We remind the reader that the bootstrap does not need to assume that the per replicate variance in serum cholesterol is the same for each individual.

The second part of the table shows results from running the rcal function in STATA. The regression calibration estimates are equivalent to the moment corrected estimates. In the first set of STATA results, the default standard errors are used. These correspond to the information-based estimate of the covariance, $\widehat{\mathbf{\Sigma}}_{RC}$, the fourth estimate described in Section 5.4.2. The semi-robust option produces standard errors very similar to our robust estimate $\widehat{\mathbf{\Sigma}}_{Rob}$. As discussed in the illustration with the corn yield example in Chapter 4, the rcal bootstrap analysis trims 2% of the sample from either end and the normal based intervals use the bootstrap standard errors.

- As may be expected, the biggest effect of correcting for measurement error is in the adjustment to the coefficient for the mismeasured sc50.
- There is a very minor change in the coefficient for age due to a weak correlation between age and sc50. As noted in the description of biases, the influence of the measurement error on the coefficient of a perfectly measured predictor will be small if the mismeasured predictor is weakly correlated with the other predictor. The observed correlation between age and the error prone sc50 (the mean of the three replicates) is $-.024$. We need to be careful interpreting this, however, as this estimate is influenced by the measurement error. The sample variance of age, $S_{W1}^2 = 116.91$, and the sample covariance of age and observed sc50,$S_{W12} = -18.80$, estimate the true variance in age and covariance of age and true sc50, respectively, because there is no measurement error in age. An estimate of the variance in true sc50 is given by $\widehat{\sigma}_2^2 = S_{W2}^2 - (\sum_i \widehat{\sigma}_{ui}^2/n) = 3100.24$. Using the corrected estimate of the correlation from Chapter 4 leads to $\widehat{\rho} = -18.80/(116.92 * 3100.24)^{1/2} = -.03$. There is very little change from the naive estimate.
- The robust and bootstrap estimates of standard errors are quite similar and similar to the standard errors based on $\widehat{\mathbf{\Sigma}}_{\beta,RC}$, the default in STATA. The normal based standard error for sc50 is deflated compared to these other two

Table 5.3 *Analysis of LA data. Cor-R uses robust covariance estimate; Cor-N uses normal based covariance.*

Method	Intercept	Age	SC50	σ^2
Naive: Est (SE)	215.83 (17.45)	$-.602$ (.279)	.250 (.042)	1812
Cor-R:	171.11 (28.78)	$-.576$ (.279)	.415 (.095)	1601
Cor-N:	171.11 (25.27))	$-.576$ (.289)	.415 (.080)	1601
Bootstrap				
Mean(SE)	165.65(31.26)	-0.569 (.280)	.435 (.104)	1561(222)
C.Intervals				
Naive				
Cor-R	(114.69, 227.54)	$(-1.122, -0.029)$	(0.228, 0.602)	
Cor-N	(121.59, 220.64)	$(-1.142, -.009)$	(.258, .572)	
Bootstrap	(90.10, 218.29)	$(-1.134, -0.011)$	(.256, .686)	

```
  rcal(sc62=age) (w2:sc501 sc502 sc503)
---------------------------------------------------------------
  sc62 |  Coef.  Std.Err.  t     P>|t|    [95% Conf. Interval]
-------+-------------------------------------------------------
   age | -.576   .291    -1.98  0.049    -1.149   -.0018
    w2 |  .4150  .094     4.44  0.000      .231     .599
 _cons | 171.11   17.45   9.80  0.000    136.69   205.53
---------------------------------------------------------------
  rcal(sc62=age) (w2:sc501 sc502 sc503),robust
---------------------------------------------------------------
       |       Semi-Robust
  sc62 | Coef.  Std. Err.  t    P>|t|    [95% Conf. Interval]
-------+-------------------------------------------------------
   age | -.576   .277    -2.08   0.039   -1.12   -.030
    w2 |  .415   .097     4.30   0.000   .225     .605
 _cons | 171.11  16.81   10.18   0.000    137.97  204.254
---------------------------------------------------------------
 rcal(sc62=age) (w2:sc501 sc502 sc503), bstrap brep(5000)
---------------------------------------------------------------
       |        Bootstrap
  sc62 | Coef.   Std. Err.   t    P>|t|    [95% Conf. Interval]
-------+-------------------------------------------------------
   age | -.576    .270    -2.13  0.034   -1.108   -.0428
    w2 | .415    .095      4.37   0.000     .228      .602
 _cons | 171.11   28.56    5.99   0.000   114.79  227.43
---------------------------------------------------------------
```

and remains so even if we incorporate the additional term in (5.15) to try and account for uncertainty due to estimating the measurement error variance (although that correction is not exactly justified with changing measurement error variances).

We also ran the weighted analysis from Section 5.5, with the weights ranging from .00023 to .00062. The results are a bit puzzling. The corrected weighted estimates are 278.20, $-.555$ and $-.00003$, respectively. The estimate for the coefficient on sc50 is dramatically different than the other corrections. In addition, the estimated variance (and covariances) associated with this estimate is near 0. Without further investigation, it is unclear just how valid this analysis is. There are also other options that can be used but were not implemented here. This includes running regression calibration using imputed values based on different reliability matrices for each observation; see Section 5.5.

5.6.3 House price example

The data used here were obtained from DASL (Data and Story Library) in StatLib where it is described as *"a random sample of records of resales of homes from Feb 15 to Apr 30, 1993 from the files maintained by the Albuquerque Board of Realtors. This type of data is collected by multiple listing agencies in many cities and is used by realtors as an information base."* Here we consider regressing the sale price on square footage and the yearly tax using the 107 (out of 117) cases that have all three variables. Information on the characteristics of a house obtained from an appraisal is often subject to measurement error. For example square footage may simply rely on initial building plans and, if there were any additions, information on associated permits. Summary statistics on all three variables along with the sample covariance $\mathbf{S}_{WW}$ for tax and sqft are given below. The house price is in hundreds of dollars so the mean price is \$10,770 (the good old days). The correlation between recorded square footage and tax level is .8586.

Variable	Mean	SD	Min	Max
PRICE	1077	383.99	540.00000	2150
SQFT	1667	531.34	837.00000	3750
TAX	793.49	308.18	223.00000	1765

$$\mathbf{S}_{WW} = \begin{bmatrix} 282319.67 & 140591.03 \\ 140591.02 & 94975.08 \end{bmatrix}.$$

Error in measured square footage only.

We carry out the analysis assuming measurement error in square footage only. To illustrate the impact of the measurement error, Figure 5.2 and Figure 5.3 demonstrate the change in the estimates of the coefficients for tax and square footage, the error variance σ^2 and the correlation between tax and square footage as the measurement error standard deviation σ_u ranged from 0 to 200 in steps of 10. This corresponds to σ_u^2 ranging from 0 to 40000 with an associated reliability ratio (for square footage alone) of 1 down to .876.

Even at the higher measurement error variance of 40000 for square footage there was no problem with negative estimates of the matrices involved (see Section 5.4.4). However at the larger measurement error variances there are increasing problems with the occurrence of negative estimates when using the bootstrap. Table 5.4 provides an analysis assuming the measurement error standard deviation in square footage is 175. This leads to a measurement error variance of 30625 and an estimated reliability for square footage by itself of $282319.7/(30625 + 282319.7) = .902$. In the analysis given 52 out of 5000 bootstrap samples required modification as described in Section 5.4.4.

This example is interesting in that there is a fairly big difference between the standard errors, and associated confidence intervals, from the robust and the "normal" based estimates of $\boldsymbol{\Sigma}_\beta$. This has to do with possible heterocedasticity in the error in the equation as the difference between these reflects the differences between robust and the usual standard errors from an analysis without measurement error. For this reason, the robust or bootstrap methods should be used rather than the more conservative "normal" based ones.

With constant measurement error variance and a constant variance for the error in the equation, a weighted analysis does not offer anything since all of the weights are equal. An alternative approach, mimicking what is done without measurement error (see for example Carroll and Ruppert (1988)), is to use modified squared residuals, as defined in 4.7, to build a model for the variance of ϵ_i. Then a weighted analysis could be used with weights $\widehat{\pi}_i = 1/(\widehat{\sigma}_i^2 + \widehat{\sigma}_q^2 - 2\widehat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\Sigma}}_{uq} + \widehat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\Sigma}}_u\widehat{\boldsymbol{\beta}}_1)$, where $\widehat{\sigma}_i^2$ is an estimate of $V(\epsilon_i)$.

5.7 Instrumental variables

This section continues to assume that the measurement error is additive but now corrects for measurement error by making use of what is known as instrumental variables. These are variables that are correlated with the mismeasured x's but not correlated with the measurement errors or the errors in the equations. Hence, they carry "independent" information about the mismeasured x's which can be utilized to obtain estimates of the coefficients. Here are some examples.

- Greene (1990, p. 300) considers an example where x is household income, measured through a self reported income (W). The error in W as an estimator of x is assumed additive. The number of checks written by the household is used as an instrumental variable.

Table 5.4 *Analysis of house price data with error in square footage assuming measurement error standard deviation in square footage of 175. Cor-R uses robust covariance estimate; Cor-N uses normal based covariance. The first portion of the table gives estimates with standard errors in parentheses.*

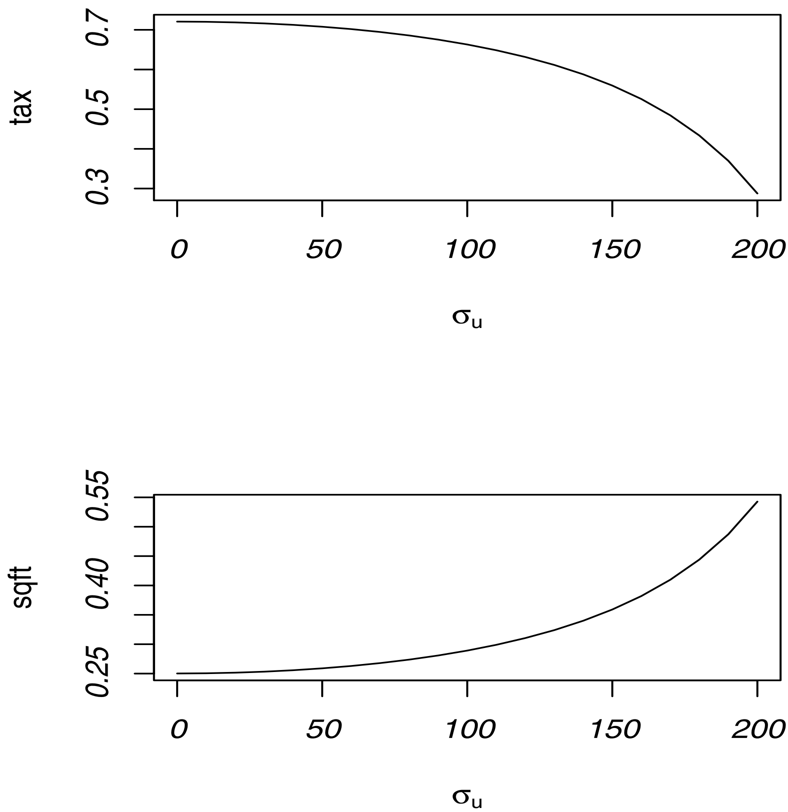
Method	Intercept	Tax	SqFt	σ^2
Naive:	88.34 (55.81)	.721 (.107)	.250 (.062)	30314.0
Naive: RobSE	88.34 (87.70)	.721 (.178)	.250 (.131)	
Cor-R:	1.71 (171.05)	.46 (.41)	.43 (.30)	27094.1
Cor-N:	1.71 (74)	.46 (.18)	.43 (.11)	27094.1
Bootstrap				
Mean(SE)	$-104.15(241.84)$	.21 (.59)	.61 (.43)	
Confidence intervals				
Naive: CI	$(-22.34, 199.01)$	(.508, .933)	(.127, .373)	
Cor-R: CI	$(-333.55, 336.98)$	$(-0.34, 1.26)$	$(-0.16, 1.01)$	
Cor-N: CI	$(-143.3, 146.75)$	(0.11, 0.81)	(0.20, 0.65)	
Boot: CI	$(-628.66, 195.33)$	$(-1.08, 0.96)$	(0.09, 1.56)	

STATA analyses using rcal function.

```
----------------------------------------------------------------
price |  Coef.   Std. Err  t     P>|t|     [95% Conf. Interval]
-------+--------------------------------------------------------
  tax | .46   .3218453     1.43   0.156   -.1777726     1.09869
    w | .43   .2236257     1.90   0.060   -.0174522    .8694642
_cons | 1.71   55.80885    0.03   0.976    -108.9563    112.3858
-----------------------------------------------------------------
      |          Semi-Robust
price |  Coef.   Std. Err   t     P>|t|     [95% Conf. Interval]
-------+---------------------------------------------------------
  tax | .46   .4149749     1.11   0.270    -.362452    1.283369
    w | .43   .2979442     1.43   0.156    -.1648285    1.016841
_cons | 1.71  87.70255     0.02   0.984    -172.2027    175.6322
-----------------------------------------------------------------
      |          Bootstrap
price |  Coef.   Std. Err.  t     P>|t|      [95% Conf. Interval]
-------+---------------------------------------------------------
  tax | .46   .5837385     0.79   0.432    -.6971165    1.618034
    w | .43   .4241468     1.00   0.318    -.415093    1.267105
_cons | 1.71   239.2451    0.01   0.994    -472.7173    476.1468
-----------------------------------------------------------------
```

- If we have two independent replicates for x then we can take one of the replicates as W and the other as an instrumental variable.
- In many cases we cannot get true replicates. This is true with many questionnaires or diet records when we want to focus on the true value at a fixed

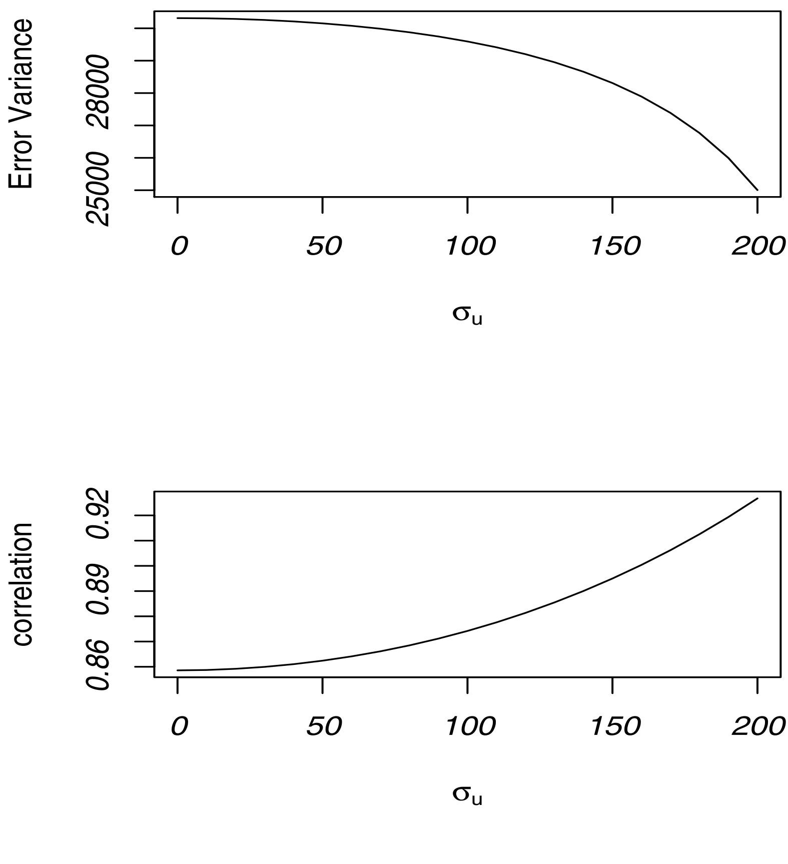
Figure 5.2 *House price example. Plot showing corrected estimates of coefficient for tax and square footage as a function of* σ_u = *standard deviation of the measurement error in square footage.*



point in time. This is because repeats over short periods of time will be correlated. In this case, an instrumental variable is helpful. This context is illustrated further in our later example.

In general, the instrumental variables are a collection of additional variables contained in an $r \times 1$ vector $\mathbf{R}_i$ where $r \geq p =$ number of elements of $\mathbf{x}_{i*}$.

Figure 5.3 *House price example. Plot showing corrected estimates of variance of the error in the equation and the correlation between square footage and tax as a function of σ_u = standard deviation of the measurement error in square footage.*



Variables in $\mathbf{x}_{i*}$ measured without error are included in $\mathbf{R}_i$, so they are instruments for themselves. For $\mathbf{R}_i$ to be instrumental for $\mathbf{x}_{i*}$ requires:

1. $\mathbf{R}_i$ is uncorrelated with $\mathbf{u}_i$, q_i and ϵ_i.
2. $\mathbf{R}_i$ is correlated with $\mathbf{x}_{i*}$.

The first assumption can be shown to be met as long as $E(\mathbf{u}_i|\mathbf{R}_i) = \mathbf{0}$, $E(\epsilon_i|\mathbf{R}_i) = \mathbf{0}$ and, if there is error in the response, $E(q_i|\mathbf{R}_i) = 0$.

The covariance between $\mathbf{R}$ and $\mathbf{x}_*$ can be defined in the way we defined other quantities in Section 5.8.2 (which accommodates fixed or random predictors). That is, with $\boldsymbol{\mu}_R = \sum_i E(\mathbf{R}_i)/n$ and $\boldsymbol{\mu}_{X*} = \sum_i E(\mathbf{X}_{i*})/n$, then

$$\boldsymbol{\Sigma}_{RX} = \frac{\sum_i E(\mathbf{R}_i - \boldsymbol{\mu}_R)(\mathbf{X}_{i*} - \boldsymbol{\mu}_{X*})}{(n-1)} = \frac{\sum_i E(\mathbf{R}_i \mathbf{x}'_{i*})}{n-1} - \frac{n}{n-1}\boldsymbol{\mu}_R\boldsymbol{\mu}'_{X*}.$$

It is necessary that $\boldsymbol{\Sigma}_{RX}$ be nonzero.

To simplify the presentation of the main idea here we'll only consider the case with $r = p$. This is not essential, however. See Section 2.4 of Fuller (1987) and Section 6.2 of Carroll et al. (2006) for generalization when $r > p$.

Define the matrix $\mathbf{R}$ to have ith row equal to $\mathbf{R}'_i$ and, as before, $\mathbf{X}$ has ith row $\mathbf{x}'_{i*}$ then $\mathbf{R}'\mathbf{X} = \sum_i \mathbf{R}_i\mathbf{x}'_{i*}$ and similarly $\mathbf{R}'\mathbf{W} = \sum_i \mathbf{R}_i\mathbf{W}'_{i*}$. The latter is assumed to be nonsingular which is tied to the assumptions that $\mathbf{R}_i$ is instrumental. With $r = p$ and $\mathbf{R}'\mathbf{W}$ nonsingular (of rank p) then the instrumental variable estimator is

$$\hat{\boldsymbol{\beta}}_{IV} = (\mathbf{R}'\mathbf{W})^{-1}\mathbf{R}'\mathbf{D}.$$

For simple linear regression, the instrumental variable estimators become

$$\widehat{\beta}_1 = S_{DR}/S_{WR} \text{ and } \widehat{\beta}_0 = \bar{D} - \widehat{\beta}_1\bar{W},$$

where S_{WR} is the sample covariance of the univariate W and the univariate instrument R and S_{DR} is similarly defined.

Motivation for the instrumental variables estimator as well as the estimated covariance matrices below appears in Section 5.8.4.

Estimating the covariance of $\widehat{\boldsymbol{\beta}}_{IV}$.

- For simple linear regression with constant measurement error variance Fuller (1987, p. 54) provides an estimated covariance matrix of the form

$$\widehat{\boldsymbol{\Sigma}}_\beta = \begin{bmatrix} v_{00} & v_{01} \\ v_{01} & v_{11} \end{bmatrix},$$

  where v_{11} = estimated variance of $\widehat{\beta}_1 = MSE_c S_{RR}/(n-1)S^2_{RW}$, where S_{RR} is the sample variance of the univariate instrumental variable R (so not including the intercept), S_{RW} is the sample covariance of the univariate W and R and as before $MSE_c = \sum_i (D_i - \widehat{\boldsymbol{\beta}}'\mathbf{W}_{i*})^2/(n-p)$. Also $v_{00} = (MSE_c/n) + \bar{W}^2 v_{11}$ and $v_{01} = \bar{W}v_{11}$.
- In general, we can argue for estimating the covariance of $\widehat{\boldsymbol{\beta}}$ using either

$$\widehat{\boldsymbol{\Sigma}}_{\beta,IV} = MSE_c(\mathbf{R}'\mathbf{W})^{-1}\mathbf{R}'\mathbf{R}(\mathbf{R}'\mathbf{W})^{-1} \tag{5.22}$$

or a robust/sandwich type estimator of the covariance given by

$$\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{\beta},rob} = (\mathbf{R}'\mathbf{W})^{-1}\mathbf{R}'\mathbf{Q}\mathbf{R}(\mathbf{R}'\mathbf{W})^{-1} \tag{5.23}$$

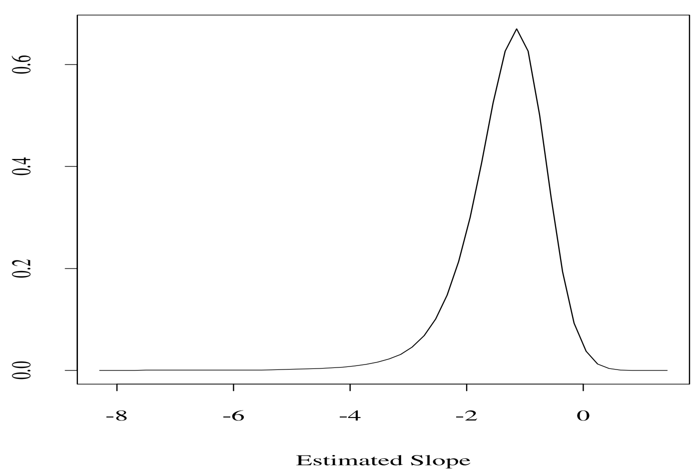
where $\mathbf{Q}$ is a diagonal matrix with the ith diagonal element equal to r_i^2, where $r_i = D_i - \mathbf{W}'_{i*}\widehat{\boldsymbol{\beta}}_{IV}$ and $MSE_c = \sum_i r_i^2/(n-p)$. See Section 5.8.4 for a heuristic argument for these expressions. We also refer the interested reader to Fuller (1987, Ch. 2) for his developments on estimating the covariance matrix of $\widehat{\boldsymbol{\beta}}_{IV}$ in simple linear regression while Carroll et al. (2006, Appendix B6) have a fairly extensive treatment of instrumental variables and techniques for estimating the covariance for multiple linear regression. For simple linear regression $\widehat{\boldsymbol{\Sigma}}_{\beta,IV}$ produces the estimated covariance matrix in Chapter 1 of Fuller (1987). We could also modify the robust estimator above by multiplying by $n/(n-p)$ which adjust for degrees of freedom. Doing this yields the robust estimator produced in the instrumental variable analysis in STATA.

5.7.1 Example

This example uses some data from the control arm of a study on beta-carotene intake and skin cancer (Greenberg et al. (1990)). These data are also used in a longitudinal context in Section 11.3.2. The data actually involve six year of data. Here we use only data from the last time point and consider fitting a linear regression model for a fat index (quetelet) on true beta-carotene intake (x) at that time. The observed beta-carotene intake W, from a diet record, is assumed to be unbiased for the true intake, i.e., the measurement error is additive. The data also contain a serum measure of beta-carotene, which, without any replicate measures of intake, is used as the instrumental variable.

Table 5.5 presents the naive fit and an instrumental variable analysis where the lab beta-carotene in year 5 is used as the instrumental variable. Analytical standard errors for the instrumental variable estimates have been computed three ways. The ones labeled IV-Full (designating agreement with Fuller's covariance matrix in the case of simple linear regression) come from the use of (5.22), while IV-Rob uses (5.23). The IV-RobA standard errors use an adjustment, multiplying IV-Rob by $n/(n-p) = 156/154$. The bootstrap standard errors and associated percentile intervals are based on 5000 bootstrap samples. We have also shown analyses from the use of SAS-SYSLIN (the instrumental variable problem is a special case of the use of simultaneous linear equations) and STATA's ivreg. The standard errors from SYSLIN and the default analysis in STATA's ivreg are identical to those labeled IV-Full. Notice that the confidence intervals from STATA, which are computed based on a t distribution with 154 degrees of freedom, differ in a minor way from ours, based on the use of a standard normal. Focusing on the coefficient for diet intake, β_1, the percentile interval from the use of the bootstrap is quite different than those from the Wald-based intervals using various standard errors. In fact from the perspective of testing $H_0 : \beta_1 = 0$, we would not reject based on the bootstrap, while we would reject based on the other analyses. The difference is due to the skewed nature of the distribution as represented by the empirical bootstrap distribution, represented in Figure 5.4. Given the lack of normality, based on the bootstrap results, the bootstrap percentile intervals should be used here rather than the Wald intervals.

Figure 5.4 *Instrumental variable example. Smoothed density estimate for the estimated slope from 5000 bootstrap samples.*



5.8 Mathematical developments

5.8.1 Motivation for moment corrections

The motivation for the corrected estimators and the bias expressions come from the fact that if we condition on the x's and y's then $E(\mathbf{S}_{WW}) = \mathbf{S}_{xx} + \mathbf{\Sigma}_u$, $E(\mathbf{S}_{WD}) = \mathbf{S}_{xy} + \mathbf{\Sigma}_{uq}$, $E(\mathbf{S}^2_d) = \mathbf{S}^2_y + \sigma^2_q$, where $\mathbf{S}_{xx}$, $\mathbf{S}_{xy}$ and S^2_d are similar to $\mathbf{S}_{XX}$, $\mathbf{S}_{XY}$ and S^2_D, but involve the fixed values. Unconditionally, using

Table 5.5 *Analysis of the beta-carotene data using instrumental variables.* v_{01} *is the estimated covariance of* $\widehat{\beta}_0$ *and* $\widehat{\beta}_1$. *See text for additional notation.*

Method	$\widehat{\beta}_0$ (SE)	$\widehat{\beta}_1$ (SE)	v_{01}
Naive:	25.55 (.600)	$-.203$(.122)	-0.065
IV-Full:	30.06(2.456)	-1.24 (.562)	-1.366
IV-Rob:	30.06(2.248)	-1.24 (.511)	-1.14
IV-RobA:	30.06(2.263)	-1.24 (.514)	-1.15
Bootstrap			
Mean(SE)	30.06(3.38)	-1.37 (.765)	
Confidence Intervals			
Naive	(24.37, 26.74)	($-.444$, .039)	
IV-Full	(25.24, 34.87)	(-2.345, $-.1419$)	
IV-RobA	(25.62, 34.49)	(-2.251, -0.236)	
Bootstrap	(26.43, 37.41)	(-2.94, .418)	

```
 SAS                        The SYSLIN Procedure
                      Two-Stage Least Squares Estimation
                           Parameter    Standard
 Variable      DF      Estimate         Error     t Value     Pr > |t|
 Intercept      1      30.05561     2.456473       12.24       <.0001
 dietcar5       1      -1.24326     0.561933       -2.21       0.0284

              ****** STATA ANALYSIS USING IVREG *******
-------------------------------------------------------------------
 quetelet |  Coef.   Std. Err.  t    P>|t|    [95% Conf. Interval]
----------+--------------------------------------------------------
 dietcar5 | -1.243    .5619   -2.21   0.028   -2.3533   -.1332
    _cons |  30.056   2.457   12.24   0.000    25.203    34.908
-------------------------------------------------------------------
          |         Robust
 quetelet |  Coef.   Std. Err.  t    P>|t|     [95% Conf. Interval]
----------+--------------------------------------------------------
 dietcar5 | -1.243   .5140    -2.42   0.017    -2.2588   -.2278
    _cons | 30.056    2.263   13.28   0.000     25.585   34.526
-------------------------------------------------------------------
Instrumented:  dietcar5
Instruments:   labcar5
```

double expectations $E(\mathbf{S}_{WW}) = E(\mathbf{S}_{XX}) + \Sigma_u = \mathbf{\Sigma}_{XX} + \mathbf{\Sigma}_u$, $E(\mathbf{S}_{WD}) = E(\mathbf{S}_{XY)} + \Sigma_{uq} = \mathbf{\Sigma}_{XY} + \mathbf{\Sigma}_{uq}$ and $E(\mathbf{S}_d^2) = E(\mathbf{S}_Y^2) + \sigma_q^2 = \sigma^2 + \boldsymbol{\beta}_1'\mathbf{\Sigma}_{XX}\boldsymbol{\beta}_1 + \sigma_q^2$.

Similarly, in terms of the uncorrected sums of squares, conditionally $E(\mathbf{W}'\mathbf{W}) = \mathbf{X}'\mathbf{X} + \Sigma_{u*}$ and $E(\mathbf{W}'\mathbf{D}) = \mathbf{X}'\mathbf{Y} + \Sigma_{uq*}$.

5.8.2 Defining terms for general combinations of predictors

In multiple linear regression some of the predictors may be fixed and some may be random. Even if all are random, they may not be identically distributed over different observations. To define approximate biases and correction methods in a general way, without having to separately isolate the functional and structural settings, it is useful to have a broad definition of $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_{XX}$ which covers all settings.

First define the "random vector" $\mathbf{X}_i = (X_{i1}, \ldots, X_{i,p-1})$. The jth variable X_{ij} could be random in the usual sense of the word, that is in having nonzero variance, which we will refer to as "truly random." Or, if the jth predictor is fixed at x_{ij}, then $X_{ij} = x_{ij}$ (fixed) with 0 variance. When we refer to the expected value and variance of a truly random X_{ij} it is actually the conditional mean and variance of X_{ij} given the fixed x's.

Defining $\bar{\mathbf{X}} = \sum_i \mathbf{X}_i/n$, we define $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_{XX}$ via

$$\boldsymbol{\mu}_X = \frac{\sum_i E(\mathbf{X}_i)}{n},$$

and

$$\boldsymbol{\Sigma}_{XX} = \frac{\sum_{i=1}^n E[(\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})']}{n-1} = \frac{\sum_{i=1}^n E(\mathbf{X}_i\mathbf{X}_i') - nE(\bar{\mathbf{X}}\bar{\mathbf{X}}')}{n-1}.$$

To be more explicit, suppose that the predictors are arranged so that $\mathbf{X}_i' = (\mathbf{X}_{i1}', \mathbf{x}_{i2})$, where $\mathbf{x}_{i2}$ is a set of fixed values, and the components of $\mathbf{X}_{i1}$ are "truly random." With the $\mathbf{x}_{i2}$ being fixed the moments of $\mathbf{X}_{i1}$ are conditional and denoted by $E(\mathbf{X}_{i1}|\mathbf{x}_{i2}) = \mu_{1|2i}$ and covariance matrix $\boldsymbol{\Sigma}_{1|2i} = Cov(\mathbf{X}_{i1}|\mathbf{x}_{i2})$. Define

$\bar{\boldsymbol{\Sigma}}_{1|2} = \sum_i \boldsymbol{\Sigma}_{1|2i}/n$

$\bar{\boldsymbol{\mu}}_{1|2} = \sum_i \boldsymbol{\mu}_{1|2i}/n$

$\mathbf{S}_{\mu\mu} = \sum_i (\boldsymbol{\mu}_{1|2i} - \bar{\boldsymbol{\mu}}_{1|2})(\boldsymbol{\mu}_{1|2i} - \bar{\boldsymbol{\mu}}_{1|2})'/(n-1)$

$\mathbf{S}_{\mu 2} = \sum_i (\boldsymbol{\mu}_{1|2i} - \bar{\boldsymbol{\mu}}_{1|2})(\mathbf{x}_{2i} - \bar{\mathbf{x}}_2)'/(n-1)$ and

$\mathbf{S}_{22} = \sum_i (\mathbf{x}_{2i} - \bar{\mathbf{x}}_2)(\mathbf{x}_{2i} - \bar{\mathbf{x}}_2)'/(n-1)$.

With these items defined straightforward calculations yield

$$\boldsymbol{\mu}_X = \begin{bmatrix} \bar{\boldsymbol{\mu}}_{1|2} \\ \bar{\mathbf{x}}_2 \end{bmatrix} \quad \text{and} \quad \boldsymbol{\Sigma}_{XX} = \begin{bmatrix} \bar{\boldsymbol{\Sigma}}_{1|2} + \mathbf{S}_{\mu\mu} & \mathbf{S}_{\mu 2} \\ \mathbf{S}_{\mu 2}' & \mathbf{S}_{22} \end{bmatrix}.$$

Notice that in the fully structural case with all variables truly random then $\mathbf{x}_2$ is empty. However, the $\mathbf{X}_i$ can have a changing expected value or covariance.

If the $\mathbf{X}_i$ are identically distributed then $\boldsymbol{\mu}_X = E(\mathbf{X}_i)$, $Cov(\mathbf{X}_i) = \boldsymbol{\Sigma}_{XX}$ and $\mathbf{S}_{\mu\mu} = \mathbf{0}$.

In the functional case with all variables fixed then $\mathbf{X}_{i1}$ is empty, and $\mathbf{x}_{2i} = \mathbf{x}_i$. This results in $\bar{\boldsymbol{\mu}} = \sum_i \mathbf{x}_i/n = \bar{\mathbf{x}}$ and $\mathbf{S}_{xx} = \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})')/(n-1)$.

5.8.3 Approximate covariance of estimated coefficients

Here we provide more detail concerning the approximate covariance of $\widehat{\boldsymbol{\beta}}$, discussed in Section 5.4.2. The emphasis is on presenting the general strategy in trying to find and estimate the covariance matrix. Most of the technical results used in this section are from Fuller (1987). Cheng and Van Ness (1999) also provide results for a number of special cases, most involving normality assumptions.

Under fairly general conditions (see Fuller (1987, Ch. 3)), $\hat{\boldsymbol{\beta}}$ is consistent for $\boldsymbol{\beta}$ and $\widehat{\boldsymbol{\beta}} \approx \boldsymbol{\beta} + \mathbf{M}_{XX}^{-1} \sum_i \boldsymbol{\Delta}_i/n$. This leads to $\hat{\boldsymbol{\beta}}$ being approximately normal with mean $\boldsymbol{\beta}$ and approximate covariance matrix

$$\boldsymbol{\Sigma}_\beta = \mathbf{M}_{XX}^{-1}\mathbf{H}\mathbf{M}_{XX}^{-1}$$

with

$$\mathbf{H} = Cov(\sum_{i=1}^{n} \boldsymbol{\Delta}_i)/n^2,$$

where

$$\boldsymbol{\Delta}_i = \mathbf{W}_{i*}(D_i - \mathbf{W}_{i*}'\boldsymbol{\beta}) - (\hat{\boldsymbol{\Sigma}}_{uqi*} - \hat{\boldsymbol{\Sigma}}_{ui*}\boldsymbol{\beta}) = \mathbf{T}_i - \mathbf{Z}_i,$$

$$e_i = \epsilon_i + q_i - \mathbf{u}_i'\boldsymbol{\beta}_1, \qquad T_i = \mathbf{W}_{i*}e_i \quad \text{and} \quad \mathbf{Z}_i = (\hat{\boldsymbol{\Sigma}}_{uqi*} - \hat{\boldsymbol{\Sigma}}_{ui*}\boldsymbol{\beta}).$$

Since $\mathbf{W}_{i*} = \mathbf{x}_{i*} + \mathbf{u}_{i*}$, $E(\mathbf{T}_i) = E(\mathbf{Z}_i) = \boldsymbol{\Sigma}_{uqi*} - \boldsymbol{\Sigma}_{ui*}\boldsymbol{\beta}$ so $E(\boldsymbol{\Delta}_i) = \mathbf{0}$.

There are two things of interest here. The first is to evaluate $\mathbf{H}$ so an analytical expression for $Cov(\widehat{\boldsymbol{\beta}})$ is available. This would produce explicit expressions that isolate the contribution of the measurement error as well as the uncertainty arising from estimating the measurement error variances and covariances. The second is to obtain an estimate of $\boldsymbol{\Sigma}_\beta$, which may or may not rely on explicit expression for $\mathbf{H}$. The estimated covariance matrix will have the general form of

$$\hat{\boldsymbol{\Sigma}}_\beta = \widehat{\mathbf{M}}_{XX}^{-1}\hat{\mathbf{H}}\widehat{\mathbf{M}}_{XX}^{-1}, \tag{5.24}$$

with special cases resulting from assumptions about the various quantities involved. A list of the questions that enter into consideration was given in Section 5.4.2.

Note that we can also write $\mathbf{H} = Cov(\bar{\mathbf{T}}) + Cov(\bar{\mathbf{Z}}) - 2Cov(\bar{\mathbf{T}}, \bar{\mathbf{Z}})$, where $\bar{\mathbf{T}}$ and $\bar{\mathbf{Z}}$ are the mean of the $\mathbf{T}_i$'s and $\mathbf{Z}_i$'s, respectively.

For convenience let $\boldsymbol{\theta}_i = (\boldsymbol{\Sigma}_{uqi}, \boldsymbol{\Sigma}_{ui})$ be the collection of measurement error parameters for the ith observation and $\widehat{\boldsymbol{\theta}}_i = (\widehat{\boldsymbol{\Sigma}}_{uqi}, \widehat{\boldsymbol{\Sigma}}_{ui})$. If there is only a single set of estimates $\widehat{\boldsymbol{\Sigma}}_{uq}$ and $\widehat{\boldsymbol{\Sigma}}_u$ then we refer simply to $\widehat{\boldsymbol{\theta}}$.

1. Robust estimate of $\boldsymbol{\Sigma}_\beta$.
   If a separate $\widehat{\boldsymbol{\theta}}_i$ is available on each unit or if $\boldsymbol{\theta}_i$ is assumed known (and possibly the same for all i), then the $\mathbf{Z}_i$, and the $\boldsymbol{\Delta}_i$, are uncorrelated over i. Recalling that $\boldsymbol{\Delta}_i$ has mean $\mathbf{0}$ this leads to
   $$\mathbf{H} = \frac{1}{n^2}\sum_{i=1}^{n} E(\boldsymbol{\Delta}_i \boldsymbol{\Delta}_i'),$$
   and results in the robust estimate given in (5.12). See Theorem 3.1.1 of Fuller (1987) for further justification. As Fuller does we use a divisor of $n(n-p)$ in place of n^2.
2. Estimated measurement error variances and covariances treated as known with normal measurement errors.
   If the estimated measurement error variances and covariances are treated as known, then the $\mathbf{Z}_i$ are constants, so there is no uncertainty associated with them and $\mathbf{H} = \sum_{i=1}^{n} Cov(\mathbf{T}_i)/n^2$. The robust estimate is still applicable but other estimates can be formed under additional assumptions. If we assume that the measurement errors are normal, then an exact expression for $Cov(\mathbf{T}_i)$ can be obtained leading to an estimate of $\boldsymbol{\Sigma}_\beta$ as given in equation (3.1.23) of Fuller (1987) and in (5.13). This is referred to as the *normal based estimate*.
3. Estimated measurement error variances and covariances assumed independent of $\mathbf{T}_i$.
   In this case $\mathbf{H} = \sum_i (Cov(\mathbf{T}_i) + Cov(\mathbf{Z}_i))/n^2 = Cov(\bar{\mathbf{T}}) + Cov(\bar{\mathbf{Z}})$ and we can use
   $$\hat{\mathbf{H}} = \frac{\sum_{i=1}^{n}(\widehat{\boldsymbol{\Sigma}}_{Ti} + \widehat{\boldsymbol{\Sigma}}_{Zi})}{n^2},$$
   where $\widehat{\boldsymbol{\Sigma}}_{Ti}$ is an estimate of $Cov(\mathbf{T}_i)$ and $\widehat{\boldsymbol{\Sigma}}_{Zi}$ an estimate of $Cov(\mathbf{Z}_i)$. We can estimate $Cov(\mathbf{T}_i)$ and $Cov(\mathbf{Z}_i)$ individually with additional assumptions. For example if we have normal replicates then the distribution of the components of the estimated measurement error variances and covariances is based on the Wishart distribution and $\widehat{\boldsymbol{\Sigma}}_{Zi}$ can be determined. Similarly if we assume the measurement errors are normally distributed an estimator $\widehat{\boldsymbol{\Sigma}}_{Ti}$ can be obtained.
4. Using common estimated measurement error parameters.
   Suppose here that we use an overall $\widehat{\boldsymbol{\Sigma}}_u$ and $\widehat{\boldsymbol{\Sigma}}_{uq}$ so for each i, $\mathbf{Z}_i = \mathbf{Z} = \hat{\boldsymbol{\Sigma}}_{uq*} - \hat{\boldsymbol{\Sigma}}_{u*}\boldsymbol{\beta}$. This means that the $\boldsymbol{\Delta}_i$ are no longer uncorrelated over i (with estimated rather than known measurement error parameters). In general,

$\mathbf{H} = Cov(\bar{\mathbf{T}}) + Cov(\mathbf{Z}) - 2Cov(\bar{\mathbf{T}}, \mathbf{Z})$, but this is not useful without further assumptions.
Under normality assumptions with the measurement error parameters treated as known

$$\hat{\mathbf{H}} = \sum_{i=1}^{n} (\mathbf{W}_{i*}\mathbf{W}_{i*}\widehat{\sigma}_{ei}^2 + \widehat{\mathbf{Z}}\widehat{\mathbf{Z}}')/n^2. \tag{5.25}$$

This is equivalent to (5.13) with $\widehat{\mathbf{Z}}_i$ replaced by the common $\widehat{\mathbf{Z}}$.
Theorem 1.2.1 in Fuller (1987) gives an explicit expression in the special case of simple linear regression with no error in Y (so σ_{uq} and σ_q^2 equal 0) and normality (as is assumed here) and known common σ_u^2.

5. $\mathbf{Z}$ independent of the $\mathbf{T}_i$'s.
In this case, $\mathbf{H} = Cov(\bar{\mathbf{T}}) + Cov(\mathbf{Z})$ and similar to the earlier case, we can use

$$\hat{\mathbf{H}} = \frac{\sum_{i=1}^{n} \widehat{\boldsymbol{\Sigma}}_{Ti} + \widehat{\boldsymbol{\Sigma}}_Z}{n^2},$$

where $\widehat{\boldsymbol{\Sigma}}_{Ti}$ is an estimate of $Cov(\mathbf{T}_i)$ and $\widehat{\boldsymbol{\Sigma}}_Z$ an estimate of $Cov(\mathbf{Z})$.
A special case of this result is Theorem 2.2.1 in Fuller, who combines assumptions leading to explicit forms for $\widehat{\boldsymbol{\Sigma}}_{Ti}$ and $\widehat{\boldsymbol{\Sigma}}_Z$. $\widehat{\boldsymbol{\Sigma}}_{Ti}$ is obtained under the assumption of a normal structural model with normal measurement error and the matrix of estimated measurement error variances and covariances having a distribution that is proportional to that of a Wishart distribution (as happens with normal replicates and common measurement error variances and covariances). In this case the estimated covariance matrix in Theorem 2.2.1 of Fuller can be expressed as $\hat{\boldsymbol{\Sigma}}_\beta = \widehat{\mathbf{M}}_{XX}^{-1}\hat{\mathbf{H}}_{NC}\widehat{\mathbf{M}}_{XX}^{-1}$, where $\mathbf{H}_{NC}$ is defined following (5.15).

5.8.4 *Instrumental variables*

- One way to motivate the instrumental variable estimator is to assume $E(\mathbf{W}_i|\mathbf{R}_i) = \mathbf{B}\mathbf{R}_i$, where $\mathbf{B}$ is a $p \times p$ matrix of coefficients of rank p. This is a multivariate regression model. The fact that that $\mathbf{R}_i$ is instrumental also leads to
$$E(D_i|\mathbf{R}_i) = \boldsymbol{\beta}'\mathbf{B}\mathbf{R}_i = \boldsymbol{\pi}'\mathbf{R}_i, \tag{5.26}$$
where $\boldsymbol{\pi} = \mathbf{B}'\boldsymbol{\beta}$. This implies $\boldsymbol{\beta} = \mathbf{B}'^{-1}\boldsymbol{\pi}$. Using results from multivariate regression, an estimate of $\mathbf{B}'$ is $(\mathbf{R}'\mathbf{R})^{-1}(\mathbf{R}'\mathbf{W})$ and from multiple linear regression an estimate of $\boldsymbol{\pi}$ is $(\mathbf{R}'\mathbf{R})^{-1}\mathbf{R}'\mathbf{D}$. Combining these two
$$\widehat{\boldsymbol{\beta}} = \hat{\mathbf{B}}'^{-1}\widehat{\boldsymbol{\pi}} = (\mathbf{R}'\mathbf{W})^{-1}\mathbf{R}'\mathbf{D},$$
which is the instrumental variable estimator.

- Another way to view the estimator is to note that $\mathbf{R}'\mathbf{W} = \sum_i \mathbf{R}_i \mathbf{W}'_{i*}$ and $\mathbf{R}'\mathbf{D} = \sum_i \mathbf{R}_i D_i$. Using the fact that $\mathbf{R}_i$ is uncorrelated with the measurement errors and writing $D_i = Y_i + q_i = \mathbf{X}'_{i*}\boldsymbol{\beta} + \epsilon_i + q_i$ leads to $E(\mathbf{R}'\mathbf{W}) = E(\sum_i \mathbf{R}_i \mathbf{W}'_{i*}) = E(\sum_i \mathbf{R}_i \mathbf{X}'_{i*}) = (n-1)\boldsymbol{\Sigma}_{RX} + n\boldsymbol{\mu}_R\boldsymbol{\mu}_{X*}$ $= \mathbf{H}$, say and
$$E(\mathbf{R}'\mathbf{D}) = E(\sum_i \mathbf{R}_i \mathbf{X}'_{i*}\boldsymbol{\beta}) = \mathbf{H}\boldsymbol{\beta}.$$
So, under suitable conditions as the sample size increases $\hat{\boldsymbol{\beta}}_{IV}$ is approximately $(E(\mathbf{R}'\mathbf{W}))^{-1}E(\mathbf{R}'\mathbf{D}) = \mathbf{H}^{-1}\mathbf{H}\boldsymbol{\beta} = \boldsymbol{\beta}$.

Notice that the above argument requires $\mathbf{H}$ to be nonsingular and that this will not work if $\boldsymbol{\Sigma}_{RX}$ is $\mathbf{0}$ since then $\mathbf{H} = n\boldsymbol{\mu}_R\boldsymbol{\mu}_{X*}$ which is a matrix of rank 1 and hence singular.

Covariance matrices.

The estimates of $Cov(\widehat{\boldsymbol{\beta}}) = \boldsymbol{\Sigma}_{\boldsymbol{\beta}}$, given in section 5.7, come from the fact that $\boldsymbol{\Sigma}_\beta \approx (\mathbf{R}'\mathbf{W})^{-1}Cov(\mathbf{R}'\mathbf{D})(\mathbf{R}'\mathbf{W})^{-1}$ and using the fact that $Cov(\mathbf{R}'\mathbf{D}) = \sum_i Cov(\mathbf{R}_i D_i)$. Since $Cov(\mathbf{R}_i D_i) = E[(\mathbf{R}_i D_i - E(\mathbf{R}_i D_i))(\mathbf{R}_i D_i - E(\mathbf{R}_i D_i))'] = E[(\mathbf{R}_i D_i - E(\mathbf{R}_i X'_{i*}\boldsymbol{\beta}))(\mathbf{R}_i D_i - E(\mathbf{R}_i \mathbf{X}'_{i*}\boldsymbol{\beta}))'] = E[\mathbf{R}_i(D_i - X'_{i*}\boldsymbol{\beta})^2\mathbf{R}'_i]$ we can estimate $\sum_i Cov(\mathbf{R}_i D_i)$ with $\sum_i \mathbf{R}_i r_i^2 \mathbf{R}'_i = \mathbf{R}'\mathbf{Q}\mathbf{R}$.

- Development of (5.26).
$D_i = \boldsymbol{\beta}'\mathbf{x}_{i*} + q_i + \epsilon_i$, so $E(D_i|\mathbf{R}_i) = E(\boldsymbol{\beta}'\mathbf{x}_{i*} + q_i + \epsilon_i|\mathbf{R}_i) = E(\boldsymbol{\beta}'(\mathbf{W}_{i*} - \mathbf{u}_{i*}) + \epsilon_i|\mathbf{R}_i) = \boldsymbol{\beta}'E(\mathbf{W}_{i*}|\mathbf{R}_i) - \boldsymbol{\beta}'E(\mathbf{u}_{i*}|\mathbf{R}_i) + E(\epsilon_i|\mathbf{R}_i)$. But, the last two terms are 0 as a result of $\mathbf{R}_i$ being uncorrelated so $E(D_i|\mathbf{R}_i) = \boldsymbol{\beta}'E(\mathbf{W}_{i*}|\mathbf{R}_i) = \boldsymbol{\beta}'\mathbf{B}_2\mathbf{R}_i$.

CHAPTER 6

Measurement Error in Regression: A General Overview

6.1 Introduction

The preceding four chapters were dedicated to the treatment of specific problems dealing with misclassification in categorical variables and additive measurement error in linear regression problems. Those settings allowed us to take an initial look at the key features in treating measurement error problems in relatively simple settings. This chapter provides a broader look at measurement error in general regression problems with subsequent chapters dedicated to application of the tools developed here to specific problems. The term regression is used here in a broad sense as the models presented can accommodate categorical or analysis of variance models, among others. In this chapter we:

1. Provide a quick review of regression models and methods of analysis without measurement error (Sections 6.2 and 6.3).
2. Describe measurement error models in a general fashion, including non-additive models and models with changing measurement error parameters (Section 6.4).
3. Describe extra types of data that are often used to estimate measurement error parameters, including replication, and external or internal validation data (Section 6.5).
4. Outline general approaches available for assessing biases in naive estimators (Sections 6.5–6.7). These include the use of "induced models," which are models for the observed rather than true values, as well as methods using estimating equations. We characterize biases in a number of settings beyond those addressed in the preceding chapters. The discussion here is also important in motivating some of the correction techniques.
5. Provide a broad overview of a number of techniques that can be used to correct for measurement error (Sections 6.9–6.16).

Some mathematical developments are presented in Section 6.17.

Within the chapter some specific models are used for illustration to break up the general discussion. The main goal here, however, is to provide the big ideas regarding models, additional data and correction methods, and then isolate specific problems in more detail in later chapters. This should allow the reader to go right to a specific chapter of interest (e.g., binary regression) and return to this chapter as needed.

While this chapter is fairly broad, it certainly does not cover everything. This is due to both space and time considerations but also in keeping with the goal of the book to be introductory in nature and focus on some basic models and methods. See the comments in the preface also. With respect to correction methods, the focus here and in subsequent chapters is on a few, relatively easy to use techniques. These include moment based corrections, regression calibration, SIMEX, modified estimating equations and, to a lesser extent, likelihood based methods. Likelihood methods are described in Section 6.12 in fairly general terms. The emphasis is primarily on the different formulations that can be used. There are some, but limited, applications of the likelihood methods in later chapters, due to computational availability. There are a number of other potentially important methods that are not covered here, for the reasons mentioned above. These include, among others, mean-score methods (e.g., Pepe et al. (1994)), expected estimating equations (e.g., Wang et al. (2008)) methods based on the empirical likelihood (e.g., Stute et al. (2007)) and a variety of other parametric and semiparametric methods (e.g., Huang and Wang (2006) and Holcroft et al. (1997)). See Carroll et al. (2006) also. There are many more references that could be listed, but these should provide access to some of the associated literature.

The other notable omission in this and subsequent chapters is further discussion of the use of instrumental variables. This was discussed in Section 5.7 for multiple linear regression. Chapter 6 of Carroll et al. (2006) discusses this in detail. A few good starting points in the journal literature are Buzas and Stefanski (1996) and Thoresen and Laake (1999).

6.2 Models for true values

For the true values, the general regression model is

$$Y_i|\mathbf{x}_i = m(\mathbf{x}_i, \boldsymbol{\beta}) + \epsilon_i, \tag{6.1}$$

$$E(\epsilon_i) = 0 \text{ and } V(\epsilon_i|\mathbf{x}_i) = v(\mathbf{x}_i, \boldsymbol{\beta}, \boldsymbol{\sigma}),$$

with the ϵ_i assumed to be independent.

The function $m(\mathbf{x}_i, \boldsymbol{\beta})$ defines the regression model and the variance function $v(\mathbf{x}_i, \boldsymbol{\beta}, \boldsymbol{\sigma})$ allows the variance to be some function of the predictors, $\mathbf{x}_i$, the parameters entering into the mean function $\boldsymbol{\beta}$, and some additional parameters contained in $\boldsymbol{\sigma}$. In the constant variance case $v(\mathbf{x}_i, \boldsymbol{\beta}, \boldsymbol{\sigma}) = \sigma^2$.

Special cases include linear regression, polynomial regression, analysis of variance models, general nonlinear models, and generalized linear models, including logistic, probit and Poisson regression.

- Linear models.
  The linear here refers to linear in the parameters, which includes models with polynomial terms and products/interactions. Models that are linear in the distinct variables with additive measurement error have already been treated in Chapters 4 and 5. If, however, the model involves nonlinear functions of the variables measured with additive errors then the methods from those chapters do not immediately apply. For example consider the model with a variable x with $E(Y|x) = \beta_0 + \beta_1 x + \beta_2 x^2$, so $x_1 = x$ and $x_2 = x^2$, and with W_i unbiased for x_i; that is, $W_i = x_i + u_i$ with $E(u_i) = 0$. W_i^2, however, is not unbiased for x_i^2 since $E(W_i^2) = x_i^2 + \sigma_{ui}^2$. Similarly if two predictors x_1 and x_2 are measured with error and the measurement errors are correlated then $E(W_{i1}W_{i2}|x's) = x_{i1}x_{i2} + \sigma_{ui12}$, where σ_{ui12} is the covariance of the two measurement errors. Additional examples include other nonlinear functions of a variable measured with error, such as $E(Y|x) = \beta_0 + \beta_1 log(x)$, where W is unbiased for x. These problems are explicitly covered in Chapter 8.
- Nonlinear models.

  Nonlinear models arise in almost every discipline; see Seber and Wild (1989) and Ratkowsky (1990), and many other regression books for numerous examples. Here are a just a couple of settings using nonlinear models, where measurement error would arise in the predictors. These will be used for illustration in Chapter 9.

  1. *A nonlinear consumption model.*
     Greene (1990, p.337) considers data from 1950 to 1985 in the U.S. with Y = consumption and X = income and fits the consumption model
     $$E(Y|x) = m(\mathbf{x}, \boldsymbol{\beta}) = \beta_0 + \beta_1 x^{\beta_2}.$$
     The data were obtained from the "Economic Report of the President." From the President, or not, the values for consumption and income are clearly estimates rather than exact values.
  2. Stromberg (1993) considered data from a number of lakes, collected as part of the National Eutrophication survey by the U.S. Environmental Protection Agency. These data also appear in exercise 13.2 of Ryan (1997). The nonlinear model of interest is
     $$E(Y_i|\mathbf{x}_i) = \frac{x_{i1}}{1 + \beta_1 x_{i2}^{\beta_2}},$$

where Y_i = mean annual total nitrogen concentration, x_{i1} = average influent nitrogen concentration and x_{i2} = water retention time. At a minimum, Y and x_1 need to be estimated and so are subject to measurement error.

- Generalized Linear Model (GLIM).

  Generalized linear models are very popular with a number of important regression models as special cases. These start with an assumption about the distribution of $Y|\mathbf{x}$ along with a regression function that depends on the predictors in a linear manner, which together yield $V(Y|\mathbf{x})$. See, for example, McCullagh and Nelder (1989) or McCulloch and Searle (2001).
  The regression part of the generalized linear model assumes

  $$m(\mathbf{x}, \boldsymbol{\beta}) = g^{-1}(\mathbf{x}_*'\boldsymbol{\beta}), \qquad \text{or} \qquad g(m(\mathbf{x}, \boldsymbol{\beta})) = \mathbf{x}_*'\boldsymbol{\beta},$$

  where the function g is called the link function. This model is linear in the sense that the expected value of Y given $\mathbf{x}$ is some function (via g^{-1}) of a linear component $\mathbf{x}_*'\boldsymbol{\beta}$. As in the multiple linear regression section, $\mathbf{x}_*$ denotes all of the predictors including a constant for the intercept if present, so $x_*'\boldsymbol{\beta} = \beta_0 + \boldsymbol{\beta}_1'\mathbf{x}$.
  The second part of the GLIM formulations assumes the distribution for $Y|\mathbf{x}$ falls into a class of distributions known as the exponential family. This includes, for example, the Poisson, negative binomial, exponential, Bernoulli and Binomial distributions. Specifying the distribution determines $V(Y|\mathbf{x})$, as seen in later examples.
  The distributional assumption is sometimes abandoned, leading to so-called quasi-likelihood approaches based on just the regression and variance model, with the variance often modified to allow for an arbitrary scaling constant.
  The most popular applications of generalized linear models arise in treating outcomes that are binary or counts, as we now describe.

- Binary Regression.
  Binary regression is concerned with the case where the outcome falls into one of two categories (success/failure, disease/no disease, presence/absence, for/against, etc.). We can, without any loss of generality, label the two possible values of Y by 1 and 0, so

  $$E(Y|\mathbf{x}) = P(Y = 1|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta}) = \mu \text{ and } V(Y|\mathbf{x}) = \mu(1 - \mu).$$

  There are two things worth noting. First, for notational convenience, we sometimes use μ in place of $m(\mathbf{x}, \boldsymbol{\beta})$. Secondly, the expression for $V(Y|\mathbf{x})$ is automatically determined by the fact that Y has possible values of 0 or 1. Different link functions yield different regression models. We describe the two most popular choices, the logistic and probit models.

$$\textbf{Logistic:} \quad P(Y=1|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta}) = \frac{e^{\mathbf{x}_*'\boldsymbol{\beta}}}{1+e^{\mathbf{x}_*'\boldsymbol{\beta}}} = \frac{1}{1+e^{-\mathbf{x}_*'\boldsymbol{\beta}}}. \quad (6.2)$$

The logistic model (not to be confused with a class of logistic models used in other nonlinear problems) corresponds to the link function

$$g(\mu) = log(\mu/(1-\mu)),$$

referred to as the logit function.

$$\textbf{Probit:} \quad P(Y=1|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta}) = \Phi(\mathbf{x}_*'\boldsymbol{\beta}), \quad (6.3)$$

where Φ denotes the cumulative distribution function (CDF) of the standard normal distribution. One motivation of the probit distribution is through the use of tolerance distributions in a dose-response context or through the use of latent variables more generally. For the probit model the link function is the inverse of the standard normal CDF, Φ^{-1} since $m(\mathbf{x}, \boldsymbol{\beta}) = \Phi^{-1}(m(\mathbf{x}, \boldsymbol{\beta})) = \mathbf{x}_*'\boldsymbol{\beta}$.

The logistic and probit models are often similar except when the probabilities are close to 0 or 1. See the discussion in Demidenko (2004, Ch. 7). One useful approximation relating the two is

$$e^a/(1+e^a) \approx \Phi(a/1.7), \quad (6.4)$$

which means a logistic can be approximated by the probit with the coefficients in the logistic model equal to coefficients in the probit model divided by 1.7. This is a useful connection since although the logistic model is the one of choice in many disciplines, the probit model is somewhat easier to handle when addressing measurement error.

See Hosmer and Lemeshow (2000) and references therein for treatment of binary regression without measurement error. Correcting for measurement error is discussed in Chapter 7.

- Regression for count data.

  If Y is a count then an appropriate discrete model should be used. Possible distributions to use include the Binomial (which can be treated through binary regression), the Poisson and the negative Binomial.

  The **Poisson regression** model assumes that $Y|\mathbf{x}$ is distributed Poisson, a distribution where the variance equals the mean so

  $$E(Y|\mathbf{x}) = \mu = m(\mathbf{x}, \boldsymbol{\beta}) \text{ and } V(Y|\mathbf{x}) = \mu.$$

  The variance model can be relaxed to $V(Y|\mathbf{x}) = \sigma^2\mu$, allowing for so-called over- or under-dispersed Poisson models. If used in the GLIM framework a link function is also needed, the most popular choices being the identity link with $\mu = m(\mathbf{x}, \boldsymbol{\beta}) = \mathbf{x}_*'\boldsymbol{\beta}$ or the log-link with $log(\mu) = \mathbf{x}_*'\boldsymbol{\beta}$, so $\mu = e^{x_*'\boldsymbol{\beta}}$.

6.3 Analyses without measurement error

There are many books that treat the analysis of general regression models. In addition to those mentioned in the preceding section, see Kutner et al. (2005), Seber and Wild (1989), and the numerous references therein. We summarize here a few of the key ideas and formulas that enter into the analyses without measurement error, and are most germane to accommodating measurement error.

Regression models are most commonly analyzed via least squares and maximum likelihood. The latter requires an assumption about the distribution of $Y|x$. The least squares approach may be either unweighted or weighted. Weighting in the case of nonconstant variance often leads to iteratively reweighted least squares. It is convenient that for a number of models, the maximum likelihood approach yields the same estimates as iteratively re-weighted least squares based on the variance function. Hence the likelihood approach is frequently robust to the exact distribution specified, but is dependent on the variance model used.

In most problems, the estimator $\widehat{\boldsymbol{\beta}}$ can be characterized as the solution for $\boldsymbol{\beta}$ to a system of equations of the form

$$S(\mathbf{t};\boldsymbol{\beta}) = \sum_i S(y_i, \mathbf{x}_i; \boldsymbol{\beta}) = \mathbf{0}, \tag{6.5}$$

where $\mathbf{t}$ represents all of the data. These are referred to as the **estimating equations.** $S(\mathbf{t}, \boldsymbol{\beta})$ is a $p \times 1$ vector, where p is the number of components in $\boldsymbol{\beta}$, which usually arises from differentiation of some objective function with respect to the coefficients. For example, with maximum likelihood, $S(\mathbf{t}, \beta)$ comes from differentiating the log-likelihood and is usually referred to as the **score vector**. In many cases the estimating equations are of the form

$$\sum_i^n (y_i - m(\mathbf{x}_i, \boldsymbol{\beta}))\boldsymbol{\Delta}(\mathbf{x}_i, \boldsymbol{\beta}) = \mathbf{0}, \tag{6.6}$$

where

$$\boldsymbol{\Delta}(\mathbf{x}_i, \boldsymbol{\beta}) = \begin{bmatrix} \Delta_1(\mathbf{x}_i, \boldsymbol{\beta}) \\ \Delta_2(\mathbf{x}_i, \boldsymbol{\beta}) \\ . \\ . \\ \Delta_p(\mathbf{x}_i, \boldsymbol{\beta}) \end{bmatrix},$$

and each $\Delta_j(\mathbf{x}, \boldsymbol{\beta})$ is a scalar. Special cases include:

Multiple linear and binary logistic regression: $\Delta(\mathbf{x}_i, \boldsymbol{\beta}) = \mathbf{x}_{i*}$.

Generalized linear models: $\Delta(\mathbf{x}_i, \boldsymbol{\beta}) = v_i^{-1}(\partial g(a)/\partial a|_{a=\mathbf{x}'_{i*}\boldsymbol{\beta}})\mathbf{x}_{i*}$, where $v_i = V(Y_i|\mathbf{x}_i)$. For a number of models, including logistic regression, $\Delta(\mathbf{x}_i, \boldsymbol{\beta}) = \mathbf{x}_i^*$.

Nonlinear, unweighted least squares: $\Delta_j(\mathbf{x}_i, \boldsymbol{\beta}) = \partial m(\mathbf{x}_i, \boldsymbol{\beta})/\partial \beta_j$.

Large sample inferences for the coefficients are usually based on $\hat{\boldsymbol{\beta}}$ being $AN(\boldsymbol{\beta}, \boldsymbol{\Sigma}_\beta)$, where $\sim AN$ stands for approximately normally distributed and the approximate covariance matrix of $\widehat{\boldsymbol{\beta}}$ is $\boldsymbol{\Sigma}_\beta = \mathbf{I}_\beta^{-1}\mathbf{Q}\mathbf{I}_\beta^{-1}$, where

$$\mathbf{I}_\beta = \sum_i E(\partial S(Y_i, \mathbf{x}_i, \boldsymbol{\beta})/\partial \boldsymbol{\beta}), \text{ and } \mathbf{Q} = \sum_i E(S(Y_i, \mathbf{x}_i, \boldsymbol{\beta})S(Y_i, \mathbf{x}_i, \boldsymbol{\beta})').$$

The notation $\mathbf{I}_\beta$ is used since in a likelihood context this is the information matrix.

There are a variety of ways to estimate $\boldsymbol{\Sigma}_\beta$ which depend on the model used and what assumptions are made. Most software provide approximate Wald type confidence intervals as well as chi-square tests for individual coefficients or linear hypotheses. These methods are described in the referenced regression books and in software documentation.

6.4 Measurement error models

Section 1.4 provided a quick introduction to measurement error models while Chapters 2 - 5 dealt with specific measurement error models arising from misclassification and additive measurement error. This section provides a more complete look at measurement error models and lays out some additional notation that is used throughout the remainder of the book.

6.4.1 General concepts and notation

The true and observed values are denoted by:

$\mathbf{T}_i = (Y_i, \mathbf{X}_i)$ = true values for observation i

$\mathbf{O}_i$ = collection of "observable" variables on observation i.

The $\mathbf{O}$ stands for observable. This is defined to include any variables that are never subject to measurement error plus those variables observed in place of variables measured without error. We do not include in $\mathbf{O}$ any variables that are subject to measurement error even if they are observed on a subset of units, as with internal validation. The notation for handling internal validation is described in Section 6.5. As will be illustrated in later examples, there are many different forms that $\mathbf{O}$ can take, depending on the application.

Notational note: Similar to earlier uses, the $\mathbf{O}$ *and* $\mathbf{T}$ *are random. The realized values of these are denoted* $\mathbf{o}$ *and* $\mathbf{t}$. *The notation* $|\mathbf{t}$ *denotes conditioning and is shorthand for "given* $\mathbf{T} = \mathbf{t}$." *Similar conventions apply to other quantities.*

- If the response is measured with error, and D denotes the error-prone version of Y then $\mathbf{O}_i = (D_i, \mathbf{W}_i)$, where $\mathbf{W}_i$ is observed in place of $\mathbf{X}_i$. If there is no error in the response then $\mathbf{O}_i = (Y_i, \mathbf{W}_i)$.
- As in previous chapters, some components of $\mathbf{X}$ may be measured without error, in which case some components of $\mathbf{W}$ equal the corresponding components of $\mathbf{X}$. It is often convenient to split up $\mathbf{X}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \text{ with measurement error} \\ \mathbf{X}_2 \text{ no measurement error} \end{bmatrix}$$

  with

$$\mathbf{W} = \begin{bmatrix} \mathbf{W}_1 \\ \mathbf{W}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{W}_1 \\ \mathbf{X}_2 \end{bmatrix},$$

  where $\mathbf{W}_1$ is observed in place of $\mathbf{X}_1$. The measurement error model for $\mathbf{W}|\mathbf{x}, y$ can be specified in terms of the behavior of $\mathbf{W}_1|\mathbf{x}_1, \mathbf{x}_2, y$. If the model is nondifferential with respect to y, then the model can be expressed in terms of the distribution of $\mathbf{W}_1|\mathbf{x}_1, \mathbf{x}_2$. If it is also nondifferential with respect to $\mathbf{x}_2$, the measurement error model can be given solely in terms of the behavior of $\mathbf{W}_1|\mathbf{x}_1$.
- The number of observed and true values involved may differ, so $\mathbf{O}$ and $\mathbf{T}$ (or more specifically $\mathbf{W}$ and $\mathbf{X}$) can be of a different dimension. Typically this occurs when there are more quantities observed than there are true values. See Weller et al. (2007) for an example involving multiple surrogates for exposure to metal working fluids in an occupational epidemiology example and the discussion of the Harvard Six Cities Study in Section 7.3.1.
- The variables involved can be a mix of quantitative and qualitative/ categorical variables.

As in earlier chapters, the "measurement error model" can be specified in one of two ways, summarized in the table below. The column labeled parameters denotes the parameters involved in specifying either the moments or distribution of the error model.

Name	Model for:	parameters
Classical ME model	Observed given truth ($\mathbf{O}|\mathbf{T} = \mathbf{t}$)	$\boldsymbol{\theta}$
Berkson Error Model	True given observed ($\mathbf{T}|\mathbf{O} = \mathbf{o}$)	$\boldsymbol{\lambda}$

- From here on the term measurement error model will usually refer to classical measurement error model or measurement error generally, while the Berkson model is referred to by name.
- Either model might specify just the mean and variance/covariance structure or it may involve distributional assumptions.
- The model will often have further structure. For example rather than just referring to the distribution of $\mathbf{O}|\mathbf{t}$, the model may be given in terms of $D|y$ (for error in the response) and/or $\mathbf{W}|\mathbf{x}$ (for error in the predictors).
- Recall from Section 1.4 that the measurement error in $\mathbf{x}$ is said to be **non-differential** with respect to Y if the distribution of $\mathbf{W}|\mathbf{x}, y$ does not depend on y. This is equivalent to $\mathbf{W}$ being a **surrogate** for $\mathbf{x}$ (the distribution of $\mathbf{W}|\mathbf{x}, y$ is the same as the distribution of $\mathbf{W}|\mathbf{x}$) and **conditional independence** ($\mathbf{W}$ and Y are independent, given $\mathbf{x}$).

Chapters 2 and 3 used misclassification (measurement error) and reclassification (Berkson) models for categorical variables while Chapters 4 and 5 were limited to additive measurement error of quantitative variables in linear regression. Misclassification models, by their very nature, are nonadditive models. There are two goals in this and subsequent chapters: to bring additive measurement error models into nonlinear regression problems and to allow nonadditive measurement error models.

The remainder of this section discusses a number of specific measurement error models and related issues.

6.4.2 Linear and additive measurement error models

There are many measuring methods that exhibit systematic biases of some sort, requiring extending the additive error model which was used in Chapters 4 and 5. Considering just error in $\mathbf{x}$, the easiest generalization is a nondifferential linear measurement error model with

$$\mathbf{W}_i = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1 \mathbf{x}_i + \mathbf{u}_i, \tag{6.7}$$

with $E(\mathbf{u}_i) = 0$ and $cov(\mathbf{u}_i|\mathbf{x}_i) = \Sigma_{ui}$, so $E(\mathbf{W}_i|\mathbf{x}_i) = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{x}_i$ and $Cov(\mathbf{W}_i|\mathbf{x}_i) = \boldsymbol{\Sigma}_{ui}$.

Additive measurement error, where $E(\mathbf{W}|\mathbf{x}) = \mathbf{x}$, is a special case with $\boldsymbol{\theta}_0 = \mathbf{0}$ and $\boldsymbol{\Theta}_1 = \mathbf{I}$ (the identity matrix). A very important result, addressed in the next section, is that linear measurement error model in combination with random true values and normality assumptions leads to a linear Berkson error model.

If either $\boldsymbol{\theta}_0$ or $\boldsymbol{\Theta}_1$ are allowed to change with y, i.e.,

$$E(\mathbf{W}|\mathbf{x}, y) = \boldsymbol{\theta}_{0y} + \boldsymbol{\Theta}_{1y}\mathbf{x},$$

then this would be a linear measurement error model with differential bias.

Linear models with error in the response also can be accommodated by replacing $\mathbf{W}$ with $\mathbf{O}$ and $\mathbf{X}$ with $\mathbf{T}$, so $E(\mathbf{O}_i|\mathbf{t}_i) = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{t}_i$. In its most general form this does not separate out a model for the response and predictors, but this can be done by setting parts of $\boldsymbol{\Theta}_1$ to 0.

In some applications, separate measurement error models are used for each of the variables subject to measurement error. In this case, the measurement error model for the kth predictor is

$$W_{ik} = \theta_{0k} + \theta_{1k}x_{ik} + u_{ik} \tag{6.8}$$

and if there were error in the response $D_i = \theta_{0d} + \theta_{1d}y_i + q_i$, where all of the u_{ik}'s (and, if present, q_i) are independent with $V(u_{ik}) = \sigma^2_{uk}$. Working just with M predictors, for illustration this leads to a diagonal $\boldsymbol{\Theta}_1$ and $\boldsymbol{\Sigma}_u$,

$$\boldsymbol{\Theta}_1 = \begin{bmatrix} \theta_{11} & 0 & . & 0 \\ 0 & \theta_{12} & . & 0 \\ . & . & . & . \\ . & . & . & \theta_{1M} \end{bmatrix} \qquad \boldsymbol{\Sigma}_u = \begin{bmatrix} \sigma^2_{u1} & 0 & . & 0 \\ 0 & \sigma^2_{u2} & . & 0 \\ . & . & . & . \\ . & . & . & \sigma^2_{uM} \end{bmatrix}.$$

Examples of linear measurement error abound. Section 8.5.3 presents an example examining the relationship between pH and alkalinity in water samples, both measured with linear measurement error, as seen in Figure 8.6. That example uses separate linear regression models for the predictor and response, as described immediately above. Ferrari et al. (2007) use linear models in relating measures of physical activity from questionnaires and accelerometers to true activity levels. See Section 6.5.5 also. Similar models are used in relating measures of diet intake from food frequency questionnaires to true diet intake. See Section 7.3.1 for an illustration using validation data from the Nurses Health Study. Another illustration of linear measurement error occurs in the example in Section 8.5.5, where the true values consist of the water and protein content of a wheat sample and the observed values are reflectance measures, which are linearly related to the true values. Although the example uses only two reflectance measures, there were four reflectance measures in the original data. Using them would be a case where $\mathbf{O}$ has more components than $\mathbf{T}$.

6.4.3 The linear Berkson model

Focusing on just errors in the predictors, the linear Berkson model assumes the true values are linearly related to the observed values. That is

$$\mathbf{X}_i|\mathbf{w}_i = \boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}_i + \boldsymbol{\delta}_i, \tag{6.9}$$

with $E(\boldsymbol{\delta}_i) = \mathbf{0}$. The additive Berkson model has $\boldsymbol{\lambda}_0 = \mathbf{0}$ and $\boldsymbol{\Lambda}_1 = \mathbf{I}$.

Equation (6.9) plays a prominent role in the measurement error literature, especially in the use of regression calibration techniques to correct for measurement error; see Section 6.10. There is a relationship between the linear Berkson model and the linear measurement error model as described in the following result.

A normal structural model for the true values in combination with a linear normal measurement error with constant variance/covariance yields a normal linear Berkson model.

More precisely, with additive Berkson error, if $\mathbf{X} \sim N(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$ and $\mathbf{W}|\mathbf{x} \sim N(\mathbf{x}, \boldsymbol{\Sigma}_u)$ then

$$\mathbf{X}|\mathbf{w} \sim N(\boldsymbol{\lambda}_o + \boldsymbol{\Lambda}_1\mathbf{w}, \boldsymbol{\Sigma}_{X|w}), \tag{6.10}$$

where

$$\boldsymbol{\Lambda}_1 = \boldsymbol{\Sigma}_X(\boldsymbol{\Sigma}_X + \boldsymbol{\Sigma}_u)^{-1}, \qquad \boldsymbol{\lambda}_0 = \boldsymbol{\mu}_x - \boldsymbol{\Lambda}_1\boldsymbol{\mu}_X \tag{6.11}$$

and $\boldsymbol{\Sigma}_{X|w} = (\mathbf{I} - \boldsymbol{\Lambda}_1)\boldsymbol{\Sigma}_X$. In this setting the linear Berkson model can be fit using replicate measures of the error prone measure as described in Section 6.5.1.

For the general linear case with $\mathbf{W}|\mathbf{x} \sim N(\boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{x}, \boldsymbol{\Sigma}_u)$, (6.10) still holds but with

$$\boldsymbol{\Lambda}_1 = \boldsymbol{\Sigma}_X\boldsymbol{\Theta}_1'(\boldsymbol{\Theta}_1\boldsymbol{\Sigma}_X\boldsymbol{\Theta}_1' + \boldsymbol{\Sigma}_u)^{-1}, \qquad \boldsymbol{\lambda}_0 = \boldsymbol{\mu}_X - \boldsymbol{\Lambda}_1(\boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\boldsymbol{\mu}_x) \tag{6.12}$$

and $\boldsymbol{\Sigma}_{X|w} = \boldsymbol{\Sigma}_X - \boldsymbol{\Lambda}_1\boldsymbol{\Theta}_1\boldsymbol{\Sigma}_X$.

It is important to point out that the linear Berkson model does not always follow from a structural model with linear measurement error model. Either non-normality or heteroscedasticity in the measurement errors (see Section 6.4.5) destroys this result.

It is true, however, that even if (6.10) does not hold exactly, assuming a structural model with linear measurement error, the best linear predictor of $\mathbf{X}$ given $\mathbf{w}$ is $\boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}$. With validation data, one can bypass building the Berkson model from the measurement error model and build the Berkson model directly. See Section 7.3.1 for one illustration.

One can also consider linear Berkson models involving both the predictors and the response. With $\mathbf{T}_i = (Y_i, X_i)'$ denoting the true values, the multivariate linear Berkson model assumes that $\mathbf{T}_i|\mathbf{o}_i = \boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{o}_i + \boldsymbol{\delta}_i$ with $E(\boldsymbol{\delta}_i) = \mathbf{0}$. This is used in the wheat example in Section 8.5.5.

6.4.4 Nonlinear measurement error models

Nonlinear measurement error models arise with many measurement techniques, including the use of spectroscopy and radioimmunoassays (RIA). In general for a mismeasured variable W corresponding to true value x the measurement error model is nonlinear if $E(W|x) = g(\boldsymbol{\theta}, x)$ where g in nonlinear in the θ's. In radioassays, the observed response is a count or standardized count and the true value a concentration. Huet et al. (2004) provide an example on the calibration of the level of cortisol using RIA based on the "Richards function," where if x is the cortisol concentration $E(W|x) = \theta_1 + (\theta_2 - \theta_1)/(1 + (e^{\theta_3 + \theta_4 log(x)})^{\theta_5})$. Section 10.4 presents an example with measurement error in the response, where the true outcome is y = urinary neopterin level, measured by a radioassay. The observed value is a standardized count D and the measurement error model used is a four parameter logistic with

$$E(D|y) = \theta_1 + (\theta_2 - \theta_1)/(1 + (e^y/\theta_3)^{\theta_4}).$$

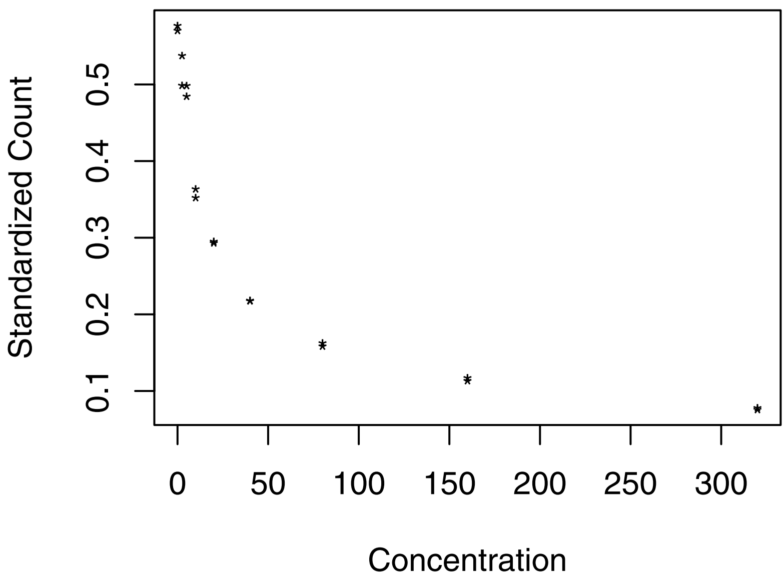
An added feature in that example (and many similar problems) is that the coefficients change across observations since the assay is recalibrated for each new batch of reagants. Figure 6.1 displays the calibration data for one of the "batches," illustrating the nonlinear nature of the measurement error model.

6.4.5 Heteroscedastic measurement error

In Chapters 4 and 5 and in discussing the linear measurement error model above we allowed for the fact that the measurement error variances, and possibly covariances, could change across observations. In those earlier chapters we bypassed a careful discussion about the nature of the heteroscedasticity so as not to distract from the main applied issues. In particular, as noted in Section 4.3, we avoided distinguishing between conditional (given the true values) and unconditional measurement error variances. The notion of the unconditional measurement error is relevant in structural models with random true values. Here, we address this model more carefully, describing the influence of both sampling effort and the possible dependence of the measurement error variance on the true values. This section is designed for those readers wanting a deeper understanding of the underlying nature of changing measurement error variances. Those more interested in the applications can bypass it without compromising their understanding of the examples.

To focus the discussion we only discuss the case of a single variable measured with additive error, but this can be readily extended to handle multiple mismeasured values and nonadditive models. As used earlier, the additive model has $W_i|x_i = x_i + u_i$, with the assumptions that $E(u_i|x_i) = 0$. Notice that if X_i is random then this model, which is conditional given $X_i = x_i$, is not

Figure 6.1 *Calibration data for urinary neoperin.*



the same as the unconditional model $W_i = X_i + u_i$. In the earlier discussion the variance of u_i was denoted (with some imprecision) by simply σ^2_{ui}. Here we distinguish the conditional measurement error variance

$$\sigma^2_{ui(c)} = V(u_i|x_i)$$

from the unconditional measurement error variance

$$\sigma^2_{ui} = V(u_i) = V(W_i - X_i).$$

The first point to address is exactly what is $\sigma^2_{ui(c)}$ and why it might change with i. In the functional case, X_i is not random, and is equal to x_i, and the conditional and unconditional variance are equal. Then we just need to address how this conditional variance might be affected by the true value x_i (or other quantities associated with the ith unit) and sampling effort, e.g., the number of replicates. There is an additional subtlety in the structural case, where an "observation" is associated with some selected "unit." In this case conditioning on the actual unit resulting in the ith position in the sample is not necessarily the same as conditioning on the true value x_i, since more than one unit in the population may have the same true value.

In either case, assume the unit carries information, what we call **inherent variation** and denoted v_i, which will influence the variance of u_i for that unit. In addition, there is some **sampling effort** applied to unit i, denoted by m_i. The notions of inherent variation and sampling effort are completely general at this point, and each of v_i and m_i could be multidimensional. In the simplest case with replication, the inherent variation is the per replicate variance for a unit and the sampling effort is simply the number of replicates. This is illustrated later.

The variance of u_i, conditional on the inherent variation associated with the unit in position i and the sampling effort, is

$$V(u_i|x_i, v_i, m_i) = t_i^2 = h(v_i, m_i). \tag{6.13}$$

As indicated, t_i^2 (with t_i temporarily having a different meaning than earlier in the chapter) depends in some way on the inherent variation v_i, which may, in turn, be a function of x_i. Knowing the actual form of h is not required in correcting for measurement error, although if known there are ways that information could be utilized. The conditional variance might also involve additional parameters, but these have been suppressed here.

In the functional model the unit and the associated x_i are fixed and the t_i^2 is the same as both the conditional and unconditional measurement error variance, i.e., $\sigma^2_{ui(c)} = \sigma^2_{ui} = t_i^2$.

In the structural case, the fact that units are random leads to the inherent variation being random, reflected by the use of a random quantity V_i, with realized value of V_i being v_i. The sampling effort could be random or fixed and can result from various mechanisms. If random, the random sampling effort is denoted by M_i. The sampling effort can generally be viewed as falling in one of three categories:

1. **Fixed sampling effort:** The sampling effort m_i is fixed and attached to the ith position in the sample.
2. **Sampling effort attached to the unit**. Here the sampling effort is associated with the selected unit. This can mean either that the effort is determined by the unit, as is the case when sampling proportional to unit size in some way, or is random with a distribution that depends on characteristics of the unit. Either way, with an overall random sample of units, unconditionally the sets (X_i, V_i, M_i) are i.i.d.
3. **Random sampling effort not associated with the unit**. Here the sampling

effort might be random on a unit, but in a way that is unrelated to the characteristics of the unit. Typically, in these cases, the distribution of the sampling effort would be common for all observations.

With fixed sampling effort, most often $m_i = m$. It is rare for there to be unequal sampling efforts attached strictly to the position in the sample. A more common occurrence is that either the sampling effort is unequal because it is attached to the unit as in Definition 2 or, as in Definition 3, the effort varies for some reasons not specific to the unit. Examples of the latter include having data missing completely at random, or a change over i in the resources available to obtain an estimate of x_i.

Equal sampling effort is defined to include either: i) fixed sampling effort with $m_i = m$ for each i, or ii) random sampling effort identically distributed for each selected unit.

In the structural model define the random quantity

$$T_i^2 = h(V_i, M_i), \tag{6.14}$$

so t_i^2 is the realization of T_i^2. Formally, the conditional measurement error variance is

$$\sigma^2_{ui(c)} = V(u_i|x_i) = E_c[T_i^2|x_i] = E_c[h(V_i, M_i)|x_i], \tag{6.15}$$

where E_c denotes expectation with respect to the conditional distribution given x_i. Equation (6.15) follows from the conditional variance identity (e.g., Casella and Berger (2002, p.167)). We point out the somewhat subtle distinction between t_i^2 and $\sigma^2_{ui(c)}$.

Although not explicit in the notation, $\sigma^2_{ui(c)}$ can depend on x_i, and so a more precise notation would be $\sigma^2_{ui(c)}(x_i)$. Using double expectation (Casella and Berger (2002, p.164)) leads to an expression for the unconditional measurement error variance of

$$\sigma^2_{ui} = E[h(V_i, M_i)]. \tag{6.16}$$

Notice that this unconditional measurement error variance averages out the inherent variation over the sampling of units. With an overall random sample the only reason that this unconditional measurement error variance may change with i is due to changes in sampling effort attached to the position i in the sample.

If the sampling effort is equal (as defined above) then in the structural case with an overall random sample, unconditionally the measurement error variances are constant; that is, $\sigma^2_{ui} = \sigma^2_u$. Notice that this result allows for conditional heteroscedasticity in the measurement errors (as a result of dependence *on the true values) but unconditionally the measurement error variances are constant.*

Illustration with replication.

Suppose x_i is estimated using $W_i = \sum_{j=1}^{m_i} W_{ij}/m_i$, which is the mean of m_i replicates with $W_{ij}|x_i = x_i + u_{ij}$ and

$$V(u_{ij}|x_i) = g(x_i, \boldsymbol{\theta}).$$

For concreteness, we have just allowed the per-replicate variance to depend on x_i and some other parameters. For example, with the power of the mean model (discussed more in Section 10.2.2), $g(x_i, \boldsymbol{\theta}) = \theta_1 |x_i|^{\theta_2}$. Since u_i is the mean of the u_{ij}'s, then conditionally,

$$\sigma^2_{ui(c)} = V(u_i|x_i, m_i) = \frac{g(x_i, \boldsymbol{\theta})}{m_i},$$

while unconditionally

$$\sigma^2_{ui} = E\left[\frac{g(X_i, \boldsymbol{\theta})}{M_i}\right].$$

If the sampling effort is fixed at m_i and the X_i are identically distributed then $\sigma^2_{ui} = E(g(X_i, \boldsymbol{\theta})/m_i) = \tau^2/m_i$, where $\tau^2 = E(g(X_i, \boldsymbol{\theta}))$. With equal sampling effort, even if the per-replicate measurement error variances changes with the true value, then unconditionally the σ^2_{ui} are constant, and equal to τ^2/m.

For some further discussion of heteroscedastic models see Buonaccorsi (2006) (upon which the discussion above is based) and Staudenmayer and Buonaccorsi (2005).

6.4.6 Multiplicative measurement error

Many authors have found it useful to view the error on a multiplicative scale. An excellent overview appears in Section 4.5 of Carroll et al. (2006). Here we simply make a few points.

Consider a single predictor x_i and suppose

$$W_i|x_i = x_i Z_i$$

with $E(Z_i|x_i) = 1$ and $V(Z_i|x_i) = \sigma^2_Z$. This means that $E(W_i|x_i) = x_i$ (so the error is additive in this sense) and $V(W_i|x_i) = x_i^2\sigma^2_Z$. Equivalently $W_i = x_i + u_i$ where $E(u_i|x_i) = 0$ and $V(u_i|x_i) = \sigma^2_{ui} = x_i^2\sigma^2_Z$. In this case the multiplicative measurement error model with W_i being unbiased for x_i leads to an additive error model with a particular form of heteroscedasticity. Most of our developments allow for heteroscedasticity so the multiplicative model is not treated separately. There may be times, however, where we want to exploit the nature of the heteroscedasticity implied by the multiplicative model. Note also that in the structural case with the X_i being i.i.d. with mean μ_X and variance σ_X^2, the unconditional variance of u_i is $\sigma_u^2 = \sigma_Z^2 E(X_i^2) = \sigma_Z^2(\mu_X^2 + \sigma_X^2)$, which is constant. This reinforces the point made in Section 6.4.5 that the measurement error can be conditionally heteroscedastic but unconditionally homoscedastic.

More generally if $E(Z_i) = \mu_Z \neq 1$ then $E(W_i|x_i) = x_i\mu_Z$, which leads to proportional bias in W as an estimate of x. This is a special case of a linear measurement error model. On the other hand $E(log(W_)|x_i) = log(x_i) + E(log(Z_i))$ which equals $log(x_i)$ if $E(log(Z_i)) = 0$. This raises the question of on which scale the measurement error should be treated as additive. The log transformation is discussed in the next section. In general if W is unbiased for x then $g(W)$ is unbiased for $E(g(W)|x) = h(x)$. This only equals $g(x)$ if g is linear. If we have replicate measures for x, and the investigator's interest is in a model involving $g(x)$, then simply transforming replicate measures is not really correct. However, as seen in the next section proceeding in this way may be just fine and it is certainly the right thing to do if the interest is in the regression model in terms of $h(x)$.

6.4.7 Working with logs

The use of logarithmic transformation is ubiquitous in regression analyses across many disciplines. They are often used to achieve linearity and/or equal variances in the errors. Their widespread use makes the questions raised at the end of the previous section particularly important when the model of interest is expressed in terms of $log(x)$ for some predictor x. Later we are faced with the following questions. If $W_1, \ldots W_m$ are replicates for x, that is $E(W_j) = x$, and $V_j = log(W_j)$, what happens if we estimate $log(x)$ using $\bar{V}$, the mean of $V_1, \ldots V_m$? What happens if we use $log(\bar{W})$? A number of the correction techniques developed later make use of replicate measures with additive error and the analysis simplifies if we can treat the $log(W_j)$'s as approximately unbiased for $log(x)$.

The general question then is: if $E(W) = x$ and $V(W) = \sigma_W^2$ and $V = log(W)$, what is $E(V)$? We have the following results:

- $E(V) > log(x)$; that is $log(W)$ overestimates $log(x)$.
  This follows from Jensen's inequality.
- If $log(W)$ is normally distributed, then

$$E(V) = log(x) + B, \text{ where } B = log[(\sigma_W^2 + x^2)/x^2]/2. \qquad (6.17)$$

  This follows from properties of the log-normal distribution.

- $$E(V) \approx log(x) + B, \text{ where } B = \sigma^2_W/2x^2. \tag{6.18}$$

  This approximation comes from the use of a second order Taylor series approximation. While the bias term here looks quite different than that in (6.17), a first order expansion of the log component in (6.17) around $\sigma^2_W = 0$ results in a bias term that agrees with the B here.

If we have replicates $W_j = x + u_j$, where u_j has per-replicate variance $\sigma^2_{u(1)}$, then the expected value of $V_j = log(W_j)$ is given by the above with $\sigma^2_W = \sigma^2_{u(1)}$. On the other hand $E(log(\bar{W}))$ is given by the case with $\sigma^2_W = \sigma^2_{u(1)}/m$.

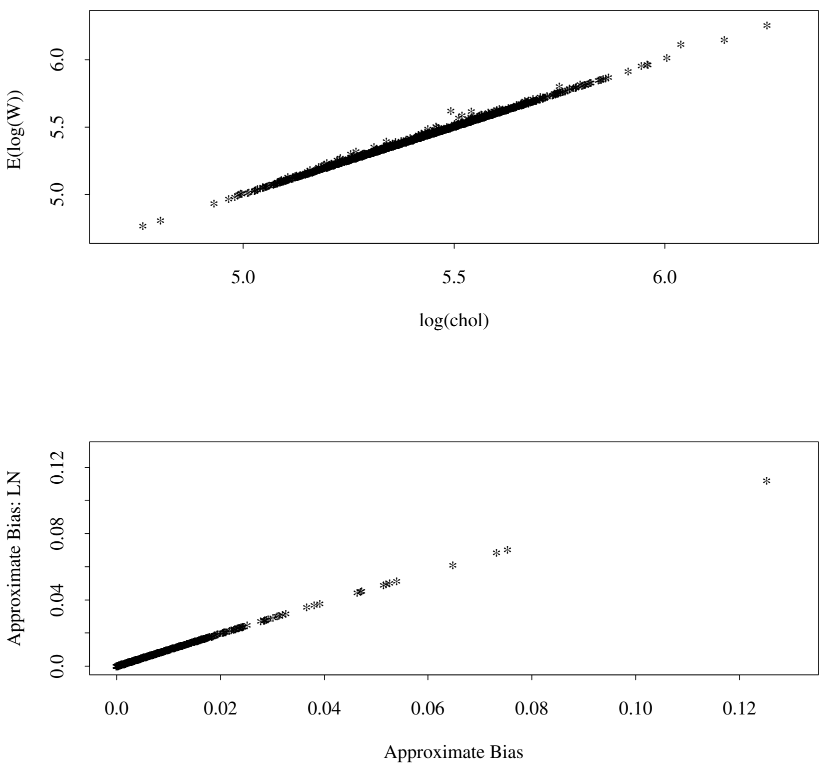
The bias in $\bar{V}$ is the same as the bias in each V_j and does not change as the number of replicates increases. The bias in $log(\bar{W})$ will go to zero as the number of replicates increases. This argues in general for using the log of the mean rather than the mean of the logs if the regression model of interest is in terms of $log(x)$. But, as demonstrated below, there are cases where use of the logs of the replicates works fine.

To illustrate the issue we consider some data from the Framingham Heart Study, which is discussed in more detail in Section 7.2.5. Here we consider the measure of cholesterol where for each of 1615 individuals we have two replicate measures. Indexing the individual by i, we will treat the mean of the two as a true x_i and the per-replicate variance as the true $\sigma^2_{ui(1)}$. Using these values we calculate biases using (6.17) and (6.18) for a single replicate. Figure 6.2 shows a plot of the bias versus $log(x)$ for each of the bias expressions and then a plot of one bias versus the other. Two things are clear here. One is that there is negligible bias in the log of a single rep as an estimate of $log(x)$. The other is that the exact bias under a log-normal assumption and the approximate bias are almost the same. The first point argues for using the logs of the replicates even if the model of interest is in terms of $log(x)$. This is illustrated in Section 7.2.5 in examining the relationship of cholesterol to coronary heart disease.

6.4.8 Misclassification from categorizing a quantitative variable

Although there are reasons to argue against it (see Royston et al. (2006)), it is common practice in many disciplines to categorize quantitative variables. This is often done using fixed cutpoints, as with categorization of body mass index (BMI) into "underweight," "normal," "overweight," or "obese" or categorizing smoking based on the number of cigarettes smoked per day. With other variables (e.g., nutritional intake) the categorization is more frequently based on percentiles from the data with quartiles, quantiles and deciles being popular

Figure 6.2 *Illustration of bias in use of log of a replicate using cholesterol data. Top panel: plot of approximate expected value of $log(W_{ij})$ versus $log(x_i)$. Bottom panel: plot of bias under log normal model versus approximate bias based on Taylor series approximation.*



choices. See Pan et al. (2005), Engeset et al. (2007) and other studies cited in Dalen et al. (2009) for examples of categorization in epidemiologic studies.

Let X denote the quantitative value and X_c its categorized version, with $X_c = j$ if X is in category j. Denote the measured version of X by W and the categorized version of W by W_c. If there is error in X, suppose $W|x = x + u$ where $E(u) = 0$.

- If there is no measurement error in X, then with fixed cutpoints, there is clearly no misclassification. On the other hand, if the categorization is done by sample percentiles there is potentially some misclassification. Fortunately, this is often negligible, especially in large epidemiologic studies involving thousands of observations. It is a potential source of concern in smaller studies.

- Misclassification probabilities.
  Consider categorizing by fixed cutpoints $C_0, \ldots, C_J$, where $X_c = j$ if $X \in A_j$, where $A_j = (C_{j-1}, C_j)$, and let $f(w|x)$ denote the density of W given x, which we assume has mean x but might have a variance that depends on x. With X random, having density f_X, then (unconditionally) the misclassification probabilities are given by $P(W_c = k|X_c = j)$

$$= \frac{\int_{x \in A_j} P(W \in A_k|x) f_X(x) dx}{\pi_j} = \frac{\int_{x \in A_j} \int_{w \in A_k} f(w|x) dw f_X(x) dx}{\pi_j},$$

  where $\pi_j = P(X \in A_j) = \int_{x \in A_j} f_X(x) dx$ is the probability that X is in category j.
  This ignores any other variables involved. The next item considers what happens when we condition on an outcome Y.

- *Nondifferential measurement error in W, with respect to Y, does not imply the misclassification is nondifferential.*

  Consider an outcome Y with $f(x|y)$ denoting the distribution of X given $Y = y$. Assume that the measurement error model for W is nondifferential with respect to Y; that is, $f(w|y, x) = f(w|x)$. Fixing y, then

$$P(W_c = k|X_c = j, y) = \frac{\int_{x \in A_j} \int_{w \in A_k} f(w|x, y) dw f_{X|y}(x|y) dx}{P(X_c = j|y)}$$
$$= \frac{\int_{x \in A_j} \int_{w \in A_k} f(w|x) dw f_X(x|y) dx}{\int_{x \in A_j} f_{X|y}(x|y) dx}.$$

  This will depend on y, unless the distribution of $X|y$ does not depend on y. Hence, the misclassification is typically differential. This was first observed in a particular context by Flegal et al. (1991) and in the case of two categories has been referred to as DDD (differential due to dichotomization) misclassification by Gustafson and Le (2002). See also the discussion in Gustafson (2004).

Table 6.1 illustrates the result above in the case where Y is a binary outcome variable. The true X is generated through $X = a + bC$, where C is distributed chi-square with d degrees of freedom (this distribution becomes more symmetric as d increases) and a and b are chosen so that $E(X) = \mu_X = 180$ and $V(X) = \sigma_X^2 = 20^2$. The distribution of $Y|X = x$ follows a logistic model with $logit(P(Y = 1|x)) = \beta_0 + \beta_1 x$, where $\beta_0 = -5$ and β_1 was chosen to alter the odds ratio, based on a change in x of 100 units. We chose to express the odds ratio on this scale since the parameters were based on the example done in Chapter 7 where x is serum cholesterol and Y indicates whether coronary heart disease is present. The cutpoints used to categorize are taken to be 180, 200 and 220. The resulting misclassification probabilities demonstrate the complex way in which they can be effected by the various inputs: the skewness of the distribution of X, the cutpoint used, the measurement error variance and the magnitude of β_1. The fact that the misclassification is differential is also demonstrated, although in many of the scenarios it is approximately nondifferential. There are exceptions to this, most notably for $P(W_c = 0|X_c = 1, y)$, in situations with $\sigma_u = 20$.

If the outcome Y is categorical and there is a single X which is categorized then this is a two-way table and the results in Chapter 3 regarding the impact of nondifferential misclassification on naive analyses apply. However the correction methods given in that chapter do not apply since we have replicate measures on the original X rather than validation data. See Section 7.4 for some discussion on correcting for the misclassification based on replicate measures of the quantitative variable.

6.5 Extra data

While there are a few exceptions, the majority of problems require some type of extra data in order to correct for measurement error. In treating misclassification problems in Chapters 2 and 3, both internal and external validation data were used, while the linear regression problems in Chapters 4 and 5 made use of replicates and instrumental variables. In this section we expand on the discussion that appeared in the linear regression chapters concerning the use of replicates and provide a broader discussion on validation data. Other types of data are touched on briefly in Section 6.5.5

6.5.1 Replicate values

Sections 4.5.3 and 5.4.3 provide a fairly detailed treatment of the use of replicate values with additive errors. The treatment there did allow for the possibility of separate replication on the response and predictors, but otherwise assumed that all mismeasured predictors were measured together on each replicate. Here we expand that discussion to allow that the replication may be unbalanced in various ways. This section also provides further discussion on borrowing information from replications on other units and the fitting of the Berkson error model from the replicates. The issue of borrowing information is of particular interest when some observations have only one replicate. This was

Table 6.1 *Misclassification rates from categorizing. Category set to 1 if value* $>$ *cut and 0 otherwise.* $W = x + u$ *with* $V(u) = \sigma_u^2$. $Y|X = x$ *follows a logistic model with odds ratio* OR. *The true* X *is a linear function of a chi-square random variable with* d *degrees of freedom, chosen so that* $\mu_X = 180$ *and* $\sigma_X = 20$. *The resulting misclassification probabilities are* $\theta_{k|j} = P(W_c = k|X_c = j)$ *and* $\theta_{k|j,y} = P(W_c = k|X_c = j, Y = y)$. *Based on 100,000 simulations.*

OR	σ_u	d	cut	$\theta_{0\|1}$	$\theta_{0\|1,0}$	$\theta_{0\|1,1}$	$\theta_{1\|0}$	$\theta_{1\|0,0}$	$\theta_{1\|0,1}$
7.4	10	2	180	0.150	0.163	0.114	0.155	0.153	0.168
7.4	10	10	180	0.152	0.159	0.142	0.146	0.144	0.150
7.4	10	100	180	0.148	0.155	0.142	0.146	0.141	0.152
7.4	10	2	200	0.148	0.166	0.115	0.043	0.040	0.060
7.4	10	10	200	0.178	0.190	0.164	0.056	0.053	0.062
7.4	10	100	200	0.195	0.208	0.186	0.067	0.061	0.073
7.4	10	2	220	0.144	0.156	0.129	0.014	0.011	0.023
7.4	10	10	220	0.186	0.202	0.171	0.019	0.016	0.024
7.4	10	100	220	0.240	0.249	0.235	0.020	0.017	0.023
7.4	20	2	180	0.236	0.252	0.194	0.286	0.284	0.295
7.4	20	10	180	0.243	0.258	0.222	0.265	0.263	0.268
7.4	20	100	180	0.249	0.259	0.240	0.252	0.248	0.257
7.4	20	2	200	0.239	0.258	0.204	0.123	0.119	0.140
7.4	20	10	200	0.275	0.282	0.267	0.139	0.134	0.150
7.4	20	100	200	0.307	0.312	0.303	0.150	0.142	0.159
7.4	20	2	220	0.248	0.271	0.221	0.045	0.041	0.065
7.4	20	10	220	0.285	0.298	0.273	0.056	0.050	0.067
7.4	20	100	220	0.334	0.359	0.318	0.061	0.055	0.068
2.7	10	2	180	0.152	0.153	0.129	0.157	0.157	0.165
2.7	10	10	180	0.148	0.149	0.142	0.146	0.145	0.155
2.7	10	100	180	0.148	0.148	0.148	0.149	0.150	0.146
2.7	10	2	200	0.151	0.151	0.143	0.042	0.042	0.053
2.7	10	10	200	0.176	0.175	0.180	0.058	0.057	0.060
2.7	10	100	200	0.198	0.198	0.191	0.069	0.068	0.073
2.7	10	2	220	0.155	0.155	0.153	0.014	0.014	0.019
2.7	10	10	220	0.194	0.193	0.206	0.017	0.017	0.022
2.7	10	100	220	0.243	0.242	0.254	0.019	0.019	0.023
2.7	20	2	180	0.238	0.240	0.203	0.289	0.289	0.309
2.7	20	10	180	0.245	0.246	0.228	0.266	0.265	0.269
2.7	20	100	180	0.250	0.251	0.248	0.250	0.249	0.259
2.7	20	2	200	0.232	0.234	0.195	0.124	0.124	0.142
2.7	20	10	200	0.277	0.280	0.241	0.140	0.139	0.150
2.7	20	100	200	0.302	0.304	0.285	0.150	0.150	0.151
2.7	20	2	220	0.224	0.225	0.216	0.044	0.044	0.058
2.7	20	10	220	0.292	0.293	0.285	0.055	0.055	0.062
2.7	20	100	220	0.344	0.342	0.361	0.062	0.062	0.066

also mentioned in Section 5.4.3. The coverage here is in terms of predictors, but this is easily extended to handle error in the response, as was done in the linear regression chapters. One way to easily accommodate this in the following discussion is to allow an index $a = 0$ to correspond to error in the response variable.

For the ith observation assume there are m_{ia} replicates on the ath variable, with replicate measures $W_{i1a}, \ldots, W_{im_{ia}a}$. We assume

$$W_{ija}|x_{ia} = x_{ia} + u_{ija},$$

where x_{ia} is the true value of the ath predictor on observation i and u_{ija} has mean 0 and variance $V(u_{ija}) = \sigma^2_{uia(1)}$. This last quantity is per-replicate variance for variable a, on the ith observation. For the ath variable on the ith observation the error-prone measure is the mean of the replicates

$$W_{ia} = \sum_j W_{ija}/m_{ia},$$

where $m_{ia} \geq 1$. This leads to $W_{ia}|x_{ia} = x_{ia} + u_{ia}$, where

$$V(u_{ia}) = \sigma^2_{uia} = \sigma^2_{uia(1)}/m_{ia}.$$

Similarly, if we assume that $Cov(u_{ija}, u_{ijb}) = \sigma_{uiab(1)}$, then

$$Cov(u_{ia}, u_{ib}) = \sigma_{uiab} = \sigma_{uiab(1)}/m_{iab},$$

where m_{iab} is the number of replicates containing both the ath and bth variables. With $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$, these components lead to

$$\mathbf{\Sigma}_{ui} = Cov(\mathbf{u}_i).$$

(See Section 6.4.5 for a discussion about the measurement error variances and covariances possibly changing with i. Here, from the conditional perspective, it is because of either changes in the per-replicate variance/covariance from unit to unit or changes in sample sizes. See also the later discussion on using information across units.)

When $m_{ia} > 1$, the per-replicate variance for variable a is estimated by

$$\widehat{\sigma}^2_{uia(1)} = S^2_{ia} = \sum_j (W_{ija} - W_{ia})^2/(m_{ia} - 1)$$

and the estimated variance of u_{ia} is $\widehat{\sigma}^2_{uia} = \widehat{\sigma}^2_{uia(1)}/m_{ia}$.

If the measurement errors for two variables are correlated then there needs to be at least some paired replicate values for those two variables, say $m_{iab} > 1$ for variables a and b. This leads to

$$\widehat{\sigma}_{uiab(1)} = S_{iab} = \sum_j (W_{ija} - W_{ia})(W_{ijb} - W_{ib})/(m_{iab} - 1),$$

where the sum is over those j for which W_{ija} and W_{ijb} are both available. An estimate of the abth element of $\boldsymbol{\Sigma}_{ui}$ is then

$$\widehat{\sigma}_{uiab} = S_{iab}/m_{iab}.$$

Taken together the quantities above lead to estimates of the pieces of $\widehat{\boldsymbol{\Sigma}}_{ui}$ for which sufficient replication is available, i.e., for the variance of variables with $m_{ia} > 1$ and covariances of pairs with $m_{iab} > 1$. For variables for which this is not the case then one has to somehow borrow information from the other units.

Using information from other units.

If there are some observations with $m_{ia} = 1$, then in order to estimate σ^2_{uia} we need to use information from other observations. One approach is to assume that the per-replicate variance is the same. In this case the average per-replicate variance is

$$S^2_a = \sum_{i:m_{ia}>1} S^2_{ia}/N_a$$

where N_a is the number of observations for which $m_{ia} > 1$. Assuming the per-replicate variance is the same across observations, then we can use

$$\widehat{\sigma}^2_{uia} = S^2_a/m_{ia}. \tag{6.19}$$

- Notice that under the assumption of constant per-replicate variance one can use this approach even if $m_{ia} > 1$ for all i (and so $N_a = n$). This is more efficient than using the individual variances when in fact the per-replicate variance is constant.
- In the structural case if the per-replicate variance is changing with the true value there is an argument for using (6.19) based on the discussion in Section 6.4.5 about changing measurement error variances.
- This same approach can be used in estimating the measurement error covariances.

Using a mean-variance model.

Another approach to using information from other units is to assume that the per-replicate variance changes as some function of i. For example, for a single predictor, suppose the conditional per-replicate variance is some function of the unobserved x_i, say $\sigma^2_{ui(1)} = g(x_i, \boldsymbol{\theta})$. If we have an estimate $\widehat{\boldsymbol{\theta}}$ then the measurement error for the ith observation can be estimated using $g(W_i, \widehat{\boldsymbol{\theta}})/m_i$. Of course, estimating $\boldsymbol{\theta}$ based on per-replicate sample variances (the S^2_i above) and the sample means (the W_i) leads to another measurement error problem which limits the value of this approach. This is discussed in Section 10.2.2. This approach becomes quite a bit more involved in trying to also model the covariances.

Estimating the structural parameters

Assuming the $\mathbf{X}_i$ are a random sample with mean $\boldsymbol{\mu}_X$ and covariance $\boldsymbol{\Sigma}_X$, there are various ways to estimate these structural parameters. Even with unequal measurement variances/covariances we could estimate $\boldsymbol{\mu}_X$ with

$$\widehat{\boldsymbol{\mu}}_X = \bar{\mathbf{W}} = \sum_{i=1}^{n} \mathbf{W}_i/n \text{ and } \widehat{\boldsymbol{\Sigma}}_X = \mathbf{S}_{WW} - \widehat{\boldsymbol{\Sigma}}_u, \tag{6.20}$$

where $\mathbf{S}_{WW}$ is the sample covariance matrix of the $\mathbf{W}_i$' and $\widehat{\boldsymbol{\Sigma}}_u = \sum_{i=1}^{n} \widehat{\boldsymbol{\Sigma}}_{ui}/n$. This simple approach was discussed in Chapter 5 and given its computational ease we use it in many of our applications. With sufficiently large measurement error it is possible that $\mathbf{S}_{WW} - \widehat{\boldsymbol{\Sigma}}_u$ can be negative. In that case some adjustment must be made; see Bock and Peterson (1975) for example. With unbalanced data the above is not necessarily the most efficient way to proceed and there are alternate approaches for estimating $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_X$. This is a mixed model problem; see for example Searle et al. (2006) for a single variable and Demidenko (2004) or other mixed models books for more general treatment. We won't provide further details here but see the end of Section 5.4.3 for some further discussion.

Besides estimating $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_X$, one might want to estimate the distribution of $\mathbf{X}$. If this distribution is normal then estimating the mean and covariance matrix is all that is needed. With other specified distributions, maximum likelihood techniques can be used, as long as the distribution of the replicates is specified. A harder problem is to estimate the distribution using semiparametric/nonparametric approaches allowing one or both of the distributions to be unspecified. This is discussed in Section 10.2.3.

Estimating the Berkson model using the replicates.

As noted in equation (6.10), with $\mathbf{X}_i$ random and additive measurement error, the best linear predictor of $\mathbf{X}_i$ given $\mathbf{w}_i$ is $\boldsymbol{\lambda}_{0i} + \boldsymbol{\Lambda}_{1i}\mathbf{w}_i$, where $\boldsymbol{\Lambda}_{1i} = \boldsymbol{\Sigma}_X(\boldsymbol{\Sigma}_X + \boldsymbol{\Sigma}_{ui})^{-1}$ and $\boldsymbol{\lambda}_{0i} = \boldsymbol{\mu}_x - \boldsymbol{\Lambda}_{1i}\boldsymbol{\mu}_X$. This is exactly $E(\mathbf{X}_i|\mathbf{w}_i)$ under normality and constant measurement error variances/covariances. Using $\widehat{\boldsymbol{\Sigma}}_X$ and $\widehat{\boldsymbol{\mu}}_X$, linear Berkson model can be estimated either with a model that is observation specific

$$\boldsymbol{\Lambda}_{1i} = \widehat{\boldsymbol{\Sigma}}_X(\widehat{\boldsymbol{\Sigma}}_X + \widehat{\boldsymbol{\Sigma}}_{ui})^{-1} \text{ and } \widehat{\boldsymbol{\lambda}}_{0i} = \widehat{\boldsymbol{\mu}}_X - \widehat{\boldsymbol{\Lambda}}_{1i}\widehat{\boldsymbol{\mu}}_X,$$

or, using a common fit, with

$$\widehat{\mathbf{\Lambda}}_1 = \widehat{\mathbf{\Sigma}}_X(\widehat{\mathbf{\Sigma}}_X + \widehat{\mathbf{\Sigma}}_u)^{-1} \text{ and } \widehat{\boldsymbol{\lambda}}_0 = \widehat{\boldsymbol{\mu}}_X - \widehat{\mathbf{\Lambda}}_1\widehat{\boldsymbol{\mu}}_X$$

where $\widehat{\mathbf{\Sigma}}_u = \sum_{i=1}^n \widehat{\mathbf{\Sigma}}_{ui}/n$.

The first leads to an estimate of $E(\mathbf{X}_i|\mathbf{w}_i)$ of $\widehat{X}_i = \widehat{\boldsymbol{\lambda}}_{0i} + \widehat{\mathbf{\Lambda}}_{1i}\mathbf{w}_i$ and the second $\widehat{X}_i = \widehat{\boldsymbol{\lambda}}_0 + \widehat{\mathbf{\Lambda}}_1\mathbf{w}_i$ with corresponding estimates of $Cov(\mathbf{X}_i|\mathbf{w}_i)$ of $\widehat{\mathbf{\Sigma}}_{\mathbf{X}_i|\mathbf{w}_i} = (\mathbf{I} - \widehat{\mathbf{\Lambda}}_{1i})\widehat{\mathbf{\Sigma}}_X$ and $\widehat{\mathbf{\Sigma}}_{\mathbf{X}_i|\mathbf{w}_i} = (\mathbf{I} - \widehat{\mathbf{\Lambda}}_1)\widehat{\mathbf{\Sigma}}_X$, respectively.

In later applications, the emphasis is on finding the best linear predictor of $\mathbf{X}$ given $\mathbf{w}$, which requires estimating the mean and covariance structure. This may not be adequate in many cases and the more challenging general problem is to estimate $E(\mathbf{X}_i|\mathbf{w}_i)$ or more generally the distribution of $\mathbf{X}_i|\mathbf{w}_i$, from the replicates. These will not be discussed here. Note that in general this requires estimating a distribution for $\mathbf{X}$ and/or for $\mathbf{W}|\mathbf{x}$. See Pierce and Kellerer (2004) for one example of this approach.

6.5.2 External replicates: Are reliability ratios exportable?

The discussion in the previous section was in the context where the replicate measures are taken on units in the main study. For many measurement instruments there are independent studies in which replicate measures are taken over a number of subjects to assess the "reliability" of the instrument. See Ginns and Barrie (2009) for one of many possible examples.

For a single variable the data consist of m_i replicates $\tilde{W}_{ij}, j = 1, \ldots, m_i$ on the ith individual. The ~ is used to distinguish these values from internal replicates. The underlying model is usually the standard one-way random effects model with $\tilde{X}_1, \ldots, \tilde{X}_n$ i.i.d. with mean $\tilde{\mu}_X$ and variance $\tilde{\sigma}_X^2$ and $\tilde{W}_{ij}|\tilde{x}_i = \tilde{x}_i + \tilde{u}_{ij}$. If the variance of $\tilde{u}_{ij}$ is constant, say $\sigma^2_{u(1)}$, then the reliability of the mean of m replicates is $\tilde{\kappa}_m = \tilde{\sigma}_X^2/(\tilde{\sigma}_X^2 + (\sigma^2_{u(1)}/m))$. The reliability ratio for independent studies is usually reported in terms of a single replicate $\tilde{\kappa}_1 = \tilde{\sigma}_X^2/(\tilde{\sigma}_X^2 + \sigma^2_{u(1)})$. These two are related through $\tilde{\kappa}_m = \tilde{\kappa}_1/(\tilde{\kappa}_1 + (1 - \tilde{\kappa}_1)/m)$, so the reliability ratio for the mean of m replicates can be obtained from that for one replicate.

With the same per-replicate measurement error variance applying to the main study, this reliability ratio (appropriately adjusted for the number of replicates) only applies to the main study if the variance among the X values is the same in the main study as it is in this external data, i.e., $\sigma_X^2 = \tilde{\sigma}_X^2$. Even if this is true, it only allows to correct for measurement error if the regression only involves this single variable.

What if the measurement error is changing? This is related to the discussion in Section 6.4.5. If the per-replicate variance of the measurement error is a function of the true x, say $g(x, \boldsymbol{\theta})$, then in the main study unconditionally $V(u_{ij}) = E(g(X, \boldsymbol{\theta})) = \sigma_U^2$ and the reliability ratio (for one replicate) is $\kappa = \sigma_X^2/(\sigma_X^2 + E(g(X, \boldsymbol{\theta})))$. If the measurement error variance has constant coefficient of variation CV then $g(x, \boldsymbol{\theta}) = CV^2 x^2$ and $E(g(X, \theta)) = CV^2(\sigma_X^2 + \mu_X^2)$ and $\kappa = 1/(2 + CV^2)$. In this case, the reliability ratio is transportable, although, as noted above, this is only useful in a single variable regression.

In summary, knowing the reliability ratio, or ratios, by themselves from external data is of limited usefulness in correcting for measurement error. It is knowledge about the measurement error variances(s) that is needed. Since the replicate data allow us to estimate the measurement error variance (and possibly how it changes with true values) it is this information that should be extracted from external studies for use elsewhere rather than reliability ratios themselves.

6.5.3 Internal validation data

With internal validation data, the true values for the mismeasured variables are observed on a subset of size n_V. This was illustrated in Section 2.4.3 in estimating a single proportion and in Section 3.6 in the context of two-way tables. For simplicity here, we assume that the true values for whatever variables are mismeasured are obtained on each observation in the validation subset. With multiple mismeasured values, it is possible to generalize this to allowing for different subsets for different variables, but this would unnecessarily complicate the presentation.

Using our earlier notation $\mathbf{O}_i$ contains the observed values that are obtained on each unit. Here we now decompose the true values into $\mathbf{T}$ into $\mathbf{T}_1$ and $\mathbf{T}_2$ where

- $\mathbf{T}_1$ contains the true values that are subject to measurement error
- $\mathbf{T}_2$ contains the true values that are not subject to measurement error.

The perfectly measured variables in $\mathbf{T}_2$ are then part of the observed $\mathbf{O}$.

The data consist of $\{(\mathbf{O}_i, \mathbf{T}_{i1}), i = 1, \ldots, n_V\}$ and $\{\mathbf{O}_i, i = n_V+1, \ldots n\}$, displayed in tabular form in Table 6.2, where - indicates missing. Here $\mathbf{T}_{i1}$ contains the true values observed on the ith validation sample for those true values that are not subject to measurement error, i.e., are not part of the always observed$\mathbf{O}_i$.

Some special cases are illustrated in Table 6.3. As elsewhere, in these examples the predictors are split into $\mathbf{X}_1$ and $\mathbf{X}_2$ where $\mathbf{X}_1$ is subject to measurement error but $\mathbf{X}_2$ is not. In these cases $\mathbf{W}$ consists of $\mathbf{W}_1$ and $\mathbf{X}_2$, where

Table 6.2 *Data layout with internal validation.*

Sample number	1	...	i	...	n_V	n_V+1	...	n
	$\mathbf{O}_1$	...	i	...	$\mathbf{O}_{n_V}$	$\mathbf{O}_{n_V+1}$	...	$\mathbf{O}_n$
	$\mathbf{T}_{11}$	...	$\mathbf{T}_{i1}$	...	$\mathbf{T}_{n_V,1}$	-	...	-

$\mathbf{W}_1$ is observed in place of the unobservable $\mathbf{X}_1$. Recall also that D is the error-prone version of Y if there is error in the response.

Table 6.3 *Internal validation examples.* $\mathbf{O}_i$ *is observed on all units.* $\mathbf{T}_{i1}$ *is observed on units in validation sample.*

Mismeasured	$\mathbf{O}_i$	$\mathbf{T}_{i1}$
Y only	$(\mathbf{X}_i, D_i)$	Y_i
$\mathbf{X}_1$ only	$(\mathbf{W}_i, Y_i)$	$\mathbf{X}_{i1}$
$\mathbf{X}_1$ and Y	$(\mathbf{W}_i, D_i)$	$(\mathbf{X}_{i1}, Y_i)$ or $\mathbf{X}_{i1}$

In the last example in Table 6.3, the validation sample could just contain the error-prone true values in $\mathbf{X}_{i1}$ since there are cases where we may not need to ever obtain Y_i to correct for error. This happens, for example, with inferences for the coefficients in linear regression with additive error in the response that is uncorrelated with the error in the predictors. It is necessary, however, to observe the true response in the validation samples if differential measurement error is allowed or the error in the response is correlated with that in the predictors.

We discuss two ways in which internal validation data is obtained.

Random/double sampling. Here a random sample of n_V out of the n units is chosen for validation. In the sampling literature this is referred to as double sampling. In this context the problem can also be viewed as a missing data problem with missing completely at random, but the measurement error context adds some additional structure to the problem.

Designed/two-phase subsampling. Here one first observes $\mathbf{o}_i$ for $i = 1$ to n. At the second phase we then select the n_V units, on which $\mathbf{T}_1$ is obtained, based in some way on the values observed in the first phase. The term two-phase sampling usually includes random subsampling but, for convenience, we will use it to refer to the case where the second stage sampling depends on the observed values in the first phase. Typically, the observed values determining the second stage sampling are categorical, but there is nothing to prohibit one from designing based on a quantitative variable.

A good example of a two-phase design appears in Tosteson and Ware (1987). This involves a case-control study of presence or absence of respiratory illness (y) and how it relates to indoor pollution (x). The variable W indicates presence or absence of a gas stove in the home, available for all homes in the study. This serves as a surrogate for x. They allow sampling at arbitrary rates within any of the four cells corresponding to combinations of Y and W. The true exposure X is measured on the selected subsample using individual monitors. This example is also interesting in that x is a continuous quantitative measure while w is a binary surrogate for x.

- With internal validation data we can fit the Berkson model for $\mathbf{T}$ given the observed $\mathbf{o}$. This is true whether the validation sample is random or designed.
- If the validation sample is a random sample then we can also estimate the measurement error model for observed given truth, i.e., $\mathbf{O}|\mathbf{t}$.
- With a designed subsample it is usually not possible to estimate the measurement error model, or in the structural case, the distribution or moments of the error-prone true values, since some portions of $\mathbf{O}$ are fixed at the second stage.

See Zhao and Lipsitz (1992), Tosteson and Ware (1990), Reilly (1996), Schill et. al. (1993) and references therein for discussion of designed two-phase studies. Notice that we have used the phrase two-phase sampling here. This is consistent with the terminology in the sampling literature, but this design is also often referred to as two-stage sampling.

See Section 6.15.2 for a discussion of correcting for measurement error using internal validation data and references to examples.

6.5.4 External validation data

Here the set of units on which we obtain true values for the mismeasured variables is independent of those used in the main study. When using external validation data, we need to keep the following points in mind.

- If the external data are collected with fixed true values, then the measurement error model (for observed given truth) must be fit. This is true with the calibration of measuring instruments using standards with known true values. This is true in the examples in Sections 8.5.3, 10.3 and 10.4.3.
- If the error-prone measures are fixed in the validation sample then the Berkson error model must be fit.

- With a random sample of units making up the external data, one could entertain fitting either of the models. As discussed in Chapters 2 and 3, however, the critical issue is which model is transportable to the main study. This is not easy to answer, although it is often easier to argue that measurement error model is more exportable when the error arises from instrument or sampling error. An important point is

  For a given measurement error model, the Berkson model will be different in the main study than in the validation sample if the distribution of true values changes.

Other examples on the use of external validation data appear in Sections 7.3.1 and 7.3.3.

6.5.5 Other types of data

Our main focus is on the use of replication and validation data to fit additive and nonadditive measurement error models, respectively. There are various other types of data which, with some assumptions, allow estimation of the measurement error model. In some cases this is necessary since neither true replication or validation is possible. Generally, this is done using latent variable/structural equation/mixed models; see for example Mueller (1996), Bollen (1989) and Dunn (2004). We do not provide details on this expansive area but describe a few examples.

Plummer and Clayton (1993) examine four different measurements of dietary intake (including four day weighed records, seven day diaries, food frequency questionnaire and recall) with one assumed to be unbiased for true intake and each of the others having a constant bias. Their focus is on estimation of the bias terms as well as the measurement error variances. In related work, Kipnis et al. (2003) attack similar problems with the use of biomarker and more general linear measurement error models, but with one measure having additive error. Ferrari et al. (2007) explored three measures of physical activity (PA) arising from a PA questionnaire (Q), four 7-day physical activity logs (R) and four sets of accelerometer measures (A). This setting can be used to describe a typical model in a little more detail. Blending their notation with ours, for subject i with true PA value x_i, and j denoting a replicate measure, the model used is $Q_i = \theta_{0Q} + \theta_{1Q}x_i + u_{Qi}$, $A_{ij} = \theta_{0A} + \theta_{1A}x_i + u_{Aij}$ and $R_{ij} = x_i + u_{Rij}$. Notice that the replicate measures from the activity log are assumed unbiased for the true activity level, while both the questionnaire and accelerometer follow linear measurement error models. In this, as well as the Plummer and Clayton (1993) paper the true values $X_1, \ldots, X_n$ are assumed to be a random sample from a population with mean μ_X and variance σ_X^2.

6.6 Assessing bias in naive estimators

The goal of the next two sections is to outline how to characterize the behavior of naive analyses that ignore measurement error. This is important for a couple of reasons. For one, measurement error is still often ignored, even when it is acknowledged to exist. This happens in part because of the lack of software to implement the corrections and the effort needed to customize the correction method to the problem at hand. Another reason is difficulty in obtaining the estimates of measurement error parameters needed to correct. For these reasons it is important to have a sense of when we can ignore measurement error without serious consequences. As seen earlier, the methods used in assessing bias due to measurement error also often lead to methods to correct for it.

The discussion here focuses primarily on the properties of estimators rather than on tests of hypotheses. This is partly a matter of space, but also a function of the somewhat limited value of hypothesis testing. The properties of tests can usually be derived from that of the naive estimators. Among others, Tosteson and Tsiatis (1998) and Cheng and Tsai (2004) provide some discussion about certain naive tests.

In general, naive estimators are typically inconsistent. As demonstrated in the preceding four chapters, however, the direction and magnitude of the bias can be quite complex. In those chapters, we were able to determine the bias of the naive estimators, either exactly or approximately/asymptotically, based on the fact that either an explicit expression for the estimators existed or one could write down an exact model for the observed values. An example of the latter was in linear regression where for the normal structural case with normal error in the equation and normal measurement errors with a constant covariance matrix, the behavior of D given $\mathbf{w}$ followed another linear model. This suggests the general strategy of trying to find a model for $D|\mathbf{w}$. If this is in the same form as the original model for $Y|\mathbf{x}$ then we can identify the biases by examining the parameters involved. This can be referred to as the **induced model approach** as it uses the model induced by the original model for the true values plus the measurement error model. This approach, which is the subject of Section 6.7, does require a structural model in which the mismeasured X's are random. This does limit its applicability.

The approach of using explicit expressions for estimators or using the induced model often fails for any number of reasons. These include the fact that the model is functional rather than structural or that the induced model is not in the same form as the original model. In these cases other strategies need to be employed. Of course, one option is to use simulations to assess the bias of the naive estimators. This is a useful tool even in cases where we have expressions for approximate/asymptotic biases as the small sample biases can differ. This was illustrated in Section 4.4. It is difficult, however, to fully characterize the biases based on simulations, except in the simplest of problems. Another fruitful approach is to examine the problem through the estimating equations. As noted in Section 6.3, the estimators used in most regression problems are solutions to a system of equations, called estimating equations. These estimating equations can be used to characterize approximate bias and to obtain corrected estimators. This approach is described in Section 6.8. Corrected estimators based on the estimating equations are discussed in Section 6.13.

6.7 Assessing bias using induced models

This section examines the bias in naive estimators using an induced model approach.

Unless noted otherwise the assumption here is that there is no measurement error in Y.

The assumption of no response error is important for some of the developments of induced models. In other cases, response error can be accommodated, as was done in the preceding four chapters and occurs in some later applications.

The induced model refers to the distribution, or sometimes just the moments, of the response Y given the observed $\mathbf{w}$. Working with just the induced regression model for $E(Y|\mathbf{w})$, with no error in the response and $\mathbf{x}$ assumed continuous for convenience,

$$\begin{aligned} E(Y|\mathbf{w}) &= \int_x E(Y|\mathbf{w},\mathbf{x})f(\mathbf{x}|\mathbf{w})d\mathbf{x} \qquad (6.21)\\ &= \text{under surrogacy}\\ &= \int_x E(Y|\mathbf{x})f(\mathbf{x}|\mathbf{w})d\mathbf{x} = \int_x m(\mathbf{x},\boldsymbol{\beta})f(\mathbf{x}|\mathbf{w})d\mathbf{x}\\ &= E_{\mathbf{X}|\mathbf{w}}[m(\mathbf{X},\boldsymbol{\beta})|\mathbf{w}], \end{aligned}$$

where for convenience we have suppressed the parameters involved in many places.

- The quantity $E_{\mathbf{X}|\mathbf{w}}[m(\mathbf{X},\boldsymbol{\beta})|\mathbf{w}]$ represents the expected value of the function $m(\mathbf{X},\boldsymbol{\beta})$ (which is random through the random $\mathbf{X}$) where the expectation being taken is with respect to the conditional distribution of $\mathbf{X}$ given $\mathbf{w}$.
- The result depends critically on the assumption that $\mathbf{W}$ is a surrogate for $\mathbf{X}$; that is, that $E(Y|\mathbf{w},\mathbf{x}) = E(Y|\mathbf{x})$. Recall that the surrogacy assumption is that the distribution of Y given $\mathbf{x}$ and $\mathbf{w}$ is the same as that of Y

given $\mathbf{x}$. This is equivalent to assuming nondifferential measurement error or conditional independence; see Section 1.4.3.

- We have expressed the intermediates steps above as integrals for convenience. If some or all of the X's are discrete then sums or a combination of sums and integrals is used.
- The development leading to (6.21) assumes there is some distribution for $\mathbf{X}$ given $\mathbf{w}$, as indicated through $f(\mathbf{x}|\mathbf{w})$. From a non-Bayesian perspective, the mismeasured part of $\mathbf{X}$, say $\mathbf{X}_1$, must be random. If $\mathbf{X}_1 = \mathbf{x}_1$ is fixed then $E(Y|\mathbf{w})$ involves $\mathbf{x}_1$, which is unobservable, and so the result is not helpful. Hence, *the induced model approach, as described here, does not work if the mismeasured values are fixed.*
- We have used the device of designating what distribution is being used by the notation of the argument, such as $f(y|\mathbf{x})$ denoting the density or mass function of Y given $\mathbf{X} = \mathbf{x}$ while $f(y|\mathbf{w})$ denotes the density or mass function of Y given $W = \mathbf{w}$. This is of course imprecise, but should not cause confusion. Where needed the distribution could be indicated, i.e., writing $f_{Y|\mathbf{w}}(y|\mathbf{w})$ to avoid confusion.
- The extended regression calibration methods, which are used later, also rely on the variance of the induced model, which can be calculated via
$$V(Y|\mathbf{w}) = V(m(\mathbf{X}, \boldsymbol{\beta})|\mathbf{w}) + E(v(\mathbf{X}, \boldsymbol{\beta}, \boldsymbol{\sigma})|\mathbf{w}). \tag{6.22}$$
As in working with the mean, the variance and expectation on the right side are with respect to the random $\mathbf{X}$ given $\mathbf{w}$.

Instead of just the moments, we could consider the actual induced distribution of $Y|\mathbf{w}$. With the remarks above still applying (most importantly the surrogacy assumption), the conditional distribution of Y given $\mathbf{w}$

$$f(y|\mathbf{w}) = E_{\mathbf{X}|\mathbf{w}}[f(y|\mathbf{X})], \tag{6.23}$$

where once again the expectation is with respect to the conditional distribution of $\mathbf{X}$ given $\mathbf{w}$.

The remainder of this section uses induced models to present bias results for some important regression models. Some of these are exact, while others are approximate. We repeat that unless noted otherwise, *it is assumed that Y is measured without error and the surrogacy assumption holds.*

6.7.1 Linear regression with linear Berkson error

If the original model is linear with $E(Y|\mathbf{x}) = \beta_0 + \boldsymbol{\beta}_1'\mathbf{x}$, then

$$E(Y|\mathbf{w}) = \beta_0 + \boldsymbol{\beta}_1' E(\mathbf{X}|\mathbf{w}).$$

If the model for $\mathbf{X}|\mathbf{w}$ is a linear Berkson model with $E(\mathbf{X}|\mathbf{w}) = \boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}$, then

$$E(Y|\mathbf{w}) = \beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1 + \boldsymbol{\beta}_1'\boldsymbol{\Lambda}_1\mathbf{w}. \tag{6.24}$$

This means that

$$E(\widehat{\beta}_{0,naive}) = \beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1 \quad \text{and} \quad E(\widehat{\boldsymbol{\beta}}_{1,naive}) = \boldsymbol{\Lambda}_1'\boldsymbol{\beta}_1. \tag{6.25}$$

These agree with our earlier result for multiple linear regression with additive error where $\boldsymbol{\Lambda}_1' = (\boldsymbol{\Sigma}_X + \boldsymbol{\Sigma}_u)^{-1}\boldsymbol{\Sigma}_X$ corresponded to the reliability matrix $\boldsymbol{\kappa}$; see (5.7), (5.3) and (5.4).

Notice that $\boldsymbol{\Lambda}_1'\boldsymbol{\beta}_1$ equal $\mathbf{0}$ if and only if $\boldsymbol{\beta}_1 = \mathbf{0}$ and so naive tests for $\boldsymbol{\beta}_1 = \mathbf{0}$ will be correct.

In the additive Berkson model, $\boldsymbol{\lambda}_0 = \mathbf{0}$ and $\boldsymbol{\Lambda}_1 = \mathbf{I}$ and $E(Y|\mathbf{w}) = \beta_0 + \boldsymbol{\beta}_1'\mathbf{w}$ and naive estimate of $\boldsymbol{\beta}$ is unbiased for $\boldsymbol{\beta}$, as observed in the earlier chapters on linear regression.

6.7.2 Second order models with linear Berkson error

A second order model based on two original predictors x_1 and x_2 assumes

$$E(Y|x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2.$$

With $\mathbf{w}$ containing w_1 and w_2, the linear Berkson model has

$$E(X_j|\mathbf{w}) = \lambda_{j0} + \lambda_{j1}w_1 + \lambda_{j2}w_2, \qquad V(X_j|\mathbf{w}) = \sigma^2_{j|w}$$

for $j = 1$ or 2 and $Cov(X_1, X_2|\mathbf{w}) = \sigma_{12|w}$. In addition, $E(X_j^2|\mathbf{w}) = \sigma^2_{j|w} + (E(X_j|\mathbf{w}))^2$ and $E(X_1X_2|\mathbf{w}) = \sigma_{12|w} + E(X_1|\mathbf{w})E(X_2|\mathbf{w})$. This results in the induced model

$$E(Y|\mathbf{w}) = \beta_0^* + \beta_1^* w_1 + \beta_2^* w_2 + \beta_3^* w_1^2 + \beta_4^* w_2^2 + \beta_5^* w_1 w_2,$$

where

$\beta_0^* = \beta_0 + \beta_1\lambda_{10} + \beta_2\lambda_{20} + \beta_3(\sigma^2_{1|w} + \lambda_{10}^2) + \beta_4(\sigma^2_{2|w} + \lambda_{20}^2) + \beta_5(\sigma_{12|w} + \lambda_{10}\lambda_{20})$,

$\beta_1^* = \beta_1\lambda_{11} + \beta_2\lambda_{21} + 2\beta_3\lambda_{10}\lambda_{11} + 2\beta_4\lambda_{20}\lambda_{21} + \beta_5(\lambda_{11}\lambda_{20} + \lambda_{21}\lambda_{10})$,

$\beta_2^* = \beta_1\lambda_{12} + \beta_2\lambda_{22} + 2\beta_3\lambda_{10}\lambda_{12} + 2\beta_4\lambda_{20}\lambda_{22} + \beta_5(\lambda_{12}\lambda_{20} + \lambda_{22}\lambda_{10})$,

$\beta_3^* = \beta_3\lambda_{11}^2 + \beta_4\lambda_{21}^2 + \beta_5\lambda_{11}\lambda_{21}$,

$\beta_4^* = \beta_3\lambda_{12}^2 + \beta_4\lambda_{22}^2 + \beta_5\lambda_{12}\lambda_{22}$,

$\beta_5^* = 2\beta_3\lambda_{11}\lambda_{12} + 2\beta_4\lambda_{21}\lambda_{22} + \beta_5(\lambda_{12}\lambda_{21} + \lambda_{11}\lambda_{22})$.

This relatively simple setting leads to very complex consequences on the bias of the naive estimators, with the bias in a particular coefficient depending on that coefficient as well as other coefficients and various measurement error parameters. In addition while a naive test of the hypothesis that all of the non-intercept coefficients equal 0 will be okay, this is not necessarily the case when testing for certain subsets.

Recall that the linear Berkson model used above will hold if there is additive normal measurement error with constant measurement error variances and covariances and normal random true values, with parameters obtained as described in Section 6.4.3.

Additive Berkson error.

Second order models like this often arise in designed experiments where the w's are target values. If the errors are additive then $\lambda_{10} = \lambda_{20} = \lambda_{12} = \lambda_{21} = 0$ and $\lambda_{11} = \lambda_{22} = 1$. This leads to all on the nonintercept coefficients in the induced model being equal to the true values (i.e., $\beta_k^* = \beta_k$ for $k = 1$ to 5) and the measurement error only leads to bias in the intercept term. However, one can show that the variance in the induced model is quadratic in the mismeasured predictors. So, naive analyses assuming constant variance are incorrect, even for the nonintercept terms. Further discussion of these second order models appears in Section 8.3.

6.7.3 *Exponential models with normal linear Berkson error*

The model for the true values is

$$E(Y|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta}) = e^{\beta_0 + \boldsymbol{\beta}_1' \mathbf{x}}.$$

This is often used when Y is a count, assumed to have a Poisson distribution, in which case this is a generalized linear model with log link, but its usefulness is not limited to that case. Under the linear normal Berkson error model $\mathbf{X}|\mathbf{w} \sim N(\boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}, \boldsymbol{\Sigma}_{X|w})$, and (based on the use of moment generating functions)

$$E(Y|\mathbf{w}) = E(m(\mathbf{X}, \boldsymbol{\beta})|\mathbf{w}) = e^{\beta_0^* + \boldsymbol{\beta}_1^{*\prime} \mathbf{w}},$$

where

$$\beta_0^* = \beta_0 + .5(\boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1) + \boldsymbol{\beta}_1'\boldsymbol{\lambda}_0, \text{ and } \boldsymbol{\beta}_1^* = \boldsymbol{\Lambda}_1'\boldsymbol{\beta}_1. \tag{6.26}$$

Under a normal linear Berkson error model, the induced model is of the same form as the original model.

This means the naive estimators are consistent for β_0^* and β_1^* rather than β_0

and β_1. Once again, the bias in the nonintercept terms is exactly like that for linear regression in (5.7) with $\mathbf{\Lambda}_1 = \boldsymbol{\kappa}$.

In more general situations where the linear Berkson model is not applicable the approximate biases in the naive estimators can be developed through the estimating equations, as described in Section 6.8.

6.7.4 *Approximate induced regression models*

Often $E(Y|w) = E_{\mathbf{X}|\mathbf{w}}[m(\mathbf{X}, \boldsymbol{\beta})|\mathbf{w}]$, as given in (6.21), cannot be determined exactly. One suggestion is to use a first order approximation

$$E(Y|\mathbf{w}) \approx m(h(\mathbf{w}), \boldsymbol{\beta}), \qquad \text{where} \qquad h(\mathbf{w}) = E(\mathbf{X}|\mathbf{w}). \tag{6.27}$$

Formally this approximation depends on $Cov(\mathbf{X}|\mathbf{w}) = \mathbf{\Sigma}_{X|w}$ being "small," in some relative sense. The next two sections utilize this approximation to assess bias in generalized linear models and, more specifically, binary regression. As discussed later in this chapter, the approximation above also suggests the regression calibration method for correcting for measurement error in which we replace $\mathbf{w}$ with an estimate of $E(\mathbf{X}|\mathbf{w})$, and more generally the use of quasi-likelihood methods, which also involve use of $V(Y|\mathbf{w})$.

The approximation in (6.27) may not always help in assessing bias. For example, suppose $m(x_i, \beta) = \beta_0 + \beta_1 x_i^{\beta_3}$, a nonlinear consumption model used in econometrics, for which

$$E(Y|w_i) = \beta_0 + \beta_1 E(X_i^{\beta_3}) \approx \beta_0 + \beta_1(\lambda_0 + \mathbf{\Lambda}_1 w)^{\beta_3}.$$

Since this is not in the same form as the original model, the bias of the naive estimators cannot be readily determined. Instead we are led to the problem of assessing the performance of the least squares estimators under model misspecification, a problem better treated through estimating equations, as discussed in Section 6.8.

6.7.5 *Generalized linear models*

The approximation in (6.27) is of immediate help in assessing bias in generalized linear models, with linear Berkson error. In that case, since $E(Y|\mathbf{x}) = g^{-1}(\mathbf{x}_*'\boldsymbol{\beta})$, using (6.27) leads to

$$E(Y|\mathbf{w}) \approx g^{-1}(\beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1 + \boldsymbol{\beta}_1'\mathbf{\Lambda}_1\mathbf{w}) = g^{-1}(\beta_{0*} + \boldsymbol{\beta}_{1*}'\mathbf{w}), \tag{6.28}$$

where $\beta_{0*} = \beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_{1*} = \mathbf{\Lambda}_1'\boldsymbol{\beta}_1$. This results in

For a normal structural model with normal additive measurement error and constant variance the approximate biases of the naive estimators of the coefficients in the generalized linear model are the same as the biases for additive error in multiple linear regression.

Accepting the approximation, all of the discussion in Section 5.3 for linear regression, including both analytical and numerical results, carry over to the generalized linear model. Recall in particular the fact that measurement error in one predictor causes bias in the naive estimators of the coefficient of perfectly measured predictor unless they are uncorrelated with the mismeasured predictors. The numerical illustration of biases in Section 5.3 are still applicable if Y is binary or a count modeled using a generalized linear model, such as for logistic, probit or Poisson regression.

The bias expression for $\boldsymbol{\beta}_1$ here also agrees with the exact expression (exact in the sense of giving the exact limiting value) for the Poisson model in Section 6.7.3, where the Berkson errors were also assumed normal. The bias for the naive estimator of β_0 given there differs from the approximate bias here by the term $.5(\boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1)$. This is due to the fact that the approximation in (6.27) is based in part on assuming $\boldsymbol{\Sigma}_{X|w}$ is small.

6.7.6 Binary regression

Here we consider some other approximations, in addition to (6.28), which apply in binary regression model. We first note that for a probit model certain conditions lead to the induced model being another probit model. This leads to an exact expression for the asymptotic bias and also leads to another approximation for logistic regression. We then consider the normal discriminant model, which models $\mathbf{X}$ given y rather than vice versa, and leads to a logistic model for $Y|\mathbf{x}$. This is one modeling option in case-control studies. The normal discriminant model with a normal linear measurement error is seen to induce another logistic model, allowing us to characterize bias.

- Probit Regression.
  Recall the probit model from (6.3), where $P(Y = 1|\mathbf{x}) = \Phi(\beta_0 + \mathbf{x}'\boldsymbol{\beta}_1)$. Assume$\mathbf{W}|\mathbf{x}$ follows the linear normal Berkson model with constant covariance, so
  $$\mathbf{X}|\mathbf{w} \sim N(\boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}, \Sigma_{X|w}). \tag{6.29}$$
  As first noted by Carroll et al. (1984), and frequently used since, the induced model is exactly another Probit regression model with
  $$P(Y = 1|\mathbf{w}) = \Phi(\beta_0^* + \boldsymbol{\beta}_1^{*'}\mathbf{w}) \tag{6.30}$$

where

$$\beta_0^* = \frac{\beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1}{\eta}, \qquad \text{and } \boldsymbol{\beta}_1^* = \frac{\boldsymbol{\Lambda}_1'\boldsymbol{\beta}_1}{\eta}, \tag{6.31}$$

with $\eta = (1 + \boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1)^{1/2}$.
The naive estimators are consistent for β_0^* and $\boldsymbol{\beta}_1^*$. Here we have exact asymptotic biases as opposed to approximations. Notice that as η gets closer to 1, the result of "small" effects (i.e., small $\boldsymbol{\beta}_1$) and/or small measurement error (i.e., small $\boldsymbol{\Sigma}_{X|w}$), the bias in the naive estimator of $\boldsymbol{\beta}_1$ approaches that arising from (6.28).

- Probit regression with additive Berkson error in designed experiments.
  Probit regression is often employed in dose-response studies (see, for example, Finney (1971)), where w is the target dose. Under the additive Berkson error model, $\boldsymbol{\lambda}_0 = 0$ and $\boldsymbol{\Lambda}_1 = I$ and the coefficients for the induced model are $\beta_0^* = \beta_0/\eta$ and $\boldsymbol{\beta}_1^* = \boldsymbol{\beta}_1/\eta$, where η is defined following (6.31).

  Unlike the case where the true values follow a linear model, with additive Berkson error the naive estimators from Probit regression are inconsistent.

  In dose-response studies with a single x one of the goals is to estimate the dose x that leads to a certain probability P of success; that is x such that $P(Y = 1|x) = m(x, \beta) = P$. This leads to what is often denoted $LD_P = (\Phi^{-1}(P) - \beta_0)/\beta_1$, where LD stands for "lethal dose," or alternately ED_P for "effective dose." If $P = .5$, this is also called the median effective dose and equals $-\beta_0/\beta_1$, since $\Phi^{-1}(P) = 0$. The naive estimator of LD_P is a consistent estimator of

$$\frac{\Phi^{-1}(P) - (\beta_0/\eta)}{\beta_1/\eta}.$$

  If $P = .5$, then $\Phi^{-1}(.5) = 0$ so:

  In probit regression with additive Berkson error the naive estimator of the median effective dose is robust, i.e., provides a consistent estimator of the median effective dose. For other percentiles the naive estimators are inconsistent.

  If $(\boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1)^{1/2}$ is relatively small, then η will be close to 1 and the biases of the naive estimators of the percentiles are negligible.

- Logistic regression.
  The logistic regression model is given in (6.2). Note that the β's in the logistic specification are not the same as those in the probit specification.
  We continue to assume (6.29). Using (11.6), $P(Y = y|\mathbf{w})$ equals $\int_{\mathbf{x}} P(Y = y|\mathbf{x})f(\mathbf{x}|\mathbf{w})d\mathbf{x}$, which cannot be simplified. There are two approximations that typically get used. The first, from (6.28), is

$$logit(P(Y = 1|\mathbf{w})) \approx \beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1 + \boldsymbol{\beta}_1'\boldsymbol{\Lambda}_1\mathbf{w}. \tag{6.32}$$

This approximation can be justified by assuming small $\boldsymbol{\beta}_1$, small $P(Y = 1|\mathbf{x})$ (e.g., rare disease) or small measurement error.
The second approximation comes from using the probit approximation to the logistic given in (6.4), which allows us to move back and forth from the logistic to the probit model. Using this and (6.30) leads to

$$logit(P(Y = 1|\mathbf{w})) \approx \beta_0^* + \boldsymbol{\beta}_1^{*'}\mathbf{w} \tag{6.33}$$

where

$$\beta_0^* = \frac{\beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1}{(1 + .346\boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1)^{1/2}}, \qquad \boldsymbol{\beta}_1^* = \frac{\boldsymbol{\Lambda}_1'\boldsymbol{\beta}_1}{(1 + .346\boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1)^{1/2}},$$

where $.346 = 1/1.7^2$. (This is equation (4.25) from Carroll et al. (2006) in slightly different form.) The biases in the naive estimators differ from those arising from the simple approximation in (6.28) due to the term $.346\boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{X|w}\boldsymbol{\beta}_1$.

- Normal discriminant model.
  The normal discriminant model provides another approach to treating a binary outcome Y with a vector of quantitative covariates in $\mathbf{X}$. It implies a logistic regression model for $P(Y = 1|\mathbf{x})$; see for example Hosmer and Lemeshow (2000, p. 43). Early work on measurement error in the normal discriminant setting includes Wu et al. (1986) and Armstrong et al. (1989).
  The normal discriminant model assumes

$$\mathbf{X}|Y = y \sim N(\boldsymbol{\mu}_y, \boldsymbol{\Omega}), \tag{6.34}$$

which has a mean vector that changes with y (= 0 or 1) but a constant covariance matrix $\boldsymbol{\Omega}$. This implies a logistic model for $P(Y = 1|\mathbf{X} = \mathbf{x})$ with

$$\boldsymbol{\beta}_1 = \boldsymbol{\Omega}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0).$$

To start, we allow a differential normal linear measurement error model in which the intercept vector changes with y. That is

$$\mathbf{W}|\mathbf{x}, y \sim N(\boldsymbol{\theta}_{0y} + \boldsymbol{\Theta}_1\mathbf{x}, \boldsymbol{\Sigma}_u). \tag{6.35}$$

It can be shown (Buonaccorsi(1990b)) that together (6.34) and (6.35) imply $\mathbf{W}|y \sim N(\boldsymbol{\theta}_{0y} + \boldsymbol{\Theta}_1\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_u + \boldsymbol{\Theta}_1\boldsymbol{\Omega}\boldsymbol{\Theta}_1')$. This in turn implies a logistic model for Y given $\mathbf{w}$,

$$logit(P(Y = 1|\mathbf{w})) = \beta_{0*} + \boldsymbol{\beta}_1^{*'}\mathbf{w},$$

where

$$\boldsymbol{\beta}_1^* = (\boldsymbol{\Sigma}_u + \boldsymbol{\Theta}_1\boldsymbol{\Omega}\boldsymbol{\Theta}_1')^{-1}(\boldsymbol{\theta}_{01} - \boldsymbol{\theta}_{00} + \boldsymbol{\Theta}_1(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)).$$

The naive estimator of $\boldsymbol{\beta}_1$ is a consistent estimator of $\boldsymbol{\beta}_1^*$.
Additive measurement error is a special case of the above with $\boldsymbol{\Theta}_1 = \mathbf{I}$ and

$\boldsymbol{\theta}_{0y} = \mathbf{0}$. A slight extension of the additive measurement error model allows a constant bias with $\boldsymbol{\theta}_{0y} = \boldsymbol{\theta}_0$, in which case $\boldsymbol{\beta}_1^* = (\boldsymbol{\Sigma}_u + \boldsymbol{\Omega})^{-1}\boldsymbol{\Omega}\boldsymbol{\beta}_1$. This gives an exact asymptotic bias, which looks similar to that from (6.28) but note that $\boldsymbol{\Omega}$ is the covariance of $\mathbf{X}$ given y, which is different than $Cov(\mathbf{X}) = \boldsymbol{\Sigma}_X$.

6.7.7 Linear regression with misclassification of a binary predictor

To motivate the model in this section, we consider the birth weight data from Hosmer and Lemeshow (2000) where the response y is birth weight of a child, x_1 is the smoking status of the mother and x_2 is the mother's weight at the time of the last menstrual cycle. Smoking status is assumed to be subject to misclassification, while the mother's weight is taken to be measured without error. The model of interest is

$$E(Y|X_1 = x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

where with x_1 equal to 0 or 1 and β_1 is the difference in expected birth weight for a smoker and nonsmoker at a common pre-birth weight of the mother.

The developments below (and the example) are based on Buonaccorsi et al. (2005). See Section 3.1 of Gustafson (2004), Christopher and Kupper (1995) and references therein for additional discussion.

The general model of interest here is

$$E(Y|X_1 = x_1, \mathbf{x}_2) = \beta_0 + \beta_1 x_1 + \boldsymbol{\beta}_2' \mathbf{x}_2,$$

where $x_1 = 0$ or 1 is a binary predictor, subject to misclassification, and $\mathbf{x}_2$ contains additional covariates measured without error. We condition on $\mathbf{x}_2$ throughout the discussion. The categorical error-prone measure of X_1 is denoted W, assumed also to be binary. The coefficient β_1 represents the difference in the expected response for $X_1 = 1$ or $X_1 = 0$ with $\mathbf{x}_2$ held fixed and so is the effect of the binary predictor. If there are multiple categories, say J, for a single predictor then we could extend the model above by either having $\mathbf{x}_1$ containing $J - 1$ dummy variables or replacing $\beta_0 + \beta_1 x_1$ with $\boldsymbol{\beta}_1' \mathbf{x}_1$, where $\mathbf{x}_1$ contains J dummy variables. We stick to treating a binary X_1 for ease of presentation but the extension to multiple categories is straightforward.

The measurement error model is given by the misclassification probabilities, $P(W_1 = w_1|X_1 = x_1, \mathbf{x}_2)$ for w_1 and x_1 each equal to 0 or 1. Notice that these can depend on $\mathbf{x}_2$. With X_1 random, the induced model is

$$E(Y|w_1, \mathbf{x}_2) = \beta_0 + \beta_1 P(X_1 = 1|w_1, \mathbf{x}_2) + \boldsymbol{\beta}_2' \mathbf{x}_2, \qquad (6.36)$$

for w_1 equal to 0 or 1. We have used the fact that $E(X_1|w_1, \mathbf{x}_2) = P(X_1 = 1|w_1, \mathbf{x}_2)$ since X_1 is binary.

In general, the induced model in (6.36) is not in the same form as the original model since $P(X_1 = 1|w_1, \mathbf{x}_2)$ is not linear in w_1 and $\mathbf{x}_2$.

In fact,

$$P(X_1 = 1|w_1, \mathbf{x}_2) = \frac{P(W_1 = w_1|X_1 = 1, \mathbf{x}_2)P(X_1 = 1|\mathbf{x}_2)}{\sum_{x_1=0,1} P(W_1 = w_1|X_1 = x_1, \mathbf{x}_2)P(X_1 = x_1|\mathbf{x}_2)}. \tag{6.37}$$

Consider the case where the misclassification in X_1 is nondifferential with respect to $\mathbf{x}_2$; that is, $P(W_1 = w_1|X_1 = x_1, \mathbf{x}_2) = \theta_{w_1|x_1}$, free of $\mathbf{x}_2$. If this occurs *and* X_1 is independent of $\mathbf{X}_2$, then $P(X_1 = 1|\mathbf{x}_2)$ is a constant, say π_1, so $P(X_1 = 1|w_1, \mathbf{x}_2) = \pi_1\theta_{w_1|1}/(\pi_1\theta_{w_1|1} + (1 - \pi_1)\theta_{w_1|0})$. This is free of $\mathbf{x}_2$ leading to the following result:

If the misclassification in X_1 is nondifferential with respect to both $\mathbf{x}_2$ and y, and X_1 is independent of $\mathbf{X}_2$, then there is no bias in the naive estimates of coefficients of the perfectly measured $\mathbf{x}_2$. However if X_1 and $\mathbf{X}_2$ are "correlated" then this result is no longer true.

This result parallels the one in Section 5.3 where for additive error there was no bias in the naive estimate of the coefficient of a perfectly measured predictor only if it was not correlated with the mismeasured predictor. It is important to note that if X_1 and $\mathbf{X}_2$ are dependent, the misclassification rates being free of $\mathbf{x}_2$ do not imply that the reclassification rates $P(X_1 = x_1|\mathbf{w}_1, \mathbf{x}_2)$ are free of $\mathbf{x}_2$. So, there will be bias in the coefficients for the perfectly measured $\mathbf{x}_2$.

Approximate biases.

The rest of this section continues to explore the bias further, assuming random X_1 (binary), and, for illustrative purposes, allowing one additional variable x_2 measured without error. With appropriate notational modifications, similar results hold for multivariate $\mathbf{x}_2$. See Section 8.6 for the case with a categorical X_1 but no X_2. The error prone measure for X_1 is W_1.

Conditioning on the observed error-prone measure $W_1 = w_1$ (0 or 1),

$$E(Y|W_1 = w_1, x_2) = \beta_0 + \beta_1\lambda_{1|w_1,x_2} + \beta_2 x_2,$$

where $\lambda_{1|w_1,x_2} = P(X_1 = 1|W_1 = w_1, x_2)$ is given in (6.37). Define $n \times 3$ matrices $\mathbf{G}_W$ and $\mathbf{G}$ with kth row $(1, w_{i1}, x_{i2})$ and $(1, \lambda_{1|w_{i1},x_{i2}}, x_{i2})$, respectively. The naive estimator is $\hat{\boldsymbol{\beta}}_{naive} = (\mathbf{G}_W'\mathbf{G}_W)^{-1}\mathbf{G}_W'\mathbf{Y}$. Given the $\mathbf{w}$'s, $E(\hat{\boldsymbol{\beta}}_{naive}|\mathbf{w}_1, \mathbf{x}_2) = \boldsymbol{\beta} + \mathbf{B}$, where the bias is

$$\mathbf{B} = (\mathbf{G}_W'\mathbf{G}_W)^{-1}(\mathbf{G} - \mathbf{G}_W)\boldsymbol{\beta} = \beta_1(\mathbf{G}_W'\mathbf{G}_W)^{-1}\mathbf{d},$$

with $\mathbf{d}' = (d_1, \ldots, d_n)$ and $d_i = \lambda_{1|w_{i1},x_{i2}} - w_{i1}$. See Section 3.1 of Christopher and Kupper (1995) for a related result.

The quantity $\mathbf{B}$ provides the conditional bias, given the observed error prone w_1's and the perfectly measured x_2 values. Since X_1 is random, we can consider the unconditional bias, either for X_1 and W_1 random with x_2's fixed, or over X_1, W_1 and X_2 all random. Either way it is difficult to calculate the exact finite sample bias.

Since the model for Y is linear in the x's, the coefficients and their estimators can be expressed in terms of variances and covariances. This, in turn, leads to the asymptotic bias of the naive estimators. Consider the case with X_1, W_1 and X_2 all random with

$$\mathbf{\Sigma}_X = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}, \mathbf{\Sigma}_W = Cov \begin{bmatrix} W_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} \sigma^2_{W1} & \sigma_{W_1,X_2} \\ \sigma_{W_1,X_2} & \sigma_2^2 \end{bmatrix}$$

and $b = Cov(X_1, Y) - CovW_1, Y = Cov(X_1, Y)(1 - \theta_{1|1} - \theta_{1|0})$. Then the naive estimators are consistent for

$$\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} + (\mathbf{\Sigma}_W^{-1}\mathbf{\Sigma}_X - \mathbf{I})\boldsymbol{\beta} + \mathbf{\Sigma}_W^{-1} \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

The second and third terms combined provide the asymptotic bias. If X_1 and X_2 are uncorrelated then both σ_{12} and $\sigma_{W_1X_2}$ equal 0 and the bias for $\hat{\beta}_{2,naive}$ becomes $\mathbf{0}$.

Using the fact that X_1 and W_1 are both binary, the asymptotic biases can also be expressed using $\sigma_1^2 = p(1-p)$, $\sigma_{12} = p(1-p)(\mu_1 - \mu_0)$, $\sigma^2_{W_1} = \theta(1-\theta)$, $\sigma_{W_1X_2} = \sigma_{12}(\theta_{1|1} - \theta_{1|0})$, $Cov(X_1, Y) = \beta_1\sigma_1^2 + \beta_2\sigma_{12}$ and $Cov(W_1Y) = (\theta_{1|1} - \theta_{1|0})Cov(X_1, Y)$, where $p = P(X_1 = 1)$, $\mu_j = E(X_2|X_1 = j)$ and $\theta = \theta_{1|1}p + \theta_{1|0}(1-p)$.

Buonaccorsi et al. (2005) used the birth weight example introduced at the start of this section to illustrate the bias implications via simulations. Four factors are varied, the amount of dependence between X_1 and X_2, the coefficients β_1 and β_2, the marginal probability of the mother being a smoker and the misclassification probabilities. Throughout, based on the original data, $\beta_0 = 5.5$ and $V(Y|x_1, x_2) = \sigma^2 = 1.6^2$.

1. Dependence between X_1 and x_2: We assume that $X_2|X_1 = j$ is distributed $N(\mu_j, \sigma^2_{2|1})$. This means the dependence between X_1 and X_2 can be characterized through $\Delta = (\mu_0 - \mu_1)/\sigma_{2|1}$. Note that under this normal discriminant model, $P(X_1 = 1|x_2)$ follows a logistic model with the coefficient for x_2 being $(\mu_1 - \mu_0)/\sigma^2_{2|1}$. In the original data, the average mother's weight was 131 and 128 for nonsmokers and smokers, respectively, with standard deviations of 28.43 and 33.79. In our simulations we fix $\mu_0 = 130$ and $\sigma_2 = 30$ and let $\mu_1 = \mu_0 - \sigma\Delta$ for $\Delta = 1/3, 2/3, 1$ and $4/3$.

2. Coefficients: $\beta_1 =$ -.6 or -1.2; $\beta_2 =$ .01 or .1.

In the original analysis, the estimate of β_1 was -.6 and for β_2 was about .01. With other things held constant, a change in the mother's weight of 10 pounds results in an expected change in the birth weight of .1 pounds when $\beta_2 = .01$, and an expected change of 1 pound when $\beta_2 = .1$.

3. Probability of a smoker: $p = P(X_1 = 1)$ = .3 or .5. In the original data 39% of the women were smokers.

4. Misclassification probabilities: The amount of misclassification is specified through $\theta_{1|1} = P(W_1 = 1|X_1 = 1)$ and $\theta_{0|0} = P(W_1 = 0|X_1 = 0)$. *Note that these are free of x_2, but the reclassification probabilities depend on x_2 unless* $\Delta = 0$. The probability of misclassifying a smoker is $1 - \theta_{1|1}$ and of misclassifying a nonsmoker is $1 - \theta_{0|0}$. Three cases are used:

Case 1. $\theta_{1|1} = .9$ and $\theta_{0|0} = .9$

Case 2. $\theta_{1|1} = .9$ and $\theta_{0|0} = .7$

Case 3. $\theta_{1|1} = .95$ and $\theta_{0|0} = .6$

The top panel of Figure 6.3 displays the simulated mean of the naive estimate of β_2 with $n = 1000$ and $\beta_2 = .01$. The general story in terms of direction and the relative magnitude of the bias is similar with $\beta_2 = .1$ and similar in the cases with $n = 100$. For each misclassification case and value of β_2, $16 = 4 \times 2 \times 2$ simulation scenarios with different combinations of Δ, p and β_1 were considered. For each setting the same 16 results are presented in three different ways; labeled according to the value of Δ (labels 1, 2, 3 and 4 for $\Delta = 1/3, 2/3, 1$ and $4/3$, respectively), according to the values of p (labels 5 and 6 for $p = .3$ and .5, respectively) and according to the values of β_x (labels 7 and 8 on the right for $\beta_1 = -.6$ and -1.2, respectively). The bias is seen to be increasing in Δ, β_1 and p. The worst scenario is presented by the row with a combination labeled 4, 6 and 8, which indicates $\Delta = 4/3$, $\beta_1 = -1.2$ and $p = .5$. For this scenario, in Case 3 (the highest misclassification rates) the bias is close to .007 and was similar for both $\beta_2 = .01$ and $\beta_2 = .1$, corresponding to relative biases of 70 and 7 percent, respectively.

The importance of the bias in β_2 depends on the application. For our example, in the worst case scenarios just described, when $\beta_2 = .01$ (and the relative bias is 70 percent) an expected change in the birth weight of .1 pounds per 10 pound change in the mother's weight is estimated on average to be .17 pounds. Similarly, when $\beta_2 = .1$ an expected change in the birth weight of 1 pound per 10 pound change in the mother's weight is estimated on average to be 1.07 pounds. Qualitatively, the biases in β_2 would not appear to be very important in this context.

The bottom panel of Figure 6.3 demonstrates the more severe, and well known, impact of the misclassification on the bias in the naive estimate of β_1 (the smoking effect) for settings with $\beta_1 = -1.2$. This represents a true difference in expected birth weight of 1.2 pounds for smokers compared to nonsmokers with mother's weight held fixed. In many of the combinations the misclassification leads to an estimate of the smoking effect that is approximately half of its true value.

6.8 Assessing bias via estimating equations

This approach is more complicated to describe and usually requires approximations, which does limit its usefulness somewhat. It does, however, offer some distinct advantages over the use of induced models, including: i) It does not rely on needing the mismeasured predictors to be random and so handles the functional setting. ii) It allows us to works directly with the measurement error model for the behavior of $\mathbf{W}|\mathbf{x}$, rather than go through the Berkson model. iii) It accommodates heteroscedastic measurement errors. iv) Measurement error in the response is more easily handled.

The use of estimating equations to determine the properties of estimators has a long history in statistics in general. Stefanski (1985) and Stefanski and Carroll (1985) were among the first to employ these techniques in measurement error problems. We treat the estimating equations in (6.5), which apply for many regression problems. Using d in place of y and $\mathbf{w}$ in place of $\mathbf{x}$, the naive estimates solve $\sum_i S(d_i, \mathbf{w}_i; \boldsymbol{\beta}) = \mathbf{0}$ as a function of $\boldsymbol{\beta}$. These are the naive estimating equations. As always with no error in the response, $y = d$ while if there is no error in the predictors, $\mathbf{w} = \mathbf{x}$.

The key result is the following:

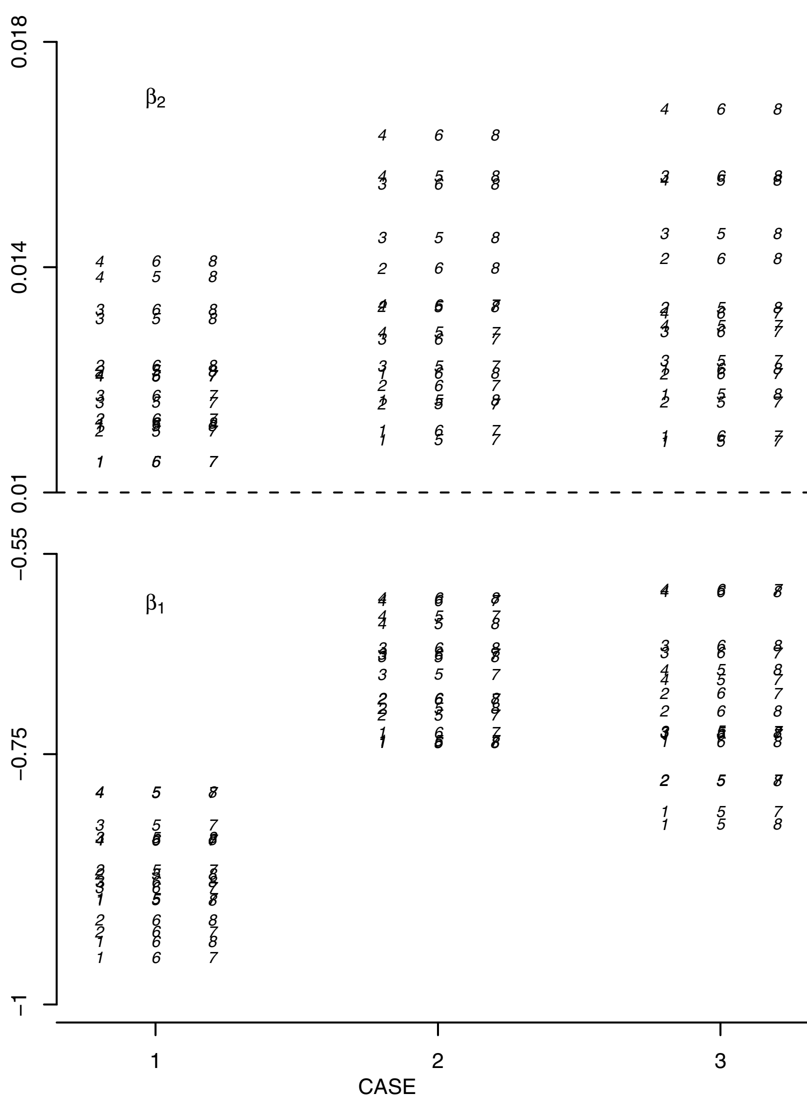
If $\boldsymbol{\beta}^*$ is such that $\sum_i E_\beta(S(D_i, \mathbf{W}_i, \boldsymbol{\beta}^*))/n \rightarrow \mathbf{0}$, then under some general conditions, $\hat{\boldsymbol{\beta}}_{naive}$ converges in probability to $\boldsymbol{\beta}^*$ and $\boldsymbol{\beta}^* - \boldsymbol{\beta}$ is the asymptotic bias of the naive estimator.

We won't go into the technicalities associated with the conditions here. Stefanski and Carroll (1987) provide a rigorous treatment of the asymptotic behavior of the naive estimators in the case of logistic regression. See Foutz (1977) and Section A.6 of Carroll et al. (2006) also for more general discussion.

For finite samples if $\sum_i E_{\boldsymbol{\beta}}(S(D_i, \mathbf{W}_i, \boldsymbol{\beta}^*))/n \approx \mathbf{0}$, then we can view $\boldsymbol{\beta}^* - \boldsymbol{\beta}$ as the approximate bias.

In general, there is no closed form solution to $\sum_i E_{\boldsymbol{\beta}}(S(D_i, \mathbf{W}_i, \boldsymbol{\beta}^*))/n = \mathbf{0}$. With distributional assumptions and specific values of the parameters, one

Figure 6.3 *Effects of misclassifying X_1 with X_2 measured without error. Plot of mean of estimate of β_2 (top plot) and β_1 (bottom plot) over 500 simulations with $n = 1000$. Cases 1, 2 and 3 as indicated in text. In top plot $\beta_2 = .01$ and for each case there are 16 scenarios with results presented with three different labelings. Within each case, the left labels 1, 2, 3 and 4 correspond to Δ = 1/3, 2/3, 1 and 4/3, respectively; the middle labels 5 and 6 correspond to $p = .3$ and .5, respectively; the right labels 7 and 8 correspond to $\beta_1 = -.6$ and -1.2, respectively. In the bottom plot the labeling is similar except that $\beta_1 = -1.2$ throughout and the labels 7 and 8 refer to $\beta_2 = .01$ and .1, respectively. From Buonaccorsi et al. (2005). Used with permission of Blackwell Publishing.*



could solve the equations numerically, but this does not lead to an analytical characterization and offers limited insight into the bias issue. One remedy is to use a first order Taylor Series expansion of the estimating equations around $\boldsymbol{\beta}^* = \boldsymbol{\beta}$. This leads to

$$\boldsymbol{\beta}^* \approx \boldsymbol{\beta} - (\sum_i E(\dot{S}(D_i, \mathbf{W}_i, \boldsymbol{\beta})))^{-1} \sum_i E(S(D_i, \mathbf{W}_i, \boldsymbol{\beta})), \tag{6.38}$$

where $\dot{S}(D_i, \mathbf{W}_i, \boldsymbol{\beta}) = \partial S(Y_i, \mathbf{W}_i, \boldsymbol{\beta})/\partial \boldsymbol{\beta}$. The expected values in (6.38) will themselves often need further approximations since they are usually nonlinear functions of the random variables involved.

Special case. Equation (6.6) covers a number of important settings. In that case, with $D_i = y_i + q_i$, $\boldsymbol{\beta}^*$ solves

$$\sum_i E\left[(m(\mathbf{W}_i, \boldsymbol{\beta}^*) - q_i)\boldsymbol{\Delta}(\mathbf{W}_i, \boldsymbol{\beta}^*)\right] = \sum_i m(\mathbf{x}_i, \boldsymbol{\beta}) E(\boldsymbol{\Delta}(\mathbf{W}_i, \boldsymbol{\beta}^*)).$$

The expectations are over q_i and W_i random and the $\mathbf{x}_i$ fixed. We note that $\boldsymbol{\beta}^*$ is involved in three places and $\boldsymbol{\beta}$ in one part of the right hand term. This still does not usually yield a closed form solution but it does provide a shortcut for exploring the bias.

- It is easy to see that with no measurement error $q_i = 0$ and $\mathbf{W}_i = \mathbf{x}_i$ the equation is satisfied at $\boldsymbol{\beta}^* = \boldsymbol{\beta}$.
- If we apply this approach to the linear case with $m(\mathbf{x}, \boldsymbol{\beta}) = \boldsymbol{\beta}'\mathbf{x}_*$ and $\boldsymbol{\Delta}(\mathbf{W}_i, \boldsymbol{\beta}) = \mathbf{W}_{i*}$ then the solution is
$$\boldsymbol{\beta}^* = \left(\frac{\mathbf{X}'\mathbf{X}}{n} + \boldsymbol{\Sigma}_u\right)^{-1} \left(\frac{\mathbf{X}'\mathbf{X}}{n}\boldsymbol{\beta} + \boldsymbol{\Sigma}_{uq}\right)$$
This agrees with the earlier results for multiple linear regression in (5.5).
- In general if the estimating equations are linear in $\boldsymbol{\beta}$ (but perhaps nonlinear in the x's) the naive estimator has a closed form; $\widehat{\boldsymbol{\beta}}_{naive} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\mathbf{D}$ for appropriately defined $\mathbf{W}$. In this case assessing the bias can be attacked more directly; see Section 8.4.
- *Logistic, Poisson and other GLIMs.* For a number of generalized linear models, including logistic and Poisson with log-link regression, $\boldsymbol{\Delta}(\mathbf{W}_i, \boldsymbol{\beta}) = \mathbf{W}_{i*}$. Recall that $\mathbf{W}'_{i*} = (1, \mathbf{W}_i)$. With no error in Y, then the problem reduces to finding $\boldsymbol{\beta}^*$ such that
$$\sum_i E(m(\mathbf{W}_i, \boldsymbol{\beta}^*)\mathbf{W}_{i*}) = \sum_i m(\mathbf{x}_i, \boldsymbol{\beta})\mathbf{x}_{i*}. \tag{6.39}$$

While the estimating equation approach does suggest a strategy for getting an analytical expression for the approximate bias when other methods are not available, it can get bogged down a bit in many nonlinear problems due to the need for approximations. Consider, for example, logistic regression with no error in the response and $\pi(\mathbf{x}, \boldsymbol{\beta}) = P(Y = 1|\mathbf{x}) = 1/(1 + e^{-\mathbf{x}'\boldsymbol{\beta}})$. Using (6.39), if we write $h(\mathbf{W}_i, \boldsymbol{\beta}^*)$ for $\pi(\mathbf{W}_i, \boldsymbol{\beta}^*)\mathbf{W}_i^*$, use a second order expansion around $\mathbf{W}_i = \mathbf{x}_i$, first order expansion around $\boldsymbol{\beta}^* = \boldsymbol{\beta}$ and assume additive error, then

$$\boldsymbol{\beta}^* \approx \boldsymbol{\beta} - \left[\sum_i \frac{\partial h(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right]^{-1} \left[\sum_i (\frac{\partial^2 h(\mathbf{x}_i, \boldsymbol{\beta})}{\partial^2 \mathbf{x}_i})\boldsymbol{\Sigma}_{u_i}\right].$$

Defining $\gamma_i = \pi_i(1 - \pi_i)$, the bias term is approximately

$$-\begin{bmatrix} \sum_i \gamma_i & \sum_i x_i\gamma_i \\ \sum_i x_i\gamma_1 & \sum_i x_i^2\gamma_i \end{bmatrix}^{-1} \begin{bmatrix} \beta_0\beta_1 \sum_i \sigma_{ui}^2 \gamma_i(1 - 2\pi_i) \\ \sum_i \sigma_{ui}^2 \{\beta_1^2 \gamma_i(x_i - 2\pi_i) + 2\beta_1\gamma_i \end{bmatrix}.$$

Log linear model.

Suppose $E(Y_i|\mathbf{x}_i) = e^{\mathbf{x}_i'\boldsymbol{\beta}}$, $V(Y_i|\mathbf{x}_i) = \sigma^2 e^{\mathbf{x}_i'\boldsymbol{\beta}}$. If $Y_i|\mathbf{x}_i$ is distributed Poisson, then $\sigma^2 = 1$. The estimating equations here are $\sum_i (Y_i - e^{\mathbf{W}_i'\boldsymbol{\beta}})\mathbf{W}_i = \sum_i S(Y_i, \mathbf{W}_i, \boldsymbol{\beta}) = \mathbf{0}$ and the approximate bias is

$$-\left[\sum_i E(\mathbf{W}_i\mathbf{W}_i' e^{\mathbf{W}_i'\boldsymbol{\beta}})\right]^{-1} \sum_i \left[e^{\mathbf{x}_i'\boldsymbol{\beta}} - E(e^{\mathbf{W}_i'\boldsymbol{\beta}})\right].$$

As in the logistic setting this needs further approximation.

6.9 Moment-based and direct bias corrections

This section, the first of a sequence describing some of the available ways to correct for measurement, examines how to correct using some direct bias corrections.

Broadly, these techniques obtain corrected estimators based on the properties of summary measures obtained from the observed $(D_i, \mathbf{W}_i)$ values, along with estimates of the measurement error parameters obtained from additional data. These methods are useful whenever the expected values of summary statistics can be written exactly, or approximately, in terms of the original parameters of interest and the measurement error parameters. The summary statistics may be means, variances and covariances or simply the naive estimators themselves. This approach was used in Chapters 2–5. For the misclassification problems in Chapters 2 and 3 the observed proportions along with the estimated misclassification or reclassification rates were used. In the linear regression problems of Chapters 4 and 5, the corrected estimators exploited the properties of sample means, variances and covariances along with estimated or known measurement error variances and covariances.

The rest of this section describes the details for the case where the corrected estimators can be obtained by linearly transforming the naive estimators.

6.9.1 *Linearly transforming the naive estimates*

One of the simplest correction techniques linearly transforms the naive estimates of the coefficients. This is used under the assumption of a linear Berkson model with $E(\mathbf{X}|\mathbf{w}) = \boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}$. As discussed earlier in this chapter, for a number of settings the expected value of the naive estimator of $\boldsymbol{\beta}_1$ is exactly or approximately $\boldsymbol{\Lambda}_1'\boldsymbol{\beta}_1$. This is an exact result in linear regression (see (6.25)) and was used as an approximation for logistic, probit and Poisson regression and, more generally, for generalized linear models; see equations (6.26), (6.28), (6.30) and (6.32). Estimates of $\boldsymbol{\Lambda}_1$ and $\boldsymbol{\lambda}_0$ may be obtained from either replicate measures or validation data, as outlined in Sections 4.5.3, 5.4.3 and 6.5.1.

Assuming $\mathbf{w}$ and $\mathbf{x}$ are of the same dimension, then a corrected estimator of the nonintercept coefficients is given by

$$\widehat{\boldsymbol{\beta}}_1 = \widehat{\boldsymbol{\Lambda}}_1'^{-1}\widehat{\boldsymbol{\beta}}_{1naive}. \tag{6.40}$$

This technique (referred to as the RSW method in Thurston et al. (2003)) was first proposed in Rosner et al. (1989), Rosner et al. (1990) and Rosner et al. (1992).

"Multiple surrogates."

This approach can also handle the case where $\mathbf{W}$ has more elements than $\mathbf{X}$ Suppose $\mathbf{W}$ is $q \times 1$ and $\mathbf{X}$ is $p \times 1$ with $q \geq p$. Assuming $\widehat{\boldsymbol{\Lambda}}_1$ (which is $p \times q$) is of rank p, then a corrected estimator is

$$\widehat{\boldsymbol{\beta}}_1 = (\widehat{\boldsymbol{\Lambda}}_1\widehat{\boldsymbol{\Lambda}}_1')^{-1}\widehat{\boldsymbol{\Lambda}}_1\widehat{\boldsymbol{\beta}}_{1naive}. \tag{6.41}$$

With $q = p$ this reduces to the estimator in 6.40. (With $q > p$ the use of the expression naive in $\widehat{\boldsymbol{\beta}}_{1naive}$ is a really a misnomer as this "naive estimator" is not of the same dimension of $\boldsymbol{\beta}_1$.) Weller et al. (2007) describe another corrected estimator that also transforms the naive estimators for the case with $\mathbf{x}' = (x_1, \mathbf{x}_2)$ and $\mathbf{W}' = (\mathbf{W}_1, \mathbf{W}_2)$, where x_1 is univariate, $\mathbf{W}_1$ is multivariate and $\mathbf{x}_2 = \mathbf{W}_2$ is measured without error.

Estimating the covariance of $\widehat{\boldsymbol{\beta}}$.

We will focus here on the case where $\mathbf{W}$ and $\mathbf{X}$ are the same dimension and outline the main points. See Thurston et al. (2003) for additional detail.

If $\widehat{\boldsymbol{\Lambda}}_1$ is considered known then $Cov(\widehat{\boldsymbol{\beta}}_1) = \widehat{\boldsymbol{\Lambda}}_1'^{-1} Cov(\widehat{\boldsymbol{\beta}}_{1,naive})\widehat{\boldsymbol{\Lambda}}_1^{-1}$, where $\boldsymbol{\Sigma}_{\beta_1,naive} = Cov(\widehat{\boldsymbol{\beta}}_{1,naive})$ is the covariance matrix of the naive estimators.

This can be estimated by $\widehat{\mathbf{\Lambda}}_1^{'-1}\widehat{\mathbf{\Sigma}}_{\beta_1,naive}\widehat{\mathbf{\Lambda}}_1^{-1}$, where $\widehat{\mathbf{\Sigma}}_{\beta_1,naive}$ is an estimate of $Cov(\widehat{\boldsymbol{\beta}}_{1,naive})$ obtained from the naive analysis. In cases where the linear correction agrees with the regression calibration technique, this version of the estimate covariance can be obtained by simply running a regression on imputed values; see Section 6.10.

Allowing for uncertainty in $\widehat{\mathbf{\Lambda}}_1$ requires more care.

- $\widehat{\boldsymbol{\beta}}_{naive}$ independent of $\widehat{\mathbf{\Lambda}}_1$.
  This will apply if $\widehat{\mathbf{\Lambda}}_1$ is calculated from external data or if we have normally distributed replicates. In this case, the approximate covariance matrix of $\widehat{\boldsymbol{\beta}}_1$ is
  $$Cov(\widehat{\boldsymbol{\beta}}_1) = \mathbf{\Sigma}_{\boldsymbol{\beta}_1} \approx \mathbf{\Lambda}_1^{'-1}\mathbf{\Sigma}_{\beta_1,naive}\mathbf{\Lambda}_1^{-1} + \mathbf{V}_2,$$
  where the additional term, $\mathbf{V}_2$, arises from uncertainty associated with the estimation of $\mathbf{\Lambda}_1$. This is estimated by
  $$\widehat{\mathbf{\Sigma}}_{\boldsymbol{\beta}_1} = \widehat{\mathbf{\Lambda}}_1^{'-1}\widehat{\mathbf{\Sigma}}_{\beta_1,naive}\widehat{\mathbf{\Lambda}}_1^{-1} + \widehat{\mathbf{V}}_2, \tag{6.42}$$
  where the $(j,k)th$ element of $\widehat{\mathbf{V}}_2$ is given by $\widehat{\boldsymbol{\beta}}'_{naive}\mathbf{A}_{jk}\widehat{\boldsymbol{\beta}}_{naive}$ and the calculation of the matrix $\mathbf{A}_{jk}$ is described in Section 6.17.2.
- $\widehat{\boldsymbol{\beta}}_{naive}$ and $\widehat{\mathbf{\Lambda}}_1$ not necessarily independent.
  This will be the case with $\mathbf{\Lambda}_1$ estimated from internal validation data or with the use of nonnormal replicates. One approach to getting $Cov(\widehat{\boldsymbol{\beta}}_1)$ is using the estimating equation approach described in Section 6.17.3. Note that if there is internal validation data there are other strategies to estimating $\boldsymbol{\beta}_1$ besides estimating $\mathbf{\Lambda}_1$ and using (6.40); see Section 6.15.

We can also correct the estimate of the intercept, if so desired. In many of the models noted above this leads to $\widehat{\beta}_0 = \widehat{\beta}_{0,naive} - \widehat{\lambda}'_0\widehat{\boldsymbol{\beta}}_1$. The discussion above on estimating covariance terms can be extended to handle the joint covariance of $\widehat{\beta}_0$ and $\widehat{\boldsymbol{\beta}}_1$, but we will omit the details here.

6.10 Regression calibration and quasi-likelihood methods

We will assume for this section that there is no error in the response. These techniques are based on an induced model for $Y|\mathbf{w}$, which is built in turn on a model for $\mathbf{X}|\mathbf{w}$. As such, these methods depend formally on the assumption that the unobserved X's are random, i.e., we are in a structural setting.

Recalling our earlier notation, for the true values, $E(Y|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta})$ and $V(Y|\mathbf{x}) = v(\mathbf{x}, \boldsymbol{\beta}, \boldsymbol{\sigma})$. Appealing to the discussion on induced models in Section 6.7, assuming nondifferential measurement error and no error in y, $E(Y|\mathbf{w}) = E(m(\mathbf{X}, \boldsymbol{\beta})|\mathbf{w})$. This expectation will depend on the parameters in the model for the true values and also on parameters in the model for $\mathbf{X}|\mathbf{w}$. To implement this approach it is assumed that

$$E(\mathbf{X}|\mathbf{w}) = h(\mathbf{w}, \mathbf{\Lambda}) \text{ and } Cov(\mathbf{X}|\mathbf{w}) = \mathbf{\Sigma}_{X|w}.$$

We will collect the parameters of the Berkson error model for $\mathbf{X}$ given $\mathbf{w}$ into $\boldsymbol{\xi} = (\mathbf{\Lambda}, \mathbf{\Sigma}_{X|w})$. These quantities will be indexed by i if they change across observations.

The general quasi-likelihood method can be described as follows:

- Find an exact or approximate expression for $E(Y_i|\mathbf{w}_i) = g(\mathbf{w}_i, \boldsymbol{\beta}, \boldsymbol{\xi}_i)$, where we have allowed the Berkson error parameters in $\boldsymbol{\xi}$ to change across observations.
- Get estimated values for the $\boldsymbol{\xi}_i$ from additional data, e.g., replication or validation data.
- Use a general regression technique to fit the model $E(Y_i) = g(\mathbf{w}_i, \boldsymbol{\beta}, \widehat{\boldsymbol{\xi}}_i)$.

The fit at the last stage might use just the assumed model for $E(Y|\mathbf{w})$ (perhaps in combination with a robust estimate of the covariance of the estimated coefficients) or it might account explicitly for the variance of $Y_i|\mathbf{w}_i$. The variance model may be automatically implied by the type of response. This is true, for example, in binary regression where $V(Y_i|\mathbf{w}_i) = g(\mathbf{w}_i, \boldsymbol{\beta}, \widehat{\boldsymbol{\xi}}_i)(1 - g(\mathbf{w}_i, \boldsymbol{\beta}, \widehat{\boldsymbol{\xi}}_i))$. In other cases an expression for the variance is needed. This can be obtained via equation (6.22), although often this expression cannot be expressed in closed form and some approximations may be needed. This variance potentially depends on $\boldsymbol{\beta}$, $\boldsymbol{\sigma}$ and $\boldsymbol{\xi}$.

With the Berkson error parameters set to estimates, these models, often referred to as QVF (quasi-likelihood and variance function) models, can be fit using general nonlinear/quasi-likelihood methods. An excellent overview for these methods is Carroll and Ruppert (1988). If the measurement error parameters can be treated as known these methods also yield standard errors, etc. for the estimates of $\boldsymbol{\beta}$ and, if needed, the variance parameters in $\boldsymbol{\sigma}$. In general, however, these standard errors are incorrect since they do not account for estimation of the measurement error parameters. The challenge here is how to obtain an additional "term" which accounts for estimation of the measurement error parameters. This is addressed later in Section 6.17.3.

Early work on this approach includes Fuller (1987, Section 3.3.1), Carroll and Stefanski (1990) and Schafer (1990). Section 4.7 in Carroll et al. (2006) provides a broader overview and additional discussion and references. When the approximation for $E(Y|\mathbf{w})$ has more terms than the one given in (6.43) below they refer to this as *expanded regression calibration.* Liu and Liang (1992) also describe this procedure in some detail for generalized linear models with a normal structural model for the mismeasured X's with a normal additive measurement error model using equal numbers of replicates per individual with constant measurement error covariance matrix. They provide an expression for the asymptotic covariance of $\widehat{\boldsymbol{\beta}}$, in order to examine the role of the number of replicates, but we note that their equation (3) has an error in that it is missing a term; see Section 6.17.3.

Regression calibration approach.

A special case of the approach outlined above is based on the simple first order approximation

$$E(Y_i|\mathbf{w}_i) \approx m(h(\mathbf{w}_i, \boldsymbol{\xi}_i), \boldsymbol{\beta}), \tag{6.43}$$

where $h(\mathbf{w}_i, \boldsymbol{\xi}_i) = E(\mathbf{X}_i|\mathbf{w}_i)$ and $\boldsymbol{\xi}_i$ denotes the parameters in the Berkson error model. These need to first be estimated with other data. In many applications $\boldsymbol{\xi}_i$ is assumed the same for all i.

What has come to be known as **regression calibration** fits the model by regressing Y_i on

$$\widehat{\mathbf{x}}_i = h(\mathbf{w}_i, \widehat{\boldsymbol{\xi}}_i),$$

using whatever procedure would be used to fit without measurement error. This amounts to doing regression with imputed values for the missing true values. The resulting estimator of $\boldsymbol{\beta}$ will be denoted by $\widehat{\boldsymbol{\beta}}_{RC}$.

With generalized linear models, (6.43) determines $V(Y_i|\mathbf{w}_i)$, but in other models this may not be so. In those cases an **extended regression calibration** could be used, based on some approximate model for $V(Y_i|\mathbf{w}_i)$. This can offer some improvements (see some of the references in Chapter 8). Out of computational convenience, however, our illustrations use simple regression calibration based on direct use of imputed values.

Regression calibration has become fairly popular for its simplicity. Early work on regression calibration and quasi-likelihood methods include Gleser (1990), Whittemore (1989), Carroll and Stefanski (1990), Armstrong (1985), Rosner, Willett and Spiegelman (1989, 1990, 1992) and Whittemore and Keller (1988). Carroll et al. (2006) provide a comprehensive summary and further details and their Appendix B.3 provides a detailed computational formula for calculation of corrected standard errors when using regression calibration with replicates. They describe both a "sandwich/robust" approach and an "information" based approach, both of which are implemented in the rcal function in STATA.

- We already encountered regression calibration in Chapters 4 and 5 in cases where $\widehat{\boldsymbol{\beta}}_{RC}$ was equivalent to the moment based corrected estimators. If we assume the linear Berkson model with $\widehat{\lambda}_0$ and $\widehat{\boldsymbol{\Lambda}}_1$ the same for all individuals, and use $\hat{x}_i = \hat{\lambda}_0 + \hat{\Lambda}_1 W_i$, then the regression calibration estimator for the nonintercept terms is equivalent to the $\widehat{\boldsymbol{\beta}}_1$ in (6.40) obtained by linearly transforming the naive estimates. For that reason the linear adjustment in (6.40) is also sometimes referred to as regression calibration. In this case we can estimate the covariance matrix of $\widehat{\boldsymbol{\beta}}_1$ in the manner described following (6.40). See also Thurston et al. (2003).

- The regression calibration estimators are not necessarily consistent. They depend on random true values with the linear Berkson model holding as well as an approximation. They have, however, been found to be rather robust, even in functional cases.

6.11 Simulation extrapolation (SIMEX)

This very creative, simulation based, method to correct for measurement error was first introduced by Cook and Stefanski (1994) and Stefanski and Cook (1995). It was initially developed for the case with additive error, but it has been extended (e.g., Kuchenoff et al. (2006)) for use in misclassification problems. Chapter 5 of Carroll et al. (2006) provides an extensive and excellent discussion of the SIMEX technique including theoretical derivations, a discussion we certainly will not try to duplicate here. Instead this section describes just the basic workings of SIMEX, which will be implemented for various examples in later chapters. Among the advantages of the SIMEX are that it removes the need for approximations and it can be used in the functional cases.

Here, the description of the SIMEX method is limited to the case with additive error in the predictors but it can easily accommodate additive error in the response. As before with additive error in the predictors $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$ with $Cov(\mathbf{u}_i) = \boldsymbol{\Sigma}_{ui}$. We begin by describing the simplest form of SIMEX, with some modifications noted in the subsequent remarks.

To motivate the estimator define

$$\beta_j(\lambda) = \text{expected (or limiting value) of the naive estimator of } \beta_j \text{ if } Cov(\mathbf{u}_i) = (1+\lambda)\boldsymbol{\Sigma}_{ui} \text{ rather than } \boldsymbol{\Sigma}_{ui}.$$

Then the true value of the jth coefficient is $\beta_j = \beta_j(-1)$.

In short, the SIMEX method estimates $\beta_j(\lambda)$ for a collection of different values of λ, say $\lambda_1, \ldots, \lambda_M$. It then fits some model for $\hat{\beta}_j(\lambda)$ as a function of λ, say $g_j(\lambda)$ and then extrapolates back to -1 to obtain the SIMEX estimator of β_j.

More precisely, the implementation is as follows:

1. For each λ_m, generate:
$$\mathbf{W}_{bi}(\lambda_m) = \mathbf{W}_i + \lambda_m^{1/2}\mathbf{U}_{bi}, \tag{6.44}$$
for $b = 1, \dots B$ and $i = 1, \dots, n$, where B is a large number and the $\mathbf{U}_{bi}$ are i.i.d. with mean $\mathbf{0}$ and covariance $\widehat{\boldsymbol{\Sigma}}_{ui}$.
Since $\mathbf{W}_i$ already has covariance $\boldsymbol{\Sigma}_{ui}$, the generated $\mathbf{W}_{bi}$ would have exactly covariance $(1+\lambda_m)\boldsymbol{\Sigma}_{ui}$ if $\widehat{\boldsymbol{\Sigma}}_{ui} = \boldsymbol{\Sigma}_{ui}$. In practice, this is never exactly the case and with the use of an estimate of $\boldsymbol{\Sigma}_{ui}$, this covariance is only approximate. See the remarks for variations on this step.
2. Find $\hat{\boldsymbol{\beta}}(\lambda_m, b)$, which is the naive estimator of $\boldsymbol{\beta}$ based on $(Y_i, \mathbf{W}_{bi}), i = 1, \dots, n$ and define
$$\bar{\boldsymbol{\beta}}(\lambda_m) = \sum_b \hat{\boldsymbol{\beta}}(\lambda_m, b)/B.$$
3. For each j, fit a model $g_j(\lambda)$ for $\bar{\beta}_j(\lambda_m)$, the jth component of $\bar{\boldsymbol{\beta}}(\lambda_m)$, as a function of λ_m
4. Get the SIMEX estimator of β_j using
$$\hat{\beta}_{j,SIMEX} = g_j(-1).$$

Some remarks:

- We have used λ here to conform to usage in most of the SIMEX literature. This should not be confused with the use of λ's for the parameters in Berkson models.
- The choice of extrapolation function can be important. The two most popular choices are the so-called rational extrapolant $g(\lambda) = c_1 + (c_2/(c_3 + \lambda))$ and the quadratic extrapolant, $g(\lambda) = c_1 + c_2\lambda + c_3\lambda^2$, where the c's are some constants. These are both available in the SIMEX function in STATA (as is a linear extrapolant) with the quadratic being the default.
- When applied to multiple linear regression with the rational extrapolant, the SIMEX methods lead to method of moments estimators.
- The covariance matrix of the SIMEX estimators can be obtained via bootstrapping, when appropriate; see the comments in Sections 4.5.2 and 6.16. The covariance and standard errors can also be obtained via analytical expressions based on the use of estimating equations, but the method is complicated; Carroll et al. (2006) for details. Both types of standard errors are available in the SIMEX function in STATA.
- There are two ways that step 1 of the algorithm is modified. This is helpful since when an estimate $\widehat{\boldsymbol{\Sigma}}_{ui}$ is used in step 1, $Cov(\mathbf{W}_{bi})$ is only approximately $(1 + \lambda_m)\boldsymbol{\Sigma}_{ui}$.
The first modification is referred to as **empirical SIMEX**; see Devanarayan

and Stefanski (2002). It can be used when there are at least two replicates on each individual and a different per replicate covariance matrix is allowed per individual. In step 1 the term $\lambda_m^{1/2}\mathbf{U}_{bi}$ is replaced by a linear combination of the replicates on unit i with coefficients chosen so that $Cov(\mathbf{W}_{bi}) = (1+\lambda_m)\mathbf{\Sigma}_{ui}$. To do this, Step 1 is now

$$\mathbf{W}_{bi}(\lambda_m) = \mathbf{W}_i + (\lambda_m^{1/2}/m_i)^{1/2}\sum_{j=1}^{m_i} c_{bij}\mathbf{W}_{ij},$$

where m_i is the number of replicates of $\mathbf{W}$ on the ith observation, $Z_{bi1}, \ldots, Z_{bim_i}$ are generated independent $N(0,1)$ random variables and $c_{bij} = (Z_{bij} - \bar{Z}_{bi.})/(\sum_j (Z_{bij} - \bar{Z}_{bi.})^2)^{1/2}$.

Non-i.i.d. pseudo errors. The second modification alters how the $\mathbf{U}_{bi}$ are obtained. In step 1 above they were chosen to be independent and identically distributed. An alternate approach, which reduces variability of the SIMEX estimator due to the simulation step, is to choose $\mathbf{U}_{b1}, \ldots, \mathbf{U}_{bn}$ so that empirically they have mean 0 and variance $\widehat{\mathbf{\Sigma}}_{ui}$. That is, for a fixed b, $\sum_{i=1}^n \mathbf{U}_{bi} = \mathbf{0}$ and $\sum_{i=1}^n (\mathbf{U}_{bi} - 0)^2/n = \widehat{\mathbf{\Sigma}}_{ui}$. See Section 5.3.4.1 of Carroll et al. (2006) for details on the calculations.

6.12 Correcting using likelihood methods

This section provides an overview of likelihood based methods based on distributional assumptions. While these can be difficult to implement (and for that reason are used sparingly in the examples appearing later in the book) as in many other areas of statistics, they provide efficient methods of inference when the assumptions are met. The goal here is primarily to describe the different likelihoods and pseudo-likelihoods that can be used.

Recalling our earlier notation, $E(Y|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta})$, $V(Y|\mathbf{x}) = v(\mathbf{x}, \boldsymbol{\beta}, \boldsymbol{\sigma})$ and the true values are contained in $\mathbf{T}$ and the observed values in $\mathbf{O}$. There are many special cases that can be developed depending on which variables are measured with error, the nature of the measurement error, and what, if any, additional data are available. The likelihood formulation involves further specification of up to three different pieces: the model for Y given $\mathbf{x}$, the model for $\mathbf{X}$ in the structural setting and the measurement error model. Each of these are discussed in turn. As elsewhere, at the risk of some mathematical imprecision, densities or mass functions are denoted by $f()$ with the symbols in the arguments denoting which distribution is being specified.

I. Model for $Y|\mathbf{x}$. This is denoted using $Y|\mathbf{x} \sim f(y|\mathbf{x}, \boldsymbol{\omega})$, where $\sim$ denotes "is distributed as," $f(y|\mathbf{x}, \boldsymbol{\omega})$ denotes the density or probability mass function (henceforth referred to as simply the density or distribution) and $\boldsymbol{\omega} = (\boldsymbol{\beta}, \boldsymbol{\sigma})$ contains the parameters. It may be that $\boldsymbol{\omega}$ consists only of $\boldsymbol{\beta}$, without any additional variance parameters, as happens in binary and Poisson regression.

II. Model for $\mathbf{X}$'s. In general, we write

$$\mathbf{X}_i \sim f_{X_i}(\mathbf{x}; \boldsymbol{\omega}_X).$$

The $\boldsymbol{\omega}_X$ can be viewed as the collection of parameters associated with the joint distribution of $\mathbf{X}_1, \ldots, \mathbf{X}_n$. It may be that the parameters associated with the distribution of X_i change with i, so only part of $\boldsymbol{\omega}_X$ may be involved in the distribution of a particular X_i. If the X_i are i.i.d. then we write $f_X(\mathbf{x}; \boldsymbol{\omega}_X)$.

In the functional case with $\mathbf{X}_i = \mathbf{x}_i$ fixed then the parameters are the $\mathbf{x}$'s themselves, so $\boldsymbol{\omega} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$.

In the fully structural case with the $\mathbf{X}_i$ assumed i.i.d. $N(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_x)$, then $\boldsymbol{\omega}_X = (\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$.

If $\mathbf{X}$ consists of $\mathbf{X}_1$ and $\mathbf{X}_2$ where $\mathbf{X}_2$ is measured without error, then we could treat $\mathbf{x}_2$ as fixed and work with the conditional model; that is with $f(\mathbf{x}_1|\mathbf{x}_2, \boldsymbol{\omega}_X)$. This also could be indexed by i to allow for changes across observations.

Nonparametric formulation.

Here the distribution of X is not specified by a parametric family. In the structural case with a single predictor X, the assumption is that X simply has some unknown distribution, denoted by F_X. In this case the "parameter," $\boldsymbol{\omega}_X$, is the distribution F_X itself.

III. Measurement error or Berkson error model.

This piece specifies either

$f(\mathbf{o}|\mathbf{t}, \boldsymbol{\theta})$ = the density of $\mathbf{O}$ given $\mathbf{T} = \mathbf{t}$ (observed given truth)

or

$f(\mathbf{t}|\mathbf{o}, \boldsymbol{\xi})$ = density of $\mathbf{T}$ given $\mathbf{O} = \mathbf{o}$ (truth given observed).

The first is measurement error model, the second, a Berkson error model. Once again, the distribution might change with i. So, for example, $f(\mathbf{o}_i|t_i, \boldsymbol{\theta}_i)$ would denote the measurement error model if the parameters change with i.

Depending on which variables are and are not measured with error, there may be more structure in specifying these distributions. For example, with no error in Y and nondifferential measurement error, the models would involve just the distribution of $\mathbf{W}$ given $\mathbf{x}$, denoted $f(\mathbf{w}|\mathbf{x}, \boldsymbol{\theta})$. Or, if $\mathbf{x}_2$ is measured without error and $\mathbf{W}_1$ is the error prone measures for $\mathbf{x}_1$ the measurement error model would specify the distribution of $\mathbf{W}_1|\mathbf{x}_1, \mathbf{x}_2$; denoted $f(\mathbf{w}_1|\mathbf{x}_1, \mathbf{x}_2)$.

The parameters in the various models are summarized in the table below.

Model for	Parameters
Y given $\mathbf{x}$	$\boldsymbol{\omega} = (\boldsymbol{\beta}, \boldsymbol{\sigma})$
$\mathbf{X}$	$\boldsymbol{\omega}_X$
ME model for observed given true	$\boldsymbol{\theta}$
Berkson model for true given observed	$\boldsymbol{\xi}$

6.12.1 Likelihoods from the main data

This section provides the likelihood functions based on just the observed values. See the discussion in Section 6.4 where what constitutes the observed values in $\mathbf{O}$ is specified. Most importantly, if true values of some mismeasured variables are obtained only in some validation data, then they are not part of $\mathbf{O}$. In most applications $\mathbf{O} = (D, \mathbf{W})$, with the possibility that $D = Y$ (no error in response) or $\mathbf{W} = \mathbf{X}$ (no error in predictors).

Since this chapter assumes independence across observations, the likelihood function is the joint density of $\mathbf{O}_1, \ldots \mathbf{O}_n$ viewed as a function of the parameters involved; that is

$$L_O(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X) = \prod_{i=1}^{n} f(\mathbf{o}_i; \boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X), \tag{6.45}$$

where

$$f(\mathbf{o}_i; \boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X) = \int_y \int_{\mathbf{x}} f(\mathbf{o}_i|\mathbf{t}_i, \boldsymbol{\theta}_i) f(y|\mathbf{x}, \boldsymbol{\omega}) f_{X_i}(\mathbf{x}, \boldsymbol{\omega}_X) d\mathbf{x} dy. \tag{6.46}$$

This likelihood involves the measurement error parameters in $\boldsymbol{\theta}$, the parameters in $\boldsymbol{\omega}$ (from the regression model for the true values) and $\boldsymbol{\omega}_X$, parameters in the model for the $\mathbf{X}$'s. As noted earlier in the fully functional case $\boldsymbol{\omega}_X$ is the $\mathbf{x}_i$'s themselves.

We describe some special cases where there is no measurement error in the response.

No error in Y, nondifferential measurement error, functional case.

In the functional case the x_i's are fixed and treated as parameters. The likelihood function associated with $\{Y_i, \mathbf{W}_i, i = 1 \text{ to } n\}$ is

$$L_{YW}(\boldsymbol{\omega}, \boldsymbol{\theta}, \mathbf{x}_1, , \ldots \mathbf{x}_n) = \prod_i f(y_i|\mathbf{x}_i; \boldsymbol{\omega}) f(\mathbf{w}_i|\mathbf{x}_i, \boldsymbol{\theta}),$$

where the $\mathbf{x}_i$ are unknown parameters.

Usually all of the parameters are not identifiable and some restrictions on the parameters are needed in order to even maximize the likelihood. Surprisingly knowing the measurement error variance by itself may not work. For example in simple linear regression with just X measured with error and known measurement error variance σ_u^2, the likelihood is unbounded so maximum likelihood breaks down; see Cheng and Van Ness (1999, p. 24).

Fuller (Sections 2.3.1, 3.1.6 and 3.2.3) and Seber and Wild (1989, Chapter 10) discuss the functional approach for linear and nonlinear problems, respectively, under the assumption that $Y_i = m(\mathbf{x}_i, \boldsymbol{\beta}) + \epsilon_i$, $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$, with

$$\begin{bmatrix} \epsilon_i \\ \mathbf{u}_i \end{bmatrix} \sim N(\mathbf{0}, \sigma^2 \mathbf{D}_i).$$

The $\mathbf{D}_i$ is assumed known, often with $\mathbf{D}_i = \mathbf{D}$ for all i. If there is error in Y with $D_i = y_i + q_i$ then ϵ_i above can be replaced by $\epsilon_i + q_i$. If there is no error in the equation then ϵ_i above is q_i. See Amemiya and Fuller (1988) in particular for treatment of the case with no error in the equation. In any of these cases the assumption of knowing the matrix $\mathbf{D}$ up to a proportionality constant is a very strong one, and one that is hard to justify in practice. Under these assumptions it is often possible to maximize the likelihood, but even then obtaining inferences can be difficult. See, for example, the discussion in Seber and Wild (1989). Carroll and Stefanski (1985) discuss functional MLE's for logistic regression with normal additive measurement error.

Given the potential problems, functional MLE's are not discussed further in this book when there is error in the predictors. However in the case when the error is only in the response and the predictors are observable, a functional maximum likelihood approach can be used, as is done in parts of Chapter 10.

No error in Y, nondifferential measurement error, structural case.

Here the likelihood based on the observed values is

$$\begin{aligned} L_{YW}(\boldsymbol{\omega}, \boldsymbol{\theta}, \boldsymbol{\omega}_X) &= \prod_i f(y_i, \mathbf{w}_i; \boldsymbol{\omega}, \boldsymbol{\theta}, \boldsymbol{\omega}_X) \\ &= \prod_i \int_x f(y_i|\mathbf{x}; \boldsymbol{\omega}_i) f(\mathbf{w}_i|\mathbf{x}; \boldsymbol{\theta}) f_{X_i}(\mathbf{x}; \boldsymbol{\omega}_X) dx. \end{aligned} \tag{6.47}$$

When we refer to the structural setting, we are assuming that $\boldsymbol{\omega}_X$ is of smaller dimension than the $\mathbf{x}_1, \ldots \mathbf{x}_n$, so we are not in the functional case.

- If $\mathbf{X}' = (\mathbf{X}_1, \mathbf{X}_2)$ where $\mathbf{X}_1$ is measured with error and $\mathbf{X}_2$ is perfectly measured, then often it is more convenient to rewrite $f_X(\mathbf{x}; \omega_X)$ as $f(\mathbf{x}_1|\mathbf{x}_2, \boldsymbol{\omega}_{1|2}) f(\mathbf{x}_2; \boldsymbol{\omega}_2)$. Also, in this case we might want to condition on the perfectly measured $\mathbf{x}_2$ throughout and use $f(\mathbf{x}_1|\mathbf{x}_2, \boldsymbol{\omega}_{1|2})$ for the last term in the likelihood.
- As before, it is often necessary to put some restrictions on parameters or to assume some parameters are known in order to find the MLE's. In some cases, however, all of the parameters are identifiable without further assumptions. This happens in linear models with normal ϵ and normal additive measurement error as long as X_i is *NOT normally distributed* (Reirersol (1950)) and for logistic regression with normal structural model and normal additive error (Kuchenoff (1990)). The MLE's in these cases however have been found to not perform well and are not used.
- One might want to treat the distribution of X nonparametrically in which case $f_{X_i}(x; \omega_X)dx$ is replaced by $F_X(dx)$, where F_X specifies the common distribution and the integral is with respect to the associated measure. In this case we view $\boldsymbol{\omega}_X$ as the distribution F_X.

Structural case using the conditional model for $Y|w$.

We continue to assume there is no error in Y and the measurement error is nondifferential, but now condition on the observed $\mathbf{w}$'s and utilize the Berkson model for $\mathbf{X}|\mathbf{w}$. The resulting likelihood is

$$L_{Y|W}(\boldsymbol{\omega}, \boldsymbol{\xi}) = \prod_i f(y_i|\mathbf{w}_i) = \prod_i \int_{\mathbf{x}} f(y_i|\mathbf{x}; \boldsymbol{\omega}) f(\mathbf{x}|\mathbf{w}_i; \boldsymbol{\xi}) d\mathbf{x}. \tag{6.48}$$

By rewriting $f(\mathbf{w}_i|\mathbf{x}; \boldsymbol{\theta}) f_{X_i}(\mathbf{x}; \boldsymbol{\omega}_X) = f(\mathbf{x}|\mathbf{w}_i; \boldsymbol{\xi}) f_{W_i}(\mathbf{w}_i, \boldsymbol{\omega}_w)$, we also see that $L_{Y|W}(\boldsymbol{\omega}, \boldsymbol{\xi}) = L_{YW} / \prod_i f_{W_i}(\mathbf{w}_i, \boldsymbol{\omega}_w)$, where L_{YW} is the likelihood from (6.47).

Similar to the use of the likelihood L_{YW}, some assumptions about, or estimates of either $\boldsymbol{\theta}$ or $\boldsymbol{\xi}$ are usually needed in order to estimate $\boldsymbol{\beta}$.

6.12.2 Likelihood methods with validation data

The structural likelihood functions in (6.46), (6.47) and (6.48) are often not useful by themselves since the parameters are unidentifiable without additional assumptions or data. This and the next section show some of the ways in which to use additional data. Here we assume that there is validation data, either external or internal.

External validation data were described in Section 6.5.4. Depending on how the data were obtained we can work with either the measurement error or the Berkson error model. The corresponding likelihood is either

$$L_E(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X) = L_O(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X) L_2(\boldsymbol{\theta})$$

or

$$L_E(\xi, \boldsymbol{\omega}, \boldsymbol{\omega}_X) = L_O(\boldsymbol{\xi}, \boldsymbol{\omega}, \boldsymbol{\omega}_X) L_2(\boldsymbol{\xi}),$$

depending on which of the error models is used, with L_2 denoting the likelihood associated with the external data only.

This extra data could be used to obtain full maximum likelihood estimators, which maximizes $L_E(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X)$ over all parameters. While easy in principle the computations can become challenging. An easier approach is the use of *pseudo-maximum likelihood estimators* (PMLEs). For these we first estimate the measurement or Berkson error parameters from extra data and then we work just with the likelihood of the observed data with the measurement error parameters held fixed at their estimates. The pseudo-MLE's are often easier to compute than full maximum likelihood estimates. This is especially true when the induced model for $Y|\mathbf{w}$ is the same as the original model for $Y|\mathbf{x}$. There are settings where the pseudo maximum likelihood estimators are in fact the full maximum likelihood estimators. This was true when estimating proportions with misclassification and internal validation in Chapter 2.

Different versions of the pseudo MLE's are presented below.

- Allowing error in the response.
  Here we only describe the use of the measurement error parameters. The PMLE maximizes either $L_O(\widehat{\boldsymbol{\theta}}, \boldsymbol{\omega}, \boldsymbol{\omega}_X)$ or $L_O(\widehat{\boldsymbol{\theta}}, \boldsymbol{\omega}, \widehat{\boldsymbol{\omega}}_X)$, where the likelihood L_O is defined in (6.45). The first version uses just the estimated measurement error parameters $\widehat{\boldsymbol{\theta}}$ and maximizes over $\boldsymbol{\omega}$ and $\boldsymbol{\omega}_X$. The second also incorporates an estimate of the parameters in the marginal distribution of the $\mathbf{X}$'s. This would be formed using information about the measurement error model from the external data and the observed values in the main study.
- No error in the response.
  For the case with no error in the response, three options are described. The first makes use of the Berkson model and the other two, the measurement error model.
  1. PMLE I: Maximize $PL_1(\boldsymbol{\omega}) = L_{Y|W}(\boldsymbol{\omega}, \widehat{\boldsymbol{\xi}})$ over $\boldsymbol{\omega}$ with $\widehat{\boldsymbol{\xi}}$ fixed.
  2. PMLE II: Maximize $PL_2(\boldsymbol{\omega}, \boldsymbol{\omega}_X) = L_{YW}(\boldsymbol{\omega}, \widehat{\boldsymbol{\theta}}, \boldsymbol{\omega}_X)$ over $\boldsymbol{\omega}$ and $\boldsymbol{\omega}_X$ with $\widehat{\boldsymbol{\theta}}$ fixed.
  3. PMLE III: Maximize $PL_3(\boldsymbol{\omega}) = L_{YW}(\boldsymbol{\omega}, \widehat{\boldsymbol{\theta}}, \widehat{\boldsymbol{\omega}}_X)$ over $\boldsymbol{\omega}$ and $\boldsymbol{\omega}_X$ with $\widehat{\boldsymbol{\theta}}$ and $\widehat{\boldsymbol{\omega}}_X$ fixed.

- Error in the response only.
  With error in the response only, we can use the functional likelihood. This amounts to the use of method 3 above where $\widehat{\boldsymbol{\omega}}_X$ is the x values themselves. This approach is used in Sections 10.3 and 10.4.

Section 6.17.3 lays out a general method to obtain the approximate covariance matrix of pseudo-estimators, whether pseudo-moment or pseudo-MLE's. The result has the form of $\boldsymbol{\Sigma}_{\omega K} + \mathbf{V}$, where $\boldsymbol{\Sigma}_{\omega K}$ is the covariance if the measurement error parameters were known and $\mathbf{V}$ is an additional term which accounts for the uncertainty arising from estimation of the measurement error parameters. Section 6.17.4 provides further detail on the properties of pseudo and full MLEs.

Internal validation data was described in detail in Section 6.5.3. Combining the main data and the validation data, and suppressing parameters in some places, the full likelihood is

$$L_I = \prod_{i=1}^{n_V} f(\mathbf{o}_i, \mathbf{t}_{1i}) \prod_{n_V+1}^{n} f(\mathbf{o}_i). \tag{6.49}$$

Recall that $\mathbf{t}_1$ contains the portion of the true values that are never observed in the main study (but are observed in the validation data) and n_V is the size of the validation sample. More explicitly $f(\mathbf{o}_i)$ is $f(\mathbf{o}_i; \boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X)$ as given by (6.46). There are different ways to parameterize the likelihood depending on whether we work with the Berkson or the measurement error model.

Using the Berkson model for $\mathbf{T}_{1i}|\mathbf{o}_i$, which has density $f(\mathbf{t}_{1i}|\mathbf{o}_i; \boldsymbol{\xi})$, the likelihood is

$$L_I(\boldsymbol{\phi}, \boldsymbol{\xi}) = \prod_{i=1}^{n} f(\mathbf{o}_i; \boldsymbol{\phi}) \prod_{n_V+1}^{n} f(\mathbf{t}_{1i}|\mathbf{o}_i; \boldsymbol{\xi}),$$

where we have used $\boldsymbol{\phi}$ to index the parameters in the marginal distribution of $\mathbf{O}$. The $\boldsymbol{\phi}$ is some function of $\boldsymbol{\omega}$, $\boldsymbol{\omega}_X$ and $\boldsymbol{\xi}$. Often we can maximize the two pieces separately and then using the estimate of $\boldsymbol{\xi}$ and the relationship of $\boldsymbol{\phi}$ to the other parameters, solve to get the MLE of $\boldsymbol{\omega}$. This method was used in Sections 2.4.3 and 3.6 for misclassification problems.

With no error in the response and $\mathbf{W}$ measured in place of $\mathbf{x}$, then (omitting parameters in the notation), the likelihood is

$$L_I = \prod_{i=1}^{n_V} f(\mathbf{y}_i, \mathbf{w}_i, \mathbf{x}_i) \prod_{n_V+1}^{n} f(y_i, \mathbf{w}_i). \tag{6.50}$$

There are many ways to rewrite this depending on what assumptions are made.

For example, with conditional independence/surrogacy,

$$L_I = \prod_{i=1}^{n_V} f(y_i|\mathbf{x}_i;\boldsymbol{\omega}) f(\mathbf{w}_i, \mathbf{x}_i) \prod_{n_V+1}^{n} f(y_i, \mathbf{w}_i).$$

We can rewrite $f(\mathbf{w}_i, \mathbf{x}_i)$ and $f(y_i, \mathbf{w}_i)$, which depend on parameters that have been omitted in the notation, further depending on whether the measurement error or Berkson model is being used.

With internal validation data and random subsampling this is a missing data problem with true values missing at random on $n - n_V$ observations. So, standard missing data techniques (e.g., Little and Rubin(1987)) can be employed, but the measurement error may impose additional structure that should be accounted for.

6.12.3 Likelihood methods with replicate data

The definition of observed values used in the preceding discussion does not contain actual replicates but just the estimate of the true value based on means. To accommodate replication explicitly, $\mathbf{O}_i$ can be replaced by $\mathbf{O}_{i,rep}$, where the latter contains the replicates. For example with error in both the response and all predictors,

$$\mathbf{O}_{i,rep} = (\{D_{ik}, k = 1 \text{ to } m_i\}, \{\mathbf{W}_{ij}, j = 1, \ldots, n_i\}),$$

while with no error in the response

$$\mathbf{O}_{i,rep} = (Y_i, \{\mathbf{W}_{ij}, j = 1, \ldots, n_i\}).$$

If $\mathbf{X}_1$ is measured with error and $\mathbf{X}_2$ is not, then we can also write $\mathbf{O}_{i,rep} = (Y_i, \mathbf{X}_{i2}, \{\mathbf{W}_{i1j}, j = 1, \ldots, n_i\})$, where the $\mathbf{W}_{i1j}$ denote the replicate values for $\mathbf{X}_{i1}$. In these examples, for simplicity, we have assumed that there are an equal number of replicates, n_i on the ith observation for each predictor. This does not have to be the case as was discussed in Section 6.5.1.

The likelihood here is identical to that in (6.45) except that $\mathbf{O}_i$ is replaced by $\mathbf{O}_{i,rep}$, yielding

$$L_{REP}(\boldsymbol{\omega}, \boldsymbol{\theta}, \boldsymbol{\omega}_X) = \prod_i \int_y \int_{\mathbf{x}} f(\mathbf{o}_{i,rep}|y, \mathbf{x}; \boldsymbol{\theta}) f(y|\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\sigma}) f_{X_i}(\mathbf{x}; \boldsymbol{\omega}_X) d\mathbf{x} dy. \tag{6.51}$$

With *no error in the response assuming conditional independence holds*, the likelihood is

$$L_{REP}(\boldsymbol{\omega}, \boldsymbol{\theta}, \boldsymbol{\omega}_X) = \prod_i \int_{\mathbf{x}} f(y_i|\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\sigma}) f(\mathbf{w}_{i,rep}|\mathbf{x}; \boldsymbol{\theta}) f_{X_i}(\mathbf{x}; \boldsymbol{\omega}_X) d\mathbf{x}. \tag{6.52}$$

Schafer (1987, 1993), Schafer and Purdy (1996) and Rabe-Hesketh et al. (2003a) contain details on the full MLE approach with a normal structural model, normal additive measurement error with constant measurement error variances and covariances and with a generalized linear model for the true values.

Rabe-Hesketh et al. (2003a) condition on any perfectly measured predictors, say $\mathbf{x}_2$, and assume $\mathbf{X}_1|\mathbf{x}_2$ is distributed $N(\boldsymbol{\alpha}_0 + \boldsymbol{\alpha}_1'\mathbf{x}_2, \boldsymbol{\Gamma})$. Denoting the corresponding density by $f(\mathbf{x}_1|\mathbf{x}_2, \boldsymbol{\omega}_X)$, where $\boldsymbol{\omega}_X = (\boldsymbol{\alpha}_0, \boldsymbol{\alpha}_1, \boldsymbol{\Gamma})$, the likelihood becomes

$$L_{REP}(\boldsymbol{\omega}, \boldsymbol{\theta}, \boldsymbol{\omega}_X) = \prod_i \int_{\mathbf{x}_1} f(y_i|\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\sigma}) f(\mathbf{w}_{i,rep}|\mathbf{x}; \boldsymbol{\theta}) f_{\mathbf{X}_{i1}}(\mathbf{x}_1|\mathbf{x}_2, \boldsymbol{\omega}_X) d\mathbf{x}_1. \tag{6.53}$$

This model can be fit using the STATA CME command, but only for X_1 univariate.

Seber and Wild (1989, Section 10.3) discuss functional MLE's for nonlinear problems using replication.

Finally, we note that a semi-parametric approach can be used where the distribution of X is estimated nonparametrically. See, for example, Schafer (2001) and Rabe-Hesketh et al. (2003ab) and the related discussion in Section 10.2.3.

6.13 Modified estimating equation approaches

6.13.1 Introduction

As noted in the introductory section of this chapter, many estimators arise from solving a set of estimating equations, which in the absence of measurement error have mean zero, either exactly or in a limiting sense. This leads (under suitable regularity conditions) to consistent estimators for the parameters. With measurement error, modified estimating equation approaches attempt to create new estimating equations based on the available data ("observed values" plus any additional data for estimating measurement error parameters) which have mean zero in the general sense above. Chapter 7 of Carroll et al. (2006) provides an excellent and complete discussion of the use of two such methods, corrected scores and conditional scores. Here, we describe an approach we refer to here as the MEE (modified estimating equation) method (Buonaccorsi (1996a)) which is similar to, and in some cases identical to, the corrected score method. It is a general way to attack a variety of measurement error problems. It is treated here in the context of additive measurement error to simplify the presentation but it can be modified to handle nonadditive errors including misclassification and linear measurement error. In fact, many of the estimators in the preceding chapters can be viewed as coming from modifying estimating equations. For example, looking back to Chapter 2, with a single proportion the naive estimator of π can be viewed as the solution in π to $p_w - \pi = 0$. With misclassification, the expected value of $p_w - \pi$ is $\pi(\theta_{11} + \theta_{00} - 2) + 1 - \theta_{00}$. The modified estimating equation approach removes this bias and solves $p_w - \pi - (\pi(\widehat{\theta}_{11} + \widehat{\theta}_{00} - 2) + 1 - \widehat{\theta}_{00})$, which yields the corrected estimator for π in (2.3).

To describe the method in detail requires a bit of notation. We begin with a basic summary of the main features of the procedure. The reader more interested in applications can read this initial overview and skim (or even skip) the more detailed description. The method is illustrated in a number of the examples that appear in later chapters.

The method is essentially a pseudo type approach in that we first obtain estimated measurement error parameters and then use these in trying to modify the naive estimating equations to obtain new estimating equations. Some key features are:

- It handles the functional case.
- It allows unequal measurement error variances and covariances.
- It does not make distributional assumptions.
- When applied to a variety of previously studied problems (e.g., linear and quadratic regression, Poisson regression and logistic regression) the MEE method leads to estimators that have been motivated in other ways. The equivalence to moment corrected estimators in linear problems is discussed later in this section.
- Although the modified estimating equation appears difficult to work with, a key result is that:

  The MEE estimates can usually be obtained by iterative use of whatever method would be used without measurement error, where on each iteration the response vector is updated!

  This is an important consequence since the estimation can then be carried out with iterative use of standard fitting routines. Note that in contrast to regression calibration, it is the response that is modified (iteratively), as explained below. (In the sense of using and updated response, the algorithm used here is similar to a method decribed by Fuller (1987, p. 264) for handling general nonlinear functional models).
- In working with the modified estimating equations there are cases where exact solutions are available, but often some approximations may be needed. In such cases, the resulting estimators are not necessarily consistent. In nonlinear problems, this need for approximations, and the resulting lack of ex-

act consistency, is a problem with many correction techniques that try to be robust to distributional assumptions.

6.13.2 *Basic method and fitting algorithm*

We sketch the main ingredients of the method first. Let $\widehat{\boldsymbol{\theta}}_i$ denote the estimated measurement error parameters associated with the ith observation. With additive error these contain estimated variances and covariances. If common estimates are used then $\widehat{\boldsymbol{\theta}}_i = \widehat{\boldsymbol{\theta}}$ for all i. Let $\mathbf{z}_i = (\mathbf{o}_i, \widehat{\boldsymbol{\theta}}_i)$ contain the observed values and estimating measurement error parameters for the ith observation. As elsewhere the random observed values are denoted $\mathbf{O}_i$, while $\mathbf{o}_i$ denotes the realized values. Similar notation applies to other quantities.

Suppose with no measurement error the estimating equations are of the form $\sum_i S_i(\mathbf{t}_i, \boldsymbol{\beta}) = \mathbf{0}$ and so the naive estimating equations, using error-prone measures, are $\sum_i S_i(\mathbf{o}_i, \boldsymbol{\beta}) = \mathbf{0}$. The modified estimating equations are of the form

$$\sum_i \Psi_i(\mathbf{z}_i, \boldsymbol{\beta}) = \sum_i (S(\mathbf{o}_i, \boldsymbol{\beta}) - A_i(\mathbf{z}_i, \boldsymbol{\beta})) = \mathbf{0}, \tag{6.54}$$

where $A_i(\mathbf{Z}_i, \boldsymbol{\beta})$ is chosen to try and satisfy

$$E_{\boldsymbol{\beta}}(\sum_i \Psi_i(\mathbf{Z}_i, \boldsymbol{\beta})) = \mathbf{0}. \tag{6.55}$$

The estimator resulting from solving (6.54) is denoted $\widehat{\boldsymbol{\beta}}_{MEE}$. Note that the expectation in (6.55) is over random $\mathbf{O}_i$ and $\widehat{\boldsymbol{\theta}}_i$ and the $\boldsymbol{\beta}$ subscript to E is a reminder that the expectation is under the true value $\boldsymbol{\beta}$ and with the argument of the estimating equation set at $\boldsymbol{\beta}$. If the expectation equals $\mathbf{0}$ exactly then we have unbiased estimating equations. A weaker condition is to have $Lim_{n\to\infty}E_{\boldsymbol{\beta}}(\sum_i \Psi(\mathbf{O}_i, \widehat{\boldsymbol{\theta}}_i, \boldsymbol{\beta}))/n = \mathbf{0}$. Estimators arising from equations satisfying this condition are typically consistent. The challenge is to find the correction term A_i.

As earlier $\mathbf{t}_i = (y_i, x_i)$ is the collection of true values and from here on we assume, as is usually the case, that the observed values separate out as $\mathbf{O}_i = (D_i, \mathbf{W}_i)$, where D_i might be Y_i. We concentrate on the cases where, without measurement error, the estimating equations are of the form given in (6.6). These can be written as

$$\mathbf{Q}(\boldsymbol{\beta})'[\mathbf{D} - \mathbf{m}(\boldsymbol{\beta})] = \sum_i^n (y_i - m(\mathbf{x}_i, \boldsymbol{\beta}))\boldsymbol{\Delta}(\mathbf{x}_i, \boldsymbol{\beta}) = \sum_i^n S(\mathbf{o}_i, \boldsymbol{\beta}) = \mathbf{0},$$

where $\mathbf{Q}(\boldsymbol{\beta})' = [\Delta(\mathbf{W}_1, \boldsymbol{\beta}), \Delta(\mathbf{W}_2, \boldsymbol{\beta}), \ldots, \Delta(\mathbf{W}_n, \boldsymbol{\beta})]$,
$\mathbf{m}(\boldsymbol{\beta})' = (m(\mathbf{W}_1, \boldsymbol{\beta}), \ldots, m(\mathbf{W}_n, \boldsymbol{\beta}))$ and $\mathbf{D}' = (D_1, \ldots, D_n)$.

As noted in Section 6.3 this covers many important regression models. Notice that since $E(Y_i) = m(\mathbf{x}_i, \boldsymbol{\beta})$ the estimating equations have mean $\mathbf{0}$ when there is no measurement error.

Assume given $y_i, \mathbf{x}_i$, the measurement error is additive with

$$\mathbf{O}_i = \begin{bmatrix} D_i \\ \mathbf{W}_i \end{bmatrix} = \begin{bmatrix} y_i \\ \mathbf{x}_i \end{bmatrix} + \begin{bmatrix} q_i \\ \mathbf{u}_i \end{bmatrix}$$

with

$$Cov(\mathbf{O}_i | y_i, \mathbf{x}_i) = \begin{bmatrix} \sigma^2_{qi} & \boldsymbol{\Sigma}'_{uqi} \\ \boldsymbol{\Sigma}_{uqi} & \boldsymbol{\Sigma}_{ui} \end{bmatrix}.$$

Conditioning on the $\mathbf{x}_i$'s, the expected value of the naive estimating equations can be shown to be

$$E(S(\mathbf{O}_i, \boldsymbol{\beta})|\mathbf{x}_i) = C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}) = E[(q_i + m(\mathbf{x}_i, \boldsymbol{\beta}) - m(\mathbf{W}_i, \boldsymbol{\beta}))\Delta(\mathbf{W}_i, \boldsymbol{\beta})]. \tag{6.56}$$

This suggests one approach, namely choosing A_i so that

$$E(A_i(\mathbf{Z}_i, \boldsymbol{\beta})|\mathbf{x}_i) = C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}), \tag{6.57}$$

which if this held would imply (6.55).

A second approach, which also leads to satisfying (6.55), is to find a correction term to meet the stronger condition that

$$E(A_i(\mathbf{Z}_i, \boldsymbol{\beta})|\mathbf{x}_i, y_i) = F(\mathbf{t}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}), \tag{6.58}$$

where $E(S(\mathbf{O}_i, \boldsymbol{\beta})|\mathbf{x}_i, y_i) = F(\mathbf{t}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta})$. Notice that here we condition on y also. This condition leads to what Stefanski (1989) and Nakamura (1990) referred to as corrected scores. See also Chapter 7 of Carroll et al. (2006) for further discussion.

We return to the question of how to determine $\mathbf{A}$ shortly. Before going into that discussion, however, we describe how the modified estimating equations can often be solved for in a relatively straightforward way by iterative use of the naive method (which would apply without measurement error) but with updating of the responses. For notational convenience we drop the dependence of $\mathbf{A}$ and $\mathbf{A}_i$ on $\mathbf{Z}_i$ in describing the fitting algorithm.

1. Start with the naive estimator, denoted $\hat{\boldsymbol{\beta}}_{(1)}$.
2. After the kth step, compute $\mathbf{A}_{(k)} = \sum_i^n \mathbf{A}_i(\widehat{\boldsymbol{\beta}}_{(k)})$ and $\mathbf{Q}_{(k)} = \mathbf{Q}(\widehat{\boldsymbol{\beta}}_{(k)})$.
3. Compute an updated vector of responses,
$$\tilde{\mathbf{D}}_{(k)} = \mathbf{D} - \mathbf{Q}_{(k)}(\mathbf{Q}'_{(k)}\mathbf{Q}_{(k)})^{-1}\mathbf{A}_{(k)}. \tag{6.59}$$
4. Obtain $\hat{\boldsymbol{\beta}}_{(k+1)}$ using the naive method with the observed predictors, $\mathbf{W}$, but with responses given by $\tilde{\mathbf{D}}_{(k)}$. These are solving for $\boldsymbol{\beta}$ in the equations $\mathbf{Q}(\boldsymbol{\beta})'(\tilde{\mathbf{D}}_{(k)} - \mathbf{m}(\boldsymbol{\beta})) = \mathbf{0}$.

The algorithm is repeated until convergence is achieved, if possible. Just as in some problems with no measurement error there can be data for which there is no solution exists. In other cases, this algorithm can get caught oscillating among values. While this is easily addressed for a single data analysis by using some other numerical method of solving the estimating equations, it could be problematic if this method is imbedded in a bootstrapping routine.

6.13.3 Further details

We return to now the issue of finding an explicit expression for the correction term $\mathbf{A}$ needed above. Recall that we are trying to satisfy (6.57) with $C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta})$ given in (6.56). This can be written in various ways including

$$C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}) = \Omega_{\Delta qi} - \Omega_i - \Omega_{\Delta mi} \tag{6.60}$$

with $\Omega_{\Delta mi} = cov(\Delta(\mathbf{W}_i, \boldsymbol{\beta}), m(\mathbf{W}_i, \boldsymbol{\beta}))$,
$\Omega_{\Delta qi} = E[cov(\Delta(\mathbf{W}_i, \boldsymbol{\beta}), q_i)|Y_i]$ and $\Omega_i = \Delta(\mathbf{x}_i, \boldsymbol{\beta})G_i(\boldsymbol{\beta}) + G_i(\boldsymbol{\beta})B_i(\boldsymbol{\beta})$, where $G_i(\boldsymbol{\beta}) = E(m(\mathbf{W}_i, \boldsymbol{\beta})) - m(\mathbf{x}_i, \boldsymbol{\beta})$ and $B_i(\boldsymbol{\beta}) = E(\Delta_i(\mathbf{W}_i, \boldsymbol{\beta})) - \Delta(\mathbf{x}_i, \boldsymbol{\beta})$. These terms can depend in general on any of the true values, $\boldsymbol{\beta}$, and the measurement error parameters. There can be closed form expression (as in multiple linear regression, as described below) but usually some approximations are needed. This suggests the use of

$$\mathbf{A}_i(\widehat{\boldsymbol{\beta}}_{(k)}) = \widehat{\boldsymbol{\Omega}}_{\Delta qi} - \widehat{\boldsymbol{\Omega}}_i - \widehat{\boldsymbol{\Omega}}_{\Delta mi} \tag{6.61}$$

to go into computing the correction term $\mathbf{A}_{(k)}$ for use in the iterative scheme.

Consider multiple linear regression with unweighted least squares. This corresponds to use of estimating equations with $S(y_i, \mathbf{x}_i^*, \boldsymbol{\beta}) = (y_i - \mathbf{x}_i'\boldsymbol{\beta})\mathbf{x}_i^*$, where $\mathbf{x}_i^*$ includes a 1 for an intercept if needed. This leads to $G_i(\boldsymbol{\beta}) = 0$, $B_i(\boldsymbol{\beta}) = \mathbf{0}$, $\boldsymbol{\Omega}_{\Delta qi} = \boldsymbol{\Sigma}_{uqi}$ and $\boldsymbol{\Omega}_{\Delta mi} = \Sigma_{ui}\boldsymbol{\beta}$. Using the notation in (5.6) this leads to modified estimating equations $\sum_i \{D_i \mathbf{w}_i^* - \widehat{\boldsymbol{\Sigma}}^*_{uqi} - (\mathbf{w}_i^* \mathbf{w}_i^{*'} - \widehat{\boldsymbol{\Sigma}}^*_{ui})\boldsymbol{\beta}\} = \mathbf{0}$ and exactly the moment correct estimator given in (5.10). This also agrees with estimators proposed in Freedman et. al. (1991) and Wang (1993).

In general the correction term in (6.61) can be estimated analytically using exact or approximate expressions for the terms on the right side of (6.60). This can be problematic as it can involve approximations and then possibly estimating some function of the true values. Another strategy is to use a simulation based approach. At the kth step this would use $\mathbf{A}_i(\widehat{\boldsymbol{\beta}}_{(k)}) = \sum_{r=1}^R \widehat{C}_{ir}/R$, where R is a large number and

$$\widehat{C}_{ir} = (q_{ir}^* + m(\mathbf{w}_i, \widehat{\boldsymbol{\beta}}_{(k)}) - m(\widehat{\mathbf{W}}_{ir}, \widehat{\boldsymbol{\beta}}_{(k)}))\Delta(\widehat{\mathbf{W}}_{ir}, \widehat{\boldsymbol{\beta}}_{(k)}),$$

and $\widehat{\mathbf{W}}_{ir} = \mathbf{w}_i + \mathbf{u}_{ir}^*$ with $(q_{ir}^*, \mathbf{u}_{ir}^*)$ generate to have mean $\mathbf{0}$ and covariance matrix

$$\begin{bmatrix} \widehat{\sigma}_{qi}^2 & \widehat{\boldsymbol{\Sigma}}_{uqi}' \\ \widehat{\boldsymbol{\Sigma}}_{uqi}' & \widehat{\boldsymbol{\Sigma}}_{ui} \end{bmatrix}.$$

The performance of this simulation based strategy has not been fully investigated, but it is illustrated in an example in Section 8.4.3.

A second correction term to consider is based on equation (6.58) and leads to $\mathbf{A}_i(\widehat{\boldsymbol{\beta}}_{(k)}) = \widehat{R}_i(\widehat{\boldsymbol{\beta}}_{(k)}) + \widehat{\boldsymbol{\Omega}}_{\Delta qi} - \widehat{\boldsymbol{\Omega}}_i - \widehat{\boldsymbol{\Omega}}_{\Delta mi}$, where $\widehat{R}_i(\boldsymbol{\beta}) = (D_i - m(\mathbf{W}_i, \boldsymbol{\beta}))\widehat{\mathbf{B}}_i(\boldsymbol{\beta})$. Of course with $\mathbf{B}_i(\boldsymbol{\beta}) = \mathbf{0}$ as occurs for linear and generalized linear models, this gives the same correction as the first approach.

Assuming the estimating equations have mean zero using the general theory of estimating equations, "under suitable conditions," $\widehat{\boldsymbol{\beta}}_{MEE}$ is consistent and asymptotically normal with approximate covariance matrix

$$\Sigma_{\beta,MEE} = I(\boldsymbol{\beta})^{-1} H(\boldsymbol{\beta}) I(\boldsymbol{\beta})^{'-1} \tag{6.62}$$

where $I(\boldsymbol{\beta}) = \sum_i E\{\partial \Psi_i(\mathbf{Z}_i, \boldsymbol{\beta})/\partial \boldsymbol{\beta}\}$, and $H(\boldsymbol{\beta}) = \sum_i cov\{\Psi_i(\mathbf{Z}_i, \boldsymbol{\beta})\}$. With $\widehat{\boldsymbol{\theta}}_i$ available for each i, the $\mathbf{Z}_i$ are independent and a robust estimator of Σ_β is

$$\hat{\Sigma}_{\beta,MEE} = \hat{\mathbf{I}}^{-1} \sum_i \Psi_i(\mathbf{Z}_i, \hat{\boldsymbol{\beta}}) \Psi_i(\mathbf{Z}_i, \hat{\boldsymbol{\beta}})' \hat{\mathbf{I}}^{-1}, \tag{6.63}$$

where $\hat{\mathbf{I}} = \sum_i \partial \Psi_i(\mathbf{Z}_i, \boldsymbol{\beta})/\partial \boldsymbol{\beta} \mid_{\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}}$. This can be rewritten further and then simplified when additional assumptions are made (e.g., normal measurement error); see Nakumaru (1990) and Buonaccorsi (1996a) for some details.

When there are common measurement error parameters getting used for each i, then the terms in the equations for different i are not independent. In this case modifications are needed in order to estimate the covariance of $\widehat{\boldsymbol{\beta}}$. In principle this can be done using the methods below in Section 6.17.3, although the implementation is nontrivial. One solution is to use bootstrap techniques, which is done in many of the later applications.

6.14 Correcting for misclassification

Our emphasis in the later chapters is on errors in quantitative predictors. This is mainly due to the need to limit the topics but also because the mismeasurement of quantitative predictors is, in general, the more pressing problem. So, only some broad comments are made here. Correcting for misclassification was treated in Chapters 2 and 3 in purely categorical contexts and Section 6.7.7 provided an in depth look at bias induced by misclassification of a predictor in linear regression in the presence of other perfectly measured predictors.

With only misclassified predictors, a number of the strategies described earlier in this chapter, and illustrated in later chapters, still work. Bias can be assessed by working directly with the estimators if they have closed form, by working with an induced model, if available, or by examining the expected value of the naive estimating equations. To correct for measurement error we can consider direct bias corrections, modified estimating equations and corrected scores (which work from the naive estimating equations), regression calibration (where now the conditional expected values for the true values are conditional probabilities), maximum likelihood and the misclassification SIMEX (Kuchenoff et al. (2006)). Some of these are discussed in more detail in Section 8.6 for a model with one misclassified predictor and a quantitative response.

The problem is more difficult with a combination of misclassified categorical and mismeasured quantitative predictors. In this case measurement error (or similarly Berkson error) models need to be built for the joint behavior of the mismeasured categorical and quantitative variables given the true values. This is easy if the different measured quantities are conditionally independent (given the true value) but is more challenging otherwise. Spiegelman et al. (2000) approach this through the use of a hierarchical model and attack getting corrected estimates through the use of maximum likelihood. More generally once a joint model for the measured variables is specified we can use the techniques listed above for bias assessment and correcting. Similar remarks apply if there is misclassification of the response also (or by itself) in the binary regression context. A couple of brief points on the issues here.

- If a regression calibration approach is going to be taken by using an approximate induced model, see equation (6.43), then all that is needed is $E(X_j|\mathbf{w})$ where X_j is one of the mismeasured values (it may be quantitative or a dummy variable associated with a categorical) and $\mathbf{w}$ is the vector of observed values. With validation data this is pretty sraightforward since it just means building a regression model for each X_j as a function of the elements in $\mathbf{W}$. The joint covariance structure associated with the conditional distribution of $\mathbf{X}|\mathbf{w}$ does not enter into finding the estimators. We are faced with a different set of issues, however, if the extra information on mismeasured variables is through replication.
- The modified estimating equation approach (closely connected to the corrected score method) involves finding the expected values of the functions of the mismeasured variables that enter into the naive estimating equations. As was seen in Section 6.13, even when dealing with additive error in quantitative variables in multiple linear regression, the variances of the measurement errors as well as their covariances came into play. In general, we will need information about the measurement error variances and covariances.

This gets trickier to work with a combination of quantitative and categorical predictors.

Shieh (2009) provides a fairly general overview on correcting for misclassification in regression problems and explores the use of corrected score and other approaches with misclassified predictors in linear regression. Other references that provide access to the literature include Akazawa et al. (1998), Davidov et al. (2003), van den Hout and Kooiman (2006), Veierød and Laake (2001), Wang et al. (2008), Yucel and Zaslasky (2005) and White et al. (2001).

6.15 Overview on use of validation data

The use of validation data has been mentioned in a number of places throughout this chapter. Here we summarize the strategies for using validation data to correct for measurement error. These are not the only possible methods, as noted in the opening to this chapter.

6.15.1 Using external validation data

The layout for external validation data along with some broad comments appear in Section 6.5.4. The primary approach with external data is to first use it to fit either a Berkson or classical measurement error model (the choice between the two is discussed in Section 6.5.4) and then use the estimated parameters to correct for measurement error using the main data. These pseudo approaches are surveyed below. One could also consider full likelihood or moment approaches that work simultaneously with the main and external data; see Section 6.12.2 for example. However, these full techniques are typically harder to implement and the gains over pseudo approaches may be modest. This is not to say that they should be totally discounted.

- Using a measurement error model.
  Here the external validation data is used to fit a model for the observed values on the true values. If the measurement error model is nondifferential with respect to the response or other perfectly measured predictors (in $\mathbf{x}_2$), then this would involve just fitting the observed $\mathbf{w}_1$ to the missing true values in $\mathbf{x}_1$, and y and $\mathbf{x}_2$ are not needed in the external validation data. If the measurement error model is differential with respect to either the response or $\mathbf{x}_2$ then the values involved must also be part of the validation data. The estimated measurement error parameters are denoted by $\widehat{\boldsymbol{\theta}}$. These can be used in different ways.

  1. Corrected naive estimates.

This approach uses an expression for the bias of the naive estimator and the estimated measurement error parameters to form the correction.

2. Pseudo maximum likelihood.

   This can be used in the structural setting. It is not generally viable in the functional case. There are actually two versions here. The first maximizes the joint pseudo-likelihood, $L_{YW}(\boldsymbol{\omega}, \widehat{\boldsymbol{\theta}}, \boldsymbol{\omega}_X)$, given in (6.47) over $\boldsymbol{\omega}$ (the parameters in the model for $Y|\mathbf{x}$) and $\boldsymbol{\omega}_X$ (the parameters in the structural model for $\mathbf{X}$). The second estimates the parameters in the structural model for X first, say using $\widehat{\boldsymbol{\omega}}_X$, and then maximizes $L_{YW}(\boldsymbol{\omega}, \boldsymbol{\theta}, \widehat{\boldsymbol{\omega}}_X)$ over just $\boldsymbol{\theta}$. With a parametric model for $\mathbf{X}$, such as assuming the $\mathbf{X}_i$ are i.i.d. $N(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$, either of these methods is straightforward in principle, although not always easily implemented. There are more challenges in applying these methods when the model for $\mathbf{X}$ is nonparametric.

3. Modified Estimating Equations.

   Section 6.13 described a modified estimating equation approach under additive error, but as noted there it can be extended to handle nonadditive errors. In general $\boldsymbol{\theta}$ involves any parameters involved in the error model for observed given truth. The corrected equations are of the same general form as in (6.54), but the nonadditivity will change the approximations and the nature of the correction term involved.

- Using a Berkson model.

  We first fit a regression of the true values on the observed values using the external data. The parameters in the resulting Berkson model are denoted by $\boldsymbol{\xi}$ and the estimates, from the validation data, by $\widehat{\boldsymbol{\xi}}$. A linear model is most frequently used, but that is by no means required. The resulting model can be used in regression calibration, pseudo-maximum likelihood estimation and other correction techniques.

  1. Regression calibration.

     Use the resulting fit to get imputed values for the unobserved functions of the predictors and then run the main regression model of interest using these imputed values. With some perfectly measured predictors (in $\mathbf{x}_2$), parts of $\mathbf{x}$ are known exactly and we only need to impute the unobserved $\mathbf{x}_1$. In this case, the fit in the validation data will come from regressing $\mathbf{x}_1$ on $\mathbf{w}$, where $\mathbf{w}$ consists of $\mathbf{w}_1$ (those quantities observed in place of $\mathbf{x}_1$) and $\mathbf{x}_2$. This approach requires that the external validation data contain $\mathbf{x}_2$.

     If a linear Berkson model is used, with common coefficients for all observation, for a number of models the RC fit is equivalent to linearly transforming the naive estimates; see Section 6.9.1.

     One can also utilize the external estimates in quasi-likelihood/expanded regression calibration methods; see Section 6.10.

2. Pseudo maximum-likelihood.

   This approach treats the Berkson parameters as known and maximizes the likelihood for the induced model; that is, maximizes $L_{Y|W}(\omega, \widehat{\xi})$, where $L_{Y|W}$ is defined in (6.48) and $\boldsymbol{\omega}$ contains $\boldsymbol{\beta}$ and any variance parameters in $\boldsymbol{\sigma}$. This may require customized programming, but otherwise standard approaches to likelihood problems can be used, including methods to obtain solutions and to estimate the covariance matrix if the estimated Berkson parameters are treated as known.

The standard errors and associated inferences coming directly from the analyses above are usually too small since they do not account for uncertainty in the estimated "measurement error" parameters. This is not much of an issue with large external validation studies. For a discussion on accounting for this additional uncertainty analytically, see Sections 6.9.1, 6.17.2 and 6.17.3 plus the comments near the end of Section 6.10. Bootstrap methods are addressed in Section 6.16.

Further details and examples using external calibration appear in Sections 7.3.1, 7.3.3, 8.5.2, 8.5.3, 10.3 and 10.4.

6.15.2 Using internal validation data

The general layout for internal validation data is given in Section 6.5.3. Recall that the validation data are of size n_V. As usual $\widehat{\boldsymbol{\theta}}$ and $\widehat{\boldsymbol{\xi}}$ broadly denote estimated measurement or Berkson error parameters, respectively, obtained from the validation data. Generally the best strategy is to make use of the Berkson model, which can be fit whether the validation sample is chosen at random, or is chosen in a two-phase fashion conditional on the observed values. (We assume throughout this section that the sampling is one of those two designs.) There are cases, however, where there are restrictions on the measurement error model and it may be better to use the fitted measurement error model since the fitted Berkson model does not necessarily capture the restrictions.

Here are some of the options:

1. Use $\widehat{\boldsymbol{\beta}}_V$ = estimator of β using the complete cases.
   Since we have the true versions of the mismeasured values on the n_V validation cases, this uses standard methods as long as the validation data is a random sample. It can also be used for the nonintercept coefficients in logistic regression for case-control studies; see the discussion in Section 7.1. For further comment on general two-phase studies see the comments at the end of this section. Since n_V is often small relative to n, this is generally inefficient since it makes no use of the other data.

2. Use the estimated measurement error or Berkson error parameters from the validation data and then correct $\widehat{\boldsymbol{\beta}}_{naive}$, the naive fit from all the data, as is done with external validation data. Under likelihood models these are pseudo MLEs. Finding standard errors and associated inferences can be complicated (see Section 6.17.3) since some of the same data is involved in estimating the measurement error parameters and in the original naive estimate.
3. Use the estimated measurement error or Berkson error parameters from the validation data and correct $\widehat{\boldsymbol{\beta}}_{naive,I}$ to obtain a corrected estimator $\widehat{\boldsymbol{\beta}}_c$, where $\widehat{\boldsymbol{\beta}}_{naive,I}$ is the naive estimator of $\boldsymbol{\beta}$ from the $n - n_V$ "incomplete" cases.
Notice that $\widehat{\boldsymbol{\beta}}_c$ comes from simply treating the validation data as external and this estimator is also inefficient when the validations data are internal. A better strategy is to use the weighted estimator
$$\widehat{\boldsymbol{\beta}}_W = \mathbf{A}_1\widehat{\boldsymbol{\beta}}_V + \mathbf{A}_2\widehat{\boldsymbol{\beta}}_c,$$
where $\mathbf{A}_1$ and $\mathbf{A}_2$ are weight matrices with $\mathbf{A}_1 + \mathbf{A}_2 = \mathbf{I}$. One option is to use $\mathbf{A}_1 = (\widehat{\boldsymbol{\Sigma}}_V^{-1} + \widehat{\boldsymbol{\Sigma}}_c^{-1})^{-1}\widehat{\boldsymbol{\Sigma}}_V^{-1}$ and $\mathbf{A}_2 = (\widehat{\boldsymbol{\Sigma}}_V^{-1} + \widehat{\boldsymbol{\Sigma}}_c^{-1})^{-1}\widehat{\boldsymbol{\Sigma}}_c^{-1}$, where $\widehat{\boldsymbol{\Sigma}}_V$ and $\widehat{\boldsymbol{\Sigma}}_c$ are estimates of $Cov(\widehat{\boldsymbol{\beta}}_V)$ and $Cov(\widehat{\boldsymbol{\beta}}_c)$, respectively. With $\widehat{\boldsymbol{\beta}}_V$ and $\widehat{\boldsymbol{\beta}}_c$ being asymptotically uncorrelated, an estimate of the approximate covariance of $\widehat{\boldsymbol{\beta}}_W$ is given by $[\widehat{\boldsymbol{\Sigma}}_V^{-1} + \widehat{\boldsymbol{\Sigma}}_c^{-1}]^{-1}$. See Spiegelman et al. (2001) for details in the case of logistic regression.
4. Regression calibration. With no measurement error in Y and nondifferential measurement error, we can use regression calibration by regressing Y_i on $\widehat{\mathbf{X}}_i$, where $\widehat{\mathbf{X}}_i = \mathbf{X}_i$ for the validated cases and $\widehat{\mathbf{X}}_i$ is an imputed value based on the estimated Berkson model for the nonvalidated cases, i.e., for $i \geq n_V + 1$. One advantage of this method is that it is easy to implement. It generally requires that the validation data are a random sample of the main study units in order to justify simply using Y_i and $\mathbf{x}_i$ in the usual way for the validated cases.
5. Modified estimating equations.
Suppose the estimating equations in terms of true values are $\sum_i(S(\mathbf{t}_i, \boldsymbol{\beta}) = \mathbf{0}$. We adopt the approach of Section 6.13 of modifying the estimating equations using estimated measurement error parameters, but we only need to modify the $n - n_V$ cases for which we do not have all of the true values. The contribution from the n_V observations with true values will just use the "score" value in terms of the true values. Hence, the modified estimating equations are
$$\sum_{i=1}^{n_V}(S(\mathbf{t}_i, \boldsymbol{\beta}) + \sum_{i=n_V+1}^{n} (S(\mathbf{o}_i, \boldsymbol{\beta}) - A_i(\mathbf{Z}_i, \boldsymbol{\beta})) = \mathbf{0}, \tag{6.64}$$

where as in Section 6.13 $\mathbf{A}_i$ is a correction term and $\mathbf{Z}_i$ contains o_i and $\widehat{\theta}_i$. This assumes the validation data is a random sub-sample.

The regression calibration approach above is really a special case of this where the modification for the incomplete cases just involves replacing the observed w_i with $\widehat{\mathbf{X}}_i$.

6. Backsolve using the Berkson model. Here we use the relationship among the properties of the true values, the properties of the observed values and the Berkson model to express the parameters in the model for $\mathbf{T}$ as a function of the parameters for $\mathbf{O}$ and $\mathbf{T}|\mathbf{o}$. Using the data we get $\widehat{\xi}$ from the validation data and separately estimate the parameters associated with the distribution of the observed values using all of the data. We can then solve to get an estimate of the parameters for the true values.

   Method 6 was used in Section 2.4.3 in estimating a single proportion and in Section 3.6 in correcting in two by two tables. Its use in the case of linear measurement error is described in Section 8.5.4; see (8.32) in particular. It is one of a number of methods used in the example in Section 8.5.5. Some of the other methods are illustrated in the context of logistic regression in Section 7.3.2.

There are other approaches including use of a full maximum likelihood approach. In many cases, method 6 above will produce the full MLEs. As noted above, we have assumed that the validation data comes from a random subsample or a two-phase design conditional on the observed values. Methods 2 and 6 can be used in either case. If the sampling is two-phase then the dependence of the second stage on the observed mismeasured values means that $\widehat{\boldsymbol{\beta}}_V$ in method 1, which is also used in method 3, may need modification, as will part of the estimating equations corresponding to the validation data in method 5. See Breslow and Holubkov (1997), Schill et al. (1993) and Spiegelman et al. (2001) for some discussion of this issue in the context of logistic regression. The first two also discuss other approaches.

6.16 Bootstrapping

This section provides a general discussion on bootstrap strategies for obtaining standard errors, estimates of bias and confidence intervals. Some of the issues related to bootstrapping were touched on in Chapters 2–4. The discussion below is fairly general, with implementation of the techniques illustrated in many of the examples in later chapters. Bootstrapping with additive error is addressed in Section 6.16.1 while bootstrapping with validation data is discussed in Section 6.16.2. The description below concentrates on describing the resampling with the assumption that the reader has some familiarity with the bootstrapping. Good general references include Efron and Tibshirani (1993), Davison and Hinkley (1997) and Manly (2000). We do note that the bootstrap standard error is defined to be the sample standard deviation of the estimates over the bootstrap samples and that the bootstrap confidence intervals used in our examples are simple percentile intervals. These use the appropriate percentiles (e.g., 2.5th and 97.5th percentiles for a 95% confidence interval) of the bootstrap estimates. There are other, more sophisticated bootstrap intervals, such as bias-corrected interval, that can be considered. See the aforementioned references.

6.16.1 Additive error

There are two bootstrapping strategies that can be employed, referred to here as one and two-stage bootstrapping. There are advantages and disadvantages to each. In general two-stage bootstrapping is preferable, because of the ability to assess bias, but it requires more model specifications than are needed for the one-stage bootstrap. Given the extra structure needed to implement the two-stage bootstrap, a reasonable strategy may be to first use it to get a general assessment of the bias. Then, if the bias is negligible and *assuming the one-stage bootstrap is applicable* use the one-stage bootstrap, which is more robust, to get standard errors, etc.

One-stage bootstrapping.

In each bootstrap sample, the resampling is simply sampling with replacement from the observations, where an observation carries with it y_i, $\mathbf{w}_i$ and the estimated measurement error variances and covariances. This method was employed in simple and multiple linear regression; see Sections 4.5.2 and 5.4.2.

This method is formally justified with an overall random sample and with a common technique used to obtain the mismeasured values and estimated measurement error parameters for each observation, although the per replicate variances and covariances can change with the true values. When applicable, this method is generally easy to implement, but a disadvantage is that it usually does not provide a bootstrap estimate of bias. This is since the resampling is not done under a regression model with a fixed set of coefficients so there is nothing to compare the mean of the bootstrap samples to in order to assess bias. In the bootstrap parlance, the problem is that the estimators used are not plug-in estimators; see Efron and Tibshirani (1993).

Two-stage bootstrapping.

Here the bootstrap is implemented by explicitly mimicking both steps leading to the data. That is we generate a response from a regression model as well as generate mismeasured predictors. This is similar to the parametric bootstrap for the functional case described in Section 4.5.2 for simple linear regression. In the linear regression setting, employing a nonparametric bootstrap in the functional setting was problematic since we needed to know, or have a nonparametric estimate of, the distribution of the error in the equation. This same problem arises in nonlinear problems if the error distribution is not just a function of the coefficients. It does not occur, however, when the response has a distribution that is completely determined by $\boldsymbol{\beta}$. This is true for binary and Poisson regression and some other generalized linear models.

We describe the method first assuming that the measurement errors are normally distributed and there is replication. For the bth bootstrap sample, generate $\{Y_{bi}, W_{bij}, j = 1 \text{ to } k_i\}$, where

$$Y_{bi} = m(\widehat{\boldsymbol{\beta}}, \widehat{\mathbf{x}}_i) + e_{bi}$$

and

$$\mathbf{W}_{bij} = \widehat{\mathbf{x}}_i + \mathbf{U}_{bij} \tag{6.65}$$

where the $\mathbf{U}_{bij}$ are i.i.d. $N(\mathbf{0}, k_i \widehat{\boldsymbol{\Sigma}}_{ui})$ and k_i is the number of replicates for the ith observation. Note that each replicate has covariance $k_i \widehat{\boldsymbol{\Sigma}}_{ui}$ so that $\mathbf{W}_{bi} = \sum_j \mathbf{W}_{ij}/k_i$ had covariance $\widehat{\boldsymbol{\Sigma}}_{ui}$. The bootstrap error e_{bi} used to generate the response depends on the particular model assumptions. As noted earlier, it may be difficult to carry out this piece nonparametrically. In other cases, the generation of Y_{bi} can be expressed in an easier fashion. For example, in binary regression Y_{bi} takes on a 0 or 1 with $P(Y_{bi} = 1) = m(\widehat{\boldsymbol{\beta}}, \widehat{x}_i)$.

For $\widehat{\mathbf{x}}_i$, one option is to simply use $\mathbf{W}_i$. This avoids any need to appeal to a structural formulation. With random true values there are other choices for $\widehat{\mathbf{x}}_i$. These include using an estimate of $E(\mathbf{X}_i|\mathbf{w}_i)$ as in regression calibration, or resampling from an estimate of the distribution of $\mathbf{X}_i$. The choice of $\widehat{\mathbf{x}}_i$'s can possibly have an impact on the bootstrap standard errors.

In general, we need to generate $\mathbf{W}_{bi}$ and $\widehat{\boldsymbol{\Sigma}}_{bui}$. With replication these are formed using the replicates $\mathbf{W}_{bij}$. Without replication we can generate $\mathbf{W}_{bi}$ and $\widehat{\boldsymbol{\Sigma}}_{bui}$ based on how the original $\mathbf{W}_{bi}$ and $\widehat{\boldsymbol{\Sigma}}_{ui}$ were obtained. That information is often not available, however. In that case we would generate $\mathbf{W}_{bi} = \widehat{\mathbf{x}}_i + \mathbf{U}_{bi}$ directly, where $\mathbf{U}_{bij}$ has mean $\mathbf{0}$ and variance $\widehat{\boldsymbol{\Sigma}}_{ui}$, and either treat the $\widehat{\boldsymbol{\Sigma}}_{ui}$ as known (i.e., set each $\widehat{\boldsymbol{\Sigma}}_{bui}$ equal to $\widehat{\boldsymbol{\Sigma}}_{ui}$) or generate $\widehat{\boldsymbol{\Sigma}}_{bui}$ based on some assumption about its precision.

The two-stage bootstrap has the advantage of providing a bootstrap estimate of bias. If $\widehat{\boldsymbol{\beta}}_b$ is the estimate for the bth bootstrap sample and $\widehat{\boldsymbol{\beta}}$ the original estimate then the bootstrap estimate of bias is

$$Bias(boot) = \frac{\sum_{b=1}^{B} \widehat{\boldsymbol{\beta}}_b}{B} - \widehat{\boldsymbol{\beta}}.$$

Bias estimates for other functions of the parameters can be computed in similar fashion. This works here since we are explicitly resampling from a model with parameter $\widehat{\boldsymbol{\beta}}$.

There are a number of variations on (6.65) that can be considered. As given, (6.65) assumes normality plus that there is joint replication on all of the mismeasured covariates. It does allow some of the variables to be perfectly measured in which case components of $\widehat{\mathbf{x}}_i$ equal a perfectly measured predictor and the corresponding components of the $\mathbf{Z}$'s equal 0. As described in Section 6.5.1 there may be separate and independent replication for different mismeasured covariates, which can be accommodated by modifying (6.65) accordingly. Other options include continuing with the normal assumption but using a $\widehat{\boldsymbol{\Sigma}}_{ui}$ based on modeling the variance (rather than just using the replicates for the ith observation) or employing a nonparametric version of the step in (6.65) by estimating the distribution of the replicates.

6.16.2 *Bootstrapping with validation data*

External validation data.

Each bootstrap sample involves the following steps.

1. Generate the main data. This can be done in one of two ways.
   One-stage: Resample the main data in a manner that reflects how it was originally collected. If it is an overall random sample this would just mean sampling with replacement n times, but in other cases this may mean resampling using some stratification (including a case-control study) that was used initially.
   Two-stage: As in the earlier discussion with replicates, resampling as described in the first step does not always allow for a bootstrap estimate of bias since we are not necessarily resampling from a model with parameters equal to the original corrected estimates. To get a bootstrap estimate of bias, we first have to work with the measurement error model for observed given true, rather than the Berkson model. Then the resampling would proceed like the two-stage bootstrap described in the previous section except that equation (6.65) would be replaced by generating $\widehat{W}_{bi}$ based on the measurement error model fit using the external validation data.
2. Generate a bootstrap sample from the external validation data. This is done using standard bootstrap techniques (see the earlier bootstrap references) suitable to the nature of the validation data. If the external validation data are a random sample we would resample observations with replacement. More typically, however, the external data are viewed with the true values fixed (as in calibration studies) or observed values fixed (when fitting a Berkson

model) with some assumed regression model. In these cases, techniques that resample from the residuals are employed.

3. Obtain corrected estimators using the method of interest (moment correction, regression calibration, etc.). This leads to corrected estimators, e.g., $\widehat{\boldsymbol{\beta}}_b$ for the bth bootstrap sample, and these are analyzed in the usual fashion.

Internal validation data.

With internal validation data, the manner of bootstrapping depends on how the validation data are selected. Here are a couple of one-stage type options.

- Initial random sample with a random subsample for validation.
  This is the easiest case. One can resample n_V times with replacement from the n_V validated observations and independently resample $n_I = n - n_V$ times with replacement from the n_I nonvalidated observations.
- Random sample with "designed double sampling."
  Usually this occurs when the subsample for validation is chosen in a stratified manner, with the stratification based on categories formed from values in $\mathbf{W}$ observed at the first stage. This can be based on the use of either error-prone measures or variables that are never measured with error and are part of $\mathbf{W}$ (or a combination thereof). Suppose that C categories are formed at the first stage with n_c observations in category c, and n_{Vc} are chosen for validation (and $n_c - n_{Vc}$ are not). Then one could resample by sampling independently within each category and within category c sampling n_{Vc} times with replacement from the n_{Vc} validated observations and sampling $n_c - n_{Vc}$ times with replacement from the $n_c - n_{Vc}$ nonvalidated observations.

One two-stage approach is to first generate all of the observed values in the same way two-stage resampling was used to generate the main data in the external validation case above. Then on the n_V validation samples the true values can be generated using the observed values and the estimated Berkson model.

6.17 Mathematical developments

6.17.1 Justifying the MEE fitting method

To see why the fitting algorithm described in Section 6.13 works note that the vector going into the naive estimating equations can be written as $\sum_i S(\mathbf{O}_i, \boldsymbol{\beta}) = \mathbf{Q}(\boldsymbol{\beta})'(\mathbf{D} - \mathbf{m}(\boldsymbol{\beta}))$ where

$$\mathbf{Q}(\boldsymbol{\beta})' = \left[\begin{array}{cccc} \Delta(\mathbf{W}_1, \boldsymbol{\beta}), & \Delta(\mathbf{W}_2, \boldsymbol{\beta}), & \ldots & \Delta(\mathbf{W}_n, \boldsymbol{\beta}) \end{array} \right]$$

$\mathbf{m}(\boldsymbol{\beta})' = (m(\mathbf{W}_1, \boldsymbol{\beta}), \ldots, m(\mathbf{W}_n, \boldsymbol{\beta}))$ and $\mathbf{D}' = (D_1, \ldots, D_n)$.

The modified estimating equations can be expressed as $\mathbf{Q}(\boldsymbol{\beta})'(\tilde{\mathbf{D}} - \mathbf{m}(\boldsymbol{\beta})) = \mathbf{0}$, where $\tilde{\mathbf{D}} = \mathbf{D} - \mathbf{Q}(\boldsymbol{\beta})(\mathbf{Q}(\boldsymbol{\beta})'\mathbf{Q}(\boldsymbol{\beta}))^{-1}\mathbf{A}(\boldsymbol{\beta})$ and $\mathbf{A}(\boldsymbol{\beta}) = \sum_i^n A_i(\mathbf{O}_i, \boldsymbol{\beta})$. Note that $\tilde{\mathbf{D}}$ is a function of $\boldsymbol{\beta}$ and the observed values.

6.17.2 The approximate covariance for linearly transformed coefficients

Consider the estimator $\widehat{\boldsymbol{\beta}}_1 = \widehat{\boldsymbol{\Lambda}}_1^{\prime -1}\widehat{\boldsymbol{\beta}}_{1naive}$, which arose in a variety of contexts. The $\widehat{\boldsymbol{\Lambda}}_1$ is obtained from either replicates or validation data. Given the explicit form of $\widehat{\boldsymbol{\beta}}_1$ the approximate covariance of $\widehat{\boldsymbol{\beta}}$ is derived here using conditioning. This can also be obtained as a special case of the next subsection, but in this case the development here is a bit more transparent. Using conditioning (See Section 13.2)

$$Cov(\widehat{\boldsymbol{\beta}}_1) = E(Cov(\widehat{\boldsymbol{\beta}}_1|\widehat{\boldsymbol{\Lambda}}_1)) + Cov(E(\widehat{\boldsymbol{\beta}}_1|\widehat{\boldsymbol{\Lambda}}_1))$$

If we *assume* $\widehat{\boldsymbol{\beta}}_{1naive}$ *is independent of* $\widehat{\boldsymbol{\Lambda}}_1$, then the covariance becomes $E(\widehat{\boldsymbol{\Lambda}}_1^{\prime -1}Cov(\widehat{\boldsymbol{\beta}}_{1naive})\widehat{\boldsymbol{\Lambda}}_1^{-1}) + Cov(\boldsymbol{\Lambda}_1^{\prime -1}\boldsymbol{\gamma}_1)$ so

$$Cov(\widehat{\boldsymbol{\beta}}_1) \approx \boldsymbol{\Lambda}_1^{\prime -1}Cov(\widehat{\boldsymbol{\beta}}_{1naive})\boldsymbol{\Lambda}_1^{-1} + \mathbf{V}_2, \qquad (6.66)$$

where $\mathbf{V}_2 = Cov(\boldsymbol{\Lambda}_1^{\prime -1}\boldsymbol{\gamma}_1)$ and $\boldsymbol{\gamma}_1 = E(\widehat{\boldsymbol{\beta}}_{1naive})$. Writing

$$\widehat{\boldsymbol{\Lambda}}_1^{\prime -1} = \begin{bmatrix} \mathbf{c}_1' \\ \mathbf{c}_2' \\ . \\ \mathbf{c}_{p-1}' \end{bmatrix}$$

and $\mathbf{A}_{jk} = Cov(\mathbf{c}_j, \mathbf{c}_k)$, then the $(j,k)th$ component of $\mathbf{V}_2$ is equal to $\boldsymbol{\gamma}_1'\mathbf{A}_{jk}\boldsymbol{\gamma}_1$.

If we collect all of the components of the matrix $\widehat{\boldsymbol{\Lambda}}_1$ into a vector $\widehat{\boldsymbol{\lambda}}_1$, then we can use the delta method and approximate $\mathbf{A}_{jk}$ via

$$\mathbf{A}_{jk} = \mathbf{r}_j' Cov(\widehat{\boldsymbol{\lambda}}_1)\mathbf{r}_k,$$

where $\mathbf{r}_j = \partial\mathbf{c}_j/\partial\boldsymbol{\lambda}_1$ is the vector of derivative of $\mathbf{c}_j$ (the jth column of $\widehat{\boldsymbol{\Lambda}}_1^{-1}$) with respect to the elements of $\widehat{\boldsymbol{\Lambda}}_1$. These components can be computed using the following result; see for example Fuller (1987, p. 390).

The derivative of $\boldsymbol{\Lambda}_1^{-1}$ *with respect to the* $(k,m)th$ *element of* $\widehat{\boldsymbol{\Lambda}}_1$ *is the* $(k,m)th$ *element of the matrix* $\widehat{\boldsymbol{\Lambda}}_1^{-1}\mathbf{E}_{km}\widehat{\boldsymbol{\Lambda}}_1^{-1}$ *where* $\mathbf{E}_{km}$ *is* $p-1 \times p-1$ *matrix with all elements equal to 0 except the* $(km)th$ *element which is equal to 1.*

6.17.3 The approximate covariance of pseudo-estimators

A number of the methods described in this chapter first obtain estimates of either $\boldsymbol{\theta}$ (the parameters in the measurement error model) or $\boldsymbol{\xi}$ (the parameters in the Berkson model) then use these values to obtain estimates for the other parameters. This includes the linear transformed estimator discussed in the previous example, quasi-likelihood and regression calibration methods, pseudo-maximum likelihood estimation and correcting using modified estimating equations. This section outlines the general approach to obtain the covariance matrix of the corrected estimators in such settings. It is described assuming the measurement error parameters in $\boldsymbol{\theta}$ are first estimated and then used to obtain the estimates for the parameters of interest. The same technique works when it is the Berkson error parameters that are estimated, with $\boldsymbol{\theta}$ replaced by $\boldsymbol{\xi}$.

Let $\boldsymbol{\omega}$ be the parameter vector of interest. Often this is simply $\boldsymbol{\beta}$ in the regression model but it can also include the variance parameters in $\boldsymbol{\sigma}$. Suppose that $\hat{\boldsymbol{\theta}}$ comes from solving equations $S_2(\boldsymbol{\theta}) = \mathbf{0}$ and the $\hat{\boldsymbol{\omega}}$ arises from solving $S_1(\boldsymbol{\omega}, \widehat{\boldsymbol{\theta}}) = \mathbf{0}$. This is not much of a limitation as almost all of our estimators can be expressed in this way. With pseudo-maximum likelihood estimation S_1 would come from the appropriate score equations. Note that both S_1 and S_2 depend on random quantities which have been suppressed in the notation. For example S_2 will involve the validation data or replicates, while S_1 will depend on the observed values, along with the $\widehat{\boldsymbol{\theta}}$.

Define $\boldsymbol{\Sigma}_K$ = approximate covariance of $\hat{\boldsymbol{\omega}}$ if θ were known and $\boldsymbol{\Sigma}_{\widehat{\theta}}$ = the approximate covariance of $\widehat{\boldsymbol{\theta}}$. Applying the general theory of estimating equations and results on the inverse of a partitioned matrix, the approximate covariance matrix of $\widehat{\boldsymbol{\omega}}$ is

$$\boldsymbol{\Sigma}_{\widehat{\omega}} = \boldsymbol{\Sigma}_K + \mathbf{H}_{11}^{-1}\mathbf{H}_{12}\boldsymbol{\Sigma}_{\hat{\boldsymbol{\theta}}}\mathbf{H}_{12}'\mathbf{H}_{11}^{-1'} - \mathbf{P}$$

where $\mathbf{H}_{11} = E(\partial \mathbf{S}_1(\boldsymbol{\omega}, \boldsymbol{\theta})/\partial \boldsymbol{\omega})$, $\mathbf{H}_{12} = E(\partial \mathbf{S}_1(\boldsymbol{\omega}, \boldsymbol{\theta})/\partial \boldsymbol{\theta})$, $\mathbf{H}_{22} = E(\partial \mathbf{S}_2(\boldsymbol{\theta})/\partial \boldsymbol{\theta})$, $\mathbf{C}_{12} = E(\mathbf{S}_1\mathbf{S}_2')$ and

$$\mathbf{P} = \mathbf{H}_{11}^{-1}(\mathbf{H}_{12}\mathbf{H}_{22}^{-1}\mathbf{C}_{12}' + \mathbf{C}_{12}\mathbf{H}_{22}^{-1}\mathbf{H}_{12}')\mathbf{H}_{11}^{-1}.$$

This result can be specialized for specific situations. Liu and Liang (1982, equation (3)) have a similar result, but they inadvertently omitted one of the two terms that enter into the center portion of $\mathbf{P}$.

- With $\widehat{\boldsymbol{\theta}}$ independent of the observed $\mathbf{O}_i$ values, which includes (but is not limited to) the case of external validation data, then $\mathbf{C}_{12} = 0$ and $\mathbf{P} = 0$, so

$$\boldsymbol{\Sigma}_{\widehat{\omega}} = \boldsymbol{\Sigma}_K + \mathbf{Q}\boldsymbol{\Sigma}_{\widehat{\theta}}\mathbf{Q}', \text{ with } \mathbf{Q} = \mathbf{H}_{11}^{-1}\mathbf{H}_{12}. \qquad (6.67)$$

If $\widehat{\boldsymbol{\omega}}$ can be written explicitly as $f(\mathbf{O}, \widehat{\boldsymbol{\theta}})$, with jth element $\hat{\omega}_j = f_j(\mathbf{O}, \widehat{\boldsymbol{\theta}})$, then $\mathbf{Q} = E(\partial f(\mathbf{O}, \boldsymbol{\theta})/\partial \boldsymbol{\theta})$, so the (j,k) element of $\mathbf{Q}$ is $E(\partial f_j(\mathbf{O}, \boldsymbol{\theta})/\partial \theta_k)$.

- With internal validation data the additional P term will come into play. If the statistics involved in S_2 are independent of the statistics involved in S_1 then $\mathbf{C}_{12} = \mathbf{0}$ and the covariance in (6.67) applies.

6.17.4 Asymptotics for ML and pseudo-ML estimators with external validation.

Here, we can provide further detail on the asymptotic properties of the MLE's when using external validation data and the measurement error model. Let $\boldsymbol{\omega}_T$ denote the combination of $\boldsymbol{\omega}$ and $\boldsymbol{\omega}_X$. This contains all of the parameters associated with the true values. In the functional case with error in the response only then $\boldsymbol{\omega}_X$ is known (it is the x's themselves) and $\boldsymbol{\omega}_T$ is just $\boldsymbol{\omega}$. Under suitable conditions (involving increasing both n and the amount of external validation data in the right way), both the full MLE ($\hat{\boldsymbol{\omega}}_{T,ML}$) and the pseudo-MLE ($\hat{\boldsymbol{\omega}}_{T,PML}$) are consistent and asymptotically normal.

Letting $\boldsymbol{\rho}' = (\boldsymbol{\omega}', \boldsymbol{\theta}')$, the information matrix associated with the main observed data $\mathbf{O}_1, \ldots, \mathbf{O}_n$ is

$$I(\boldsymbol{\rho}) = -E\left[\frac{\partial^2}{\partial^2 \boldsymbol{\rho}} log(L_O(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}_X))\right] = \begin{bmatrix} I(\boldsymbol{\omega}_T) & I(\boldsymbol{\omega}_T, \boldsymbol{\theta}) \\ I(\boldsymbol{\omega}_T, \boldsymbol{\theta})' & I(\boldsymbol{\theta}) \end{bmatrix}.$$

The asymptotic covariance of the pseudo-MLE is

$$A(\hat{\boldsymbol{\omega}}_{T,PML}) = I(\boldsymbol{\omega}_T)^{-1} + I(\boldsymbol{\omega}_T)^{-1} I(\boldsymbol{\omega}_T, \boldsymbol{\theta}) I_c(\boldsymbol{\theta})^{-1} I(\boldsymbol{\theta}, \boldsymbol{\omega}_T) I(\boldsymbol{\omega}_T)^{-1}, \tag{6.68}$$

where $I_c(\boldsymbol{\theta})$ is the information matrix for $\boldsymbol{\theta}$ from the external validation data.

The asymptotic covariance matrix of the full MLE is

$$A(\hat{\boldsymbol{\omega}}_{T,ML}) = I(\boldsymbol{\omega}_T)^{-1} + I(\boldsymbol{\omega}_T)^{-1} I(\boldsymbol{\omega}_T, \boldsymbol{\theta}) \mathbf{Q}^{-1} I(\boldsymbol{\theta}, \boldsymbol{\omega}_T) I(\boldsymbol{\omega}_T)^{-1}$$

where $\mathbf{Q} = I(\boldsymbol{\theta}) + I_c(\boldsymbol{\theta}) - I(\boldsymbol{\theta}, \boldsymbol{\omega}_T) I(\boldsymbol{\omega}_T)^{-1} I \boldsymbol{\omega}_T, \boldsymbol{\theta})$.

Notice that estimating the asymptotic covariance matrix of the pseudo-MLE does not require estimating $I(\boldsymbol{\theta})$, which can be tedious to compute. This is needed in using the full MLE.

The loss in asymptotic efficiency with the use of pseudo rather than full maximum likelihood estimators is the difference in the second terms in these two expressions.

CHAPTER 7

Binary Regression

7.1 Introduction

This chapter provides an isolated treatment of measurement error in binary regression, probably the most important of the nonlinear regression models. This model, described in detail in Section 6.2, has an outcome Y which takes on two values, 0 or 1, with

$$E(Y|\mathbf{x}) = P(Y=1|\mathbf{x}) = m(\mathbf{x}, \boldsymbol{\beta}) = \mu \text{ and } V(Y|x) = \mu(1-\mu).$$

As elsewhere, with an intercept in the model we write $\mathbf{x}'_* = (1, \mathbf{x}')$ and $\mathbf{x}'_*\boldsymbol{\beta} = \beta_0 + \mathbf{x}'\boldsymbol{\beta}_1$.

The primary goal is usually estimation of the individual β_j's and functions of them. Of particular interest in epidemiology is the odds ratio associated with an increase of c units in the predictor x_j with other values held fixed. For logistic regression this is $e^{c\beta_j}$ and $c\beta_j$ is the log odds-ratio. With a rare outcome the odds-ratio approximates the relative risk. Additional problems of interest include estimating $P(Y=1|\mathbf{x})$ for a specified $\mathbf{x}$ and estimating $\mathbf{x}$ such that $m(\mathbf{x}, \boldsymbol{\beta}) = P$ for a specified probability P. The latter, referred to sometimes as **regulation**, is often the objective when estimating dose-response relationships in quantal bioassay.

The bias of the naive estimators was discussed in some detail in Section 6.7.6. The main goal of this chapter is to describe the correction techniques from Chapter 6 in more detail as they apply in binary regression, and illustrate their use.

The other issue that arises with a binary outcome is the different types of study designs that can be employed. These include structural models with an overall random sample, a functional model with predictors treated as fixed, or a combination with some predictors fixed and some random. Since the response is binary, a case-control study can also be used, in which the response is observed first. In the unmatched case-control, independent samples are then taken from units with $y = 1$ and $y = 0$, respectively, while in matched case-control studies the sampling is within different strata. See, for example, Chapter 7 of Hosmer and Lemeshow (2000). Case-control designs are especially common in epidemiology but also appear in other disciplines. For example, Charalambous et al. (2000) describe an example predicting bankruptcy. It is well known that without measurement error, logistic regression, carried out based on the model for Y given $\mathbf{x}$, provides valid inferences for the nonintercept coefficients under case-control sampling.

Section 7.2 treats the case where the measurement error is additive. A variety of methods are described and illustrated before Section 7.2.5 turns to fitting a model involving $log(x)$ when there are replicates for x. In Section 7.3 we turn our attention to situations where the measurement error is nonadditive but there is internal or external validation data allowing estimation of the measurement or Berkson error model.

7.2 Additive measurement error

As in Chapter 5, the additive measurement error model for the predictors is

$$\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i, \quad E(\mathbf{u}_i) = \mathbf{0}, \quad \text{and } Cov(\mathbf{u}_i) = \mathbf{\Sigma}_{ui}.$$

Also, as before, some of the variables may be measured exactly, in which case certain components of $\mathbf{\Sigma}_{ui}$ equal 0.

For those variables that are measured with error, we assume the availability of replicates or other data leading to $\widehat{\mathbf{\Sigma}}_{ui}$. This was discussed in Sections 5.4.3 and 6.5.1, where it was noted that there may be replication for some but not all of the units in the study. If unit i has no replication, then some assumptions need to be made about the per replicate variances in order to construct $\widehat{\mathbf{\Sigma}}_{ui}$. More generally, there may be unequal numbers of replicates for different mismeasured variables. The average estimated measurement error covariance matrix is denoted by $\widehat{\mathbf{\Sigma}}_u = \sum_i \widehat{\mathbf{\Sigma}}_{ui}/n$. (See also Thoresen and Laake (2003) for a discussion on the use of correlated replicates.)

7.2.1 Methods

There is a plethora of correction techniques that can be used as outlined in Chapter 6. Here we comment on the use of regression calibration, corrections to naive estimates and modified estimating equation (MEE) approaches when there is an overall random sample. In addition we discuss a correction based on the normal discriminant models and comment briefly on dealing with case-control studies. The discussions in Chapter 6 concerning likelihood methods (Section 6.12.3) and SIMEX (6.11) will not be repeated here but they will be used in some of the examples. We also refer the reader to Section 7.2.2 of Carroll et al. (2006) for a description of conditional and corrected score methods in logistic regression. The MEE approach is very similar to the corrected score method. SIMEX, MEE and other corrected estimating equation approaches are applicable in the functional case. It is hard to say that a particular method is best. There have been a number of assessments via simulations. For example, Thoresen and Laake (2000) compare regression calibration (RC), the correction based on the probit correction (corrected naive II below), an exact maximum likelihood estimator, and the naive method, and find that the RC estimator does quite well. In our examples the two-stage bootstrap provides for some comparisons.

Except for case-control studies, where the outcome variable is fixed, inferences can be based on the bootstrap methods described in Section 6.16. Recall that the one-stage bootstrap simply resamples observations while the two-stage bootstrap generates predictors and a response. For the latter, on the bth bootstrap sample the outcome for the ith observation is Y_{bi}, generated by $P(Y_{bi} = 1) = m(\widehat{\boldsymbol{\beta}}, \widehat{x}_i)$, where $m(\widehat{\boldsymbol{\beta}}, x)$ is the fitted regression function. See the examples for illustration of these bootstrap methods and further discussion of them. For case-control studies, the bootstrap is based on resampling independently from the cases ($Y = 1$) and controls ($Y = 0$), or for matched case-control studies, from resampling within each stratum. We describe this bootstrap further when discussing the normal discriminant model below, but this type of bootstrapping should be used in any case-control setting.

Corrected naive I / Simple regression calibration.

The simplest regression calibration estimator relies on the assumption that $E(\mathbf{X}|\mathbf{w}) = \boldsymbol{\lambda}_o + \boldsymbol{\Lambda}_1\mathbf{w}$, where $\boldsymbol{\Lambda}_1 = \boldsymbol{\Sigma}_X(\boldsymbol{\Sigma}_X + \boldsymbol{\Sigma}_u)^{-1}$ and $\boldsymbol{\lambda}_0 = \boldsymbol{\mu}_X - \boldsymbol{\Lambda}_1\boldsymbol{\mu}_X$. The estimators, denoted with an RC subscript, can be obtained in one of two ways. The first is by fitting the assumed regression model using imputed predictors $\widehat{\mathbf{x}}_i = \widehat{\boldsymbol{\lambda}}_o + \widehat{\boldsymbol{\Lambda}}_1\mathbf{w}_i$, where $\widehat{\boldsymbol{\Lambda}}_1 = \widehat{\boldsymbol{\Sigma}}_X(\widehat{\boldsymbol{\Sigma}}_X + \widehat{\boldsymbol{\Sigma}}_u)^{-1}$ and $\widehat{\boldsymbol{\lambda}}_0 = \widehat{\boldsymbol{\mu}}_X - \widehat{\boldsymbol{\Lambda}}_1\widehat{\boldsymbol{\mu}}_X$. Notice that a common $\widehat{\boldsymbol{\Sigma}}_u$ is used throughout.

Equivalently, the regression calibration estimators can be computed via

$$\widehat{\boldsymbol{\beta}}_{1RC} = \widehat{\boldsymbol{\Lambda}}_1^{\prime -1}\widehat{\boldsymbol{\beta}}_{1naive} \text{ and } \widehat{\beta}_{0RC} = \widehat{\beta}_{0naive} - \widehat{\boldsymbol{\lambda}}_0'\widehat{\boldsymbol{\beta}}_{1RC}. \tag{7.1}$$

The RC estimators are motivated by the expression for $P(Y = 1|\mathbf{w})$ given in (6.28) and can be applied to either logistic or probit regression. Since the expression in (6.28) is only approximate the resulting estimators are not necessarily consistent, although they have been found to perform well in a large number of settings.

If the estimated $\widehat{\boldsymbol{\Sigma}}_u$ is independent of the $(Y_i, \mathbf{W}_i)$, then an estimate of approximate covariance matrix of $\widehat{\boldsymbol{\beta}}_{1RC}$ is given by (6.42). This would be applicable, for example, if the replicates are normally distributed since $\widehat{\boldsymbol{\Sigma}}_u$ involves sample variances and covariances of the replicates which are independent of the mean values going into $\mathbf{W}_i$. Even with the use of (6.42), it is still nontrivial to get $Cov(\widehat{\boldsymbol{\beta}}_{1RC})$ since the second term requires a covariance matrix for the elements of $\widehat{\boldsymbol{\Lambda}}_1$. This is not difficult to obtain when there is validation data and the $\widehat{\boldsymbol{\Lambda}}_1$ contains coefficients from regressing $\mathbf{W}$ on $\mathbf{X}$. It is more tedious to obtain, however, under replication since we first need to obtain the variance-covariance structure of the various components entering into $\widehat{\boldsymbol{\Sigma}}_u$ and $\mathbf{S}_{WW}$ (this can be done either using normality assumptions or robustly) and then use the delta method to get an approximate covariance of $\widehat{\boldsymbol{\lambda}}_1$. See Rosner et al. (1992) for details.

Use of just the first part of (6.42) is equivalent to using standard errors and associated inferences from running a logistic regression using the imputed predictors. As seen in the examples, there are situations where this simple approach appears to be adequate. More generally, the covariance of $\widehat{\boldsymbol{\beta}}$ can be obtained using the results in Section 6.17.3 and estimated through the bootstrap.

Regression calibration II.

A second version of regression calibration uses separate measurement error variances and covariances and regresses Y_i on $\widehat{\mathbf{x}}_i$ where

$$\hat{x}_i = \hat{\boldsymbol{\lambda}}_{0i} + \hat{\Lambda}_{1i} W_i, \tag{7.2}$$

with $\widehat{\boldsymbol{\Lambda}}_{1i} = \widehat{\boldsymbol{\Sigma}}_X(\widehat{\boldsymbol{\Sigma}}_X + \widehat{\boldsymbol{\Sigma}}_{ui})^{-1}$ and $\widehat{\boldsymbol{\lambda}}_{0i} = \widehat{\boldsymbol{\mu}}_X - \widehat{\boldsymbol{\Lambda}}_{1i}\widehat{\boldsymbol{\mu}}_X$. This uses observation specific estimates, $\widehat{\boldsymbol{\Sigma}}_{ui}$, of the measurement error covariance. The $\widehat{\boldsymbol{\Sigma}}_{ui}$ might arise from using the replicates associated with the ith observation or from constructing $\widehat{\boldsymbol{\Sigma}}_{ui}$ using a pooled estimate of the per replicate variances/covariances combined with the numbers of replicates observation i. The latter approach was used by Kim and Seleniuch-Jacquotte (1997). See Section 6.5.1 for some important details. Notice that, unlike the previous version, we cannot compute this version of the RC estimator by simply transforming the naive estimators.

For estimating the covariance of $\widehat{\boldsymbol{\beta}}$, much of the discussion for the first regression calibration estimator above applies here. In particular, one can bootstrap or consider using approximate inferences based on results from the logistic regression with imputed values. More generally, these are pseudo-type estimators and as such the estimated covariance will be of the form $\boldsymbol{\Sigma}_{\widehat{\beta},K} + \widehat{V}_2$, where $\boldsymbol{\Sigma}_{\widehat{\beta},K}$ is the covariance matrix obtained from the regression of Y_i on $\widehat{\mathbf{x}}_i$, and $\widehat{V}_2$ is an additional term accounting for uncertainty from estimating the measurement error parameters; see Sections 6.10 and 6.17.3.

Before leaving the RC estimators, we reemphasize a critical point made elsewhere, namely:

The model used to obtain the $\widehat{\mathbf{x}}_i$*'s is built from the model for* $\mathbf{X}$ *given* $\mathbf{w}$*; that is using all of the variables simultaneously. It does NOT come from just imputing individual predictors by building a model for that predictor given its corresponding measured value.*

Corrected naive II/Pseudo MLE.

As seen in Section 6.7.6, under the probit model for $Y|\mathbf{x}$ along with a normal linear Berkson model, we obtain another probit model. Also, see the bias results in (6.31. This leads to a pseudo-model using the estimated Berkson parameters of

$$P(Y_i = 1|\mathbf{w}_i; \boldsymbol{\beta}, \widehat{\boldsymbol{\xi}}) = \Phi\left[\frac{\beta_0 + \widehat{\mathbf{x}}_i'\boldsymbol{\beta}_1}{(1 + \boldsymbol{\beta}_1'\widehat{\boldsymbol{\Sigma}}_{X|W}\boldsymbol{\beta}_1)^{1/2}}\right],$$

where $\widehat{\mathbf{x}}_i = \hat{\boldsymbol{\lambda}}_0 + \hat{\boldsymbol{\Lambda}}_1\mathbf{w}_i$ and $\widehat{\boldsymbol{\xi}} = (\widehat{\boldsymbol{\Sigma}}_{X|w}, \hat{\boldsymbol{\lambda}}_0, \hat{\boldsymbol{\Lambda}}_1)$. The pseudo-MLE, denoted $\widehat{\boldsymbol{\beta}}_{probit}$, maximizes the likelihood $L(\boldsymbol{\beta}) = \prod_i f(y_i|\mathbf{w}_i, \boldsymbol{\beta}, \widehat{\boldsymbol{\xi}})$. There is a closed form solution (see below) with a single predictor but in general a system of nonlinear equations needs to be solved. Converting to the logistic model leads to a corrected estimator of

$$\widehat{\boldsymbol{\beta}}_{logistic} = 1.7\widehat{\boldsymbol{\beta}}_{probit}.$$

These estimators are based on (6.30), (6.4) and (6.33). See Carroll et al. (1984), Liang and Liu (1991), Liu and Liang (1992), Burr (1988) and Tosteson et al. (1989) for some of the original work on the pseudo-MLE's for the probit model. The last two papers discuss the fact that in the case of a single predictor, the resulting estimators are

$$\begin{aligned}
\widehat{\beta}_{1probit} &= \frac{\widehat{\beta}_{1RC(probit)}}{(1 - \widehat{\beta}^2_{1RC(probit)}\hat{\sigma}^2_{X|w})^{1/2}} \text{ and} \\
\widehat{\beta}_{0probit} &= \widehat{\beta}_{0RC(probit)} - \widehat{\boldsymbol{\lambda}}_0'\widehat{\boldsymbol{\beta}}_{1probit},
\end{aligned}$$

where $\widehat{\beta}_{RC(probit)}$ denotes the regression calibration estimators in (7.1) based on probit regression. This gives the pseudo-MLE if $\hat{\sigma}^2_{X|W}\hat{\beta}^2_{1RC(probit)} < 1$.

For the logistic model, even with the normal linear Berkson model, $f(y_i|w_i)$ does not exist in closed form and obtaining the exact pseudo-MLE requires specialized programming. The regression calibration estimators are approximate pseudo-MLE's obtained under (6.32) while if we accept the approximation in (6.33) the estimator $\widehat{\boldsymbol{\beta}}_{logistic}$ above is the approximate pseudo-MLE. Rosner, Willett and Spiegelman (1989) and Schafer (1987) discuss other approximations to the pseudo-MLE.

Modified estimating equations

This section describes how to apply the modified estimating approach of Section 6.13 to binary regression. There are no distributional assumptions on either the unobserved true values, which may be fixed, or the measurement error.

The naive estimating equations for binary regression are $\sum_{i=1}^n \{Y_i - m(\mathbf{w}_i, \boldsymbol{\beta})\}\mathbf{w}_{i*} = \mathbf{0}$, where $m(\mathbf{x}_i, \boldsymbol{\beta}) = P(Y_i = 1|\mathbf{x}_i)$. Equivalently these estimators can also be found using weighted nonlinear least squares with $E(Y_i|\mathbf{x}_i) = m(\mathbf{x}_i, \boldsymbol{\beta}) = \pi_i$ and weight is $1/(\pi_i(1-\pi_i))$. As described in Section 6.13, MEE estimators can usually be fit by iteratively fitting the naive model where the predictors are the observed $\mathbf{w}_i$'s but the responses are updated. Notice that here the updated responses are no longer binary. This is convenient, since the naive method can be easily programmed up to be fit using iteratively reweighted linear least squares or a maximization routine using the log-likelihood.

The updated responses to use on the $(k+1)st$ are

$$\mathbf{Y}_{(k)} = \mathbf{Y} - \mathbf{W}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{A}_{(k)},$$

where $\mathbf{W}$ is the design matrix with ith row equal to $(1, \mathbf{w}_i')$ and $\mathbf{A}_{(k)} = \sum_i \widehat{\mathbf{A}}_{i(k)}$ with $\widehat{\mathbf{A}}_{i(k)}$ an estimate of the bias $\sum_i C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta})$ for the ith contribution to the naive estimating equations. Applying the methods in 6.13 with $\Delta(\mathbf{x}_i, \boldsymbol{\beta}) = \mathbf{x}_{i*}$, $B_i(\boldsymbol{\beta}) = \mathbf{0}$ and $\Omega_{\Delta qi} = 0$ leads to

$$C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}) = \begin{bmatrix} -G_i(\boldsymbol{\beta}, \mathbf{x}_i) \\ -G_i(\boldsymbol{\beta}, \mathbf{x}_i)\mathbf{x}_i \end{bmatrix} - \begin{bmatrix} 0 \\ cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta})) \end{bmatrix},$$

with $G_i(\boldsymbol{\beta}, \mathbf{x}_i) = E[m(\mathbf{W}_i, \boldsymbol{\beta})] - m(\mathbf{x}_i, \boldsymbol{\beta})$. Using an analytical expression requires approximating $G_i(\boldsymbol{\beta}, \mathbf{x}_i)$ and $cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta}))$ since they do not have a closed form. For the logistic case, using a second order approximation for $E(m(\mathbf{W}_i, \boldsymbol{\beta}))$ and a first order approximation for $Cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta}))$ leads to

$$G_i(\boldsymbol{\beta}, \mathbf{x}_i) \approx m(\mathbf{x}_i, \boldsymbol{\beta})\{1 - m(\mathbf{x}_i, \boldsymbol{\beta})\}\{1 - 2m(\mathbf{x}_i, \boldsymbol{\beta})\}\boldsymbol{\beta}'\boldsymbol{\Sigma}_{ui}\boldsymbol{\beta}/2$$

and $Cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta})) \approx \Sigma_{ui} h_i(\boldsymbol{\beta})$, where

$$h_i(\beta) = m(\mathbf{x}_i, \boldsymbol{\beta})\{1 - m(\mathbf{x}_i, \boldsymbol{\beta})\}\boldsymbol{\beta}.$$

This points out the challenge of using the analytical method in that even after the use of the approximations, we are faced with estimating nonlinear functions of the unobservable predictors. A simple approach, implemented in the examples, is to use

$$\widehat{\mathbf{A}}_{i(k)} = -\mathbf{w}_{i*}\hat{G}_i(\widehat{\boldsymbol{\beta}}_{(k)}) - \hat{\Sigma}_{ui}\hat{h}_i(\widehat{\boldsymbol{\beta}}_{(k)}), \tag{7.3}$$

where $\hat{G}_i(\boldsymbol{\beta}) = m(\mathbf{w}_i, \boldsymbol{\beta})\{1 - m(\mathbf{w}_i, \boldsymbol{\beta})\}\{1 - 2m(\mathbf{w}_i, \boldsymbol{\beta})\}\boldsymbol{\beta}'\hat{\boldsymbol{\Sigma}}_{ui}\boldsymbol{\beta}/2$, $\hat{h}_i(\boldsymbol{\beta}) = m(\mathbf{w}_i, \boldsymbol{\beta})\{1 - m(\mathbf{w}_i, \boldsymbol{\beta})\}\boldsymbol{\beta}$ and $\widehat{\boldsymbol{\beta}}_{(k)}$ is the estimator of $\boldsymbol{\beta}$ from the kth iteration.

With a single predictor, the resulting estimator agrees with the one proposed by Stefanski (1989). One could try to improve on this analytical approach by using further approximations to try and remove bias in $\widehat{G}_i(\boldsymbol{\beta})$ and $\widehat{h}_i(\boldsymbol{\beta})$, but this doesn't necessarily improve the performance.

The approximate covariance of $\widehat{\boldsymbol{\beta}}_{MEE}$ can be estimated using (6.63). This requires differentiation of the modified estimation with respect to the components of $\boldsymbol{\beta}$, which is easy, but tedious. One and two-stage bootstraps are used for inferences in the examples that follow. As noted earlier, at the final step the MEE estimates are equivalent to least squares estimates from a weighted nonlinear regression analysis with modified responses. While it would be convenient to use standard errors and inferences from that final fit, our later examples show that the resulting standard errors are misleading in being too small by a significant amount.

Corrections based on the normal discriminant model.

This model assumes that conditional on the response the covariates are normally distributed with changing mean vector and common covariance matrix $\boldsymbol{\Omega}$; that is, $\mathbf{X}|Y = y \sim N(\boldsymbol{\mu}_y, \boldsymbol{\Omega})$. This implies that $P(Y = 1|\mathbf{X} = \mathbf{x})$ follows the logistic model with $\boldsymbol{\beta}_1 = \boldsymbol{\Omega}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$ and $\beta_0 = log(\pi_1/\pi_0) - \boldsymbol{\beta}_1'(\boldsymbol{\mu}_0 + \boldsymbol{\mu}_1)/2$, with $\pi_y = P(Y = y)$.

The normal discriminant model has fallen out of favor as a way to handle binary regression and so our treatment is brief. It is useful with a small number of predictors when the assumptions are approximately true, since the problem can be attacked via estimating mean vectors and covariance matrices. It also afforded some "exact" result concerning bias of naive estimators as seen in Section 6.7.6. Because the model conditions on y, it also provides a setting for discussing how to bootstrap in case-control studies.

We continue to use the additive measurement error model, $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$. This was first investigated in the normal discriminant model by Wu et al. (1986) and Armstrong et al. (1989), the latter allowing for a constant measurement error bias in each group.

Estimation can be carried out by first estimating $\boldsymbol{\mu}_0$, $\boldsymbol{\mu}_1$ and $\boldsymbol{\Omega}$. This can be done by first splitting the data into two groups according to whether the observed Y is 0 or 1. In the group with observed outcome y, we make use of the replicates to obtain $\widehat{\boldsymbol{\mu}}_y$ and $\widehat{\boldsymbol{\Omega}}_y$, the latter an estimate of the covariance of $\mathbf{X}$ given $Y = y$. These estimates are obtained with the methods described in Section 6.5.1 but applied to just the observations with outcome $Y = y$. Since $Cov(\mathbf{X}|y)$ is assumed to be the same for each y, a pooled estimate $\widehat{\boldsymbol{\Omega}} = (n_0\widehat{\boldsymbol{\Omega}}_0 + n_1\widehat{\boldsymbol{\Omega}}_1)/n$ is obtained. The nonintercept coefficients are then estimated using

$$\widehat{\boldsymbol{\beta}}_1 = \widehat{\boldsymbol{\Omega}}^{-1}(\widehat{\boldsymbol{\mu}}_1 - \widehat{\boldsymbol{\mu}}_0). \tag{7.4}$$

Since we have an explicit expression and the estimator depends on the data through the per unit mean vector and the sample covariance matrices, both among and within observations, the covariance of $\widehat{\boldsymbol{\beta}}_1$ can be approximated using the delta method.

Notice that in a case-control scenario we cannot estimate the intercept β_0 without knowledge of π_0 and π_1. With a random sample, these can be estimated with the sample proportions, say p_0 and p_1 and an estimate of β_0 obtained using $\widehat{\beta}_0 = log(p_1/p_0) - \widehat{\boldsymbol{\beta}}_1'(\widehat{\boldsymbol{\mu}}_0 + \widehat{\boldsymbol{\mu}}_1)/2$.

There are a variety of ways to bootstrap under this model. We describe three below. If a case-control design is used then the third bootstrapping method should be used. For an overall random sample any of the three methods can be used. The advantage of the first two is they allow for inferences on β_0 in addition to β_1 and hence for inferences on $P(Y = 1|\mathbf{X} = \mathbf{x})$ for a specified $\mathbf{x}$. Further, with the model assumptions holding, the parametric bootstrap is preferred since it allows for a bootstrap estimate of bias.

- One-stage bootstrap. This applies for overall random samples and just resample individuals.
- Two-stage bootstrap. Also designed for overall random samples. This is a little different than the two-stage bootstrap used elsewhere. Here the y values are generated using the estimates of π_0 and π_1, with y_{bi} equal 1 with probability $\widehat{\pi}_y$. Then the predictors are generated, which can be done either parameterically or nonparameterically. In the parametric version the predictors $\mathbf{X}_{bi}$ are generated from a multivariate normal with mean $\widehat{\boldsymbol{\mu}}_{y_{bi}}$ and covariance $\widehat{\boldsymbol{\Omega}}$ and then replicate mismeasured values are generated using a normal with mean $\mathbf{X}_{bi}$ and covariance $\widehat{\boldsymbol{\Sigma}}_u$. This version implies equal measurement error variance-covariances. It can be readily modified if $\boldsymbol{\Sigma}_{ui} = \boldsymbol{\Sigma}_{u(1)}/k_i$ where k_i is the number of replicates but otherwise we cannot handle changing $\boldsymbol{\Sigma}_{ui}$ without some model for how this changes with $\mathbf{X}_i$. The nonparametric version resamples from the observed $\mathbf{W}$ values from the data with $Y = y_{bi}$.
- Stratified (on y) resampling. This is the method that must be used for a case-control study. It is similar to the preceding method except the y values

are fixed at their observed values, i.e., in the bth bootstrap sample y_{bi} is always y_i. The predictors are then generated as described in the second part of the preceding method. The nonparametric version resamples n_0 times with replacement from the observations with $Y = 0$ and independently samples n_1 times with replacement from the observations with $Y = 1$. When an observation is selected we use all of the information from the replicates, both the mean and variances/covariances from that observation. If the design is a matched case-control, then the resampling is done within each stratum, as discussed and illustrated by Kim and Seleniuch-Jacquotte (1997).

For additional discussion on correcting for measurement error in case-control settings see the recent work of Guolo (2008) and references therein.

7.2.2 *Example: Cholesterol and heart disease*

The Framingham Heart Study is one of the longest running epidemiologic studies and one which has been used in many contexts to illustrate measurement error effects and corrections. Here we start with data on 1615 individuals (obtained with appreciation from Ray Carroll). The data for each individual consist of age, smoking status, the presence or absence of coronary heart disease (Y) and two measures of serum cholesterol and systolic blood pressure, from two different visits. These are treated as replicates.

For the illustration in this section, we use that part of the data consisting of 1475 individuals for which the estimated mean cholesterol level is between 175 and 325 mg.Dl, inclusive. We examine the relationship between coronary heart disease and "true" mean cholesterol, ignoring the other variables. The data were subsetted for this illustration for two reasons. First, there is less of an issue here than with the full data set as to whether a logistic model with untransformed cholesterol is appropriate. This avoids the question of transforming which is addressed later in Section 7.2.5. Secondly, the narrower range on the predictor provides a better illustration of the changes from correcting with measurement error. We explore this example in great detail in order to provide a comprehensive look at the inner workings of the various methods in a relatively simple setting.

The two measures of cholesterol intake are denoted W_{i1} and W_{i2} for the ith individual. These are treated as replicates with $W_i = (W_{i1} + W_{i2})/2$ being the observed mean cholesterol, and s_i^2 and $\widehat{\sigma}_{ui}^2 = s_i^2/2$ denoting the sample per-replicate variance and estimated measurement error variance, respectively. Table 7.1 shows descriptive statistics on the estimated mean cholesterol and estimated measurement error variance for those with and without heart disease.

Table 7.1 *Descriptive statistics on observed average cholesterol (W) and estimate measurement error variance ($\widehat{\sigma}_u^2$) using subsetted data. S.D. = standard deviation.*

Heart disease?	Sample Size	Variable	Mean	S.D.
No ($Y = 0$)	1354	W	229.37	30.999
		$\widehat{\sigma}_u^2$	228.68	438.59
Yes ($Y = 1$)	121	W	239.73	30.751
		$\widehat{\sigma}_u^2$	218.18	341.60
Combined	1475	W	230.22	31.10
		$\widehat{\sigma}_u^2$	227.82	431.37

The logistic model is fit using the naive method, which ignores measurement error, and corrections using regression calibration (two types), SIMEX, the modified estimating equation approach, the normal discriminant model and maximum likelihood estimation. A summary of the resulting analyses appear in Tables 7.5 - 7.7. The standard errors labeled B and B2 and associated bootstrap percentile confidence intervals were obtained using one-stage and two-stage resampling, respectively. See Section 6.16 and some of the comments in Section 7.1. These analyses were computed using customized programs (using SAS-IML) and involve no trimming of the bootstrap samples. The associated confidence interval are 95% bootstrap percentile intervals. Here the two-stage bootstrap was only implemented for the RC and MEE methods.

We first provide some additional details on the methods used and related output.

- Naive estimates.
  These come from running a standard logistic regression analysis along with Wald confidence intervals. We also computed bootstrap standard errors and confidence intervals based on one and two-stage bootstrap resampling for comparison to results from the correction methods.
- Regression calibration.
  Two type of regression calibration estimators are used. The one labeled RC uses the same $\widehat{\lambda}_0$ and $\widehat{\lambda}_1$ in forming imputed values $\hat{x}_i$ (see equation (7.1)), while RC-I uses imputed values based on individual measurement error variances (see equation (7.2)).

  The standard RC estimates can also be obtained using STATA's RCAL command, which obtains standard errors in one of two ways, bootstrapping by resampling observations (B:STATA) with trimmed samples (2.5% on either end) or using an analytical expression (A:STATA). In each case the 95% confidence intervals use a normal approximation, i.e., estimate $\pm 1.96 SE$.

As discussed in applications for linear regression, the analytical standard errors for β_0 from this procedure are incorrect, since they are not properly adjusted.

For the RC estimates, the standard errors labeled "imputed" and associated Wald confidence intervals result from simply running logistic regression on the imputed $\widehat{x}_i$ values. We also obtained nonsignificant lack of fit tests from these analyses. The lack of fit test for the first RC estimator, which uses a common linear imputation for all observations, is identical to the lack of fit test from the naive analysis.

- Modified estimating equations.
  The MEE estimator arises from the modified estimating equation approach using (7.3). As with the RC estimators, inferences are presented based on both the one and two-stage bootstrap resampling. We also give the standard errors (labeled NLIN) that result from the final weighted nonlinear regression that produces the MEE estimators.
- Normal Discriminant.
  Estimates of μ_0 and μ_1 are given by the sample means in Table 7.1. Table 7.2 shows individual moment estimates of $V(X|Y = y) = \Omega_y$ for each y and a pooled estimate assuming $\Omega_y = \Omega$. That table also provides estimates of $\mu_X = E(X)$ and σ^2_X. The estimate of the variance in the true cholesterol values is $\widehat{\sigma}^2_X = 31.1^2 - 227.82 = 739.4$, which is the variance in the cholesterol means minus the average measurement error variance. Using (7.4), the corrected estimate of β_1 is $\widehat{\beta}_1 = (229.37 - 239.73)/731.835 =$ .01415.

Table 7.2 *Cholesterol/heart disease example: Estimates of $\Omega_y = V(X|Y = y)$ and pooled estimate plus estimate of mean and variance in true cholesterol values.*

Parameter	Ω_0	Ω_1	Ω (pooled)	μ_X	σ^2_X
Estimate	732.229	727.429	731.835	230.224	739.4

As described earlier the bootstrapping for the normal discriminant model is different than for the other methods. Here we have provided results from fixing the y's and resampling within each group either nonparametrically (BN) or parameterically (BP). The nonparametric approach simply resamples observations in each group using the mean and estimated measurement error variance for the selected observation. The parametric approach generates true values and then replicates in each group under normality and the assumption of constant per replicate measurement error variances (within each group).

- SIMEX
  The Simex estimates and associated bootstrap standard errors and confidence intervals can be obtained via the SIMEX function in STATA. See the comments above on the output from using the STATA RCAL function. We used the default values in STATA which take $\lambda = 0$, .5, 1, 1.5 and 2 and use a quadratic extrapolant. We also programmed SIMEX using a slightly richer grid for λ, still using quadratic extrapolation and also using a version allowing for separate measurement error variances; the results for the latter indicated by *Ind* (for individual variances) in the output. Figure 7.1 shows the SIMEX fits and extrapolation for the two coefficients assuming constant measurement error variances.

- Maximum Likelihood under normality.
  Finally, we have a full maximum likelihood analysis under the assumption of normallly distributed true values and normal replicates with constant measurement error variance along with the assumed logistic model for Y given x; see Section 6.12.3. This is carried out using the CME function in STATA, with the commands given in Table 7.3 and output on the fitted structural and measurement error model appearing in Table 7.4. The true covariate part of the output gives the MLE'S for μ_X (230.224) and σ^2_X (738.619), the latter differing only slightly from the estimate of 739.913 in Table 7.2 arising from an unbiased moment approach. The measurement model part of the output provides an estimate (455.644) of the per-replicate measurement error variance, $\sigma^2_{u(1)}$ say. The measurement error variance σ^2_u for the mean of two replicates is $\sigma^2_{u(1)}/2$. This leads to a maximum likelihood estimate of $\widehat{\sigma}^2_{u,mle} = 445.644/2 = 222.822$, compared to the moment estimate of 227.82 in Table 7.1.
  The results for the corrected logistic fit are contained in Tables 7.5 and 7.7. The standard errors produced are obtained using the information matrix and the confidence intervals are Wald intervals.

The normal discriminant approach and the ML method assume normality of the underlying true cholesterol values: overall in the case of using ML and within each disease category for the use of the discriminant model. Figure 7.3 shows nonparametric estimates of these distributions to provide a general sense of their shape. These were obtained using the NPMLE technique described in Section 10.2.3, based on a fixed grid of 50 points. The top two panels show the estimated distributions for the cases and noncases, respectively. These use separate measurement error variances for each individual. With a limited number of cases the estimated distribution is a bit rough. The bottom two panels estimated the cholesterol distribution for all data combined, the one to the left using individual measurement error variances, the one to the right using the average measurement error variance for each observation. Throughout there are signs of asymmetry, raising questions about the validity of the normal discriminant and MLE approaches. We have retained those analyses, however, in order to demonstrate their usage.

In terms of fitting the logistic model, another option is to use a semiparametric method in that the measurement errors are still assumed normal, but the distribution of the true cholesterol values is estimated nonparametrically using the nonparametric MLE. This is not implemented here but it can be carried out using the gllamm function in STATA. See Rabe-Hseketh et al. (2003a) for details on implementation and a numerical example and Rabes-Hesketh et al. (2003b) for additional discussion.

Table 7.3 *STATA commands for fitting logistic regression with measurement error. The response is chd, while c1 and c2 are the replicate measures of cholesterol.*

Method	Command
RC, bootstrap	rcal (chd=) (w:c1 c2), fam(bin) bstrap brep(1000)
RC, analytical	rcal (chd=) (w:c1 c2), fam(bin)
SIMEX, bootstrap	simex (chd=) (w:c1 c2), fam(bin) bstrap brep(1000)
Maximum Likelihood	cme chd (w: c1 c2), l(logit) f(binom)

Table 7.4 *Part of the output from the STAT CME command*

TRUE COVARIATE MODEL

w	Coef.	Std. Err.	[95% Conf. Interval]	
cons	230.2237	.8094531	228.6372	231.8102
res. var.	738.6194	36.56271	668.6959	812.0191

MEASUREMENT MODEL

	Coef.	Std. Err.	[95% Conf. Interval]	
error var.	455.6435	16.77823	423.9174	489.7439
reliability	.618473	.0160782	.586601	.649561

As expected, all of the correction methods lead to larger estimates of β_1 and the associated odds ratio. The changes from correcting for measurement error appear to be modest at first, but this is a bit deceptive due to the scale involved. The odds ratios are for a change of 1mg/100ml of cholesterol, which is not of major clinical importance. The consequences of the corrections for measurement error are better illustrated through the fitted response curves, given in Figure 7.2. We can also examine the estimated odds ratios and associated confidence intervals for clinically important changes in cholesterol; see Table 7.6 for changes of 10 and 100 mg/100ml. The corrected fits show substantially different estimates for the probability of heart disease at higher values of mean cholesterol using corrected versus naive estimates. Similarly, the impact of the corrections on odds ratio is more obvious when computed over meaningful changes in cholesterol.

Overall, there are no dramatic differences in the results for the various methods, even though the techniques vary considerably in their approach to the problem and the underly assumption (e.g., structural or functional, constant or changing measurement error variance). Still there are some differences to note. The MEE and normal discriminant methods lead to slightly larger estimates of β_1 than the RC estimates, which in turn are slightly larger than those from RC-I and SIMEX (assuming constant measurement error variance). Interestingly the estimate of β_1 from regular SIMEX (assuming constant M.E. variance) is closer to the result from RC-I (which allows unequal M.E. variances) while the SIMEX-I allowing unequal M.E. variances gives a result closer to that of the RC method (which uses an assumption of common M.E. variance). This may be related to bias issues, discussed below. Notice that the estimates of the odds ratios associated with changes of 10 and 100 in cholesterol show a bit more of a distinction between the MEE and other methods.

The two-stage bootstrap estimates of standard error are very similar for RC, RC-I and MEE and similar to the one-stage bootstrap standard error from SIMEX. The normal discriminant analysis stands out in having a larger standard error for the slope, with larger bootstrap confidence intervals but, as noted above, this analysis is a bit suspect. Notice that for the RC estimates the standard errors from the two-stage bootstrap intervals are generally smaller than the one-stage bootstrap and there are some differences in the confidence intervals. Notice also that for estimating β_1 using regression calibration, the standard error from simply running logistic regression on imputed values is essentially the same as the analytical standard error (obtained here from STATA), and so from that perspective nothing is lost by simply analyzing the imputed values. The key reason for this is that the correction really involves the average estimated measurement error variance, which is estimated rather precisely given the large sample size.

Given the varying nature of the estimation techniques and assumptions used, bias is a potential issue. Table 7.8 provides further information on the bootstrap analyses to address this question for some of the estimation methods. Recall that the one-stage bootstrap does not provide a bootstrap estimate of bias; see Section 6.16. A comparison of the bootstrap mean to the estimate for the one-stage bootstrap provides the misleading impression that bias is not of any concern. The two-stage bootstrap explicitly resamples from a model with coefficients equal to the estimates and does provide a bootstrap estimate of bias via the bootstrap mean − original corrected estimate. Note: Each two-stage bootstrap was tailored to generate responses using the corrected estimate associated with the method under consideration. For this particular example, the MEE and normal discriminant estimates have minimal bias concerns compared to the regression calibration estimates, which have indications of some downward bias.

Figure 7.1 *Illustration of determination of SIMEX estimates in cholesterol example.*

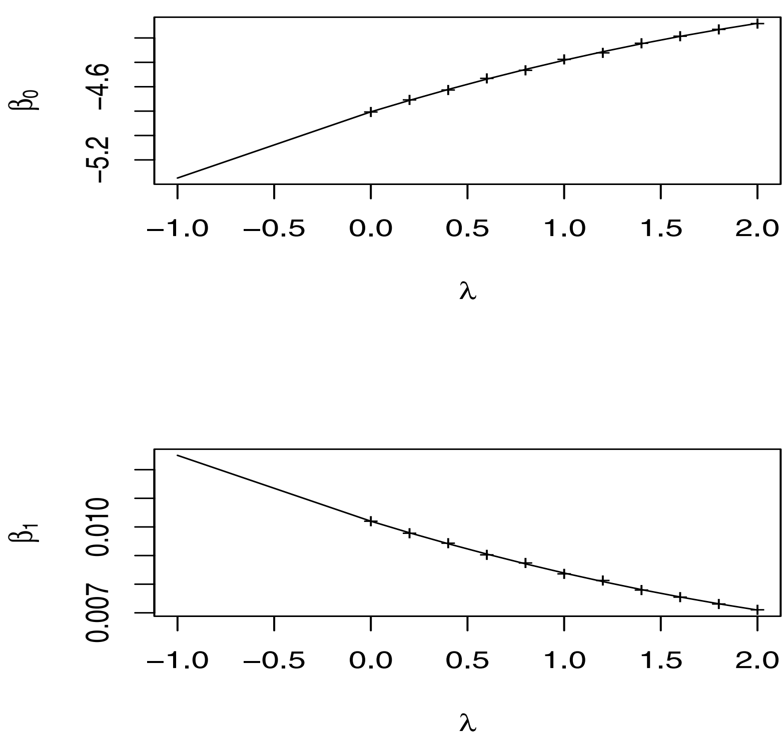


Table 7.5 *Cholesterol/heart disease example. Estimation of* β_1 *and odds ratio. All confidence intervals are 95%. See text for additional discussion on labels.*

Method		Estimate	SE	CI
Naive	β_1	.0102	0.00293	(0.0045, 0.0159)
			0.0027 (B)	(0.0053, 0.0155)
			0.0027 (B2)	(0.0036, 0.0140)
	OR	1.010	0.0021	(1.004, 1.016)
			0.0027 (B)	(1.005, 1.016)
			0.0027 (B2)	(1.004 ,1.014)
RC	β_1	.0133	0.0035 (B)	(0.0069, 0.0206)
			0.0031 (B2)	(0.0041, 0.0159)
			0.0038 (Impute)	(0.0058, 0.0209)
			0.0034 (B:STATA)	(.0066, .0201)
			0.0038 (A:STATA)	(0.0058, 0.0209)
	OR	1.0134	0.0036 (B)	(1.007, 1.021)
			0.0031 (B2)	(1.004, 1.016)
			(Impute)	(1.006,1.021)
RC-I	β_1	.0123	0.0032 (B)	(0.0063, 0.0189)
			0.0030 (B2)	(0.0039, 0.0155)
			0.0035 (Impute)	(0.0054, 0.0192)
	OR	1.0124	0.0033 (B)	(1.006, 1.019)
			0.0030 (B2)	(1.004, 1.016)
			(Impute)	(1.005, 1.019)
MEE	β_1	.0135	0.0037 (B)	(0.0065, 0.0209)
			0.0031 (B2)	(0.0074, 0.0194)
			.0029 (Nlin)	
	OR	1.0143	0.0037 (B)	(1.007, 1.021)
			0.0031 (B2)	(1.007, 1.020)
N.Disc	β_1	.01415	0.0040 (BN)	(0.0061, 0.0218)
			0.0042 (BP)	(0.0059, 0.0218)
SIMEX	β_1	.0125	0.0031 (B:STATA)	(.0065, .0185)
		.0138 (Ind)		
MLE	β_1	.0134	0.0039	(0.0058, 0.0210)

Table 7.6 *Cholesterol/heart disease example. Estimated odds ratio and 95% confidence intervals for changes of 10 (OR-10) and 100 (OR-100) in cholesterol. Using two-stage bootstrap intervals.*

	OR-10		OR-100	
Method	Estimate	CI	Estimate	CI
Naive	1.105	(1.037, 1.150)	2.705	(1.432, 4.056)
RC	1.142	(1.042, 1.172)	3.785	(1.506, 4.891)
RCII	1.131	(1.040, 1.167)	3.429	(1.476, 4.702)
MEE	1.153	(1.077, 1.214),	4.137	(2.090, 6.970)

Table 7.7 *Cholesterol/heart disease example. Estimate, standard errors and 95% confidence intervals for β_0.*

Method	Estimate	SE	CI
Naive	-4.8072	0.7058	(-6.1905, -3.4239)
		0.6494 (B)	(-6.1052, -3.6472)
		0.6515 (B2)	(-5.7280, -3.2751)
RC	-5.531	0.8427 (B)	(-7.2229, -4.0035)
		0.7341(B2)	(-6.1577, -3.3809)
		0.8087 (B:STATA)	(-7.1174, -3.9448)
		0.7049 (A:STATA)	(-6.9120, -4.1402)
		0.9121 (Impute)	(-7.3188, -3.7434)
RC-I	-5.283	0.7723 (B)	(-6.8612, -3.9024)
		0.7100 (B2)	(-6.0238, -3.3087)
		.8349 (Impute)	(-6.9195, -3.6468)
SIMEX	-5.352	0.7351(B:STATA)	(-6.794, -3.910)
	-5.423 (Ind)		
MEE	-5.5840	0.8848 (B)	(-7.4254, -3.9305)
		0.7498(B2)	(-7.0482, -4.1397)
		.7115 (Nlin)	
MLE	-5.554	.9292	(-7.3758, -3.7335)

Table 7.8 *Cholesterol/heart disease example. Further bootstrap analysis of parameters using one and two-stage bootstrap sampling with bias and relative bias estimated from the two-stage bootstrap.*

β_0

Method	Estimate	BMean-1	BMean -2	Bias	Rel.Bias
Naive	-4.8072	-4.854	-4.5385	0.269	-5.590
RC	-5.531	-5.593	-4.8327	0.698	-12.625
RC-I	-5.283	-5.328	-4.734	0.549	-10.392
MEE	-5.584	-5.623	-5.575	0.009	-0.161
MEE(G=0)	-4.817	-4.172	-4.307	0.510	-10.588

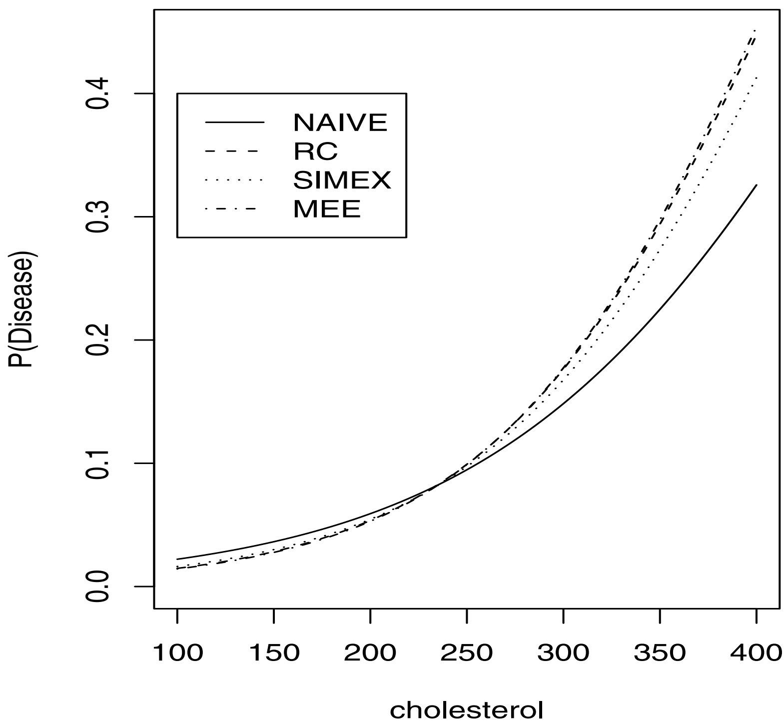
β_1

Method	Estimate	BMean-1	BMean -2	Bias-2	Rel.Bias
Naive	.0102	.01037	.0090	-0.0012	-11.7647
RC	.0134	.0136	.0103	-0.0031	-23.1343
RC-I	.0123	.0125	.0098	-0.0025	-20.3252
MEE	.0135	.0136	.0135	0.0000	0.0000
MEE(G=0)	.0103	.0076	.0082	-0.0021	-20.3883
ND	.0142	.0139(N)	.0141(N)	-0.0001	-0.7042

Odds Ratio

Method	Estimate	BMean-1	BMean -2	Bias-2	Rel.Bias
Naive	1.0103	1.0104	1.0091	-0.0012	-0.1188
RC	1.0134	1.0137	1.0103	-0.0031	-0.3059
RC-I	1.0124	1.0126	1.0099	-0.0025	-0.2469
MEE	1.0143	1.0137	1.0136	-0.0007	-0.0690
MEE(G=0)	1.0103	1.0076	1.0082	-0.0021	-0.2079

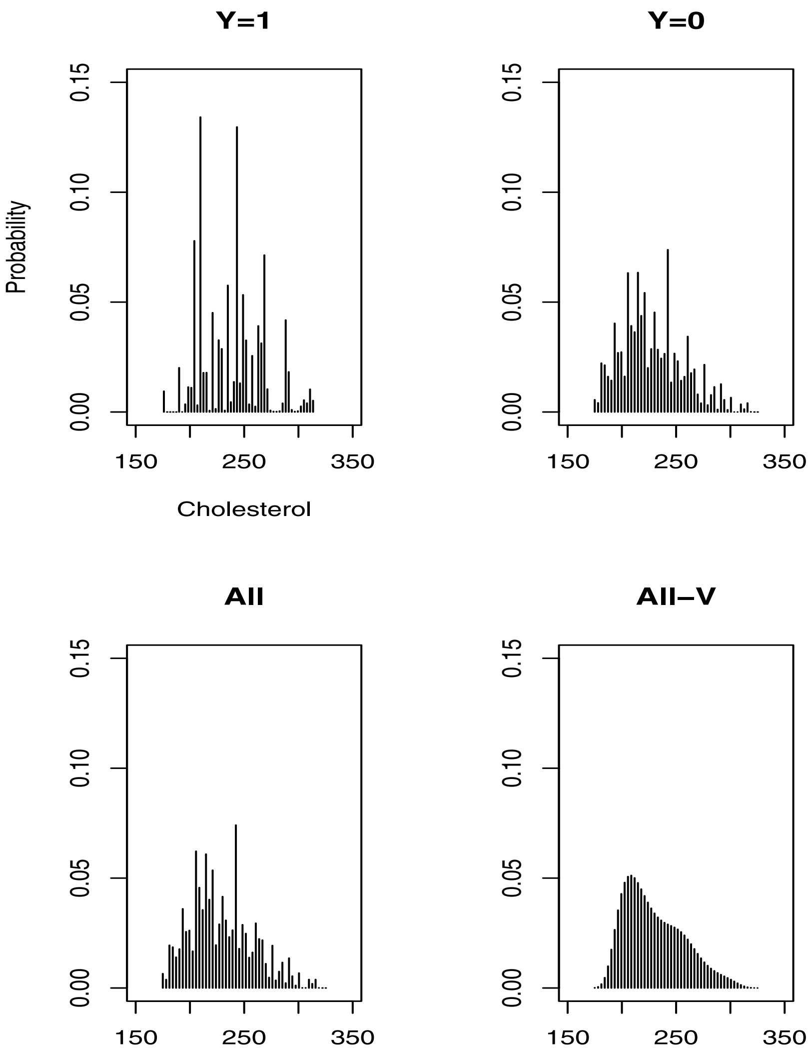
Figure 7.2 *Naive and corrected estimators of the regression function for the cholesterol example.*



7.2.3 *Example: Heart disease with multiple predictors*

Here we reconsider the Framingham data, still using 1475 selected observations, but now consider a logistic model with age, smoking (a binary variable indicating smoker or not), systolic blood pressure and cholesterol. The last two are both measured with error with two replicates for each. If these are taken as the first two variables then $\mathbf{x}_i' = (x_{i1}, x_{i2}, x_{i3}, x_{i4})$ and $\mathbf{w}_i' = (w_{i1}, w_{i2}, x_{i3}, x_{i4})$. For illustration we fit a logistic model with cholesterol and systolic blood pressure on their original scale using RC, SIMEX and MEE with

Figure 7.3 *Nonparametric estimation of distribution of cholesterol. Based on individual measurement error variances except All-V which uses fitted variances.*



results given in Table 7.9. The RC and SIMEX results are directly from STATA. The three methods yield very similar corrected inferences in this case. As expected, the biggest changes are in the coefficients for systolic blood pressure and cholesterol. As in the previous example, the impact of the correction on the odds ratios looks modest because these are a change of a single unit. As noted elsewhere, the corrections also alter the estimates of the perfectly measured predictors, in this case leading to a slight increase in the coefficient (and odds ratio) for smoking and a slight decrease for age.

7.2.4 Notes on ecological applications

While our examples above are epidemiologic, another very important area where measurement error occurs in binary regression is in ecological settings. In many studies there is a binary outcome observed (nesting success, presence or absence of a species, etc.) at some location and the goal is to relate this outcome to associated habitat (ground cover, canopy cover, etc.) or weather variables. Habitat variables are often characterized by some sampling of the surrounding area, while weather variables are usually obtained from nearby monitoring stations or other reports. Both scenarios introduce measurement error in the predictors. Klaus and Buehler (2001) is just one of a large number of examples that related bird breeding behavior to habitat variables. Clark et al. (2001) provide another example where measurement error might be a concern. They examined the presence or absence of establishment of the insect *Agapeta zoegana* which was introduced to control an invasive plant, spotted knapweed. In addition to other variables, precipitation was one of the predictors. It was measured at the nearest monitoring station to the site and that value will obviously differ from the actual value at the location of interest. This is an example where one could consider modeling either the truth given the observed, or vice versa.

As with all of our applications, the impact of the measurement error in these contexts may or may not be important. It depends on the range of the true predictors involved. In general, the larger the spread of the true values, the less the impact. For correcting, when there is sampling involved, estimates of the measurement error variance and covariances can be obtained. With replication this would proceed as in the egg mass/defoliation example used in Chapters 4 and 5. In other cases these estimates, as well as the estimates of the true values, would be built based on the sampling technique used.

7.2.5 Fitting with logs

How do we correct for measurement error if we have additive error for the variables in $\mathbf{x}$ but the model of interest involves nonlinear functions of $\mathbf{x}$, such

Table 7.9 *Framingham data. Analysis of coefficients with both systolic blood pressure and cholesterol both measured with error. CI = (Lower,Upper).*

Coefficient	Estimate	B-Mean	SE	Lower	Upper
NAIVE					
intercept	-10.1257		1.1003		
SBP	0.0173		0.00443		
Chol	0.0101		0.00304		
Smoke	0.5440		0.2572		
Age	0.0537		0.0122		
MEE: One-stage bootstrap					
intercept	-11.14979	-11.134	1.2558	-13.612	-8.8217
SBP	0.0196677	0.0199	0.0051	0.0096	0.0296
Chol	0.01336	0.0130	0.0039	0.0052	0.0205
Smoke	0.560203	0.5781	0.2587	0.1166	1.0926
Age	0.0521569	0.0524	0.0111	0.0310	0.0756
MEE: Two-stage bootstrap					
intercept	-11.14979	-11.805	1.2407	-14.163	-9.4421
SBP	0.0196677	0.0229	0.0051	0.0131	0.0329
Chol	0.01336	0.0146	0.0034	0.0079	0.0220
Smoke	0.560203	0.5932	0.2678	0.0977	1.1616
Age	0.0521569	0.0498	0.0127	0.0254	0.0742
RC- bootstrap					
intercept	-11.08264		1.10135	-13.24302	-8.922253
SBP	.0196986		.0046981	.0104829	.0289143
Chol	.013217		.0038089	.0057456	.0206884
Smoke	.5544538		.2482588	.0674746	1.041433
Age	.0518879		.0107299	.0308404	.0729354
RC- analytical					
intercept	-11.08264				
SBP	.0196986		.0049281	.0100317	.0293655
Chol	.013217		.003833	.0056983	.0207357
Smoke	.5544538		.2478094	.068356	1.040552
Age	.0518879		.011885	.0285745	.0752014
SIMEX - bootstrap					
intercept	-10.98222		1.13487	-13.20835	-8.756077
SBP	.0191628		.0045044	.0103271	.0279986
Chol	.0129372		.0036502	.0057772	.0200973
Smoke	.5502957		.2542386	.0515865	1.049005
Age	.0523052		.0107563	.0312059	.0734044

Table 7.10 *Framingham data. Analysis of odds ratios with systolic blood pressure and cholesterol both measured with error. B2 denotes two-stage bootstrap.*

Method Variable	Estimate	SE	Lower	Upper
Naive				
SBP	1.017		1.009	1.026
Chol	1.010		1.004	1.016
Smoke	1.723		1.041	2.852
Age	1.055		1.030	1.081
MEE- B2				
SBP	1.0232	0.0052	1.0131	1.0335
Chol	1.0147	0.0035	1.0080	1.0223
Smoke	1.8781	0.5417	1.1026	3.1950
Age	1.051	2 0.013	3 1.0257	1.0771

as squares, products and logs? Section 8.4 provides a full discussion of this issue in the context of linear models. A lot of the discussion there is useful in nonlinear models. We will limit the discussion here to the use of logs and illustrate the problem by revisiting the cholesterol/heart disease example. The issues in working with logs were explored in some detail in Section 6.4.7 and is also treated further in Section 8.4.3. One of the points made earlier was that with $E(W_{ij}) = x_i$, $E(log(W_{ij}))$ is often approximately $log(x_i)$. If this is the case, this suggests the first two methods.

Method 1: Take the log of each original replicate and treat these as new replicates. Proceed as before using regression calibration, SIMEX, MEE, etc.

An alternative is

Method 2: Treat $log(W_i)$ as as an estimate of $\log(x_i)$, where W_i is the mean of the original replicates. The $log(W_i)$ will have smaller bias than the mean of the $log(W_{ij})$'s as an estimator of $log(x)$. If the bias of the latter is small to begin with, this is a moot point. If σ^2_{ui} is the measurement error variance of $\bar{W}_i$ as an estimate of x_i then the approximate variance of $log(W_i)$ is σ^2_{ui}/x_i^2. Notice that if the measurement error coefficient of variation is constant then the approximate measurement error variance on the log scale is constant. This can be estimated using $\widehat{\sigma}^2_{ui}/(W_i^2 - \widehat{\sigma}^2_{ui})$, which is now treated as the measurement error variance attached to $log(W_i)$. (One could encounter problems with $W_i^2 - \widehat{\sigma}^2_{ui}$ being negative if the measurement error variance is large enough.) With more than one variable measured with error, then the measurement error covariance, which involves the covariance of $log(W_i)$ with other mismeasured values, needs to be modified; see Section 8.4 for details. With an estimated covariance matrix, regression calibration, SIMEX, MEE, etc. can be used as before.

If we are unsure if either the log of a replicate or the log of the mean is approximately unbiased for $log(x)$ and it is the model in terms of $log(x)$ that is of interest, then some other strategy is needed. Here are a couple of possibilities.

Method 3: Use SIMEX by generating replicates on the original scale and then using the log of the mean of the replicates in fitting. Notice that in general any nonlinear function of the original predictors (measured with additive error) can be immediately accommodated by SIMEX.

Method 4: Use the modified estimating equation approach, allowing for bias in $log(W_i)$ as an estimator of $log(x_i)$. This is described in more detail in linear models in Section 8.4.3.

The cholesterol example revisited.

Here we'll revisit the example modeling the probability of heart disease but assume the model is now logistic in $log(chol)$, i.e. $logit(P(Y = 1|chol)) = \beta_0 + \beta_1 log(chol)$. We'll continue to work with the subsetted data set of 1475 individuals. Naive analyses suggest a slightly better fit using log(chol) rather than chol itself, but the difference is fairly small. In Section 6.4.7, we showed that there is negligible bias in the log of a replicate as an estimate of the log of the true value. So, an analysis using the logs of the replicates is justified, but we also present the analysis using the log of the mean for comparison. Results are given in Table 7.11. Estimates are obtained using methods 1, 2 and 3 listed above. Under method 1, the log of each replicate is taken and then we proceed as before. This is done using SIMEX, regression calibration (RC) and MEE. The first two are computed using STATA with the RC inferences given using both the bootstrap and the analytical results. In using the MEE both one and two-stage bootstrap sampling is used. For the two-stage bootstraps the mean is given in parentheses next to the estimate. Analyses that use the log of the mean are indicated by an M, including the naive analysis, MEE-M (which corresponds to method 2 listed above) and SIMEX-M (method 3 above). Only the estimates were obtained for using SIMEX in the latter fashion. There are small differences between analyses using the mean of the replicates and those using the logs of the replicates. This is expected given the bias exploration in Section 6.4.7. There is general agreement among the estimation methods. The two-stage bootstrap provides an estimate of bias. This is done here just for the MEE estimates for illustration. The potential bias in the coefficients is relatively small. The bias looks like more of an issue with the odds ratio, but this does not dilute the usefulness of the confidence intervals for the odds ratio, which, since we are using the bootstrap, are just a transformation of the intervals for β_1.

Table 7.11 *Analysis of β_1 for heart disease-cholesterol data with logistic model in log(cholesterol). Naive-M, SIMEX-M and MEE-M use the log of the mean. The others use the log of replicates. Boot1 indicates one-stage bootstrap. Boot2 indicates two-stage bootstrapping. Bootstrap mean in parentheses.*

	β_1		
Method	Estimate	SE	CI
Naive	2.5199	.7098	(1.1287, 3.9111
Naive-M	2.5106	.7100	(1.1190, 3.9022
RC-Boot	3.3208	.8490	(1.6554, 4.9861
RC-A	3.3207	.9344	(1.4878, 5.1536
SIMEX	3.1706	.8359	(1.5309, 4.8104
SIMEX-M	3.2805		
MEE-Boot1	3.2734	0.9157	(1.5138, 5.0923)
MEE-Boot2	3.2734 (3.2322)	0.7612	(1.7999, 4.6925)
MEE-M-Boot2	3.2963 (3.2969)	0.7673	(1.8335, 4.9335)
	Odds Ratio		
Naive	12.312		3.062 49.509
MEE-Boot2	26.402 (33.7392)	28.5163	(6.0490, 109.1256)
	β_0		
Naive	-16.1403	3.8806	(-23.74618, -8.5343)
Naive-M	-16.0946	3.8831	(-23.7053, -8.4838)
RC-Boot	-20.4873	4.6379	(-29.5849, -11.3897)
RC-A	-20.4873	3.8709	(-28.0804, -12.8943)
SIMEX	-19.6859	4.5682	(-28.6467, -10.7250)
SIMEX-M	-20.2991		
MEE-Boot1	-20.2443 (-20.0780)	4.999	(-30.14421, -10.5551)
MEE-Boot2	-20.2443 (-20.0264)	4.1655	(-28.1111, -12.2650)
MEE-M-Boot2	-20.3763 (-20.3804)	4.1994	(-29.2990, -12.4521)

7.3 Using validation data

This section uses a combination of real and simulated data to illustrate the use of validation data, as summarized in Section 6.15. If we have external validation data and the Berkson model is assumed to be exportable to the main study, then regression calibration type methods can be applied. This is illustrated in Section 7.3.1 using two previously published studies, the Nurses Health Study and the Harvard Six Cities Study. Section 7.3.2 describes some ways to correct for measurement error using internal validation data and the linear Berkson model with an illustration using simulated data. Section 7.3.3 returns to the use of external validation data, but now only assumes that the measurement error model is transportable. This section also describes how (with some assumptions) we can use the measurement error model from the external data plus the data from the main study to estimate the Berkson error model for the main study.

7.3.1 Two examples using external validation and the Berkson model

Here we present published results from two epidemiologic studies, the Nurses Health Study and the Harvard Six Cities Study. Both of these utilize external validation data, with a linear Berkson model relating the true values to the error-prone measures.

The Nurses Health Study.

This analysis, taken from Rosner, Spiegelman and Willett (1990), looks at a cohort study of $n = 89538$ women from the Nurses Health Study and examines the influence of dietary intakes on the occurrence of breast cancer. The outcome Y is an indicator of breast cancer, with 601 cases ($Y = 1$). The predictor variables involved are x_1 = saturated fat (g/day), x_2 = total calories (kg/day), x_3 = alcohol (g/day), and x_4 to x_7 are four dummy variables associated with age group membership. There are five age groups with the coefficients and odds ratios for these variables defined relative to the reference group, which is ages 35-39. The goal is to fit the logistic model

$$logit(P(Y = 1|\mathbf{x})) = \beta_0 + \beta_1'\mathbf{x},$$

where $\mathbf{x}_i' = (x_{i1}, x_{i2}, x_{i3}, x_{i4}, x_{i5}, x_{i6}, x_{i7})$ and x_{i1}, x_{i2} and x_{i3} are measured with error. The age group variables are assumed known without error.

The error-prone measures of daily fat, calories and alcohol, w_{i1},w_{i2} and w_{i3}, are obtained from a food frequency questionnaire (FFQ) leading to

$$\mathbf{w}_i' = [w_{i1}, w_{i2}, w_{i3}, x_{i4}, x_{i5}, x_{i6}, x_{i7}].$$

A linear Berkson model is assumed with $E(\mathbf{X}_i|\mathbf{w}_i) = \boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}_i$. This models the true values as linear functions of the FFQ values, with each true value allowed to depend on any of the FFQ values. Since the last four components are measured without error,

$$\boldsymbol{\lambda}_0 = \begin{bmatrix} \boldsymbol{\lambda}_{01} \\ 0 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{\Lambda}_1 = \begin{bmatrix} \boldsymbol{\Lambda}_{11} \\ \boldsymbol{\Lambda}_{12} \end{bmatrix},$$

where $\boldsymbol{\lambda}_{01}$ is a 3 by 1 vector, $\boldsymbol{\Lambda}_{11}$ is a 3 by 7 matrix and $\boldsymbol{\Lambda}_{12}$ is a 4×7 matrix equal to = $(\mathbf{0}, \mathbf{I}_4)$.

An independent validation sample, with 173 individuals, has both "true" x values, obtained from average weighed food records and the FFQ values. The food records are not real true values, but in fitting the Berkson model any additive noise in these values will not bias estimation of the Berkson coefficients. The resulting estimates are $\hat{\boldsymbol{\lambda}}_{01}^{'} = (21.0, 1300.7, 4.7)$ and

$$\hat{\boldsymbol{\Lambda}}_{11} = \begin{bmatrix} .38 & -.0003 & .09 & -.08 & -3.5 & -2.5 & -3.1 \\ .29 & .23 & 4.8 & -15.6 & -141.4 & -58.8 & -36.5 \\ .003 & -.001 & .672 & -.519 & -.001 & .11 & .14 \end{bmatrix}.$$

They use regression calibration, equivalently linear transforming the naive estimators (see Sections 6.9.1 and 6.10), to obtain corrected estimates and obtain an estimate of $\Sigma_{\hat{\beta}}$ using the method described in Section 6.9. The analyses are presented in Tables 7.12 and 7.13. The impact of the correction for measurement error is greatest on the fat and alcohol coefficient and their associated odds ratios. We also see some changes in the coefficients for the age values, which recall were measured without error.

Table 7.12 *Naive and corrected estimates and standard errors for Nurses Health Study. From Rosner et al. (1990). Used with permission of Oxford University Press.*

	Naive		Corrected	
Variable	$\hat{\beta}_{naive}$	SE(naive)	$\hat{\beta}$	SE
intercept	-5.58	.	5.586	.
1(fat)	-.00063	.00063	-.0169	.0176
2 (cal)	.000023	.00015	.000025	.000553
3 (alcohol)	.0115	.0031	.0192	.0055
4 (age1: 40-44)	.350	.157	.358	.160
5 (age2: 45-49)	.827	.143	.772	.155
6 (age3: 50-54)	.813	.144	.770	.152
7 (age4: 55 +)	.978	.145	.923	.157

Table 7.13 *Naive and Corrected odds ratios and approximate 95% confidence intervals for Nurses Health Study. From Rosner et al. (1990). Used with permission of Oxford University Press.*

Variable	Naive	CI	Corrected	CI
Fat (increment of 10g)	.94	(.83, 1.06)	.84	(.59, 1.2)
Calories (increment of 800)	1.02	(.81, 1.28)	1.02	(.46, 2.24)
alcohol (increment of 25g)	1.33	(1.14, 1.55)	1.62	(1.23, 2.12)

The Harvard Six Cities Study. The Harvard Six Cities Study assessed the influence of air pollutants on childhood respiratory illnesses. Here we present an analysis from Tosteson et al. (1989). The goal was to fit a probit model $P(Y = 1|x) = \Phi(\beta_0 + \beta_1 x)$, where Y was an indicator of whether a child had wheezing (or persistent wheezing; a separate outcome) and x was the child's exposure to nitrous dioxide (NO_2). Measuring the true value x is difficult and expensive. This example is interesting for its use of multiple surrogates for a univariate true value and also making use of the multiple surrogates to develop a test for conditional independence/surrogacy (something we do not discuss here). The surrogate for x is two dimensional, consisting of W_1 and W_2, which are measures of NO_2 measured in the child's bedroom and the kitchen in the house. The main study consisted of $n = 231$ children. The validation data were external, coming from two different studies, one in Wisconsin and one in the Netherlands, with the "true value" measured with the use of a personal lapel monitor worn by the subject. A Berkson model is fit regressing x on w_1 and w_2, with very similar results from the two studies. A combined analysis was used based on $E(X|w_1, w_2) = 1.22 + .3w_1 + .33w_2$ and an estimate for $V(X|w_1, w_2)$ of $\widehat{\sigma}^2_{X|\mathbf{w}} = .06$. Under normality assumptions and the probit model, the pseudo-MLE's can be computed in closed form as described in Section 7.2.1 and standard errors obtained treating the Berkson parameters as known. For the wheezing outcome this resulted in $\widehat{\beta}_1 = -.08$ with a standard error of .21, with the large uncertainty due in part to the small sample size. This analysis turns out to be very close to a regression calibration approach which just fits a probit model of Y on $\widehat{X}_i = \widehat{\lambda}_0 + \widehat{\lambda}_1' \mathbf{w}_i$. As noted by the authors, and emphasized elsewhere in this book, the analysis depends critically on the Berkson model being transportable from the external validation studies to the main study.

7.3.2 Fitting with internal validation data and the Berkson model

Here we implement some of the strategies outlined in Section 6.15.2 for the use of internal validation data. To illustrate this we use simulated data with a binary disease outcome modeled as a function of saturated fat and caloric intake. This is connected to the Nurses Health Study only in that the generated data was based in some part on the values associated with the validation data from that study. *The regression coefficients in the binary regression model have nothing to do with the original analysis.* Specifically the generated true values for saturated fats and daily caloric intake (drsat and drcal) and the corresponding measured values from the FFQ (fqsat and fqcal) maintained some of the correlation structure and the measurement error model that was present in the validation data. 3000 observations were generated by sampling from the diet record and FFQ values in the validation data but over a limited range of the caloric intake.

For each observation a response Y_i was generated with $P(Y = 1)$ based on a logistic model using the diet values and $\boldsymbol{\beta}' = (-3.75, .0262, .000867)$, which corresponds to odds ratios of 1.3 (per 25g) for saturated fats and 2 (per 800 kg) for daily caloric intake. β_0 was chosen to ensure a sufficient number of cases to avoid fitting problems. In the validation sample 51 of the 300 observations have $Y = 1$, while 410 of the full 3000 have $Y = 1$. This is to avoid fitting problems which can occur with a "small" sample size and rare outcomes. This is a general problem in binary regression, not just when correcting for measurement error. In practice the sample size for rare events would usually be larger than 3000, but 3000 is used here because of the computational time involved in running the analyses. The first 300 observations will serve as the validation data while the other 2700 are treated as having only the FFQ values and outcome.

For the 300 validated samples the estimated Berkson coefficients from regressing the drsat and drcal on fqsat and fqcal are $\hat{\boldsymbol{\lambda}}_{01}' = (20.47, 1411.4)$ and

$$\hat{\boldsymbol{\Lambda}}_{11} = \begin{bmatrix} .419 & -.004 \\ .07 & 5.1 \end{bmatrix},$$

where we have used the same notation as in (7.3.1).

Five analyses were run with results given in Table 7.14.

1. An analysis based on the validation sample of 300 only.
2. Naive-NV. A naive analysis on the 2700 unvalidated observations.
3. RC-ext. This treats the 300 validated cases as external validation, fits the Berkson model and then corrects the estimates from the 2700 nonvalidated observations, as done in the preceding section.
4. Weighted: This uses a weighted average of the estimates from items 1 and 2.
   This is the estimator proposed by Spiegelman et al. (2001) for internal validation data. This and the RC-EXT fit were obtained using the SAS macro Blinplus8 (see the preface).
5. RC-all. This fits a logistic regression using the 2700 imputed values (based on the Berkson model) for the nonvalidated observations and the true values for the 300 validated observations. The standard errors are as given directly by the logistic analysis, which does not account for the uncertainty from estimating the Berkson parameters.

There are obvious gains in precision from using all 3000 observations rather than just the 300 validated ones. Treating the validation data as external rather than internal is also seen to be inefficient.

The analysis from the weighted estimates and the estimates computed using

Table 7.14 *Analysis of simulated binary regression data, fitting a logistic model as a function of saturated fat and caloric intake and using internal validation data. Overall sample size is $n = 3000$ with $n_V = 300$ composing the internal validation data. See text for labels on methods.*

Method	Validated only		RC-Ext		Naive-NV	
Predictor	Estimate	SE	Estimate	SE		
Intercept	-5.335	1.2694			-1.9834	.1640
fat	-.0468	.0427	.0183	.0251	.0078	.0010
calories	.003	.0011	.00097	.00099	.00007	.0002

Method	Weighted		RC-all	
Predictor	Estimate	SE	Estimate	SE
Intercept			-4.654	.8752
fat	.00049	.0216	.0011	.0209
calories	.0018	.0007	.0018	.0007

true values for the validated observations and imputed predictors for the nonvalidated observations are identical with respect to estimation of the coefficient for calories, but differ a bit with the estimate of the coefficient for saturated fats, which has a large amount of uncertainty. Theoretically, the weighted estimator has smaller asymptotic variances than the other estimators, as shown by Thurston et al. (2005). They also investigated an estimator that is similar to RC-all but uses a general quasi-likelihood approach which weights each observation.

The analyses given rely on the linear Berkson model holding, which was reasonable with this data. In general, this should be checked with the validation data. If necessary, other models can be used to regress $\mathbf{X}$ on $\mathbf{w}$. Then the easiest way to proceed is as in method 4, imputing missing predictors for the nonvalidated observations and then running a logistic analysis. This will lead to standard errors that are too small but this can be compensated for in various ways, including through bootstrapping.

7.3.3 Using external validation data and the measurement error model

We describe three methods that could be used with external validation data but employing the measurement, rather than Berkson, error model. As noted in Section 6.15.1 if the Berkson error model is transportable, then the methods based on that model should be used. Even then, there may be some advantages to using the measurement error model if there are restrictions on it, but it remains to be seen what advantages that offers. The techniques are described here under the assumption of a linear measurement error model with constant covariance matrix; that is $E(\mathbf{W}|\mathbf{x}) = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{x}$ and $Cov(\mathbf{W}|\mathbf{x}) = \boldsymbol{\Sigma}_{W|x}$. (Note: We have used $\boldsymbol{\Sigma}_{W|x}$ here rather than $\boldsymbol{\Sigma}_u$, since the latter will get used in a different way in the correction methods that follow.)

Broadly the three approaches are: 1) Get adjusted values with approximately additive error and apply the methods for additive error; 2) Get imputed values by combining the measurement error model from the external data and the data in the main study to build a Berkson model for the true values given the observed values; 3) Modify the estimating equations. We'll see that there is a connection between methods 1 and 2 in that using regression calibration in 1 is equivalent to the method in 2.

1. Using adjusted values. This method obtains adjusted values which have approximate additive error, along with an associated measurement error covariance and then uses the correction methods for additive error. Using the linearity of the measurement error leads to the adjusted values

$$\mathbf{x}_{Ai} = \widehat{\boldsymbol{\Theta}}_1^-(\mathbf{W}_i - \widehat{\boldsymbol{\theta}}_0),$$

where $\widehat{\boldsymbol{\Theta}}_1^- = (\widehat{\boldsymbol{\Theta}}_1'\widehat{\boldsymbol{\Theta}}_1)^{-1}\widehat{\boldsymbol{\Theta}}_1'$ is a "generalized inverse" of $\widehat{\boldsymbol{\Theta}}_1$. This allows $\mathbf{W}$ to have more values than $\mathbf{X}$ (it can not have less). If $\mathbf{W}$ and $\mathbf{X}$ contain an equal number of elements then $\widehat{\boldsymbol{\Theta}}_1^- = \widehat{\boldsymbol{\Theta}}_1^{-1}$.

If the measurement error parameters were known then $\mathbf{x}_{Ai}$ is unbiased for $\mathbf{x}_i$ with covariance $\boldsymbol{\Sigma}_{ui} = \widehat{\boldsymbol{\Theta}}_1^-\boldsymbol{\Sigma}_{W|x}\widehat{\boldsymbol{\Theta}}_1^{-'}$. The methods for additive error (SIMEX, RC, MEE, etc.) can now be applied with $\mathbf{x}_{Ai}$ playing the role of $\mathbf{W}_i$ in the additive formulation and an estimate of $\boldsymbol{\Sigma}_{ui}$ obtained using an estimate of $\boldsymbol{\Sigma}_{W|x}$. For regression calibration an explicit expression for the resulting imputed values is given in (7.5) below.

This method can be vulnerable with small or even moderately size validation samples. First, given the uncertainty in $\widehat{\boldsymbol{\Theta}}_1$ and $\widehat{\boldsymbol{\theta}}_0$, $\mathbf{x}_{Ai}$ is not unbiased for $\mathbf{x}_i$ and there is additional uncertainty in $\boldsymbol{\Sigma}_{ui}$ beyond what is given in the expression above. In addition, the measurement errors associated with these adjusted values are now correlated since they are all calculated using common coefficients. Analytical calculation of the covariance of the resulting estimators will be difficult, but bootstrapping (with resampling of both the external data and the main study) is always an option.

2. Imputation/regression calibration. First we show how one can fit the Berkson model for the main study, based on the measurement error model from the external data and the observed values in the main study.

With the linear measurement error for $\mathbf{W}|\mathbf{x}$ and $\mathbf{X}$ having mean $\boldsymbol{\mu}_X$ and covariance $\boldsymbol{\Sigma}_X$, the best linear predictor of $\mathbf{X}$ given $\mathbf{w}$ (and $E(\mathbf{X}|\mathbf{w})$ under normality is $\boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\mathbf{w}$, where from (6.12),

$$\boldsymbol{\Lambda}_1 = \boldsymbol{\Sigma}_X\boldsymbol{\Theta}_1'(\boldsymbol{\Theta}_1\boldsymbol{\Sigma}_X\boldsymbol{\Theta}_1' + \boldsymbol{\Sigma}_{W|x})^{-1}, \qquad \boldsymbol{\lambda}_0 = \boldsymbol{\mu}_X - \boldsymbol{\Lambda}_1(\boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\boldsymbol{\mu}_X).$$

Using estimates of $\boldsymbol{\theta}_0$ and $\boldsymbol{\Theta}_1$ from the validation data and the covariance matrix of the $\mathbf{W}$ values from the main study (assuming an overall random sample), the structural parameters for $\mathbf{X}$ can be estimated by

$$\widehat{\boldsymbol{\Sigma}}_{XX} = \widehat{\boldsymbol{\Theta}}_1^-(\mathbf{S}_{WW} - \hat{\boldsymbol{\Sigma}}_{W|x})\widehat{\boldsymbol{\Theta}}_1^{\prime -} \text{ and } \widehat{\boldsymbol{\mu}}_X = \widehat{\boldsymbol{\Theta}}_1^-(\bar{\mathbf{W}} - \widehat{\boldsymbol{\theta}}_0).$$

(See the discussion and example in Section 8.5 for related discussion.) These can be used to estimate $\boldsymbol{\lambda}_0$ and $\boldsymbol{\Lambda}_1$ which yields an estimated model for $\mathbf{X}|\mathbf{w}$ in the main study.

We can now consider using this fitted Berkson model to get imputed values

$$\widehat{\mathbf{x}}_i = \widehat{\lambda}_0 + \widehat{\boldsymbol{\Lambda}}_1\mathbf{w}_i = \widehat{\boldsymbol{\Theta}}_1^-(\mathbf{S}_{WW} - \widehat{\boldsymbol{\Sigma}}_{W|x})\mathbf{S}_{WW}^{-1}\mathbf{w}_i \tag{7.5}$$

and running the analysis using these imputed values. These imputed values turn out to be the same as what results when we apply regression calibration in the context of method 1.

3. Modified estimating equations. This allows for a functional setting and does not rely on the approximate additivity of the adjusted values. The expected value of the naive estimating equations under additive measurement error was given in Section 7.2.1. The only difference here is that $E(\mathbf{W}_i|\mathbf{x}_i) = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{x}_i$, instead of $\mathbf{x}_i$. With some substitution and simplification the expected value of the naive estimating equations is $\sum_i C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta})$ where

$$C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}) = -G_i\begin{bmatrix} 1 \\ \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{x}_i \end{bmatrix} - \begin{bmatrix} 0 \\ Cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta})) \end{bmatrix}.$$

As in the additive case, $G_i = E(m(\mathbf{W}_i, \boldsymbol{\beta})) - m(\mathbf{x}_i, \boldsymbol{\beta})$, while $\boldsymbol{\theta}_i$ contains the measurement error parameters, $\boldsymbol{\theta}_0$, $\boldsymbol{\Theta}_1$ and $Cov(\mathbf{W}_i|\mathbf{x}_i)$. The quantity $m(\mathbf{W}_i, \boldsymbol{\beta})$ is the binary regression model (e.g., the logistic) given in terms of the random $\mathbf{W}_i$. Both G_i and $Cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta}))$ depend on $\boldsymbol{\beta}$, $\boldsymbol{\theta}_i$ and $\mathbf{x}_i$.

As in our other uses of this method we proceed by iteratively fitting a nonlinear weighted regression model with updated responses. The updated responses are calculated using (6.59) with $\widehat{\mathbf{A}}_i(\widehat{\boldsymbol{\beta}}_{(k)})$ being an estimate of $C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta})$ at the kth step. This needs estimates of G_i and $Cov(\mathbf{W}_i, m(\mathbf{W}_i, \boldsymbol{\beta}))$, which can be computed using approximations, along with the $\widehat{\boldsymbol{\theta}}$'s, and either the adjusted or imputed value for $\mathbf{x}_i$.

If the measurement error is significantly noisy, that is $\boldsymbol{\Sigma}_{W|x}$ is too big, this method may not work well at all. This is the case if we try to use the validation data from the Nurses Health Study in this way. This points out one of the big advantages of working with the Berkson rather than the measurement error model.

Example.

Here is a small, partially toy example, illustrating a couple of the methods just described. Consider the example from Section 7.2.2 but now suppose we want to model the probability of coronary heart disease as a function of cholesterol at a fixed point in time, let us say the time corresponding to the first replicate in that sample (this is labeled cholest2 in the data set). In this case, the measurement error is essentially instrument error. Instruments that measure quantities like cholesterol are re-calibrated on a routine basis, using standards. (This is similar to examples in Sections 10.3 and 10.4.) I actually have an old set of calibration data from a friend who was a medical technologist. It used three replicates at each of three standards (50, 200 and 400) yielding a fitted regression model of $E(W_i|x_i) = 10.78 + .9374x_i$ with an estimated variance of $\sigma^2_{W|x} = 28.9$, or an estimated standard deviation for an observation of 5.38. Using this, we will examine correcting the logistic regression for the Framingham data using observations where cholest2 is limited to be between 175 and 325. This involves 1444 observations of which 120 are cases. We assume for illustration that the linear measurement error model associated with the calibration data holds. This is an example where the true values in the external data are fixed standards and so using the Berkson model to correct is not an option.

Using the methods above leads to an estimated measurement error variance associated with an adjusted value of $\widehat{\sigma}^2_u = 5.735$. Working with the adjusted values and using this estimated measurement error variance we correct as done earlier with additive error using the RC and MEE method. (This MEE is on the adjusted values, not method 3 above.) In bootstrapping we used a two-stage approach as used earlier but also resampled the calibration data (with the standards fixed) as part of each bootstrap sample. Result are given in Table 7.15. In this case, the measurement error associated with the adjusted values is so small, relatively, that the adjustment is fairly minor, and the results are very similar for both RC and MEE. Skipping the resampling of the validation data had very little impact on the solution. This happened since the coefficients were estimated fairly precisely, relative to the other terms involved. All of this is a result of fairly small noise around the regression line relating the measured value to the true value. As this increases we would see both larger adjustments and more of an effect from including the uncertainty associated with estimating the measurement error model.

Table 7.15 *Logistic regression of coronary heart disease on cholest2 for the Framingham data assuming linear measurement error as represented by model from calibration data. Naive analysis is from standard logistic regression. Inferences for the RC and MEE estimates are based on two-stage bootstrap that samples both from a regression model in the main study and resamples the calibration data.*

Naive	Estimate	SE	CI
Intercept	-4.7363	.6722	
Chol	.00091	.00277	
RC			
Intercept	-4.689	.6941	(-3.516,-4.762)
Chol	.0095	.0028	(.0153, .0098)
Adj-MEE			
Intercept	-4.684	.6667	(-3.427, -4.750)
Chol	.0095	.0027	(.0150, .0098)

7.4 Misclassification of predictors

Section 6.14 provided a brief overview on handling misclassified predictors in multiple predictor setting. We won't say more than that here but instead focus on the particular problem with a single misclassified predictor.

If the misclassified categorical predictor has J categories, with either external or internal validation data, then we are in the setting of Chapter 3 with a $2 \times J$ table. This section addresses the somewhat different problem where the categorical predictor comes from categorizing a quantitative variable, and there are replicate measures on the original quantitative variable. Categorizing is a common practice and as discussed in detail in Section 6.4.8, it leads to differential misclassification, even if the original error on the quantitative scale was nondifferential.

Suppose the categorical predictor is X_c with J categories and

$$\pi_j = P(Y = 1|X_c = j).$$

Interest may lie in the π_j's, differences among them, or in odds ratio, such as

$$OR_{jk} = \pi_j(1 - \pi_k)/\pi_k(1 - \pi_j)$$

which is the odds ratio for category j relative to category k. As noted in Chapter 3 this odds ratio can also be expressed as

$$OR_{jk} = \alpha_{j|1}(1 - \alpha_{k|0})/\alpha_{k|1}(1 - \alpha_{j|0}), \tag{7.6}$$

where

$$\alpha_{j|y} = P(X_c = j|Y = y).$$

Suppose that the quantitative variable X is assigned to category j if it falls in the region A_j. So,

$$\alpha_j = P(X_c = j) = P(X \in A_j) = \int_{x \in A_j} f_X(x)dx,$$

where $f_X(x)$ is the "density" function of X. Sometimes the A_j correspond to fixed cutpoints, other times to percentiles of the underlying distribution, which have to be estimated from the data. We also have

$$\pi_j = P(Y = 1|X_c = j) = \int_{x \in A_j} m(x, \boldsymbol{\beta}) f_X(x)dx/\alpha_j$$

where $m(x, \boldsymbol{\beta}) = P(Y = 1|X = x)$.

NOTE: We will use the word density and also use integrals, but X can be discrete in which case f_X is a mass function and the integral is a sum over possible values of X.

How do we correct for measurement error when we have replication? The replicates are on the original quantitative variable with the categorized value on the ith observation coming from categorizing W_i, the mean of the replicates. We assume the original quantitative variable is subject to additive error so as before $W_{ij}|x_i = x_i + u_{ij}$ where the u_{ij} have mean 0 and variance σ^2_{ui}. Various aspects of this problem are addressed by Dalen et al. (2006) and Dalen et al. (2009).

If we assume a model for $P(Y = 1|x) = m(x, \boldsymbol{\beta})$, then one approach is to estimate $\boldsymbol{\beta}$ using a correction techniques from earlier in this chapter, and then use

$$\widehat{\pi}_j = \int_{x \in A_j} m(x, \widehat{\boldsymbol{\beta}}) \widehat{f}_X(x)dx/\widehat{\alpha}_j$$

where $\widehat{f}_X(x)$ is an estimate of the density of X and $\widehat{\alpha}_j = \int_{x \in A_j} \widehat{f}_X(x)dx$. We would assume a random sample in order to estimate f_X, which can be done either parametrically or nonparametrically, as discussed in Section 10.2.3. If the A_j come from percentiles these would be estimated from the fitted distribution.

One disadvantage of the method above, in addition to requiring an overall random sample, is that it depends on a model for $P(Y = 1|x)$. One of the reasons practitioners give for categorizing is the desire to estimate odds ratios associated with categories without specifying a model for $P(Y = 1|x)$. (Another is that the impacts of misclassification on the fitting may be less of an issue than measurement error in the original X.)

From the perspective of estimating the odds-ratios, an alternative approach,

Table 7.16 *Framingham Heart Study. Estimates of odds ratios for pre-hypertension (P-Hyp) and hypertesion (Hyp) as defined in text. Boot-CI is the percentile interval. Adapted from Dalen et al. (2009).*

	Naive		Corrrected	
	Est.	Boot-CI	Est.	Boot - CI
P-Hyp	2.22	(1.29, 4.29)	3.29	(1.63, 11.50)
Hyp	4.03	(2.36, 7.94)	5.91	(3.27, 21.05)

which does not require a model for $m(x, \boldsymbol{\beta})$, was proposed by Dalen et al. (2009). It uses the expression for OR_{jk} given in (7.6) and obtains $\widehat{\alpha}_j = \int_{x \in A_j} \widehat{f}_{X|Y}(x|j)dx$, where $f_{X|Y}(x|j)$, estimated by $\widehat{f}_{X|Y}(x|j)$, is the conditional "density" of X given $Y = j$ for $j = 0$ or 1. As with f_X above, $f_{X|Y}$ may be discrete and the integral replaced by a sum. Using this approach, Dalen et al. (2009) estimate $f_{X|y}$ nonparametrically for each y using the smoothed NPMLE of Eggermont and LaRiccia (1997) (see Section 10.2.3) and assuming the replicates are normally distributed. They provide simulations of the proposed method under a variety of choices for the distribution of X given y and found the method to be fairly robust. Notice that this method can be used with case-control data as it just needs to estimate the distribution of X for $Y = 0$ and $Y = 1$. One potential problem is having a small number of cases with which to estimate $f_{X|1}$, in which case further investigation of parametric forms for $f_{X|y}$ is useful.

Dalen et al. (2009) also provide an example using the 1615 observations from the Framingham heart data, as described in Section 7.2.2. They examined the odds ratios associated with pre-hypertension (SBP between 129 and 139 mmHG) and hypertension (SBG ≥ 140) relative to normal (SBB < 129) with respect to the occurrence of heart disease, where SBP denotes systolic blood pressure. The naive estimates are based on categorizing SBP based on the mean of the two replicate measures. The correction techniques estimate the distribution of SBP in the cases (128 observations) and controls separately, using the estimated distributions to get estimated odds ratios, as described above. Standard errors and confidence intervals were obtained using the bootstrap. The results, shown in Table 7.16, show a modest shift upwards in the estimated odds ratios and the lower end of the confidence intervals, but also a huge increase in the upper bound of the confidence intervals. These reflect the uncertainty associated with accommodating misclassification due to noisy individual replicates. The estimated distribution of SBP in each group (not shown) is skewed, as others have noted. Dalen et al. (2009) also discuss the strategy of working with the log of the replicates directly.

CHAPTER 8

Linear Models with Nonadditive Error

8.1 Introduction

This chapter reconsiders linear models but now with nonadditive measurement error. Recall that linear here means linear in the parameters, so in terms of the true values and a suitably defined $\mathbf{x}$ (which may be functions of other predictors), $Y_i = \beta_0 + \mathbf{x}_i'\boldsymbol{\beta}_1 + \epsilon_i$, where ϵ_i has mean 0.

With additive error, treated in Chapters 4 and 5, the elements in $\mathbf{x}_i$ were either known exactly or estimated unbiasedly. Nonadditive error, the topic of this chapter, arises in one of two general ways. The first is when there is additive measurement error in some original variables but some nonlinear functions of those variables enter into the regression model. That is, there are some original predictors contained in $\tilde{\mathbf{x}}_i$ for which we have measured values $\tilde{\mathbf{W}}_i$, such that

$$\tilde{\mathbf{W}}_i = \tilde{\mathbf{x}}_i + \tilde{\mathbf{u}}_i$$

with $E(\tilde{\mathbf{u}}_i) = \mathbf{0}$. However the $\mathbf{x}_i$ which enters into the regression model contains some nonlinear functions, $g(\tilde{\mathbf{x}}_i)$, so that $g(\tilde{\mathbf{W}}_i)$ is not unbiased for $g(\tilde{\mathbf{x}}_i)$.

The problems here can be further subdivided. First there are those involving squares and products (i.e., quadratic models and models with interactions) for which some exact expressions are available. In those problems, treated in Sections 8.2 through 8.3, the problems can be converted to one with additive error. The other problems of this type involve functions such as logs, reciprocals or higher order powers, for which exact expressions for $E(g(\tilde{\mathbf{W}}_i))$ are not available. These are discussed in Section 8.4.

The second broad category is where the measurement error is nonadditive from the beginning. This includes linear or nonlinear measurement error models or misclassification of predictors. These problems are treated in Sections 8.5 and 8.6.

Section 8.7.1 provides a general expression for obtaining the approximate/asymptotic bias of naive estimators, under broad conditions, and isolates details for a number of special cases. These are necessary since the approach of using an induced model to assess bias does not always work. Section 8.7.2 describes how one could use likelihood methods under normality, an efficient approach if the distributional assumptions are met.

Returning to linear models at this point may seem a little narrow. The models considered here, however, are very important ones in their own right. Equally important, the methodology laid out here provides a useful base for handling other nonlinear models and dealing with nonadditive error using validation data.

8.2 Quadratic regression

Quadratic, and more generally polynomial, models are popular in part because of their ability to approximate certain nonlinear behaviors, while lending themselves to easier analysis than models that are nonlinear in the parameters. This is true both with and without measurement error. We begin with a single predictor x and the quadratic model

$$Y_i|x_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i,$$

where, as always, $E(\epsilon_i) = 0$.

The observed value W_i follows the additive measurement error model

$$W_i|x_i = x_i + u_i, \tag{8.1}$$

where u_i has mean 0 and variance σ_{ui}^2. In this case, $\tilde{x}_i = x_i$ and $\mathbf{x}_i' = (1, x_i, x_i^2)$.

Since $E(W_i^2|x_i^2) = x_i^2 + \sigma_{ui}^2$, although we have additive error in the single predictor,

$$E\begin{bmatrix} W_i \\ W_i^2 \end{bmatrix} = \begin{bmatrix} x_i \\ x_i^2 + \sigma_{ui}^2 \end{bmatrix} \neq \begin{bmatrix} x_i \\ x_i^2 \end{bmatrix}.$$

This means the results of Chapter 5 are not directly applicable, at least not until we reformulate the problem.

The next section investigates the effects of measurement error on naive analyses. This is explored first in terms of the linear Berkson model for $X|w$ and then through the use of the explicit form of the naive estimators and associated estimating equations. This latter strategy is more flexible since it does not carry the same assumptions that are needed for the linear Berkson model to hold. It can also accommodate functional models and is readily expanded to allow for error in the response. As in other contexts, the bias results provide an understanding of when the measurement errors are of consequence while also laying the groundwork for correction methods.

Section 8.2.2 discusses the use of moment corrected and regression calibration methods. In Section 8.2.3, those methods and SIMEX are applied to an example. Sections 8.2.4 and 8.2.5 discuss extensions to account for error in the response and additional covariates, respectively. Kuha and Temple (2003) provide an excellent overview and bibliography on measurement error in quadratic regression.

8.2.1 Biases in naive estimators

Linear Berkson model.

With X_i random, the induced model is

$$E(Y_i|w_i) = \beta_0 + \beta_1 E(X_i|w_i) + \beta_2[E(X_i|w_i)^2 + V(X_i|w_i)]. \qquad (8.2)$$

Using the linear Berkson model with $E(X_i|w_i) = \lambda_0 + \lambda_1 w_i$ and $V(X_i|w_i) = \sigma^2_{X|w}$, (8.2) becomes

$$E(Y_i|w_i) = \beta_0^* + \beta_1^* w_i + \beta_2^* w_i^2, \qquad (8.3)$$

where

$$\begin{aligned}
\beta_0^* &= \beta_0 + \beta_1\lambda_0 + \beta_2(\sigma^2_{X|w} + \lambda_0^2)\\
\beta_1^* &= \beta_1 + \beta_1(\lambda_1 - 1) + 2\beta_2\lambda_0\lambda_1 = \lambda_1\beta_1 + 2\beta_2\lambda_0\lambda_1 \text{ and}\\
\beta_2^* &= \beta_2 + \beta_2(\lambda_1^2 - 1) = \beta_2\lambda_1^2.
\end{aligned}$$

The naive least squares estimators are unbiased for the β^*'s and hence are biased for the β's with the biases appearing in the expressions for β_0^*, β_1^* and β_2^* above. Notice that the bias in the linear coefficient β_1 depends on β_2, as well as other quantities.

For estimating the mean at x_0, $E(Y|X = x_0) = \beta_0 + \beta_1 x_0 + \beta_2 x_0^2$, the naive estimator has a bias of

$$\beta_1\lambda_0 + \beta_2(\sigma^2_{X|w} + \lambda_0^2) + [\beta_1(\lambda_1 - 1) + 2\beta_2\lambda_0\lambda_1]x_0 + \beta_2(\lambda_1^2 - 1)x_0^2.$$

This is complicated and depends on the value x_0; see Figure 8.1 for an illustration.

Recall that the linear Berkson model results from the normal structural model for X_i, and a normal additive measurement error with constant variance, σ_u^2, for which

$$\lambda_1 = \sigma_X^2/(\sigma_X^2 + \sigma_u^2), \ \lambda_0 = \mu_X(1 - \lambda_1) \text{ and } \sigma^2_{X|w} = (1 - \lambda_1)\sigma_X^2.$$

This leads to bias expressions for the coefficients in terms of the structural parameters for the X's and the measurement error variance. Figure 8.1 and

Table 8.1 illustrate the biases in this context using a setting with $\beta_0 = 0$, $\beta_1 = 1$, $\beta_2 = 2$, $\mu_X = 0$ and $\sigma_X = 2$. The reliability ratio is λ_1 which varies from .5 to 1,with the case of 1 corresponding to no measurement error. The impacts of the measurement error are larger, relatively, for the quadratic coefficient than they are for the linear coefficient.

Table 8.1 *Expected value of naive estimators in quadratic regression assuming normality for the true X and for the measurement error. True values are $\beta_0 = 0$, $\beta_1 = 1$ and $\beta_2 = 2$, σ_u^2 = the measurement error variance, the reliability ratio is $\lambda_1 = \sigma_X^2/(\sigma_X^2 + \sigma_u^2)$, where $\sigma_X^2 = 4$.*

λ_1	σ_u^2	β_0^*	β_1^*	β_2^*
0.5	4	4	0.5	0.5
0.6	2.67	3.2	0.6	0.72
0.7	1.7	2.4	0.7	0.98
0.8	1	1.6	0.8	1.28
0.9	.444	0.8	0.9	1.62
1 (true)	0	0	1	2

Designed experiments with additive Berkson error.

Quadratic models are a popular choice in "dose-response" type experiments. This is the pure Berkson setting where the w_i are fixed target "doses" and the true value, X_i, is assumed random with $X_i = w_i + e_i$, with e_i having mean 0 and variance σ_e^2. In the paper tensile strength example considered later, this model would apply if the reported hardwood concentration was a target value for a batch. Recall from Section 4.2 that in simple linear regression additive Berkson error was of no consequence in estimating the coefficients. Here, with the w_i fixed the induced model is

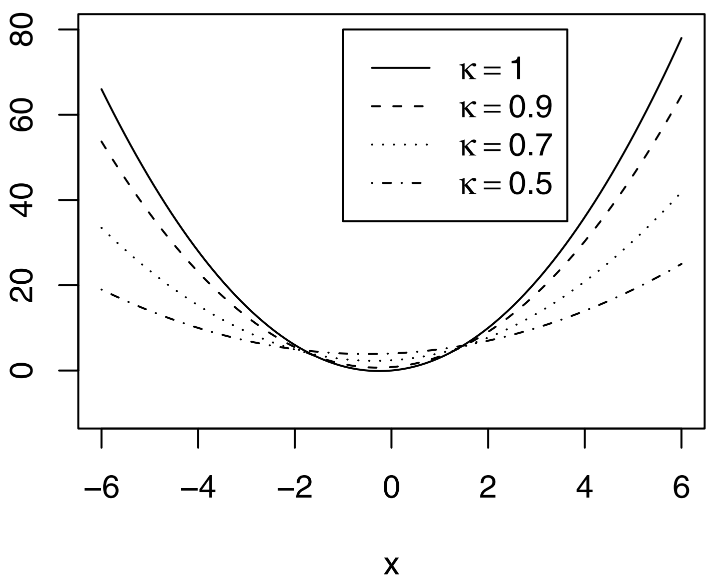
$$Y_i|w_i = \beta_0^* + \beta_1 w_i + \beta_2 w_i^2 + \eta_i^*,$$

where $\beta_0^* = \beta_0 + \beta_2\sigma_e^2$ and $V(\eta_i^*) = V(\beta_1 e_i + \beta_2 2e_i w_i + \beta_2 e_i^2 + \epsilon_i)$. This leads to the following conclusions:

- The variance is a quadratic function of w_i, unless there is no measurement error.
- The naive estimators of β_1 and β_2 are unbiased but the naive estimator of the intercept is biased. This also implies that the estimated expected response at a fixed x is biased, with a constant bias of $\beta_2\sigma_e^2$.

If the goal of the problem is either inverse prediction (estimating the un-

Figure 8.1 *Illustration of impacts of additive measurement error in a quadratic regression model. True regression line, denoted by solid line at $\kappa = 1$. The other lines indicate the expected value of naive fit as the reliability ratio goes from .5 to .9.*



observed x) from an observed response, or regulation (estimating the x for which $E(Y|x) = c$, a fixed value), then the bias in β_0 is important.

On the other hand, if the objective is estimation of $-\beta_1/2\beta_2$ which is the value of x at which the expected value of Y is maximized or minimized (depending on the sign of β_2), the bias in β_0 is of no consequence. However the heteroscedasticity mentioned above may be.

- Even if interest focuses solely on β_1 and β_2, the usual analyses assuming constant variance are incorrect due to the fact that the variance of η_i^* is quadratic in w_i. The remedy for this is to use naive least squares but with a robust covariance estimate (available in most statistical software packages) or to consider iterative reweighted least squares, exploiting the form of $V(\eta_i^*)$.

More general bias expressions.

If the assumptions of a normal structural model with normal additive measurement error with constant variance apply, the linear Berkson model holds and the bias results of the preceding section can be used. This was discussed and illustrated in Figure 8.1 and Table 8.1. The treatment below provides the approximate bias under weaker assumptions and allows the x values to be fixed.

The design matrix for the true values, $\mathbf{X}$, has ith row equal to $(1, x_i, x_i^2)$. Using the results in Section 8.7.1

$$E(\widehat{\boldsymbol{\beta}}_{naive}) \approx \boldsymbol{\beta}^* = \boldsymbol{\beta} + \left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{Q}\right)^{-1} (\mathbf{HX} - \mathbf{Q})\,\boldsymbol{\beta} \tag{8.4}$$

where

$$\mathbf{Q} = \frac{1}{n}\begin{bmatrix} 0 & 0 & \sum_i \sigma_{ui}^2 \\ 0 & \sum_i \sigma_{ui}^2 & \sum_i (3x_i\sigma_{ui}^2 + E(u_i^3)) \\ \sum_i \sigma_{ui}^2 & \sum_i (3x_i\sigma_{ui}^2 + E(u_i^3)) & \sum_i (6x_i^2\sigma_{ui}^2 + 4x_iE(u_i^3) + E(u_i^4)) \end{bmatrix} \tag{8.5}$$

and

$$\mathbf{HX} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \sum_i \sigma_{ui}^2 & \sum_i \sigma_{ui}^2 x_i/n & \sum_i \sigma_{ui}^2 x_i^2/n \end{bmatrix}.$$

The expression above provides approximate biases for the naive estimators allowing the x's to be fixed, possibly changing measurement error variances and arbitrary measurement error distributions. If there is error in the response that is correlated with the error in x then an additional term would be added. If the measurement error distribution is symmetric then $E(u_i^3) = 0$, while if normality is assumed then in addition $E(u_i^4) = 3\sigma_{ui}^4$, both leading to some simplification of the approximate bias expressions.

In the above, $x_1, \ldots, x_n$ and the σ_{ui}^2, which in turn may be functions of the x_i, are considered fixed. In the structural model an unconditional approximation is obtained by replacing quantities involving x_i with X_i and then taking expected values. If the measurement error variances do not depend on the true values and the X_i's i.i.d. then in the expression for $\mathbf{HX}$, x_i is replaced by $E(X_i)$, x_i^2 is replaced by $V(X_i) + E(X_i)^2$ and

$$\mathbf{X}'\mathbf{X}/n = \boldsymbol{\Sigma}_{XX*} + \boldsymbol{\mu}_{X*}\boldsymbol{\mu}'_{X*} \tag{8.6}$$

where with $\mathbf{X}_i^{*'} = (1, X_i, X_i^2)$, $\boldsymbol{\Sigma}_{XX*} = Cov(\mathbf{X}_i^*)$ and $\boldsymbol{\mu}_{X*} = E(\mathbf{X}_i^*)$. In the normal structural setting with normal measurement error having constant variance, the biases obtained here will agree with those from the linear Berkson model of the previous section.

8.2.2 Correcting for measurement error

This section describes how to correct for measurement using moment corrected and regression calibration methods. SIMEX is also used in our example but the general discussion in Section 6.11 does not need repeating. Of course there are other options, such as the use of fully parametric or semi-parametric likelihood methods in the structural setting; see Sections 6.12 and 6.12.3 and the discussion on likelihood methods in 8.7.2.

Moment corrections.

Moment corrected estimators are obtained by converting the problem to one that lets us apply the methods in Chapter 5. To this end the vector $\mathbf{W}_i$ is now defined as

$$\mathbf{W}_i = \begin{bmatrix} W_i \\ W_i^2 - \widehat{\sigma}_{ui}^2 \end{bmatrix}, \tag{8.7}$$

so $W_{i2} = W_i^2 - \sigma_{ui}^2$ and $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$ with $\mathbf{u}_i' = (u_i, 2x_i u_i + u_i^2 - \widehat{\sigma}_{ui}^2)$. As long as $\widehat{\sigma}_{ui}^2$ is unbiased for σ_{ui}^2, then $E(\mathbf{u}_i) = \mathbf{0}$, i.e., the measurement error is additive in terms of the redefined $\mathbf{W}_i$.

Including the uncertainty from estimating σ_{ui}^2 (although this is typically ignored) the covariance of $\mathbf{u}_i$ is

$$\mathbf{\Sigma}_{ui} = \begin{bmatrix} \sigma_{ui}^2 & 2x_i\sigma_{ui}^2 + E(u_i^3) - Cov(u_i, \widehat{\sigma}_{ui}^2) \\ 2x_i\sigma_{ui}^2 + E(u_i^3) - Cov(u_i, \widehat{\sigma}_{ui}^2) & c_{22} \end{bmatrix}, \tag{8.8}$$

where

$$c_{22} = 4x_i^2\sigma_{ui}^2 + V(u_i^2) + 4x_i[E(u_i^3) - Cov(u_i, \widehat{\sigma}_{ui}^2)] - 2Cov(u_i^2, \widehat{\sigma}_{ui}^2) + V(\widehat{\sigma}_{ui}^2).$$

The correction method in (5.9) leads to

$$\hat{\boldsymbol{\beta}}_1 = \widehat{\mathbf{\Sigma}}_{XX}^{-1}\mathbf{S}_{WY} \qquad \widehat{\beta}_0 = \bar{Y} - \widehat{\boldsymbol{\beta}}_1'\bar{\mathbf{W}}, \tag{8.9}$$

where $\widehat{\mathbf{\Sigma}}_{XX} = \mathbf{S}_{WW} - \hat{\mathbf{\Sigma}}_u$ and $\widehat{\mathbf{\Sigma}}_u = \sum_{i=1}^n \widehat{\mathbf{\Sigma}}_{ui}/n$. As in Chapter 5, $\bar{\mathbf{W}}$ and $\mathbf{S}_{WW}$ are the sample mean vector and covariance matrix of the $\mathbf{W}_i$ (as defined in (8.7) above), $\bar{Y}$ is the average response and $\mathbf{S}_{WY}$ is the sample covariance of $\mathbf{W}_i$ with Y_i. That chapter also provides an alternative computational formula for $\widehat{\boldsymbol{\beta}}$, given in equation (5.10), as well as methods for estimating the covariance of $\widehat{\boldsymbol{\beta}}$.

In addition to x_i, x_i^2 and σ_{ui}^2, which can be estimated by W_i, $W_i^2 - \widehat{\sigma}_{ui}^2$ and $\widehat{\sigma}_{ui}^2$, respectively, $\mathbf{\Sigma}_{ui}$ also depends on $E(u_i^3)$, $E(u_i^4)$, $V(\widehat{\sigma}_{ui}^2)$, as well as the covariance of $\widehat{\sigma}_{ui}^2$ with some powers of u_i. If there are replicates available then there are various ways to estimate all of these quantities, without further distributional assumptions. For example one could use $\sum_j (W_{ij} - W_i)^k/m_i$ (or some variation on this) to estimate $E(u_i^k)$.

The approach above allows us to handle very general measurement errors (as well as error in the response; see Section 8.2.4). The performance of this correction under the most general setting has not received much attention. Instead, the estimation usually proceeds with some assumptions for which $\mathbf{\Sigma}_{ui}$ simplifies. For example, if the measurement errors are assumed symmetrically distributed around 0, then $E(u_i^3) = 0$ and this term can be eliminated. With normally distributed replicates then $E(u_i^3) = 0$, $\widehat{\sigma}_{ui}^2$ is independent of u_i, $V(u_i^2) = 2\sigma_{ui}^4$ and $V(\widehat{\sigma}_{ui}^2) = 2\sigma_{ui}^4/(m_i - 1)$. This simplifies $\mathbf{\Sigma}_{ui}$ considerably and with x_i^2 estimated by $W_i^2 - \widehat{\sigma}_{ui}^2$ this results in

$$\widehat{\mathbf{\Sigma}}_{ui} = \begin{bmatrix} \widehat{\sigma}_{ui}^2 & 2W_i\widehat{\sigma}_{ui}^2 \\ 2W_i\widehat{\sigma}_{ui}^2 & 4W_i^2\widehat{\sigma}_{ui}^2 - 2\widehat{\sigma}_{ui}^4 + 2\widehat{\sigma}_{ui}^4/(m_i - 1) \end{bmatrix}. \tag{8.10}$$

It is important to note that this form requires the normality assumption on the replicates. If the $\widehat{\sigma}_{ui}^2$ are treated as known, so the last element of the (2,2) term is set to 0, and a common $\widehat{\sigma}_u^2$ is used for all i, this expression leads to

$$\widehat{\mathbf{\Sigma}}_{ui} = \begin{bmatrix} \widehat{\sigma}_u^2 & 2W_i\widehat{\sigma}_u^2 \\ 2W_i\widehat{\sigma}_u^2 & 4W_i^2\widehat{\sigma}_u^2 - 2\widehat{\sigma}_u^4 \end{bmatrix}.$$

This yields the "method-of-moment estimators" on page 129 of Kuha and Temple (2003).

As discussed in Chapter 5 the moment corrected estimators may need to be modified to account for negative estimates of a matrix that should be positive. See Section 5.4.4 for details. Our later examples, including the use of the bootstrap, all incorporate those modifications where needed. Kuha and Temple (2003) discuss some other small sample modifications.

Finally, we note that the moment corrected estimators could be obtained in a different way using the approach in Section 6.13 based on modified estimating equations. The one advantage to this approach is that the estimate can be computed sequentially using quadratic regression but with an updated response vector.

Regression calibration estimators.

There are various types of regression calibration estimators that can be used based on the induced model in (8.2). All of these regress Y_i on $\widehat{X}_i$ and $\widehat{X}_{i2}$ where $\widehat{X}_i$ is an estimate of $E(X_i|w_i)$ and $\widehat{X}_{i2}$ is an estimate of $E(X_i^2|w_i)$. The different versions result according to what is used for $\widehat{X}_{i2}$, whether separate measurement error variances are used for "imputation" and whether the fit is simple unweighted least squares or uses expanded regression calibration, involving weighted least squares based on a variance model for $V(Y|w)$.

In general the replicates are first used to obtain $\widehat{\mu}_X$ and $\widehat{\sigma}^2_X$ (see the discussion in Section 6.5.1) and $\widehat{\sigma}^2_u = \sum_i \widehat{\sigma}^2_{ui}/n$. From these,

$$\widehat{\lambda}_1 = \widehat{\sigma}^2_X/(\widehat{\sigma}^2_X + \widehat{\sigma}^2_u), \ \ \widehat{\lambda}_0 = \widehat{\mu}_X(1 - \widehat{\lambda}_1), \ \ \widehat{\sigma}^2_{X|w} = (1 - \widehat{\lambda}_1)\widehat{\sigma}^2_X,$$

and

$$\widehat{\lambda}_{1i} = \widehat{\sigma}^2_X/(\widehat{\sigma}^2_X + \widehat{\sigma}^2_{ui}), \ \ \widehat{\lambda}_{0i} = \widehat{\mu}_X(1 - \widehat{\lambda}_{1i}), \ \ \widehat{\sigma}^2_{X_i|w_i} = (1 - \widehat{\lambda}_{1i})\widehat{\sigma}^2_X.$$

The basic regression calibration approach that results from (6.43) would use $\widehat{X}^2_i$ for $E(X^2_i|w_i)$, but that can be improved on in this case since we know that $E(X^2_i|w_i) = V(X_i|w_i) + E(X_i|w_i)^2$. This leads us to two estimators, where I stands for use of individual variances.

1. Standard RC: $\widehat{X}_i = \widehat{\lambda}_0 + \widehat{\lambda}_1 w_i$ and $\widehat{X}_{i2} = \widehat{X}^2_i + \widehat{\sigma}^2_{X|w}$.
2. Standard RC-I: $\widehat{X}_i = \widehat{\lambda}_{0i} + \widehat{\lambda}_{1i} w_i$ and $\widehat{X}_{i2} = \widehat{X}^2_i + \widehat{\sigma}^2_{X_i|w_i}$,

Expanded regression calibration estimators fit the regression using iteratively reweighted least squares, based on an expression for $V(Y_i|w_i)$. See the discussion in Section 6.10. Kuha and Temple (2003) provide details on this approach.

8.2.3 *Paper example*

To illustrate, we consider an example from Montgomery and Peck (1992) for which a quadratic model was used to model the tensile strength in Kraft paper as a function of the hardwood concentration in the batch of pulp used. We treat the concentration of hardwood going into the batch as not known exactly (as is realistic) and instead is estimated through sampling. We used the reported concentrations as true values and generated estimate concentrations (the W_i's), response values ($Y_i's$), and estimated measurement error standard deviations ($\widehat{\sigma}_{ui}$'s) as given in Table 8.2. The underlying measurement error variance was allowed to increase with concentration, with a true coefficient of variation of .15. This is reflected in the estimated measurement error standard deviations.

Table 8.3 shows results from a naive analysis and corrected estimates using the moment correction, the two forms of regression calibration itemized above and SIMEX. SIMEX was implemented using normal measurement errors, with different measurement error variances for each observation and 500 simulations per λ. (Notice there are two sets of estimates for SIMEX, corresponding to where the two different types of bootstrapping were used. This represents variation from the use of 500 simulations at each λ.) The moment estimates were corrected using the correction in (8.10) corresponding to assuming the measurement errors are symmetric and normally distributed, but

Table 8.2 *Simulated data for quadratic regression example.*

Strength (Y)	percent (W)	$\widehat{\sigma}_u$
4.07	1.3168	0.16566
8.99	1.4568	0.19220
18.13	2.0142	0.23254
17.59	3.7964	0.42964
32.64	3.0619	0.61739
35.65	5.2092	0.59729
28.70	6.2867	0.73140
39.27	4.7119	0.87010
47.64	5.7388	0.90761
44.43	6.0952	0.83353
49.66	5.5888	0.83913
45.30	6.3794	1.09284
49.64	12.1366	0.99204
44.28	8.8247	1.35212
34.48	10.5993	1.73473
37.31	12.9195	1.76275
37.26	13.3474	1.61300
27.99	13.9513	1.94470
29.87	17.3876	2.51410

with $2\widehat{\sigma}_{ui}^4/(m_i - 1)$ set equal to 0. As noted earlier one could expand on these methods by estimating higher order moments and the variance of $\widehat{\sigma}_{ui}^2$, based on replication, and using the more general form $\widehat{\mathbf{\Sigma}}_{ui}$, but we have not implemented any of these more general corrections here. For all of the estimation methods standard errors and confidence intervals were calculated using one and two-stage bootstrap sampling (see Section 6.16), each using 500 bootstrap samples. For the moment estimators, we also used robust and "normal based" estimators of $Cov(\widehat{\boldsymbol{\beta}})$ (see Chapter 5) with the associated confidence intervals computed using $estimate \pm 1.96SE$. The fitted lines for the five methods are displayed in Figure 8.2.

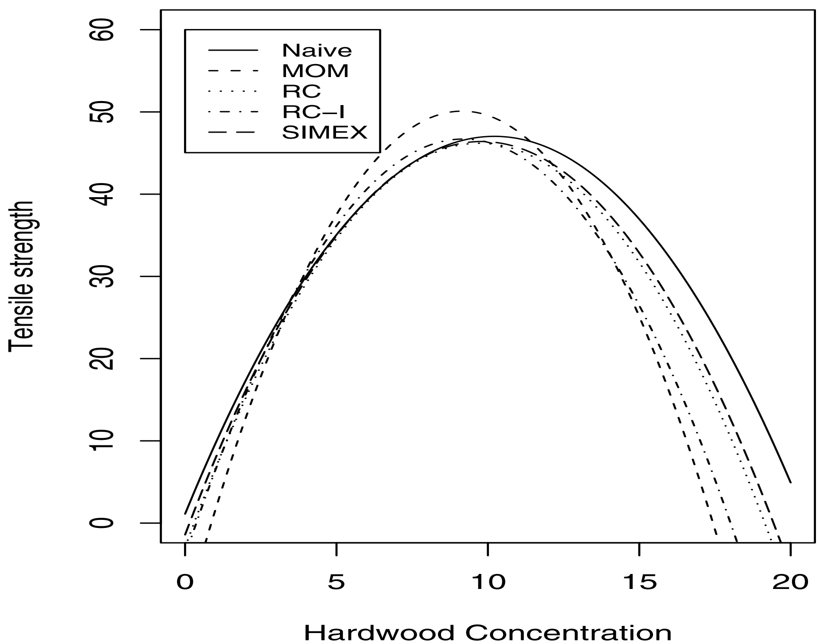
Here are some comments on the analysis.

- The naive estimate of σ^2 is 70.32, while the corrected estimate, computed using (5.11), is $\widehat{\sigma}^2 = 30.03$. This change in the variance would have important implications in predicting tensile strength given a true hardwood concentration.
- The most severe modification to the naive estimates results from the moment correction.
- The RC and SIMEX estimates are fairly similar.

Table 8.3 *Paper example, fitting a quadratic regression model. Boot-1 and Boot-2 indicate one- and two-stage bootstrap respectively. RC-I is regression calibration using individual measurement error variances. 95% Confidence interval given by Lower, Upper.*

Method	Par.	Estimate	B-Mean	SE	Lower	Upper
Naive	β_0	1.11		6.25	-12.14	14.37
	β_1	8.99		1.72	5.34	12.63
	β_2	-0.45		0.10	-.66	-2.5
Moment-Robust	β_0	-11.03		6.81	-24.37	2.31
	β_1	13.36		2.48	8.50	18.23
	β_2	-0.73		0.16	-1.04	-0.43
Moment-Boot-1	β_0	-11.03	-13.88	9.28	-40.34	-0.95
	β_1	13.36	14.37	2.90	9.85	22.11
	β_2	-0.73	-0.81	0.20	-1.25	-0.53
	σ^2	20.51	15.56	13.53	3.19	49.72
Moment-Boot-2	β_0	-11.03	-13.90	9.10	-34.24	2.77
	β_1	13.36	14.53	3.45	8.34	22.84
	β_2	-0.73	-0.82	0.25	-1.43	-0.41
	σ^2	30.03	25.89	37.22	3.07	140.72
RC Boot-1	β_0	-.2.95	-4.80	7.55	-17.74	10.95
	β_1	10.11	10.96	2.50	6.57	16.05
	β_2	-0.52	-0.58	0.17	-0.94	-0.33
RC Boot-2	β_0	-2.95	-3.49	6.15	-15.81	7.44
	β_1	10.11	10.09	1.99	6.54	14.23
	β_2	-.52	-0.51	0.13	-0.79	-0.29
RC-I Boot-1	β_0	-3.99	-4.13	7.87	-17.42	12.80
	β_1	11.03	10.78	2.59	6.27	15.85
	β_2	-.60	-0.57	0.17	-0.93	-0.32
RC-I Boot-2	β_0	-3.99	-3.10	5.57	-13.86	7.33
	β_1	11.03	11.01	1.95	7.30	14.76
	β_2	-.60	-0.62	0.14	-0.91	-0.37
SIMEX Boot-1	β_0	-.453	-5.21	9.28	-23.58	10.60
	β_1	9.39	11.32	3.22	6.47	18.52
	β_2	-.475	-0.61	0.21	-1.09	-0.34
SIMEX Boot-2	β_0	-1.42	-1.59	6.71	-15.17	10.82
	β_1	9.77	9.83	2.36	5.71	14.86
	β_2	-. 499	-0.50	0.16	-0.85	-0.24
	σ^2	58.36	32.01	23.76	-5.25	90.28

Figure 8.2 *Fitted curves for paper tensile strength example.*



- The RC-I using individual measurement error variances falls between the moment and the other two in terms of correcting the coefficients.
- The impact of the correction on the fitted line depends on the concentration level. At the lower levels of concentration the differences are minor, while the methods diverge considerably at the higher concentration, with the RC-I behaving more like the moment estimator and estimating lower tensile strengths. See Figure 8.2.
- The two-stage bootstrap, which allows us to assess bias, requires specifying coefficients, a variance for the error in the equation and true values. The original observed W_i's and $\widehat{\sigma}^2_{ui}$ are used as the true values and measurement error variances, respectively. Separate coefficients were used for each of the estimators to generate the true responses, so the mean of the bootstrap sample can be compared to the original estimates. The variance for the error in the equation was taken to be $\sigma^2 = 30.03$, which is the corrected estimate

above. The error in the equation is assumed normal as is the distribution of the replicate values. Five replicates were used in generating the W and the estimated measurement error variances.

- Section 5.4.4 describes how to fix the moment estimator of the coefficients and σ^2 when a certain estimated matrix is negative. This was not a problem with the original data, but this fix was often needed when using the bootstrap; 283 out of 500 times in the one-stage bootstrap and 328 out of 500 times for the two-stage bootstrap. From the perspective of estimating the coefficients one could ignore this correction. Running the two-stage without the correction led to bootstrap means (and SE's) of -13.78 (8.66), 14.55 (3.35) and -.82 (.24) for the three coefficients, respectively. These are very similar to results with the correction. The estimate of σ^2 is more erratic however.
- The bootstrap standard errors and confidence intervals from the two-stage bootstrap are generally smaller. The one-stage bootstrap estimates of standard errors and associated confidence intervals are more robust, but they do require that the original sample was an overall random sample. Recall also that the one-stage bootstrap should not be used for assessing bias.
- A SIMEX estimate of σ^2 was computed using the same technique that is used for the coefficients. The estimate of 58.3 is quite different than the moment based estimator. In looking at the two-stage bootstrap, the bootstrap mean should be compared to the 30.03 rather than to the 58.3, since it was the former that was used in generating responses. From this perspective, the SIMEX estimator does a reasonable job in terms of bias. Whether SIMEX of the moment method is used, it is clear from the standard errors and size of the confidence intervals, that there is limited information about the variance of the error in the equation.
- The MEE estimator is more variable than either the RC or SIMEX estimators. The analyses based on analytical rather than bootstrap estimates of the covariance of $\widehat{\boldsymbol{\beta}}$ showed a big difference between robust and "normal-based" standard errors. As elsewhere, we put normal-based in quotes since it is not just normality that drives this estimate (see the discussion in Chapter 5). The normal based standard errors (which were 20.94, 7.84 and .52, respectively) are considerably larger than with all of the other methods The robust estimates are more similar to the bootstrap, although a bit smaller. Some simulations based on this setting reinforce the tendency for the robust and normal-based procedures to under and over-estimate the true standard errors, respectively, given the relatively small sample size.
- Using the two-stage results, the RC and SIMEX estimators have less bias than the moment correction. Given that the RC estimators are motivated from a normal structural setting and the two-stage bootstrap is basically providing an internal simulation in a functional setting, this robustness of

the RC estimator is a pleasant surprise. The moment estimators have some modest bias issues. This is not surprising given what is known about the moment estimators at small sample sizes and the fact that there are only twenty observations here. As noted elsewhere there are small sample modifications that could be considered.

- Based on both the bias and variability assessments, either the RC or SIMEX analysis is preferred here over the use of moment estimators.

8.2.4 Additive error in the response

As in linear problems, error in the response is a common occurrence. For example, in the paper example, the tensile strength of the paper from a batch may not be known but rather estimated by sampling from the batch. As in the earlier chapters, assume we observe $D_i = y_i + q_i$ with q_i having mean 0 and variance σ^2_{qi} and with q_i possibly correlated with u_i. If the error in the response is correlated with the error in the mismeasured predictor then, as was the case in the linear problems, this will alter the bias expressions for naive estimators.

In correcting, the error in the response is easily accommodated using either the moment correction or SIMEX methods, but not with regression calibration. The SIMEX method would simply simulate by generating measurement errors (q_i, u_i) using the estimated covariance matrix of the two errors. For using a moment correction, recall that $\mathbf{u}_i' = (u_i, 2x_i u_i + u_i^2 - \widehat{\sigma}^2_{ui})$, where u_i is the measurement error (univariate) in x_i, so

$$Cov(q_i, \mathbf{u}_i) = \mathbf{\Sigma}_{qui} = (cov(q_i, u_i), 2x_i cov(q_i, u_i) + cov(q_i, u_i^2) - cov(q_i, \widehat{\sigma}^2_{ui})).$$

With the uncertainty in $\widehat{\sigma}^2_{ui}$ ignored, the last term is zero. With estimates of σ^2_{qi} and $\mathbf{\Sigma}_{qui}$, inferences proceed as in Section 5.4. This results in a slight modification of 8.9, where with $\widehat{\mathbf{\Sigma}}_{qu} = \sum_i \widehat{\mathbf{\Sigma}}_{qui}/n$,

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{S}_{WW} - \widehat{\mathbf{\Sigma}}_u)^{-1}(\mathbf{S}_{WD} - \widehat{\mathbf{\Sigma}}_{qu}) \tag{8.11}$$

and $\widehat{\beta}_0 = \bar{D} - \widehat{\boldsymbol{\beta}}_1' \bar{\mathbf{W}}$. Alternatively equation (5.10) can be used. Recall that in these corrections $\mathbf{W}_i$ is defined as in (8.7).

8.2.5 Quadratic models with additional predictors

The preceding methods can be extended to handle additional predictors measured without error, or measured with additive error and entering the model linearly. Consider the model

$$Y_i|\mathbf{x}_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \boldsymbol{\beta}_2' \mathbf{x}_{i2} + \epsilon_i, \tag{8.12}$$

where $\mathbf{x}_{i2}$ contains the additional predictors. Notice that in an attempt to retain consistency with previously used notations, the scalar β_2 is used for the coefficient of the quadratic term while at the same time using the vector $\boldsymbol{\beta}_2$ for the vector of coefficients for the additional predictors.

Ancestry example. Divers et al. (2007) provide an example of this situation where x_{i1} is a measure of an individual's ancestry (proportion of that individual's ancestors who come from a particular population), Y is a phenotypic measure and the additional predictors are three genetic markers denoted $g280$, $g690$ and $g870$. The model they employ is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \beta_3 g280_i + \beta_4 g690_i + \beta_5 g870_i + \epsilon_i.$$

In our notation $\mathbf{x}'_{i2} = (g280_i, g690_i, g870_i)$ and $\boldsymbol{\beta}'_2 = (\beta_3, \beta_4, \beta_5)$. The measurement error is in the ancestry value which is estimated using a derived value W_{i1}, as described by Divers et al. (2007). The main objective is inference for the genetic markers and in particular testing for no effect of a marker, e.g., testing $\beta_j = 0$ for $j \geq 3$.

Returning to the general problem we start with

$$\tilde{\mathbf{W}}_i = \begin{bmatrix} W_{i1} \\ \mathbf{W}_{i2} \end{bmatrix} = \begin{bmatrix} x_{i1} \\ \mathbf{x}_{i2} \end{bmatrix} + \begin{bmatrix} u_{i1} \\ \mathbf{u}_{i2} \end{bmatrix} = \tilde{\mathbf{x}}_i + \begin{bmatrix} u_{i1} \\ \mathbf{u}_{i2} \end{bmatrix}, \qquad (8.13)$$

where $E(u_{i1}) = 0$, $E(\mathbf{u}_{i2}) = \mathbf{0}$, $V(u_{i1}) = \sigma^2_{ui}$, $Cov(\mathbf{u}_{i2}) = \mathbf{\Sigma}_{ui2}$ and $Cov(u_{i1}, \mathbf{u}_{i2}) = \tilde{\mathbf{\Sigma}}_{ui12}$.

Both $\mathbf{\Sigma}_{ui2}$ and $\tilde{\mathbf{\Sigma}}_{ui12}$ are all zeros if there is no measurement error in the additional predictors (as is the case in the ancestry example). We assume no error in Y in the discussion here, but it can be accommodated in the same way as was done without additional covariates in the preceding section.

Bias in naive estimators.

This relatively simple setting also provides us with another situation in which we cannot quickly rely on an induced model to asses the bias of naive estimators, even with normality assumptions and constant measurement error variances and covariances. With no error in Y and recalling the definitions for $\tilde{\mathbf{W}}_i$ and $\tilde{\mathbf{x}}_i$ in (8.13), the induced model is

$$E(Y_i|\tilde{\mathbf{W}}_i) = \beta_0 + \beta_1 E(X_{i1}|\tilde{\mathbf{W}}_i) + \beta_2 E(X_{i1}^2|\tilde{\mathbf{w}}_i) + \boldsymbol{\beta}'_2 E(\mathbf{X}_{i2}|\tilde{\mathbf{w}}_i). \quad (8.14)$$

Since $E(X_{i1}^2|\tilde{\mathbf{w}}_i) = V(X_{i1}|\tilde{\mathbf{w}}_i) + (E(X_{i1}|\tilde{\mathbf{w}}_i))^2$, and $E(X_{i1}|\tilde{\mathbf{w}}_i))^2$ is usually linear in $\mathbf{w}_{i2}$, the induced model is usually quadratic in the components of $\mathbf{w}_{i2}$ while the original model was linear in these terms. This is true even when there is no measurement error in $\mathbf{x}_{i2}$.

In general, approximate biases need to be developed using the expression for the naive estimators. These are given in Section 8.7.1. As in the linear problems in Chapter 5, the error in X_{i1} will induce bias in the estimates of the coefficients of $\mathbf{x}_{i2}$, even if there is no measurement error in those additional variables.

Testing: One question of interest is whether the naive estimators of $\boldsymbol{\beta}_2$ are 0 when $\boldsymbol{\beta}_2 = \mathbf{0}$. Unless X_{i1} is uncorrelated with $\mathbf{X}_{i2}$, the answer is generally no. This means that naive tests for the components of $\boldsymbol{\beta}_2$ equaling zero are invalid. This is of special interest in the ancestry example, since one of the main objectives is to test whether the coefficients for the genetic markers are 0. Divers et al. (2007) show through simulations that the naive test can produce rejection rates quite different than the nominal rates, even with what might be considered moderate amounts of measurement error. They also show that compared to SIMEX and regression calibration, modified tests based on the moment corrected techniques below do the best job of attaining the nominal level, across a variety of settings.

Correction methods.

Once again we provide details on only some of the correction options. The use of SIMEX is straightforward. The measurement error u_{i1} is generated along with $\mathbf{u}_{i2}$, if there is error in $\mathbf{x}_{i2}$, and/or q_i, if there is error in the response. These are generated accounting for any correlation structure that may exist among the measurement errors.

Moment correction: Define $\mathbf{W}'_i = (\mathbf{W}'_{i1}, \mathbf{W}'_{i2})$, where $\mathbf{W}'_{i1} = (W_{i1}, W_{i1}^2 - \widehat{\sigma}^2_{ui1})$. So, $\mathbf{W}_{i1} = \mathbf{x}_{i1} + \mathbf{u}_{i1}$, with $\mathbf{u}'_{i1} = (u_{i1}, 2x_{i1}u_{i1} + u_{i1}^2 - \widehat{\sigma}^2_{ui1})$ and $\mathbf{x}'_{i1} = (x_{i1}, x_{i1}^2)$. With $\mathbf{x}'_i = (\mathbf{x}'_{i1}, \mathbf{x}'_{i2})$, then $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$, with

$$\boldsymbol{\Sigma}_{ui} = \begin{bmatrix} \boldsymbol{\Sigma}_{ui1} & \boldsymbol{\Sigma}_{ui12} \\ \boldsymbol{\Sigma}'_{ui12} & \boldsymbol{\Sigma}_{ui2} \end{bmatrix}$$

where $Cov(\mathbf{u}_{i1}) = \boldsymbol{\Sigma}_{ui1}$ is as given in (8.8), and

$$\boldsymbol{\Sigma}_{ui12} = \begin{bmatrix} Cov(u_{i1}, \mathbf{u}_{i2}) \\ 2x_{i1}Cov(u_{i1}, \mathbf{u}_{i2}) + Cov(u_{i1}^2, \mathbf{u}_{i2}) - Cov(\widehat{\sigma}^2_{ui1}, \mathbf{u}_{i2}) \end{bmatrix}.$$

The last term, $Cov(\widehat{\sigma}^2_{ui1}, \mathbf{u}_{i2})$ would typically be set to $\mathbf{0}$.

With *no error in* $\mathbf{x}_{i2}$, $\boldsymbol{\Sigma}_{ui2}$ and $\boldsymbol{\Sigma}_{ui12}$ are both $\mathbf{0}$ and so we only need $\boldsymbol{\Sigma}_{ui1}$. Estimation of $\boldsymbol{\Sigma}_{ui1}$ is discussed in Section 8.2.2. If $\mathbf{x}_{i2}$ is measured with error but those errors are uncorrelated with the error in X_{i1}, then $\boldsymbol{\Sigma}_{ui12} = \mathbf{0}$. Otherwise the terms in $\boldsymbol{\Sigma}_{ui12}$ need to be estimated. Once an estimate of $\boldsymbol{\Sigma}_{ui}$ is obtained the correction proceeds as in Section 8.2.2 when there is no error in Y and as in Section 8.2.5 (with an extended definition of $\boldsymbol{\Sigma}_{qui}$) if there is error in Y.

Regression calibration. The induced model (under some assumptions) was given in (8.14). To use regression calibration here we first fit a model for $E(\tilde{\mathbf{X}}_i|\tilde{\mathbf{W}}_i)$ which in the linear Berkson case is given by $E(\tilde{\mathbf{X}}_i|\tilde{\mathbf{W}}_i) = \boldsymbol{\lambda}_0 + \boldsymbol{\Lambda}_1\tilde{\mathbf{W}}_i$. After fitting this model (e.g., based on replicates; see Section 6.5.1) we directly obtain imputed values for X_{i1} and $\mathbf{X}_{i2}$, say $\hat{X}_{i1}$ and $\hat{\mathbf{X}}_{i2}$. The imputed value for $\mathbf{X}_{i2}$ equals $\mathbf{x}_{i2} = \mathbf{w}_{i2}$ if there is no error in the additional covariates. The imputed value for X_{i1}^2 is $\widehat{X}_{i1}^2 + \hat{V}(X_{i1}|\tilde{\mathbf{w}}_i)$, where $\hat{V}(X_{i1}|\tilde{\mathbf{w}}_i)$ is an estimate of $V(X_{i1}|\tilde{\mathbf{w}}_i)$. Once again, see Section 6.5.1 for details. As elsewhere, the coefficients used in the imputation may be allowed to change with i, although it is not clear that this is always advantageous with equal numbers of replicates per individual.

Basic regression calibration now fits the linear model using the imputed values. Extended regression calibration would utilize $V(Y_i|\tilde{\mathbf{W}}_i)$ and iteratively reweighted least squares. See Divers et al. (2007) for details on the use of the extended RC method in the ancestry example with no error in the additional covariates.

It is important to note that even if there is no measurement error in $\mathbf{X}_{i2}$, so $\mathbf{W}_{i2} = \mathbf{X}_{i2}$, the imputation for X_{i1} and X_{i1}^2 depend on the additional predictors, in addition to W_{i1}. It is incorrect, in general, to impute X_{i1} and X_{i1}^2 by developing just the model for X_{i1} given W_{i1}! This same theme appears elsewhere, in particular in the use of regression calibration techniques in linear or generalized linear models.

8.3 First order models with interaction

Interaction models involve products of predictors, allowing the effect of one variable to depend on the level of other variables. For motivation we consider examples in two recent papers addressing measurement error in models with interaction.

The first example, from Murad and Freedman (2007), uses data from the Israeli GOH (Glucose Intolerance, Obsesity and Hypertension) Study to fit a model relating fasting homocystiene to levels of folate and vitamin B12 in a model with interaction. Both predictors are measured with error. This paper also provides a good list of additional references treating measurement error in interaction models. In their example the model is actually expressed in terms of the log of folate and B12 and the measurement error was taken to be additive on this scale with constant variance.

A second example comes from Huang et al. (2005). They consider data from the Seychelles Study in which the response is the Bender visual motor gestalt measure and the predictors include (among others) pre-natal and post-natal exposure to mercury (x_1 and x_2, respectively), gender of the child ($x_3 = 0$ or 1), maternal age (x_4) and birth weight (x_5). The model has an interaction of pre and post-natal exposure with gender, which allows changing coefficients in the exposure variables for males and females. The pre-natal measure of exposure comes from a biomarker obtained from maternal hair during pregnancy. This serves as the surrogate for exposures in the fetal brain. They also considered error in post-natal exposure, which was measured through the child's hair, and even allowed a small amount of measurement error in maternal age and birth weight, based on variation in responses on different occasions. All measurement errors were treated as additive and uncorrelated with each other. Values for the measurement error variances were obtained from external sources. The only interaction term they allow is with gender and gender is not subject to misclassification. In this case the methods of Chapter 5 can be directly applied, with slight modifications as described in what follows. If products involving the mismeasured variables are present, as was the case in Murad and Freedman (2007), then we need the more general methods described below.

The main points in treating interaction can be illustrated in the simple setting with two variables. See Section 8.3.4 for some comments on more general settings. In the two variable model,

$$E(Y|x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2.$$

For a fixed x_2, the regression of Y on x_1 changes with x_2, having intercept $\beta_0 + \beta_2 x_2$ and slope $\beta_1 + \beta_3 x_2$. If x_2 is binary then β_2 and β_3 represent the change in the intercept and slope over the two groups indicated by x_2.

Consider an additive measurement error, possibly in both variables, satisfying

$$\tilde{\mathbf{W}}_i = \begin{bmatrix} W_{i1} \\ W_{i2} \end{bmatrix} = \begin{bmatrix} x_{i1} \\ x_{i2} \end{bmatrix} = \tilde{x}_i + \begin{bmatrix} u_{i1} \\ u_{i2} \end{bmatrix},$$

where the u_{ij} have mean 0 with $V(u_{ij}) = \sigma^2_{uij}$ and $cov(u_{i1}, u_{i2}) = \sigma_{ui12}$.

If one or the other variables is measured without error, or if both are measured with error, the measurement errors are uncorrelated, then the overall model is an additive one and the methods from Chapter 5 can be used.

The reason for this is that in either case $\sigma_{ui12} = 0$, so $E(W_{i1}W_{i2}) = x_{i1}x_{i2}$. This means that all of the discussion and methods in Chapter 5 are applicable with $\mathbf{x}_i' = (x_{i1}, x_{i2}, x_{i1}x_{i2})$ and $\mathbf{W}_i' = (W_{i1}, W_{i2}, W_{i1}W_{i2})$. The only thing to do is modify the covariance of $\mathbf{u}_i$, which becomes

$$\mathbf{\Sigma}_{ui} = \begin{bmatrix} \sigma^2_{ui1} & 0 & x_{i2}\sigma^2_{ui1} \\ 0 & \sigma^2_{ui2} & x_{i1}\sigma^2_{ui2} \\ x_{i2}\sigma^2_{ui1} & x_{i1}\sigma^2_{ui2} & x^2_{i1}\sigma^2_{ui2} + x^2_{i2}\sigma^2_{ui1} + \sigma^2_{ui1}\sigma^2_{ui2} \end{bmatrix}. \tag{8.15}$$

With correlated measurement error in both variables, $\sigma_{ui12} \neq 0$ leading to $E(W_{i1}W_{i2}) = x_{i1}x_{i2} + \sigma_{ui12} \neq x_{i1}x_{i2}$. This requires some further developments. Even if $\sigma_{ui12} = 0$, the discussion below is important from the perspective of assessing the impact of the measurement error on naive estimators, since these models were not addressed explicitly in Chapter 5.

8.3.1 Bias in naive estimators

This fairly simple setting generates a rather complex story in terms of bias in the naive estimators when both variables are measured with error.

Consider first the linear Berkson model with

$$E(X_j|w_1, w_2) = \lambda_{j0} + \lambda_{j1}w_1 + \lambda_{j2}w_2 \text{ and } V(X_j|w_1, w_2) = \sigma^2_{j|w} \quad (8.16)$$

for $j = 1$ or 2 and $Cov(X_1, X_2|w_1, w_2) = \sigma_{12|w}$. Using the results from Section 6.7.2, with $\beta_3 = 0$, $\beta_4 = 0$ and relabeling the β_5 there as β_3 here, the induced model is

$$E(Y|w_1, w_2) = \beta_0^* + \beta_1^* w_1 + \beta_2^* w_2 + \beta_3^* w_1 w_2 + \beta_4^* w_1^2 + \beta_5^* w_2^2$$

where

$$\beta_0^* = \beta_0 + \beta_1\lambda_{10} + \beta_2\lambda_{20} + \beta_3\lambda_{10}\lambda_{20},$$

$$\beta_1^* = \beta_1\lambda_{11} + \beta_2\lambda_{21} + \beta_3(\lambda_{11}\lambda_{20} + \lambda_{21}\lambda_{10}),$$

$$\beta_2^* = \beta_1\lambda_{12} + \beta_2\lambda_{22} + \beta_3(\lambda_{12}\lambda_{20} + \lambda_{22}\lambda_{10}),$$

$$\beta_3^* = \beta_3(\lambda_{12}\lambda_{21} + \lambda_{11}\lambda_{22}),\ \beta_4^* = \beta_3\lambda_{11}\lambda_{21}, \text{ and } \beta_5^* = \beta_3\lambda_{12}\lambda_{22}.$$

This leads to:

Even though the original model is linear in x_1, x_2 and x_1x_2, the induced model may be quadratic in w_1 and w_2. Hence, the presence of measurement error can lead to an incorrect conclusion about the order of the model.

If the induced model is quadratic then it does not tell us anything directly about the bias of naive estimators. Instead the approximate bias must be assessed using the expression for $\widehat{\boldsymbol{\beta}}_{naive}$. These bias expressions, and related discussion, appear in Section 8.7.1.

If the Berkson errors are additive, as is often assumed in designed experiments, then $\lambda_{21} = \lambda_{12} = \lambda_{10} = \lambda_{20} = 0$ and $\lambda_{11} = \lambda_{22} = 1$. The induced model is the same as the true model, except for a change in the intercept term. This was seen more generally in Section 6.7.2.

Simulations of biases.

The expressions in 8.7.1 provide a fully general way to approximate biases under a wide array of settings, accommodating the functional case and allowing nonnormal distributions, either for the measurement error or the X's. Unfortunately there are usually no easy final expressions that provide real easy insight into the nature of the biases. Here we provide some simulations illustrating the bias of the naive estimators for the normal structural model with normal measurement errors having constant variances and covariance. Throughout, the true coefficients are $\beta_0 = 0$, $\beta_1 = \beta_2 = 1$ and $\beta_3 = 2$, and the two predictors, X_1 and X_2, are random, each having mean 0 and variance 1 = $\sigma^2_{X_1} = \sigma^2_{X_2}$. We present the results for $n = 100$, the figures for other sample sizes being similar. The correlation between the predictors, the two measurement error variances and the correlation between the measurement errors were varied as follows:

- $\rho_X =$ correlation between X_1 and X_2: 0 to .8 by .2.
- $\kappa_1 = \sigma^2_{X_1}/(\sigma^2_{X_1} + \sigma^2_{u1})$: .2 to 1 by .2.
- $\kappa_2 = \sigma^2_{X_2}/(\sigma^2_{X_2} + \sigma^2_{u2})$: .2 to 1 by .2.
- $\rho_u =$ correlation between u_1 and u_2: 0 to .8 by .2.

Note that the reliability ratios, κ_1 and κ_2, determine the measurement error variances since the variance of X_1 and X_2 are fixed at 1.

Figure 8.3 shows the average estimated values for β_0, β_1 and β_3 over 1000 simulations. We omit β_2 since it equals β_1 and X_1 and X_2 are exchangeable here. In each figure, the cases correspond to varying ρ_X, κ_1, κ_2 and ρ_u in the order above. The first 125 cases correspond to $\rho_X = 0$, the next 125 to $\rho_X = .2$, etc. Within these the cases vary according to the ordering above, e.g., the first 25 are for $\kappa_1 = .2$, etc. The cases with no bias (mean near 0 for β_0, near 1 for β_1, near 2 for β_3) correspond to the cases where both reliability ratios equal 1 and so there is no measurement error. Here are some specific observations:

- The intercept is prone to underestimation when the predictors are uncorrelated but otherwise tends to be overestimated.
- The coefficient of X_1 is generally underestimated with the exception of some cases with high correlation between the two predictors.
- The coefficient for the interaction coefficient β_3 is always underestimated. The bias can be rather severe even when the measurement error is somewhat moderate. For example, consider the case with $\kappa_1 = 1$ (no error in X_1) and $\kappa_2 = .8$ for the measurement error variance in X_2. The mean values for the estimate of β_3 range (over 25 combinations for ρ_X and ρ_u) from .99 to 1.33, or biases from -1.01 to -.67 for estimating a value of 2, the smaller

biases corresponding to smaller correlation between the predictors. *This is a somewhat dramatic result in terms of bias in the interaction coefficient and serves as a strong warning about the consequences of measurement error in the presence of interaction.* The reason for this impact on estimation of β_3 is due to the fact that the measurement error associated with the product $W_1 W_2$ involves terms like $W_1^2 \sigma_{u1}^2$ and $W_2^2 \sigma_{u2}^2$ (see the next section), so the measurement error variances get inflated by the square of the predictors.

8.3.2 *Correcting for measurement error*

We adopt the same main strategies used in the quadratic model: modifying the problem to an additive error problem and using a moment correction, using regression calibration and using SIMEX. As elsewhere, the use of SIMEX simulates measurement errors that account for any potential covariance structure. Murad and Freedman (2007) examine the use of both moment and RC methods in a structural setting, while Huang et al. (2005) consider the moment method only. Both assume constant measurement error variances and covariances.

Moment corrections.

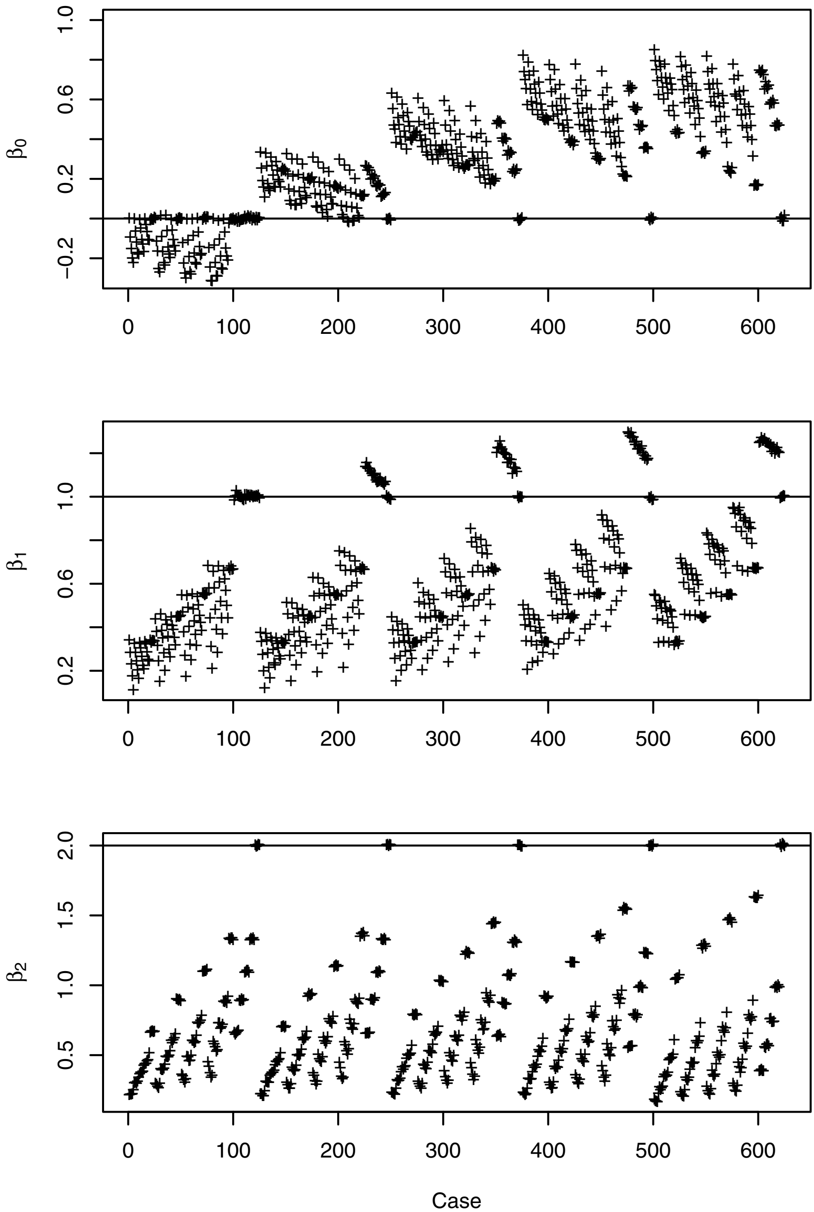
For the moment corrections, define $\mathbf{W}_i' = (W_{i1}, W_{i2}, W_{i1}W_{i2} - \widehat{\sigma}_{ui12})$ and $\mathbf{x}_i' = (x_{i1}, x_{i2}, x_{i1}x_{i2})$. Then $\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i$, with $E(\mathbf{u}_i) = \mathbf{0}$ (assuming $\widehat{\sigma}_{ui12}$ is unbiased for σ_{ui12}) and

$$\mathbf{\Sigma}_{ui} = Cov(\mathbf{u}_i) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \sigma_{ui1}^2 & \sigma_{ui12} & a_{24} \\ 0 & \sigma_{ui12} & \sigma_{ui2}^2 & a_{34} \\ 0 & a_{24} & a_{34} & a_{44} \end{bmatrix}, \tag{8.17}$$

where
$a_{24} = x_{i2}\sigma_{ui1}^2 + x_{i1}\sigma_{ui12} + E(u_{i1}^2 u_{i2})$, $a_{34} = x_{i1}\sigma_{ui2}^2 + x_{i2}\sigma_{ui12} + E(u_{i2}^2 u_{i1})$,
and
$a_{44} = x_{i1}^2\sigma_{ui2}^2 + x_{i2}^2\sigma_{ui1}^2 + 2x_{i1}x_{i2}\sigma_{ui12} + 2x_{i2}E(u_{i1}^2 u_{i2}) + 2x_{i1}E(u_{i2}^2 u_{i1}) + E(u_{i1}^2 u_{i2}^2) - \sigma_{ui12}^2$.

With these definitions for $\mathbf{W}_i$, $\mathbf{x}_i$ and the associated $\mathbf{W}$ and $\mathbf{X}$ matrices, along with an estimate of $\mathbf{\Sigma}_{ui}$, the methods in Chapter 5 can now be used. Note that if both variables are measured with error, estimation of $\mathbf{\Sigma}_{ui}$ requires estimation of some higher moments of the measurement errors. This can be done using replicates. However, if one only has information about the measurement error variances from other sources, then some assumptions would be needed. For example if bivariate normality of the measurement errors is assumed then $E(u_{i1}^2 u_{i2}) = E(u_{i2}^2 u_{i1}) = 0$ and $E(u_{i1}^2 u_{i2}^2) = \sigma_{ui1}^2\sigma_{ui2}^2 + 2\sigma_{ui12}^2$ (e.g., Anderson (1984) p. 49). So, in this case, $a_{24} = x_{i2}\sigma_{ui1}^2 + x_{i1}\sigma_{ui12}$, $a_{34} = x_{i1}\sigma_{ui2}^2 + x_{i2}\sigma_{ui12}$, $a_{44} = x_{i1}^2\sigma_{ui2}^2 + x_{i2}^2\sigma_{ui1}^2 + 2x_{i1}x_{i2}\sigma_{ui12} + \sigma_{ui1}^2\sigma_{ui2}^2 + \sigma_{ui12}^2$.

Figure 8.3 *Illustration of impacts of additive measurement error in a model with interaction. Plot is of mean value of estimate over 1000 simulations versus case number (see text). Top panel is for $\beta_0 = 0$, middle panel for $\beta_1 = 1$ and bottom panel for $\beta_3 = 2$. Horizontal line is at the true value.*



If there is error in Y, the same approach as in Section 8.2.5 can be used with an appropriately modified definition of $\mathbf{\Sigma}_{qui}$.

Regression calibration.

We describe regression calibration under the linear Berkson model given in (8.16), under which $E(X_1X_2|w_1,w_2) = \sigma_{12|w}+E(X_1|w_1,w_2)E(X_2|w_1,w_2)$. Using replicates (or validation data) to estimate the Berkson model, the basic regression calibration approach fits the original model but using imputed values given by:

Quantity	Imputed value
X_{i1}	$\widehat{X}_{i1} = \widehat{\lambda}_{10} + \widehat{\lambda}_{11}w_{i1} + \widehat{\lambda}_{12}w_{i2}$
X_{i2}	$\widehat{X}_{i2} = \widehat{\lambda}_{20} + \widehat{\lambda}_{21}w_{i1} + \widehat{\lambda}_{22}w_{i2}$
$X_{i2}X_{i2}$	$\widehat{\sigma}_{12\vert w_i} + \widehat{X}_{i1}\widehat{X}_{i2}$

Notice that here the RC approach requires an estimate of $Cov(X_1, X_2|w_1,w_2) = \sigma_{12|w}$, and not just the conditional expectations. Murad and Freedman (2007) omit the $\widehat{\sigma}_{12|w_i}$ and use just $\widehat{X}_{i1}\widehat{X}_{i2}$ as the imputed value for $X_{i1}X_{i2}$. If $\widehat{\sigma}_{12|w_i}$ is constant in i this is not an issue in making inferences for the nonintercept term, although its omission will bias the estimate of the intercept as well as the estimate of $E(Y|x_1,x_2)$.

As elsewhere, there are variations on this that could be used. One would be to observation specific $\widehat{\lambda}$'s (see Section 8.2.2), another to use an extended RC method using a model for $V(Y_i|\mathbf{w}_i)$.

Murad and Freedman (2007) provides some simulations comparing the moment and regression calibration methods for structural models with both X_1 and X_2 measured with error, with uncorrelated normal measurement errors with constant variance. They find that the RC method performs better when the true values are normally distributed (the setting under which the expected value of the true values is linear in the observed values, which motivates the RC method) but the reverse is true with nonnormal distribution for the true values.

8.3.3 Example

This example uses the data from a sample of 200 employees from Los Angeles county, which was described in Section 5.6.2. Among other variables it contains measures of cholesterol and weight in 1950 and 1962. Here, for illustration, we consider regressing serum cholesterol in 1962 (sc62) on serum cholesterol in 1950 (sc50) and weight in 1962 (wt62). For convenience in terms of decimal places involved, results are given with weight expressed in 100's of pounds, i.e., the original weight in pounds is divided by 100. A naive analysis, assuming no measurement error in any variables, leads to a significant interaction. See Table 8.4 and Figure 8.4, which shows the fitted lines for sc62 as a function of sc50 at a weight of 100, 150 and 200 pounds, respectively.

To illustrate the effects of measurement error we assume that both sc50 and wt62 are measured with error, with measurement error standard deviations of 15 for sc50 and 5 lbs for the original weight measure (.05 on the transformed scale). Any additive error in the response, sc62, will not alter the inferences on the coefficients, as long as that measurement error is not correlated with the other measurement errors. It will influence estimation of σ^2, as discussed later. The measurement errors in wt62 and sc50 are assumed uncorrelated. However, there is measurement error in their product and that error is correlated with the two original measurement errors. Expressions for these are given following (8.17). These quantities, which are used in the moment correction, are calculated assuming the measurement errors are normally distributed. Regression calibration and SIMEX estimates are also obtained. Throughout the measurement error variances are treated as known. When the measurement error variances are estimated, as they usually are, there is additional uncertainty that would be accounted for. This is not illustrated here, but was in the paper example in Section 8.2.3, where it was assumed that the estimated variances each came from five replicates.

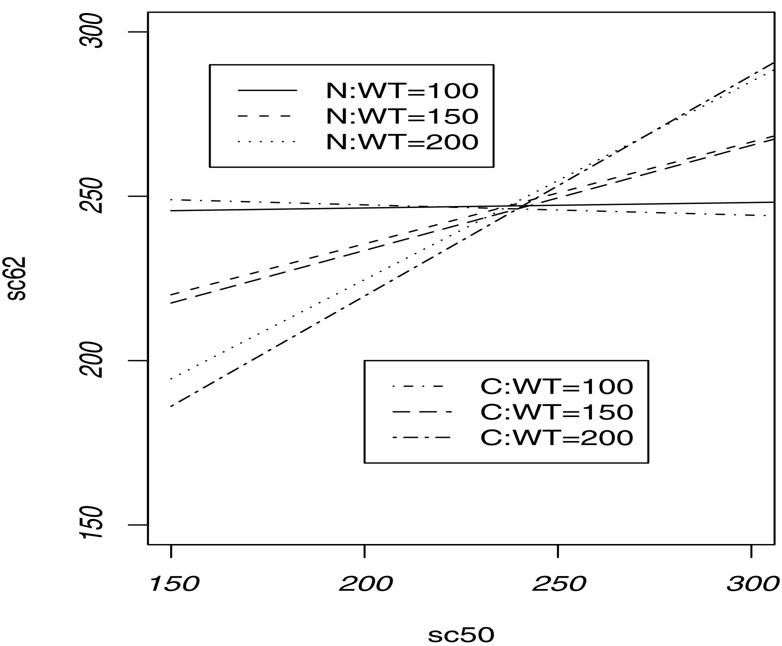
Table 8.4 shows the various estimates, standard errors and confidence intervals with corrected, weight specific, regression lines, from the moment correction, given in Figure 8.4. The general nature of the corrections is similar for all three methods. As a relative percent of the naive estimates the changes in all of the coefficients are somewhat substantial. When viewed in terms of the fitted lines in Figure 8.4, the impact of the correction for measurement error is not quite as dramatic as the changes in the coefficients might imply. While the various corrected estimators are similar there are differences. An examination of the bootstrap means and medians also indicates that there some potential bias concerns here with the RC and SIMEX methods, with an indication of bias away from 0 for the former and towards 0 for the latter. As in all of our two-stage bootstrap analyses, responses are generated for each of the different methods using the estimates for that method, so the bootstrap means and medians can be directly compared to the corrected estimates for that method.

Table 8.4 also shows inferences for σ^2 based on moment and SIMEX corrected estimates using the two-stage bootstrap. The naive estimate is 1655.06 with corrected estimates of 1603 (moment based) and 1610 (using SIMEX). Allowing measurement error in the response will change this. For example, if the sc62 has a measurement error standard deviation of 10, the moment based estimator of σ^2 drops to 1593.

Table 8.4 *Analysis of LA county data, with interaction, fitting* $E(Y|x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$, *where* $Y = sc62$, $x_1 = sc50$ *and* $x_2 = wt62$ *(in 100's of pounds). Measurement error standard deviation is 15 for sc50 and .05 for wt62. Naive and Mom-Robust use analytical based standard errors and confidence intervals. All other standard errors and confidence intervals (Lower, Upper) are based on two-stage bootstrapping. B-Mean and B-Med denote bootstrap mean and median, respectively.*

	Estimate	B-Mean	SE	Lower	Upper	B-Med
Naive						
β_0	382.36	93.81				
β_1	-.5756	.3586				
β_2	-139.11	56.32				
β_3	.5864	.2158				
MOM-R						
β_0	421.87		95.85	234.00	609.73	
β_1	-.7337		0.3798	-1.4782	0.0107	
β_2	-168.20		55.01	-276.02	-60.38	
β_3	.7023		0.2196	0.2720	1.1327	
MOM-N						
β_0	421.87		109.88	206.51	637.22	
β_1	-.7337		0.4219	-1.5606	0.0931	
β_2	-168.20		66.45	-298.45	-37.95	
β_3	.7023		0.2557	0.2012	1.2045	
MOM-B2						
β_0	421.87	426.484	109.76	234.96	645.32	426.58
β_1	-.7337	-.7522	0.4191	-1.5933	-.0189	-.7369
β_2	-168.20	-171.45	66.24	-310.81	-56.96	-170.37
β_3	.7023	0.7153	0.2533	0.2810	1.2374	0.7218
σ^2	1603.02	1604.17	156.54	1319.43	1913.82	1599.32
RC-B2						
β_0	402.67	426.422	108.55	224.44	643.12	424.35
β_1	-.6583	-.7538	0.4127	-1.5341	-.0020	-.7460
β_2	-156.20	-175.92	65.09	-305.03	-57.57	-174.78
β_3	.6551	0.7337	0.2480	0.2947	1.2300	0.7233
SIMEX-B2						
β_0	413.89	400.853	107.44	207.66	615.00	398.21
β_1	-.7009	-.6513	0.4100	-1.4680	0.0769	-.6403
β_2	-162.66	-155.25	64.72	-288.89	-44.43	-154.65
β_3	.6797	0.6516	0.2474	0.2168	1.1556	0.6629
σ^2	1610.03	1610.33	156.56	1320.26	1918.01	1606.88

Figure 8.4 *Interaction example. Plots of expected value in sc62 versus sc50 at weight = 100, 150 and 200. N indicates naive fit. C indicates corrected fit based on moment corrected estimates.*



8.3.4 More general interaction models

The results of the previous section naturally extend to models involving additional predictors and multiple interaction terms involving products of two variables. SIMEX and the regression calibration methods extend in a natural way. For the latter the imputed value for $X_{ij}X_{ik}$ is $\widehat{X}_{ij}\widehat{X}_{ik} + \widehat{\sigma}_{jk|\mathbf{w}_i}$, where $\widehat{\sigma}_{jk|\mathbf{w}_i}$ is an estimate of the covariance of X_{ij} and X_{ik} given $\mathbf{w}_i$. For the moment corrections, if all of the products involve just one variable measured with error, or if all of the measurement errors are independent of one another, then $E(W_{ij}W_{ik}|\mathbf{x}_i) = x_{ij}x_{ik}$ and the problem can be treated in the context of additive errors. However, the covariance matrix of the measurement errors associated with the $\mathbf{W}$ vector needs to be modified, along the lines of (8.15). In other cases terms like $W_{ij}W_{ik} - \widehat{\sigma}_{uijk}$ are introduced into $\mathbf{W}_i$ and additional terms are added to $\mathbf{\Sigma}_{ui}$, similar to those given in (8.17) for two variables.

Finally, we comment briefly on models that include both interaction and quadratic terms. The general strategies for attacking this problem are similar to those used for quadratic and first order models with interaction. For illustration, consider the simple case

$$E(Y|x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2.$$

Section 6.7.2 provided a complete discussion of the the properties of naive estimators for this model under the linear Berkson model. In cases where the induced approach is not applicable, Section 8.7.1 provides expressions for the approximate bias of the naive estimators.

For correcting, the use of SIMEX is identical to other problems with multiple predictors. For regression calibration we combine the expressions given for quadratic regression and models with interaction, leading to the imputations below. Again, with changing variances the $\widehat{\lambda}$'s can be made observation specific.

Quantity	Imputed value
X_{i1}	$\widehat{X}_{i1} = \widehat{\lambda}_{10} + \widehat{\lambda}_{11} w_{i1} + \widehat{\lambda}_{12} w_{i2}$
X_{i2}	$\widehat{X}_{i2} = \widehat{\lambda}_{20} + \widehat{\lambda}_{21} w_{i1} + \widehat{\lambda}_{22} w_{i2}$
X_{i1}^2	$\widehat{\sigma}^2_{1\|w_i} + \widehat{X}_{i1}^2$
X_{i2}^2	$\widehat{\sigma}^2_{2\|w_i} + \widehat{X}_{i2}^2$
$X_{i2}X_{i2}$	$\widehat{\sigma}_{12\|w_i} + \widehat{X}_{i1}\widehat{X}_{i2}$

To use the moment method of correcting, with $\mathbf{x}_i' = (x_{i1}, x_{i2}, x_{i1}^2, x_{i2}^2, x_{i1}x_{i2})$, define

$$\mathbf{W}_i = \begin{bmatrix} W_{i1} \\ W_{i2} \\ W_{i1}^2 - \widehat{\sigma}^2_{ui1} \\ W_{i2}^2 - \widehat{\sigma}^2_{ui2} \\ W_{i1}W_{i2} - \widehat{\sigma}_{ui12} \end{bmatrix}.$$

Then $\mathbf{W}_i|\mathbf{x}_i = \mathbf{x}_i + \mathbf{u}_i$ with $E(\mathbf{u}_i) = \mathbf{0}$ and

$$\mathbf{u}_i = \begin{bmatrix} u_{i1} \\ u_{i2} \\ 2x_{i1}u_{i1} + u_{i1}^2 - \widehat{\sigma}^2_{ui1} \\ 2x_{i2}u_{i2} + u_{i2}^2 - \widehat{\sigma}^2_{ui2} \\ x_{i1}u_{i2} + x_{i2}u_{i1} + u_{i1}u_{i2} - \widehat{\sigma}_{ui12} \end{bmatrix}.$$

We now obtain $\mathbf{\Sigma}_{ui} = Cov(\mathbf{u}_i)$, estimate it and proceed as in the earlier illustrations; see for example (8.9) and, if there is error in the response, (8.11).

8.4 General nonlinear functions of the predictors

This section considers models that are linear in the parameters but involve nonlinear functions of variables that are measured with additive error. That is,

$$E(Y_i|\mathbf{x}_i) = \beta_0 + \sum_{j=1}^{p} \beta_j g_j(\mathbf{x}_i) = \beta_0 + \beta_1' \boldsymbol{g}_i, \tag{8.18}$$

where

$$\mathbf{x}_i' = (x_{i1}, \ldots, x_{ik}) \text{ and } \boldsymbol{g}_i' = [g_1(\mathbf{x}_i), \ldots, g_p(\mathbf{x}_i)].$$

Here, we have adopted a different notation from earlier in the chapter where now $\mathbf{x}$ contains the original predictors measured with additive error and $\boldsymbol{g}$ is composed of the functions of them that enter into the linear regression model. The g_j are known functions of the original predictors and at least one of the functions is nonlinear. If the g functions are quadratic and/or use products then the developments in Sections 8.2 and 8.3 apply. The concern here is with other nonlinear functions, such as $log(x)$, $1/x$ or e^x. Some of the predictors may still enter linearly.

Models like those in (8.18) enjoy widespread usage with the use of log transformations being especially popular in biological, economic and ecological applications. Here are a few examples.

- The Cobb-Douglas production model (see Greene (1990, p216-217)) assumes
$$E(log(Q)) = \beta_0 + \beta_1 log(L) + \beta_2 log(C),$$
where Q, L and C denotes output, labor and capital, respectively. Note that this is not exactly equivalent to $E(Q|L,C) = \tilde{\beta}_0 L^{\beta_1} C^{\beta_2}$.
An extension of this model is the "translog function" which adds the terms $\beta_3 log(L)^2 + \beta_4 log(C)^2 + \beta_5 log(L) log(C)$. If we had additivity on the log-scale then this could be handled using the earlier methods for additive error in models involving quadratic and/or interaction terms.
- Schennach (2004) considers an example from the Consumer Expenditure Survey where with Y denoting the consumer's expenditure for a group and x the total expenditure, $E(Y|x) = \beta_0 + \beta_1 x + \beta_1 x log(x)$.
- For a final example, we point to Example 3.7 in Montgomery and Peck (1992) in which the power output of a windmill (Y) is modeled as function of wind speed (x) using $E(Y|x) = \beta_0 + \beta_1(1/x)$.

The original observed values are $\mathbf{W}_i' = (W_{i1}, \ldots, W_{ik})$ and we denote

$$\mathbf{Z}_i' = [g_1(\mathbf{W}_i), \ldots, g_p(\mathbf{W}_i)].$$

On the original scale, the measurement error is assumed additive with

$$\mathbf{W}_i = \mathbf{x}_i + \mathbf{u}_i, \quad E(\mathbf{u}_i) = \mathbf{0}, \quad Cov(\mathbf{u}_i) = \mathbf{\Sigma}_{ui}.$$

If there is error in the response, it is assumed additive with

$$D_i = y_i + q_i, \quad E(q_i) = 0, \quad V(q_i) = \sigma^2_{qi}, \quad Cov(\mathbf{u}_i, q_i) = \mathbf{\Sigma}_{uqi}.$$

This is primarily for simplifying the presentation, but can be modified to allow a nonlinear function in the response.

8.4.1 Bias of naive estimators

The naive estimators result from regressing Y_i (or D_i if there is error in the response) on the observed z'_{ij}s, where $z_{ij} = g_j(\mathbf{w}_i)$. The induced model under a structural assumption, and with no error in the response, is

$$E(Y_i|\mathbf{w}_i) = \beta_0 + \sum_j \beta_j E[g_j(\mathbf{X}_i)|\mathbf{w}_i].$$

Typically $E[g_j(\mathbf{X}_i|\mathbf{w}_i)]$ cannot be written down exactly and even if it could be $E(Y_i|\mathbf{w}_i)$ is usually not of the form in (8.18). So, this does not help in assessing the bias of the naive estimators. Defining the design matrix as

$$\mathbf{W} = \begin{bmatrix} 1 & g_1(\mathbf{W}_1) & . & . & g_p(\mathbf{W}_1) \\ . & . & . & . & . \\ 1 & g_1(\mathbf{W}_n) & . & . & g_p(\mathbf{W}_n) \end{bmatrix}, \tag{8.19}$$

then $\widehat{\boldsymbol{\beta}}_{naive} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}\mathbf{D}$. With $\mathbf{X}$ defined in a similar manner to $\mathbf{W}$, and treated as fixed, $E(\mathbf{W}'\mathbf{D}) = E(\mathbf{W}')\mathbf{X}\boldsymbol{\beta} + \sum_i E(\mathbf{W}_i q_i)$. This implies that the naive estimators are approximately estimating

$$\boldsymbol{\beta}^* = \left[(E(\mathbf{W}'\mathbf{W}))^{-1}[E(\mathbf{W}')\mathbf{X}\boldsymbol{\beta} + \sum_i E(\mathbf{W}_i q_i)]\right]$$

leading to an approximate bias of

$$(E(\mathbf{W}'\mathbf{W}))^{-1}\{[E(\mathbf{W}')\mathbf{X} - E(\mathbf{W}'\mathbf{W})]\boldsymbol{\beta} + \sum_i E(\mathbf{W}_i q_i)\}. \tag{8.20}$$

With no error in the response, or error uncorrelated with errors in the predictors, $E(\mathbf{W}_i q_i) = \mathbf{0}$, $\boldsymbol{\beta}^* = (E(\mathbf{W}'\mathbf{W}))^{-1}E(\mathbf{W}')\mathbf{X}\boldsymbol{\beta}$ and the bias is approximately $(E(\mathbf{W}'\mathbf{W}))^{-1}[E(\mathbf{W}')\mathbf{X} - E(\mathbf{W}'\mathbf{W})]\boldsymbol{\beta}$. This requires expressions for the terms $E[g_j(\mathbf{W}_i)]$, $E[g_j(\mathbf{W}_i)g_m(\mathbf{W}_i)]$ and $E[g_j(\mathbf{W}_i)q_i]$, which usually depend on the true x's, as well as the measurement error variances/covariances. For most nonlinear functions some approximations are needed.

8.4.2 Correcting for measurement error

We concentrate primarily on the same options used elsewhere: moment corrections, regression calibration, SIMEX and modified estimating equations. Likelihood methods, which we do not implement here, are discussed broadly in Section 8.7.2. SIMEX proceeds as described in many other places and requires no further discussion here.

Using transformed replicates.

If W is unbiased for x then $g(W)$ is unbiased for $E(g(W)|x) = h(x)$. This only equals $g(x)$ if g is linear. If we have replicate measures for x, and the investigator's interest is in a model involving $g(x)$, then simply transforming replicate measures is not always the correct way to proceed. However, that is fine if the regression model of interest is stated in terms of $h(x)$ or, if for a replicate W_j, $E(g(Wj)) \approx g(x)$. This was discussed in detail for the log function in Section 6.4.7 where it was seen that this approximation is often reasonable. The same often happens for the reciprocal; that is $E(1/W_j) \approx 1/x$. This happens since the approximate bias is σ_u^2/x^3 and in many applications the x's are large enough to lead to negligible bias. Of course, this is not true for small x. So, there are many settings where working with the transformed replicates and an additive error model is satisfactory. However, this should be examined closely on a case by case basis. If this fails to work, or there are no replicates, other strategies need to be considered.

Using an approximate additive model.

Since $E(g_j(\mathbf{W}_i)|\mathbf{x}_i)$ is not $g_j(\mathbf{x}_i)$, the measurement error model on the transformed values is not additive. One attempt to fix the problems is to adjust $g(\mathbf{W}_i)$ to get an approximately unbiased estimator for $g(x_i)$, get the associated measurement error covariance matrix and use the methods from Chapter 5 assuming additive error. With replicates the $\mathbf{W}_i$ will use a mean over replicates, so this assumption is weaker than the assumption that a replicate is essentially unbiased.

This does not help much, however, unless $g_j(\mathbf{W}_i)$ is taken to be essentially unbiased for $g_j(\mathbf{x}_i)$. Consider using an approximation arising from a second order Taylor series expansion, $E(g_j(\mathbf{W}_i)|\mathbf{x}_i) \approx g_j(\mathbf{x}_i) + b_{ij}$. Then b_{ij} will usually depend on the x's and the variances/covariances of the original measurement errors. However, in order to estimate the bias to remove it, other nonlinear functions of the x's need to be estimated. For example, $E[log(W_i)] \approx log(x_i) - \sigma_{ui}^2/2x_i^2$, requiring an estimate of $\sigma_{ui}^2/2x_i^2$ to remove the approximate bias. A simple approach would be to just substitute the W's for the x's in making this correction, but even then we are faced with the task of finding the covariance of the measurement errors associated with these corrected values.

If we do assume $E(g_j(\mathbf{W}_i)) \approx g_j(\mathbf{x}_i)$, then, with $\mathbf{Z}_i$ and $\boldsymbol{g}_i$ as defined earlier,

$$\mathbf{Z}_i = \boldsymbol{g}_i + \mathbf{u}_{gi} \text{ with } E(\mathbf{u}_{gi}) \approx \mathbf{0} \text{ and } Cov(\mathbf{u}_{gi}) = \mathbf{\Sigma}_{ugi}.$$

Using the delta method

$$\mathbf{\Sigma}_{ugi} \approx \begin{bmatrix} c_{11i} & c_{12i} & . & c_{1pi} \\ c_{12i} & c_{22i} & . & c_{2pi} \\ . & . & . & . \\ c_{1pi} & c_{p2i} & . & c_{ppi} \end{bmatrix}, \tag{8.21}$$

where $c_{jmi} = Cov(g_j(\mathbf{W}_i), g_m(\mathbf{W}_i)) \approx \mathbf{d}'_{ji}\mathbf{\Sigma}_{ui}\mathbf{d}_{mi}$ and the kth element of the vector $\mathbf{d}_{ji}$ is $\partial g_j(\mathbf{x}_i)/\partial x_{ik}$.

So, one approach here is the following:

i) Determine $\mathbf{\Sigma}_{ugi}$. This usually requires some approximations.

ii) Estimate $\mathbf{\Sigma}_{ugi}$. This can be done using an estimate for $\mathbf{\Sigma}_{ui}$, obtained from replicates on the original variables, but in addition any functions of the x's that occur in the $\mathbf{d}$ vectors must be estimated.

One can see the obvious problems with using i) and ii) above. An alternative is to estimate $\mathbf{\Sigma}_{ugi}$ through simulation. That is first generate $\mathbf{W}_{Sim} = \mathbf{W}_i + \mathbf{u}_{im}$ where $\mathbf{u}_{im}$ has mean 0 and covariance $\widehat{\mathbf{\Sigma}}_{ui}$ and $m = 1$ to M (large). Then form $\mathbf{Z}_{Sim}$ for each m and take $\widehat{\mathbf{\Sigma}}_{ugi}$ to be the sample covariance of $\mathbf{Z}_{Si1}, \ldots, \mathbf{S}_{SiM}$.

iii) Apply the methods from Chapter 5, where $\mathbf{W}_i$ and $\widehat{\mathbf{\Sigma}}_{ui}$ in Chapter 5 are replaced with $\mathbf{Z}_i$ and the estimate of $\mathbf{\Sigma}_{ugi}$, respectively.

If there is also additive error in the response that is correlated with the error in the predictors then $\mathbf{\Sigma}_{uqi}$ in Chapter 5 is replaced by the vector whose jth component is an estimate of $Cov(g_j(\mathbf{W}_i), q_i) \approx d'_{ji}\mathbf{\Sigma}_{uqi}$. Recall that $\mathbf{\Sigma}_{uqi}$ contains the covariances of the errors on the original scale with the error in the response.

Regression calibration.

As elsewhere this depends formally on a structural setting and no error in the response. With no error in the response, the induced model is

$$E(Y_i|\mathbf{w}_i) = \beta_0 + \sum_j \beta_j E[g_j(\mathbf{X}_i)|\mathbf{w}_i]$$

and regression calibration proceeds by fitting a linear regression of Y_i on $\widehat{g}_{i1}, \ldots, \widehat{g}_{ip}$ where $\widehat{g}_{ij}$ is an estimate of $E(g_j(\mathbf{X}_i)|\mathbf{w}_i)$. Replication or validation data can be used to first fit a model for $E(\mathbf{X}_i|\mathbf{w}_i)$ and $Cov(\mathbf{X}_i|\mathbf{w}_i)$, as used elsewhere.

One strategy then is to simply use $\widehat{g}_{ij} = g_j(\widehat{\mathbf{X}}_i)$. This is RC in its simplest form. An extension of simple RC comes from writing $E(g_j(\mathbf{X}_i)|\mathbf{w}_i) \approx g_j(E(\mathbf{x}_i|\mathbf{w}_i)) + B_{ij}$ where B_{ij} is an additional term and use $\widehat{g}_{ij} = g_j(\widehat{\mathbf{X}}_i) + \widehat{B}_{ij}$. With squares and/or product terms we had exact expressions for the B terms, but otherwise some approximation is needed. With either strategy, expanded regression calibration would incorporate a model for $V(Y_i|\mathbf{w}_i)$ into the model fitting.

Finding an analytical expression for the covariance of the RC estimators is complicated (see Section 6.10 for some general discussion), but bootstrapping is always an option for inference.

Modified estimating equations.

This setting is one where modifying the estimating equations is an attractive approach since the correction ends up being relatively simple. Define $\boldsymbol{g}_i^{*'} = (1, \ \boldsymbol{g}_i')$ and $\mathbf{Z}_i^{*'} = (1, \ \mathbf{Z}_i')$. Then $m(\mathbf{x}_i, \boldsymbol{\beta}) = \boldsymbol{g}_i^{*'}\boldsymbol{\beta}$ and in the notation of Section 6.13 (where we note that $\mathbf{Z}_i$ has a different meaning than here) $\boldsymbol{\Delta}(\mathbf{W}_i, \boldsymbol{\beta}) = \mathbf{Z}_i^*$. A simple calculation shows that the expected value of the naive estimating equation, given in (6.56), is $C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}) = \mathbf{H}_i\boldsymbol{\beta}$, where

$$\mathbf{H}_i = \begin{bmatrix} 0 & -\mathbf{B}_i' \\ 0 & -Cov(\mathbf{Z}_i) + E(\mathbf{Z}_i)\mathbf{B}_i' \end{bmatrix},$$

and $\mathbf{B}_i = E(\mathbf{Z}_i) - \boldsymbol{g}_i$ is the bias in $\mathbf{Z}_i$ as an estimator of $\boldsymbol{g}_i$. This bias arises from the use of nonlinear functions.

With $\mathbf{W}$ defined as in (8.19) (this has the nonlinear functions in it), the naive estimating equations are $\mathbf{W}'\mathbf{Y} - \mathbf{W}'\mathbf{W}\boldsymbol{\beta} = \mathbf{0}$. The corrected equations are

$$\mathbf{W}'\mathbf{Y} - \mathbf{W}'\mathbf{W}\boldsymbol{\beta} - \mathbf{H}\boldsymbol{\beta} = \mathbf{0} = \mathbf{W}'\mathbf{Y} - (\mathbf{W}'\mathbf{W} + \mathbf{H})\boldsymbol{\beta} = \mathbf{0},$$

where $\mathbf{H} = \sum_i \mathbf{H}_i$. Given an estimate $\widehat{\mathbf{H}} = \sum_i \widehat{\mathbf{H}}_i$ this leads to a closed form solution for the MEE estimator of

$$\widehat{\boldsymbol{\beta}}_{MEE} = \mathbf{W}'\mathbf{Y}(\mathbf{W}'\mathbf{W} + \widehat{\mathbf{H}})^{-1}\mathbf{W}'\mathbf{Y}. \tag{8.22}$$

To estimate $\mathbf{H}_i$, $E(\mathbf{Z}_i)$ is replaced by $\mathbf{Z}_i$ and estimates $\widehat{\mathbf{B}}_i$ and $\widehat{\boldsymbol{\Sigma}}_{ugi}$ (which estimates $Cov(\mathbf{Z}_i)$) are needed. These latter two can be obtained either using approximations or via simulations. The estimation of $\boldsymbol{\Sigma}_{ugi}$ via simulation was described above for the moment correction method. Using the notation there the simulated estimate of $\mathbf{B}_i$ would be $\widehat{\mathbf{B}}_i = \bar{\mathbf{Z}}_{Si} - \mathbf{Z}_i$, where $\bar{\mathbf{Z}}_{Si}$ is the mean of the simulated $\mathbf{Z}_{Sik}$.

If B_i is taken to be 0 then the MEE method is equivalent to the moment correction above. Notice that the MEE method is simply based on corrected estimates of $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{Y}$. There are other corrected estimating equation approaches that can be considered, as mentioned in Chapter 6. We also point the interested reader to a method proposed by Schennach (2004), which employs another technique (see Stefanski (1989) for related work) for estimating $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{Y}$ under some weak assumptions on the measurement errors. This is an elegant approach, but complicated to both describe and implement and not treated further here.

8.4.3 Linear regression in log(x)

This section makes the preceding discussion more explicit by considering a model that is linear in $log(x)$ rather than x, with no error in the response. This fairly simple setting is useful for providing insight into the main issues in treating models involving nonlinear functions of predictors measured with additive error, and so we work through it in some detail and provide a comprehensive look at an example.

Biases in naive estimators.

If X is random and conditional independence holds, the induced model is $E(Y|x) = \beta_0 + \beta_1 E(log(X)|w))$. This is not in the same form as the original model, even if we are willing to approximate it by $\beta_0 + \beta_1 log(E(X|w))$. To examine the bias we use (8.20), where with $Z_i = log(W_i)$ and the x_i's treated as fixed the approximate bias in $\widehat{\boldsymbol{\beta}}_{naive}$ is

$$\beta_1 \begin{bmatrix} n & \sum_i E[Z_i] \\ E[Z_i] & \sum_i E[Z_i^2] \end{bmatrix}^{-1} \begin{bmatrix} \sum_i (log(x_i) - E[Z_i]) \\ (\sum_i (log(x_i)E[Z_i]) - \sum_i E[Z_i^2]) \end{bmatrix}.$$

This needs further approximations. For example using a second order Taylor series, $E[Z_i] \approx log(x_i) - \sigma_{ui}^2/2x_i^2$ and $V[Z_i] \approx \sigma_{ui}^2/x_i^2$. This in turn leads to an approximate expression for $E[Z_i^2] = V[Z_i] + E[Z_i]^2$. This is referred to as approximation 1 in the simulations below.

Another approximation to the bias comes from simply taking $E(log(W_i)) \approx log(x_i)$. This brings us back to an additive error situation but with a measurement error variance on the log-scale of approximately σ_{ui}^2/x_i^2. This is approximation 2 in the simulations below.

Finally, if W_i were distributed log-normal with mean x_i and variance σ_{ui}^2, then $E[log(W_i)] = log(x_i) - V(log(W_i)/2$ and $V[log(W_i)] = log((\sigma_{ui}^2 + x_i^2)/x_i^2)/2$. These are exact expressions.

Notice that if we want to approximate the bias for a structural model with random X then x_i is replaced by X_i and expectations are taken over both X_i and W_i. This adds another level of difficulty since X_i^2 appears in the denominator.

Simulations.

To illustrate the bias in naive estimators we present some simulations using values motivated by the egg mass/defoliation example in Chapter 4 but now assuming the model for defoliation is linear in log(x), where x = egg mass density. For this illustration we fixed the x's using 12 plots with true x ranging from 21 to 72. For each setting 10000 simulations were run with $\beta_0 = -30$, $\beta_1 = 25$ and σ (the standard deviation of the error in the equation) = 5. Two versions of measurement error are used. The first set is $\sigma_{ui} = cv * x_i$, which models the measurement error as having a fixed coefficient of variation (cv) relative to the true value. In the second setting σ_{ui} is constant, either 3, 4 or 5. In this application the assumption of the measurement error having constant coefficient is probably more reasonable. With constant cv, the approximate bias for the slope is the same using the two approximations. As expected the biases increase as the amount of measurement error increases and are somewhat severe at what could be viewed as moderate amounts of measurement error. The analytical approximations to the bias (described above) also start to do worse as the amount of measurement error increases. Note that this is a functional setting with only 12 observations having fixed true values, chosen to actually test the approximations in challenging setting, and so the failure of the approximations at larger measurement error variance is not terribly surprising. Approximate biases based on the assumption that $log(W)$ is normal (when in the simulations it is W that is normal) are quite similar to those given for approximation 1. It is interesting to note that the approximate biases based on assuming that $log(W)$ is approximately unbiased for $log(x)$ does better than the one trying to include an additional term.

Correcting for measurement error.

In applying the *moment correction*, we suppose that the measurement error is additive on the log scale; that is $E(log(W_i)) = log(x_i)$ (only approximately true) with variance $\sigma^2_{ugi} \approx \sigma^2_{ui}/2x_i^2$. The correction for simple linear regression with additive error is then applied (see Chapters 4 and 5), where x and W are now $log(x)$ and $log(W)$, respectively, and σ^2_{ui} there is now σ^2_{ugi}. This latter variance must be estimated, which in our applications is done analytically using $\widehat{\sigma}^2_{ugi} = \widehat{\sigma}^2_{ui}/(W_i^2 - \widehat{\sigma}^2_{ui})$, based on the fact that $W_i^2 - \widehat{\sigma}^2_{ui}$ estimates x_i^2. Of course, this does not mean that $1/(W_i^2 - \widehat{\sigma}^2_{ui})$ is a good estimate of $1/x_i^2$. This illustrates a weak point in the analytical approach to the moment correction: the need to estimate the approximate variance, which often involves nonlinear functions of the true values. An alternative is to estimate this variance via simulation as described in the preceding section.

Table 8.5 *Simulated mean and bias = Mean - true value of naive estimators based on 10000 simulations for model linear in* $log(x)$ *with* $\beta_0 = -30$ *and* $\beta_1 = 25$ *and* $\sigma = 5$. *Measurement error standard deviation is* $\sigma_{ui} = cv{*}x_i$ *for cv =.1, .2, .3 or .4 or* σ_u *= 3, 5 or 7. App1 and App2 are analytical approximations 1 and 2 to the bias, as described in text.*

Parameter	σ_{ui}	Mean	Bias	App1	App2
$\beta_0 = -30$					
	$0.1x_i$	-30.2700	-0.270	0.000	0.000
	$0.2x_i$	-25.3442	4.656	5.932	5.815
	$0.3x_i$	-12.1870	17.813	19.986	19.591
	$0.4x_i$	24.3487	54.349	49.019	48.050
	3	-26.239	3.761	4.821	4.181
	4	-23.874	6.125	8.210	7.182
	5	-20.319	9.680	12.172	10.755
$\beta_1 = 25$					
	$0.1x_i$	25.0703	0.070	0.000	0.000
	$0.2x_i$	23.7745	-1.225	-1.560	-1.560
	$0.3x_i$	20.3272	-4.673	-5.257	-5.257
	$0.4x_i$	10.5995	-14.401	-12.894	-12.894
	3	24. 0064	-0.994	-1.271	-1.122
	4	23.3755	-1.625	-2.165	-1.927
	5	22.4435	-2.556	-3.209	-2.886

For *regression calibration*, we first impute a value for X_i, use that to create an imputed predictor $\widehat{g}_i$, for $log(x_i)$, and then regressing Y_i on $\widehat{g}_i$. Table 8.6 shows four variations of regression calibration that can be considered, depending on whether a bias term or individual measurement error variances are used.

Table 8.6 *Regression calibration options for model linear in* $log(x)$.

Method	$\widehat{X}_i$	$\widehat{g}_i$
1	$\widehat{\lambda}_0 + \widehat{\lambda}_1 W_i$	$log(\widehat{X}_i)$
2	$\widehat{\lambda}_{0i} + \widehat{\lambda}_{1i} W_i$	$log(\widehat{X}_i)$
3	$\widehat{\lambda}_0 + \widehat{\lambda}_1 W_i$	$log(\widehat{X}_i) - \widehat{\sigma}^2_X/2\widehat{X}^2_i$
4	$\widehat{\lambda}_{0i} + \widehat{\lambda}_{1i} W_i$	$log(\widehat{X}_i) - \widehat{\sigma}^2_X/2\widehat{X}^2_i$

The *MEE estimator* is computed by using (8.22) where

$$\widehat{\mathbf{H}}_i = \begin{bmatrix} 0 & \widehat{B}_i \\ 0 & \widehat{V}(log(W_i)) + log(W_i)\widehat{b}_i' \end{bmatrix},$$

where $\widehat{B}_i$ is an estimate of $E(log(W_i)) - x_i$ and $\widehat{V}(log(W_i))$ is an estimate of $V(log(W_i))$. One option is based on using the approximations $B_i \approx -\sigma^2_{ui}/(2x_i^2)$ and $V(log(W_i)) \approx \sigma^2_{ui}/x_i^2$. These require estimating $1/x_i^2$, which was discussed in the moment correction above. The other option is to estimate these using simulation.

Capital/output example.

To illustrate fitting with a log term as a predictor we consider production data from Griffiths et al. (1993, p. 724) for which we will consider a model in which $y = log(output)$ is linear in $z = log(x)$, where x is capital. We will assume output is measured without error, and there is additive error in the measure of capital, $W_i = x_i + u_i$. The estimated measurement error variance is set to be 5 percent of the capital value, i.e., $\widehat{\sigma}^2_{ui} = .05w_i$. The data are shown in Table 8.7, where we have eliminated a few outlying values from the original data. Also given are approximate values for $\sigma^2_{ugi} = V(log(W_i)|x_i)$ and the bias in $log(W_i)$ as an estimator of $log(x_i)$. See also Figure 8.5. While we have assumed that log(output) is linear in log(capital) for illustration, there is some hint that this breaks down as capital increases and other nonlinear models might be entertained.

Table 8.8 show results from fitting in a variety of ways including naive, moment correction (MOM), the four versions of regression calibration described above, SIMEX using both an average measurement error variances and individual measurement error variances (the latter denoted by I), and the MEE approach. Each of the MOM and MEE were implemented two ways. The first, which uses analytical expressions for $V(log(W_i)$ and bias in $log(W_i)$, is denoted with -A, and the second, using simulated values for these quantities (based on 5000 simulations), is denoted with -S. The Simex uses 500 simulations at each λ. The bootstrapping is done using a two-stage bootstrap with 500 bootstrap samples. This is a little low due to the computational time involved. As in our other examples, separate outcomes are generated for each method based on the original estimates for that method so that bias can be assessed by comparing the bootstrap mean (or median to avoid potential problems with outliers) to the original estimates.

This is a relatively challenging situation to handle because of the small sample size of 26. Here are a few observations on the results:

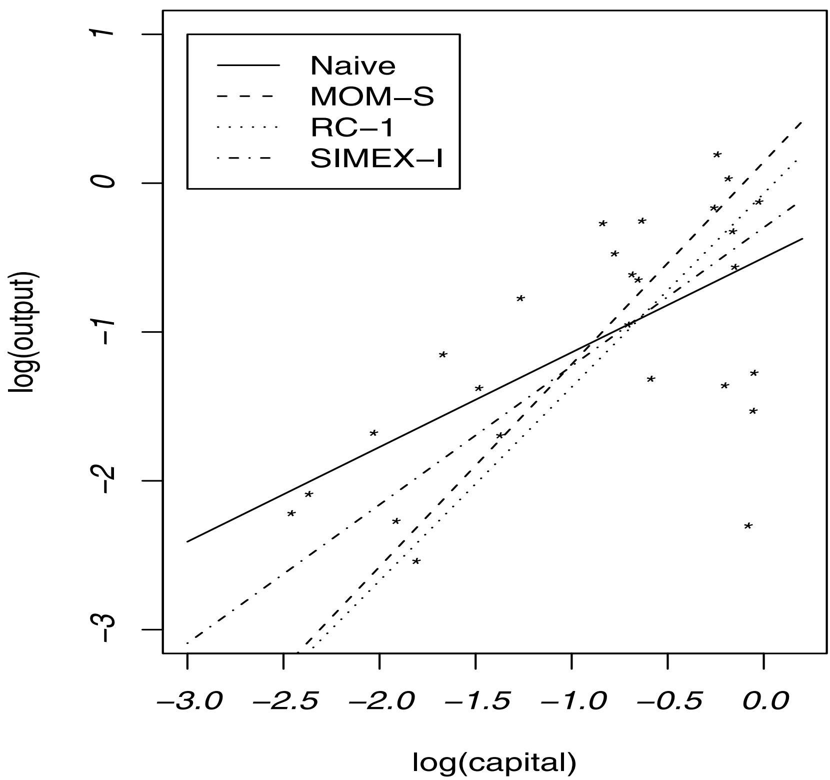
- The moment based approach using the simulation technique appears to have smaller bias and variability than that based on analytical expressions.

Table 8.7 *Data for capital-output example. First two columns extracted from Griffiths et al. (1993). Bias is an estimate of bias in estimate of log(capital).*

log(output)	capital	$log(capital)$	$\widehat{\sigma}^2_{ui}$	$\widehat{\sigma}^2_{ugi}$	Bias
-1.359	0.802	-0.221	0.040	0.066	-0.033
0.193	0.771	-0.260	0.039	0.069	-0.035
-0.165	0.758	-0.277	0.038	0.071	-0.035
-0.473	0.452	-0.794	0.023	0.124	-0.062
-0.563	0.845	-0.168	0.042	0.063	-0.031
-2.218	0.084	-2.477	0.004	1.471	-0.735
-1.315	0.546	-0.605	0.027	0.101	-0.050
-1.377	0.223	-1.501	0.011	0.289	-0.145
-2.539	0.161	-1.826	0.008	0.450	-0.225
-0.324	0.836	-0.179	0.042	0.064	-0.032
-1.530	0.930	-0.073	0.047	0.057	-0.028
-1.151	0.185	-1.687	0.009	0.370	-0.185
-0.951	0.485	-0.724	0.024	0.115	-0.057
-1.695	0.249	-1.390	0.012	0.251	-0.126
-0.649	0.511	-0.671	0.026	0.108	-0.054
-0.270	0.425	-0.856	0.021	0.133	-0.067
0.031	0.817	-0.202	0.041	0.065	-0.033
-0.125	0.958	-0.043	0.048	0.055	-0.028
-0.773	0.277	-1.284	0.014	0.220	-0.110
-1.678	0.129	-2.048	0.006	0.633	-0.316
-2.301	0.906	-0.099	0.045	0.058	-0.029
-2.270	0.145	-1.931	0.007	0.526	-0.263
-0.253	0.521	-0.652	0.026	0.106	-0.053
-0.614	0.495	-0.703	0.025	0.112	-0.056
-2.089	0.092	-2.386	0.005	1.190	-0.595
-1.275	0.934	-0.068	0.047	0.057	-0.028

- RC-2 and SIMEX-I, both of which make use of the individual measurement error variances, are preferred, both in terms of bias and variability, to their counterparts (RC-1 and SIMEX) which use an average measurement error variance. This is probably not surprising given that the measurement error variances differ considerably over observations.
- The RC-3 and RC-4 methods perform pretty badly. Both of these try to account for potential bias in $log(W_i)$ as an estimator of $log(x)$ in doing imputation. As noted earlier, the approximate bias involves $1/x_i^2$. This can be difficult to estimate and that fact appears to manifest itself in the behavior of these estimates.

Figure 8.5 *Capital-output example. Plot of log(output) versus log(capital) and fitted regression lines using naive method, MOM-S, RC-1 and SIMEX-I as described in text.*



- The MEE approach fails badly, from both a bias and variance perspective. In principle, it should be preferred over the MOM method (which assumes no bias in $log(W_i)$, since it tries to account for how bias in $log(W_i)$ enters into creating bias in the $\mathbf{W}'\mathbf{W}$ matrix. However, since the estimate of $\mathbf{W}'\mathbf{W}$ is inverted in calculating $\widehat{\boldsymbol{\beta}}_{MEE}$, it appears that the extra variability in this estimate leads to bias in the estimate of $(\mathbf{W}'\mathbf{W})^{-1}$ and the resulting poor behavior of the MEE estimates.
- In summary, the estimates from MOM-S, RC-2 and SIMEX-I appear to be the most reliable. All of these lead to substantially different fits than that based on the naive estimates; see Figure 8.5.

Table 8.8 *Analysis of capital-output data. Besides naive and the R(robust) results for MOM estimators, all standard errors and confidence intervals (Lower, Upper) are from two-stage bootstrap. B-Mean and B-Median are the bootstrap mean and median, respectively. A refers to making use of analytical expressions for bias and variance associated with* $log(W)$*, while - S refers to estimating these quantities via simulation. See text for further discussion. B-mean and B-Med denote bootstrap mean and median, respectively.*

Method		Est.	B-Mean	SE	Lower	Upper	B-Med
Naive	β_0	-0.501		0.1954			
	β_1	0.636		0.1667			
MOM-A-R	β_0	-0.068		0.4135	-0.8789	0.7421	
	β_1	1.122		0.3286	0.4783	1.7664	
MOM-S-R	β_0	.143		0.5358	-0.9075	1.1929	
	β_1	1.360		0.4526	0.4725	2.2468	
MOM-A	β_0	-.068	0.1039	0.2197	-0.3339	0.5747	0.0999
	β_1	1.122	1.2218	0.2270	0.7974	1.7106	1.2011
MOM-S	β_0	.143	0.2570	0.1673	-0.0335	0.6218	0.2389
	β_1	1.360	1.3859	0.1877	1.0621	1.7753	1.3637
RC-1	β_0	-.324	-0.2140	0.2162	-0.5965	0.2314	-0.2211
	β_1	0.988	1.2393	0.2374	0.8366	1.7159	1.2206
RC-2	β_0	-.069	-0.0507	0.2119	-0.4544	0.3881	-0.0581
	β_1	1.301	1.2729	0.1985	0.9299	1.6686	1.2532
RC-3	β_0	-.335	-0.7083	0.5661	-1.3375	0.4175	-1.0034
	β_1	0.896	0.4741	0.6049	-0.0619	1.7233	0.0667
RC-4	β_0	-.101	-0.4312	0.6156	-1.3297	0.4819	-0.2315
	β_1	1.170	0.6927	0.5940	-0.0417	1.5889	0.8458
SIMEX	β_0	-.178	0.0456	0.2470	-0.4085	0.5432	0.0339
	β_1	1.078	1.3256	0.2635	0.8612	1.8357	1.2959
SIMEX-I	β_0	-.298	-0.1995	0.2146	-0.5958	0.2224	-0.2106
	β_1	0.931	1.0389	0.2166	0.6644	1.4779	1.0249
MEE-A	β_0	.297	1.0235	0.6401	0.1248	2.5376	0.8983
	β_1	1.799	2.8207	0.8616	1.6475	4.9297	2.6239
MEE-S	β_0	.437	1.4484	1.1523	0.4098	3.8266	1.2520
	β_1	1.915	3.2170	1.4513	1.8464	6.2656	2.8943

8.5 Linear measurement error with validation data

We continue to work with the linear model for true values, with $E(Y_i|x_i) = \beta_0 + \boldsymbol{\beta}_1'\mathbf{x}_i$, but move away from the additive measurement error model. This section allows systematic biases through a linear measurement or Berkson error model. These models were presented in Sections 6.4.2 and 6.4.3. We first provide a motivating example.

Ph/Alkalinity Example. This example examines the relationship between alkalinity and pH based on 24 water samples from Barnstable county. This was part of the Acid Rain Monitoring Project (Godfrey et al., 1985), with statistical aspects addressed in Buonaccorsi (1989). Both variables are subject to some measurement error. External calibration/validation data were collected based on blind samples sent to the lab with known pH and alkalinity. These true values are considered as fixed. Figures 8.6 show the observed pH and alkalinity values for the 24 observations making up the main study, along with the calibration data for each of the measures. Notice that the measure of alkalinity is particularly noisy and there are systematic biases present. For illustration, we have chosen a part of the data where the measurement error is potentially important. The question is how to correct the naive inferences using the calibration data. The naive fit is the solid line in Figure 8.6, while the others are corrected fits, as explained in Section 8.5.3 when we return to this example.

8.5.1 *Models and bias in naive estimators*

Collectively the true values are in $\mathbf{T}_i = (Y_i, \mathbf{X}_i')'$ and the observed values are in $\mathbf{O}_i$. It is possible for $\mathbf{O}_i$ to be of a different dimension than $\mathbf{T}_i$. If we have $\mathbf{W}_i$ and D_i as error-prone versions of $\mathbf{X}_i$ and Y_i, respectively, then $\mathbf{O}_i = (D_i, \mathbf{W}_i')'$, of the same size as $\mathbf{T}_i$.

The general linear measurement error model is

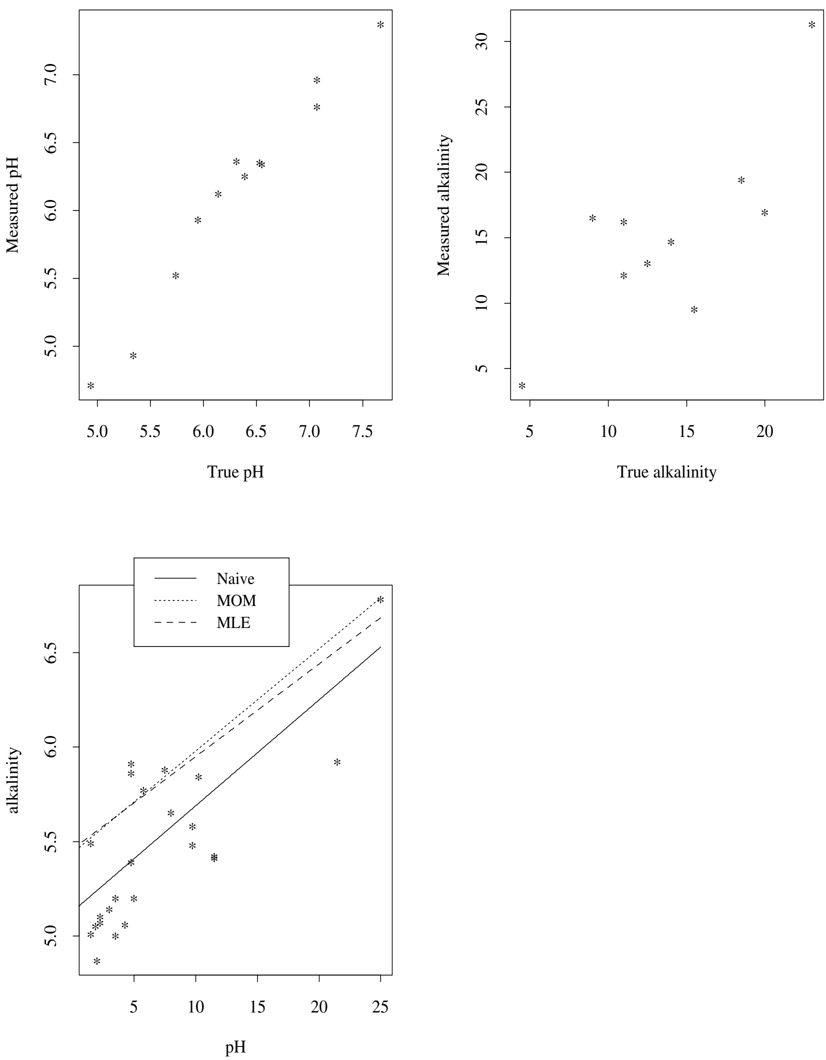
$$\mathbf{O}_i|\mathbf{t}_i = \boldsymbol{\theta}_{0t} + \boldsymbol{\Theta}_{1t}\mathbf{t}_i + \boldsymbol{\delta}_i \tag{8.23}$$

where $E(\boldsymbol{\delta}_i) = \mathbf{0}$, $Cov(\boldsymbol{\delta}_i) = \boldsymbol{\Sigma}_{\delta i}$ and $\mathbf{t}_i$ denotes the given true values. The matrix $\boldsymbol{\Theta}_{1t}$ may not be square, but it is assumed to be be of full column rank, requiring that $\mathbf{O}$ has at least as many components as $\mathbf{T}$. The model in (8.23) allows any of the error-prone variables to depend on any of the true values. To estimate it in this general form the validation data must have all of the observed and true values for each unit.

Often there is some simplifying structure on the measurement error model. For example, if $\mathbf{W}_i$ and D_i are error-prone versions of $\mathbf{X}_i$ and Y_i, one assumption is that

$$\mathbf{W}_i|\mathbf{x}_i = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\mathbf{x}_i + \mathbf{u}_i \text{ and } D_i|y_i = \theta_{0d} + \theta_{1d}y_i + q_i \tag{8.24}$$

Figure 8.6 *pH Example. Top panels are calibration data. Bottom panel plots alkalinity versus pH for main study with associated fits.*



where

$$E\begin{bmatrix} q_i \\ \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{0} \end{bmatrix}, Cov\begin{bmatrix} q_i \\ \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} \sigma^2_{qi} & \mathbf{\Sigma}'_{uqi} \\ \mathbf{\Sigma}_{uqi} & \mathbf{\Sigma}_{ui} \end{bmatrix}.$$

This allows error in the response and/or the predictors, but each is modeled only in terms of their corresponding true values. With no error in the response, $D_i = Y_i$, so $\sigma^2_{qi} = 0$, $\mathbf{\Sigma}_{uqi} = \mathbf{0}$, $\theta_{0d} = 0$ and $\theta_{1d} = 1$. Similar adjustments are made to parts of $\boldsymbol{\theta}_0$, $\mathbf{\Theta}_1$, $\mathbf{\Sigma}_{ui}$ and $\mathbf{\Sigma}_{uqi}$ if some of the predictors are measured without error.

Often (8.24) is simplified further by assuming $\mathbf{\Theta}_1$ is diagonal so

$$E(W_{ij}|\mathbf{x}_i) = \theta_{0j} + \theta_{1j}x_{ij}. \tag{8.25}$$

In this case each observed error-prone value has its own linear regression model involving (conditionally) only its corresponding true value.

The linear Berkson models work in the other direction and assume

$$\mathbf{T}_i|\mathbf{o}_i = \boldsymbol{\lambda}_{0t} + \mathbf{\Lambda}_{1t}\mathbf{o}_i + \mathbf{e}_i \tag{8.26}$$

with $E(\mathbf{e}_i) = \mathbf{0}$. See Section 6.4.3 for a description of how the linear Berkson model relates to the linear measurement error model under normality and a structural setting.

In assessing bias and correction approaches, we use the notation in Chapter 5 with $\bar{\mathbf{W}}$ and $\mathbf{S}_{WW}$ denoting the sample mean and covariance matrix of the $\mathbf{W}_i$ values, with similar notation for other random vectors. The sample covariance matrix and mean vector of the observed values have expected values of

$$E(\mathbf{S}_{OO}) = \mathbf{\Theta}_{1t}\mathbf{\Sigma}_{TT}\mathbf{\Theta}'_{1t} + \mathbf{\Sigma}_\delta, \text{ and } E(\bar{\mathbf{O}}) = \boldsymbol{\theta}_{0t} + \mathbf{\Theta}_{1t}\mu_T.$$

Also, as in Chapter 5, quantities such as $\boldsymbol{\mu}_T = \sum_i E(\mathbf{T}_i)/n$, $\mathbf{\Sigma}_{TT} = E(\mathbf{S}_{TT})$, etc. can be interpreted as "population" mean and covariance matrices for either functional or structural models (or models in between).

Bias in naive estimators. Assuming we have $\mathbf{W}_i$ and D_i as measured versions of $\mathbf{x}_i$ and Y_i, respectively, the naive estimators are $\hat{\boldsymbol{\beta}}_{1naive} = \mathbf{S}^{-1}_{WW}\mathbf{S}_{WD}$ and $\hat{\beta}_{0naive} = \bar{D} - \hat{\boldsymbol{\beta}}'_{1naive}\bar{\mathbf{W}}$. Using the model in (8.24), $E(\mathbf{S}_{WW}) = \mathbf{\Theta}_1\mathbf{\Sigma}_{XX}\mathbf{\Theta}'_1 + \mathbf{\Sigma}_u$, $E(\mathbf{S}_{WD}) = \mathbf{\Theta}_1\mathbf{\Sigma}_{Xy}\theta_{1d} + \mathbf{\Sigma}_{uq}$ and $E(S^2_d) = \theta^2_{1d}\sigma^2_Y + \sigma^2_q$, so the naive estimators have approximate expected values of

$$E(\hat{\boldsymbol{\beta}}_{1naive}) \approx \boldsymbol{\gamma}_1 = (\mathbf{\Theta}_1\mathbf{\Sigma}_{XX}\mathbf{\Theta}'_1 + \mathbf{\Sigma}_u)^{-1}(\mathbf{\Theta}_1\mathbf{\Sigma}_{XX}\boldsymbol{\beta}_1\theta_{1d} + \mathbf{\Sigma}_{uq}) \tag{8.27}$$

$$E(\hat{\boldsymbol{\beta}}_{0naive}) \approx \gamma_0 = \beta_0 + (\boldsymbol{\beta}_1 - \boldsymbol{\gamma}_1)'\boldsymbol{\mu}_X. \tag{8.28}$$

With no error in Y, then (8.27) becomes

$$E(\hat{\boldsymbol{\beta}}_{1naive}) \approx \boldsymbol{\gamma}_1 = (\mathbf{\Theta}_1\mathbf{\Sigma}_{XX}\mathbf{\Theta}'_1 + \mathbf{\Sigma}_u)^{-1}\mathbf{\Theta}_1\mathbf{\Sigma}_{XX}\boldsymbol{\beta}_1 = \boldsymbol{\kappa}_1\boldsymbol{\beta}_1. \tag{8.29}$$

These generalize the results for additive error given in Chapter 5. As in those chapters, these are exact for the normal structural models with normal measurement error and constant measurement error covariance; that is $E(D_i|\mathbf{W}_i) = \gamma_0 + \boldsymbol{\gamma}_1'\mathbf{W}_i$, exactly. Otherwise the biases are approximate. Of course these biases can be much more complicated than with additive error. See Section 6.4.3 for more discussion of the induced model under linear Berkson error.

8.5.2 *Correcting with external validation data*

We first correct using external validation data to fit a measurement error model, regressing observed on true values. With fixed true values in the validation data, as is the case in the pH/alkalinity example, this is the approach that should be used.

Assuming $\mathbf{O}$ is of the same dimension as $\mathbf{T}$, then using the estimated measurement error parameters,

$$\widehat{\mu}_T = \widehat{\boldsymbol{\Theta}}_{1t}^{-1}(\bar{\mathbf{O}} - \widehat{\boldsymbol{\theta}}_{0t}) \text{ and } \widehat{\boldsymbol{\Sigma}}_{TT} = \widehat{\boldsymbol{\Theta}}_{1t}^{-1}(\mathbf{S}_{OO} - \hat{\boldsymbol{\Sigma}}_\delta)(\widehat{\boldsymbol{\Theta}}_{1t}^{-1})'.$$

If $\mathbf{O}$ has more elements than $\mathbf{T}$ then $\widehat{\boldsymbol{\Theta}}_{1t}^{-1}$ above is replaced by $\widehat{\boldsymbol{\Theta}}_{1t}^{-} = (\widehat{\boldsymbol{\Theta}}_{1t}'\widehat{\boldsymbol{\Theta}}_{1t})^{-1}\widehat{\boldsymbol{\Theta}}_{1t}'$, which is a "generalized inverse" of $\widehat{\boldsymbol{\Theta}}_{1t}$.

From $\widehat{\boldsymbol{\mu}}_T$ and $\widehat{\boldsymbol{\Sigma}}_T$, we can extract estimates of $\boldsymbol{\mu}_X$, $\boldsymbol{\Sigma}_{XX}$, $\boldsymbol{\Sigma}_{XY}$, μ_Y and σ_Y^2. Under (8.24) these can be written explicitly as

$$\widehat{\boldsymbol{\Sigma}}_{XX} = \widehat{\boldsymbol{\Theta}}_1^{-1}(\mathbf{S}_{WW} - \hat{\boldsymbol{\Sigma}}_u)\widehat{\boldsymbol{\Theta}}_1^{\prime -1}, \qquad \widehat{\boldsymbol{\Sigma}}_{XY} = \widehat{\boldsymbol{\Theta}}_1^{-1}(\mathbf{S}_{WD} - \widehat{\boldsymbol{\Sigma}}_{uq})/\widehat{\theta}_{1d},$$

$$\widehat{\mu}_Y = (\bar{D} - \widehat{\theta}_{0y})/\widehat{\theta}_{1d} \quad \text{and} \quad \widehat{\boldsymbol{\mu}}_X = \widehat{\boldsymbol{\Theta}}_1^{-1}(\bar{\mathbf{W}} - \widehat{\boldsymbol{\theta}}_0).$$

Of course the estimator of $\boldsymbol{\Sigma}_{TT}$ must be nonnegative and a modification is needed if it is not. See the discussion in Chapter 5 as well as Bock and Peterson (1975) and related discussion in Buonaccorsi (1989) for the special case with separate measurement error models for each variable.

The assumed model for $Y|\mathbf{x}$ implies that $\boldsymbol{\beta}_1 = \Sigma_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$, $\beta_0 = \mu_Y - \boldsymbol{\beta}_1'\boldsymbol{\mu}_X$ and $\sigma_Y^2 = \sigma^2 + \boldsymbol{\beta}_1'\boldsymbol{\Sigma}_{XX}\boldsymbol{\beta}_1$. This leads to corrected estimators of the regression coefficients

$$\widehat{\boldsymbol{\beta}}_1' = \widehat{\boldsymbol{\Sigma}}_{XX}^{-1}\widehat{\boldsymbol{\Sigma}}_{XY} = \widehat{\boldsymbol{\Theta}}_1'(\mathbf{S}_{WW} - \hat{\boldsymbol{\Sigma}}_u)^{-1}(\mathbf{S}_{WD} - \hat{\boldsymbol{\Sigma}}_{uq})/\widehat{\theta}_{1d} \tag{8.30}$$

$\widehat{\beta}_0 = \widehat{\mu}_Y - \widehat{\boldsymbol{\beta}}_1'\widehat{\mu}_X$ and $\widehat{\sigma}^2 = \widehat{\sigma}_Y^2 - \widehat{\boldsymbol{\beta}}_1'\widehat{\boldsymbol{\Sigma}}_{XX}\widehat{\boldsymbol{\beta}}_1$. These can also be expressed as

$$\hat{\boldsymbol{\beta}} = \widehat{\mathbf{M}}_{XX}^{-1}\widehat{\mathbf{M}}_{XY} \tag{8.31}$$

where

$$\widehat{\mathbf{M}}_{XX} = \begin{bmatrix} 1 & \widehat{\boldsymbol{\mu}}_x \\ \widehat{\boldsymbol{\mu}}_X' & \widehat{\boldsymbol{\Sigma}}_{XX} + \widehat{\boldsymbol{\mu}}_X\widehat{\boldsymbol{\mu}}_X' \end{bmatrix} \text{ and } \widehat{\mathbf{M}}_{XY} = \begin{bmatrix} \widehat{\mu}_Y \\ \widehat{\boldsymbol{\Sigma}}_{XY} + \widehat{\boldsymbol{\mu}}_X'\widehat{\mu}_Y \end{bmatrix}.$$

The easiest approach to obtaining an analytical expression for $Cov(\widehat{\boldsymbol{\beta}})$ is to view $\widehat{\boldsymbol{\beta}}$ as the solution for $\boldsymbol{\beta}$ in $\mathbf{S}_1(\boldsymbol{\beta}, \widehat{\boldsymbol{\theta}}) = \widehat{\mathbf{M}}_{XX}\boldsymbol{\beta} - \widehat{\mathbf{M}}_{XY} = \mathbf{0}$ where $\widehat{\boldsymbol{\theta}}$ contains all of the measurement error parameters estimated from the external validation data. These are in turn obtained by solving a set of equations $\mathbf{S}_2(\boldsymbol{\theta}) = \mathbf{0}$ and an approximation to $Cov(\widehat{\boldsymbol{\beta}})$ can be derived using the method in Section 6.17.3. This is equivalent to the use of the delta method. While relatively straightforward in principle, the algebra is messy. Some general developments and specific formulas for the case where each of the variables are calibrated separately can be found in Buonaccorsi (1989). Bootstrapping, as described in Section 6.16, is also an alternative.

An alternative correction method is to assume normality throughout and obtain maximum likelihood estimators. Since the validation data is independent of the main study, the pseudo-maximum likelihood estimators, obtained by first getting the MLE's for the measurement error parameters and then correcting, are the overall MLE's so long as the estimates are in the parameter space. When that happens the moment-corrected estimators above are almost the same as the MLEs.

Berkson model correction.

If the Berkson model in (8.26) can be fit from the external validation data, and that model is exportable to the main data, then we can correct in a different way, as described briefly in Chapter 6. This assumes that the validation data have true and observed values obtained simultaneously.

Allowing error in any variables and assuming the $\mathbf{T}_i$ are a random sample with mean $\boldsymbol{\mu}_T$ and covariance $\boldsymbol{\Sigma}_T$, then $\boldsymbol{\mu}_T = \boldsymbol{\lambda}_{0t} + \boldsymbol{\Lambda}_{1t}\boldsymbol{\mu}_O$ and $\Sigma_T = \boldsymbol{\Lambda}_{1t}\boldsymbol{\Sigma}_O\boldsymbol{\Lambda}_{1t}'$. Using estimates from the validation data leads to

$$\widehat{\boldsymbol{\mu}}_T = \widehat{\boldsymbol{\lambda}}_{0t} + \widehat{\boldsymbol{\Lambda}}_{1t}\bar{\mathbf{O}} \text{ and } \widehat{\boldsymbol{\Sigma}}_T = \widehat{\boldsymbol{\Lambda}}_{1t}\mathbf{S}_{OO}\widehat{\boldsymbol{\Lambda}}_{1t}'.$$

From these estimates of $\boldsymbol{\Sigma}_{XX}$, $\boldsymbol{\mu}_X$, etc. and then of the regression parameters can be calculated in the same way as above. An approximation to $Cov(\widehat{\boldsymbol{\beta}})$ here could be derived using the covariance of matrix of the estimated coefficients in the Berkson model, the variance/covariance structure of $\bar{\mathbf{O}}$ and $\mathbf{S}_{OO}$, and the delta method.

Regression calibration. With no error in Y, $\mathbf{X}$ random, $\mathbf{W}$ the same size as $\mathbf{X}$ and the measurement error being nondifferential with respect to Y then the induced model is $E(Y|\mathbf{w}) = \beta_0 + E(\mathbf{X}|\mathbf{w})'\boldsymbol{\beta}_1 = \beta_0 + \boldsymbol{\lambda}_0'\boldsymbol{\beta}_1 + \boldsymbol{\beta}_1'(\boldsymbol{\Lambda}_1\mathbf{w})$. In this case the external validation only needs to have $\mathbf{W}$ and $\mathbf{X}$ and the coefficients can be estimated via regression calibration, regressing Y_i on $\widehat{\mathbf{X}}_i = \widehat{\lambda}_0 + \widehat{\boldsymbol{\Lambda}}_1 w_i$ or by linearly transforming the naive estimators; see Section 6.9.1.

8.5.3 External validation example

We return to the water quality example introduced earlier, where the goal is to regress pH (Y) on alkalinity (x) assuming $E(Y|x) = \beta_0 + \beta_1 x$ and $V(Y|x) = \sigma^2$. The measured values are W and D, respectively, with

$$W_i|x_i = \theta_0 + \theta_1 x_i + u_i \text{ and } D_i|y_i = \theta_{0d} + \theta_{1d} y_i + q_i$$

where u_i has mean 0 and variance σ_u^2, q_i has mean 0 and variance σ_q^2 and u_i and q_i are assumed uncorrelated.

The estimated measurement error parameters are given in Table 8.9. For the alkalinity measure, $\widehat{\theta}_0 = 4.67$, $\widehat{\theta}_1 = .602$ and $\widehat{\sigma}_u^2 = 13.44$. Similarly the first row gives estimated measurement error parameters for the response, pH. As is typical the error variances are computed using a divisor of $m - 2$ where m is the number of calibration samples. (See the later discussion.)

Table 8.9 *Summary of calibration of pH and alkalinity measures*

Variable	Sample size	Intercept	Slope	MSE
pH	12	-.1916	1.003	.0199
Alkalinity	10	4.67	.602	13.438

A common strategy with data like this is to use the calibration curves to obtain "adjusted/imputed" values and then analyze those quantities as if they are the true values. For the ith observation in the main study the imputed pH and alkalinity measures are $\widehat{Y}_i = (D_i - \widehat{\theta}_{0d})/\widehat{\theta}_{1d}$ and $\widehat{X}_i = (W_i - \widehat{\theta}_0)/\widehat{\theta}_1$, with the corresponding "naive" analysis of these imputed values given in the second line of Table 8.10. While using the imputed values may be better than using raw values, in that they may have approximately additive error, we know from our earlier results that simply analyzing these will still lead to inconsistent estimators of the regression coefficients. Further the measurement errors in the imputed values are now correlated since the imputed values use common estimated measurement error parameters. Finally, any uncertainty from estimating the measurement error parameters is being ignored. In general this approach cannot be recommended unless there is very little variability around the fitted calibration curves so that the measurement error in the adjusted values is essentially negligible.

The analysis given here is based on Buonaccorsi (1989) (with a few numerical corrections). The estimate of $Cov(\widehat{\boldsymbol{\beta}})$ is computed assuming normality throughout. This simplifies some of the computations. Adjustments can be made when the normality assumption is dropped or bootstrapping used. The moment estimators in Table 8.10 come from using (8.30) directly. This leads to an estimated covariance matrix of the true values of

$$\widehat{\mathbf{\Sigma}}_{TT} = \begin{bmatrix} 0.1706 & 3.305 \\ 3.305 & 62.104 \end{bmatrix}.$$

This matrix is negative, with a corresponding estimate of the variance of the error in the equation of $\widehat{\sigma}^2 = \widehat{\sigma}_Y^2 - \widehat{\beta}_1^2 \widehat{\sigma}_X^2 = -.0008$, and so is inadmissible. To avoid this problem the MLEs given in the last line are computed using a divisor of m for estimating variances in the calibration data and a divisor of $n = 24$ in computing the sample covariance of the observed values. This leads to a positive definite estimate of $\mathbf{\Sigma}_{TT}$ and an estimate of $\widehat{\sigma}^2 = .010$. In this case the corrections for the linear measurement errors have a relatively small effect on the estimated slope, but more of an effect on the estimated intercept. The two corrected lines are given in the bottom panel of Figure 8.6.

Table 8.10 *Analysis of the pH-alkalinity example. $C(\widehat{\beta}_0, \widehat{\beta}_1)$ is the estimated covariance of $\widehat{\beta}_0$ and $\widehat{\beta}_1$.*

Method	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\sigma}^2$	$SE(\hat{\beta}_0)$	$SE(\hat{\beta}_1)$	$C(\hat{\beta}_0, \hat{\beta}_1)$
Naive	5.08	.056	.0826	.0905	.00999	-.0007
Adjusted	5.52	.034	.0822	.063	.006	-.0001
Moment	5.44	.054	-.0008	.502	.0006	-.007
MLE	5.46	.049	.010	.479	.0004	-.0005

The example in the next section provides another illustration of fitting with external validation data with use of the bootstrap.

8.5.4 Correcting with internal validation

Here we observe $\mathbf{o}_i$, which contains the true values for variables without error and error-prone versions of variables measured with error, on all n observations. The vector of true values, $\mathbf{t}_i$, is subdivided into $\mathbf{t}_{1i}$ and $\mathbf{t}_{2i}$ where $\mathbf{t}_{2i}$ contains the true values measured without error (and so is part of $\mathbf{o}_i$) and $\mathbf{t}_{1i}$ contains the true values subject to measurement error. The latter are observed only on the n_V units that make up the validation subsample.

Assuming the linear model $E(Y|\mathbf{x}) = \beta_0 + \boldsymbol{\beta}_1'\mathbf{x}$ implies $\boldsymbol{\beta}_1 = \mathbf{\Sigma}_X^{-1}\mathbf{\Sigma}_{XY}$ and $\beta_0 = \mu_Y - \boldsymbol{\beta}_1'\boldsymbol{\mu}_X$ and so we can begin by focusing on estimation of μ_T and $\mathbf{\Sigma}_T$. In many problems, $\boldsymbol{\mu}_T$ and/or $\mathbf{\Sigma}_T$ may be the main objects of interest. From that perspective, some of the discussion and the exam below connect to the next chapter, where the interest is in errors in responses only.

The general strategies for the use of internal validation data were laid out in

Section 6.15.2 and demonstrated in the context of logistic regression in Section 7.3.2. So, the discussion here is brief. In the case of random subsampling for the validation data, the theory, computational techniques and further details associated with the discussion below can be found in Buonaccorsi (1990a)(but that paper uses a notation that is in conflict with what is used here).

Fitting using the linear Berkson model.

Assuming $E(\mathbf{T}|\mathbf{o}) = \boldsymbol{\lambda}_{0t} + \boldsymbol{\Lambda}_{1t}\mathbf{o}$, then

$$\boldsymbol{\mu}_T = \boldsymbol{\lambda}_{Ot} + \boldsymbol{\Lambda}_{1t}\boldsymbol{\mu}_O \text{ and } \boldsymbol{\Sigma}_T = \boldsymbol{\Lambda}_{1t}\boldsymbol{\Sigma}_O\boldsymbol{\Lambda}'_{1t} + \boldsymbol{\Sigma}_{T|o}.$$

The validation sample can be used to fit the Berkson model, from which we obtain

$$\widehat{\boldsymbol{\mu}}_T = \widehat{\boldsymbol{\lambda}}_{Ot} + \widehat{\boldsymbol{\Lambda}}_{1t}\bar{\mathbf{O}} \text{ and } \widehat{\boldsymbol{\Sigma}}_T = \widehat{\boldsymbol{\Lambda}}_{1t}\mathbf{S}_{OO}\widehat{\boldsymbol{\Lambda}}'_{1t} + \widehat{\boldsymbol{\Sigma}}_{T|o}. \tag{8.32}$$

Note that this does not require that the validation sample be a random subsample. Under normality these are the full maximum likelihood estimators if $\widehat{\boldsymbol{\Sigma}}_{T|o}$ is computed using a divisor of n_V and $\mathbf{S}_{OO}$ is computed using a divisor of n. Typically however modified divisors based on the associated degrees of freedom involved would be used and these are approximate MLEs.

With random subsampling for the validation data, this is a missing data problem with missing completely at random and the asymptotic properties of the resulting estimators can be obtained from the missing data literature; see Anderson (1984) or Little and Rubin (1987) for example and details in Buonaccorsi (1990a).

Fitting using the measurement error model.

Often the measurement error specification is given in terms of the distribution of $\mathbf{O}|\mathbf{t}$. If this is a general linear measurement error model with no restrictions then it is better to work with the Berkson error model, assuming it is linear. (Under normality we can move from the unrestricted linear measurement error model to the unrestricted linear Berkson error model, and viceversa.) Often, however, there is some structure on the measurement error model and it may (but not always) be more efficient to work with the measurement error model. For example, in the pH-alkalinity example from the preceding section, it may be reasonable to assume that the measured pH or alkalinity depends only on its corresponding true values. In general, under the linear measurement error model and random sampling

$$E(\mathbf{O}_i) = \boldsymbol{\mu}_O = \boldsymbol{\theta}_0 + \boldsymbol{\Theta}_1\boldsymbol{\mu}_T \text{ and } \boldsymbol{\Sigma}_O = Cov(\mathbf{O}_i) = \boldsymbol{\Theta}_1\boldsymbol{\Sigma}_T\boldsymbol{\Theta}'_1 + \boldsymbol{\Sigma}_{O|t}.$$

The measurement error model can be fit using the validation data and then corrected estimators of the population mean and covariance matrix of the true values calculated using

$$\widehat{\boldsymbol{\Sigma}}_T = \widehat{\boldsymbol{\Theta}}_1^-(\mathbf{S}_O - \hat{\boldsymbol{\Sigma}}_{O|t})\widehat{\boldsymbol{\Theta}}_1^{'-} \text{ and } \widehat{\boldsymbol{\mu}}_T = \widehat{\boldsymbol{\Theta}}_1^-(\bar{\mathbf{O}} - \widehat{\boldsymbol{\theta}}_0), \tag{8.33}$$

where $\widehat{\mathbf{\Theta}}_{1t}^{-} = (\widehat{\mathbf{\Theta}}_{1t}^{'}\widehat{\mathbf{\Theta}}_{1t})^{-1}\widehat{\mathbf{\Theta}}_{1t}^{'}$. As earlier, this is a "generalized inverse" of $\widehat{\mathbf{\Theta}}_{1t}$, allowing $\mathbf{O}$ to have more values than $\mathbf{T}$, and equals $\widehat{\mathbf{\Theta}}_{1t}^{-} = \widehat{\mathbf{\Theta}}_{1t}^{-1}$ if $\mathbf{O}$ and $\mathbf{T}$ have the same number of values.. These estimators are pseudo-MLE's under normality. Full MLEs under normality can be computed using the EM-algorithm. Obtaining the covariance matrix of these estimates is a bit difficult since the same data get used in estimating the measurement error model and again in getting the corrected estimates. The problem is a little easier under normality where general tools for handling likelihood estimators are applicable.

With estimates of $\boldsymbol{\mu}_T$ and $\mathbf{\Sigma}_T$, estimation of the regression parameters proceeds as in (8.30) and the discussion following it.

8.5.5 Internal validation example

Wheat content example. To illustrate the use of internal validation data, we consider a problem discussed by Brown (1982, p. 299) in a multivariate calibration framework. Brown considered the problem of predicting percent water and percent protein in a wheat sample based on four infrared reflectance measures. Here we consider the problem of making inferences about the characteristics of water (X) and protein (Y) and their relationship, assuming a simple linear regression for Y given x with constant variance. The reflectance measures are cheaper to obtain than true water and protein values. These will be available on a large sample, while the true values are determined on a subset. For simplicity, we utilize only two of the four reflectance measures (Y_3 and Y_4 in Brown), denoted by w_1 and w_2 here. Note that although not implemented here, the methods can handle the use of all four reflectance measures, in which case there are more observed than true values.

There are $n = 100$ total observations of which $n_V = 21$ make up the internal validation sample. These 21 observations, containing $t = (y, x, w_1, w_2)'$, are the values given in Table 1 of Brown (1982). The other 79 data points contain only the reflectance measures, $\mathbf{o}' = (w_1, w_2)$. These were generated by a multivariate normal with moments essentially the same as the sample moments from the validation data.

Using the validation data, linear measurement and Berkson error models are both fit. Results for the first, which regress the reflectance measures on the true values, are shown in Table 8.11. The first two columns are from unrestricted fits for which the estimated error covariance matrix is

$$\widehat{\mathbf{\Sigma}}_{O|t} = \begin{bmatrix} 1.97453 & 1.43442 \\ 1.43442 & 3.10577 \end{bmatrix}.$$

The coefficient for protein is nonsignificant when w_2 is regressed on both pro-

Table 8.11 *Fitted measurement error models for the wheat example, regressing reflectance measures w_1 and w_2 on protein (x) and water (y).*

Response	w_1		w_2		w_2	
Predictor	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	334.1477	6.36373	456.7516	7.98113	456.7516	7.98113
Water	-24.2321	0.68494	-23.8376	0.85903	-23.8376	0.85903
Protein	-1.8944	0.22460	0.136663	0.28168		
MSE	1.975		3.106		2.981	

Table 8.12 *Fitted Berkson error models for the wheat example, regressing protein (x) and water (y) on the reflectance measures w_1 and w_2.*

Response	x		y	
Predictor	Estimate	SE	Estimate	SE
Intercept	17.562	0.8265	-46.121	8.040
w_1	-0.0091	0.0047	-0.3915	0.0454
w_2	-0.0315	0.0051	0.3870	0.0501
MSE	.0045		.4272	

tein and water. The last column fit w_2 as a function of y only. Using this the matrix above is modified to have an off-diagonal term of 1.396 and the (2,2) element is 2.981.

Results for the fitted Berkson error model are given in Table 8.12 with an associated error covariance matrix of

$$\widehat{\mathbf{\Sigma}}_{T|o} = \begin{bmatrix} 0.0045 & -.0244 \\ -.0244 & 0.427280 \end{bmatrix}.$$

Estimation is carried out using essentially four methods, numbers 1, 2, 3 and 6 in Section 6.15.2. Results are given in Table 8.13. For each we use the bootstrap by resampling with replacement from the validation data and then independently resampling with replacement from the 70 observations with the reflectance measures.

- VALID: Using the 21 validated cases only.
- ME-EXT: Treating the 21 validation observations as external and the other 79 is the main data. The correction uses (8.33) but with $\bar{\mathbf{O}}$ and $\mathbf{S}_{OO}$ obtained from the 79 unvalidated observations. This is just an additional illustration of the techniques for using external validation data in the previous section.

Table 8.13 *Analysis of wheat data. One-stage bootstrap mean, standard error and confidence intervals.*

Parameter	Method	Estimate	Boot Mean	Boot SE	Boot-CI Lower	Boot-CI Upper
μ_X	Valid	9.541	11.242	0.314	10.629	11.825
	ME-EXT	9.570	11.092	0.302	10.609	11.591
	ME-INT	9.564	11.123	0.241	10.756	11.520
	BERK-INT	9.562	11.172	0.192	10.815	11.566
μ_Y	Valid	11.26	11.242	0.314	10.629	11.825
	ME-EXT	11.115	11.092	0.302	10.609	11.591
	ME-INT	11.145	11.123	0.241	10.756	11.520
	BERK-INT	11.175	9.560	0.050	9.461	9.656
σ_X^2	Valid	.226	1.994	0.392	1.257	2.789
	ME-EXT	.231	1.898	0.994	0.946	3.242
	ME-INT	.228	1.921	0.976	1.095	2.974
	BERK-INT	.2284	1.943	0.348	1.335	2.683
	BERK-MLE	.2255	1.873	0.342	1.284	2.611
σ_Y^2	Valid	1.103	0.212	0.057	0.115	0.336
	ME-EXT	1.739	0.230	0.036	0.160	0.300
	ME-INT	1.782	0.226	0.031	0.167	0.290
	BERK-INT	1.971	0.226	0.030	0.168	0.287
	BERK-MLE	1.895	0.223	0.030	0.166	0.284
β_0	Valid	3.603	4.406	5.865	-5.011	17.877
	ME-EXT	4.152	4.114	5.555	-6.542	15.622
	ME-INT	4.030	4.060	4.662	-4.163	13.854
	BERK-INT	4.259	4.195	3.865	-2.741	12.149
	BERK-MLE	4.103	4.053	3.860	-2.887	12.040
β_1	Valid	.8026	0.715	0.625	-0.730	1.708
	ME-EXT	.7276	0.729	0.578	-0.452	1.815
	ME-INT	.7440	0.735	0.485	-0.275	1.578
	BERK-INT	.7233	0.730	0.401	-0.087	1.464
	BERK-MLE	.7397	0.745	0.401	-0.075	1.476

- ME-INT: Using the measurement error model and correcting using all 100 observations. This uses (8.33) with $\bar{\mathbf{O}}$ and $\mathbf{S}_{OO}$ obtained from all 100 observations.
- BERK-INT AND BERK-MLE. These are very similar. They treat the validation data as internal and correct using the Berkson error model based on (8.32). They only differ in the divisors used in computing the covariance matrices involved, as described following (8.32). The BERK-MLE are the full maximum likelihood estimates under normality. The two versions are the same for estimating the means so only BERK-INT is listed.

Analytical based standard errors for the MLE's of $\widehat{\mu}_X$, $\widehat{\mu}_Y$ and $\widehat{\beta}_1$ under normality are (from Buonaccorsi (1990a)) .1411, .048 and .2890, respectively. The first two are similar to the bootstrap standard errors here but the last is quite a bit smaller than .401. This is in part due to the bootstrap standard error being inflated by some outliers but also due to the fact that the bootstrap standard error does not utilize a normality assumption.

Looking at the confidence intervals in combination with the standard errors when the data is treated as internal, the Berkson based methods are more efficient than using the validation data only or employing the measurement error model. This is expected from the theory. The methods using the measurement error model are seen to be generally inefficient. This could change if we fit measurement error models with further structure on them; for example, if we assume the model for w_1 depends on x only (see Table 8.11).

8.6 Misclassification of a categorical predictor

8.6.1 Introduction and bias of naive estimator

This section provides a brief introduction to the problem of misclassification of a single categorical predictor, which denotes group membership. For motivation we consider data from an exercise in Kutner et al. (2005) in which rehabilitation times after knee surgery for young males are compared across three fitness categories: below average, average and above average. The observed data yielded means and variances as given in Table 8.14.

Classifying fitness category is known to be difficult and the question is what effect misclassification would have on estimating group means. We address this first and then provide an overview on how to correct for misclassification.

The setting here is the so-called "one way" model, where if the group membership is known without error for the kth observation from the jth group,

$$Y_{jm} = \mu_j + \epsilon_{jm},$$

Table 8.14 *Means and variance in rehabilitation times (in days) following near surgery in young adult males, grouped by fitness category.*

Group	Sample size	mean	variance
Below	8	38.00	30.00
Average	10	32.00	12.00
Above	6	24.00	19.60

where ϵ_{jm} has mean 0 and so μ_j is the population mean for group j. We'll use μ's rather than β's here, in keeping with the traditional usage and denote the vector of means by $\boldsymbol{\mu}' = (\mu_1, \ldots \mu_J)$, where J is the number of groups.

It is easier for our purpose to index the data from $i = 1$ to n, where n is the total sample size and define a categorical variable X_{ci}, which denotes group membership, and write $E(Y_i|X_{ci} = j) = \mu_j$.

Chapter 3 treated the case where Y is categorical and μ_j was a probability/proportion. Here Y is assumed quantitative. The model for the true values can also be expressed in terms of dummy variables as $E(Y_i|\mathbf{x}_i) = \mathbf{x}_i'\boldsymbol{\mu}$, where $\mathbf{x}_i' = (x_{i1}, \ldots, x_{iJ})$ with $x_{ij} = 1$ if the ith observation is from group j and equals 0 otherwise. Notice there is no intercept here.

With no misclassification, the number of true observations in category j is $n_j = \sum_{i=1}^n x_{ij}$ and the estimate of μ_j is simply the mean of the Y's for the n_j observations from group j. That is,

$$\widehat{\mu}_j = \frac{\sum_{i=1}^n Y_i x_{ij}}{n_j} = \frac{T_j}{n_j},$$

where T_j is the sum of the observations from group j.

The misclassification probabilities are given by

$$P(W_{(c)i} = j|x_{(c)i} = k) = \theta_{jk},$$

where $x_{(c)i} = k$ if the ith observation is from group k and $W_{(c)i}$ is similarly defined. This denotes the probability that an individual really in group k is classified into group j.

Besides a few comments at the end, we assume that the misclassification is nondifferential, so the misclassification probabilities still apply if we also condition on $Y = y$. This also means (see Section 1.4.3) that $E(Y|X_c = j, W_c = k) = E(Y|X_c = j)$; that is W_c is a surrogate for X_c. This may not be reasonable in scenarios where the classification is associated with an underlying, but unobserved, latent variable. This could be the case in the fitness example. The fact that the resulting misclassification may be differential is related to the discussion in Section 6.4.8.

Similar to x_{ij}, define W_{ij} to be the error-prone dummy variable indicating group membership. Based on this, the number of observations in group j is $n_{Wj} = \sum_i W_{ij}$ and the naive estimator is $\widehat{\mu}_{naive,j} = \sum_{i=1}^{n} Y_i w_{ij}/n_{wj}$. As discussed further at the end of this section, the naive estimator of μ_j can be shown to be consistent for

$$\mu_j^* = \frac{\sum_{k=1}^{J} \theta_{jk} n_k \mu_k}{\sum_{k'=1}^{J} \theta_{jk'} n_{k'}}. \tag{8.34}$$

This result, which is only approximate since the n_{Wj} are random, is conditional on the true group sample sizes $n_1, \ldots, n_J$. With a random sample the n_k are not fixed but the same result applies if we replace n_k by the expected count $n\pi_k$, where π_k is the proportion of the population in group k. (The n will cancel so this is the same as replacing n_k with π_k.) Notice that these biases are functions of the misclassification rates, the true group means and the true allocations across the groups.

If the X values are random, and so the grouping comes from post-stratification, we can also arrive at the bias through

$$\mu_j^* = \sum_k E(Y|X_c = k)P(X_c = k|W_c = j) = \sum_k \mu_k \lambda_{k|j}, \tag{8.35}$$

where $\lambda_{k|j} = P(X_c = k|W_c = j)$ is a "reclassification" probability. This is the conditional probability of an observation really belonging in group k given it is classified into group j. This formulation is helpful for correcting with internal validation.

To illustrate the biases we consider the fitness-rehabilitation example with two sets of misclassification rates

$$\text{Case I: } \mathbf{\Theta} = \begin{bmatrix} .95 & .03 & .02 \\ .03 & .9 & .07 \\ .03 & .07 & .9 \end{bmatrix} \text{Case II: } \mathbf{\Theta} = \begin{bmatrix} .8 & .15 & .05 \\ .03 & .9 & .07 \\ .02 & .18 & .8 \end{bmatrix}.$$

These could be viewed as moderate and severe misclassification, respectively. We take the means in Table 8.14 as the true μ_j's and use $\sigma^2 = 19.8$ (the estimate from the data). Table 8.15 shows the approximate expected values, of the naive estimators, from (8.34), and the mean values over 10,000 simulations. The sample sizes used were the original of 8, 10 and 6. As expected the biases are more important under the fairly severe misclassification in Case II. Even there, however, the bias is somewhat modest. This could change with different means. The expression for the approximate expected value does not work as well in Case II (e.g., compare 36.4 to 37.56), as expected. Recall that this is only approximate since the number in the denominator is random because of the misclassification. The accuracy of this approximation will improve with larger sample sizes. It is surprising, in a way, that the approximation does as well as it does here with such small sample sizes.

Table 8.15 *Effects of misclassification on estimation of group means. μ_j is true mean, μ_j^* is from analytical expression for approximate/limiting value of naive estimator. Sim-mean is simulated mean based on 10,000 simulations.*

		Case I		Case II	
Group	μ_j	μ_j^*	Sim-mean	μ_j^*	Sim-mean
1	38	37.57	37.50	36.40	37.56
2	32	31.80	31.81	31.80	31.87
3	24	25.41	25.11	26.46	25.70

8.6.2 Correcting for misclassification

Our treatment of correction techniques is much less in depth here than in earlier sections, mostly an overview of possible methods. See Mak and Li (1988) for some additional discussion and references.

In terms of the dummy variables, $P(W_{ij} = 1|\mathbf{x}_i) = \boldsymbol{\theta}_j'\mathbf{x}_i$, where $\boldsymbol{\theta}_j' = (\theta_{j1}, \ldots, \theta_{jJ})$ and

$$E(n_{Wj}) = \boldsymbol{\theta}_j'\mathbf{n}_x.$$

In vector form $E(\mathbf{n}_W) = \boldsymbol{\Theta}\mathbf{n}$, where $\mathbf{n}$ and $\mathbf{n}_W$ contain the group counts for the true and misclassified values, respectively.

With the true group counts fixed, the limiting value of the vector of naive estimates can be written as $\boldsymbol{\mu}^* = \mathbf{D}_w^{-1}\boldsymbol{\Theta}\mathbf{D}_x\boldsymbol{\mu}$ where $\mathbf{D}_x = diag(n_1, \ldots, n_J)$, $\mathbf{D}_w = diag(E(n_{W1}), \ldots, E(n_{WJ}))$, $diag$ denotes a diagonal matrix and

$$\boldsymbol{\Theta} = \begin{bmatrix} \theta_{11} & \ldots & \theta_{1J} \\ \theta_{21} & \ldots & \theta_{2J} \\ . & \ldots & . \\ \theta_{21} & \ldots & \theta_{2J} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\theta}_1' \\ \boldsymbol{\theta}_2' \\ . \\ \boldsymbol{\theta}_J' \end{bmatrix}.$$

To correct for misclassification with known or estimated misclassification rates, the unobserved counts for the true x's can first be estimated using $\widehat{\mathbf{n}}_x = \widehat{\boldsymbol{\Theta}}^{-1}\mathbf{n}_w$, where $\mathbf{n}_W' = (n_{W1}, \ldots, n_{WJ})$. From these an estimate of $\mathbf{D}_x$,
$\widehat{\mathbf{D}}_x = diag(\widehat{n}_{x1}, \ldots, \widehat{n}_{xJ})$ can be obtained and then the corrected estimator

$$\widehat{\boldsymbol{\mu}} = (\widehat{\boldsymbol{\Theta}}\widehat{D}_x)^{-1}\mathbf{D}_w\widehat{\boldsymbol{\mu}}_{naive} = (\widehat{\boldsymbol{\Theta}}\widehat{D}_x)^{-1}\mathbf{W}'\mathbf{Y}.$$

These corrected estimators can also be viewed as solving the modified estimating equations $(\widehat{\boldsymbol{\Theta}}\widehat{D}_x)\widehat{\boldsymbol{\mu}} - \mathbf{W}'\mathbf{Y} = \mathbf{0}$, which would have mean $\mathbf{0}$ if the misclassification matrix $\boldsymbol{\Theta}$ were known. An approximate covariance matrix of $\widehat{\boldsymbol{\mu}}$ can be developed using the method in Section 6.17.3 along with the delta method,

Table 8.16 *Simulated performance of naive and corrected estimates of group means. Small sample refers to* $n_1 = 8$, $n_2 = 10$, $n_3 = 6$*; large sample size refers to* $n_1 = n_2 = n_3 = 100$. *Misc. refers to misclassification scenarios as given above. True means are* $\mu_1 = 38$, $\mu_2 = 32$, $\mu_3 = 24$. *Based on 10000 simulations.*

		Naive		Corrected	
Sample sizes	Misc.	mean	s.d.	mean	s.d.
small	Case I				
		37.52	1.737	37.96	1.860
		31.81	1.536	32.03	1.689
		25.12	2.148	23.48	2.842
large	Case I				
		37.40	0.506	37.85	0.542
		31.61	0.496	32.00	0.566
		24.83	0.516	23.82	0.604
small	Case II				
		37.55	1.997	41.2 13	11.590
		31.87	1.5778	32.04	1.735
		25.71	2.415	20.18	35.937
large	Case II				
		37.47	0.548	40.24	0.848
		31.54	0.485	31.81	0.546
		25.36	0.569	22.80	0.926

We do not explore these correction methods further but do note that even with known misclassification rates, correcting in this manner may not be better than ignoring misclassification. This is illustrated through simulation in Table 8.16. When the misclassification is moderate, the corrected estimators are a slight improvement over the naive estimators in terms of bias, but with slightly more variability. On the other hand, with more severe misclassification, the corrected estimators are substantially worse than the naive approach, even at the larger sample sizes with 100 in each group. This performance is largely a function using the inverse of $\widehat{D}_x$ with random estimated group sizes on the diagonal. It suggests the need for some other type of bias correction or using a weighted average of the naive estimator and the corrected estimator. This last idea was explored by Schafer (1986) in estimating the slope in linear regression and by Shieh (2009) in estimating a proportion when there is misclassification. Another strategy that can be tried is the misclassification SIMEX as proposed by Kuchenoff et al. (2006).

Using estimated reclassification rates.

Suppose we have estimated reclassification rates, most likely coming from internal validation data. Two of our general strategies for correcting for measurement error are moment corrections and regression calibration. The two are equivalent here. For the moment approach, if we define $\mathbf{\Lambda}$ to have jth row $(\lambda_{1|j}, \lambda_{2|j}, \ldots, \lambda_{J|j})$, then from (8.35), $\mu^* = \mathbf{\Lambda}\mu$ which suggests using $\widehat{\boldsymbol{\mu}} = (\widehat{\mathbf{\Lambda}})^{-1}\widehat{\mu}_{naive}$. With a little bit of algebra this can be shown to be equivalent to a regression calibration approach where we fit a regression model $E(Y_i) = \widehat{\mathbf{x}}_i'\boldsymbol{\mu}$, where $\widehat{\mathbf{x}}_i' = \mathbf{w}_i'\widehat{\mathbf{\Lambda}}$ and $\mathbf{w}_i'$ is the vector of dummy variables associated with error prone measure for the ith observation. If the ith observation was classified into category j then $\widehat{\mathbf{x}}_i'$ is simply $(\widehat{\lambda}_{1|j}, \ldots, \widehat{\lambda}_{J|j})$. Notice that this model has no intercept.

The general strategies for getting the covariance of $\widehat{\mu}$ are outlined in Chapter 6. See also Section 6.14 for some related general comments. Shieh (2009) has a more detailed treatment of this problem, especially in the case of $J = 2$ where she also explores the use of Fieller's method for getting a confidence interval on the difference in the two means.

The problem gets more complicated if the misclassification is differential; equivalently, W_c is not a surrogate for X_c. In that case a model is needed for either W_c given X_c and Y or for X_c given W_c and Y. With Y being quantitative this means specifying a polytomous regression model (a binary model if there are only two categories). This could be estimated from validation data, which must now include Y.

8.6.3 Further details

Using the true values the estimating equations for $\boldsymbol{\beta}$ are $n_j\beta_j - T_j = 0$ ($j = 1$ to J) or $(\mathbf{X}'\mathbf{X})\boldsymbol{\beta} = \mathbf{X}'\mathbf{Y}$, where $\mathbf{X}'\mathbf{X} = D_x = diag(n_1, \ldots, n_J)$. This means $E(\mathbf{W}_i|\mathbf{x}_i) = \mathbf{\Theta}\mathbf{x}_i$, where $\mathbf{\Theta}$ has jth row $\boldsymbol{\theta}_j'$. If $\mathbf{n}_W$ and $\mathbf{n}_x$ denote the sample sizes using W and x, respectively, then $E(\mathbf{n}_W|\mathbf{n}_x) = \mathbf{\Theta}\mathbf{n}_x$ and $E(\mathbf{W}'\mathbf{W}) = \mathbf{D}_w = diag(E(n_{W1}), \ldots, E(n_{WJ})$, so $E(n_{Wj}) = \boldsymbol{\theta}_j'\mathbf{n}_x$.

The expected value of the naive estimating equations (with $\boldsymbol{\beta}^*$ as the argument and $\boldsymbol{\beta}$ denoting the true parameters) is

$$E(\mathbf{W}'\mathbf{W}\boldsymbol{\beta}^* - \mathbf{W}'\mathbf{Y}) = \mathbf{D}_w\boldsymbol{\beta}^* - \mathbf{\Theta}\mathbf{D}_x\boldsymbol{\beta}.$$

8.7 Miscellaneous

8.7.1 Bias expressions for naive estimators

This section lays out the general way to evaluate approximate biases for the models in this chapter, where we condition on the true values.

In the original form of $\mathbf{W}_i$ suppose $\mathbf{W}_i^* = \mathbf{x}_i^* + \mathbf{u}_i^*$, where as before the * version includes the 1 for the intercept (if present), e.g., $\mathbf{w}_i^{*'} = (1, \mathbf{w}_i')$. Hence $\mathbf{W}'\mathbf{W} = \sum_i \mathbf{W}_i^*\mathbf{W}_i^{*'}$ and $\mathbf{X}'\mathbf{X} = \sum_i \mathbf{x}_i^*\mathbf{x}_i^{*'}$.

Conditioning on the $x's$ **throughout**, we allow for an offset in the expected value of $\mathbf{u}_i^*$ as arises with the use of squares, products, etc. That is, we assume $E(\mathbf{u}_i^*|\mathbf{x}_i^*) = \mathbf{h}_i$ and $Cov(\mathbf{u}_i^*|\mathbf{x}_i^*) = \mathbf{\Sigma}_{ui}^*$. Then

$$E(\mathbf{W}'\mathbf{W}) = \sum_i \mathbf{E}(\mathbf{W}_i^*\mathbf{W}_i*') = \mathbf{X}'\mathbf{X} + \sum_i (\mathbf{\Sigma}_{ui}^* + \mathbf{h}_i\mathbf{x}_i^{*'} + \mathbf{x}_i^*\mathbf{h}_i' + \mathbf{h}_i\mathbf{h}_i')$$

and $E(\mathbf{W}'\mathbf{Y}) = \sum_i \mathbf{E}(W_i^*Y_i) = (\mathbf{x}_i^* + \mathbf{h}_i)\mathbf{x}_i^{*'}\boldsymbol{\beta} = \mathbf{X}'\mathbf{X}\boldsymbol{\beta} + \sum_i \mathbf{h}_i\mathbf{x}_i^{*'}\boldsymbol{\beta}$.

Hence without error in Y,

$$E(\widehat{\boldsymbol{\beta}}_{naive}) \approx \boldsymbol{\beta}^* = \left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{Q}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{H}\mathbf{X}\right)\boldsymbol{\beta}$$

$$= \boldsymbol{\beta} + \left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{Q}\right)^{-1}(\mathbf{H}\mathbf{X} - \mathbf{Q})\,\boldsymbol{\beta}$$

where

$$\mathbf{Q} = \sum_i (\mathbf{\Sigma}_{ui}^* + \mathbf{h}_i\mathbf{x}_i^{*'} + \mathbf{x}_i\mathbf{h}_i' + \mathbf{h}_i\mathbf{h}_i')/n \text{ and } \mathbf{H} = \frac{1}{n}\left[\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_n\right].$$

If there is additive error in Y_i, with $D_i = Y_i + q_i$ with $Cov(\mathbf{u}_i^*, q_i) = \mathbf{\Sigma}_{uqi}^*$ then $E(\mathbf{W}'\mathbf{D})/n = \sum_i E(\mathbf{x}_i^* + \mathbf{u}_i^*, Y_i + q_i)/n = ((\mathbf{X}'\mathbf{X})/n + \mathbf{H}\mathbf{X})\boldsymbol{\beta} + \mathbf{\Sigma}_{uq}^*$, where $\mathbf{\Sigma}_{uq}^* = \sum_i \mathbf{\Sigma}_{uqi}^*/n$ and

$$E(\widehat{\boldsymbol{\beta}}_{naive}) \approx \boldsymbol{\beta}^* = \left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{Q}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{H}\mathbf{X}\right)\boldsymbol{\beta} + \left(\frac{\mathbf{X}'\mathbf{X}}{n} + \mathbf{Q}\right)^{-1}\mathbf{\Sigma}_{uq}^*. \tag{8.36}$$

Quadratic model with additional predictors.

Here the ith row of $\mathbf{X}$ is $\mathbf{x}_i^* = (1, x_{i1}, x_{i1}^2, \mathbf{x}_{i2}')$. In the bias expressions

$$\mathbf{Q} = \begin{bmatrix} Q_1 & Q_{12} \\ Q_{12}' & Q_2 \end{bmatrix},$$

with Q_1 as in (8.5) with x_{i1} in place of x_i, σ^2_{ui1} in place of σ^2_{ui}, etc.,

$$\mathbf{Q}_{12} = \frac{1}{n}\begin{bmatrix} \mathbf{0} \\ \sum_i \tilde{\mathbf{\Sigma}}_{ui12} \\ \sum_i 2x_{i1}\tilde{\mathbf{\Sigma}}_{ui12} + Cov(u^2_{i1}, \mathbf{u}_{i2}) \end{bmatrix},$$

$\mathbf{Q}_2 = \sum_{i=1}^n \mathbf{\Sigma}_{ui2}$, and

$$\mathbf{H} = \frac{1}{n}\begin{bmatrix} 0 & . & . & 0 \\ 0 & . & . & 0 \\ \sigma^2_{u1} & \sigma^2_{u2} & . & \sigma^2_{un} \\ \mathbf{0} & . & . & \mathbf{0} \end{bmatrix}.$$

As in the discussion in Section 8.2.1, with a structural model where the measurement errors do not depend on the true values the quantities in $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}$ can be replaced by population parameters arising from the distribution of random X_{i1} and $\mathbf{X}_{i2}$, see (8.6) and the related discussion, but now $\mathbf{X}^{*'}_i = (1, X_i, X^2_i, \mathbf{X}'_{i2})$.

With no error in $\mathbf{x}_{i2}$ the expressions above simplify considerably since both $\mathbf{Q}_{12}$ and $\mathbf{Q}_2$ are zero. In general, however, there will still be biases in the naive estimate of $\boldsymbol{\beta}_2$, the result of the error in x_{i1}.

Two variable models with interaction.

For this model

$$\mathbf{Q} = \frac{1}{n}\begin{bmatrix} 0 & 0 & 0 & \sum_i \sigma_{ui12} \\ 0 & \sum_i \sigma^2_{ui1} & \sum_i \sigma_{ui12} & a_{24} \\ 0 & \sum_i \sigma_{ui12} & \sum_i \sigma^2_{ui2} & a_{34} \\ \sum_i \sigma_{ui12} & a_{24} & a_{34} & a_{44} \end{bmatrix},$$

with
$a_{24} = \sum_i (x_{i2}\sigma^2_{ui1} + 2x_{i1}\sigma_{ui12} + E(u^2_{i1}u_{i2})$,
$a_{34} = \sum_i (x_{i1}\sigma^2_{ui2} + 2x_{i2}\sigma_{ui12} + E(u^2_{i2}u_{i1})$,
$a_{44} = \sum_i (x^2_{i1}\sigma^2_{ui2} + x^2_{i2}\sigma^2_{ui1} + 4x_{i1}x_{i2}\sigma_{ui12} + 2x_{i2}E(u^2_{i1}u_{i2}) + 2x_{i1}E(u^2_{i2}u_{i1})$
$+ E(u^2_{i1}u^2_{i2}))$ and

$$\mathbf{HX} = \frac{1}{n}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sum_i \sigma_{ui12} & \sum_i \sigma_{ui12}x_{i1} & \sum_i \sigma_{ui12}x_{i2} & \sum_i \sigma_{ui12}x_{i1}x_{i2} \end{bmatrix}.$$

- If just one of the variables, say x_1, is measured with error then the only nonzero elements of $\mathbf{Q}$ are $q_{22} = \sigma^2_{ui1}/n$, $q_{24} = \sum_i x_{i2}\sigma^2_{ui1}/n$ and $q_{44} = \sum_i x^2_{i2}\sigma^2_{ui1}/n$. In addition, $\mathbf{HX} = \mathbf{0}$.
- For the structural setting the x's are replaced by random X's and expectations taken. This can include having the measurement error parameters varying in the true values.

- Suppose the measurement errors are *normally distributed with constant variances and covariance*. Then

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & \sigma_{u12} \\ 0 & \sigma^2_{u1} & \sigma_{u12} & q_{24} \\ 0 & \sigma_{u12} & \sigma^2_{u2} & q_{34} \\ \sigma_{u12} & q_{24} & q_{34} & q_{44} \end{bmatrix},$$

and the last row of $\mathbf{HX}$ becomes

$$[\sigma_{u12}, \sigma_{u12}\bar{x}_1, \sigma_{u12}\bar{x}_2, \sigma_{u12}(S_{x12} - \bar{x}_1\bar{x}_2)],$$

with $q_{24} = \bar{x}_2\sigma^2_{u1} + 2\bar{x}_1\sigma_{u12},\ q_{34} = \bar{x}_1\sigma^2_{u2} + 2\bar{x}_2\sigma_{u12}$,
$q_{44} = (S^2_{x1} - \bar{x}^2_1)\sigma^2_{u2} + (S^2_{x2} - \bar{x}^2_2)\sigma^2_{u1} + 4\bar{x}_1\bar{x}_2\sigma_{u12} + E(u^2_{i1}u^2_{i2}))$, where $E(u^2_{i1}u^2_{i2})) = \sigma^2_{u1}\sigma^2_{u2} + 2\sigma^2_{u12}$, $S^2_{xj} = \sum_j(x_{ij} - \bar{x}_j)^2/n$ and $S_{x12} = \sum_j(x_{i1} - \bar{x}_s)(x_{i2} - \bar{x}_2)/n$.
In the structural case $\bar{x}_j$ is replaced by $\mu_j = E(X_{ij})$ and the S's can be replaced by population variances and covariances. These would also enter into replacing $\mathbf{X}'\mathbf{X}$ with its expected value.
Interaction and quadratic terms.
With interactions and quadratic terms both $\mathbf{x}_i = (x_{i1}, x_{i2}, x^2_{i1}, x^2_{i2}, x_{i1}x_{i2})'$

$$E\begin{bmatrix} W_{i1} \\ W_{i2} \\ W^2_{i1} \\ W^2_{i2} \\ W_{i1}W_{i2} \end{bmatrix} = \mathbf{x}_i + \mathbf{h}_i, \text{ where } \mathbf{h}_i = \begin{bmatrix} 0 \\ 0 \\ \sigma^2_{ui1} \\ \sigma^2_{ui2} \\ \sigma_{ui12} \end{bmatrix}.$$

The terms in $\mathbf{Q}$ and $\mathbf{H}$ can be obtained with a combination of the results for the earlier models.

8.7.2 Likelihood methods in linear models

Here we address the general computational approach to using likelihood methods in the structural setting with replicates (or known measurement error parameters). This method has some connections to the modified estimating equation approach. The true predictors are random and for illustration here assume that $Y_i|\mathbf{x}_i$ is distributed normal with mean $\mathbf{x}_i'\boldsymbol{\beta}$ and variance σ^2 and Y_i is measured without error. In this chapter, $\mathbf{x}_i$ usually contains some nonlinear functions of quantities measured with additive error. Without measurement error then assuming there are no restrictions relating $\boldsymbol{\beta}$ and σ^2 to the model for $\mathbf{X}$, the MLE of $\boldsymbol{\beta}$ is well known to be $\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$. With measurement error, we can view the problem as a missing data problem and use of the EM-algorithm leads to an estimate of $\boldsymbol{\beta}$ on the kth step of $\widehat{\boldsymbol{\beta}}_{(k)} = \mathbf{M}^{-1}_{(k)}\widehat{\mathbf{X}}'_{(k)}\mathbf{Y}$, where $\mathbf{M}_{(k)}$ is an estimate of $E(\mathbf{X}'\mathbf{X}|\mathbf{o})$ and $\widehat{\mathbf{X}}_{(k)}$ is an estimate of $E(\mathbf{X}|\mathbf{o})$.

The $\mathbf{o}$ denotes the observed data: Y, $\mathbf{W}$, replicates, etc. These expectation are with respect to the conditional distribution of $\mathbf{X}$ given $\mathbf{o}$ and then evaluated using estimated parameters from the previous step. The easiest approach here is a pseudo-approach where the distribution of $\mathbf{X}$ and $\mathbf{W}|\mathbf{x}$ is estimated first and not updated at each step. In essence this modifies the estimating equations by substituting corrected estimates of $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{Y}$ under specific assumptions. The challenge is first getting the conditional distribution of $\mathbf{X}|\mathbf{o}$ and then in calculating the conditional expectations. The latter usually involves nonlinear functions of the X's. While this is theoretically straightforward, implementation is certainly not trivial.

CHAPTER 9

Nonlinear Regression

9.1 Poisson regression: Cigarettes and cancer rates

This is a very short chapter which begins with a Poisson regression example in this section and then turns to a brief discussion on fitting general nonlinear models in Section 9.2.

The example uses data obtained from STATLIB-DASL (http://lib.stat.cmu.edu/, hosted by Carnegie Mellon). It contains cigarette consumption (a measure of the number of cigarettes smoked per capita) along with death rates (per 1000) for lung and other cancers, for 43 states and the District of Columbia. Two cases with consumption greater than 40 were eliminated for this analysis. The cigarette consumption is clearly an estimate rather than an exact value. Here we illustrate fitting a Poisson regression model with log-link relating the lung cancer rate, Y, to cigarette consumption, x, allowing for error in cigarette consumption.

We assume a log-link with

$$E(Y_i|\mathbf{x}_i) = e^{\boldsymbol{\beta}'\mathbf{x}_i} = m(\boldsymbol{\beta}, \mathbf{x}_i) = \mu_i,$$

and so we are in the case of a generalized linear model. Under the Poisson model the distribution of $Y_i|\mathbf{x}_i$ is assumed to be Poisson which implies that $V(Y_i|\mathbf{x}_i) = \mu_i$. Estimation of the coefficient can be carried out either using maximum likelihood (usually through a generalized linear models routine) or, equivalently, through weighted nonlinear least squares where the weight is $1/\mu_i$. In fitting through nonlinear least squares the implied variance model is $V(Y_i|\mathbf{x}_i) = \sigma^2\mu_i$ which allows over or under-dispersion. This has no effect on the estimated coefficients but does influence their standard errors and associated inferences. Many generalized linear model routines will also allow this scaling factor.

Suppose for illustration the error in estimated cigarette consumption is additive with variance $\sigma_u^2 = 4$. We correct for measurement error using regression calibration (RC), SIMEX and the modified estimating equation (MEE) methods. These proceed in a manner similar to what was done in the preceding two chapters. For SIMEX, the measurement errors are simulated assuming normality.

Buonaccorsi (1996a) provides details on the MEE approach allowing multiple predictors and possible error in the response. He gives both approximate expressions and exact expressions under normal measurement errors, for the expected value of the naive estimating equations. In the later case the modified estimating equations are equivalent to the corrected score equations of Nakamura (1990). Here with a single predictor and no error in Y, the approximate expected value of the ith contribution to the naive estimating equations becomes

$$C(\mathbf{x}_i, \boldsymbol{\theta}_i, \boldsymbol{\beta}) = \begin{bmatrix} -G_i(\boldsymbol{\beta}, x_i)\mu_i \\ -G_i(\boldsymbol{\beta}, x_i)\mu_i x_i - \mu_i \sigma^2_{ui}\beta_1 \end{bmatrix}$$

where $\mu_i = e^{\beta_0 + \beta_1 x_i}$ and $G_i(\boldsymbol{\beta}, x_i) = \beta_1^2 \sigma^2_{ui}\mu_i/2 \approx E(m(W_i, \boldsymbol{\beta})) - m(x_i, \boldsymbol{\beta})$. The estimation proceeds as discussed in Section 6.13, iteratively fitting as we would with the naive approach, but with the response vectors updated as in (6.59). On the kth step the estimated correction term $\mathbf{A}_i(\widehat{\boldsymbol{\beta}}_{(k)})$ is computed by computing the vector above replacing w_i for x_i, $\widehat{\sigma}^2_{ui}$ for σ^2_{ui} and $\boldsymbol{\beta}$ equal to the estimate from the previous step. One could try other ways to estimate the correction term here, but we have often found that this can do more harm than good with respect to both bias and variance. Computationally there are two levels of iteration here. One is the iteration based on updated responses in the MEE context. Then the naive fit itself involves an iterative fit, which we found easiest to do through iteratively reweighted linear least squares.

RC and SIMEX were run using STATA and then also as part of our own program which also computes the MEE estimates. Results appear in Table 9.1. All bootstrapping is single stage, i.e., resamples cases. The intervals from STATA are normal based using the bootstrap standard error with trimmed samples (2.5% on either end). The other analyses are not trimmed and the confidence intervals are bootstrap percentile intervals. The difference in the two SIMEX estimates is from the sampling involved. As with other analyses, the standard error for the RC estimate of the intercept is not corrected unless the bootstrap is used. The three techniques give remarkably similar results, given the difference in the assumptions that motivate them. Figure 9.1 shows a fairly dramatic change in the estimated rates due to a measurement error standard deviation of 2.

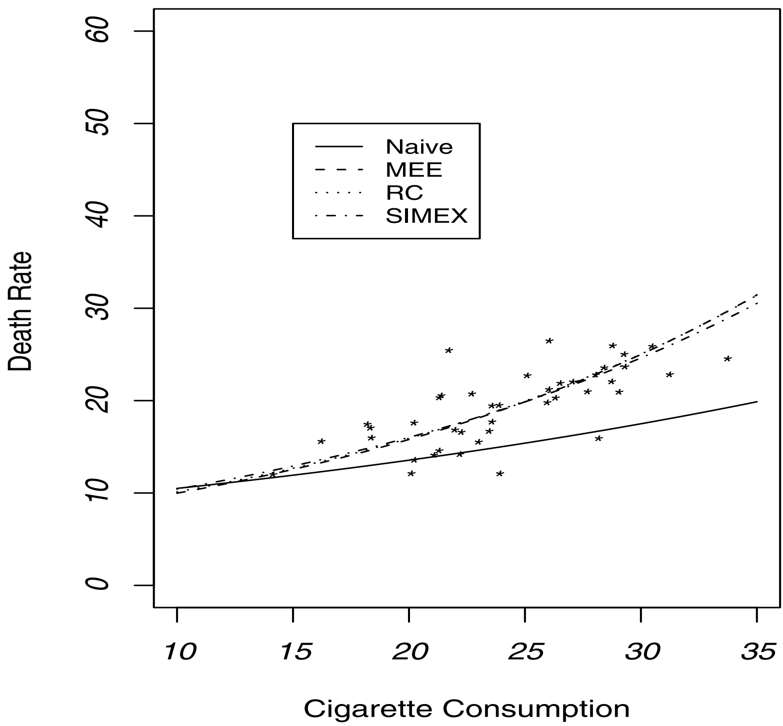
Table 9.1 *Fitting of Poisson model with measurement error regressing death rate for lung disease as a function of statewide cigarette consumption (Cig). Measurement error variance* $\sigma_u = 4$. *Except for estimates and naive SE's, other quantities are based on the one-stage bootstrap with 1000 bootstrap samples. (Low, Up) is the 95% bootstrap percentile interval.*

Naive	Estimate	Mean	SE	Low	Up	Med.
Intercept	2.096		.2087			
Cig	.0355		.0083			
MEE						
Int.	1.842	1.8000	0.2011	1.3497	2.1527	1.8321
Cig	.0459	0.0476	0.0079	0.0341	0.0653	0.0468
RC						
Int.	1.861	1.8247	0.1887	1.4040	2.1599	1.8521
Cig.	.0453	0.0468	0.0074	0.0336	0.0630	0.0460
SIMEX						
Int.	1.914	1.8770	0.1695	1.5277	2.1883	1.8912
Cig.	.0430	0.0445	0.0066	0.0321	0.0582	0.0439

```
STATA ANALYSES

Regression calibration
------------------------------------------------------------------
        lung |      Coef.   Std. Err.     [95% Conf. Interval]
-------------+----------------------------------------------------
           w |   .0452679   .0080085      .0290821    .0614536
       _cons |   1.861235   .1406126      1.577046    2.145424
-------------------------------------------------------------------
Regression calibration
-------------------------------------------------------------------
             |             Bootstrap
        lung |      Coef.   Std. Err.     [95% Conf. Interval]
-------------+----------------------------------------------------
           w |   .0452679   .0064504      .0322311    .0583046
       _cons |   1.861235   .1635974      1.530592    2.191878
-------------------------------------------------------------------
Simulation extrapolation
--------------------------------------------------------------
             |             Bootstrap
        lung |      Coef.   Std. Err.    [95% Conf. Interval]
-------------+------------------------------------------------
           w |   .0417657   .0063296     .0289731    .0545583
       _cons |   1.937465   .1596458     1.614809    2.260121
--------------------------------------------------------------
```

Figure 9.1 *Fit of lung cancer death rate as a function of cigarette consumption.*



9.2 General nonlinear models

The previous two chapters and the example above handle generalized linear models and linear models involving nonlinear functions of mismeasured predictors. The linear components involved in those models simplify matters a bit in correcting for measurement error. Fortunately, these models are pretty rich and can serve as approximations for more general nonlinear models since the latter can pose some challenges when correcting for measurement error (and, sometimes, even without measurement error).

The discussion in Chapter 6 was broad enough to handle any type of nonlinear regression model. Of course, the question is just how hard is it to correct for measurement error. Consider the model $Y_i = m(\mathbf{x}_i, \boldsymbol{\beta}) + \epsilon_i$, where ϵ_i has mean 0 and the regression function does not fall into the forms studied in Chapter 7 or 8. These problems may often be accompanied by changing variances for the errors in the equation.

The discussion of likelihood methods in Section 6.12 discussed the use of functional MLE's for certain variance structures and the more general treatment accommodates structural models with a specified distribution for the true values, the measurement error and the error in the equation. Here we comment briefly on some of the potential challenges in using the basic methods we have been employing. An example appears in Section 10.2.4. See also the discussion and example in Section 8.4 which deals with related issues. We will assume the measurement errors are additive for the discussion here.

- SIMEX is carried in the same manner as done many times before this. This can be used with weighted or unweighted nonlinear least squares estimates of the coefficients and can also be used to get estimates of variance parameters if desired. (See the interaction example in Section 8.3 for an illustration of the use of SIMEX to estimate the variance of the error in the equation.) It can also handle error in the response along with error in any predictors. Computing, however, is a potential stumbling block here. First many calls to some type of optimization or least squares routine is needed to get the SIMEX estimates. This can always be delicate, even with a single data set. At a minimum, hundreds of fits are needed. This is compounded when the bootstrap is used, in which case tens of thousands of fits are required. Just how serious of a problem this is will depend on the magnitude of the measurement error, the complexity of the model being fit and the sophistication of the fitting routine. It works just fine in the example in Section 10.2.4.
- Regression calibration also proceeds in much the same way as used previously. In the simplest form x_i in the original model is replaced by $E(X_i|\mathbf{w}_i)$, estimated as done before. In some cases there may be other functions of x_i in the model, such as squares or products. These can be handled a little differently, as was done in some sections of Chapter 8. Only one fit is needed to get the RC estimates. It is unclear just how valid the standard errors and other inferences that accompany the fit are since the uncertainty that comes from the imputation is ignored. As elsewhere, the bootstrap is an option, but once again, this will require many optimizations.
  Regression calibration does require no error in the response and nondifferential measurement error. In the manner it is usually used it depends on a linear approximation for $E(\mathbf{X}_i|\mathbf{w}_i)$ being reasonable, but this can be generalized.
- Modified Estimating Equation approach.
  Consider just unweighted least squares for which the naive estimating equa-

tions are

$$\sum_{i}^{n}(d_i - m(\mathbf{w}_i, \boldsymbol{\beta}))\boldsymbol{\Delta}(\mathbf{x}_i, \boldsymbol{\beta}) = \mathbf{0},$$

where $\Delta_j(\mathbf{w}_i, \boldsymbol{\beta}) = \partial m(\mathbf{w}_i, \boldsymbol{\beta})/\partial \beta_j$. Under normal errors in the equation with constant variance these are also the score equations for maximum likelihood. In that case, modifying the estimating equations is approximating the ML estimates.

The expected value of the naive estimating equations is given in (6.56) and a general approximate way to estimate the correction term appears in (6.61). The MEE approach may, or may not, be easy to implement. It will certainly involve approximations, with the difficulty depending on how complicated the derivatives in $\boldsymbol{\Delta}$ are. This turns out to be relatively easy to handle in the example in Section 10.2.4. Once an approximation is used the MEE is easy to use in the sense of repeatedly using the naive method, but as noted above there can be computational challenges in implementing it. It does offer the possibility of using analytical expression for the covariance matrix of the estimates. This is because once the approximations are specified there is a closed form for the new estimating equations being solved. This allows the use of some of the standard theory for estimating equations.

The MEE approach could also be modified to handle estimation of variance parameters which would have their own estimating equations. In the case where $V(\epsilon_i) = \sigma^2$, then the naive estimating equation for σ^2 is $(n-p)\sigma^2 - \sum_i (d_i - m(\mathbf{x}_i, \boldsymbol{\beta}))^2 = 0$.

CHAPTER 10

Error in the Response

10.1 Introduction

Many parts of the preceding chapters accommodated the fact that the response might be measured with error. Chapters 2 and 3 allowed misclassification of a categorical outcome, while Chapters 4 and 5, as well as parts of Chapter 8 allowed for additive, and in some cases linear, error in the response. In this chapter we isolate some additional problems with error in a quantitative response.

Section 10.2 considers three general problems associated with measurement error in the response in a single sample: estimation of the mean and variance, estimation of the distribution of the true values and estimation of the relationship between the mean and per-replicate variance in problems with replication. Sections 10.3 and 10.4 turn to the problem of estimating and comparing group means when the measurement error is not additive and possibly changing over observations, but with external validation/calibration data which allows us to estimate the measurement error models. The last section is more general in that it allows nonlinear measurement error models, but we treat the case of linear measurement error models separately in Section 10.3 since it allows more explicit results. If the validation is internal then this opens up some other strategies, but that is not discussed in this chapter. See Section 6.15.2 for general comments and Section 8.5.4 for some closely related examples.

10.2 Additive error in a single sample

This section is concerned with measurement error in a simple one-sample problem where $X_1, \ldots, X_n$ are a random sample from some "population." That is, the X's are i.i.d. with some distribution having a mean μ and variance σ^2, with the objective being estimation of μ and σ^2 and more generally the distribution of X and other functions of it, such as percentiles. The case where X is categorical and the measurement error is misclassification was already treated in Chapter 2.

Note: For this and the next section we temporarily abandon our convention of using Y for the response variable and instead use X.

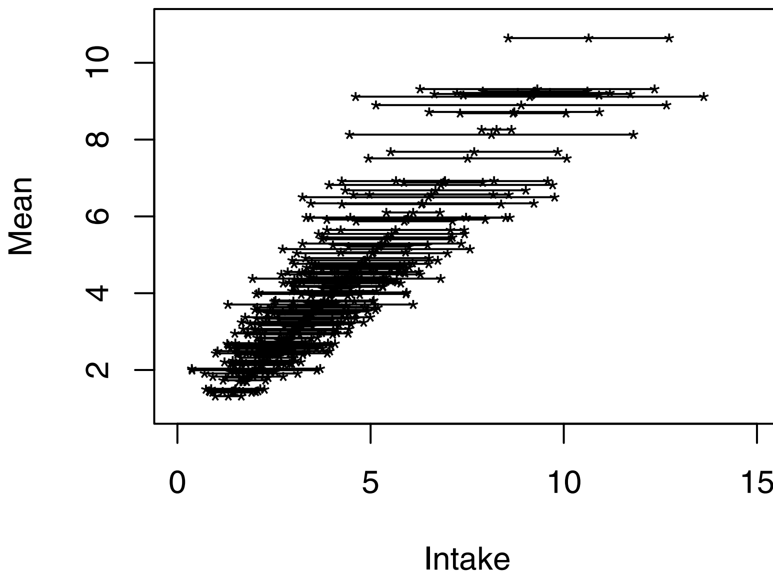
Table 10.1 *Data Structure in one-sample problems.*

Observation	1	...	i	...	n
True value (unobserved)	x_1	...	x_i	...	x_n
Observed	w_1	...	w_i	...	w_n
Estimated standard error	$\widehat{\sigma}_{u1}$	...	$\widehat{\sigma}_{ui}$	...	$\widehat{\sigma}_{un}$

The generic set-up is described in Table 10.1 The realized value of X_i is denoted by x_i, but this is unobservable. Instead, there are data collected at a "second stage" which leads to an estimator W_i, assumed to be unbiased for x_i; that is, $W_i = x_i + u_i$ where $E(u_i|x_i) = 0$ and $V(u_i|x_i) = \sigma^2_{ui}$. As elsewhere, u_i is measurement error in the broad sense of the word since it may just be sampling error. The estimate W_i is often, but not always, accompanied by an estimated standard error, $\widehat{\sigma}_{ui}$. The issue of why the variance of W_i may change over units was discussed in detail in Section 6.4.5. That section also made a careful distinction between the conditional measurement error variance (the variance of u_i given $X_i = x_i$) and the variance of $W_i - X_i$. For notational ease, we gloss over this and simply write σ^2_{ui}, but more precisely $\sigma^2_{ui|x}$ could be used for the conditional variance given x_i.

The setting described above encompasses a large number of statistical problems, many not always labeled as measurement error problems, per se. This includes the one-way random effects models with replication, random coefficient models, meta-analysis and multi-stage sampling of finite populations. Buonaccorsi (2006) presents examples for each of these. See also the last example of Section 1.2 for a case where this problem arises in analyzing ROC curves in the presence of measurement error. Of particular interest to us here is where W_i is the mean of the replicates. Any of the examples appearing elsewhere in the book involving replicates could be used for illustration here. The numerical example presented later is based on the beta-carotene intake data, first introduced in Section 5.7, and treated again later in Section 11.3.2. This involves 158 individuals from the control arm of a skin cancer study with each individual having 6 years of data. We will work with just the first four years for illustration with values from different years treated as replicates. We note there was no evidence of any time trends in the data. Treating these as replicates, the true value here is the expected/average value of this diet measured over the four year period. Figure 10.1 shows the mean of the four values plus or minus one standard error for each of the 158 individuals. This shows the amount of uncertainty associated with each of the means, and, since the standard error of the mean is the square root of the per replicate variance divided by 4, this il-

Figure 10.1 *Beta-carotene data. Individual mean ± one standard error. Based on four replicates per individual.*



lustrates how the per-replicate variability tends to increase with the mean. See also Figure 10.2. This is discussed further in Section 10.2.4.

The next three sections address estimation of the population mean and variance, estimation of the mean-variance relationship and nonparametric estimation of the distribution of X, respectively. These are all illustrated in the example in Section 10.2.4.

10.2.1 Estimating the mean and variance

While this problem has been extensively treated, some interesting questions arise when the measurement error variances are changing over observations. From the earlier assumptions, $E(W_i) = \mu$ and $V(W_i) = \sigma^2 + \sigma^2_{ui}$, where $\sigma^2_{ui} = V(W_i - X_i)$ represents the unconditional measurement error variance. See equation (6.16) and related discussion. We also observe that even if X_i is normally distributed and $W_i|x_i$ is normal, the unconditional distribution of W_i is not normal unless $V(W_i|x_i)$ is constant. This is due to the fact that mixtures of normals with unequal variances are not normal.

Let $\bar{W} = \sum_{i=1}^{n} W_i/n$ and $S_W^2 = \sum_i (W_i - \bar{W})^2/(n-1)$ denote the sample mean and variance, respectively, of the W_i. The mean $\bar{W}$ is obviously unbiased for μ. In addition we have the following (proved and discussed further in Buonaccorsi (2006)):

Result 1 *Under only the assumption that the W_i are uncorrelated and $E(W_i|x_i) = x_i$, S_W^2/n is an unbiased estimator of the variance of $\bar{W}$.*

Result 2 *If $\widehat{\sigma}_{ui}^2$ is unbiased for $V(W_i|x_i)$ then an unbiased estimate of σ^2 is*

$$\hat{\sigma}^2 = S_W^2 - \sum_i \hat{\sigma}_{ui}^2/n. \tag{10.1}$$

Result 3 *With either equal sampling effort or sampling effort attached to the unit (see Section 6.4.5), $\bar{W}$ is the best linear unbiased estimator of μ, even if the conditional measurement error variances change with i.*

Result 4 *Under the assumptions of the previous results and with either equal sampling effort or sampling effort attached to the unit, the sets $(W_i, \hat{\sigma}_{ui}^2)$ are i.i.d. This means the variance of $\hat{\sigma}^2$ and the covariance of $\bar{W}$ and $\hat{\sigma}^2$ can be estimated in a fairly straightforward manner using the delta method. In general, the bootstrap can be used by resampling the $(W_i, \widehat{\sigma}_{ui}^2)$ sets.*

Result 1 states that if the goal is estimation of the mean we can treat the W_i as if they were the X_i's, in that the sample mean is unbiased for μ and its variance can be estimated unbiasedly by S_W^2/n. This holds *regardless of the type of heteroscedasticity* allowing any type of changing inherent variation and/or changing sampling effort. This conclusion seems surprising at first glance in that it shows that potentially complicated sources of heteroscedasticity can be ignored and there is no need to make any model assumptions about the within unit data. All that is required is that the W_i are uncorrelated and that W_i is conditionally unbiased for x_i. This result is actually well known in the finite population sampling literature when first stage sampling is carried out with replacement (possibly with unequal probabilities); see for example Cochran (1977, p. 307) or Lohr (1999, p.221). What is surprising is that the applicability of this result to other contexts is not more greatly appreciated. Result 1 is also related to the method of White (1980) for obtaining robust standard errors in a linear model. In this context, White's estimator (known to be consistent) for the variance of $\bar{W}$ is $[(n-1)/n]S_W^2/n$. The result here is a little stronger, providing an unbiased estimator of $V(\bar{W})$. Result 1 also extends to the case where $\mathbf{X}$ is multivariate, which is very useful in the analysis of random coefficient models.

Result 3 is not as powerful as it first appears. It says that $\bar{W}$ is optimal among unbiased estimators of the form $\sum_i a_i W_i$, where the a_i are fixed constants. It does not cover estimators where the a_i may be random, such as the commonly used weighted average $\sum_i a_i W_i$, with $a_i = (\hat{\sigma}^2 + \hat{\sigma}^2_{ui})^{-1} / \sum_{i'} (\hat{\sigma}^2 + \hat{\sigma}^2_{ui'})^{-1}$, and $\hat{\sigma}^2$ is some estimator of σ^2. This weighted estimator is used in many contexts, although the motivation for it is usually under the assumption that the σ^2_{ui} are fixed.

The last two results do not cover the case where there are unequal sampling efforts attached to the positions in the sample. This includes either fixed sampling effort with unequal m_i or random sampling effort with a distribution specific to the position in the sample, both of which occur infrequently in practice. Even in those settings, however, one can still estimate μ and σ^2 unbiasedly and get an unbiased estimate of the variance of $\bar{W}$.

The estimation of σ^2 is challenging if the measurement error is severe enough, in that $S^2_W - \sum_i \hat{\sigma}^2_{ui}/n$ can be negative. This is a well known problem in the one-way random effects setting. There are a number of strategies for dealing with this problem, ranging from modifying the estimation method, truncating the estimator to 0 or bypassing point estimation completely and using other strategies to obtain confidence intervals for σ^2. A full discussion of this issue is omitted since it is a bit tangential to our primary focus. One example of a modified estimator is to use the variance associated with an estimate of the distribution of X; see Section 10.2.3. Other estimators arise in the case where there is replication in the one-way random effects setting, for which a comprehensive summary can be found in Chapter 3 of Searle et al. (1992). These approaches treat the sample sizes $m_1, \ldots, m_n$ as fixed and usually assume constant within unit variance. With all $m_i = m$, μ is estimated using $\bar{W}$. Both the ANOVA and REML (restricted maximum likelihood under normality) estimators of σ^2 are identical to $\hat{\sigma}^2$ in equation (10.1). With unequal m_i, the ANOVA, ML and REML estimators (the latter two under normality) of σ^2 all differ from (10.1).

10.2.2 Estimating the mean-variance relationship

The relationship of the per-replicate variance to the mean is important in a variety of contexts. In treating additive measurement error with heteroscedasticity in both linear and nonlinear models, we often used individual estimated measurement error variances, $\widehat{\sigma}^2_{ui}$. As suggested in Section 6.5.1, rather than just use S^2_i/m_i to estimate σ^2_{ui}, it may be better to smooth the estimated per-replicate variances; that is, replace S^2_i by $g(\widehat{x}_i)$ for some function g. This is especially helpful in situations where there may be outliers in the S_i^2 and/or few replicates go into S_i^2. In the case where a unit has no replication, some way is needed to impute a per-replicate variance for those units. The simplest, but clearly not always appropriate, way is to use the average per-replicate variance from the units with replication. Estimation of the mean-variance relationship is also receiving increasing attention in the analysis of microarray data (see Carroll and Wang (2008) and Wang, Ma and Carroll (2009)) and has a long history in ecological applications. Finally, this relationship can be helpful in the estimation of the underlying distribution of X, a topic discussed in the next section.

The ith unit in the sample has some true unobserved mean x_i and an associated per-replicate variance, which it is more convenient to now denote t_i^2. These can be viewed as the realizations of a random pair (X_i, T_i^2). The objective is to explore how t_i^2 relate to x_i. If every unit with true value x had exactly the same variance then $t^2 = g(x, \boldsymbol{\beta})$ exactly, for some function g and parameters $\boldsymbol{\beta}$. More plausible is to view this as an expected value over individuals with mean x; that is, $E(T^2|x) = g(x, \boldsymbol{\beta})$. Either way we are back to another measurement error problem with additive error in both the response and predictor. The sample mean and variance, W_i and S_i^2, are such that $E(W_i|x_i, t_i^2) = x_i$ and $E(S_i^2|x_i, t_i^2) = t_i^2$ and we are interested in the possibly nonlinear function g, which models $E(T^2|x)$. In this case S_i^2 plays the role of D_i (mismeasured response) and W_i (the mean of the replicates) is the error-prone version of x_i. If the replicates are normally distributed then S_i^2 is independent of W_i and so we only need to worry about error in x so far as estimation of $\boldsymbol{\beta}$ goes. However, if this is not the case, then the covariance between S_i^2 and W_i comes into play.

One model of special interest is the **power of the mean model** for which $E(T^2|x) = g(x, \boldsymbol{\beta}) = e^{\beta_0 + \beta_1 log(x)} = e^{\beta_0} x^{\beta_1}$ or

$$T_i^2|x_i = e^{\beta_0 + \beta_1 log(x_i)} + \epsilon_i, \tag{10.2}$$

where $E(\epsilon_i) = 0$. The exponential parameterization is used to guarantee a positive value for the variance. This is a popular and flexible model, which enjoys widespread use in ecology (see, for example, Kilpatrick and Ives (2003)), where it is referred to as Taylor's power law. When $\beta_1 = 2$, there is constant coefficient of variation; this is the quadratic model with no intercept in (10.3) below. The constant CV model along with the model with $g(x, \boldsymbol{\beta}) = \beta_0 + \beta_1 x^2$ are examined in detail by Wang et al. (2009). If the mean-variance relationship is assumed deterministic, then there is no *error in the equation*, $T_i^2|x_i = e^{\beta_0 + \beta_1 log(x_i)}$ exactly and $log(T_i^2) = \beta_0 + \beta_1 log(x_i)$. On the other hand if there is error in the equation, then $E(log(T^2|x))$ is not $\beta_0 + \beta_1 log(x)$ and it is better to view the model in terms of (10.2). The power of the mean model is often estimated (naively) by linearly regressing $log(S_i^2)$ on $log(W_i)$;

see Clark and Perry (1994) for example. This both ignores measurement error and assumes that the linearity holds on the log scale.

We comment on three general strategies that can be used in estimating $g(x, \boldsymbol{\beta})$: a fully parametric approach, SIMEX, and modifying the estimating equations. For some models, the latter leads to a direct moment correction method. This is not intended to be exhaustive, but concentrates on some of the basic methods used throughout the book. Methodologies are still emerging for treating this problem. See Carroll and Wang (2008) and Wang et al. (2009) for further discussion of the permutation SIMEX mentioned below and of a semi-parametric approach for the structural setting with normal replicates and an unknown distribution for the true x's. Staudenmayer et al. (2008) also develop a Bayesian approach which simultaneously estimates the power of the mean model and the distribution of the latent X.

- Assume a parametric form for the distribution of X_i and for the distribution of $W_{ij}|x_i$, (the latter required to have mean x_i and variance $g(x_i, \boldsymbol{\beta})$) and use likelihood based methods. This implicitly assumes that x completely determines the per-replicate variance. In this form we can employ all of the standard likelihood tools, with the normal-normal case being particularly easy to treat. See Finney and Phillips (1977) for the normal case and Carroll and Ruppert (1988) for some general tools for estimating variance functions.
- SIMEX
  The use of SIMEX is not as straightforward here as in earlier contexts. The issue is that the same replicates enter into calculation of the mean W_i and the sample variance S_i^2. The problem is discussed in detail by Wang et al. (2009). They show that for certain quadratic models,
  $$g(\boldsymbol{\beta}, x) = \beta_1 x^2 \text{ or } g(\boldsymbol{\beta}, x) = \beta_1 + \beta_2 x^2, \tag{10.3}$$
  one can implement SIMEX by just generating $W_{bi}(\lambda)$, using the empirical SIMEX method, as described in Section 6.11 and leaving S_i^2 unchanged. *They note, however, this only works if the naive method used is a method of moments approach rather than standard least squares.* To overcome this problem they develop what they call a permutation SIMEX method for use with standard least squares estimators and general regression models (in addition to a semi-parametric approach).
- Moment correction approach
  If we use either of the quadratic models in (10.3), then the methods in Sections 8.2.4 and 8.2.2 can be used but with modification to account for the fact that the linear term and possibly the intercept are not present. Note that these correct unweighted least squares estimates. These moment correction methods use $\widehat{\sigma}_{ui}^2 = S_i^2/m_i$ but may also need an estimate of

$Cov(S_i^2, W_i|x_i)$. This covariance can be estimated from the replicates, although not very well if estimated separately for each observation with a small number of replicates. If W_i and S_i^2 are conditionally uncorrelated then the correction for the coefficients only needs $\widehat{\sigma}_{ui}^2$. The estimated variance of S_i^2 would only be needed in order to estimate the variance of the error in the equation, but this is typically not constant. If the replicates are assumed to be normally distributed then W_i and S_i^2 are independent and the estimated conditional variance of $S_i^2 = t_i^2 + q_i$ is $\widehat{\sigma}_{qi}^2 = 2S_i^4/(m_i - 1)$.

- Modifying the estimating equations

In general, the MEE approach of Section 6.13 can be used. With the quadratic models and the use of unweighed least square, this reduces to the moment approach above. Here, we describe this approach for the power of the mean model, again using unweighted least squares. (Often the variance of T_i^2 given x_i will be changing with x_i (see Figure 10.2) suggesting the use of weighted least. That changes the naive estimating equations, but the MEE method can still be used. We omit any details for that case, however.)

Using the estimating equations for nonlinear regression in Section 6.3, the naive estimating equations are $\sum_i (S_i^2 - g(x_i, \boldsymbol{\beta}))\Delta(x_i, \boldsymbol{\beta})$, where for the power of the mean model, $\Delta(x_i, \boldsymbol{\beta})' = [g(x_i, \boldsymbol{\beta}), g(x_i, \boldsymbol{\beta})log(x_i)]$. Applying the results in Section 6.13, after some manipulation, the expected value of the naive estimating equations *conditional on the x_i's* is

$$\mathbf{C}_i = \left[\begin{array}{l} Cov(g(W_i), S_i^2) - E(g(W_i)^2) + E(g(W_i))g(x_i) \\ Cov(P(W_i), S_i^2) - E(g(W_i)P(W_i)) + E(P(W_i))g(x_i) \end{array} \right],$$

where $P(W_i) = g(W_i)log(W_i)$. In most places we have suppressed $\boldsymbol{\beta}$ for notational convenience and written $g(W_i)$ and $g(x_i)$ instead of the more precise $g(W_i, \boldsymbol{\beta})$ and $g(x_i, \boldsymbol{\beta})$.

Recall that in computing the MEE estimator we fit the nonlinear model iteratively. Here the predictor is always the observed W_i but on the $(k+1)st$ step the new vector of responses is $\tilde{\mathbf{D}}_{(k)} = \mathbf{D} - \mathbf{Q}_{(k)}(\mathbf{Q}'_{(k)}\mathbf{Q}_{(k)})^{-1}\mathbf{A}_{(k)}$, where $\mathbf{D}' = (S_1^2, \ldots, S_n^2)$ is the vector with the observed sample variances, the ith row of $\mathbf{Q}_{(k)}$ is $[g(w_i, \widehat{\boldsymbol{\beta}}_{(k)}), g(w_i, \widehat{\boldsymbol{\beta}}_{(k)})log(w_i)]$ and $\widehat{\boldsymbol{\beta}}_{(k)}$ is the estimate of $\boldsymbol{\beta}$ from the previous step. Finally, with

$$\widehat{\mathbf{A}}_{i(k)} = \left[\begin{array}{l} \widehat{Cov}(g(W_i), S_i^2) - g(W_i)^2 + g(W_i)\widehat{g}(x_i) \\ \widehat{Cov}(P(W_i), S_i^2) - g(W_i)P(W_i) + P(W_i)\widehat{g}(x_i) \end{array} \right]$$

$\mathbf{A}_{(k)} = \sum_i \widehat{\mathbf{A}}_{i(k)}$ is an estimate of the bias term $\mathbf{C}_i$. This will depend potentially on $\widehat{\boldsymbol{\beta}}_{(k)}$ through the two estimated covariance terms and the estimate of $g(x_i)$.

To proceed analytically some approximation is needed to get estimates of

$g(x_i)$, $Cov(g(W_i), S_i^2)$ and $Cov(P(W_i), S_i^2)$. Using earlier notation, with $G_i = E(g(W_i)) - g(x_i)$, then $\widehat{g}(x_i) = g(W_i) - \widehat{G}_i$, where $\widehat{G}_i$ is an estimate of the approximate bias in $g(W_i)$ as an estimator of $g(x_i)$. Using a second order Taylor series approximation, $G_i \approx \beta_1(\beta_1 - 1)g(x_i, \boldsymbol{\beta})/2x_i^2$. In our example, we estimate this using $\widehat{\boldsymbol{\beta}}_{(k)}$ and replacing $g(W_i, \boldsymbol{\beta})$ for $g(x_i, \boldsymbol{\beta})$ and $W_i^2 - \sigma_{ui}^2$ for x_i^2.

If we assume the replicates are normal then the first two covariances can be set to 0. Otherwise, the covariance of W_i and S_i^2 must be first estimated and then the covariances in the correction term approximated with the delta method and then estimated.

10.2.3 Nonparametric estimation of the distribution

Estimation of the distribution of X, when X is not observed, is important from numerous perspectives. Often the distribution is of interest in its own right, as in estimating the distribution of individual's dietary intake, or some exposure variable. A similar problem arises in estimating the distribution of individual nutritional requirements (which in turn is used to determine population requirements) where an individual's requirement cannot be observed exactly but is estimated through an intake/balance study involving repeated measures on an individual. See Atinmo et al. (1988) for an application and Section 11.2 for an associated problem. In structural measurement error problems considered elsewhere in this book, the distribution of the true values is something of a nuisance "parameter" but using an estimate of it can contribute to more efficient likelihood based corrections for measurement error. For example, Carroll et al. (1999) model the latent variable using a mixture of normals. Regression calibration methods, described in Section 6.10, require at least knowledge of $E(X|w)$, often modeled linearly based on normality assumptions. A richer approach, however, uses a nonparametric estimate of the distribution of X in estimating the distribution of $X|w$; see for example Pierce and Kellerer (2004). Nonparametric estimates of some distributions were presented (without discussion of the methodology) in Section 7.2.2. Additional applications arise in estimating the distribution of X separately for two groups for use in nonparametric approaches to dealing with measurement error in ROC curves and in the estimation of odds ratios when a continuous exposure variable has been categorized; see Section 7.4. Finally, consider the problem of estimating the distribution of the error in the equation in a linear regression problem with measurement error. This is needed to employ a two-stage nonparametric bootstrap, as discussed in Section 4.5.2 in the context of simple linear regression. In the multiple regression setting the ith residual, after correcting for measurement error, is approximately $r_i = \epsilon_i + q_i - \boldsymbol{\beta}'\mathbf{u}_i$, where q_i is the error in the response and $\mathbf{u}_i$ the error in predictors. Estimating the distribution of ϵ_i involves removing the effects of the term $q_i - \boldsymbol{\beta}'\mathbf{u}_i$ which has mean 0.

In general the problem here falls under the heading of unmixing, or in the case of additive error, deconvolution, and is worthy of a book in its own right. Our goal here is not a complete discussion but a quick glimpse at the general problem, an overview of some of the methods, and a description of a fairly simple method using the nonparametric maximum likelihood estimator (NPMLE), or an approximation to it. General discussions, with access to additional literature, can be found in Böhning (2000), Böhning and Siedel (2003) and Section 12.1 of Carroll et al. (2006).

At a minimum we observe $W_1, \ldots, W_n$. In the case of replication $\{W_{ij}, i = 1, \ldots n, j = 1, \ldots m_i\}$ is available from which the sample mean W_i and the sample variance S_i^2 are computed. More generally, W_i might be accompanied by an estimated measurement error variance $\widehat{\sigma}^2_{ui}$. Let $\mathbf{Z}_i$ denote the collection of observed values on unit i. This may consist of W_i by itself, the set of replicates on observation i, or the set $(W_i, \widehat{\sigma}^2_{ui})$.

Our discussion here is limited to semiparametric methods in which $\mathbf{Z}_i|x$ has a specified density or mass function, $f_i(\mathbf{z}|x, \boldsymbol{\theta}_i)$. The i in the subscript allows the distribution to change over i in a general manner (e.g., through changing sampling effort) while $\boldsymbol{\theta}_i$ contains "parameters." We put parameters in quotes as $\boldsymbol{\theta}_i$ may in fact involve random quantities associated with the unit going into the ith observation. Our focus is on the case where $W_i|x_i$ is assumed to be $N(x_i, \sigma^2_{ui})$. With replication, and adopting the notation of the previous section, where t_i^2 denotes the variance of W_{ij} given x_i, then $\sigma^2_{ui} = t_i^2/m_i$. The t_i^2 may depend on x_i in some known or unknown fashion. The presence of these possibly changing measurement error variances opens up a new set of problems for which methods are still being developed. See Staudenmayer et al. (2008) and references therein for some recent developments.

If $\boldsymbol{\theta}_i = \boldsymbol{\theta}$ and a parametric model is also specified for F, the distribution of X, then estimation can proceed using standard techniques based on method of moments or maximum likelihood (e.g., Maritz and Lwin, (1989), Ch. 2). See also Cordy and Thomas (1997) who use a fully parametric approach with F assumed to be a finite mixture of known distributions, but with the mixing proportions as unknown parameters. Rather than use one of the usual low dimension parametric families for F, richer families can be considered. For example, Davidian and Gallant (1993) considered this problem in a random coefficients context using a family for F that capture many smooth densities. In the replicate setting, Nusser et al. (1990) considered a parametric context which allows the variance in the replicates to depend on the true mean, while Fuller (1995) and Nusser et al. (1996) address the problem by working with normal parametric models on transformed scales and then transforming back to the original scale.

With additive i.i.d. measurement errors deconvolution can also be used to estimate the density function associated with F; see, for example, Fan and Masry (1992) and Section 12.1 of Carroll et al. (2006). Other approaches include that of Stefanski and Bay (1996), who use a SIMEX estimator when the measurement error is additive and normal with a common variance, either known or estimated and a method proposed by Eltinge (1999) based on small measurement error approximations. Aickin and Ritenbaugh (1991) and Mack and Matloff (1991) both address the problem of interest here in the context of replication. However, the latter depends on assuming the measurement becomes negligible, while the former uses a distribution based on imputed values which give a biased estimator of the variance of X.

The NPMLE, which we now focus on, estimates F by maximizing a likelihood over all distributions. There is a rich and elegant literature and theory associated with this problem. A sampling of references on the NPMLE include Laird (1978), DerSimonian (1986), Lindsay (1983ab), Lesperance and Kalbfleisch (1992), Böhning and Siedel (2003) and Böhning (2000). Section 6.5 of Skrondal and Rabes-Hesketh (2004) and Rabes-Hesketh et al. (2003a) also provide good access to the use of the NPMLE in measurement error problems. The NPMLE has also been used in estimating the distribution of random coefficients in both linear and nonlinear mixed models; see, for example, Mallet (1986) and Aitken (1999).

Both full and pseudo approaches can be used.

Full NPMLE: Maximize

$$L(F, \boldsymbol{\theta}) = \prod_i f_i(\mathbf{z}_i; F, \boldsymbol{\theta}) = \prod_i \int_x f_i(\mathbf{z}_i|x; \boldsymbol{\theta})F(dx)$$

over F and $\boldsymbol{\theta}$. Here $\boldsymbol{\theta}$ is common over all observations and $\mathbf{z}_i$ could contain W_i, $(W_i, \widehat{\sigma}^2_{ui})$ or replicate measures, depending on the situation.

Pseudo-NPMLE: Here we first estimate the measurement error parameters, $\widehat{\theta}_i$ for the ith observation, and then maximize

$$L(F) = \prod_i \int_x f_i(w_i|x; \widehat{\boldsymbol{\theta}}_i)F(dx)$$

over F.

We will concentrate on fitting models where $f_i(w_i|x; \widehat{\boldsymbol{\theta}}_i)$ is taken to be normal with mean x_i and variance $\tilde{\sigma}^2_{ui}$, some estimate of σ^2_{ui}. This may be just $\widehat{\sigma}^2_{ui}$. When there is replication $\tilde{\sigma}^2_{ui}$ is of the form $\widehat{t}^2_i/m_i$, where $\widehat{t}^2_i$ is an estimate of the per-replicate variance for the ith observation. There are a number of possible choices including $\widehat{t}^2_i = S^2_i$, $\widehat{t}^2_i = \sum_i S^2_i/n$ or $\widehat{t}^2_i = g(\widehat{x}_i, \widehat{\boldsymbol{\beta}})$ where g models the mean-variance relationship as discussed in the preceding section.

The use of S_i^2/m_i is fairly common; see for example DerSimonian (1986). Chen (1996) investigated some other choices and discusses some of the fitting issues that may arise.

This technique is ad hoc, and not pseudo-estimation in the usual sense of the word when there is no model for how the measurement error variance changes over i. In that case, it appears there are an increasing number of nuisance parameters (the so-called Neyman-Scott problem) but in fact these are usually random variables not fixed quantities.

Restricted NPMLE This uses either the full or pseudo-NPMLE as described above, but rather than fit over all distributions, the fit is constrained to discrete distributions having support on a fixed grid of points $r_1, \ldots r_K$. Some of the fitted probabilities on these specified points may be 0.

We summarize a few key points from the literature.

- The NPMLE is a discrete distribution with at nonzero probability on $J \leq n$ points. This means the likelihood can be maximized over all sets of n support points and associated probabilities. The solution is a probability mass function with probabilities $p_1, \ldots, p_J$ ($\sum_{j=1}^{J} p_j = 1$) at support point $r_1, \ldots r_J$, and the corresponding CDF, $\hat{F}$, is a step function.
- There are various strategies for computing the NPMLE. The same characterization theorems that come into play in optimal experimental design can be utilized in computing the NPMLE, often in combination with the EM algorithm.
- The asymptotic theory of the NPMLE (or pseudo-NPMLE) is difficult to establish due to the fact that the number of support points J increases and the support points change.
- The reason for considering restricted NPMLEs is two-fold. First, the biggest computational challenge in getting the NPMLE is determining the support points. These get updated iteratively using some characterization theorems and some care needs to be exercised, especially in repeated and automatic use of the method, such as in bootstrapping. The restricted NPMLE approach fixes the support points and reduces the problem to a parametric one since there are $K - 1$ unique probabilities to determine. The EM algorithm is particularly useful here. With larger data sets, the use of a fixed grid of say 30–50 support points should not result in much loss of information. The other reason is theoretical. With the restricted NPMLE the parameter space is now finite and estimating equation approaches can be employed to attack the properties of the resulting estimators. Work remains to be done on this aspect of the problem.
- There are also smoothed versions of the NPMLE; see for example Laird

and Louis (1991) and Eggermont and LaRiccia (1997), and the application in Dalen et al. (2010). In our examples and simulations, where we estimate cumulative probabilities and percentiles, we will "smooth" in a simplistic manner by "connecting the dots" to create a piecewise linear cumulative distribution function. This translates to a density with piecewise uniform distributions. There are certainly other ways to consider smoothing the discrete fit that comes from the NPMLE.

We describe two bootstrapping techniques modeled on ones proposed by Laird and Louis (1987) and employed by others (e.g., Link and Sauer (1995)) in a closely related empirical Bayes problem. These match in principle to our general one-stage and two-stage bootstrap methods described in Section 6.16 for regression problems. The "one-stage bootstrap" here resamples units and uses the information from the selected unit. In the case of replication this information is the sample mean, variance and sample size. This approach avoids modeling how the variance depends on the mean but does not provide a bootstrap estimate of bias, which can be of particular concern here. The "two-stage bootstrap" samples with replacement from the fitted distribution. These values are taken as the true means. At the next step replicates are simulated using a per-replicate variance which depends on x. This means we need to model and estimate the variance function, discussed in the previous section. A cruder alternative would be to order the W's, divide them into groups, say using deciles, assume the per-replicate variance is constant in groups (outliers should probably be excluded) and either resample from residuals in the group or, under normality, use a variance estimated by the average per-replicate variance within the group.

Simulations.

The asymptotic properties of the NPMLE have proved difficult to establish especially with changing measurement error variances. This is more of an issue when the final goal is estimation of the distribution, less of a problem when the NPMLE is used as one part of a measurement error analysis, a role in which it has enjoyed some use. One advantage is that it is relatively easy to compute and produces a mass function that is easy to work with further.

Here a few limited simulations are presented to look at the performance and potential of the estimators of cumulative probabilities and percentiles resulting from the NPMLE. The version used in the simulations treats $W_i|x_i$ treated as $N(x_i, S_i^2/m_i)$. The parameters used are based loosely on the beta-carotene example. Throughout the true distribution is normal with mean $\mu = 4$ and standard deviation $\sigma = 1$ and $n = 158$. There are $m = 6$ or $m = 4$ replicates per subject. The variance model uses either constant per-replicate variance $t_i^2 = t^2$ = .09 or .25 or variance proportional to the mean, $t_i^2 = \beta_0 x_i$, with $\beta_0 = .05$ or .10. The quantities estimated were the cumulative probabilities at 2, 4 and

6, denoted $F(2)$, etc., and the 10th, 50th and 90th percentiles, denoted by $P_{.1}$, etc. While the simulations are certainly limited, the results are encouraging in terms of both bias and approximate normality of the estimators (plots not shown). The bias is a little more noticeable, but still not major, as the per replicate measurement error standard deviation increased to .5 ($t_i^2 = .25$).

Table 10.2 *Mean estimates for cumulative probabilities and percentiles based on 500 simulations. True values are in first row.*

		$F(2)$	$F(4)$	$F(5)$	$P_{.1}$,	$P_{.5}$	$P_{.9}$
True Values		0.0228	0.5	0.9773	2.7185	4	5.28155
m	t_i^2						
4	$.05x_i$	0.0226	0.5522	0.9878	2.6808	3.8737	5.0915
6	$.05x_i$	0.0237	0.5452	0.9903	2.6904	3.8934	5.0394
4	$.10x_i$	0.0214	0.5608	0.9896	2.7092	3.8578	5.0200
6	$.10x_i$	0.0214	0.5553	0.9937	2.7273	3.8811	4.9342
4	.09	0.0221	0.5432	0.9838	2.6768	3.8920	5.1616
6	.09	0.0231	0.5361	0.9851	2.6800	3.9076	5.1439
4	.25	0.0177	0.5348	0.9888	2.7562	3.9168	5.0688
6	.25	0.0183	0.5340	0.9876	2.7717	3.9165	5.0488

10.2.4 Example

We return to the data described in the introduction, with 4 replicate measures of diet intake per subject. The mean for individual i is $W_i = \sum_{j=1}^{4} W_{ij}/4$ and $\widehat{\sigma}_{ui}^2 = S_i^2/4$, where S_i^2 is the sample variance among the four replicates. See Figure 10.1.

The average measurement error variance is $\widehat{\sigma}_u^2 = .5346$. Using the individual means, the overall mean is $\bar{W} = 4.209$ and $\widehat{\sigma}_{naive}^2$ (the sample variance of the 158 sample means) is 4.207. The moment correct estimator of the variance is $\widehat{\sigma}^2 = 4.207 - .5346 = 3.672$. The naive standard error of the mean is $.1632 = (4.209/158)^{1/2}$. As noted in Section 10.2.1 this is a valid standard error for $\bar{W}$, even with changing measurement error variances.

Methods for estimating the mean-variance relationship were outlined in Section 10.2.2. Here we fit two models, the quadratic model with no linear term, and the power of the mean (POM) model with results in Table 10.3. The quadratic model was fit using the moment corrected method, labeled as MEE, with standard errors computed using the robust analytical form and via a one-stage bootstrap. Figure 10.2 plots S_i versus W_i along with the naive and corrected fits for the quadratic and POM models. The two lines higher up are the naive quadratic fit (solid line) and the moment corrected estimator (dashed line). The two lower down in the figure, which overlap closely, are the naive and corrected POM fits. The corrected POM fit was obtained using the modified estimating equation approach. While the correction for the quadratic model leads to some substantial changes, the measurement error correction has little effect on the power of the mean model. The reason for this is that if we examine the biases entering into the naive estimating equations for the power of the mean model, they are quite small. This is, in part, due to the fact that the model involves $log(W_i)$. The SIMEX estimates in the quadratic case were included just for comparison. As noted in Section 10.2.2, SIMEX need to be modified for this setting.

Table 10.3 *Beta-carotene data. Estimation of mean-variance relationship. QUAD fits the quadratic model $E(T_i^2|x_i)\beta_1 + \beta_2 x_i^2$ while POM (power of the mean model) fits $E(T_i^2|x_i) = e^{\beta_1+\beta_2 log(x_i)}$. MEE is modified estimating equations with R indicating use of robust analytical covariance and B bootstrapping. 1000 bootstrap samples.*

QUAD.

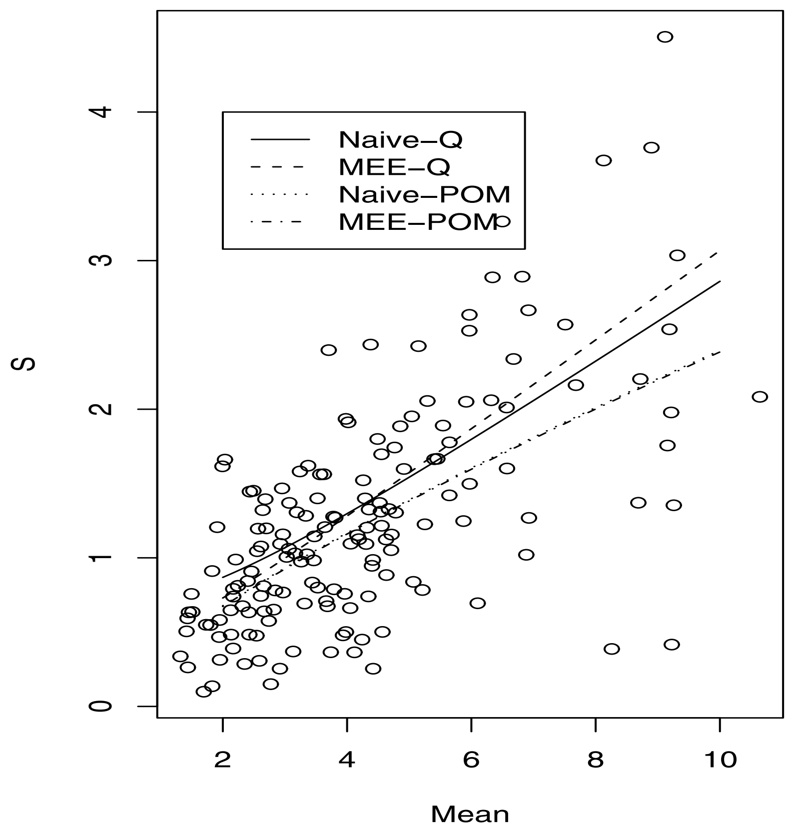
	β_1(SE)	CI	β_2(SE)	CI
Naive	.4446 (.2510)		.0774 (.0081)	
MEE-R	.1625 (.3906)	(-.603,.928)	.0925 (.0249)	(.044,.141)
MEE-B	.1625 (.4105)	(-.801,.763)	.0925 (.0256)	(.053,.153)
SIMEX	.2057		.0904	

POM

	β_1(SE)	CI	β_2(SE)	CI
Naive	-1.8655 (.3527)		1.5691 (.1808)	
MEE-B	-1.884 (.4669)	(-3.141, -1.228)	1.5733 (.2939)	(1.129, 2.346)

Finally, we consider nonparametric estimation of the distribution of individual true mean intakes, along with estimation of cumulative probabilities at 2, 4 and 6 and the 10th, 50th and 90th percentiles of the distribution, denoted $P_{.1}$, etc. Table 10.4 shows the naive estimates along with estimated values and associated inferences from using the NPMLE. The NPMLE was fit using a restricted NPMLE over 50 evenly spaced points between 0 and 10 and with individual measurement error variances. A plot of the NPMLE is shown in the second panel of Figure 10.3. The top figure is a density estimate on the original W values. The bootstrap analysis in Table 10.4 is based on the one-stage bootstrap, which resamples individuals, using 500 bootstrap samples. The naive confidence interval for σ^2 is normal based; the others are distribution free. Although there are fairly serious changes with respect to estimating of the population variance σ^2, the impact on inferences for the other quantities is somewhat

Figure 10.2 *Beta-carotene example. Plot of per-replicate standard deviation S_i versus the mean W_i and fits from naive and corrected fits for quadratic and power of the mean model. See text for details.*



muted. There was a tendency for a shift upward in the upper bound of the intervals for cumulative probabilities and a shift downward in the lower bound of the intervals for percentiles.

The bottom part of Figure 10.3 shows the NPMLE fit using smoothed variances based on the power of the mean fit, with the measurement error variance for individual i taken to be $\widehat{\sigma}^2_{ui} = e^{-1.884+1.5733 log(W_i)}/4$. This removes some of the impact of very large (or small) individual variances and generally leads to a smoother estimate. There are only minor changes in fitted values, with this fit leading to $\widehat{\mu} = 4.033$, $\widehat{\sigma}^2 = 3.6616$, estimated cumulative probabilities of .1173, .5541 and .8457, respectively, and estimated percentiles of 1.804, 3.878 and 6.407.

Table 10.4 *Analysis of beta-carotene data using four replicates. NPMLE is restricted to 50 points between 0 and 11 and used individual measurement error variances. Bootstrap analysis based on one-stage bootstrap.*

	Estimates		One-stage Boot	
Parameter	Naive(SE)	NPMLE	Mean	SE
Mean μ	4.209 (0.1632)	4.0329	4.0494	0.1662
Variance σ^2	4.2068	3.6563	3.6781	0.5529
$P(X \leq 2)$	0.1076 (0.0247)	0.1202	0.1194	0.0355
$P(X \leq 4)$	0.5316 (0.0398)	0.5720	0.5780	0.0603
$P(X \leq 6)$	0.8354 (0.0296)	0.8524	0.8549	0.0357
$P_{.10}$	1.9541	1.7879	1.8933	0.1976
$P_{.5}$	3.7998	3.8233	3.7999	0.2129
$P_{.9}$	6.9270	6.3542	6.8008	0.9289

Confidence Intervals

Parameter	Naive	Bootstrap
Mean μ	(3.887,4.532)	(3.7154, 4.3993)
Variance σ^2	(3.412,5.318)	(2.6017, 4.7542)
$P(X \leq 2)$	(.0587,.1564)	(0.0552, 0.1910)
$P(X \leq 4)$	(.4530,.6103)	(0.4557, 0.6941)
$P(X \leq 6)$	(.7770,.8939)	(0.7799, 0.9222)
$P_{.10}$	(1.732,2.208)	(1.5940, 2.2880)
$P_{.5}$	(3.463,4.260)	(3.0506, 4.1161)
$P_{.9}$	(6.340,8.898)	(5.2972, 8.9048)

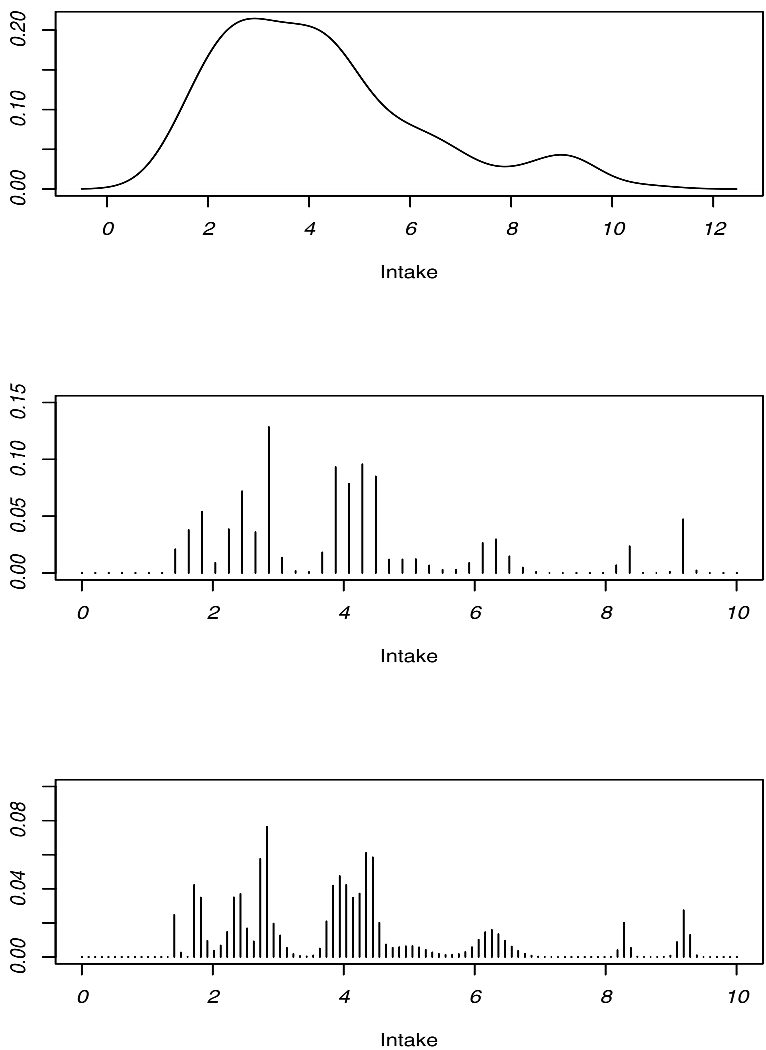
10.3 Linear measurement error in the one-way setting

Here we consider the standard "one-way" setting in which the goal is estimation and comparison of means across different groups/treatments but there is measurement error in a quantitative response. The case with a categorical response was covered in Chapter 3 while the setting with no error in a quantitative response but misclassification in group membership was discussed in Section 8.6.

The model for the true values is

$$X_{jk} = \mu_j + \epsilon_{jk}, \qquad j = 1 \text{ to J, k = 1 to } n_j \tag{10.4}$$

Figure 10.3 *Beta-carotene example. Top panel is naive, based on density estimate using observed means. Middle panel is NPMLE using individual measurement error variances. Bottom panel is NPMLE using smoothed estimates of measurement error variances.*



where the ϵ_{jk} are independent with mean 0 and $V(\epsilon_{jk}) = \sigma_j^2$. The group membership is assumed known.

What is observed is W_{jk}. As described in Chapter 6, there are many quantities where a nonadditive model relating the measured and true value is needed. There is a long history on the estimation of such relationships based on standards, and use of the resulting "calibration curve" to estimate true values (here the realizations of the X_{jk}). See, for example, Osborne (1991), Brown (1982) and references and discussion therein.

The emphasis here is on the use of external validation data, but allowing for the fact that the measurement error model may change across observations. This can occur for many reasons including the use of different labs, re-calibration of an instrument, etc. These are referred to broadly as different "methods," even though it may be re-calibration of the same method. Each observation is analyzed by only one method. For now the measurement error model is assumed linear, with

$$E(W|x) = \theta_{0m} + \theta_{1m}x \quad \text{and} \quad V(W|x) = \sigma_{um}^2$$

for method m. The methods and associated parameters are treated as fixed, although a random coefficients approach (Vecchia et al. (1989)) may sometimes be appropriate. With additive error and a single method the effect is to add a variance component and inferences for means are unaffected. With constant biases, the methods play essentially the role of blocks; see Cochran (1968) for some discussion. Mulroe et al. (1988) examined the performance of naive confidence intervals and tolerance intervals for a single mean in the presence of only one measuring method.

The complexity of this problem depends on the assignment of observations to measuring methods. We first treat the case where a single method is used for each group, which leads to a relatively straightforward analysis. The case of "overlapping design," where some groups have data analyzed by more than one method, is then discussed. The methods and results discussed in this section are taken from Buonaccorsi (1991).

Acid Rain Monitoring Example. For motivation, we consider some data from the Massachusetts Acid Rain Monitoring Project. Godfrey et al. (1985) describe the project and examine data from March 1983 to April 1984, during which water samples were collected from approximately 1800 water bodies and chemical analyses were carried out by 73 laboratories. We focus on just a small piece of the data arising from Barnstable county in April of 1983. Table 10.6 displays summary statistics for measured pH values for four districts in the county along with corrected inferences that are explained later. Over the course of the experiment, the labs involved were sent blind samples with "known" pH values. Figure 10.4 contains plots of observed versus true values for each of four labs while Table 10.5 shows estimated measurement error parameters for the four different labs. The question we address is how to incorporate this calibration/quality assurance data if the main objective is estimation and comparison of the mean pH for the four districts. The behavior of lab 4 would be unsatisfactory to many from a quality assurance perspective since the intercept and slope appear quite different than 0 and 1, respectively. This may well result in the elimination of all data from that lab, but this clearly discards useful information. It is the measurement error variance rather than the coefficients that is of the most importance.

Figure 10.4 *Calibration of measure of water pH.*

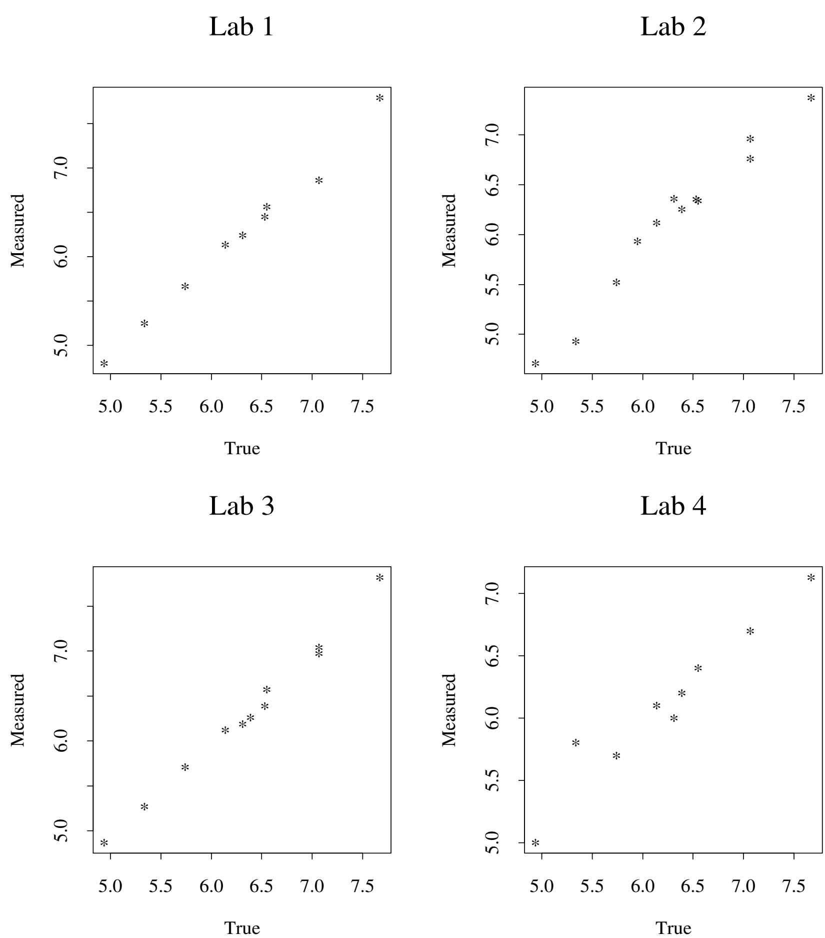


Table 10.5 *Estimates from simple linear regression of observed pH on true pH for four labs.*

LAB	1	2	3	4
Sample Size (R)	9	12	11	9
Intercept ($\widehat{\theta}_0$)	-.3675	-.1916	-.3216	1.7029
Slope ($\widehat{\theta}_1$)	1.0492	1.0026	1.0430	.7071
Variance ($\hat{\sigma}_u^2$)	.0081	.0199	.0064	.0258

10.3.1 One measuring method per group

Consider first a single measuring method where, dropping the m subscript,

$$W_{jk} = \theta_0 + \theta_1 x_{jk} + u_{jk}, \tag{10.5}$$

with $E(u_{jk}) = 0$, $V(u_{jk}) = \sigma_u^2$, and the u_{jk} are assumed independent and independent of the ϵ's in (10.4). This leads to independent W_{jk} with

$$E(W_{jk}) = \theta_0 + \theta_1 \mu_j = \mu_{Wj} \quad and \quad V(W_{jk}) = \theta_1^2 \sigma_j^2 + \sigma_u^2 = \phi_j. \tag{10.6}$$

Some hypotheses can be tested without any calibration data. Consider the null hypothesis $H_0 : \mathbf{H}\boldsymbol{\mu} = \mathbf{0}$, where $\mathbf{H}$ is $q \times J$ of rank $q \leq J$. The vector of means associated with the raw values is $\boldsymbol{\mu}_W' = (\mu_{W1}, \ldots, \mu_{WJ})$ where $\boldsymbol{\mu}_W = \theta_0 \mathbf{1} + \theta_1 \boldsymbol{\mu}$. If $\mathbf{H}\boldsymbol{\mu}$ consists of a set of contrasts (i.e., the elements in each row of $\mathbf{H}$ sum to 0) then $\mathbf{H}\boldsymbol{\mu} = \mathbf{0}$ if and only if $\mathbf{H}\boldsymbol{\mu}_W = \mathbf{0}$. So:

Tests about contrast can be carried out using the raw data.

It is obvious, however, that confidence intervals and point estimators based on raw data are incorrect and that the μ_j's and σ_j's are not identifiable without information about the measurement error parameters.

As noted in treating the pH-alkalinity example in Section 8.5.3, it is common practice to form adjusted values

$$\widehat{X}_{jk} = (W_{jk} - \widehat{\theta}_0)/\widehat{\theta}_1 \tag{10.7}$$

and then proceed with a standard analysis using these values. Tests about contrasts are still correct, as they were for raw values, but confidence intervals based on adjusted values are typically too small. For example, under normality it can be shown (Mulroe et al. (1988)) that the coverage rate is strictly less than the desired level if $\sigma_u^2 > 0$. This is due to not accounting for the uncertainty in the estimated measurement error parameters.

Correcting for measurement error.

The external calibration data for method m consist of (W^*_{rm}, x^*_{rm}) with

$$W^*_{rm} = \theta_{0m} + \theta_{1m}x^*_{rm} + u_{mr}, \qquad r = 1 \text{ to } R_m \tag{10.8}$$

where the x^*_{rm}'s are fixed and the u_{rm} are uncorrelated with mean 0 and variance σ^2_{um}. The θ's are estimated using standard least squares. For a single method below the m subscript is dropped.

Given (10.6), for a single method the obvious moment corrected estimators are

$$\hat{\mu}_j = (\overline{W}_j - \widehat{\theta}_0)/\widehat{\theta}_1 \qquad and \qquad \hat{\sigma}^2_j = max(0, (S^2_j - \widehat{\sigma}^2_u)/\widehat{\theta}^2_1) \tag{10.9}$$

where $\overline{W}_j$ and S^2_j are the sample mean and variance of the W's in group j. If normality is assumed for all random quantities then the estimators in (10.9) would be maximum likelihood estimators if S^2_j and $\widehat{\sigma}^2_u$ are defined using divisors of n_j and R, respectively, and $\widehat{\sigma}^2_j$ is positive. If the σ^2_j are assumed equal a pooled estimator $\hat{\sigma}^2 = \max[0, (MSE_W - \widehat{\sigma}^2_u)/\widehat{\theta}^2_1]$ can be used where MSE_W is the error mean square from analyzing the raw data. The quantity $\hat{\mu}_j$ is also the average of the adjusted values.

The estimator $\hat{\mu}_j$ is a ratio and we can draw on the extensive literature on ratios in attacking this problem; see for example Chapter 6 of Miller (1986). We note that under normality the mean and variance of $\widehat{\mu}_j$ do not exist, but in practice the distribution of $\widehat{\theta}_1$ is often bounded away from 0 and the nonexistence of the moments is not a practical issue. Standard approximations lead to

$$E(\widehat{\mu}_j) \approx \mu_j + \frac{\sigma^2_u[\mu_j - \overline{x}^*]}{\theta^2_1 S_{xx}} \tag{10.10}$$

and

$$\lambda_{jj} = V(\widehat{\mu}_j) \approx \frac{1}{\theta^2_1}\left[\frac{\phi_j}{n_j} + \sigma^2_u\left(\frac{1}{R} + \frac{(\overline{x}^* - \mu_j)^2}{S_{xx}}\right)\right] \tag{10.11}$$

where $S_{xx} = \sum_r (x^*_r - \bar{x}^*)^2$ and R is the number of observations in the calibration data. From a design standpoint, we see the advantage of $\overline{x}^* = \mu_j$ since this eliminates first order bias and minimizes the approximate variance.

Define the vector of estimated means $\widehat{\boldsymbol{\mu}}' = (\hat{\mu}_1, \ldots, \hat{\mu}_J)$. Using the delta method and some standard asymptotic arguments $\widehat{\boldsymbol{\mu}}$ is asymptotically normal with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Lambda}$, where $\boldsymbol{\Lambda}$, has jth diagonal element λ_{jj} given above and off diagonal elements

$$\lambda_{jl} = \frac{\sigma^2_u}{\theta^2_1}\left[\frac{1}{R} + \frac{(\overline{x}^* - \mu_j)(\overline{x}^* - \mu_l)}{S_{xx}}\right], \qquad j \neq l. \tag{10.12}$$

Notice that the estimators of the different group means are correlated through common use of the estimated measurement error parameters.

Approximate confidence intervals and tests for linear combinations of $\boldsymbol{\mu}$ can be carried out using an estimate of $\boldsymbol{\Lambda}$ and Wald methods; see Section 13.3. As noted earlier tests about contrasts can also be carried out using raw or adjusted values.

Consider estimating a linear combination of the cell means,

$$\omega = \sum c_j \mu_j.$$

The approximate Wald confidence interval is $\sum c_j \hat{\mu}_j \pm z_{\alpha/2}(\mathbf{c}'\hat{\boldsymbol{\Lambda}}\mathbf{c})^{1/2}$, where $\mathbf{c}' = (c_1, \ldots, c_J)$. With $C = \sum_j c_j$, we can also write $\omega = E(\mathbf{c}'\bar{\mathbf{W}} - C\widehat{\theta}_0)/E(\widehat{\theta}_1)$, where $\bar{\mathbf{W}}' = (\bar{W}_1, \ldots, \bar{W}_J)$. This expresses ω as a ratio of parameters and rather than use a Wald interval it is better to obtain a confidence set for ω using Fieller's method (e.g., Read (1983), Buonaccorsi(1998)). Define $v_{11} = \sum c_j^2 s_j^2/n_j + \widehat{\sigma}_u^2 C^2(\sum_r x_r^{*2}/S_{xx})$, $v_{22} = \widehat{\sigma}_u^2/S_{xx}$ and $v_{12} = C\widehat{\sigma}_u^2\overline{x}^*/S_{xx}$. These are the estimated variance of the numerator, the estimated variance of the denominator and their estimated covariance, respectively. With $f_o = (\mathbf{c}'\bar{\mathbf{W}} - C\widehat{\theta}_0)^2 - z^2 v_{11}$, $f_1 = (\mathbf{c}'\bar{\mathbf{W}} - C\widehat{\theta}_0)\widehat{\theta}_1 - z^2 v_{12}$, $f_2 = \widehat{\theta}_1^2 - z^2 v_{22}$ and $D = f_1^2 - f_o f_2$, the Fieller confidence interval for ω is

$$\frac{f_1}{f_2} \pm \frac{D^{1/2}}{f_2}$$

as long as $f_2 > 0$. This condition corresponds to rejecting $H_0 : \theta_1 = 0$ using a Z-test and so concluding the denominator is significantly different than 0. It would be rare for the measuring instrument to have $f_2 \leq 0$ but if this did occur the confidence region is either the whole real line or the complement of a finite interval.

Multiple methods.

With multiple methods, quantities associated with method m are indexed by an m subscript. It is now more difficult to evaluate the behavior of inferences based on raw or adjusted values and unlike the single method setting, it is no longer the case that tests about contrasts in the means based on raw or adjusted data are correct.

Since here the assumption is that all data from a group comes from a common method, the results above can be used. If the data for group j are analyzed by method m, then $\hat{\mu}_j = (\overline{W}_j - \widehat{\theta}_{0m})/\widehat{\theta}_{1m}$, with similar modifications to $\widehat{\sigma}_j^2$, and inferences for μ_j by itself proceed as for a single method. For comparisons among means, let $\boldsymbol{\mu}_{(m)}$ denote the vector of means for groups analyzed by method m and arrange the means into $\boldsymbol{\mu}' = (\boldsymbol{\mu}'_{(1)}, \ldots, \boldsymbol{\mu}'_{(M)})$. Then the approximate covariance of $\widehat{\boldsymbol{\mu}}$ is $\boldsymbol{\Lambda}$, a block diagonal matrix with $\boldsymbol{\Lambda}_1, \ldots, \boldsymbol{\Lambda}_M$ on the diagonal and each $\boldsymbol{\Lambda}_m$ computed using (10.11) and (10.12).

pH example revisited. Returning to the pH example, the data from district

j were analyzed by lab j The resulting point estimators, obtained from (10.9), are given in Table 10.6 along with the estimated standard error of $\hat{\mu}$ obtained using just adjusted values or the estimated standard error using (10.11) which employs the delta method. The latter accounts for the uncertainty from estimating the measurement error parameters, but as seen from the results the contribution from that term is minor in this case. This is due to the relatively small measurement error variances.

Table 10.6 *Summary statistics of observed pH for four districts and corrected estimates.*

District	1	2	3	4
Sample Size (n_j)	14	14	20	15
Mean (W_j)	6.306	5.824	5.653	6.525
St. Dev. (S_j)	.607	.779	.896	.634
$\hat{\mu}_j$	6.361	6.00	5.728	6.820
$\widehat{\sigma}_j^2$	.5721	.7641	.8556	.8674
SE-Adjusted	.1546	.2077	.1921	.2315
SE-Delta	.1573	.2133	.1946	.2451

Table 10.7 *pH example. Confidence intervals for means.*

District	1	2	3	4
Raw	(5.96,6.66)	(5.37,6.27)	(5.23,6.07)	(6.17,6.88)
Adjusted	(6.03,6.70)	(5.55,6.45)	(5.33,6.13)	(6.32,7.32)
Delta(z)	(6.05,6.67)	(5.58,6.42)	(5.35,6.11)	(6.34,7,30)
Fieller(z)	(6.05,6.67)	(5.58,6.42)	(5.35,6.11)	(6.34,7.30)
Delta(t)	(6.02,6.70)	(5.54,6.46)	(5.32,6.14)	(6.29,7.35)
Fieller(t)	(6.02,6.70)	(5.53,6.46)	(5.32,6.13)	(6.30,7.39)

Table 10.7 displays confidence intervals computed a variety of ways. The first two use intervals from raw and adjusted values, respectively, based on a t distribution with $n - 1$ degrees of freedom. The Wald and Fieller intervals were computed using either a z table value or a t-value based on $n - 1$ degrees of freedom. The latter was used to provide a more direct comparison to the intervals from raw and adjusted values. The Fieller and delta methods are similar due to the tests for zero slope being highly significant. Since we know from theory that the adjusted value intervals are too small, for moderate sample sizes we recommend using a t value in the Fieller intervals. Here this is done using $n - 1$ degrees of freedom, but an adjusted degrees of freedom can be used. This is used by Mulroe et al. (1988) for a single mean based on a Satterthwaite approach. It can be extended to handle general linear combinations.

The estimated covariance matrix of $\hat{\mu}$ is $\hat{\Lambda}$ = diag(.0247,.0455,.0279,.0601).

Table 10.8 *pH example. Standard errors for pairwise differences in means.*

	RAW	ADJUSTED	DELTA
1 vs. 2	.263	.260	.265
1 vs. 3	.258	.246	.250
1 vs. 4	.231	.278	.291
2 vs. 3	.290	.283	.289
2 vs. 4	.265	.311	.325
3 vs. 4	.259	.301	.313

The approximate Wald test of $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ has an observed chi-square statistic of 14.21 with a P-value (based on chi-square with 3 degrees of freedom) of .0025. The F-test based on raw data yields a P-value of .0047 assuming equal variances and a P-value of .0077 for unequal variances using Welch's test. For adjusted values, the resulting P-values are .0015 (equal) and .0067 (unequal). These F-tests are provided only for comparison and they are not correct here due to the use of multiple methods.

While there is little doubt that one should reject the hypothesis of equal means, confidence intervals for the differences in means are more important. Table 10.8 gives estimated standard errors for each of the pairwise differences computed based on raw data, adjusted values and the delta method, all without assuming any equality of variances. As with the estimation of individual means, there are small increases in the estimated standard errors with the delta method versus using just adjusted values. Note that in comparison to the standard errors from the raw values the corrected standard errors can go in either direction as a result of the manner in which the measurement error parameters enter into (10.11).

10.3.2 General designs

Designs in which data from a group passes through multiple methods can readily occur when samples are analyzed over time and the measuring method changes due to re-calibration or other reasons. This problem is a special case of the general treatment of the next section which allows an arbitrary linear model for the true values, multiple methods, nonlinear measurement error models and has no restrictions on how the methods are distributed over observations in the main unit. However, we can isolate some specific results here for the case of linear measurement error.

Let $m(jk)$ denote the method used to obtain the kth observation in the jth group, so $E(W_{jk}) = \theta_{0m(jk)} + \theta_{1m(jk)}\mu_j$ and

$$V(W_{jk}) = \phi_{jm} = \theta^2_{1m(jk)}\sigma^2_j + \sigma^2_{um(jk)}. \tag{10.13}$$

There are a number of different strategies for correcting for measurement error, related to the approaches outlined in Section 6.15 for the use of external validation data. Full moment or normal based likelihood approaches fit (10.13) and (10.8) simultaneously under either just moment assumptions or assuming normality throughout. These methods were explored by Wiedie (1995). The "full" methods do offer some potential gains in efficiency; however, the gains are generally small and the methods more complicated to implement than pseudo approaches. The pseudo approaches simply replace the measurement error parameters in (10.13) with estimates from the calibration data and then estimate the μ_j's and σ_j^2's. With the measurement error parameters fixed at their estimates this is a linear model with a specific type of heteroscedasticity, which can be used in a variety of ways. Let $\overline{W}_{j(m)}$ denote the mean of n_{jm} observations from group j produced by method m, A_j denote the collection of methods producing data for group j and let $\hat{\mu}_{j(m)} = (\overline{W}_{j(m)} - \widehat{\theta}_{0m})/\widehat{\theta}_{1m}$ denote the estimator of μ_j arising from data analyzed by method m.

- **Pseudo-moment I.** Estimate the μ_j's using just the mean part of the model. This leads to simply using the adjusted values so

$$\widehat{\mu}_j = \sum_k \widehat{X}_{jk}/n_j = \sum_{m \in A_j} (n_{jm}/n_j)\widehat{\mu}_{j(m)},$$

  where $\widehat{X}_{jk} = (W_{jk} - \widehat{\theta}_{0m(jk)})/\widehat{\theta}_{1m(jk)}$ is the jkth adjusted value in group j. This is particularly easy to implement and the approximate covariance matrix of $\widehat{\boldsymbol{\mu}}$ is obtained from the variance/covariance matrix of the $\widehat{\mu}_{j(m)}$, which are computed using equations (10.11) and (10.12). See also Section 5.1 of Buonaccorsi (1996b) and the result in (10.17) of the next section.
- **Pseudo MLE**. Use both the mean and variance part of (10.13) to estimate the μ_j and σ_j^2's using pseudo-maximum likelihood under normality. In fact the pseudo-MLE's for μ_j under normality are robust to the normality assumption essentially being weighted least squares estimates.

The pseudo-MLE is most easily computed using the EM algorithm (Dempster, Laird and Rubin, 1977). The approximate covariance matrix of $\widehat{\boldsymbol{\mu}}$ can be computed using the results from the next section. Both the full and pseudo estimators of μ_j turns out to be a weighted average of the $\hat{\mu}_{j(m)}$'s.

10.4 General measurement error in the response in linear models

We return to the general linear model for true values and to our customary notation with $Y_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i$, where the ϵ_i are i.i.d. normal with mean 0 and variance σ^2. The $\mathbf{x}_i$ are treated as fixed and observable here and the measurement error is in the response only. As in the earlier chapters, the mismeasured response is denoted by D_i and as in the preceding section, there may be different measuring "methods" involved.

Urinary neopterin example. To motivate the coverage of this section, we consider an example involving a comparison of serum neopterin levels between 14 HIV-positive and 15 HIV-negative individuals enrolled in a methadone treatment program. Some aspects of this were introduced briefly in Section 6.4.4. The data come from Matthew DiFranco with certain aspects of the analysis treated in Buonaccorsi and Tosteson (1993). True neopterin levels cannot be observed exactly but instead are assessed through a radioimmunoassay which returns a standardized radioactive count. Each observation associated with one of two batches of reagents, each batch having its own calibration curve. These were fit using a four-parameter logistic models of the form

$$E(D|y) = \alpha_1 + \frac{\alpha_2 - \alpha_1}{1 + (e^y/\alpha_3)^{\alpha_4}}$$

where y is the log of the neopterin concentration (in nmol/L). This parameterization was used since for likelihood methods the assumption of normality for log(concentration) was more reasonable than for concentration itself. There are two curves of this form corresponding to re-calibration with two batches of reagents. Each calibration curve is fit using 18 data points arising from 2 replicates on each of 9 standards. See Figure 6.1 for one set of data and Table 10.12 for the estimated parameters. Although not apparent from the figure, the fitted curves do have an asymptote as concentration approaches 0. A conventional approach is to create imputed or adjusted values from the measurements by inverting using the calibration curve for the batch corresponding to the given observation. Table 10.9 shows the return standardized count, a batch indicator and the adjusted value, expressed in log(nmol/L). We return to this example in Section 10.4.3.

10.4.1 Models

For measuring method m, the measurement error model is

$$E(D|y) = g(y, \boldsymbol{\alpha}_m) \text{ and } V(D|y) = v(y, \boldsymbol{\alpha}_m, \boldsymbol{\tau}_m). \tag{10.14}$$

The function $g(y, \boldsymbol{\alpha}_m)$ is the calibration curve for method m and is assumed to be a monotonic function of y. The case where the calibration curves are linear was treated in the preceding section. In the most general form the variance can depend on the true value, the mean parameters in $\boldsymbol{\alpha}_m$ and additional parameters in $\boldsymbol{\tau}_m$. With constant measurement error variance for each method $v(y, \boldsymbol{\alpha}_m, \boldsymbol{\tau}_m) = \tau_m^2$. The measurement error parameters for method m are collected in the vector $\boldsymbol{\theta}_m = (\boldsymbol{\alpha}_m, \boldsymbol{\tau}_m)$.

Table 10.9 *Raw standardized counts and adjusted values for serum neopterin in 15 HIV- and 14 HIV+ individuals*

	HIV -			HIV +	
Stand. Count	Adjusted Value	Batch #	Stand. Count	Adjusted Value	Batch #
.321	2.8	1	.255	3.3	1
.267	3.2	1	.211	3.7	1
.291	3.1	1	.175	4.0	1
.327	2.8	1	.222	3.6	1
.315	2.9	2	.321	2.8	2
.320	2.8	2	.235	3.5	2
.352	2.6	1	.315	2.9	2
.292	3.1	1	.238	3.5	2
.306	3.0	1	.216	3.6	1
.321	2.9	1	.224	3.6	1
.362	2.5	1	.372	2.4	2
.282	3.1	1	.305	2.9	2
.229	3.5	2	.273	3.2	2
.284	3.1	1	.228	3.5	2
.205	3.7	1			

Notational note: We have switched from the usage in earlier chapters and sections where θ's denoted the coefficients in the regression model for the measurement error. This is because here we want $\boldsymbol{\theta}$ to contain all measurement error parameters, both coefficients in the ME regression model plus variance parameters.

The calibration data for method m consist of $(y^*_{mr}, D^*_{mr}), r = 1$ to R_m, where the y^*_{mr} are fixed values and the model for $D^*_{mr}|y^*_{mr}$ is the one given in (10.14). From these estimated measurement error parameters can be obtained using standard method, e.g., weighted or unweighted least squares, maximum likelihood, etc.

An observation in the main study is produced by one of the methods with $m(i) = m$ if the ith observation is obtained from method m. Given the true values

$$D_i|y_i = g(y_i, \boldsymbol{\alpha}_{m(i)}) + u_i$$

where the u_i are independent with mean 0 and variance $v(y_i, \boldsymbol{\alpha}_{m(i)}, \boldsymbol{\tau}_{m(i)})$. Unconditionally,

$$\begin{aligned} E(D_i) &= E[g(Y_i, \boldsymbol{\alpha}_{m(i)}] \text{ and} \\ V(D_i) &= V[g(Y_i, \boldsymbol{\alpha}_{m(i)}] + E[v(Y_i, \boldsymbol{\alpha}_{m(i)}, \tau_{m(i)})]. \end{aligned}$$

It is clear that in most settings, unless the measurement error parameters are known then $\boldsymbol{\omega} = (\boldsymbol{\beta}, \sigma^2)$ is not identifiable, even within a fully parametric framework. There is no universal guarantee that $\boldsymbol{\omega}$ is identifiable even with the calibration data, although this is often the case. For example, using results from Maritz and Lwin (1989, p. 38), one can show identifiability under the normal measurement error model, as long as $g(y, \boldsymbol{\alpha})$ is strictly monotonic in y and $\boldsymbol{\beta}$ is identifiable in the original model (meaning the design matrix $\mathbf{X}$ is of full column rank).

10.4.2 Correcting for measurement error

Two general methods of correcting are described here. The first makes use of the adjusted/imputed values for the unobserved Y's; the second set uses maximum likelihood. Pseudo-MLE's will be emphasized given some of the computational problems with using full maximum likelihood. Both the use of adjusted values and pseudo-MLEs can be framed in the context of using modified estimating equations, suggesting other approaches to the problem (see Buonaccorsi(1996b)) but given their complexity we do not describe them here.

In order to get the main points across easily, the two methods are described broadly here along with some limited simulations, while detailed results and computational expressions are given in Section 10.4.4.

- *Using adjusted values.* The ith adjusted value is

  $$\widehat{Y}_i = g^{-1}(D_i, \widehat{\alpha}_{m(i)}).$$

  This is simply the value of y for which the fitted calibration curve for method $m(i)$ equals the observed D_i. A "naive analysis" based on the adjusted values uses

  $$\widehat{\boldsymbol{\beta}}_A = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\widehat{\mathbf{Y}} \text{ and } \widehat{\sigma}^2_A = \sum_i (\widehat{Y}_i - \mathbf{x}_i'\widehat{\boldsymbol{\beta}}_A)^2/(n-p).$$

  Naive standard errors and other inferences follow from estimating $Cov(\widehat{\boldsymbol{\beta}}_A)$ with $\widehat{\sigma}^2_A(\mathbf{X}'\mathbf{X})^{-1}$.
  The estimators based on adjusted values are not always consistent. The approximate bias of $\widehat{\boldsymbol{\beta}}_A$ is given in result **A1** in Section 10.4.4. If $g(y, \boldsymbol{\theta}_m)$ is linear in y, then the asymptotic bias is in fact $\mathbf{0}$. With nonlinear calibration curves, $\widehat{\boldsymbol{\beta}}_A$ is not only biased for finite sample size but it is also inconsistent with asymptotic bias depending on the curvature of the calibration curves, and the relative magnitude of σ^2 and the measurement error variances. In practice the bias is often small enough to be ignored in which case $\widehat{\boldsymbol{\beta}}_A$ is attractive for its simplicity.
  When the bias in $\widehat{\boldsymbol{\beta}}_A$ is negligible, the approximate bias of $\widehat{\sigma}^2_A$ is given in

result **A2** in Section 10.4.4, and the covariance of $\widehat{\boldsymbol{\beta}}_A$ and how to estimate it are given in equations (10.17) and (10.18). The latter is given in a form that adds terms to the estimated covariance that comes from simply analyzing the adjusted values.

- *Likelihood based methods*

To describe the likelihood methods, let $f(y|\mathbf{x}_i, \boldsymbol{\omega})$ and $f(d_i|y; \boldsymbol{\theta}_m)$ denote the densities of Y_i and $D_i|Y_i = y$, respectively. With $\boldsymbol{\omega} = (\boldsymbol{\beta}, \sigma^2)$, D_i has marginal density

$$f(d_i \mid \mathbf{x}_i; \boldsymbol{\omega}, \boldsymbol{\theta}) = \int_y f(d_i|y; \boldsymbol{\theta}_{m(i)}) f(y|\mathbf{x}_i, \boldsymbol{\omega}) dy. \tag{10.15}$$

Combining the main study and the calibration data, the full likelihood is

$$L(\boldsymbol{\omega}, \boldsymbol{\theta}) = L_1(\boldsymbol{\omega}, \boldsymbol{\theta}) \cdot L_2(\boldsymbol{\theta})$$

where $L_1(\boldsymbol{\omega}, \boldsymbol{\theta}) = \prod_i^n f(d_i \mid \mathbf{x}_i; \boldsymbol{\omega}, \boldsymbol{\theta})$ is from the joint density of $D_1, \ldots, D_n$ and $L_2(\boldsymbol{\theta}) = \prod_{m=1}^M \prod_{r=1}^{R_m} f(d^*_{mr} \mid y^*_{mr}; \boldsymbol{\theta}_m)$ arises from the calibration data.

The full maximum likelihood estimators maximizes $L(\boldsymbol{\omega}, \boldsymbol{\theta})$ over $\boldsymbol{\omega}$ and $\boldsymbol{\theta}$ simultaneously. Obtaining the full MLE's can be cumbersome, especially with multiple methods, even with normal models having constant variance. Standard approaches require the evaluation of numerous integrals at each iteration and when the calibration curves are nonlinear in the parameters, the EM algorithm is not particularly helpful both since the M-step involves numerical integration, and the E-step does not have a closed form solution.

To avoid some of the computational problems associated with full maximum likelihood estimation, we turn to the pseudo-maximum likelihood estimator which maximizes $L_1(\boldsymbol{\omega}, \widehat{\boldsymbol{\theta}})$ over $\boldsymbol{\omega}$ with $\widehat{\boldsymbol{\theta}}$ fixed at the estimates obtained from the calibration data by itself. With this approach, computing is relatively easy. Full details on computing both full and pseudo-MLEs under normal models with constant variances are given by Buonaccorsi and Tosteson (1993). Assuming the ϵ_i in the model for the true values is normal, on the $(k+1)st$ step of the fitting procedure the M-step uses the usual estimates for the linear model but with Y_i and Y_i^2 replaced by estimates of $E(Y_i|d_i)$ and $E(Y_i^2|d_i)$, respectively, evaluated using estimated parameters from the previous step.

While the assumption of constant measurement error variance is reasonable in our example there are many situations where the variance is related to the mean or some other function of the true values. With the Y's being normally distributed, the EM algorithm still works in an easy manner for getting pseudo-MLEs, but the numerical integrations used at the E-step may be more difficult.

Simulations.

Table 10.10 *Simulated behavior of point estimators of μ and σ^2 = .02 based on 250 simulations. k = number of simulations used for AV methods.*

			Estimate of μ		Estimate of σ^2		
μ	n		PML	AV	PML	AV	k
2.303	20	MEAN	2.300	2.300	0.018	0.020	250
		S.D.	0.044	0.044	0.008	0.009	
2.303	100	MEAN	2.303	2.303	0.020	0.021	250
		S.D.	0.030	0.030	0.006	0.007	
2.485	20	MEAN	2.479	2.480	0.018	0.021	249
		S.D.	0.051	0.051	0.009	0.010	
2.485	100	MEAN	2.483	2.484	0.020	0.022	247
		S.D.	0.041	0.042	0.006	0.007	
2.708	20	MEAN	2.698	2.704	0.020	0.027	234
		S.D.	0.070	0.072	0.012	0.019	
2.708	100	MEAN	2.704	2.712	0.020	0.026	207
		S.D.	0.067	0.068	0.007	0.011	

Here we present some limited simulation results, a subset of those given by Buonaccorsi and Tosteson (1993). These are based on the example but here we focus on just a single group with a single measuring method. The measurement error model is the four parameter logistic curve, expressed in terms of e^y as used in the example. Throughout, the calibration parameters are $\alpha_1 = .05$, α_2 = .55, α_3 = 10, $\alpha_4 = 4$ and τ^2 = .0002. The calibration data were generated using a normal error model with 9 observations at standards e^{Y^*} = 0,1.25,2.5,5,10,20,40,80 and 160. Estimates of the α's and τ^2 were obtained using nonlinear least squares. Independently $Y_1, \dots Y_n$ were generated as i.i.d. $N(\mu, \sigma^2)$. The variance σ^2 was always .02 while the mean μ took on three values, 2.303 = log(10), 2.485 = log(12) and 2.708 = log(15). Results for sample sizes of 20 and 100 are shown based on 250 simulations.

Estimators were obtained using pseudo-MLE's (PML) and adjusted values (AV). In a few settings the AV analysis is not always based on 250 simulations since there were some cases in which the adjusted values were not defined for all of the n observations. This happens since the fitted calibration curve is bounded between a high of $\hat{\alpha}_2$ and a low of $\hat{\alpha}_1$ so if D_i falls outside of this range, the adjusted value does not exist. There are obviously some strategies for dealing with that, but the results are presented as in the original paper.

Point estimation results are presented in Table 10.10. For μ, there are rel-

Table 10.11 *Estimated coverage rates of confidence intervals for μ. Number of simulations used given in Table 10.10 .*

μ	n	PML	AV	AV-MOD
2.303	20	0.9080	0.8280	0.9200
2.303	100	0.9240	0.6520	0.9320
2.485	20	0.9120	0.7590	0.9116
2.485	100	0.9000	0.5061	0.9150
2.708	20	0.9360	0.6795	0.9444
2.708	100	0.9274	0.3623	0.9227

atively minor differences between the two estimators and signs of very little bias. For σ^2, the sample variance of adjusted values tends to overestimate σ^2, as expected from the theory, and is also a bit more variable than the pseudo-MLE. There are indications of small amounts of downward bias in the pseudo-MLE for σ^2 in some settings.

Table 10.11 shows the performance of supposed 95% Wald intervals for μ calculated using $\hat{\mu} \pm 1.96(SE)$. The naive AV approach uses the mean of the adjusted values with a standard error of $(S_A^2/n)^{1/2}$ where S_A^2 is the sample variance of the adjusted values. AV-MOD uses a modified standard error based on (10.18). Clearly, the straight use of adjusted values is unacceptable. The intervals based on unmodified adjusted value get even worse as n increases, which is expected based on theoretical results. The coverage rates associated with the pseudo-MLE will still fall a bit short as n increases if the size of the calibration data is kept fixed.

These simulations are certainly limited. The main point was simply to demonstrate that the blind use of adjusted values should generally be avoided in estimating the variance and in forming confidence intervals for the means and to show that the modified confidence intervals using adjusted values are competitive with those based on pseudo-MLE. This is good as using the adjusted values is an easier computational approach and distribution free.

10.4.3 Example

We return to neopterin example introduced at the beginning of this section. An examination of the adjusted values indicated that it makes more sense to assume Y = log(concentration) is normally distributed rather than concentration itself. All results are given in terms of the parameters for Y. (We will proceed under this model, but this use of adjusted values is not without problems. The adjusted values do not give the actual log(concentrations) because of uncertainty in the calibration parameters, which leads to both potential bias,

Table 10.12 *Estimated parameters with standard errors in parentheses based on calibration data for the two batches in the neopterin example.*

	BATCH 1	BATCH 2
$\hat{\alpha}_1$	0.0737 (.0205)	0.0669 (.0171)
$\hat{\alpha}_2$	0.524 (.0159)	0.580 (.0127)
$\hat{\alpha}_3$	20.235 (2.688)	16.325 (1.766)
$\hat{\alpha}_4$	1.243 (.1829)	1.040 (.1011)
$\hat{\tau}^2$	0.0006	0.0004

Table 10.13 *Analysis of neopterin example using pseudo-maximum likelihood under normality and based on a naive analysis of adjusted values.*

Type	$\hat{\mu}_+(SE)$	$\hat{\mu}_-(SE)$	SE(Diff)	$\hat{\sigma}^2$
Pseudo-MLE	3.312 (.1113)	3.0101(.1134)	.14199	.1162
Adj-naive	3.320(.1140)	3.011 (.0816)	.1387	.1394

variability and correlated measurement errors. This brings us back to the question of estimating a distribution in the presence of measurement error, which was addressed in Section 10.2.3, albeit in a simpler context than here.)

The calibration data were given in terms of standardized counts and working with those the assumption of a constant variance, τ_m^2 for method m, over concentrations appear reasonable. This is often not the case if one is working with raw, unstandardized counts, where the variance is often proportional to the mean. There is one other little twist here in that the raw value in Table 10.9 is actually the average of two determinations. This information could have been exploited using the two replicates. We did not do that in this analysis so as to not further complicate the issue, but use the fact that the variance of a count from the main study, using method m is $\tau_m^2/2$, where τ_m^2 is the variance of a single value in the calibration data for method m.

The linear model for the true values is a simple two sample model, with μ_+ and μ_- denoting the true means for the HIV positive and negative groups, respectively. Table 10.13 shows the analysis from the pseudo-MLE approach and from a simple two sample analysis of the adjusted values with (SE) denoting estimated standard error. The quantity SE(Diff) is the estimated standard error of $\hat{\mu}_+ - \hat{\mu}_-$. The confidence intervals for the difference were based on a normal approximation for the PML and using a t with 27 degrees of freedom for the adjusted values. In this example, the magnitude of the measurement error is such that the estimated means using adjusted values are very close to the pseudo-MLE's. Notice that the naive estimator of the variance tends to be too big and the naive standard errors for these means (and hence for the differences), which do not have the extra term to account for uncertainty from estimating the measurement error parameters, are too small.

10.4.4 Further detail

This section describes some of the details associated with correcting for measurement error.

Define $\dot{Y}_i = g^{-1}(D_i, \boldsymbol{\alpha}_{m(i)})$. This is like an adjusted value except that it uses the true measurement error parameters rather than the estimated ones. Suppose

$$E(\dot{Y}_i) = \mu_i + \delta_i \text{ and } V(\dot{Y}_i) = \sigma^2 + c_i.$$

In the linear case, with $g(y, \alpha_m) = \alpha_{0m} + \alpha_{1m}y$, $\delta_i = 0$ and $V(\dot{Y}_i) = \sigma^2 + (\tau^2/\alpha^2_{1m(i)})$. In other cases δ_i and c_i usually need to be approximated. If δ_i is negligible then using a first order approximation

$$c_i \approx h_i^2 V(v(Y_i, \boldsymbol{\alpha}_{m(i)}, \boldsymbol{\tau}_{m(i)})), \tag{10.16}$$

where $h_i = \partial g^{-1}(D, \boldsymbol{\alpha}_{m(i)})/\partial D$ evaluated at $D = g(\mu_i, \boldsymbol{\alpha}_{m(i)})$.

A1. The asymptotic bias of $\widehat{\boldsymbol{\beta}}_A$ is $\mathbf{B} = Lim_{n\to\infty}(\mathbf{X}_n'\mathbf{X}_n)^{-1}\mathbf{X}_n'\boldsymbol{\delta}_n$, where $\boldsymbol{\delta}_n' = (\delta_1, \ldots, \delta_n)$.

A2. If the δ_i are negligible then the approximate bias of the naive estimator of σ^2 is $\sum_i c_i/n$, where one approximation to c_i is given in (10.16). A corrected estimator of σ^2 is

$$\widehat{\sigma}_c^2 = \widehat{\sigma}_A^2 - \sum_i \widehat{c}_i/n.$$

A3. The covariance of $\widehat{\boldsymbol{\beta}}_A$ is $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'Cov(\widehat{\mathbf{Y}})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$. Since $Cov(\widehat{\mathbf{Y}})$ cannot be written down exactly, an approximation is needed. Assuming the bias in $\widehat{\boldsymbol{\beta}}_A$ is negligible, the approximate covariance is

$$\boldsymbol{\Sigma}_A = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{V}_2, \tag{10.17}$$

where $\mathbf{V}_2 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{C} + \mathbf{F})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$, $\mathbf{C} = diag(c_1, \ldots, c_n)$.

The matrix $\mathbf{F}$ is a $n \times n$ matrix computed in the following way. Define $\mathbf{b}_i = \partial g^{-1}(D, \boldsymbol{\alpha}_{m(i)})/\partial \boldsymbol{\alpha}_{m(i)}$ evaluated at $D = g(\mu_i, \boldsymbol{\alpha}_{m(i)})$. Letting $\boldsymbol{\Gamma}_m$ denote the approximate covariance of $\widehat{\boldsymbol{\alpha}}_m$, the elements of $\mathbf{F}$ are given by $f_{ii} = \mathbf{b}_i'\boldsymbol{\Gamma}_{m(i)}\mathbf{b}_i$, $f_{ij} = \mathbf{b}_i'\boldsymbol{\Gamma}_{m(i)}\mathbf{b}_j$, if $m(i) = m(j)$ and $f_{ij} = 0$, if $m(i) \neq m(j)$.

A4. An estimate of the covariance of $\widehat{\boldsymbol{\beta}}_A$ is given by

$$\widehat{\mathbf{\Sigma}}_A = \widehat{\sigma}^2_A(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{E} + \widehat{\mathbf{F}})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \tag{10.18}$$

where $\mathbf{E} = diag(\widehat{c}_1 - \bar{c}, \ldots, \widehat{c}_n - \bar{c})$, with $\bar{c} = \sum_i \widehat{c}_i/n$. The result has been given in a form that allows adding terms to the naive covariance matrix $\widehat{\sigma}^2_A(\mathbf{X}'\mathbf{X})^{-1}$ which is obtained from a standard regression analysis of the adjusted values.

If the δ_i are nonnegligible, a higher order approximation to c_i can be used leading to a modified corrected estimator of σ^2 and subsequently for $\mathbf{\Sigma}_A$.

Asymptotic properties of the MLE and pseudo-MLE.

The asymptotic properties of the full or pseudo-MLE can be obtained using the results in Section 6.17.4. In calculating the approximate covariance using (6.68), since the calibration data from the different methods are independent, $I_c(\boldsymbol{\theta})^{-1}$ is block diagonal $(\mathbf{\Gamma}_1, \ldots \mathbf{\Gamma}_M)$, where $\mathbf{\Gamma}_m$, is the approximate covariance of $\widehat{\boldsymbol{\alpha}}_m$. See Seber and Wild (1989, p.33) for $\mathbf{\Gamma}_m$ under normality. The second piece of (6.68) is the contribution due to uncertainty in the estimated measurement error parameters.

CHAPTER 11

Mixed/Longitudinal Models

11.1 Introduction, overview and some examples

All of the treatment to this point has assumed that the true responses are independent, or at least uncorrelated. Here we turn our attention to mixed or longitudinal models where multiple observations are associated with a main "unit," leading to correlation among observations. A unit may be an individual person, plant, etc. or a cluster, such as a hospital or a school. For convenience, the unit is referred to generically as a subject. There is a huge literature on treating such models, which comes under various names, with connections to hierarchical, structural, multilevel and latent variable models. A few representative texts are Davidian and Giltian (1995), Demidenko (2004), Diggle et al. (1994), Fitzmaurice et al. (2004), McCulloch and Searle (2001) and Skrondal and Rabe-Hesketh (2004).

The general data layout, displayed in terms of the response, is:

	Within subject				
Subject	1	.	j	.	.
1	Y_{11}	.	Y_{1j}	.	Y_{1m_1}
.	.	.	.	.	.
i	Y_{i1}	.	Y_{ij}	.	Y_{im_i}
.	.	.	.	.	.
n	Y_{n1}	.	Y_{nj}	.	Y_{nm_n}

where the vector of responses for subject i ($i = 1, \ldots n$) is denoted by

$$\mathbf{Y}_i' = (Y_{i1}, \ldots, Y_{im_i}).$$

A key component of mixed models is a vector of random effects $\mathbf{b}_i$, associated with the ith subject, where

$$E(\mathbf{b}_i) = \mathbf{0} \text{ and } Cov(\mathbf{b}_i) = \mathbf{\Sigma}.$$

A very general, possibly nonlinear, mixed model has the form

$$E(Y_{ij}|\mathbf{x}_{ij}, \mathbf{z}_{ij}, \mathbf{b}_i) = m(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \mathbf{b}_i, \boldsymbol{\beta}) \text{ and } Cov(\mathbf{Y}_i|\mathbf{X}_i, \mathbf{Z}_i, \mathbf{b}_i) = \boldsymbol{\Sigma}_{\epsilon i}, \tag{11.1}$$

where $\mathbf{x}_{ij}$ and $\mathbf{z}_{ij}$ contain predictors/covariate and $\mathbf{X}_i$ and $\mathbf{Z}_i$ have jth row equal to $\mathbf{x}'_{ij}$ and $\mathbf{z}'_{ij}$, respectively. With a generalized linear mixed model, $m = g^{-1}$ where g is a link function such that

$$g(E(Y_{ij}|\mathbf{x}_{ij}, \mathbf{z}_{ij}, \mathbf{b}_i)) = \mathbf{x}'_{ij}\boldsymbol{\beta} + \mathbf{z}'_{ij}\mathbf{b}_i.$$

In the linear mixed model g is the identity function and the model can be written as

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \boldsymbol{\epsilon}_i,$$

where $E(\boldsymbol{\epsilon}_i) = \mathbf{0}$ and $Cov(\boldsymbol{\epsilon}_i) = \boldsymbol{\Sigma}_{\epsilon i}$ and unconditionally

$$E(\mathbf{Y}_i) = \mathbf{X}_i\boldsymbol{\beta} \text{ and } Cov(\mathbf{Y}_i) = \mathbf{Z}'_i\boldsymbol{\Sigma}\mathbf{Z}_i + \boldsymbol{\Sigma}_{\epsilon i}.$$

Other links of particular interest include the logit, probit and log links.

The fixed effects part of the model is $\mathbf{X}_i\boldsymbol{\beta}$, where $\boldsymbol{\beta}$ is a fixed vector of parameters. The vector $\mathbf{b}_i$ consists of random "effects," with mean $\mathbf{0}$, associated with the ith subject. There may be common components in $\mathbf{X}$ and $\mathbf{Z}$. In fact for the random coefficients model discussed in the next section, $\mathbf{X}_i = \mathbf{Z}_i$. The covariance matrix $\boldsymbol{\Sigma}_{\epsilon i}$ is the within subject covariance matrix, i.e., given $\mathbf{X}_i$ and the subject specific random effects in $\mathbf{b}_i$. This might come from spatial or temporal factors within a subject (e.g., autoregressive errors for observations taken over time on a subject). In many mixed model applications with a quantitative outcome, the within subject covariance matrix, $\boldsymbol{\Sigma}_{\epsilon i}$, is taken to be $\sigma^2\mathbf{I}_{m_i}$ where $\mathbf{I}_{m_i}$ is the $m_i \times m_i$ identity matrix. Hence σ^2 represents a within subject variance. For generalized linear mixed models such as those involving logistic or Poisson regression, the diagonal terms of $\boldsymbol{\Sigma}_{\epsilon_i}$ will be functions of $\mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i$.

With measurement error the model can be decomposed further into

$$\mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i = \mathbf{X}_{i1}\boldsymbol{\beta}_1 + \mathbf{X}_{i2}\boldsymbol{\beta}_2 + \mathbf{Z}_{i1}\mathbf{b}_{i1} + \mathbf{Z}_{i2}\mathbf{b}_{i2},$$

where $\mathbf{X}_{i2}$ and $\mathbf{Z}_{i2}$ are observed exactly while $\mathbf{X}_{i1}$ and $\mathbf{Z}_{i1}$ are subject to measurement error.

The work on measurement error in mixed models is newer than much of the work in earlier chapters and still somewhat evolving. Many of the tools described in Chapter 6 can be used in this setting, but besides being more difficult to handle notationally, the additional complexity of the model for true values leads to some new challenges and technical issues. In addition, while previous chapters focused heavily on regression coefficients, the variance parameters often become main objects of interest in mixed models. Given these issues and the desire to keep our treatment from becoming too advanced, the objectives in this chapter are narrower than in many of the earlier ones. The rest of this section presents a broad overview and describe a few examples, but instead of providing many details, we point the interested reader to some of the relevant literature. Sections 11.2 and 11.3 then treat some specific linear mixed models in a bit more detail in order to illustrate the main points. The first of these sections is primarily concerned with the impact of additive Berkson errors in designed experiments with repeated measures, while the latter looks at additive measurement error in linear mixed models. Section 11.2 also provides a look at settings where the measurement errors may be correlated either among or within subjects due to the manner in which "doses" are prepared.

As in earlier chapters we have two questions to answer. What are the effects of measurement error and how do we correct for it?

Assessing bias of naive estimators.

To assess bias, essentially three strategies were employed in our earlier chapters. These were: i) work with an explicit expression for the estimators, ii) work with the estimating equations, and iii) find a model in terms of the induced model, which if in the same form as the original model allows for bias assessment. With mixed models these strategies are not always as productive as they were in the standard regression setting. The first strategy can be employed exactly with certain linear models as seen in the following sections.

Working with the induced model assumes a structural model where the unobserved true values are random. For bias assessment this model has to be in exactly the same form as the original model. In some cases the mean part of the model will be preserved and the bias in naive estimates of the coefficients assessed. This happens in some linear and nonlinear mixed models; see, for example Wang et al. (1998) and the discussion in Chapter 11 of Carroll et al. (2006). Often, however, the mean structure is not preserved, even in some relatively simple linear models, as seen in Section 11.3. That section also illustrates that there is a difference between what part of the mean structure needs to be preserved for bias assessment and what is needed to use a regression calibration approach to correct for measurement error. The induced covariance structure may, or may not, be preserved, as is also seen in Section 11.3. If it is then bias in the naive estimators of the variance parameters can be assessed. In general if the induced model is not of the same form, assessment of the naive biases usually needs to be pursued via an analysis of the naive estimating equations. This is not always easy and usually some approximations are need. See Wang and Davidian (1996) for a treatment covering nonlinear models.

To illustrate some of the points above, consider the linear mixed model. If we stack up all of the response vectors into a large vector $\mathbf{Y}$, the model can be written as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$, where $\boldsymbol{\delta}$ (which has contributions from the $\mathbf{Z}_i\mathbf{b}_i$ and $\boldsymbol{\epsilon}_i$) has mean $\mathbf{0}$ and covariance $\boldsymbol{\Sigma}_\delta$. The terms in $\boldsymbol{\Sigma}_\delta$ are functions of $\boldsymbol{\Sigma}$ and the $\boldsymbol{\Sigma}_{\epsilon i}$'s. The unweighted least squares estimator of $\boldsymbol{\beta}$ is $\widehat{\boldsymbol{\beta}}_{LS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\mathbf{Y}$. Usually a generalized least squares estimator, $\widehat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}'\widehat{\boldsymbol{\Sigma}}_\delta^{-1}\mathbf{X})^{-1}\mathbf{X}'\widehat{\boldsymbol{\Sigma}}_\delta^{-1}\mathbf{Y}$, is used since this takes advantage of the covariance structure. The approximate behavior of $\widehat{\boldsymbol{\beta}}_{LS}$ can be attacked directly using the measurement error model, similar to what was done for multiple linear regression in Chapter 5. The GLS, which is also the maximum likelihood estimator under normality, is obtained iteratively and needs an estimate of $\boldsymbol{\Sigma}_\delta$. The estimate of $\boldsymbol{\Sigma}_\delta$ in turn is found by solving a set of estimating equations, making use of the previous stage estimator of $\boldsymbol{\beta}$. Suppose $\mathbf{W}$ is the measured version of the matrix X. *If* the induced model has $E(\mathbf{Y}|\mathbf{W}) = \mathbf{W}\boldsymbol{\beta}^*$ then the naive GLS, or OLS, estimators will be approximately estimating $\boldsymbol{\beta}^*$. This is true for $\widehat{\boldsymbol{\beta}}_{GLS}$ regardless of what structure is assumed in forming $\widehat{\boldsymbol{\Sigma}}_\delta$. Similar results would hold for generalized linear mixed models. It may be that $E(\mathbf{Y}|\mathbf{W}) = \mathbf{Q}\boldsymbol{\beta}$, where $\mathbf{Q}$ is some function of $\mathbf{W}$, but this cannot be written as $\mathbf{W}\boldsymbol{\beta}^*$. In this case, although we cannot directly assess bias a regression calibration approach to correcting can be used.

Correcting for measurement error.

There are some mixed models where we can correct for measurement error without additional data. This can happen, for example, when the measurement error is in a time-varying covariate with some restrictions put on the longitudinal model for the unobserved true values. This is discussed and illustrated in Section 11.3 for a linear mixed model and is also used by Li et al. (2005) in a longitudinal model with a mismeasured semicontinuous covariate. In this context, the analysis options include likelihood approaches, moment approaches and regression calibration.

If the measurement error parameters are estimated or treated as known, the primary strategies employed in earlier chapters can be used. These are regression calibration, likelihood methods, SIMEX and modification of the naive estimating equations. The last two methods can be applied in the functional case. The modified estimating equation approach has received limited attention in the mixed model setting where it can be difficult to obtain, or even approximate, the expectation of the naive estimating equations. SIMEX is straightforward with additive error and estimated measurement error variances and has the advantages of being adaptable to any naive estimating method. Wang et al. (1998) present some simulations assessing the performance of SIMEX and describe an application of it to a longitudinal setting based on the Framingham Heart Study. Lin and Carroll (2000) illustrate the use of SIMEX in correcting for additive error using nonparametric techniques in a longitudinal setting.

Similar to the general regression setting, regression calibration proceeds by using a model for the unobserved true values given the observed values in order to get imputed values for the missing true values or functions of them that enter into the mean function. This can be conditional on other perfectly measured predictors. It requires the induced model is in the same general form as the original model (at least approximately) with the unobserved true values replaced by their conditional expectation. This is used for linear mixed models in Section 11.3. A robust estimate of the covariance of $\widehat{\boldsymbol{\beta}}$ is often needed since the covariance of $\mathbf{Y}$ in the induced model is usually not of the same form as the original model. This still leaves the problem of estimating the variance parameters, which is usually not done correctly by regressing with the imputed values. Instead we could use likelihood methods under distributional assumptions, modify the naive estimating equations (difficult in general) or using a least squares approach based on residuals. See the next two sections for further illustration.

In general, with distributional assumptions, likelihood approaches can be taken in the structural case assuming all of the parameters are identifiable, either through additional data or restrictions on the model. This is straightforward in principle but can certainly pose computational challenges. Johnson et al. (2005) provide a nice example of the use of likelihood methods, in addition to SIMEX and regression calibration, in a nonlinear mixed model applied to modeling the relationship of biomarkers to occupational benzene exposure in Chinese workers.

There have been a number of recent examples dealing with measurement error in mixed models. Among others, these include: logistic regression with repeated binary measures of the occurrence of LVH (left ventricular hypertrophy) with mismeasured systolic blood pressure and additional covariates in the Framingham Heart Study (Wang et al. (1998)); the relationship of daily death rates to air pollution using a Poisson model, external studies and Bayesian methods (Dominici et al. (2000)); the relationship of HIV viral load to CD4+ cells in a longitudinal mixed effects model with varying coefficients, error in CD4+ counts and the use of replicates (Liang et al. (2003)); the Wisconsin Sleep Cohort Study (Li, Shao and Palta (2005)). These articles demonstrate a variety of models and correction methods and provide access to related literature.

Finally, we describe one more example in some detail. This example comes from Follman et al. (1999), motivated by the Diet Intervention of School Age Children (DISC) Study. They describe a general two-stage approach for handling repeated measures with a probit regression and additive measurement error. The data consists of approximately 600 children with high LDL cholesterol, with randomization of children to a dietary intervention. The interest in this part of the analysis was on whether the intervention might be lowering the children's intake of essential minerals and vitamins. There are repeated binary outcomes on children in years 0, 1 and 3. For the jth visit of the ith child:

- y_{ij} = 1 if child meets the RDA for zinc, 0 otherwise;
- x_{ij1} = fat intake ($\%$ calories from fat). This is measured with W_{ij1} = mean of three 24 hour recalls within a 2-week period of the visit. The measurement error here is treated as additive.
- $\mathbf{X}_{ij2}$ contains indicators for gender and treatment group (intervention or not). These are not subject to error.

The assumption is that Y_{ij} follows a generalized linear mixed model with a probit link function and the random components including random child effects and random time effects. Similar to what happened in the uncorrelated case (see equation (6.30)) the induced model with x_{ij1} replaced by W_{ij1} is another probit model. The authors then use a pseudo-approach, first using the replicates to estimate the parameters in the model for $\mathbf{X}_{ij}|\mathbf{W}_{ij}$, fixing these and then estimating $\boldsymbol{\beta}$, using a GEE (generalized estimating equation) approach. Standard errors were obtained using the bootstrap. The results appear in Table 11.1. As expected the coefficient for fat intake increases with the correction for measurement error, but there is also almost a tripling of the estimated treatment effect.

Table 11.1 *Analysis of Diet Intervention Study from Follman et al. (1999). Used with permission of the International Biometrics Society.*

Parameter	Naive (SE)	Corrected(SE)
Intercept	-.594 (.1960)	-1.022 (.2246)
Treatment	.077 (.0694)	.198 (.0872)
Gender	.071 (.0676)	.056 (.0709)
Fat	.028 (.00565)	.068 (.0130)

The remaining two sections address some linear mixed measurement error problems.

11.2 Additive Berkson error in designed repeated measures

As described earlier in the book, pure Berkson measurement error arises when there are fixed target values such as dose, temperature, pressure, etc. but the unobserved true values can differ from the target values. Here we consider designed repeated measures studies where each "subject" is exposed to different target "doses" of a predictor variable w, and some response Y is observed. The goal of such studies is typically to estimate the overall regression model for

Y on the true dose x and/or to estimate among and within subject variability. In the absence of measurement error the analyses usually proceed through a mixed model/random coefficient formulation; see for example Crowder and Hand (1990), Longford (1993) and Gumpertz and Pantula (1989) for the linear models and Davidian and Giltian (1995) for treatment of nonlinear models.

Nitrogen intake/balance example. To motivate the problem, we consider a study from Atinmo et al. (1988), which is representative of many studies examining the relationships between nitrogen intake (x) and nitrogen balance (Y) with the objective of assessing nitrogen requirement. This is a repeated measures study in which each subject is observed under four intake levels with supposed intake values of $w_1 = 48$, $w_2 = 72$, $w_3 = 96$ and $w_4 = 120$ mg N/kg body-weight per day. Population requirements are often defined in terms of the distribution of individual requirements. An individual's requirement is defined as that intake at which the expected balance is 0. If a linear model with intercept β_{0k} and slope β_{1k} is appropriate for subject k in the population then the requirement for that subject is $-\beta_{0k}/\beta_{1k}$. (This can be extended to accommodate quadratic or other nonlinear models.) The distribution of the individual requirements depends on the distribution of the β's over the population. Berkson error arises here since the actual intakes will differ from the target values for various reasons. There are many interesting issues surrounding the determination of requirements that are beyond the scope of the discussion here. The point is that population coefficients and their variance-covariance structure play a critical role and it is important to understand the consequences of using target values rather than true doses in estimating them.

The primary issue addressed in this section is how additive Berkson errors effect the naive analyses which use the target doses. We do touch briefly on correction methods at the end. An additional complication here, which has been side-stepped in earlier chapters, is that the Berkson errors may be correlated, either within or among subjects, depending on how the doses are prepared. Racine-Poon et al. (1991) and Higgins et al. (1998) first addressed this with correlated errors arising from serial dilution. In addition, the error variance may be related to the target value. Wang and Davidian (1996) provide a comprehensive treatment of the approximate biases of naive inferences in nonlinear repeated measures models, assuming the errors are uncorrelated but allow the variance to be a function of the target dose. As they noted, and Carroll et al. (2006) summarized, moderate errors may have a substantial impact on estimation of the variance parameters involved. The results below reinforce this conclusion in the linear setting allowing a variety of measurement error covariance structures.

The rest of this section is based primarily on Buonaccorsi and Lin (2002). They provide exact results about the performance of naive estimators when the dose-response relationship is either linear or quadratic. The quadratic model is important since it can approximate some nonlinear dose-response curves, but the results for the quadratic model are not given here. In most of our discussion w is a "dose," but the results might apply when w is time, which comes under the heading of growth curves (e.g., Ksihrsagar and Smith, 1995). The measurement error in this case arises from the actual time of observation (which is not recorded) not being the scheduled one; see Wang and Davidian (1996) for an example and Nummi (2000).

Y_{ij}, x_{ij} and w_{ij} denote the response, true dose and target dose, respectively, for the jth repeated measure (j = 1 to m_i) on subject i. Further, let

$$\mathbf{w}_i' = (w_{i1}, \ldots, w_{im_i}), \mathbf{x}_i' = (x_{i1}, \ldots, x_{im_i}) \text{ and } \boldsymbol{\epsilon}_i' = (\epsilon_{i1}, \ldots \epsilon_{im_i}).$$

In the absence of any additional covariates, we use the random coefficients model with

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{X}_i\mathbf{b}_i + \boldsymbol{\epsilon}_i = \mathbf{X}_i\boldsymbol{\beta}_i + \boldsymbol{\epsilon}_i, \tag{11.2}$$

where the $\boldsymbol{\epsilon}_i$ are uncorrelated with mean $\mathbf{0}$ and covariance $\sigma^2\mathbf{I}_{m_i}$ and the $\mathbf{b}_i$ are independent "random effects" with mean $\mathbf{0}$ and covariance $\boldsymbol{\Sigma}$. Equivalently, $\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_n$ are independent $q \times 1$ vectors of random coefficients with mean $\boldsymbol{\beta}$ and covariance $\boldsymbol{\Sigma}$, where $\boldsymbol{\beta}_i = \boldsymbol{\beta} + \mathbf{b}_i$. In general $\mathbf{X}_i$ is a design matrix depending on the x_{ij}'s through the specified dose-response model. Notice that in this model $\mathbf{Z}_i = \mathbf{X}_i$. The primary objective is inferences for any of $\boldsymbol{\beta}$, $\boldsymbol{\Sigma}$ or σ^2.

As noted above, we will only consider the simple linear regression model

$$Y_{ij}|x_{ij} = \beta_0 + \beta_1 x_{ij} + b_{0i} + b_{1i}x_{ij} + \epsilon_{ij}. \tag{11.3}$$

This leads to $\mathbf{X}_i$ being an $m_i \times 2$ matrix with jth equal to $(1,\ x_{ij})$, $\boldsymbol{\beta}' = (\beta_0, \beta_1)$, $\mathbf{b}_i' = (b_{i0}, b_{i1})$ and

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_0^2 = V(b_{i0}) & \sigma_{01} = cov(b_{i0}, b_{i1}) \\ \sigma_{01} & \sigma_1^2 = V(b_{i1}) \end{bmatrix}.$$

The variance components σ_0^2 and σ_1^2 represent among subject heterogeneity in that they are the variance of the subject specific coefficients.

The matrix $\mathbf{W}_i$ is constructed in the same manner as $\mathbf{X}_i$, but with w_{ij} in place of x_{ij}. In the special case of a balanced design each subject has the same target values, so $m_i = m$ and $\mathbf{w}_i' = (w_1, \ldots, w_m)$, the same for each i, and each $\mathbf{W}_i$ equals a common $\mathbf{W}$.

The Berkson errors are assumed additive so

$$\mathbf{X}_i = \mathbf{w}_i + \mathbf{e}_i, \tag{11.4}$$

where $\mathbf{X}_i$ denotes the random true doses occurring for subject i and $E(\mathbf{e}_i) = \mathbf{0}$. The most general model is

$$cov(\mathbf{e}_i) = \boldsymbol{\Sigma}_{ei}, i = 1 \text{ to } n, \text{ and } cov(\mathbf{e}_i, \mathbf{e}_k) = \boldsymbol{\Sigma}_{eik}, i \neq k,$$

which allows the Berkson errors to be correlated either among or within subjects. The presence of nonzero off-diagonal terms in $\mathbf{\Sigma}_{ei}$ corresponds to correlation of measurement errors within subjects, while nonzero $\mathbf{\Sigma}_{eik}$ corresponds to correlation of measurement errors among different subjects. As discussed by Racine-Poon et al. (1991) and Higgins et al. (1998) correlation of the errors over different doses results from the use of serial dilution. Correlated measurement errors over subjects can result when a "batch" of material is produced to meet a target dose and then subdivided in order to deliver the dose to different individuals. See the discussion in Fuller (1987, p. 82) for an example based on applying fertilizer in agricultural settings.

The bias results in the next section are totally general, but for further discussion and specific numerical illustrations, we discuss a few of the models used by Buonaccorsi and Lin (2002). These were labeled Models 1, 3a, 4a and 4b by them and we retain that labeling here.

Model 1: The measurement errors are uncorrelated among and within subjects, with constant variance. That is, $\mathbf{\Sigma}_{ei} = \sigma_e^2\mathbf{I}$, $i = 1$ to n, $\mathbf{\Sigma}_{eik} = \mathbf{0}$, for $i \neq k$.

Model 3a: With $g = -1/(n-1)$,

$$\mathbf{\Sigma}_{ei} = (\sigma_e^2 + \sigma_a^2)\mathbf{I} \text{ and } \mathbf{\Sigma}_{eik} = (\sigma_e^2 + g\sigma_a^2)\mathbf{I}, i \neq k.$$

This model comes from preparation of an overall batch of material with target dose w_j which is then subdivided to produce a dose for each of the n individuals. The variance in the overall batch value is σ_e^2 and σ_a^2 represents within batch variability in doses.

Model 4: Measurement error from serial dilution.

Here, the target values are $w_i = log(d_i)$ where $d_1, \ldots, d_m$ are original doses, in descending order, obtained through serial dilution. The largest dose $d_1$is obtained by a K step dilution from a stock solution and subsequent doses are 1/2 of the preceding one. X_j is the log of the true dosage. Racine-Poon et al. (1991) show that $X_j \approx w_j + e_j$ or $\mathbf{X} \approx \mathbf{w} + \mathbf{e}$, where $\mathbf{e}$ has mean $\mathbf{0}$ and $Cov(\mathbf{e}) = \tau^2\mathbf{C}$, and the (i, j) element of $\mathbf{C}$ is $c_{ij} = min(i, j) + (K - 1)$. Racine-Poon et al. (1991) and Higgins et al. (1998) address Berkson errors in fitting nonlinear radioassay models under this model.

Model 4a: If a separate dilution is carried out for each subject, then $\mathbf{e}_i$ is distinct for each subject, so $\mathbf{X}_i \approx \mathbf{w} + \mathbf{e}_i$, with $\mathbf{\Sigma}_{ei} = \tau^2\mathbf{C}$ and $\mathbf{\Sigma}_{eik} = 0, i \neq k$.

Model 4b: If a single dilution used and the material within a batch with a particular target dose is homogeneous, then each subject gets the same (error-prone) dose leading to

$$\mathbf{\Sigma}_{ei} = \tau^2\mathbf{C} \text{ and } \mathbf{\Sigma}_{eik} = \tau^2\mathbf{C}, i \neq k.$$

11.2.1 Bias in naive estimators

We first need to describe what the naive analysis is, for which a number of approaches are available. With no measurement error $\mathbf{W}_i = \mathbf{X}_i$ and $\hat{\boldsymbol{\beta}}_i = (\mathbf{W}_i'\mathbf{W}_i)^{-1}\mathbf{W}_i'\mathbf{Y}_i$ is the least squares estimate of the coefficients for subject i. Unbiased, moment-based estimators for the parameters are

$$\hat{\boldsymbol{\beta}} = \sum_i \widehat{\boldsymbol{\beta}}_i/n, \quad \hat{\sigma}^2 = \sum_i \hat{\sigma}_i^2/n \text{ and } \hat{\boldsymbol{\Sigma}} = \mathbf{S}_{bb} - \hat{\sigma}^2 \sum_i (\mathbf{W}_i'\mathbf{W}_i)^{-1}/n, \tag{11.5}$$

where $\hat{\sigma}_i^2 = (\mathbf{Y}_i - \mathbf{W}_i\widehat{\boldsymbol{\beta}}_i)'(\mathbf{Y}_i - \mathbf{W}_i\widehat{\boldsymbol{\beta}}_i)/(m_i - q)$ is the residual variance from the ith individual and $\mathbf{S}_{bb} = \sum_i(\widehat{\boldsymbol{\beta}}_i - \widehat{\boldsymbol{\beta}})(\widehat{\boldsymbol{\beta}}_i - \widehat{\boldsymbol{\beta}})'/(n-1)$. As shown by Gumpertz and Pantula (1989), $\mathbf{S}_{bb}/n$ is an unbiased estimator of $Cov(\widehat{\boldsymbol{\beta}})$, even when the $\boldsymbol{W}_i$ are unequal. For a balanced design with all $\mathbf{W}_i = \mathbf{W}$, $\widehat{\boldsymbol{\beta}}$ is the MLE of $\boldsymbol{\beta}$ and $\hat{\sigma}^2$ and $\widehat{\boldsymbol{\Sigma}}$ are restricted maximum likelihood (REML) estimators of the variance and covariance parameters.

With an unbalanced design, the naive inferences are typically obtained using a general mixed model/REML approach which leads to estimated variance parameters and a generalized least squares estimator for $\boldsymbol{\beta}$. Since the interest here is mainly on balanced designs, we focus on the properties of the naive estimators in (11.5). The properties of estimators with unbalanced designs can be attached through the estimating equations (similar to what is done in Section 6.8) or developed as a special case of the general treatment in Wang and Davidian (1996).

Incorporating the Berkson error leads to

$$\mathbf{Y}_i = \mathbf{W}_i\boldsymbol{\beta} + \mathbf{W}_i\mathbf{b}_i + \boldsymbol{\eta}_i, \tag{11.6}$$

where $\boldsymbol{\eta}_i = (\beta_1 + b_{i1})\mathbf{e}_i + \boldsymbol{\epsilon}_i$ has mean $\mathbf{0}$ and covariance $\boldsymbol{\Sigma}_{\eta_i} = \boldsymbol{\Sigma}_{e_i}(\beta_1^2 + \sigma_1^2) + \sigma^2 I$. This leads to $Cov(\mathbf{Y}_i) = \mathbf{W}_i\boldsymbol{\Sigma}\mathbf{W}_i' + \boldsymbol{\Sigma}_{\eta_i}$ and $Cov(\mathbf{Y}_i, \mathbf{Y}_k) = \beta_1^2\boldsymbol{\Sigma}_{eik}, i \neq k$. Even if all random vectors involved are normally distributed, the distribution of $\mathbf{Y}_i$ is not normal when there is measurement error. This is due to the term $b_{i1}\mathbf{e}_i$ which is not normally distributed.

Using this model and defining $\mathbf{H}_i = \mathbf{W}_i(\mathbf{W}_i'\mathbf{W}_i)^{-1}\mathbf{W}_i'$ and $\mathbf{Q}_i = (\mathbf{W}_i'\mathbf{W}_i)^{-1}\mathbf{W}_i'$ we have the following exact results for the simple linear regression model:

1. $E(\widehat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$.
   This says that the naive estimator of the coefficients is unbiased. This is true regardless of the covariance structure of the Berkson errors.
2. $E(\hat{\sigma}^2) = \sigma^2 + B(\sigma^2)$ where the bias is

$$B(\sigma^2) = \frac{\beta_1^2 + \sigma_1^2}{n} \sum_{i=1}^{n} \frac{trace((\mathbf{I} - \mathbf{H}_i)\boldsymbol{\Sigma}_{ei})}{m_i - 2}, \tag{11.7}$$

This indicates positive bias in estimating σ^2.

3. $E(\hat{\mathbf{\Sigma}}) = \mathbf{\Sigma} + B(\mathbf{\Sigma})$, where the bias is given by

$$\begin{aligned} B(\mathbf{\Sigma}) &= \frac{\beta_1^2 + \sigma_1^2}{n} \\ &\quad \times \left[\sum_i \mathbf{Q}_i \mathbf{\Sigma}_{ei} \mathbf{Q}_i' - \frac{\sum_i (\mathbf{W}_i'\mathbf{W}_i)^{-1}}{n} \sum_i \frac{trace[(\mathbf{I} - \mathbf{H}_i)\mathbf{\Sigma}_{ei}]}{m_i - 2} \right] \\ &\quad - \frac{\beta_1^2}{n(n-1)} \sum_i \sum_{k \neq i} \mathbf{Q}_i \mathbf{\Sigma}_{eik} \mathbf{Q}_k'. \end{aligned} \tag{11.8}$$

The size and direction of the biases in the estimate of $\mathbf{\Sigma}$ depend on the error model. Some variance components may be underestimated and in others overestimated.

4. $Cov(\hat{\boldsymbol{\beta}}) =$

$$\frac{\mathbf{\Sigma}}{n} + \frac{\beta_1^2 + \sigma_1^2}{n^2} \sum_i \mathbf{Q}_i \mathbf{\Sigma}_{ei} \mathbf{Q}_i' + \frac{\sigma^2}{n^2} \sum_i (\mathbf{W}_i'\mathbf{W}_i)^{-1} + \frac{\beta_1^2}{n^2} \sum_i \sum_{k \neq i} \mathbf{Q}_i \mathbf{\Sigma}_{eik} \mathbf{Q}_k'. \tag{11.9}$$

5. The bias in $\mathbf{S}_{bb}/n$ as an estimator of $Cov(\hat{\boldsymbol{\beta}})$ is

$$E(\mathbf{S}_{bb}/n) - Cov(\hat{\boldsymbol{\beta}}) = -\frac{\beta_1^2}{n(n-1)} \sum_i \sum_{k \neq i} \mathbf{Q}_i \mathbf{\Sigma}_{eik} \mathbf{Q}_k'.$$

This provides the bias in the naive estimator of $Cov(\widehat{\boldsymbol{\beta}})$. An important consequence here is that:

As long as there is no correlation among measurement errors for different subjects then $\mathbf{S}_{bb}/n$ is unbiased for $Cov(\widehat{\boldsymbol{\beta}})$ and naive tests and confidence intervals for $\boldsymbol{\beta}$ are approximately correct.

This holds regardless of the covariance structure of the Berkson errors within a subject. On the other hand, if there is correlation in measurement errors among subjects these inferences are no longer valid. This can potentially be a serious problem.

Further comments can be made about particular models.

Model 1: In this case the model in (11.6) has the same structure as the original model (11.2), $\boldsymbol{\eta}_i$ replacing $\boldsymbol{\epsilon}_i$ $\sigma_\eta^2 = \sigma_u^2(\beta_1^2 + \sigma_1^2) + \sigma^2$ in place of σ^2. This means that the naive estimators for $\boldsymbol{\beta}$ and $\mathbf{\Sigma}$ and associated inferences are "correct" in the sense of having exactly the same properties as they would with no measurement error. The naive estimator of σ^2 is estimating σ_η^2 and hence is biased upward by the amount $\sigma_u^2(\beta_1^2 + \sigma_1^2)$. With unbalanced designs the ML/REML estimators of $\boldsymbol{\beta}$ and $\boldsymbol{\Sigma}$ are still consistent, even though the $\mathbf{Y}_i$ are not normally distributed. Any inferences which are asymptotically correct under (11.2) without assuming normality will still be correct (asymptotically) using the analysis based on the w's. In particular the standard confidence intervals and tests for $\boldsymbol{\beta}$ are okay.

Model 3a: The bias in $\hat{\sigma}^2$ is $B(\sigma^2) = (\beta_1^2 + \sigma_1^2)\sigma_u^2$, where $\sigma_u^2 = \sigma_e^2 + \sigma_a^2$. Recalling that the design is balanced, the bias in $\hat{\boldsymbol{\Sigma}}$ is $B(\boldsymbol{\Sigma}) = -\beta_1^2(\sigma_e^2 + g\sigma_a^2)(\mathbf{W}'\mathbf{W})^{-1}$, which is also the bias in $\mathbf{S}_{bb}/n$ as an estimator of $Cov(\hat{\boldsymbol{\beta}})$. Hence, the naive approach leads to bias in the estimate of $\boldsymbol{\Sigma}$ and a biased estimator of the covariance of the estimated coefficients. The latter also leads to incorrect inferences for the coefficients. Since $(\mathbf{W}'\mathbf{W})^{-1}$ is positive definite, the biases associated with the diagonal elements are negative leading to *underestimation* of the variance components and estimated standard errors of the coefficients which are generally too small. These biases do not change with n but decrease as $(\mathbf{W}'\mathbf{W})^{-1}$ is made smaller. The bias in the estimate of the variance of the slope as well as the bias in the naive estimate of the variance of the estimated slope is minimized by minimizing $\sum_j (w_j - \bar{w})^2$. This is achieved with a two point design taking the two distinct w values as far apart as possible, although this design is not usually employed since it prohibits assessment of the linearity assumption.

More generally for any balanced designs with a common $\boldsymbol{\Sigma}_{\mathbf{eik}}$ for each $i \neq k$ (as is true in model 3a), as n gets big $Cov(\hat{\boldsymbol{\beta}})$ converges to $\beta_1^2\mathbf{Q}\boldsymbol{\Sigma}_{eik}\mathbf{Q}'$ while the bias in $\mathbf{S}_{bb}/n$ is $-\beta_1^2\mathbf{Q}\boldsymbol{\Sigma}_{eik}\mathbf{Q}'$. That is, the asymptotic bias in $\mathbf{S}_{bb}/n$ as an estimator of $Cov(\hat{\boldsymbol{\beta}})$ is of the same size as $Cov(\hat{\boldsymbol{\beta}})$ itself. In this case, inferences for $\boldsymbol{\beta}$ can be grossly incorrect. However, as n gets large model 3a would not apply. Instead there would be multiple batches used, rather than one large batch and this would alter the covariance structure of the Berkson errors.

We illustrate these results using some parametric values based on the nitrogen intake/balance example with target intake values of $w_1 = 48$, $w_2 = 72$, $w_3 = 96$ and $w_4 = 120$. The parameters are set to $\beta_0 = -60$, $\beta_1 = .5$, $\sigma^2 = 25$, $\sigma_0^2 = 100$, $\sigma_{01} = -.5(10)\sigma_1$ and the variance of the slope is $\sigma_1^2 =$ .001, .01 or .1. The covariance term σ_{01} comes from allowing a correlation of -.5 between the random intercept and random slope. For simplicity assume $\sigma_a^2 = 0$ so there is no within batch variability. For model 3a, the error at a particular dose is common to all subjects with constant variance σ_e^2. Table 11.2 provides biases in the estimated variance parameters with σ_e = 2, 4 and 6. As expected from the theory σ^2 is overestimated while the two variance components corresponding to the random coefficients are underestimated. As noted above, $\mathbf{S}_{bb}/n$ is a biased estimator of $Cov(\hat{\boldsymbol{\beta}})$. Table 11.3 shows the standard deviation of $\widehat{\beta_1}$, denoted $\sigma(\widehat{\beta_1})$, and the approximate expected value of the naive estimator of it in the case where $\sigma_e = 4$. The resulting underestimation of the standard error of $\hat{\beta}_1$, which is fairly severe for $n = 100$ and 1000, leads

Table 11.2 *Model 3a: Biases in estimates of variance/covariance parameters.*

σ_1^2	σ_e	$B(\sigma^2)$	$B(\sigma_0^2)$	$B(\sigma_{01})$	$B(\sigma_1^2)$
0.001	2	1.0	-2.70	.0292	-.0003
0.010	2	1.0	-2.70	.0292	-.0003
0.100	2	1.4	-2.70	.0292	-.0003
0.001	4	4.0	-10.80	.1167	-.0014
0.010	4	4.2	-10.80	.1167	-.0014
0.100	4	5.6	-10.80	.1167	-.0014
0.001	6	9.0	-24.30	.2625	-.0031
0.010	6	9.4	-24.30	.2625	-.0031
0.100	6	12.6	-24.30	.2625	-.0031

Table 11.3 *Model 3a: $\sigma(\widehat{\beta}_1)$ = simulated standard deviation of $\widehat{\beta}_1$ and $\hat{\sigma}(\widehat{\beta}_1)$ = approximate expected value of naive estimate of standard deviation of $\widehat{\beta}_1$. Case with $\sigma_u = 4$.*

n	σ_1^2	$\sigma(\hat{\beta}_1)$	$\widehat{\sigma}(\hat{\beta}_1)$
10	0.001	0.049	0.032
10	0.010	0.057	0.04
10	0.100	0.111	0.104
100	0.001	0.039	0.01
100	0.010	0.04	0.014
100	0.100	0.05	0.033
1000	0.001	0.037	0
1000	0.010	0.037	0
1000	0.100	0.039	0.01

to confidence intervals that are too small and test of hypotheses that reject too often. As noted above, however, Model 3a is not very reasonable once n gets too large.

Model 4: Defining $T = tr((\mathbf{I} - \mathbf{H})\mathbf{C})$ and noting the design is balanced leads to $B(\sigma^2) = \tau^2(\beta_1^2 + \sigma_1^2)T/(m-2)$. Under Model 4a the bias in $\hat{\mathbf{\Sigma}}$ is $B(\mathbf{\Sigma}) = \tau^2(\beta_1^2 + \sigma_1^2)[\mathbf{QCQ}' - T(\mathbf{W}'\mathbf{W})^{-1}/(m-2)]$ and under Model 4b it is $B(\mathbf{\Sigma}) = \tau^2\sigma_1^2\mathbf{QCQ}' - \tau^2(\beta_1^2 + \sigma_1^2)T(\mathbf{W}'\mathbf{W})^{-1}/(m-2)$. The nature of the biases can be very different between models 4a and 4b, as seen in the numerical illustrations that follow. For Model 4a, $\mathbf{S}_{bb}/n$ is an unbiased estimator of $Cov(\hat{\boldsymbol{\beta}})$ and inferences for $\boldsymbol{\beta}$ are approximately correct. Under model 4b $\mathbf{S}_{bb}/n$ is biased by $-\beta_1^2\tau^2\mathbf{QCQ}'$.

To illustrate model 4 we use some parameters based on a portion of the data presented in Higgins et al. (1998). The response Y is a reaction rate (expressed

Table 11.4 *Measurement error standard deviations under model 4.*

	target w			
τ	4.4	3.7	3.0	2.3
.01	.022	.024	.026	.028
.03	.067	.073	.079	.084
.05	.112	.122	.132	.141

in units of change in optical density over change in time) and the dose is an antigen level. Our objectives are quite different than theirs and we only make use of their data to get reasonable values for the coefficients and the value of τ in the regression dilution model. We use the portion of the data corresponding to doses 10, 20, 40 and 80 with target values $w_1 = 2.303$ (i.e., log(10)), $w_2 = 2.996$, $w_3 = 3.689$ and $w_4 = 4.382$. We limited ourselves to this dose range, over which linearity of Y on log(dose) was reasonable. We take $\beta_0 = -279$ and $\beta_1 = 146$. In their simulations, Higgins et al. use a value of $\tau = .05$ with $K = 5$. Here we take $K = 5$ and vary τ over .01, .03 and .05. Table 11.4 shows the standard deviations in the measurement error for each target value and choice of τ. Depending on the application, the "subjects" over which we consider the random coefficients could denote different plates, different sequences within a plate, different technicians running the assay, etc. We set the residual standard deviation to $\sigma = 10$, the standard deviation in the intercepts to $\sigma_0 = 5$ and assume a correlation of $-.7$ between the intercept and slope, so $\sigma_{01} = -.7\sigma_1$. The standard deviation in slopes, σ_1, is taken to be either 1, 2 or 4.

The biases for the variance components appear in Table 11.5. The results are quite surprising in the magnitude of the biases. This is primarily due to the naive method ignoring the correlations induced by the serial dilution procedure. Under Model 4a the variances in the random coefficients are *overestimated* and the biases are quite large at even moderately sized measurement errors. For Model 4b on the other hand, the variances in the random coefficients are underestimated. Although the biases are not as large as those occurring in Model 4a, they are still quite serious. The bias in $\mathbf{S}_{bb}/n$ as estimator of $Cov(\hat{\boldsymbol{\beta}})$ is $-\beta_1^2\tau^2\mathbf{QCQ}'$. Table 11.6 displays the components of $Cov(\hat{\boldsymbol{\beta}})$ and associated biases for $\tau = .03$ as well as the standard deviation of the slope and the approximate expected value of the naive estimate of it. As in model 3, there can be serious problems with estimating the uncertainty in the estimated regression coefficients.

Table 11.5 *Biases in estimates of variance/covariance parameters for Models 4a and 4b.* $\sigma^2 = 100$, $\sigma_0^2 = 25$.

				Model 4a			Model 4b	
σ_1^2	τ	$B(\sigma^2)$	$B(\sigma_0^2)$	$B(\sigma_{01})$	$B(\sigma_1^2)$	$B(\sigma_0^2)$	$B(\sigma_{01})$	$B(\sigma_1^2)$
1	0.01	0.9	29.08	-5.39	1.154	-4.18	1.186	-.355
4	0.01	0.9	29.09	-5.39	1.154	-4.17	1.185	-.355
16	0.01	0.9	29.10	-5.40	1.154	-4.16	1.182	-.354
1	0.03	7.7	261.73	-48.5	10.38	-37.59	10.67	-3.19
4	0.03	7.7	261.77	-48.5	10.38	-37.55	10.67	-3.19
16	0.03	7.7	261.91	-48.6	10.39	-37.41	10.64	-3.19
1	0.05	21.3	727.03	-135	28.84	-104.4	29.65	-8.87
4	0.05	21.3	727.13	-135	28.84	-104.3	29.63	-8.87
16	0.05	21.3	727.54	-135	28.86	-103.9	29.56	-8.85

Table 11.6 *Model 4b: Components of* $Cov(\hat{\boldsymbol{\beta}})$ *and associated biases* (B_0, B_{01} *and* B_1) *from use of* $\mathbf{S}_{bb}/n$ *for* $\tau = .03$. $\hat{sd}(\hat{\beta}_1)$ = *approximate expected value of naive estimate of* $sd(\hat{\beta}_1)$.

n	σ_1^2	$V(\hat{\beta}_0)$	$cov(\hat{\beta}_0, \hat{\beta}_1)$	$V(\hat{\beta}_1)$	B_0	B_{01}	B_1
10	1.000	165.8	-45.8	17.84	-114.3	31.54	-13.58
10	4.000	165.8	-46.2	18.14	-114.3	31.54	-13.58
10	16.00	165.8	-46.9	19.34	-114.3	31.54	-13.58
100	1.000	119.5	-33.0	14.00	-114.3	31.54	-13.58
100	4.000	119.5	-33.0	14.03	-114.3	31.54	-13.58
100	16.00	119.5	-33.1	14.15	-114.3	31.54	-13.58
1000	1.000	114.8	-31.7	13.62	-114.3	31.54	-13.58
1000	4.000	114.8	-31.7	13.62	-114.3	31.54	-13.58
1000	16.00	114.8	-31.7	13.63	-114.3	31.54	-13.58

11.2.2 Correcting for measurement error

The main objective in this Berkson error setting was to characterize the bias of naive inferences, but we make some brief comments here on the ability to correct the deficiencies of the naive methods.

No additional data/information.

It is clear that without knowledge of the measurement error variances and covariances it is not possible to correct for biases in the naive estimators of $\boldsymbol{\Sigma}$ and σ^2. The other problem of interest is the bias in $\mathbf{S}_{bb}/n$, as an estimator of $Cov(\hat{\boldsymbol{\beta}})$ in the case when some of the $\boldsymbol{\Sigma}_{eik}$ are nonzero (e.g., models 3 and 4). This can usually be corrected even without any additional data. Consider the matrix $\mathbf{D} = \sum_i \sum_{k \neq i} \mathbf{r}_i \mathbf{r}_k'$, where $\mathbf{r}_i = \mathbf{Y}_i - \mathbf{W}_i \widehat{\boldsymbol{\beta}}_i$ is the vector of residuals for the ith subject, computed using that subject's estimated coefficients. If $\boldsymbol{\Sigma}_{eik}$ is of the form $\gamma \mathbf{C}_{ik}$ for some $\gamma > 0$ then the bias in $\mathbf{S}_{bb}/n$ is $-\beta_1^2 \gamma \sum_i \sum_{k \neq i} \mathbf{Q}_i \mathbf{C}_{ik} \mathbf{Q}_k'/(n(n-1))$ and $E(\mathbf{D}) = \beta_1^2 \gamma \sum_i \sum_{k \neq i} (\mathbf{I} - \mathbf{H}_i) C_{ik} (\mathbf{I} - \mathbf{H}_k)$. If the $\mathbf{C}_{ik}$ are known, as is often the case, then we can construct an estimator of $\beta_1^2 \gamma$ which can then be used to correct for the bias in $\mathbf{S}_{bb}/n$.

Known or estimated measurement error parameters.

Suppose the $\boldsymbol{\Sigma}_{ei}$ and $\boldsymbol{\Sigma}_{eik}$ are considered known. There are a couple of approaches that could be taken here. In a direct bias correction approach, we estimate the bias and then subtract this from the naive estimates to yield a corrected estimator.

Consider first estimating $Cov(\widehat{\boldsymbol{\beta}})$. If the $\boldsymbol{\Sigma}_{eik}$ all equal 0 then $\mathbf{S}_{bb}/n$ is unbiased for $Cov(\widehat{\boldsymbol{\beta}})$, so no correction is needed. With some $\boldsymbol{\Sigma}_{eik}$ nonzero for $i \neq k$, $E(\mathbf{r}_i \mathbf{r}_k') = (\mathbf{I} - \mathbf{H}_i) \beta_1^2 \boldsymbol{\Sigma}_{eik} (\mathbf{I} - \mathbf{H}_k)$. With $\boldsymbol{\Sigma}_{eik}$ known an unbiased estimator $\widehat{\beta_1^2}$ for β_1^2 can be constructed and $\mathbf{S}_{bb}/n - \widehat{\beta_1^2} \sum_i \sum_{k \neq i} \mathbf{Q}_i \boldsymbol{\Sigma}_{eik} \mathbf{Q}_k'/n(n-1)$ used as an estimator of $Cov(\hat{\boldsymbol{\beta}})$.

Getting corrected estimators for σ^2 or the components of $\boldsymbol{\Sigma}$ is a little more complicated since the bias involves σ_1^2. Letting c be the (2,2) element of

$$\left[\sum_i \mathbf{Q}_i \boldsymbol{\Sigma}_{ei} \mathbf{Q}_i' - \frac{\sum_i (\mathbf{W}_i' \mathbf{W}_i)^{-1}}{n} \sum_i \frac{trace[(\mathbf{I} - \mathbf{H}_i) \boldsymbol{\Sigma}_{ei}]}{m_i - 2} \right]$$

and d the (2,2) element of $\sum_i \sum_{k \neq i} \mathbf{Q}_i \boldsymbol{\Sigma}_{eik} \mathbf{Q}_k'/(n(n-1))$, then $E(\hat{\sigma}_1^2) = \sigma_1^2(1 + c/n) + \beta_1^2[(c/n) - d]$, where $\widehat{\sigma}_1^2$ is the naive estimator of σ_1^2. This suggests a corrected estimator for σ_1^2

$$\hat{\sigma}_{1c}^2 = \frac{\hat{\sigma}_1^2 - \widehat{\beta_1^2}[(c/n) - d]}{1 + (c/n)},$$

where $\widehat{\beta_1^2}$ is an estimate of β_1^2. Since $E(\hat{\beta}_1^2) = \beta_1^2 + V(\hat{\beta}_1)$, if all $\boldsymbol{\Sigma}_{eik} = 0$ we can use $\widehat{\beta_1^2} = \widehat{\beta}_1^2 - (2,2)$ element of $\mathbf{S}_{bb}/n$. With some $\boldsymbol{\Sigma}_{eik}$ nonzero an estimator of β_1^2 was given above under the discussion for no extra data. Using $\widehat{\beta_1^2} + \hat{\sigma}_{1c}^2$ to estimate $\beta_1^2 + \sigma_1^2$ we can now estimate the bias in the naive estimators of other components of $\boldsymbol{\Sigma}$ and σ^2.

Of course standard errors must be obtained for the bias corrected estimators. This can be attacked in the usual ways either through a combination of an estimating equation approach for the behavior of the naive estimators and the delta method, or through bootstrapping.

11.3 Additive measurement error in the linear mixed model

This section considers additive measurement error in a linear mixed model. The case of mismeasured variables in the random effects piece poses some difficult challenges; see Ganguli et al. (2005) for example. Among other things, one needs to deal with a product of random effects and random measurement errors. Here we will only discuss the case with no error in the variables entering into the random effects part of the model.

To simplify the notation, we will also work with the case where just one of the predictors, x_{ij1}, is subject to measurement error. So,

$$Y_{ij} = x_{ij1}\beta_1 + \mathbf{x}'_{ij2}\boldsymbol{\beta}_2 + \mathbf{z}'_{ij}\mathbf{b}_i + \epsilon_{ij},$$

or in matrix form

$$\mathbf{Y}_i|\mathbf{X}_i, \mathbf{b}_i = \beta_1\mathbf{x}_{i1} + \mathbf{X}_{i2}\boldsymbol{\beta}_2 + \mathbf{Z}_i\mathbf{b}_i + \epsilon_i, \tag{11.10}$$

where $\mathbf{X}_{i2}$ and $\mathbf{Z}_i$ are measured without error. The vector $\mathbf{x}_{i1}$ contains the true values of the mismeasured variable over repeated measures for subject i. The $\boldsymbol{\epsilon}_i$ are assumed to have $\mathbf{0}$ and covariance $\sigma^2_\epsilon\mathbf{I}$. The $\boldsymbol{\epsilon}_i$ and $\mathbf{b}_i$ are assumed independent of each other and independent over subjects with $Cov(\mathbf{b}_i) = \boldsymbol{\Sigma}$.

The measurement error for $\mathbf{x}_{i1}$ is assumed additive with

$$\mathbf{W}_i = \mathbf{x}_{i1} + \mathbf{u}_i. \tag{11.11}$$

$E(\mathbf{u}_i) = \mathbf{0}$, $Cov(\mathbf{u}_i) = \boldsymbol{\Sigma}_{ui}$ and $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are assumed independent.

If the mismeasured value does not change within a subject (e.g., in the longitudinal setting, x_{i1} is *not* a time varying covariate) then $\mathbf{x}_i = x_{i1}\mathbf{1}$, where x_{i1} is a scalar and $\mathbf{1}$ a vector of 1's. In this case if the error in x_{i1} has variance σ^2_{ui} then $\boldsymbol{\Sigma}_{ui} = \sigma^2_{ui}\mathbf{J}$, where $\mathbf{J}$ is a matrix of 1's.

11.3.1 Naive estimators and induced models

The general approaches to assessing bias were discussed in the introduction. Here we assume a structural model where the $\mathbf{X}_{i1}$ are independent with $E(\mathbf{X}_{i1}) = \boldsymbol{\mu}_{Xi}$ and $Cov(\mathbf{X}_{i1}) = \boldsymbol{\Sigma}_X$. We condition on $\mathbf{X}_{i2}$ which is why $\boldsymbol{\mu}_{Xi}$ may change with i. Unconditionally

$$E(\mathbf{W}_i) = \boldsymbol{\mu}_{Xi}, Cov(\mathbf{W}_i) = \boldsymbol{\Sigma}_W = \boldsymbol{\Sigma}_X + \boldsymbol{\Sigma}_u, \tag{11.12}$$

where, for now, we have assumed that $\boldsymbol{\Sigma}_{ui} = \boldsymbol{\Sigma}_u$. Under multivariate normality, this leads to the induced model

$$\begin{aligned} \mathbf{Y}_i \mid \mathbf{X}_{i2}, \mathbf{W}_i &= \mathbf{X}_{i2}\boldsymbol{\beta}_2 + \beta_1\mathbf{Q}_i + \boldsymbol{\epsilon}^*_i \qquad (11.13) \\ &= \mathbf{X}_{i2}\boldsymbol{\beta}_2 + \beta_1(\mathbf{I} - \boldsymbol{\Sigma}_X\boldsymbol{\Sigma}_W^{-1})\boldsymbol{\mu}_{Xi} + (\beta_1\boldsymbol{\Sigma}_X\boldsymbol{\Sigma}_W^{-1})\mathbf{W}_i + \boldsymbol{\epsilon}^*_i \end{aligned}$$

where $\mathbf{Q}_i = (\mathbf{I} - \mathbf{\Sigma}_X\mathbf{\Sigma}_W^{-1})\boldsymbol{\mu}_{Xi} + \mathbf{\Sigma}_X\mathbf{\Sigma}_W^{-1}\mathbf{W}_i$, and $\boldsymbol{\epsilon}_i^*$ has covariance

$$\mathbf{\Psi}_i = \mathbf{Z}_i\mathbf{\Sigma}\mathbf{Z}_i' + \sigma^2\mathbf{I} + \beta_1^2\mathbf{\Sigma}_X(\mathbf{I} - \mathbf{\Sigma}_W^{-1}\mathbf{\Sigma}_X). \tag{11.14}$$

Naive estimation refers to fitting (11.10) using $\mathbf{W}_i$ in place of $\mathbf{X}_{i1}$. It is clear from the induced model above that naive estimators of either $\boldsymbol{\beta}$ or the variance parameters will usually be biased. Often, however, we cannot identify the bias immediately since neither the fixed effects portion or the covariance structure is the same as in the original model. For the fixed effects part of the model to have the same form $E(\mathbf{Y}_i \mid \mathbf{X}_{i2}, \mathbf{W}_i)$ needs to be able to be expressed as $\mathbf{X}_{i2}\boldsymbol{\beta}_2^* + \beta_1^*\mathbf{W}_i$. This is clearly not always true. In particular, this would require that $\mathbf{\Sigma}_X\mathbf{\Sigma}_W^{-1}$ be proportional to the identity matrix. If this were true, the naive estimators are estimating $\boldsymbol{\beta}^*$. There are some cases where the covariance structure is preserved (see (11.18) for example) and the bias in the naive estimators of the variance parameters can be identified. In general the asymptotic biases for any of the naive estimators can also be examined through the estimating equations, an approach taken by Wang, Lin, Gutierrez and Carroll (1998). Their Sections 3.1 and 4.1 provide a bias analysis for special cases of our linear model, with $\mathbf{X}_{i2} = \mathbf{1}$ and $\mathbf{Z}_i = \mathbf{1}$.

In this section our emphasis is on correcting for measurement error, so details on the bias are not discussed further beyond a few points made in the following sections.

11.3.2 Correcting for measurement error with no additional data

This case is not as academic as it may look since with some measuring instruments it is impossible to obtain real replicates. Suppose for example the true value is a dietary intake over a narrow range of time, to be measured by a food questionnaire, record or diary. These instruments cannot generally be administered more than once over a short time period of time and treated as true replicates.

We have the induced model in (11.13) and the model for the $\mathbf{W}_i$ in (11.12) to work from. In the absence of any additional data, some restrictions must be employed to correct for measurement error. Paterno and Amemiya (1996) approach this using a general measurement error covariance $\mathbf{\Sigma}_{ui} = \mathbf{\Sigma}_u$ but with a fairly strong restriction on $E(\mathbf{X}_{i1})$. Here we describe the approach in Tosteson et al. (1998) and Buonaccorsi et al. (2000), which assumes a mixed model for the unobserved true values of the form

$$\mathbf{X}_{i1} = \mathbf{A}_i\boldsymbol{\alpha} + \mathbf{R}\boldsymbol{\phi}_i \tag{11.15}$$

where $\mathbf{A}_i$ is known $t \times r$ matrix, $\boldsymbol{\alpha}$ is an $r \times 1$ vector of parameters, $\mathbf{R}$ is a known $t \times q$ matrix of rank $q < t$ and $\boldsymbol{\phi}_i$ is a $q \times 1$ random effect with mean $\mathbf{0}$

and a nonsingular covariance $\boldsymbol{\Omega}_X$. Note that the assumption that $q < t$, which is critical, results in $\boldsymbol{\Sigma}_X = \mathbf{R}\boldsymbol{\Omega}_X\mathbf{R}'$ being singular. Also note that there is no additional error term. This model may be a bit strong in some settings, but it is one way to identify the parameters of interest without additional data. See the discussion in the example also. It can obviously be relaxed when there is additional information about the measurement error, as discussed in Section 11.3.3.

With (11.15) holding the model for $\mathbf{W}_i$ is

$$\mathbf{W}_i = \mathbf{A}_i\boldsymbol{\alpha} + \mathbf{R}\boldsymbol{\phi}_i + \mathbf{u}_i, \tag{11.16}$$

where we assume that $\boldsymbol{\Sigma}_{ui} = \sigma_u^2\mathbf{I}$, so the measurement errors are uncorrelated with constant variance. The $\boldsymbol{\alpha}$, σ_u^2 and $\boldsymbol{\Omega}_X$ can now be estimated from the W data.

The covariance of $\boldsymbol{\epsilon}_i^*$ in (11.13) can be written as

$$\boldsymbol{\Psi}_i = \mathbf{Z}_i\boldsymbol{\Sigma}\mathbf{Z}_i' + \mathbf{R}\boldsymbol{\Omega}_\tau\mathbf{R}' + \sigma_\epsilon^2\mathbf{I}, \tag{11.17}$$

where $\boldsymbol{\Omega}_\tau = \beta_1^2(\boldsymbol{\Omega}_X - \boldsymbol{\Omega}_X\mathbf{R}'\boldsymbol{\Sigma}_W^{-1}\mathbf{R}\boldsymbol{\Omega}_X)$.

If $\mathbf{Z}_i = \mathbf{Z} = \mathbf{R}$, then $\mathbf{Y}_i|\mathbf{X}_i = \mathbf{X}_{i2}\boldsymbol{\beta}_2 + \boldsymbol{\beta}_1\mathbf{Q}_i + \epsilon_i^*$ where

$$Cov(\boldsymbol{\epsilon}_i^*) = \boldsymbol{\Psi}_i = \mathbf{Z}\boldsymbol{\Sigma}^*\mathbf{Z}' + \sigma_\epsilon^2\mathbf{I}, \tag{11.18}$$

with $\boldsymbol{\Sigma}^* = \boldsymbol{\Sigma} + \boldsymbol{\Omega}_\tau$. In this case the covariance structure is the same as in the original model but with $\boldsymbol{\Omega}^*$ in place of $\boldsymbol{\Omega}$ and so a naive approach leads to overestimation of $\boldsymbol{\Omega}$.

Likelihood Methods. Let $\boldsymbol{\theta}_1$ contain the original parameters of interest ($\boldsymbol{\beta}$, the components of $\boldsymbol{\Sigma}$ and σ_ϵ^2) and let $\boldsymbol{\theta}_2$ contain the parameters in the model for $\mathbf{W}_i$ ($\boldsymbol{\alpha}$, $\boldsymbol{\Omega}_X$ and σ_u^2). Under normality $(\mathbf{Y}_i, \mathbf{W}_i)$ has likelihood $L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = f(\mathbf{y}_i|\mathbf{w}_i; \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)f(\mathbf{w}_i; \boldsymbol{\theta}_2)$, where $f(\mathbf{y}_i|\mathbf{w}_i; \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$ is the normal density based on (11.13) and $f(\mathbf{w}_i|\boldsymbol{\theta}_2)$ is the normal density for $\mathbf{W}_i$. Full maximum likelihood estimators maximize $L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$ in both $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ while pseudo maximum likelihood estimators maximize $L(\boldsymbol{\theta}_1, \widehat{\boldsymbol{\theta}}_2)$ where $\widehat{\boldsymbol{\theta}}_2$ is obtained using just the $\mathbf{W}_i$ data. The latter leads to fitting the model

$$\mathbf{Y}_i \mid \mathbf{W}_i \sim N(\mathbf{X}_{i2}\boldsymbol{\beta}_2 + \hat{\mathbf{Q}}_i\beta_1, \boldsymbol{\Psi}(\boldsymbol{\theta}_1, \widehat{\boldsymbol{\theta}}_2)), \tag{11.19}$$

where $\hat{\mathbf{Q}}_i = (\mathbf{I} - \hat{\boldsymbol{\Sigma}}_X\hat{\boldsymbol{\Sigma}}_W^{-1})\mathbf{A}_i\widehat{\boldsymbol{\alpha}} + \hat{\boldsymbol{\Sigma}}_X\hat{\boldsymbol{\Sigma}}_W^{-1}\mathbf{W}_i$ and $\boldsymbol{\Psi}(\boldsymbol{\theta}_1, \widehat{\boldsymbol{\theta}}_2) = \mathbf{Z}_i\boldsymbol{\Omega}\mathbf{Z}' + \sigma_\epsilon^2\mathbf{I} + \beta_1^2\hat{\boldsymbol{\Sigma}}_X(\mathbf{I} - \hat{\boldsymbol{\Sigma}}_W^{-1}\hat{\boldsymbol{\Sigma}}_X)$. The pseudo-ML estimators for both $\boldsymbol{\beta}$ and the variance parameters could be fit using standard mixed models software, if the structure in $\boldsymbol{\Psi}(\boldsymbol{\theta}_1, \hat{\boldsymbol{\theta}}_2)$ can be accommodated by the program.

Regression calibration approaches. The regression calibration approach is to fit $\mathbf{Y}_i = \mathbf{X}_{i2}\boldsymbol{\beta}_2 + \hat{\mathbf{Q}}_i\gamma + \boldsymbol{\epsilon}_i^*$ as a linear mixed model. Under normality, $\hat{\mathbf{Q}}_i$ is an estimate of $E(\mathbf{X}_{i1}|\mathbf{W}_i)$. Some covariance structure needs to be used in the fitting. If the exact form of $\mathbf{\Psi}$ can be used then this is equivalent to the pseudo-ML above and one could also make inferences for the variance parameters. If the mixed model software cannot accommodate the form of $\mathbf{\Psi}$, most mixed model software will produce an "empirical" or "sandwich" type estimate of the covariance of $\widehat{\boldsymbol{\beta}}$ which allows for the fact that we may have the covariance structure wrong. The estimated covariance of $\widehat{\boldsymbol{\beta}}$ from regression $\mathbf{Y}_i$ on $\mathbf{X}_{i2}$ and $\widehat{\mathbf{Q}}_i$ ignores the uncertainty arising from estimating $\boldsymbol{\theta}_2$.

With $\mathbf{Z}_i = \mathbf{R}$ the regression calibration estimators are the pseudo maximum likelihood estimators under normality, as long as the estimated variance parameters are such that the estimate of $\mathbf{\Sigma}$ is nonnegative. Buonaccorsi et al. (2000) provide detailed expressions for the asymptotic covariance of the full and pseudo-MLE's of both the coefficients and corrected estimators of variance in this case. The asymptotic properties are derived without the normality assumptions and so are robust. These are used in the example below.

Estimating the variance parameters.

Estimation of the variance terms depends on the form of $\mathbf{\Psi}$ and is more difficult.

For the special case of $\mathbf{Z}_i = \mathbf{R}$, the $\mathbf{\Psi}$ is as in (11.18) and the regression calibration approach under that covariance structure leads directly to an estimate of σ^2 and of $\mathbf{\Sigma}^*$, say $\widehat{\mathbf{\Sigma}}^*$. Using this, one estimate of $\mathbf{\Sigma}$ is

$$\widehat{\mathbf{\Sigma}} = \widehat{\mathbf{\Sigma}}^* - \hat{\beta}_1{}^2(\hat{\mathbf{\Omega}}_X - \hat{\mathbf{\Omega}}_X\mathbf{R}'\hat{\mathbf{\Sigma}}_W^{-1}\mathbf{R}\hat{\mathbf{\Omega}}_X).$$

When $\mathbf{Z}_i \neq \mathbf{R}$, estimating the variance components is less straightforward. Of course, under normality or other distribution assumptions, full or pseudo maximum likelihood can be used and the challenge is primarily computational. A moment based approach is to leave $\mathbf{\Psi}$ unstructured and then use a least squares approach based on treating $\widehat{\mathbf{\Psi}} - \widehat{\beta}_1^2\hat{\mathbf{\Sigma}}_X(\mathbf{I} - \hat{\mathbf{\Sigma}}_W^{-1}\hat{\mathbf{\Sigma}}_X)$ as if it were unbiased for $\mathbf{Z\Sigma Z}' + \sigma_\epsilon^2\mathbf{I}$. In the case where $\mathbf{X}_i = \mathbf{X}$ and $\mathbf{A}_i = \mathbf{A}$, the method of moments/nonlinear least squares approach in Section 4.2 of Fuller (1987) could be used.

Beta-carotene example.

Here we consider the beta-carotene data which were first introduced in Section 5.7 and then used again in Section 10.2. There are 158 individuals, each having six serum measures of beta carotene and six measures of beta-carotene intake from a food frequency questionnaire. The main goal is to build a regression model relating the log of the serum measure (Y) in a particular year to the corresponding true intake (x_1) defined to be the expected value of the log(FFQ) measure in that year. The true intake is unobserved and there is no replication in a year. For individual i at time $j = 1, 2, \ldots, 6$, the model used is

$$Y_{ij} = \beta_0 + \beta_1 x_{ij1} + \beta_2 x_{ij2} + b_{i1} + b_{i2} x_{ij2} + \epsilon_i,$$

where $x_{ij2} = j - 1$ is a time variable (ranging from 0 to 5) and b_{i1} and b_{i2} are random effects. The b_{i1} represents a random subject effect while b_{i2} allows a random time trend in serum level after conditioning on dietary intake. This could capture, for example, changes in the effect of diet on serum level as an individual ages. In equation (11.10), $\mathbf{X}_{i1}$ is a 6×1 vector of true intakes, measured by $\mathbf{W}_i$, and the random effects matrix $\mathbf{Z}_i = \mathbf{Z}$ is the same for all i,

$$\mathbf{Z}' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 & 5 \end{bmatrix}.$$

This also happens to be the $\mathbf{X}_{i2}$ matrix.

To identify the model,

$$\mathbf{X}_{i1} = \mathbf{R}\boldsymbol{\alpha} + \mathbf{R}\boldsymbol{\phi}_i$$

is used for (11.15), where $\mathbf{R}$ is taken to be the same as $\mathbf{Z}$ above. This implies an exact linear trend in "true" diet intake over time for each subject, with random coefficients over subjects (these are different random terms than the b_{i1} and b_{i2} in the earlier model). If only an intercept was included this would correspond to treating different years as replicates. The model here is richer than that, but might still be viewed as a bit restrictive. Tosteson et al. (1998) provide some justification for this being a reasonable approximation in this example. Fitting the longitudinal model $\mathbf{W}_i = \mathbf{R}\boldsymbol{\alpha} + \mathbf{R}\boldsymbol{\phi}_i + \mathbf{u}_i$ leads to $\hat{\boldsymbol{\alpha}}' = (1.2719, .0143)$, $\hat{\sigma}_u^2 = .119$ and

$$\hat{\boldsymbol{\Omega}}_X = \begin{bmatrix} .217 & -.011 \\ -.011 & .0044 \end{bmatrix}.$$

In this case, with $\mathbf{Z} = \mathbf{R}$, the pseudo-MLEs under normality are equivalent to the regression calibration estimators, taken to mean the RC estimates of $\boldsymbol{\beta}$ and associated corrected estimates of the variance parameters. The results appear in Table 11.7. The standard errors were obtained based on equation (18) in Buonaccorsi et al. (2000) which provides the asymptotic covariance of the estimators under normality. The most dramatic effect of the correction for measurement error here is on the estimate of β_1 which changes from .156 in the naive analysis to .454 when we correct for measurement error. The corrected estimates of σ_{11} and σ_{22} (the variance in subject effect and the variance in the random time coefficient, respectively) are smaller than the naive estimates with reductions of 9.4% and 25%, respectively. This is indicative of the fact that naive estimation of the variance components tends to produce overestimates.

Figure 11.1 plots the log(serum) value versus the log(diet) value and shows naive and corrected fits associated with time = 1 (1st year after baseline) and

Table 11.7 *Fitting of serum beta-carotene as function of diet intake and time Beta-carotene values are on log scale. Estimates with standard errors in parentheses. From Buonaccorsi et al. (2000).*

Parameter	PML/RC	Naive
$\widehat{\beta}_0$	4.73 (.11)	5.10 (.062)
$\widehat{\beta}_1$(diet)	.454 (.093)	.156 (.030)
$\widehat{\beta}_2$(time)	.0049 (.007)	.0091 (.007)
$\widehat{\sigma}_{11}$	.29 (.040)	.32 (.042)
$\widehat{\sigma}_{12}$	-.0053 (.0046))	-.0072 (.0046)
$\widehat{\sigma}_{22}$	.0018 (.00096))	.0024 (.00091)
$\widehat{\sigma}$	.097 (.0055)	.095 (.0054)

time = 5 (fifth year). The model allowed a term involving time, originally so the $\mathbf{Z}$ and $\mathbf{R}$ matrix are the same, which simplifies the computation. This figure shows that the time effect is minimal, but illustrates the large impact of the measurement

11.3.3 Correcting for measurement error with additional data

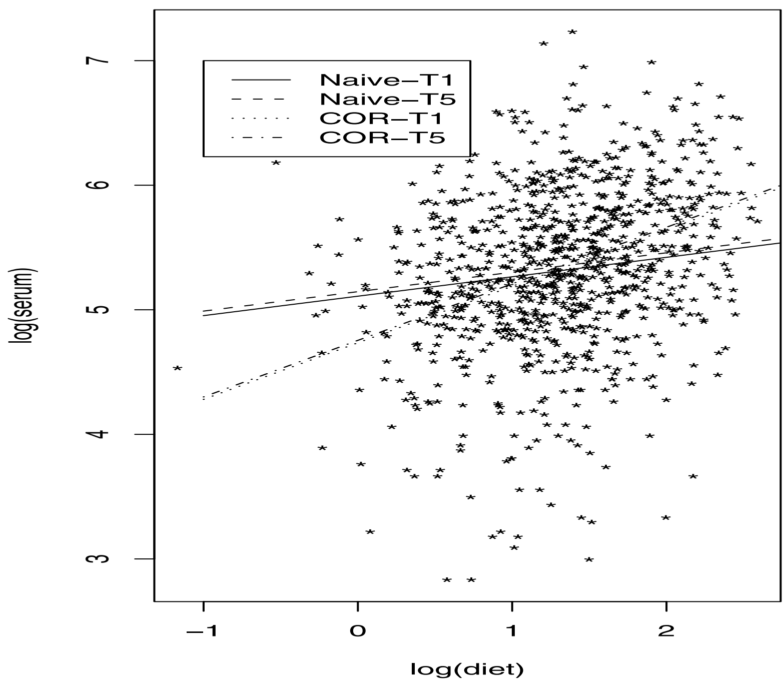
In the preceding section (11.15) was used to ensure that $\mathbf{\Sigma}_X$ was identifiable. This allowed σ_u^2 and $\mathbf{\Sigma}_X$ to both be estimated from the W data. Here we return to model for true values in (11.10) along with the measurement error model in (11.11) but assume there are data (e.g., replication) leading to estimates of the $\mathbf{\Sigma}_{ui}$'s.

Assuming normal measurement errors, SIMEX is easy to use, as discussed in the introduction. We can also use a pseudo-ML/regression calibration approach based on an induced model. Using the $\widehat{\mathbf{\Sigma}}_{ui}$'s, the $\mathbf{W}_i$ values and whatever model is given for $\mathbf{X}_{i1}$, estimates can be developed for $\boldsymbol{\mu}_{Xi}$ and $\mathbf{\Sigma}_X$. If the $\mathbf{X}_{i1}$ were assumed to be i.i.d, these would be $\widehat{\boldsymbol{\mu}}_{Xi} = \bar{\mathbf{W}}$ and $\widehat{\mathbf{\Sigma}}_X = \mathbf{S}_{WW} - (\sum_i \widehat{\mathbf{\Sigma}}_{ui}/n)$. Other models for $\mathbf{X}_{i1}$ could be used including allowing the mean $\boldsymbol{\mu}_{Xi}$ to depend on the additional predictors in $\mathbf{X}_2$. If we assumed that $\mathbf{Y}|\mathbf{X}_{i1}, \mathbf{W}_i$ is normally distributed with the model in (11.11) but with $\mathbf{\Sigma}_W$ replaced by $\mathbf{\Sigma}_{Wi} = \mathbf{\Sigma}_X + \mathbf{\Sigma}_{ui}$, then inferences for $\boldsymbol{\beta}$ can be carried out using the pseudo-ML/regression calibration estimator as described in the preceding section. This runs the regression with $\mathbf{X}_i$ replaced by

$$\hat{\mathbf{Q}}_i = (\mathbf{I} - \hat{\mathbf{\Sigma}}_X \hat{\mathbf{\Sigma}}_{Wi}^{-1})\widehat{\boldsymbol{\mu}}_{Xi} + \hat{\mathbf{\Sigma}}_X \hat{\mathbf{\Sigma}}_{Wi}^{-1} \mathbf{W}_i.$$

A robust estimator of $Cov(\widehat{\boldsymbol{\beta}})$ should be used, but this still does not account for uncertainty in the estimates of $\mathbf{\Sigma}_{ui}$. This can be done by bootstrapping, which can also be used to obtain the covariance of the SIMEX estimators. For

Figure 11.1 *Beta-carotene example. Plot of log(serum) versus log(diet) and naive and corrected lines corresponding to Time = 1 (T1) and Time = 5 (T5).*



estimating the variance parameters, either pseudo-likelihood or least squares approaches can be used.

The induced model used for PML/regression calibration relies on multivariate normality and a structural model. Without the resulting induced model some other strategy needs to be employed. This might include full maximum likelihood, incorporating the main study data and the replication, modifying the estimating equations (may be difficult) or, for estimating $\boldsymbol{\beta}$, a direct correction to the naive ordinary least squares estimator can be used. The latter is not necessarily very efficient but it is easy to implement. For example if we return to the general model with $E(\mathbf{Y}_i) = \mathbf{X}_i\boldsymbol{\beta}$, the naive OLS estimator of $\boldsymbol{\beta}$ is $\widehat{\boldsymbol{\beta}}_{naive} = (\sum_i \mathbf{W}_i'\mathbf{W}_i)^{-1}\sum_i \mathbf{W}_i\mathbf{Y}_i)$, where $\mathbf{W}_i$ is similar to $\mathbf{X}_i$ but with error-prone values replacing true values where needed. If the jth row of $\mathbf{W}_i$ is $\mathbf{W}_{ij}'$, this becomes $(\sum_i\sum_j \mathbf{W}_{ij}\mathbf{W}_{ij}')^{-1}\sum_i\sum_j \mathbf{W}_{ij}Y_{ij}$. If $\mathbf{W}_{ij} = \mathbf{x}_{ij} + \mathbf{u}_{ij}$ where $\mathbf{u}_{ij}$ has mean 0 and covariance $\boldsymbol{\Sigma}_{uij}$, then the corrected estimator is

$\widehat{\boldsymbol{\beta}} = (\sum_i \sum_j (\mathbf{W}_{ij}\mathbf{W}'_{ij} - \widehat{\boldsymbol{\Sigma}}_{uij}))^{-1} \sum_i \sum_j \mathbf{W}_{ij}Y_{ij}$. Unlike general multiple linear regression, we now need to account for the correlation structure among the Y_{ij}'s as well as possible correlations among the $\mathbf{u}_{ij}$ in obtaining $Cov(\widehat{\boldsymbol{\beta}})$ and an estimate of it.

CHAPTER 12

Time Series

12.1 Introduction

As with many other types of data, time series data are prone to measurement error. Examples include the mismeasuring of population abundances or densities, air pollution or temperatures levels, disease rates, medical indices (Scott and Smith, 1974, Scott, Smith and Jones, 1977), the labor force in Israel (Pfeffermann, Feder, and Signorelli, 1998) and retail sales figures (Bell and Hillmer, 1990, Bell and Wilcox, 1993).

The series of true values is denoted by $\{X_t\}$, where X_t is a random variable and t indexes time. We will work under the assumption that a specific dynamic model holds for the true values. The objective of the analysis may include estimation of model parameters, or functions of them, "prediction" of X_t for t in the range of observed data or forecasting X_t for t beyond range of observed data. We will not try to address the problem of model identification here.

As elsewhere, instead of the realized value x_t we observe the outcome of W_t. Typically W_t is an estimator of x_t and the error is additive sampling error, but there are cases where a bias term is included, especially when an index is used; see Section 12.2 for example. There are some issues here regarding the measurement error which, for the most part, did not arise in the standard regression contexts. One is the reason for the changing measurement error variance. In the time series context this results from changes in sampling effort over time (e.g., due to changes in funding) and/or from changes in the underlying population being sampled. The latter source often leads to the measurement error variance depending in part on the true value being estimated. This has been allowed in previous chapters, but here there is the added problem that the measurement errors are unconditionally correlated if they depend on the underlying X_t. This is true even if conditionally (given the true values) the measurement errors are uncorrelated. Further, the assumption of conditionally uncorrelated measurement errors is itself often violated due to common "units" being involved in estimating the true values at different time points. A good example of this is in block resampling techniques used in many large national surveys where households or individuals remain in the sample over multiple time periods. See Pfeffermann et al. (1998) or Bell and Wilcox (1993), for example.

The data that lead to W_t usually allow for finding $\widehat{\sigma}^2_{ut}$, an estimate of the conditional measurement error variance. Unfortunately many available time series do not include these measures of uncertainty, leading to the need for methods that correct for measurement error without this information, if possible.

As in the preceding chapter, the exposition here takes on more the form of a survey than earlier chapters in the book. The rest of this section provides a brief summary of work on measurement error in time series. Sections 12.2 and 12.3 then consider two specific classes of models, the random walk and linear autoregressive models, in a bit more detail. These will serve to illustrate the main modeling and methodological issues, and some of the remaining challenges, but even for these models we provide more of an overview. The latter section also touches on the use of additional predictors.

Beside the two set of models considered in some detail, other time series models where measurement error has been considered include autoregressive moving average (ARMA) models (Lee and Shin (1997), Wong et al. (2001), Koreisha and Fang (1999)), autoregressive integrated moving avergage (ARIMA) models (Wong and Miller (1990), Bell and Hilmer (1990), Bell and Wilcox (1993)), moving average (MA) models (Koons and Foutz (1990)) and basic structural models (Pfeffermann (1991), Pfeffermann et al. (1998), Feder (2001)). This list is meant to be representative, not exhaustive. While a large number of the papers mentioned later in treating autoregressive and random walk models assume additive measurement error with constant variance, some of the papers just listed do allow more complicated measurement error models. For example Bell and Hilmer (1990), Bell and Wilcox (1993), Pfeffermann (1991), Pfeffermann et al.(1998) and Scott and Smith (1974) and Scott et al. (1977) all allow either ARIMA, AR or ARMA models for the measurement error. These models can sometimes be fit separately first (e.g., based on the sampling design structure over time) and then treated as known in order to estimate the original model of interest through maximum likelihood estimation or some other technique. As we will also see in the later sections there are contexts in which the measurement error parameters and the original model of interest can be estimated simultaneously without additional data. The objective of a number of the papers listed above is with improved estimation of the current true value (x_t) (improved with respect to just using the estimate from the current sample) and forecasting future values. The focus of this chapter is on parameter estimation.

There has also been a flurry of recent work on nonlinear population dynamic models; see for example de Valpine, (2003), de Valpine and Hastings (2002), Calder et al. (2003) and Clark and Bjornstad (2004). Examples of the nonlinear models used include the Ricker's model with $X_t = X_{t-1} + \beta_0 + \beta_1 e^{X_{t-1}} + \epsilon_t$ and the Beaverton-Holt model with $X_t = X_{t-1} + \beta_0 + log(1 + \beta_1 e^{X_{t-1}}) + \epsilon_t$, where $X_t = log(N_t)$ and N_t is the abundance at time t. The majority of this work assumes the measurement error for X_t is normal with a constant variance, under which all of the parameters are identifiable. A combination of likelihood and Bayesian methods have been employed for inferences. Solow (1998) takes a different approach in fitting the nonlinear model $X_t = X_{t-1} + \beta_0 + log(N_t) + \epsilon_t$. The ϵ_t are assumed normal but the measurement error is additive on the abundance scale with an observed value $Z_t = cQ_t$ where given the abundance n_t, Q_t is Poisson with mean n_t/c and c is a known value, related to sampling effort. SIMEX was used for parameter estimation.

12.2 Random walk/population viability models

As noted above the assessment of population dynamics in the presence of measurement error (also called observation error in this context) has been a topic of great interest of late. Here we focus on the random walk with drift model

$$X_t = X_{t-1} + \mu + \epsilon_t. \tag{12.1}$$

The "process errors," the ϵ_t's, are assumed to be independent and identically normally distributed with mean 0 and variance σ^2. In this context $X_t = log(N_t)$ where N_t is the abundance (or density) of the population at time t. This model can also be written as $N_t|n_{t-1} = e^{\mu} n_{t-1}\delta_t$, where $\delta_t = e^{\epsilon_t}$ is distributed log-normal. So

$$E(N_t|n_{t-1}) = \lambda n_{t-1}, \tag{12.2}$$

where λ is the trend. Notice that if we condition on the first value n_0, then unconditionally $E(N_t) = n_0\lambda^t$, which is increasing for $\lambda > 1$ and decreasing for $\lambda < 1$. Under the normality assumption associated with (12.1), $\lambda = e^{\mu+(\sigma^2/2)}$. This is also referred to as the "finite rate of increase." One could drop the normality assumption and work directly with a nonlinear model for N_t, but methods for this are still under development.

This model, frequently employed in population viability analysis, is nonstationary, with a drift up or down as noted above. It has also been referred to as the diffusion approximation model (Dennis et al. (1991), Holmes (2004), Staples et al. (2004)) and is a stochastic Malthusian model. The goal is estimation of the basic parameters, μ and σ^2, and functions of them based on a series of estimated abundances. Dennis et al. (1991) and Morris and Doak (2002), among others, provide extensive discussion of this model and treat many of the statistical issues in the absence of observation error.

Data are collected at time points $t_0 < t_1 < t_2 < \ldots < t_q$, possibly unequally spaced, with X_i denoting the random abundance at time t_i. Then

$$D_i = X_i - X_{i-1} = \tau_i \mu + \epsilon_i \tag{12.3}$$

for $i = 1, \ldots q$, where $\tau_i = t_i - t_{i-1}$ is the time between observations $i-1$ and i. The ϵ_i are independent and normally distributed with mean 0 and variance $V(\epsilon_i) = \tau_i \sigma^2$.

Measurement error arises from estimating the abundance through some type of sampling. The estimated log-abundance at the ith collection time is W_i. We need to specify whether the model is specified in terms of the estimated abundance or log-abundance, whether the variance is changing over time and whether the measurement errors are correlated. Certain ones of these issues were addressed in Chapter 6, see in particular Sections 6.4.5 and 6.4.7, while others were discussed in the introduction to this chapter. A fuller discussion of the nature of the measurement errors in time series can also be found in Buonaccorsi et al. (2006) and Staudenmayer and Buonaccorsi (2006). The majority of the results and discussion in this section come from those papers plus Buonaccorsi and Staudenmayer (2009).

The assumed measurement error model is

$$W_i | x_i = m + x_i + u_i, \tag{12.4}$$

with

$$E(u_i) = 0, \quad V(u_i) = \sigma^2_{ui}, \quad Cov(u_i, u_j) = 0, \text{ for } i \neq j.$$

This allows a bias m in W_i as an estimator of the true log-abundance x_i. This would arise, for example, if W_i is the log of an index, such as unadjusted counts from trapping, aerial surveys, etc. This model also assumes the measurement errors are conditionally uncorrelated. Both of these assumptions can be relaxed, but we will not do so here in order to concentrate on the model which has been used in much of the previous work in this area.

The estimated difference in log-abundances is

$$\widehat{D}_i = W_i - W_{i-1} \tag{12.5}$$

for which

$$E(\widehat{D}_i) = \tau_i \mu, \quad V(\widehat{D}_i) = \tau_i \sigma^2 + \sigma^2_{ui} + \sigma^2_{u(i-1)}, \quad Cov(\widehat{D}_i, \widehat{D}_{i+1}) = -\sigma^2_{ui}, \tag{12.6}$$

and $Cov(\widehat{D}_i, \widehat{D}_j) = 0, \text{ if } |i - j| > 1$. These properties determine the behavior of naive analysis and also motivate the correction techniques.

12.2.1 Properties of naive analyses

The "naive" approaches use the estimated log-abundances as if they were the true log-abundances. Without measurement error the model is a standard linear model and the naive estimators are

$$\hat{\mu}_{naive} = \frac{\sum_{i=1}^{q} \widehat{D}_i}{t_q - t_0} \text{ and } \hat{\sigma}^2_{naive} = \frac{\sum_{i=1}^{q} (\widehat{D}_i \tau_i^{-1/2} - \tau_i^{1/2} \hat{\mu}_{naive})^2}{q-1}. \quad (12.7)$$

With equally spaced data these are just the sample mean and variance, respectively, of $\widehat{D}_1, \ldots, \widehat{D}_q$. Naive confidence intervals for μ and σ^2, which are exact with no measurement error and under normality, are computed based on the use of the t and chi-square distributions, respectively.

Numerous authors have documented the impacts of observation errors on naive analyses which ignore it, either through simulations or analytically; see for example, Ludwig (1999), Meier and Fagan (2000), Holmes (2001, 2004), and Holmes and Fagan (2002). Under the measurement error model specified following (12.4) we have the following exact results.

- There is no bias in the naive estimator of μ.
- $\widehat{\sigma}^2$ is biased upward with a bias of

$$b(\sigma^2) = \frac{1}{q-1}\left[\sum_{i=1}^{q} \frac{\sigma^2_{u,i-1} + \sigma^2_{ui}}{\tau_i} - \frac{\sigma^2_{u0} + \sigma^2_{uq}}{t_q - t_0}\right] > 0.$$

- The naive estimator of the trend parameter, $\widehat{\lambda} = e^{\widehat{\mu} + \widehat{\sigma}^2/2}$, is usually biased downward for λ, with a limiting value of $\lambda e^{b(\sigma^2)/2} > \lambda$. This is in contrast to our earlier results for simple linear regression where measurement error in the predictor usually lead to underestimation of the slope; see Chapter 4.

Buonaccorsi et al. (2006) provide extensive numerical illustrations and additional discussion of the biases in naive estimators.

12.2.2 Correcting for measurement error

There are two general sets of correction methods used here. One set does not use any estimated measurement error variances, while the others do. Within each of these both likelihood and moment based methods are available.

Correcting without estimated measurement error variances.

These methods have been the ones most commonly employed in the literature. They use only the estimated differences in log abundances from which all of μ, σ^2 and σ_u^2 can be estimated based just on the $\widehat{D}_i$'s, so long as the measurement error variance is constant.

Likelihood methods: If the D_i's and the measurement errors are normally distributed with constant measurement error variance, σ_u^2, then the $\widehat{D}_i$'s are also normally distributed and μ, σ^2 and σ_u^2 are all identifiable. Either maximum likelihood (ML) or restricted maximum likelihood (REML) estimation can be used, with REML being the preferred technique. Lindley (2003), Holmes (2004) and Staples et al. (2004) all used this approach for equally spaced data. As recognized by Staples et al. (2004) for equally spaced data, the model can be formulated as a linear mixed model and mixed models software can be exploited to carry out the analysis. This can be extended to handle unequally spaced data, including fitting via proc MIXED in SAS; see Buonaccorsi and Stuadenmayer (2009) for details and programs. These mixed model analyses automatically produce estimates, standard errors and confidence intervals for the three main parameters.

The REML estimators of the variance components are denoted $\widehat{\sigma}^2_{REML}$ and $\widehat{\sigma}^2_{u,REML}$. The associated estimator of the mean, $\widehat{\mu}_{REML}$, produced by most software is a generalized least squares (GLS) estimator (e.g., equation (12) in Staples et al. (2004) or (2.26) in Demidenko (2004)) which uses estimates of the two variance components. Approximate standard errors of these REML estimators are based on the large sample theory for maximum likelihood estimates; see Demidenko (2004) for example.

An approximate confidence interval for μ is given by $\widehat{\mu}_{REML} \pm t(1 - \alpha/2, d)SE(\widehat{\mu}_{REML})$ where d is an approximate degrees of freedom. There are various options for obtaining d. We use the Satterthwaite approach in our examples, although other choices can be made. An alternative for estimating μ is to use the naive estimator, $\sum_{i=1}^{q} \widehat{D}_i/(t_q - t_0)$, which we now denote $\widehat{\mu}_{mom}$, since it is a moment estimator. While asymptotically it may be less efficient than the weighted REML estimator, it has the advantage of being exactly unbiased for any sample size, while the REML estimator is not. The estimated standard error of $\widehat{\mu}_{mom}$ is

$$SE(\widehat{\mu}_{mom}) = \left[\frac{\widehat{\sigma}^2_{REML}}{t_q - t_0} + \frac{2\widehat{\sigma}^2_{u,REML}}{(t_q - t_0)^2}\right]^{1/2} \tag{12.8}$$

and the associated Wald confidence interval is $\widehat{\mu}_{mom} \pm z_{1-\alpha/2}SE(\widehat{\mu}_{mom})$.

There are multiple approaches available for obtaining confidence intervals on the variance components. Two, the Wald and chi-square based intervals, are usually available from standard mixed model analyses. To illustrate consider intervals for σ^2, with parallel results for σ_u^2. The first technique uses the Wald interval of the form $\hat{\sigma}^2_{REML} \pm z_{1-\alpha/2}SE(\hat{\sigma}^2_{REML})$. The second utilizes a chi-square approximation for the estimated variance leading to the interval

$$\left[\frac{v\hat{\sigma}^2_{REML}}{\chi^2_{1-\alpha/2,v}}, \frac{v\hat{\sigma}^2_{REML}}{\chi^2_{\alpha/2,v}}\right], \tag{12.9}$$

where $v = 2\widehat{\sigma}^4_{REML}/(SE(\hat{\sigma}^2_{REML}))^2$ is an estimated degrees of freedom. One or both of these intervals are usually produced by mixed model software. While chi-square based intervals are generally better in mixed models, in this context they often encounter problems due to small estimated degrees of freedom and can behave erratically, So, we favor the Wald intervals here.

Under the assumptions of normality and constant measurement error variance, one can also obtain confidence intervals using two more computationally intensive techniques: the parametric bootstrap or a profile-likelihood method. Dennis et al. (1991) utilized the bootstrap in the random walk model with no measurement error. To implement the bootstrap here, for $b = 1$ to B (large), the bth bootstrap sample, indexed by b, is generated by:

i) Generate true values. First set $X_{b0} = \widehat{X}_0$ which uses the estimated log-abundance at time 0 as the starting point for each bootstrap sample. The remaining true values are generated by $X_{bi} = X_{b,i-1} + \tau_i\widehat{\mu} + \epsilon_{bi}$ for $i = 1$ to q, where the ϵ_{bi} are independent normal with mean 0 and variance $\tau_i\widehat{\sigma}^2$.

ii) Generate error-prone measures: For $i = 0$ to q, $W_{bi} = X_{bi} + u_{bi}$, where the u_{bi} are i.i.d. normal with mean 0 and variance $\widehat{\sigma}^2_u$. Then $\widehat{D}_{bi} = W_{bi} - W_{b,i-1}$, for $i = 1$ to q.

Estimates are then computed for each of the B bootstrap samples, and standard errors and bootstrap percentile intervals obtained in the usual manner.

Under normality, profile-likelihood confidence intervals (e.g., Section 3.4 of Demidenko (2004)) can also be used. For the variance components we use the restricted log-likelihood, $l(\sigma^2, \sigma^2_u)$ and let $\hat{l}$ denote its maximized value, maximized under the constraint that the variance estimates are nonnegative. The profile likelihood confidence interval for σ^2_u is $C(\sigma^2_u) = \{c : -2P(c) \leq -2\hat{l} + z^2_{1-\alpha/2}\}$, where $P(c) = max_{\sigma^2} l(\sigma^2, c)$ is referred to as the profile-likelihood for σ^2_u. A profile likelihood confidence interval for σ^2 is obtained in a similar manner by swapping the roles of σ^2 and σ^2_u. This method could also be used to get an interval for μ, but it would utilize the full likelihood (which is a function of μ, σ^2 and σ^2_u) rather than the restricted likelihood.

Moment based methods.

The likelihood approach described above depends on the assumed normality of the estimated difference in log-abundances. While the assumption of normality of the X_i (and hence the D_i) is required for the expressions used for the finite rate of increase, the probability of extinction and other quantities, the measurement errors may or may not be normally distributed. If they are not normal then neither are the $\widehat{D}_i$'s. So, a moment based approach is helpful. We continue to assume the measurement error variance is constant.

For estimating the mean the naive estimator, now denoted $\widehat{\mu}_{mom}$, can be used. This is an unbiased estimator. Consistent moment-based estimators of the variance components are given by

$$\widehat{\sigma}^2_{u,mom} = \left[\widehat{\mu}^2_{mom}\sum_{i=1}^{q-1}\tau_i\tau_{i+1} - \sum_{i=1}^{q-1}\widehat{D}_i\widehat{D}_{i+1}\right]/(q-1), \tag{12.10}$$

and

$$\widehat{\sigma}^2_{mom} = \widehat{\sigma}^2_{naive} - \frac{2\widehat{\sigma}^2_{u,mom}}{q-1}\left[\left(\sum_{i=1}^{q}\tau_i^{-1}\right) - \frac{1}{t_q - t_0}\right]. \tag{12.11}$$

The standard error of $\widehat{\mu}_{mom}$ can be estimated using (12.8) with $\widehat{\sigma}^2_{mom}$ in place of $\widehat{\sigma}^2_{REML}$. Section 5.1 of Staudenmayer and Buonaccorsi (2006) provides expressions for the approximate variances of $\widehat{\sigma}^2_{mom}$ and $\widehat{\sigma}^2_{u,mom}$ as well as for the covariance among them and $\widehat{\mu}_{mom}$. While the analytical expressions for the true standard errors of the variance estimators do not depend on normality, obtaining estimated standard errors without assuming normality is general challenging, and fully robust methods need further development. One solution is to estimate the standard errors based on the normality assumption. The procedure then is only partly distribution free; the estimators and their properties do not depend on the normality assumption but the estimation of standard errors and associated confidence intervals do.

Once the standard errors for the moment estimators are obtained, Wald and chi-square confidence intervals can be obtained in the same manner as with the REML estimators of the preceding section, except that the Wald interval for μ will use a z rather than a t value. If we assumed normality but used the moment estimators for computational convenience, we can bootstrap them in the same manner as we did the REML estimators. Without normality of the measurement errors the bootstrap is not easily applied in a nonparametric manner. The difficulty is the need for a nonparametric estimate of the distribution of the measurement error. This was discussed in Section 10.2.3 but in a simpler setting than arises in the current problem.

Correcting using estimated measurement error variances.

The estimate of x_i may be accompanied by an estimated standard error, $\widehat{\sigma}_{ui}$. If this is the case we can use "pseudo approaches" in which the $\widehat{\sigma}^2_{ui}$ are substituted for the measurement error variances in (12.6), treated as known and then μ and σ^2 are estimated using either likelihood or moment techniques. If the observation error variance can be assumed to be constant, then instead of using the individual $\widehat{\sigma}^2_{ui}$, the pseudo estimators could be obtained after setting each σ^2_{ui} to the common $\widehat{\sigma}^2_u = \sum_{i=0}^q \widehat{\sigma}^2_{ui}/(q+1)$, which is simply the mean of the estimated measurement error variances. The pseudo approach has been employed in other dynamic models; see Williams et al. (2003)) and the next section.

The pseudo-REML estimators can be obtained directly with the appropriate implementation of some mixed model software, whether we use the individual measurement error variances or their average. The pseudo-MOM estimator of σ^2, which arises by a direct, explicit correction for the bias of the naive estimator, is

$$\widehat{\sigma}^2_{pmom} = \widehat{\sigma}^2_{naive} - \frac{1}{q-1}\left[\left(\sum_{i=1}^{q} \frac{\widehat{\sigma}^2_{u_{i-1}} + \widehat{\sigma}^2_{ui}}{\tau_i}\right) - \frac{\widehat{\sigma}^2_{u0} + \widehat{\sigma}^2_{uq}}{t_q - t_0}\right] \tag{12.12}$$

in general, and $\widehat{\sigma}^2_{naive} - 2\widehat{\sigma}^2_u(\sum_{i=1}^q \tau_i^{-1} - (t_q - t_0)^{-1})/(q-1)$ under the assumption of constant measurement error variance.

If we treat the estimated measurement error variances as known, then standard error and associated chi-square or Wald confidence intervals can be easily obtained for either the pseudo-REML estimators, based on output from mixed model software, or the pseudo-moment estimators. Under normality assumptions, one can also obtain profile-likelihood confidence intervals or use the bootstrap as described earlier with the obvious modifications when allowing unequal observation error variances. Again, direct implementation of these approaches assumes the measurement error variances are fixed and known.

If possible, we should account for the uncertainty due to estimating the measurement error variances. This enters into the standard error of pseudo-estimators of σ^2. If SE_K denotes the standard error obtained assuming the measurement error variances are known, the corrected standard error is $SE = (SE_K^2 + Q)^{1/2}$, where Q is an additional term based on the uncertainty in the estimated measurement error variances. Details on the calculation of Q and other aspects of using the pseudo-estimators can be found in Staudenmayer and Buonaccorsi (2006) and Buonaccorsi and Staudenmayer (2009).

Additional considerations.

As in many similar problems, negative estimates of the variance components may arise. In this case it is better to work with the profile confidence intervals, if applicable, and bypass the estimation problem directly. See Buonaccorsi and Staudenmayer (2009) for more discussion on this and illustrations of how the problem arises in assessing trends in California condor and Puerto Rico parrot populations.

Another problem of interest is how to get confidence intervals for functions of μ and σ^2, specifically for the trend $\lambda = e^{\mu+\sigma^2/2}$ or the probability of "extinction," taken to mean the population reaches a critical point x_c. The probability of extinction is $\pi = e^{-2\mu(x_c - x_0)/\sigma^2}$ if the starting value is x_0 and $\mu > 0$, and equals 1 if $\mu \leq 0$. Other probabilities about the population at specific future points in time may also be of interest. There are at least three approaches to estimating $\theta = g(\mu, \sigma^2)$ for some function g. These are i) bootstrapping; ii) using the delta method to get estimated standard errors and approximate normal based confidence intervals; iii) using simultaneous confidence intervals (or a confidence region) for μ and σ^2 and then using these to construct a confidence set for θ. In those cases where the bootstrap can be used (see earlier discussions), bootstrap standard errors and confidence intervals can be obtained for θ directly, but some care must be exercised. The bootstrap is not a panacea and special attention needs to be given to problems where one or more values of an estimate repeatedly occur in the bootstrap sample. This happens here when estimating the probability of extinction. The delta method is standard fare, but requires approximating and then estimating the variance of the estimator and relying on approximate normality (and unbiasedness) of the estimator. These can all be problematic with the short series common in ecological applications and this method can fail badly in estimating the probability of extinction.

The third approach, which can be referred to as a projection method, has not been used much in these types of problems, except for some work by Ludwig (1999). It starts with a joint two dimensional confidence region for μ and σ^2 denoted by $\mathbf{R}$ and computes a confidence set for θ via

$$C_\theta = \{g(\mu, \sigma^2), (\mu, \sigma^2) \in \mathbf{R}\}. \tag{12.13}$$

The confidence region for θ is simply the set of values that result by calculating the function of interest over all values of μ and σ^2 in $\mathbf{R}$. In most cases, the confidence set $C(\theta)$ is an interval obtained by finding the min and max of the function of interest over values in $\mathbf{R}$, and it is easy to get closed form expressions for estimating λ and π, in particular. An advantage of this projection approach is that we have a bound on the confidence coefficient. If the confidence coefficient of the region $\mathbf{R}$ is $\geq 1 - \alpha$ then the same is true for the confidence coefficient for $C(\theta)$. The easiest way to get $\mathbf{R}$ is as a rectangle obtained by using simultaneous confidence intervals for μ and σ^2 which, based on Bonferroni's inequality, are each constructed with a confidence level of $1 - \alpha/2$.

Simulations.

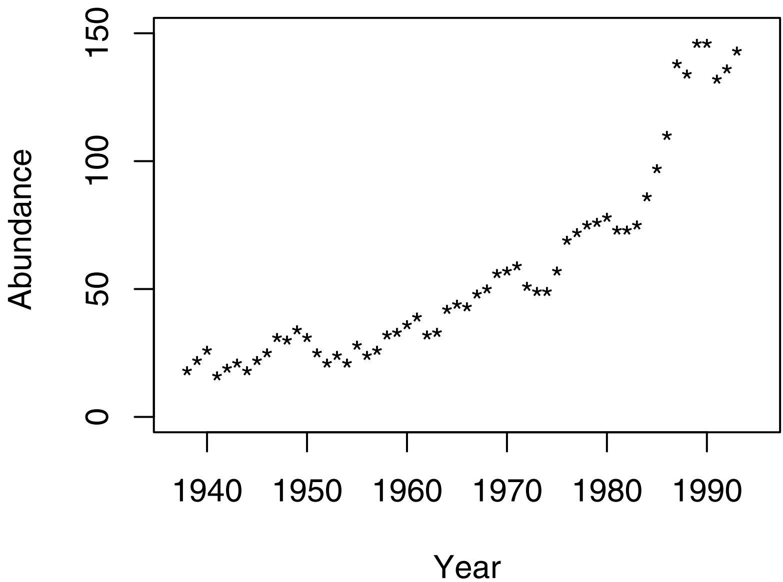
Buonaccorsi and Staudenmayer (2009) conducted a series of simulations to compare the performance of the various estimators and confidence interval across a variety of parameter combinations and measurement error models. This included both constant and unequal measurement error variances and the measurement errors being either uniform or normally distributed. Series of length 21 and 11 were used to reflect the reality of short ecological time series. They limited their investigation to moment and mixed model/REML based techniques that would be readily available to most applied ecologists (as opposed to profile likelihood and bootstrap methods). Their results document the fact that separating the dynamic model parameters from the measurement error can be a challenging problem, especially with short series and/or large process or measurement error variances. The estimators can be quite variable and in some cases many of the confidence intervals fail to achieve the desired coverage rate, sometimes by a large amount. This is not solely the fault of measurement error, but instead a common problem in complex models using relatively small sample sizes. A few general recommendations did emerge, however. One is that overall the moment based estimators and associated Wald type confidence interval appear to be the most robust and are preferred at this point. In the case where estimated measurement error variances are available, pseudo methods that incorporate them should generally be used, although they do not provide large gains in all cases. The pseudo methods assuming a constant measurement error variance are recommended. Interestingly they are fairly robust to the variances being unequal and are more stable than the pseudo estimators that use individual variances.

12.2.3 Example

To illustrate we present an analysis of the whooping crane data, used by Staples et al. (2004) and Buonaccorsi and Staudenmayer (2009) and shown in Figure 12.1. Table 12.1 provides estimates, standard errors and confidence intervals for the process mean and variance, μ and σ^2, the measurement error variance σ_u^2, the trend parameter λ and the probability of "extinction" π, using naive, likelihood and moment based procedures under the assumption of constant measurement error. Here extinction is taken to mean that population drops below 10.

- Since the naive estimator of μ is also a moment based estimator of μ, the naive and moment estimators of μ are the same. This estimator is unbiased for μ, but the naive standard error of the estimate of μ is not correct. The corrected standard error is the one attached to the moment estimator in the output.
- As expected, the corrected estimates of σ^2 (.0134 and .0137) are smaller than the naive estimate of .0193.
- The REML and moment estimators of the process variance are almost the same.

Figure 12.1 *Plot of estimated crane population versus year.*



- Based on simulation results, we tend to favor the moment based Wald confidence intervals, which for μ and σ^2 both are fairly similar to the profile likelihood intervals.
- The measurement error variance, estimated here to be approximately .0028, is of less interest than the other parameters.
- There are differences between the profile likelihood and bootstrap intervals, with the former being more similar to the moment-based Wald intervals than the latter. Both methods rely on the assumptions of normality and constant measurement error variance which may or may not hold. The profile likelihood intervals are also based on a likelihood ratio test statistic being distributed approximate chi-square, which is an asymptotic result, while the bootstrap intervals are computed using the percentile method, which is not guaranteed to always work. These are the key reasons for potential differences in the two techniques, and, of course, the data analysis itself cannot decide which method is better.

Table 12.1 *Estimation of whooping crane data.*

Method	μ	σ^2	σ_u^2
Naive (SE)	.0377 (.0187)	.0193 (.0037)	
MOM (SE)	.0377 (.0157)	.0134 (.0053)	.0029 (.0027)
REML (SE)	.0372 (.0158)	.0137 (.0055)	.0028 (.0028)
REML (BSE)	.0372 (.0176)	.0137 (.0040)	.0028 (.0013)
Naive-CI	(.0001, .0753)	(.0137, .0293)	
MOM-CI(W)	(.0069, .0684)	(.0030, .0238)	(.0000, .0082)
REML-CI(t)	(.0030, .0714)	(.0030, .0244)	(.0000, .0082)
REML-CI(B)	(.0022, .0714)	(.0067, .0231)	(.0000, .0043)
CI(Prof)	(.005, .069)	(.006, .027)	(.0000, .0094)

Method	λ	π
Naive (SE)	1.0484 (.0197)	.00002879
MOM (SE)	1.0454 (.0166)	.00000030
REML (SE)	1.0450 (.0168)	.00000048
REML (BSE)	1.0450 (.0184)	
Naive-CI(W)	(1.0098, 1.0872)	
Naive-CI(P-tC)	(1.0010, 1.1013)	
MOM-CI(W)	(1.0128, 1.0780)	
MOM-CI(P-WW))	(1.0033, 1.0893)	(.0000, .7209)
REML-CI(W)	(1.0121, 1.0780)	
REML-CI(P-tW)	(0.9978, 1.0945)	(.0000, .5892)
REML-CI(B)	(1.0097, 1.0816)	(.0000, 1.000)
CI(PP)	(1.0025, 1.0936)	(.0000, 1.000)

MOM = moment based. REML = restricted maximum likelihood. (SE) = standard error. (BSE) = bootstrap standard error. For confidence intervals: CI(t) = t based interval; CI(W) = Wald interval; CI(B) = bootstrap percentile interval; CI(Prof) = profile likelihood confidence interval. CI(P-ab) indicates a projection confidence interval based on interval a for μ and interval b interval for σ^2. CI(PP) indicates projection of the profile likelihood intervals.

- Confidence intervals for λ or π, based on the projection method, use either the Wald or profile likelihood based intervals for σ^2. All of the confidence are 95%. This means the the projection intervals for λ and π are not computed directly from the intervals in the top part of the table. Rather they are calculated from 97.5% confidence intervals for μ and σ^2 (based on the use of Bonferroni's inequality). These projection intervals are more conservative than the Wald intervals.

The confidence intervals for λ are not very dependent on the method used, ranging generally from about 1 to about 1.09. There is not much information about the probability of eventual extinction (taken here to be the population dropping below 10) as evidence by the confidence intervals. Based on the projection methods these are (0,1) and (0,.721). As noted in the general discussion, confidence intervals for π are best obtained by the projection method. The bootstrap percentile interval is questionable because of a very discrete empirical distribution.

12.3 Linear autoregressive models

This section examines measurement error in linear autoregressive models

$$X_t = \phi_0 + \phi_1 X_{t-1} + \ldots + \phi_p X_{t-p} + \epsilon_t, \tag{12.14}$$

where the ϵ_t are i.i.d. with mean 0 and variance σ^2 and $\phi_1, \ldots, \phi_p$ are such that the process X_t is stationary (e.g., Box, Jenkins, and Reinsel, 1994, Chapter 3). The stationarity assumption can be viewed as the series having a type of equilibrium. The intercept term ϕ_0 is 0 if the X_t have been centered to have mean 0. This model in (12.14) is referred to as a model of order p. Terms involving other predictors could be added. For example, Williams et al. (2003) consider an AR(1) model but with additional terms involving quadratic effects in time. We do not treat that extension explicitly, but do provide an example at the end of the section.

Autoregressive models are a popular choice in many disciplines for the modeling of time series. This is especially true in population dynamics where AR(1) or AR(2) models are often employed, e.g., Williams and Liebhold (1995) and Dennis and Taper (1994). With N_t denoting abundance at time t and $X_t = log(N_t)$, the AR(1) model is also referred to as the Gompertz model. Among others, Stenseth et al. (2003), Ives et al. (2003) and Dennis et al. (2006) consider certain aspects of measurement error in AR models in the population dynamic context.

The collection of realized values is denoted by $\mathbf{x} = (x_1, \ldots, x_T)'$. The assumption of stationary means that the covariance between observations k time points apart,

$$\gamma_k = cov(X_t, X_{t+k}),$$

is free of t. Note that $\gamma_0 = V(X_t)$. The autoregressive coefficients can be expressed in terms of these quantities as $\boldsymbol{\phi} = \boldsymbol{\Gamma}^{-1}\boldsymbol{\gamma}$, where

$$\boldsymbol{\Gamma} = \begin{pmatrix} \gamma_0 & \cdots & \gamma_{p-1} \\ \vdots & \ddots & \vdots \\ \gamma_{p-1} & \cdots & \gamma_0 \end{pmatrix} \text{ and } \boldsymbol{\gamma} = \begin{pmatrix} \gamma_1 \\ \cdot \\ \cdot \\ \gamma_p \end{pmatrix}.$$

Similar to the preceding section, given the realized values

$$W_t|\mathbf{x} = x_t + u_t,$$

but now the error is additive so $E(u_t|\mathbf{x}) = 0$. In addition $V(u_t|\mathbf{x}) = \sigma^2_{ut}$ and $cov(u_t, u_s|\mathbf{x})$ is assumed to be 0 for $t \neq s$, i.e., the measurement errors are conditionally uncorrelated. As in the earlier discussion in this chapter the conditional measurement error variance σ^2_{ut} can change over time. Unconditionally

$$V(W_t) = \gamma_0 + E(\sigma^2_{ut}) \text{ and } cov(W_t, W_{t+k}) = \gamma_k. \tag{12.15}$$

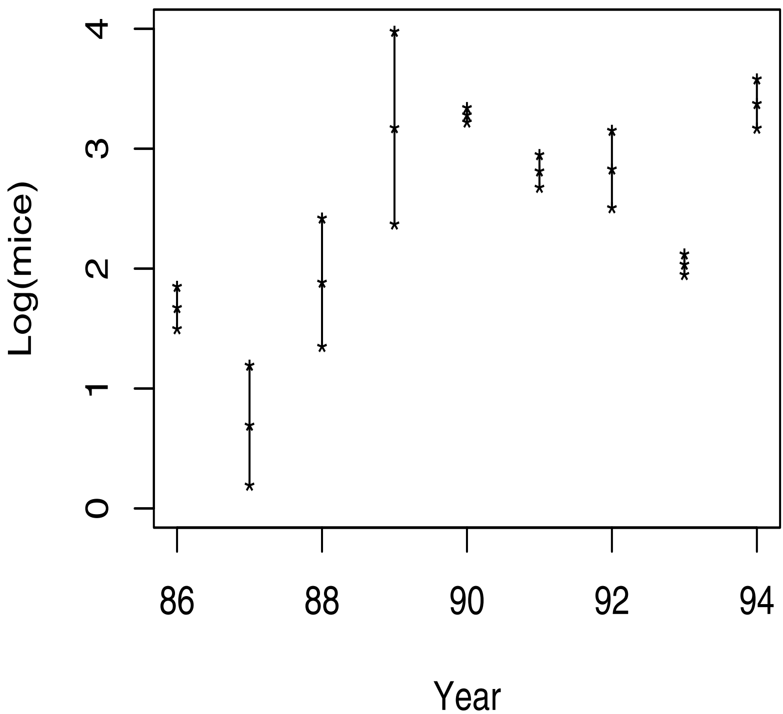
(Note: The use of an expectation in $E(\sigma^2_{ut})$ above allows for the fact that the conditional measurement error variance at time t may depend on x_t. If it does, then unconditionally it becomes random as a function of X_t, even though this is not explicit in the notation. See also the discussion in Section 6.4.5. As elsewhere we just write σ^2_{ut}.)

For a motivating example, Figure 12.2 presents estimated mouse densities (mice per hectare on the log scale) with associated standard errors for a nine year period on a stand in the Quabbin Reservoir, located in Western Massachusetts. This is one part of the data used in Elkinton et al. (1996). The sampling effort consisted of two subplots in 1987, 89 and 90 and three subplots in the other years. The density in year t was estimated using the mean of m_t subplots, denoted by A_t and the estimated measurement error variance of the estimated density in year t is S^2_t/m_t where S^2_t is the among subplot variance and m_t is the number of subplots. On the log scale the estimated log-abundance is $W_t = log(A_t)$ with an approximate variance (based on the delta method) of $\widehat{\sigma}^2_{ut} = S^2_t/(m_t A^2_t)$. The biases from the use of logs appear negligible here (see also the discussion in Section 6.4.7). Large differences in the standard errors over years are evident, due to a combination of changes in sampling effort, changes in the population being sampled and noise in estimating the standard errors.

12.3.1 *Properties of naive estimators*

Let $\gamma_k = cov(X_t, X_{t+k})$ denote the lag k autocovariance in the true series and $\widehat{\gamma}_k = \sum_{t=1}^{T-k}(W_t - \bar{W})(W_{t+k} - \bar{W})/T$. We could modify the divisor in $\widehat{\gamma}_k$, to

Figure 12.2 *Quabbin mouse data. Estimated log(densities) as a function of year, plus and minus one standard error.*



be $T - k$, but following Brockwell and Davis (1996) we use T throughout to ensure the overall estimated covariance matrix is nonnegative.

There are a variety of estimation methods employed without measurement including, among others, maximum likelihood, conditional least-squares and Yule-Walker (YW) estimates. These are equivalent asymptotically so the properties can be assessed through the naive Yule-Walker estimators. These are given by

$$\widehat{\boldsymbol{\phi}}_{naive} = \widehat{\boldsymbol{\Gamma}}^{-1}\widehat{\boldsymbol{\gamma}}. \tag{12.16}$$

When $p = 1$, $\widehat{\phi}_{naive} = \widehat{\gamma}_1/\widehat{\gamma}_0$.

Defining $\sigma_u^2 = Lim_{T\to\infty} \sum_t \sigma_{ut}^2/T$, then under some regularity conditions to ensure the convergence of sample autocovariances, $\widehat{\gamma}_k$ converges to γ_k for $k \geq 1$. However, $\widehat{\gamma}_0$ converges to $\gamma_0 + \sigma_u^2$ so the naive estimator converges in probability to

$$(\boldsymbol{\Gamma} + \sigma_u^2 \mathbf{I}_p)^{-1}\boldsymbol{\Gamma}\boldsymbol{\phi}.$$

When $p = 1$, the naive estimator is consistent for

$$\frac{\gamma_0}{\gamma_0 + \sigma_u^2}\phi_1 = \kappa\phi_1,$$

where $\kappa = \gamma_0/(\gamma_0 + \sigma_u^2)$. Somewhat surprisingly, this is the same type of attenuation that resulted in simple linear regression (under certain conditions) as seen in Section 4.4.

Similar to the case with multiple linear regression (Chapter 5), for $p > 1$, the asymptotic biases become more complicated. For instance, with $p = 2$ the naive estimators of ϕ_1 and ϕ_2 are consistent for

$$\frac{1}{1 - \kappa^2\rho_1^2}\begin{pmatrix} \kappa\rho_1 - \kappa^2\rho_1\rho_2 \\ \kappa\rho_2 - \kappa^2\rho_1^2 \end{pmatrix},$$

where $\rho_k = \gamma_k/\gamma_0$ is the lag k correlation in the true series, $\rho_1 = \phi_1/(1-\phi_2)$ and $\rho_2 = (\phi_1^2 + \phi_2 - \phi_2^2)/(1 - \phi_2)$. The asymptotic bias in either element can be either attenuating (smaller in absolute value) or accentuating (larger in absolute value), and the bias depends on both the amount of measurement error and the true values for ϕ_1 and ϕ_2.

The asymptotic bias of the naive estimator of σ^2 is, in general,

$$\sigma_u^2 + \boldsymbol{\gamma}'\left\{\boldsymbol{\Gamma}^{-1} - (\boldsymbol{\Gamma} + \sigma_u^2\mathbf{I})^{-1}\right\}\boldsymbol{\gamma} > 0. \tag{12.17}$$

12.3.2 Correcting for measurement error

Using no information about the measurement error variance.

It is possible to estimate the autoregressive parameters without estimates of the measurement error variance(s). Not surprisingly, however, for shorter series, it is usually difficult to do this efficiently. If the measurement error variance is constant, ARMA and likelihood based methods are available, while a modified Yule-Walker method can be considered with unequal variances.

The **ARMA method** is based on the fact (see, for example, Box et al. (1994), Section A4.3 or Pagano (1974)) that if X_t is an AR(p) process with autoregressive parameters $\boldsymbol{\phi}$ and the u_t are i.i.d. with mean 0 and constant variance σ_u^2, then $W_t = X_t + u_t$ follows an $ARMA(p, p)$ process where the autoregressive parameters in the ARMA model are the same as in the original autoregressive model. As a result, a simple (but not necessarily efficient) approach to estimating $\boldsymbol{\phi}$ is to simply fit an ARMA(p, p) to the observed $\mathbf{w}$ series and to use the resulting estimate of the autoregressive parameters. Call this $\widehat{\boldsymbol{\phi}}_{ARMA}$. This is part of the hybrid approach in Wong and Miller (1990). The asymptotic properties of $\widehat{\boldsymbol{\phi}}_{ARMA}$ are well known; see, for example, Chapter 8 in Brockwell and Davis (1996).

There are restrictions on the moving average parameters in the ARMA model based on how it arises from the original AR model plus measurement error. Some authors, such as Pagano (1974) and Miazaki and Dorea (1993), obtain more efficient estimators by enforcing restrictions on the moving average parameters.

ML method. One can also take a maximum likelihood approach under normality. In essence, this is what correcting for the restrictions in the ARMA approach is trying to do. Lee and Shin (1997) address this in a more general context than being treated here (allowing ARMA models for both the true values and the measurement error). They develop a computational approach based on an approximation to the log-likelihood. (Note: their use of the expression "restricted maximum likelihood" refers to the accompanying restrictions on the parameters, not the way this expression has come to be used when talking about REML estimators.) Dennis et al. (2006) provide a comprehensive discussion on obtaining ML estimators in the AR(1) setting (what they refer to as the Gompertz model), provide some examples and discuss some of the computation challenges. This includes the possibility of multiple modes in the likelihood function. They also show that the ML's can be fit easily with the appropriate implementation of proc MIXED in SAS.

Modified Yule-Walker. This approach also assumes uncorrelated measurement errors and requires no knowledge of the measurement error variances. It allows the measurement error variances to be changing. It exploits the fact that for $k \geq 1$ the lag k sample covariance of the observed W's estimate the true lag k covariances in the true series; see (12.15). For a fixed $j > p$ define $\widehat{\boldsymbol{\gamma}}_{(j)} = (\widehat{\gamma}_j, \ldots, \widehat{\gamma}_{j+p-1})'$ and

$$\widehat{\boldsymbol{\Gamma}}_{(j)} = \begin{pmatrix} \widehat{\gamma}_{j-1} & \cdots & \widehat{\gamma}_{j-p} \\ \vdots & \ddots & \vdots \\ \widehat{\gamma}_{j+p-2} & \cdots & \widehat{\gamma}_{j-1} \end{pmatrix}.$$

There are multiple modified Yule-Walker estimators, corresponding to different j, although using $j = 2$ (as we do in examples and simulations) is the most obvious choice.

For the AR(1) model, $\widehat{\phi}_{MYW_j} = \widehat{\gamma}_j / \widehat{\gamma}_{j-1}$, for $j \geq 2$. These estimators are consistent. They were first discussed in detail by Walker (1960) (his Method B) and subsequently by Sakai et al. (1979) and Sakai and Arase (1979), who provide the asymptotic covariance matrix of $\widehat{\boldsymbol{\phi}}_{MWY_j}$ for general AR models. Chanda (1996) proposed a related estimator. While the modified Yule-Walker estimators have the advantage of being easy to calculate and robust to unequal measurement error variances they can perform badly, particularly for short series.

Other methods. There are other approaches that can be taken without estimates of the measurement error variance(s), as long as original parameters are identifiable. For example, in modeling vole populations Stenseth et al. (2003) used an autoregressive model of order 2 for the true log(abundance), and used a Poisson model for the number of voles captured. This plays the role of the measurement error model. Under these assumptions they carry out inferences using a Bayesian approach.

Correction methods using estimated measurement error variances.

These methods assume the availability of an estimated measurement error variance $\widehat{\sigma}^2_{ut}$ at each time point.

Moment approach. The problem with the naive estimators arises from the bias in the estimator of $\boldsymbol{\gamma}_0$. This suggests the corrected estimator

$$\widehat{\boldsymbol{\phi}}_{CEE} = (\widehat{\boldsymbol{\Gamma}} - \widehat{\sigma}^2_u \mathbf{I}_p)^{-1}\widehat{\boldsymbol{\gamma}} \tag{12.18}$$

where $\widehat{\sigma}^2_u = \sum_{t=1}^{T} \widehat{\sigma}^2_{ut}/T$. When $p = 1$, $\widehat{\phi}_1 = \widehat{\gamma}_1/(\widehat{\gamma}_0 - \widehat{\sigma}^2_u)$. This is a moment estimator which comes from correcting the naive Yule-Walker estimating equations (hence the CEE). Other estimators would result from modifying the naive estimating equations associated with other methods (e.g., maximum likelihood or conditional least squares). This corrected estimator is trivial to compute. Like moment estimators in linear regression $\widehat{\phi}_{CEE}$ can have problems for small samples. One reason is the distribution of the corrected estimator of γ_0, $\widehat{\gamma}_c = \widehat{\gamma}_0 - \widehat{\sigma}^2_u$ can have positive probability around zero so the estimate of γ_0 may be negative. In this case some modification is needed.

Detailed expressions for the asymptotic covariance of $\widehat{\boldsymbol{\phi}}_{CEE}$, which are complicated, are given in Staudenmayer and Buonaccorsi (2005). They show that under certain conditions that $\widehat{\boldsymbol{\phi}}_{CEE}$ is consistent for $\boldsymbol{\phi}$ and asymptotically normal, provide expressions for the asymptotic covariance matrix and discuss ways to estimate it. These results do not require the measurement error or the unobserved time series to have a normal distribution, but the asymptotic variances do have slightly simpler forms when that is the case. The biggest challenge is in how to obtain an estimated standard error if we account for the uncertainty in the estimated measurement error variances. Consider the AR(1) model. Assuming the estimated measurement error variances are independent of the W_t's, the approximate variance of $\widehat{\phi}_1$ is

$$\left[\frac{(1-\phi_1^2)(2-\kappa)}{\kappa} + \frac{(1-\phi_1^2)\overline{\sigma_u^4} + \phi_1^2\overline{E(u^4)}}{\gamma_0^2}\right] + \frac{\phi_1^2 V(\widehat{\sigma}_u^2)}{\gamma_0^2}, \qquad (12.19)$$

where $\overline{\mathrm{E}(u^4)} = \sum_t E(u_t^4)/T$ and $\overline{\sigma_u^4} = \sum_t \sigma_{ut}^4/T$. Recall that $\widehat{\sigma}_u^2 = \sum_{t-1}^T \widehat{\sigma}_{ut}^2/T$. (If the $\widehat{\sigma}_u^2$ and W_t are not independent, additional terms involving the covariance of $\widehat{\sigma}_u^2$ with $\widehat{\gamma}_0$ and $\widehat{\gamma}_1$ are needed.) Under normality $E(u_t^4) = 3\sigma_{ut}^4$, which can be estimated using $3\widehat{\sigma}_{ut}^4$. Otherwise $E(u_t^4)$ must be estimated in some manner from the data at time t. If we treat $\widehat{\sigma}_u^2$ as fixed then the last term in (12.19) can be ignored. If not, estimating $V(\widehat{\sigma}_u^2)$ can be hard to do, especially if we allow heteroscedasticity without modeling exactly how it changes over time. Staudenmayer and Buonaccorsi (2005) have further discussion and this is also touched on further in the example below.

Pseudo-likelihood methods. If $\mathbf{X}$ and $\mathbf{u}$ are both normal and independent of each other then $\mathbf{W} = \mathbf{X} + \mathbf{u}$ is normal with covariance matrix $\mathbf{\Sigma}_X + \mathbf{\Sigma}_u$ and maximum likelihood (ML) techniques can be employed. Notice that $\mathbf{\Sigma}_X$ will reflect the autoregressive structure of the model for the true values. Our interest is in cases where an estimate of $\mathbf{\Sigma}_u$ is available. Treating these as fixed and maximizing the Gaussian likelihood over the rest of the parameters leads to a pseudo-ML estimation. Bell and Hilmer (1990) and Bell and Wilcox (1993) took this approach when both X_t and u_t are assumed to follow ARIMA models. Wong, Miller and Shrestha (2001) used this approach for ARMA models and employed a state space model for computational purposes. These papers illustrate how to obtain the covariance of $\widehat{\boldsymbol{\phi}}$ when $\mathbf{\Sigma}_u$ is treated as known. Obtaining the covariance of $\widehat{\boldsymbol{\phi}}$ when accounting for the uncertainty of the estimated measurement error variances is more difficult. With constant variance and the use of $\widehat{\sigma}_u^2$, Staudenmayer and Buonaccorsi (2005) provide an expression for the approximate covariance of the pseudo-ML estimator, which involves the variance of $\widehat{\sigma}_u^2$. As with the CEE estimator, accounting for the uncertainty in the estimated measurement error variances when they change in an unspecified fashion is a difficult problem that is a topic of current research.

There are other issues when the measurement error is heteroscedastic. For one, if the measurement error variances are treated as unknown parameters without additional structure (as in Wong et al., 2001), then the number of parameters increases with the sample size. In this case, the asymptotic properties of estimators in the time series setting have not been fully explored and the analysis is certainly nontrivial. An alternative is to model the measurement error variance parametrically, maybe as a function of sample effort and/or x_t and then try a pseudo or full likelihood approach. An additional problem is that with heteroscedasticity, the marginal distribution of $\mathbf{W}$ is not normal even if $\mathbf{X}$ and $\mathbf{u}$ are normally distributed.

SIMEX is applied in a straightforward manner by generating pseudo-values at each time point. If there are replicates, then the empirical SIMEX could be used.

The issue of unequal measurement error variances also raises problems in how to bootstrap here. The bootstrapping certainly has to be dynamic. That part is not a problem. However, if the measurement error variance changes over time, a model is needed for how they do so.

Given the complexity of the problem and the fact that methodology is still emerging, it is difficult to provide definitive statements about the properties of the various methods. This is especially true when the measurement error variances are changing. Stuadenmayer and Buonaccorsi (2005) provide some initial assessments of the asymptotic efficiency of the modified Yule-Walker, ARMA, CEE and PML estimators and some simulations comparing the performance of the CEE, PML and ARMA estimators over samples of size 20, 50 or 200 and various measurement error models. The modified Yule-Walker estimators had very unstable behavior. Broadly, if estimated measurement error variances are available, the PML and CEE estimators, which perform similarly, are preferred. Without these estimates the ARMA estimate is certainly preferred to the modified Yule-Walker approach, but the ARMA does require the assumption of equal measurement error variances. The ML estimators were not examined by them. They will be more efficient than the ARMA estimators.

12.3.3 Examples

Mouse Example.

We return to the mouse example, introduced earlier. See Figure 12.2. This is a very short series, although not atypical for ecological applications. Even without measurement error estimation of the parameters may be difficult.

Table 12.2 shows results from a variety of methods. The ARMA and modified Yule-Walker estimates, both of which do not use the estimated measurement error variances, are rather strange. The former is based on the assumption of constant variance, which is questionable here. In addition, the ARMA approach lacks efficiency since it does not bring in the restrictions associated with the moving average parameters. This may be particularly problematic with small sample sizes. The erratic behavior of the MYW was noted elsewhere. The ML estimate is also computed under the assumption of constant variance without using the estimated measurement error variances. This was computed using proc mixed. In this case, the estimated measurement error variance was 0 (which we know to not be true) and the estimate of ϕ_1 is just the naive ML estimate. The difference between the results for the Naive-ML and ML is simply due to the first being fit using a time series routine (proc ARIMA in SAS) while the latter was calculated via proc MIXED.

The discussion of the estimators above was for illustration and comparison, since they do not make use of the measurement error variances. The last four estimates in the table do. For the CEE estimate the standard error was computed two ways. The first, yielding the .4055, used (12.19) setting $V(\widehat{\sigma}_u^2) = 0$ and estimating $E(u_t^4)$ with $3\widehat{\sigma}_{ut}^4$. This assumes normal measurement errors (on the log-scale). The second standard error, .4212, adds in an estimate of $V(\widehat{\sigma}_u^2) = \sum_t V(\widehat{\sigma}_{ut}^2)/T^2$. Recall that $W_t = log(A_t)$ where A_t was the mean mouse density over m_t subplots and the estimate of the variance of W_t is $\widehat{\sigma}_{ut}^2 = S_t^2/(m_t W_t^2)$. Assuming normal replicates, so S_t^2 is distributed proportional to a chi-square with $m_t - 1$ degrees of freedom, and ignoring the uncertainty in W_t^2, this calculation estimates the variance of $\widehat{\sigma}_{ut}^2$ roughly using $2S_t^4/(m_t^2 W_t^4(m_t - 1)$. This is certainly a bit crude but does show that the contribution from this additional term is small relative to the other component.

The first pseudo-ML estimate used the individual measurement error variances. This actually results in a smaller estimate of ϕ_1 than the naive estimates. The second pseudo estimator, PML-c, assumes a constant coefficient of variation, CV, over years, for the replicate measures of density over subplots. In this case the estimated measurement error variance for year t (on the log-scale) was taken to be $\widehat{\sigma}_{ut}^2 = \bar{CV}/m_t$, where $\bar{CV} = \sum_t (S_t^2/W_t^2)/t$. This method is somewhat in the spirit of using the average estimated measurement error variance, even when the variances appear unequal (which was found in some simulations to be better than using the individual estimated variances). Here though, we have used the knowledge about the sampling effort in each year. The results for PML-c are more in line with those from the CEE and SIMEX. Both PML fits were obtained using implementations of proc mixed, with standard errors obtained treating the estimated measurement error variances as known. As noted earlier, obtaining additional terms to account for the extra uncertainty in the PML estimators is hard when the measurement error variances are changing. In addition, as also noted earlier, further modeling is needed about the measurement error variances and how estimates of them arise, in order to apply the bootstrap.

Childhood Respiratory Disease Study.

This example illustrates the use of an AR(1) model with addition predictors. The analysis is from Schmid et al. (1994), based on the childhood respiratory disease study. This was a longitudinal study of child risk factors for pulmonary disease. The mismeasured variables include the outcome y = log(FVC), where

Table 12.2 *Estimation of ϕ_1 for mouse data based on AR(1) model. CLS = conditional least squares; MYW = modified Yule-Walker; CEE = corrected moment estimator; PML = pseudo maximum likelihood. ML = maximum likelihood.*

Method	Estimate	SE
Naive-ML	.4216	.3697
Naive-CLS	.4375	.3764
Naive-YW	.3731	
ARMA-CLS	.1278	.9426
ARMA-ML	.1313	1.0199
Modified YW	-1.939	
ML	.4215	.3139
CEE	.4741	.4055 (.4212)
SIMEX	.4609	
PML	.3952	.3486
PML-c	.4566	.3644

FVC is forced vital capacity and x_1 = log(% change in FEV + 1), where FEV is forced expiratory volume. Both are estimated using the mean of two replicates. (Note: As discussed in numerous places elsewhere the use of the replicates of logs is formally fitting a model involving the expected value of the log rather than the log of the expected value. As discussed in Section 6.4.7, however, this often suffices for fitting the model in terms of the latter.)

The perfectly measured predictors, contained in $\mathbf{x}_2$, include age, gender and height. For the ith child in time period j the assumed model is

$$y_{ij} = \beta_0 + \phi_1 y_{i,j-1} + \mathbf{x}'_{2ij}\boldsymbol{\beta}_2 + \beta_1 x_{1ij} + \epsilon_{ij}.$$

Here, the longitudinal aspect is captured through a dynamic term for the FVC variable rather than through random effects as in the previous chapter. Normality is assumed throughout, with a constant measurement error variance. The results below show naive and corrected estimators with associated standard errors, the latter computed using maximum likelihood under normality. Not surprisingly, the biggest change is in the coefficient associated with the mismeasured predictor, log(FEV), where the naive estimate of .263 increases rather dramatically to .680. Changes in the other estimates are fairly modest.

Table 12.3 *Childhood respiratory disease example from Schmid et al. (1994). Used with permission of Elsevier Publishing.*

Parameter	Naive(SE)	Corrected (SE)
Intercept	1.263 (.009)	1.245 (.012)
γ (lag FEV)	.729 (.040)	.745 (.049)
Gender	-.05 (.0112)	-.05 (.0110)
Age	.0018 (.0026)	-.0023 (.0027)
Height	.824 (.141)	.743 (.158)
log(FEV) (x)	.263 (.106)	.680 (.212)
Residual variance	.0044	.0025

CHAPTER 13

Background Material

13.1 Notation for vectors, covariance matrices, etc.

Column vectors are denoted by boldface while row vectors will always use the transpose notation, e.g., $\mathbf{x}$ is a column vector and $\mathbf{x}'$ a row vector.

A random vector, $\mathbf{X}$, is a $p \times 1$ column vector whose elements are random variables $X_1, \ldots, X_p$.

The **covariance** of two random variables, say X_1 and X_2, is written $cov(X_1, X_2)$. This is a scalar.

If $E(X_i) = \mu_i$, $V(X_i) = \sigma_i^2$ and $cov(X_i, X_j) = \sigma_{ij}$, then the mean vector is defined by $E(\mathbf{X}) = \boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)$ and the variance-covariance or dispersion matrix of $\mathbf{X}$ is written $Cov(\mathbf{X}) = \boldsymbol{\Sigma}$, where

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & . & \sigma_{1p} \\ \sigma_{21} & \sigma_2^2 & . & \sigma_{2p} \\ . & . & . & . \\ \sigma_{p1} & \sigma_{p2} & . & \sigma_p^2 \end{bmatrix}.$$

13.2 Double expectations

The use of double expectations is a valuable modeling tool. It provides a convenient way to get unconditional means, variances and covariances of random variables by building things up in a hierarchical manner using conditional properties. In the measurement error setting the conditional model is the measurement error model given the true values. See Casella and Berger (2002) for discussion of the results below.

Let $\mathbf{X}$, $\mathbf{Y}$ and $\mathbf{W}$ be random vectors.

- $E(\mathbf{Y}) = E[E(\mathbf{Y}|\mathbf{W})]$. In words, the expected value of $\mathbf{Y}$ is the expected value of the conditional expected value.

- $Cov(\mathbf{Y}, \mathbf{X}) = E[Cov(\mathbf{Y}, \mathbf{X}|\mathbf{W})] + Cov(E(\mathbf{Y}|\mathbf{W}), E(\mathbf{X}|\mathbf{W})$. In words, the unconditional covariance is the expected value of the conditional covariance plus the covariance of the conditional expected value.
  With $\mathbf{Y} = \mathbf{X}$, the above specializes to
- $Cov(\mathbf{Y}|\mathbf{W}) = E[Cov(\mathbf{Y}|\mathbf{W})] + Cov(E(\mathbf{Y}|\mathbf{W})$.
  In the case of univariate random variables this becomes

$$V(Y|W) = E[V(Y|W)] + V(E(Y|W)).$$

13.3 Approximate Wald inferences

Wald-based methods make use of the approximate normality of an estimator, along with an estimate of its variance (or variance-covariance matrix if multivariate) to form approximate confidence intervals and tests of hypotheses.

Suppose $\boldsymbol{\theta}$ is a vector of parameters estimated by $\widehat{\boldsymbol{\theta}}$, which is approximately normal with mean $\boldsymbol{\theta}$ and covariance $\boldsymbol{\Sigma}_\theta$, estimated by $\widehat{\boldsymbol{\Sigma}}_\theta$. The approximate Wald confidence interval for $\eta = \mathbf{c}'\boldsymbol{\theta}$ is given by

$$\mathbf{c}'\widehat{\boldsymbol{\theta}} \pm z(1 - \alpha/2)(\mathbf{c}'\widehat{\boldsymbol{\Sigma}}_\theta\mathbf{c})^{1/2}.$$

If η is the jth component of $\boldsymbol{\theta}$ then this is simply $\widehat{\theta}_j \pm z(1 - \alpha/2)SE(\widehat{\theta}_j)$, where $SE(\widehat{\theta}_j)$, the square root of the $(j, j)th$ element of of $\widehat{\boldsymbol{\Sigma}}_\theta$, is the estimated standard error of $\widehat{\theta}_j$.

An approximate test of size α of $H_0 : \theta_j = b$ versus $H_A : \theta_j \neq b$, (b a specified constant) is given by treating $Z = (\widehat{\theta}_j - b)/SE(\widehat{\theta}_j)$ as distributed approximately $N(0, 1)$ under H_0. H_0 if $|Z| > z_{\alpha/2}$. Equivalently, a test of size α rejects H_0 if the confidence interval of level $1 - \alpha$ does not contain the constant b.

A **general Linear test** tests $H_0 : \mathbf{H}\boldsymbol{\theta} = \mathbf{h}$, versus $H_0 : \mathbf{H}\boldsymbol{\theta} \neq \mathbf{h}$, where $\mathbf{H}$ is a $q \times p$ matrix of rank q and $\mathbf{h}$ is a $q \times 1$ vector of constants. The Wald test uses the test statistic

$$C = (\mathbf{H}\widehat{\boldsymbol{\theta}} - \mathbf{h})'(\mathbf{H}\widehat{\boldsymbol{\Sigma}}_\theta\mathbf{H}')^{-1}(\mathbf{H}\widehat{\boldsymbol{\theta}} - \mathbf{h})$$

which under H_0 is treated as approximately chi-square with q degrees of freedom.

13.4 The delta-method: Approximate moments of nonlinear functions

The so-called delta method provides a method for approximating the mean and variance of nonlinear functions of random quantities. Consider random variables $W_1, \ldots, W_K$, with $E(W_k) = \mu_k$, $V(W_k) = \sigma_k^2$ and $cov(W_k, W_j) = \sigma_{kj}$. Write Σ_W for the variance covariance matrix and $\boldsymbol{\mu}$ for the mean vector; so σ_{jj} and σ_j^2 are the same thing.

Let

$$T = g(W_1, \ldots, W_K) = g(\mathbf{W})$$

define a new random variable. Unless g is linear we cannot get the mean and variance of T from just the given moments of the W's. Using a second order Taylor Series expansion of g around $\mu_1, \ldots, \mu_K$ leads to

$$E(T) \approx g(\boldsymbol{\mu}) + \sum_j e_j \sigma_j^2/2 + \sum_j \sum_{k>j} e_{jk}\sigma_{jk},$$

where $e_j = \partial^2 g(\boldsymbol{\mu})/\partial^2 mu_j$ and $e_{jk} = \partial^2 g(\boldsymbol{\mu})/\partial\mu_j$. This provides an approximate bias for $g(\mathbf{W})$ as an estimator of $g(\boldsymbol{\mu})$.

Also,

$$V(T) \approx \mathbf{d}'\Sigma_W \mathbf{d} = \sum_j d_j^2 \sigma_j^2 + 2\sum_j \sum_{k>j} d_j d_k \sigma_{jk}$$

where $\mathbf{d}' = (d_1, \ldots d_K)$ and $d_j = \partial g(\boldsymbol{\mu})/\partial\mu_j$

13.5 Fieller's method for ratios

Fieller's method provides a way to obtain a confidence set for a ratio of parameters. See Read (1983), Buonaccorsi (1998) and references therein for details and discussion.

Let Z_1 and Z_2 be two random variables with $E(Z_1) = \theta_1$ and $E(Z_2) = \theta_2$ and define $\rho = \theta_1/\theta_2$. Further let $\hat{V}_1$, $\hat{V}_2$ and $\widehat{V}_{12}$ be estimates of $V(Z_1)$ and $V(Z_2)$ and $cov(Z_1, Z_2)$, respectively. Motivated by the fact that $(Z_1 - \rho Z_2)/(V_1 - 2\rho V_{12} + \rho^2 V_2)^{1/2}$ is approximately standard normal leads to a confidence region for ρ. If $f_2 > 0$, then approximate Fieller confidence interval is given by

$$\frac{f_{12}}{f_2} \pm \frac{D^{1/2}}{f_2} \tag{13.1}$$

where $f_1 = Z_1^2 - z^2\widehat{V}_1$, $f_2 = Z_2^2 - z^2\widehat{V}_2$, $f_{12} = Z_1 Z_2 - z^2\widehat{V}_{12}$ and $D = f_{12}^2 - f_1 f_2$.

If $f_2 < 0$ the Fieller method yields an infinite region, rather than a finite interval; either the complement of a finite interval when $D \geq 0$, or the whole real line if $D < 0$. The case with $f_2 > 0$ and $D < 0$ cannot occur here.

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